

cardona-qft-methods: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
cardona-qft-methods_F00001	8	■ 1.000	A d S / C F T	AdS/CFT	then discusses the AdS/CFT duality and its relation to integrability in a
cardona-qft-methods_F00002	8	■ 1.000	$\mathcal{N}=4$	$\mathcal{N} = 4$	$\mathcal{N} = 4$ Supersymmetric Yang-Mills Theory, after which he describes some
cardona-qft-methods_F00003	10	■ 1.000	G	G	Index Theory for Non-compact G -manifolds
cardona-qft-methods_F00004	14	■ 1.000	A	A	tells us that any selfadjoint operator A can be seen as an integral over its
cardona-qft-methods_F00005	14	■ 0.992	$A=\int_a \in \operatorname{Sp}(a)} a \, d P_a$	$A = \int_{a \in \operatorname{Sp}(a)} adP_a$	spectral measure $A = \int_{a \in \operatorname{Sp}(a)} a \, dP_a$ if $\operatorname{Sp}(A)$ is the spectrum of A . This
cardona-qft-methods_F00006	14	■ 0.992	$\operatorname{Sp}(A)$	$\operatorname{Sp}(A)$	spectral measure $A = \int_{a \in \operatorname{Sp}(a)} a \, dP_a$ if $\operatorname{Sp}(A)$ is the spectrum of A . This
cardona-qft-methods_F00007	15	■ 0.523	$\psi(\vec{x})$	$\psi(\vec{x})$	sider fields $\psi(\vec{x})$ for $\vec{x} \in \mathbb{R}^4$ and the electric and magnetic fields $E, B \in$
cardona-qft-methods_F00008	15	■ 0.523	$\vec{x} \in \mathbb{R}^4$	$\vec{x} \in \mathbb{R}^4$	sider fields $\psi(\vec{x})$ for $\vec{x} \in \mathbb{R}^4$ and the electric and magnetic fields $E, B \in$
cardona-qft-methods_F00009	15	■ 0.523	E	E	sider fields $\psi(\vec{x})$ for $\vec{x} \in \mathbb{R}^4$ and the electric and magnetic fields $E, B \in$
cardona-qft-methods_F00010	15	■ 0.523	$B \in$	$B \in$	sider fields $\psi(\vec{x})$ for $\vec{x} \in \mathbb{R}^4$ and the electric and magnetic fields $E, B \in$
cardona-qft-methods_F00011	15	■ 0.803	$\operatorname{Function}\left(M=\mathbb{R}^4,\mathbb{R}^3\right)$	$\operatorname{Function}\left(M=\mathbb{R}^4,\mathbb{R}^3\right)$	$\operatorname{Function}(M=\mathbb{R}^4,\mathbb{R}^3)$. Einstein introduced in 1915 the gravitational field
cardona-qft-methods_F00012	15	■ 1.000	x^μ	x^μ	the physical meaning of coordinates x^μ and equations of fields? Assume
cardona-qft-methods_F00013	15	■ 1.000	$g_{\mu\nu}(x)$	$g_{\mu\nu}(x)$	the general covariance of field equations. If $g_{\mu\nu}(x)$ or the tetradfield $e_\mu^I(x)$
cardona-qft-methods_F00014	15	■ 1.000	$e_{\mu}^I(x)$	$e_\mu^I(x)$	the general covariance of field equations. If $g_{\mu\nu}(x)$ or the tetradfield $e_\mu^I(x)$
cardona-qft-methods_F00015	15	■ 1.000	I	I	is a solution (where I is a local inertial reference frame), then, for any dif-
cardona-qft-methods_F00016	15	■ 1.000	ϕ	ϕ	feomorphism ϕ of M which is active or passive (i.e. change of coordinates),
cardona-qft-methods_F00017	15	■ 1.000	M	M	feomorphism ϕ of M which is active or passive (i.e. change of coordinates),
cardona-qft-methods_F00018	15	■ 1.000	$e_{\nu}^{\prime I}(x)=\frac{\partial x^\mu}{\partial \phi(x)^\nu} e_\mu^I(x)$	$e_\nu^{\prime I}(x) = \frac{\partial x^\mu}{\partial \phi(x)^\nu} e_\mu^I(x)$	$e_\nu^{\prime I}(x) = \frac{\partial x^\mu}{\partial \phi(x)^\nu} e_\mu^I(x)$ is also a solution. As a consequence, when relativity
cardona-qft-methods_F00019	15	■ 1.000	$\left(\square+m^2\right) \psi(\vec{x})=s_b(\vec{x})$	$(\square+m^2)\psi(\vec{x})=s_b(\vec{x})$	Gordon equation $(\square+m^2)\psi(\vec{x})=s_b(\vec{x})$ and fermions satisfying $(i\rlap{\not{D}}-$
cardona-qft-methods_F00020	15	■ 1.000	$(i \not{\partial}-$	$(i \not{\partial}-$	Gordon equation $(\square+m^2)\psi(\vec{x})=s_b(\vec{x})$ and fermions satisfying $(i\rlap{\not{D}}-$
cardona-qft-methods_F00021	15	■ 1.000	$m) \psi(\vec{x})=s_f(\vec{x})$	$m)\psi(\vec{x})=s_f(\vec{x})$	$m)\psi(\vec{x})=s_f(\vec{x})$ for sources s_b,s_f . Here $\rlap{\not{D}}$ can be seen as a square root of
cardona-qft-methods_F00022	15	■ 1.000	s_b, s_f	s_b,s_f	$m)\psi(\vec{x})=s_f(\vec{x})$ for sources s_b,s_f . Here $\rlap{\not{D}}$ can be seen as a square root of

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cardona-qft-methods_ F00023	15	■ 1.000	<code>\not \partial</code>	∂	$m)\psi(\vec{x}) = s_f(\vec{x})$ for sources s_b, s_f . Here ∂ can be seen as a square root of
cardona-qft-methods_ F00024	15	■ 0.999	<code>{ }^{24,30}</code>	24,30	Noncommutative geometry, mainly pioneered by A. Connes (see ^{24,30}),
cardona-qft-methods_ F00025	15	■ 0.945	<code>(\mathcal{A}, \mathcal{H}, \mathcal{D})</code>	$(\mathcal{A}, \mathcal{H}, \mathcal{D})$	is based on a spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ where the $*$ -algebra $\mathcal{A}$ generalizes
cardona-qft-methods_ F00026	15	■ 0.945	<code>*</code>	$*$	is based on a spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ where the $*$ -algebra $\mathcal{A}$ generalizes
cardona-qft-methods_ F00027	15	■ 0.945	<code>\mathcal{A}</code>	$\mathcal{A}$	is based on a spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ where the $*$ -algebra $\mathcal{A}$ generalizes
cardona-qft-methods_ F00028	15	■ 1.000	<code>\mathcal{H}</code>	$\mathcal{H}$	product, $\mathcal{H}$ generalizes the Hilbert space of above quoted spinors ψ and
cardona-qft-methods_ F00029	15	■ 1.000	<code>\psi</code>	ψ	product, $\mathcal{H}$ generalizes the Hilbert space of above quoted spinors ψ and
cardona-qft-methods_ F00030	15	■ 1.000	<code>\mathcal{D}</code>	$\mathcal{D}$	$\mathcal{D}$ is a selfadjoint operator on $\mathcal{H}$ which generalizes ∂ via a connection on
cardona-qft-methods_ F00031	15	■ 1.000	<code>\Lambda^{-1}</code>	Λ^{-1}	scale, phase-space points are indistinguishable. Below the scale Λ^{-1} , a cer-
cardona-qft-methods_ F00032	16	■ 1.000	<code>S(\mathcal{D}, \Lambda, f)=\operatorname{Tr}(f(\mathcal{D} / \Lambda))</code>	$S(\mathcal{D}, \Lambda, f) = \operatorname{Tr}(f(\mathcal{D}/\Lambda))$	by Chamseddine and Connes ¹⁰ and defined as $S(\mathcal{D}, \Lambda, f) = \operatorname{Tr}(f(\mathcal{D}/\Lambda))$
cardona-qft-methods_ F00033	16	■ 1.000	<code>\Lambda \in \mathbb{R}^{+}</code>	$\Lambda \in \mathbb{R}^{+}$	where $\Lambda \in \mathbb{R}^{+}$ plays the role of a cut-off and f is a positive even function.
cardona-qft-methods_ F00034	16	■ 1.000	<code>f</code>	f	where $\Lambda \in \mathbb{R}^{+}$ plays the role of a cut-off and f is a positive even function.
cardona-qft-methods_ F00035	16	■ 1.000	<code>\Lambda \rightarrow \infty</code>	$\Lambda \rightarrow \infty$	The asymptotic series in $\Lambda \rightarrow \infty$ yields to an effective theory. For instance,
cardona-qft-methods_ F00036	16	■ 1.000	<code>M \times F</code>	$M \times F$	this action applied to a noncommutative model of space-time $M \times F$ with
cardona-qft-methods_ F00037	16	■ 1.000	<code>F</code>	F	a fine structure for fermions encoded in a finite geometry F gives rise from
cardona-qft-methods_ F00038	16	■ 1.000	<code>{ }^{11,20,30}</code>	11,20,30	pure gravity to the standard model coupled with gravity. ^{11,20,30}
cardona-qft-methods_ F00039	16	■ 0.997	<code>\Psi</code>	Ψ	ators (Ψ DO's) over a compact Riemannian manifold. A description of the
cardona-qft-methods_ F00040	16	■ 1.000	<code>\mathcal{L}^{1,\infty}(\mathcal{H})</code>	$\mathcal{L}^{1,\infty}(\mathcal{H})$	spaces of compact operators are described in particular, the ideal $\mathcal{L}^{1,\infty}(\mathcal{H})$.
cardona-qft-methods_ F00041	16	■ 1.000	<code>\sum_{n=1}^{\infty} n^{-1}</code>	$\sum_{n=1}^{\infty} n^{-1}$	to control the logarithmic divergency of the series $\sum_{n=1}^{\infty} n^{-1}$. We will see
cardona-qft-methods_ F00042	16	■ 0.987	<code>(M, g)</code>	(M, g)	operator on a compact Riemannian manifold (M, g) endowed with a vector
cardona-qft-methods_ F00043	16	■ 1.000	<code>E=\mathcal{C} \ell^{*} M</code>	$E = \mathcal{C}\ell^{*} M$	bundle E . An example is the (Clifford) bundle $E = \mathcal{C}\ell^{*} M$ where $\mathcal{C}\ell T_x^{*} M$
cardona-qft-methods_ F00044	16	■ 1.000	<code>\mathcal{C} \ell T_x^{*} M</code>	$\mathcal{C}\ell T_x^{*} M$	bundle E . An example is the (Clifford) bundle $E = \mathcal{C}\ell^{*} M$ where $\mathcal{C}\ell T_x^{*} M$
cardona-qft-methods_ F00045	16	■ 1.000	<code>x \in M</code>	$x \in M$	is the Clifford algebra for $x \in M$. This leads to the notion of spin structure,
cardona-qft-methods_ F00046	16	■ 0.999	<code>\nabla^S</code>	∇^S	spin connection ∇^S and Dirac operator $\mathcal{D} = -ic \circ \nabla^S$ where c is the Clifford
cardona-qft-methods_ F00047	16	■ 0.999	<code>\not D=-i c \circ \nabla^S</code>	$\mathcal{D} = -ic \circ \nabla^S$	spin connection ∇^S and Dirac operator $\mathcal{D} = -ic \circ \nabla^S$ where c is the Clifford
cardona-qft-methods_ F00048	16	■ 0.999	<code>c</code>	c	spin connection ∇^S and Dirac operator $\mathcal{D} = -ic \circ \nabla^S$ where c is the Clifford
cardona-qft-methods_ F00049	16	■ 1.000	<code>g</code>	g	multiplication. A special focus is put on the change of metrics g under

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cardona-qft-methods_F00050	16	■ 1.000	$e^{\{t \ \Delta\}, \ t \ \mathrm{in} \ \mathbb{R}^{\{+\}}$	$e^{t\Delta}, t \in \mathbb{R}^+$	namely the Green function of the heat operator $e^{t\Delta}, t \in \mathbb{R}^+$. In partic-
cardona-qft-methods_F00051	16	■ 1.000	$t \rightarrow 0^{\{+\}}$	$t \rightarrow 0^+$	ular, its expansion as $t \rightarrow 0^+$ in terms of coefficients of the elliptic operator
cardona-qft-methods_F00052	17	■ 0.990	Δ	Δ	Δ , with a method to compute the coefficients of this expansion is explained,
cardona-qft-methods_F00053	17	■ 1.000	$\mathcal{D}^2$	$\mathcal{D}^2$	the idea being to replace the Laplacian Δ by $\mathcal{D}^2$ later on.
cardona-qft-methods_F00054	17	■ 1.000	$\mathrm{Tr}(f(\mathcal{D} \ / \ \Lambda))$	$\mathrm{Tr}(f(\mathcal{D}/\Lambda))$	or Yang–Mills actions. The spectral action $\mathrm{Tr}(f(\mathcal{D}/\Lambda))$ is justified and the
cardona-qft-methods_F00055	17	■ 1.000	Λ	Λ	link between its asymptotic expansion in Λ and the heat kernel coefficients
cardona-qft-methods_F00056	17	■ 1.000	$ \mathcal{D} $	$ \mathcal{D} $	is given via noncommutative integrals of powers of $ \mathcal{D} $.
cardona-qft-methods_F00057	17	■ 1.000	$\mathbb{N}=\{1,2, \ \mathrm{ldots}\}$	$\mathbb{N} = \{1, 2, \dots\}$	$\mathbb{N} = \{1, 2, \dots\}$ is the set of positive integers and $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ the set of
cardona-qft-methods_F00058	17	■ 1.000	$\mathbb{N}_{\{0\}}=\mathbb{N} \ \mathrm{cup} \ \{0\}$	$\mathbb{N}_0 = \mathbb{N} \cup \{0\}$	$\mathbb{N} = \{1, 2, \dots\}$ is the set of positive integers and $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ the set of
cardona-qft-methods_F00059	17	■ 1.000	$\mathbb{R}^{\{d\}}$	$\mathbb{R}^d$	On $\mathbb{R}^d$, the volume form is $dx = dx^1 \wedge \dots \wedge dx^d$.
cardona-qft-methods_F00060	17	■ 1.000	$d \ x=d \ x^{\{1\}} \ \mathrm{wedge} \ \mathrm{cdots} \ \mathrm{wedge} \ d \ x^{\{d\}}$	$dx = dx^1 \wedge \dots \wedge dx^d$	On $\mathbb{R}^d$, the volume form is $dx = dx^1 \wedge \dots \wedge dx^d$.
cardona-qft-methods_F00061	17	■ 1.000	$\mathbb{S}^{\{d\}}$	$\mathbb{S}^d$	$\mathbb{S}^d$ is the sphere of radius 1 in dimension d . The induced metric $d\xi =$
cardona-qft-methods_F00062	17	■ 1.000	d	d	$\mathbb{S}^d$ is the sphere of radius 1 in dimension d . The induced metric $d\xi =$
cardona-qft-methods_F00063	17	■ 1.000	$d \ \mathrm{xi} =$	$d\xi =$	$\mathbb{S}^d$ is the sphere of radius 1 in dimension d . The induced metric $d\xi =$
cardona-qft-methods_F00064	17	■ 1.000	$\left \sum_{j=1}^{\{d\}}(-1)^{\{j-1\}} \ \mathrm{xi}_{\{j\}} \ d \ \mathrm{xi}_{\{1\}} \ \mathrm{wedge} \ \mathrm{cdots} \ \mathrm{wedge} \ \widehat{d \ \mathrm{xi}}_{\{j\}} \ \mathrm{wedge} \ \mathrm{cdots} \ \mathrm{wedge} \ d \ \mathrm{xi}_{\{d\}}\right $	$\left \sum_{j=1}^d(-1)^{j-1}\xi_j d\xi_1 \wedge \dots \wedge \widehat{d\xi_j} \wedge \dots \wedge d\xi_d\right $	$ \sum_{j=1}^d(-1)^{j-1}\xi_j d\xi_1 \wedge \dots \wedge \widehat{d\xi_j} \wedge \dots \wedge d\xi_d $ restricts to the volume form on
cardona-qft-methods_F00065	17	■ 1.000	$\mathbb{S}^{\{d-1\}}$	$\mathbb{S}^{d-1}$	$\mathbb{S}^{d-1}$.
cardona-qft-methods_F00066	17	■ 1.000	$U, \ V$	U, V	U, V are open set either in M or in $\mathbb{R}^d$.
cardona-qft-methods_F00067	17	■ 0.998	$_{\{g\}}$	g	We denote by $dvol_g$, for all positively oriented g -orthonormal basis
cardona-qft-methods_F00068	17	■ 0.997	$\left\{\mathrm{xi}_{\{1\}}, \ \mathrm{cdots}, \ \mathrm{xi}_{\{d\}}\right\}$	$\{\xi_1, \dots, \xi_d\}$	$\{\xi_1, \dots, \xi_d\}$ of $T_x M$, for $x \in M$, the unique volume element such that
cardona-qft-methods_F00069	17	■ 0.997	$T_{\{x\}} \ M$	$T_x M$	$\{\xi_1, \dots, \xi_d\}$ of $T_x M$, for $x \in M$, the unique volume element such that
cardona-qft-methods_F00070	17	■ 0.792	$d \ v \ o \ l_{\{g\}}\left(\mathrm{xi}_{\{1\}}, \ \mathrm{cdots}, \ \mathrm{xi}_{\{d\}}\right)=1$	$dvol_g(\xi_1, \dots, \xi_d) = 1$	$dvol_g(\xi_1, \dots, \xi_d) = 1$. Thus, in a local chart, $\sqrt{\det g_x} dx = dvol_g $.
cardona-qft-methods_F00071	17	■ 0.792	$\sqrt{\mathrm{operatorname{det} \ g_{\{x\}}}} d \ x =\left d \ v \ o \ l_{\{g\}} \ \mathrm{right} $	$\sqrt{\det g_x} dx = dvol_g $	$dvol_g(\xi_1, \dots, \xi_d) = 1$. Thus, in a local chart, $\sqrt{\det g_x} dx = dvol_g $.
cardona-qft-methods_F00072	17	■ 1.000	$\alpha \ \mathrm{in} \ \mathbb{N}^{\{d\}}$	$\alpha \in \mathbb{N}^d$	When $\alpha \in \mathbb{N}^d$ is a multi-index, $\partial_x^\alpha = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_d}^{\alpha_d}$, $ \alpha = \sum_{i=1}^d \alpha_i$ and
cardona-qft-methods_F00073	17	■ 1.000	$\partial_{\{x\}}^{\{\alpha\}}=\partial_{\{x_1\}}^{\{\alpha_1\}} \ \partial_{\{x_2\}}^{\{\alpha_2\}} \ \mathrm{cdots} \ \partial_{\{x_d\}}^{\{\alpha_d\}} \ \partial_{\{\alpha_d\}}, \ \mathrm{quad} \alpha =\sum_{i=1}^{\{d\}} \ \alpha_{\{i\}}$	$\partial_x^\alpha = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_d}^{\alpha_d}, \quad \alpha = \sum_{i=1}^d \alpha_i$	When $\alpha \in \mathbb{N}^d$ is a multi-index, $\partial_x^\alpha = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_d}^{\alpha_d}$, $ \alpha = \sum_{i=1}^d \alpha_i$ and
cardona-qft-methods_F00074	17	■ 1.000	$\alpha !=\alpha_{\{1\}} \ a_{\{2\}} \ \mathrm{cdots} \ \alpha_{\{d\}}$	$\alpha! = \alpha_1 \alpha_2 \dots \alpha_d$	$\alpha! = \alpha_1 \alpha_2 \dots \alpha_d$. For $\xi \in \mathbb{R}^d$, $ \xi = \left(\sum_{k=1}^d \xi_k ^2\right)^{1/2}$ is the Euclidean metric.

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cardona-qft-methods_ F00075	17	■ 1.000	$\xi \in \mathbb{R}^d, \xi = \left(\sum_{k=1}^d \xi_k ^2 \right)^{1/2}$	$\xi \in \mathbb{R}^d, \xi = \left(\sum_{k=1}^d \xi_k ^2 \right)^{1/2}$	$\alpha! = \alpha_1 \alpha_2 \cdots \alpha_d$. For $\xi \in \mathbb{R}^d$, $ \xi = \left(\sum_{k=1}^d \xi_k ^2 \right)^{1/2}$ is the Euclidean metric.
cardona-qft-methods_ F00076	17	■ 1.000	$\mathcal{B}(\mathcal{H}), \mathcal{K}(\mathcal{H}), \mathcal{L}^p(\mathcal{H})$	$\mathcal{B}(\mathcal{H}), \mathcal{K}(\mathcal{H}), \mathcal{L}^p(\mathcal{H})$	$\mathcal{H}$ is a separable Hilbert space and $\mathcal{B}(\mathcal{H}), \mathcal{K}(\mathcal{H}), \mathcal{L}^p(\mathcal{H})$ denote respectively
cardona-qft-methods_ F00077	17	■ 1.000	p	p	the set of bounded, compact and p -Schatten-class operators, so $\mathcal{L}^1(\mathcal{H})$ are
cardona-qft-methods_ F00078	17	■ 1.000	$\mathcal{L}^1(\mathcal{H})$	$\mathcal{L}^1(\mathcal{H})$	the set of bounded, compact and p -Schatten-class operators, so $\mathcal{L}^1(\mathcal{H})$ are
cardona-qft-methods_ F00079	17	■ 0.998	$\Psi DO(M)$	$\Psi DO(M)$	on the set $\Psi DO(M)$ of pseudodifferential operators on a compact manifold
cardona-qft-methods_ F00080	17	■ 1.000	$\{ \}^{99,102}$	99,102	References for this section: Classical books are. ^{99,102} For an orientation
cardona-qft-methods_ F00081	17	■ 1.000	$\{ \}^{86,87}$	86,87	more in the spirit of noncommutative geometry here we follow ^{86,87} based
cardona-qft-methods_ F00082	17	■ 1.000	$\{ \}^{3,33}$	3,33	on, ^{3,33} see also the excellent books. ^{49,82,83,104,105}
cardona-qft-methods_ F00083	17	■ 1.000	$\{ \}^{49,82,83,104,105}$	49,82,83,104,105	on, ^{3,33} see also the excellent books. ^{49,82,83,104,105}
cardona-qft-methods_ F00084	18	■ 1.000	$m \in \mathbb{C}$	$m \in \mathbb{C}$	Definition 2.1. In the following, $m \in \mathbb{C}$.
cardona-qft-methods_ F00085	18	■ 1.000	$\sigma(x, \xi)$	$\sigma(x, \xi)$	A symbol $\sigma(x, \xi)$ of order m is a C^∞ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$
cardona-qft-methods_ F00086	18	■ 1.000	m	m	A symbol $\sigma(x, \xi)$ of order m is a C^∞ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$
cardona-qft-methods_ F00087	18	■ 1.000	C^∞	C^∞	A symbol $\sigma(x, \xi)$ of order m is a C^∞ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$
cardona-qft-methods_ F00088	18	■ 1.000	$(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$	$(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$	A symbol $\sigma(x, \xi)$ of order m is a C^∞ function: $(x, \xi) \in U \times \mathbb{R}^d \rightarrow \mathbb{C}$
cardona-qft-methods_ F00089	18	■ 0.915	$\left \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi) \right \leq C_{\alpha\beta}(x) (1 + \xi)^{\Re(m) - \beta }, C_{\alpha\beta}$	$\left \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi) \right \leq C_{\alpha\beta}(x) (1 + \xi)^{\Re(m) - \beta }, C_{\alpha\beta}$	(i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi) \leq C_{\alpha\beta}(x) (1 + \xi)^{\Re(m) - \beta }$, $C_{\alpha\beta}$ bounded on U .
cardona-qft-methods_ F00090	18	■ 0.915	U	U	(i) $ \partial_x^\alpha \partial_\xi^\beta \sigma(x, \xi) \leq C_{\alpha\beta}(x) (1 + \xi)^{\Re(m) - \beta }$, $C_{\alpha\beta}$ bounded on U .
cardona-qft-methods_ F00091	18	■ 1.000	$\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$	$\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$	(ii) We suppose that $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$ where σ_k is homoge-
cardona-qft-methods_ F00092	18	■ 1.000	σ_k	σ_k	(ii) We suppose that $\sigma(x, \xi) \simeq \sum_{j \geq 0} \sigma_{m-j}(x, \xi)$ where σ_k is homoge-
cardona-qft-methods_ F00093	18	■ 1.000	k	k	neous of degree k in ξ where $\simeq$ means a controlled asymptotic behavior
cardona-qft-methods_ F00094	18	■ 1.000	ξ	ξ	neous of degree k in ξ where $\simeq$ means a controlled asymptotic behavior
cardona-qft-methods_ F00095	18	■ 1.000	$\simeq$	$\simeq$	neous of degree k in ξ where $\simeq$ means a controlled asymptotic behavior
cardona-qft-methods_ F00096	18	■ 1.000	$ \xi \geq 1$	$ \xi \geq 1$	for $ \xi \geq 1$, with $C_{N\alpha\beta}$ bounded on U . The set of symbols of order m is
cardona-qft-methods_ F00097	18	■ 1.000	$C_{N\alpha\beta}$	$C_{N\alpha\beta}$	for $ \xi \geq 1$, with $C_{N\alpha\beta}$ bounded on U . The set of symbols of order m is
cardona-qft-methods_ F00098	18	■ 1.000	$S^m(U \times \mathbb{R}^d)$	$S^m(U \times \mathbb{R}^d)$	denoted by $S^m(U \times \mathbb{R}^d)$.
cardona-qft-methods_ F00099	18	■ 1.000	$a \in C^\infty(U \times U \times \mathbb{R}^d)$	$a \in C^\infty(U \times U \times \mathbb{R}^d)$	A function $a \in C^\infty(U \times U \times \mathbb{R}^d)$ is an amplitude of order m , if for any
cardona-qft-methods_ F00100	18	■ 1.000	$K \subset U$	$K \subset U$	compact $K \subset U$ and any $\alpha, \beta, \gamma \in \mathbb{N}^d$ there exists a constant $C_{K\alpha\beta\gamma}$ such

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cardona-qft-methods_ F00101	18	■ 1.000	$\alpha, \beta, \gamma \in \mathbb{N}^d$	$\alpha, \beta, \gamma \in \mathbb{N}^d$	compact $K \subset U$ and any $\alpha, \beta, \gamma \in \mathbb{N}^d$ there exists a constant $C_{K\alpha\beta\gamma}$ such
cardona-qft-methods_ F00102	18	■ 1.000	$C_{K\alpha\beta\gamma}$	$C_{K\alpha\beta\gamma}$	compact $K \subset U$ and any $\alpha, \beta, \gamma \in \mathbb{N}^d$ there exists a constant $C_{K\alpha\beta\gamma}$ such
cardona-qft-methods_ F00103	18	■ 1.000	$A^m(U)$	$A^m(U)$	The set of amplitudes is written $A^m(U)$.
cardona-qft-methods_ F00104	18	■ 1.000	$\sigma \in S^m(U \times \mathbb{R}^d)$	$\sigma \in S^m(U \times \mathbb{R}^d)$	For $\sigma \in S^m(U \times \mathbb{R}^d)$, we get a continuous operator $\sigma(\cdot, D) : u \in C_c^\infty(U) \rightarrow$
cardona-qft-methods_ F00105	18	■ 1.000	$\sigma(\cdot, D) : u \in C_c^\infty(U) \rightarrow$	$\sigma(\cdot, D) : u \in C_c^\infty(U) \rightarrow$	For $\sigma \in S^m(U \times \mathbb{R}^d)$, we get a continuous operator $\sigma(\cdot, D) : u \in C_c^\infty(U) \rightarrow$
cardona-qft-methods_ F00106	18	■ 1.000	$C^\infty(U)$	$C^\infty(U)$	$C^\infty(U)$ given by
cardona-qft-methods_ F00107	18	■ 0.994	$\sigma(\cdot, D)$	$\sigma(\cdot, D)$	where $\hat{\cdot}$ means the Fourier transform. This operator $\sigma(\cdot, D)$ will be also
cardona-qft-methods_ F00108	18	■ 0.848	$Op(\sigma)$	$Op(\sigma)$	denoted by $Op(\sigma)$. For instance, if $\sigma(x, \xi) = \sum_\alpha a_\alpha(x) \xi^\alpha$, then $\sigma(x, D) =$
cardona-qft-methods_ F00109	18	■ 0.848	$\sigma(x, \xi) = \sum_\alpha a_\alpha(x) \xi^\alpha$	$\sigma(x, \xi) = \sum_\alpha a_\alpha(x) \xi^\alpha$	denoted by $Op(\sigma)$. For instance, if $\sigma(x, \xi) = \sum_\alpha a_\alpha(x) \xi^\alpha$, then $\sigma(x, D) =$
cardona-qft-methods_ F00110	18	■ 0.848	$\sigma(x, D) =$	$\sigma(x, D) =$	denoted by $Op(\sigma)$. For instance, if $\sigma(x, \xi) = \sum_\alpha a_\alpha(x) \xi^\alpha$, then $\sigma(x, D) =$
cardona-qft-methods_ F00111	18	■ 1.000	$\sum_\alpha a_\alpha(x) D_x^\alpha$	$\sum_\alpha a_\alpha(x) D_x^\alpha$	$\sum_\alpha a_\alpha(x) D_x^\alpha$ with $D_x = -i\partial_x$. Remark that, by transposition, there is
cardona-qft-methods_ F00112	18	■ 1.000	$D_x = -i\partial_x$	$D_x = -i\partial_x$	$\sum_\alpha a_\alpha(x) D_x^\alpha$ with $D_x = -i\partial_x$. Remark that, by transposition, there is
cardona-qft-methods_ F00113	18	■ 1.000	$\mathcal{D}'_c(U)$	$\mathcal{D}'_c(U)$	a natural extension of $\sigma(\cdot, D)$ from the set $\mathcal{D}'_c(U)$ of distributions with
cardona-qft-methods_ F00114	18	■ 0.999	$\mathcal{D}'(U)$	$\mathcal{D}'(U)$	compact support in U to the set of distributions $\mathcal{D}'(U)$.
cardona-qft-methods_ F00115	18	■ 1.000	$ \alpha = m$	$ \alpha = m$	By definition, the leading term for $ \alpha = m$ is the <i>principal symbol</i> and
cardona-qft-methods_ F00116	18	■ 1.000	$\sigma(x, D)$	$\sigma(x, D)$	the <i>Schwartz kernel</i> of $\sigma(x, D)$ is defined by
cardona-qft-methods_ F00117	18	■ 0.907	$Op(a)$	$Op(a)$	operator $Op(a)$ associated to the amplitude a is
cardona-qft-methods_ F00118	18	■ 0.907	a	a	operator $Op(a)$ associated to the amplitude a is
cardona-qft-methods_ F00119	18	■ 0.997	$P : C_c^\infty(U) \rightarrow C^\infty(U)$	$P : C_c^\infty(U) \rightarrow C^\infty(U)$	Definition 2.2. $P : C_c^\infty(U) \rightarrow C^\infty(U)$ (or $\mathcal{D}'(U)$) is said to be smooth-
cardona-qft-methods_ F00120	18	■ 1.000	$C^\infty(U \times U)$	$C^\infty(U \times U)$	ing if its kernel is in $C^\infty(U \times U)$ and $\Psi DO^{-\infty}(U)$ will denote the set of
cardona-qft-methods_ F00121	18	■ 1.000	$\Psi DO^{-\infty}(U)$	$\Psi DO^{-\infty}(U)$	ing if its kernel is in $C^\infty(U \times U)$ and $\Psi DO^{-\infty}(U)$ will denote the set of
cardona-qft-methods_ F00122	19	■ 1.000	$\Psi DO^m(U)$	$\Psi DO^m(U)$	For $m \in \mathbb{C}$, the set $\Psi DO^m(U)$ of pseudodifferential operators of order
cardona-qft-methods_ F00123	19	■ 1.000	P	P	m is the set of P such that $P : C_c^\infty(U) \rightarrow C^\infty(U)$, $Pu(x) = (\sigma(x, D) +$
cardona-qft-methods_ F00124	19	■ 1.000	$P : C_c^\infty(U) \rightarrow C^\infty(U)$, $Pu(x) = (\sigma(x, D) +$	$P : C_c^\infty(U) \rightarrow C^\infty(U)$, $Pu(x) = (\sigma(x, D) +$	m is the set of P such that $P : C_c^\infty(U) \rightarrow C^\infty(U)$, $Pu(x) = (\sigma(x, D) +$
cardona-qft-methods_ F00125	19	■ 0.914	$R)(u)$	$R)(u)$	$R)(u)$ where $\sigma \in S^m(U \times \mathbb{R}^d)$, $R \in \Psi DO^{-\infty}$. We call σ the symbol of P .
cardona-qft-methods_ F00126	19	■ 0.914	$\sigma \in S^m(U \times \mathbb{R}^d)$, $R \in \Psi DO^{-\infty}$	$\sigma \in S^m(U \times \mathbb{R}^d)$, $R \in \Psi DO^{-\infty}$	$R)(u)$ where $\sigma \in S^m(U \times \mathbb{R}^d)$, $R \in \Psi DO^{-\infty}$. We call σ the symbol of P .

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cardona-qft-methods_ F00127	19	■ 0.914	$\backslash\sigma$	σ	$R)(u)$ where $\sigma \in S^m(U \times \mathbb{R}^d)$, $R \in \Psi DO^{-\infty}$. We call σ the symbol of P .
cardona-qft-methods_ F00128	19	■ 0.861	$m \backslash\mathrm{in} \backslash\mathrm{mathbb{R}}$	$m \in \mathbb{R}$	operator whose amplitude is in $A^m(U)$ for all $m \in \mathbb{R}$: by (2), $a(x, y, \xi) =$
cardona-qft-methods_ F00129	19	■ 0.861	$(2), a(x, y, \backslash\mathrm{xi})=$	$(2), a(x, y, \xi) =$	operator whose amplitude is in $A^m(U)$ for all $m \in \mathbb{R}$: by (2), $a(x, y, \xi) =$
cardona-qft-methods_ F00130	19	■ 1.000	$e^{\{-i(x-y) \backslash\mathrm{cdot} \backslash\mathrm{xi}\}} k(x, y) \backslash\mathrm{phi}(\backslash\mathrm{xi})$	$e^{-i(x-y) \cdot \xi} k(x, y) \phi(\xi)$	$e^{-i(x-y) \cdot \xi} k(x, y) \phi(\xi)$ where the function $\phi \in C_c^\infty(\mathbb{R}^d)$ satisfies $\int_{\mathbb{R}^d} \phi(\xi) d\xi =$
cardona-qft-methods_ F00131	19	■ 1.000	$\backslash\mathrm{phi} \backslash\mathrm{in} C_{\mathrm{c}}^{\infty}(\backslash\mathrm{infty}) \backslash\mathrm{left}(\backslash\mathrm{mathbb{R}}^{\mathrm{d}} \backslash\mathrm{right})$	$\phi \in C_c^\infty(\mathbb{R}^d)$	$e^{-i(x-y) \cdot \xi} k(x, y) \phi(\xi)$ where the function $\phi \in C_c^\infty(\mathbb{R}^d)$ satisfies $\int_{\mathbb{R}^d} \phi(\xi) d\xi =$
cardona-qft-methods_ F00132	19	■ 1.000	$\backslash\mathrm{int}_{\backslash\mathrm{mathbb{R}}^{\mathrm{d}}} \backslash\mathrm{phi}(\backslash\mathrm{xi}) d \backslash\mathrm{xi}=$	$\int_{\mathbb{R}^d} \phi(\xi) d\xi =$	$e^{-i(x-y) \cdot \xi} k(x, y) \phi(\xi)$ where the function $\phi \in C_c^\infty(\mathbb{R}^d)$ satisfies $\int_{\mathbb{R}^d} \phi(\xi) d\xi =$
cardona-qft-methods_ F00133	19	■ 1.000	$(2 \backslash\mathrm{pi})^{\mathrm{d}}$	$(2\pi)^d$	$(2\pi)^d$. The main obstruction to smoothness is on the diagonal since $k^{\sigma(x, D)}$
cardona-qft-methods_ F00134	19	■ 1.000	$k^{\{\backslash\sigma(x, D)\}}$	$k^{\sigma(x, D)}$	$(2\pi)^d$. The main obstruction to smoothness is on the diagonal since $k^{\sigma(x, D)}$
cardona-qft-methods_ F00135	19	■ 1.000	$\backslash\sigma^{\mathrm{A}}(x, \backslash\mathrm{xi})=e^{\{-i x \backslash\mathrm{cdot} \backslash\mathrm{xi}\}} A \backslash\mathrm{left}(x \backslash\mathrm{rightarrow} e^{\{i x \backslash\mathrm{cdot} \backslash\mathrm{xi}\}} \backslash\mathrm{right})$	$\sigma^A(x, \xi) = e^{-ix \cdot \xi} A(x \rightarrow e^{ix \cdot \xi})$	where we used the following definition $\sigma^A(x, \xi) = e^{-ix \cdot \xi} A(x \rightarrow e^{ix \cdot \xi})$.
cardona-qft-methods_ F00136	19	■ 1.000	$k^{\mathrm{A}}(x, y, \backslash\mathrm{xi})$	$k^A(x, y, \xi)$	where $k^A(x, y, \xi)$ is the amplitude of $k_\sigma^A(x, y)$. Actually, $\sigma^A(x, \xi) =$
cardona-qft-methods_ F00137	19	■ 1.000	$k_{\{\backslash\sigma\}^{\mathrm{A}}}(x, y)$	$k_\sigma^A(x, y)$	where $k^A(x, y, \xi)$ is the amplitude of $k_\sigma^A(x, y)$. Actually, $\sigma^A(x, \xi) =$
cardona-qft-methods_ F00138	19	■ 1.000	$\backslash\sigma^{\mathrm{A}}(x, \backslash\mathrm{xi})=$	$\sigma^A(x, \xi) =$	where $k^A(x, y, \xi)$ is the amplitude of $k_\sigma^A(x, y)$. Actually, $\sigma^A(x, \xi) =$
cardona-qft-methods_ F00139	19	■ 1.000	$e^{\{i D_{\backslash\mathrm{xi}} D_{\mathrm{y}}\}} k^{\mathrm{A}}(x, y, \backslash\mathrm{xi})_{\backslash\mathrm{mid} y=x}$	$e^{iD_\xi D_y} k^A(x, y, \xi) _{y=x}$	$e^{iD_\xi D_y} k^A(x, y, \xi) _{y=x}$ and $e^{iD_\xi D_y} = 1 + iD_\xi D_y - \frac{1}{2}(D_\xi D_y)^2 + \dots$. Thus
cardona-qft-methods_ F00140	19	■ 1.000	$e^{\{i D_{\backslash\mathrm{xi}} D_{\mathrm{y}}\}}=1+i D_{\backslash\mathrm{xi}} D_{\mathrm{y}}-\backslash\mathrm{frac}{1}{2} \backslash\mathrm{left}(D_{\backslash\mathrm{xi}} D_{\mathrm{y}} \backslash\mathrm{right})^{\mathrm{2}}+\backslash\mathrm{cdots}$	$e^{iD_\xi D_y} = 1 + iD_\xi D_y - \frac{1}{2}(D_\xi D_y)^2 + \dots$	$e^{iD_\xi D_y} k^A(x, y, \xi) _{y=x}$ and $e^{iD_\xi D_y} = 1 + iD_\xi D_y - \frac{1}{2}(D_\xi D_y)^2 + \dots$. Thus
cardona-qft-methods_ F00141	19	■ 1.000	$A=0 p \backslash\mathrm{left}(\backslash\sigma^{\mathrm{A}} \backslash\mathrm{right})+R$	$A = Op(\sigma^A) + R$	$A = Op(\sigma^A) + R$ where R is a regularizing operator on U .
cardona-qft-methods_ F00142	19	■ 1.000	R	R	$A = Op(\sigma^A) + R$ where R is a regularizing operator on U .
cardona-qft-methods_ F00143	19	■ 1.000	$P_{\{1\}} \backslash\mathrm{in} \backslash\mathrm{Psi} D 0^{\mathrm{m}_{\{1\}}}$	$P_1 \in \Psi DO^{m_1}$	Theorem 2.1. (i) If $P_1 \in \Psi DO^{m_1}$ and $P_2 \in \Psi DO^{m_2}$, then $P_1 P_2 \in$
cardona-qft-methods_ F00144	19	■ 1.000	$P_{\{2\}} \backslash\mathrm{in} \backslash\mathrm{Psi} D 0^{\mathrm{m}_{\{2\}}}$	$P_2 \in \Psi DO^{m_2}$	Theorem 2.1. (i) If $P_1 \in \Psi DO^{m_1}$ and $P_2 \in \Psi DO^{m_2}$, then $P_1 P_2 \in$
cardona-qft-methods_ F00145	19	■ 1.000	$P_{\{1\}} P_{\{2\}} \backslash\mathrm{in}$	$P_1 P_2 \in$	Theorem 2.1. (i) If $P_1 \in \Psi DO^{m_1}$ and $P_2 \in \Psi DO^{m_2}$, then $P_1 P_2 \in$
cardona-qft-methods_ F00146	19	■ 1.000	$\backslash\mathrm{Psi} D 0^{\mathrm{m}_{\{1\}}+\mathrm{m}_{\{2\}}}$	$\Psi DO^{m_1+m_2}$	$\Psi DO^{m_1+m_2}$ with symbol
cardona-qft-methods_ F00147	19	■ 1.000	$P_{\{1\}} P_{\{2\}}$	$P_1 P_2$	The principal symbol of $P_1 P_2$ is $\sigma_{m_1+m_2}^{P_1 P_2}(x, \xi) = \sigma_{m_1}^{P_1}(x, \xi) \sigma_{m_2}^{P_2}(x, \xi)$.
cardona-qft-methods_ F00148	19	■ 1.000	$\backslash\sigma_{\mathrm{m}_{\{1\}}+\mathrm{m}_{\{2\}}}^{\mathrm{P}_{\{1\}} P_{\{2\}}}(x, \backslash\mathrm{xi})= \backslash\sigma_{\mathrm{m}_{\{1\}}}^{\mathrm{P}_{\{1\}}}(x, \backslash\mathrm{xi}) \backslash\sigma_{\mathrm{m}_{\{2\}}}^{\mathrm{P}_{\{2\}}}(x, \backslash\mathrm{xi})$	$\sigma_{m_1+m_2}^{P_1 P_2}(x, \xi) = \sigma_{m_1}^{P_1}(x, \xi) \sigma_{m_2}^{P_2}(x, \xi)$	The principal symbol of $P_1 P_2$ is $\sigma_{m_1+m_2}^{P_1 P_2}(x, \xi) = \sigma_{m_1}^{P_1}(x, \xi) \sigma_{m_2}^{P_2}(x, \xi)$.
cardona-qft-methods_ F00149	19	■ 1.000	$P \backslash\mathrm{in} \backslash\mathrm{Psi} D 0^{\mathrm{m}}(U)$	$P \in \Psi DO^m(U)$	(ii) Let $P \in \Psi DO^m(U)$ and $\phi \in \mathrm{Diff}(U, V)$ where V is another open
cardona-qft-methods_ F00150	19	■ 1.000	$\backslash\mathrm{phi} \backslash\mathrm{in} \backslash\mathrm{operatorname{Diff}}(U, V)$	$\phi \in \mathrm{Diff}(U, V)$	(ii) Let $P \in \Psi DO^m(U)$ and $\phi \in \mathrm{Diff}(U, V)$ where V is another open
cardona-qft-methods_ F00151	19	■ 1.000	V	V	(ii) Let $P \in \Psi DO^m(U)$ and $\phi \in \mathrm{Diff}(U, V)$ where V is another open
cardona-qft-methods_ F00152	19	■ 1.000	$\backslash\mathrm{phi}_{\{*\}} P: f \backslash\mathrm{in} C^{\infty}(V) \backslash\mathrm{rightarrow} P(f \backslash\mathrm{circ} \backslash\mathrm{phi}) \backslash\mathrm{circ} \backslash\mathrm{phi}^{\{-1\}}$	$\phi_* P : f \in C^\infty(V) \rightarrow P(f \circ \phi) \circ \phi^{-1}$	set of $\mathbb{R}^d$. The operator $\phi_* P : f \in C^\infty(V) \rightarrow P(f \circ \phi) \circ \phi^{-1}$ satisfies

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cardona-qft-methods_ F00153	19	■ 1.000	$\phi_{*} P \in \Psi D^0{}^m(V)$	$\phi_{*} P \in \Psi DO^m(V)$	$\phi_{*} P \in \Psi DO^m(V)$ and its symbol is
cardona-qft-methods_ F00154	20	■ 1.000	ϕ_{α}	ϕ_{α}	where ϕ_{α} is a polynomial of degree α in ξ . Moreover, its principal symbol
cardona-qft-methods_ F00155	20	■ 1.000	α	α	where ϕ_{α} is a polynomial of degree α in ξ . Moreover, its principal symbol
cardona-qft-methods_ F00156	20	■ 1.000	$\sigma^{\phi_{*} P}{}_m = \phi_{*} \sigma_m^P$	$\sigma^{\phi_{*} P}{}_m = \phi_{*} \sigma_m^P$	$\sigma^{\phi_{*} P}{}_m = \phi_{*} \sigma_m^P$.
cardona-qft-methods_ F00157	20	■ 1.000	$\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$	$\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$	An interesting remark is in order: $\sigma^P(x, \xi) = e^{-ix \cdot \xi} P(x \rightarrow e^{ix \cdot \xi})$, thus
cardona-qft-methods_ F00158	20	■ 1.000	$\xi \rightarrow t\xi$	$\xi \rightarrow t\xi$	the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$
cardona-qft-methods_ F00159	20	■ 1.000	$t > 0$	$t > 0$	the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$
cardona-qft-methods_ F00160	20	■ 1.000	$t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$	$t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$	the dilation $\xi \rightarrow t\xi$ with $t > 0$ gives $t^{-m} e^{-itx \cdot \xi} P e^{itx \cdot \xi} = t^{-m} \sigma^P(x, t\xi) \simeq$
cardona-qft-methods_ F00161	20	■ 1.000	$t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$	$t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$	$t^{-m} \sum_{j \geq 0} \sigma_{m-j}^P(x, t\xi) = \sigma_m^P(x, \xi) + o(t^{-1})$. Thus, if $P \in \Psi DO^m(U)$ with
cardona-qft-methods_ F00162	20	■ 1.000	$m \geq 0$	$m \geq 0$	$m \geq 0$,
cardona-qft-methods_ F00163	20	■ 1.000	$h \in C^{\infty}(U)$	$h \in C^{\infty}(U)$	where $h \in C^{\infty}(U)$ is (almost) defined by $dh(x) = \xi$.
cardona-qft-methods_ F00164	20	■ 1.000	$dh(x) = \xi$	$dh(x) = \xi$	where $h \in C^{\infty}(U)$ is (almost) defined by $dh(x) = \xi$.
cardona-qft-methods_ F00165	20	■ 1.000	$\Psi D^0{}^m(M)$	$\Psi DO^m(M)$	Definition 2.3. $\Psi DO^m(M)$ is defined as the set of operators P :
cardona-qft-methods_ F00166	20	■ 1.000	$C_c^{\infty}(M) \rightarrow C^{\infty}(M)$	$C_c^{\infty}(M) \rightarrow C^{\infty}(M)$	$C_c^{\infty}(M) \rightarrow C^{\infty}(M)$ such that
cardona-qft-methods_ F00167	20	■ 1.000	$k^P \in C^{\infty}(M \times M)$	$k^P \in C^{\infty}(M \times M)$	(i) the kernel $k^P \in C^{\infty}(M \times M)$ off the diagonal,
cardona-qft-methods_ F00168	20	■ 1.000	$f \in C_c^{\infty}(\phi(U)) \rightarrow P(f \circ \phi) \circ \phi^{-1} \in C^{\infty}(\phi(U))$	$f \in C_c^{\infty}(\phi(U)) \rightarrow P(f \circ \phi) \circ \phi^{-1} \in C^{\infty}(\phi(U))$	(ii) The map : $f \in C_c^{\infty}(\phi(U)) \rightarrow P(f \circ \phi) \circ \phi^{-1} \in C^{\infty}(\phi(U))$ is in
cardona-qft-methods_ F00169	20	■ 0.999	$\Psi D^0{}^m(\phi(U))$	$\Psi DO^m(\phi(U))$	$\Psi DO^m(\phi(U))$ for every coordinate chart $(U, \phi : U \rightarrow \mathbb{R}^d)$.
cardona-qft-methods_ F00170	20	■ 0.999	$(U, \phi : U \rightarrow \mathbb{R}^d)$	$(U, \phi : U \rightarrow \mathbb{R}^d)$	$\Psi DO^m(\phi(U))$ for every coordinate chart $(U, \phi : U \rightarrow \mathbb{R}^d)$.
cardona-qft-methods_ F00171	20	■ 1.000	$P : \Gamma_c^{\infty}(M, E) \rightarrow \Gamma^{\infty}(M, E)$	$P : \Gamma_c^{\infty}(M, E) \rightarrow \Gamma^{\infty}(M, E)$	a linear map $P : \Gamma_c^{\infty}(M, E) \rightarrow \Gamma^{\infty}(M, E)$ is in $\Psi DO^m(M, E)$ when k^P
cardona-qft-methods_ F00172	20	■ 1.000	$\Psi D^0{}^m(M, E)$	$\Psi DO^m(M, E)$	a linear map $P : \Gamma_c^{\infty}(M, E) \rightarrow \Gamma^{\infty}(M, E)$ is in $\Psi DO^m(M, E)$ when k^P
cardona-qft-methods_ F00173	20	■ 1.000	k^P	k^P	a linear map $P : \Gamma_c^{\infty}(M, E) \rightarrow \Gamma^{\infty}(M, E)$ is in $\Psi DO^m(M, E)$ when k^P
cardona-qft-methods_ F00174	20	■ 1.000	ΨD^0	ΨDO	is smooth off the diagonal, and local expressions are ΨDO 's with matrix-
cardona-qft-methods_ F00175	20	■ 1.000	σ_m^P	σ_m^P	The covariance formula implies that σ_m^P is independent of the chosen
cardona-qft-methods_ F00176	20	■ 1.000	$T^*M \rightarrow M$	$T^*M \rightarrow M$	local chart so is globally defined on the bundle $T^*M \rightarrow M$ and σ_m^P is defined
cardona-qft-methods_ F00177	20	■ 1.000	$P \in \Psi D^0{}^m$	$P \in \Psi DO^m$	for every $P \in \Psi DO^m$ using overlapping charts and patching with partition
cardona-qft-methods_ F00178	20	■ 1.000	$P \in \Psi D^0{}^m(M, E)$	$P \in \Psi DO^m(M, E)$	Definition 2.4. $P \in \Psi DO^m(M, E)$ is elliptic if $\sigma_m^P(x, \xi)$ is invertible for

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cardona-qft-methods_ F00179	20	■ 1.000	$\sigma_m^P(x, \xi)$	$\sigma_m^P(x, \xi)$	Definition 2.4. $P \in \Psi DO^m(M, E)$ is elliptic if $\sigma_m^P(x, \xi)$ is invertible for
cardona-qft-methods_ F00180	20	■ 1.000	$0 \neq \xi \in TM_x^*$	$0 \neq \xi \in TM_x^*$	all $0 \neq \xi \in TM_x^*$.
cardona-qft-methods_ F00181	21	■ 1.000	$ \sigma^P(x, \xi) \geq c_1(x) \xi ^m$	$ \sigma^P(x, \xi) \geq c_1(x) \xi ^m$	This means that $ \sigma^P(x, \xi) \geq c_1(x) \xi ^m$ for $ \xi \geq c_2(x)$, $x \in U$ where c_1, c_2
cardona-qft-methods_ F00182	21	■ 1.000	$ \xi \geq c_2(x), x \in U$	$ \xi \geq c_2(x), x \in U$	This means that $ \sigma^P(x, \xi) \geq c_1(x) \xi ^m$ for $ \xi \geq c_2(x)$, $x \in U$ where c_1, c_2
cardona-qft-methods_ F00183	21	■ 1.000	c_1, c_2	c_1, c_2	This means that $ \sigma^P(x, \xi) \geq c_1(x) \xi ^m$ for $ \xi \geq c_2(x)$, $x \in U$ where c_1, c_2
cardona-qft-methods_ F00184	21	■ 0.967	$Op(\sigma) \in \Psi DO^m(U)$	$Op(\sigma) \in \Psi DO^m(U)$	(i) $Op(\sigma) \in \Psi DO^m(U)$ is elliptic.
cardona-qft-methods_ F00185	21	■ 1.000	$\sigma' \in S^{-m}(U \times \mathbb{R}^d)$	$\sigma' \in S^{-m}(U \times \mathbb{R}^d)$	(ii) There exist $\sigma' \in S^{-m}(U \times \mathbb{R}^d)$ such that $\sigma \circ \sigma' = 1$ or $\sigma' \circ \sigma = 1$.
cardona-qft-methods_ F00186	21	■ 1.000	$\sigma \circ \sigma' = 1$	$\sigma \circ \sigma' = 1$	(ii) There exist $\sigma' \in S^{-m}(U \times \mathbb{R}^d)$ such that $\sigma \circ \sigma' = 1$ or $\sigma' \circ \sigma = 1$.
cardona-qft-methods_ F00187	21	■ 1.000	$\sigma' \circ \sigma = 1$	$\sigma' \circ \sigma = 1$	(ii) There exist $\sigma' \in S^{-m}(U \times \mathbb{R}^d)$ such that $\sigma \circ \sigma' = 1$ or $\sigma' \circ \sigma = 1$.
cardona-qft-methods_ F00188	21	■ 0.998	$Op(\sigma) Op(\sigma') = Op(\sigma') Op(\sigma) = 1$	$Op(\sigma) Op(\sigma') = Op(\sigma') Op(\sigma) = 1$	(iii) $Op(\sigma) Op(\sigma') = Op(\sigma') Op(\sigma) = 1$ modulo $\Psi DO^{-\infty}(U)$.
cardona-qft-methods_ F00189	21	■ 1.000	$Op(\sigma') \in \Psi DO^{-m}(U)$	$Op(\sigma') \in \Psi DO^{-m}(U)$	Thus $Op(\sigma') \in \Psi DO^{-m}(U)$ is also elliptic.
cardona-qft-methods_ F00190	21	■ 1.000	$P \in \Psi DO(M, E)$	$P \in \Psi DO(M, E)$	Remark that any $P \in \Psi DO(M, E)$ can be extended to a bounded operator
cardona-qft-methods_ F00191	21	■ 1.000	$L^2(M, E)$	$L^2(M, E)$	on $L^2(M, E)$ when $\Re(m) \leq 0$ but this needs an existing scalar product for
cardona-qft-methods_ F00192	21	■ 1.000	$\Re(m) \leq 0$	$\Re(m) \leq 0$	on $L^2(M, E)$ when $\Re(m) \leq 0$ but this needs an existing scalar product for
cardona-qft-methods_ F00193	21	■ 1.000	$P \in \Psi DO^{-m}(M, E)$	$P \in \Psi DO^{-m}(M, E)$	Theorem 2.2. (see ⁴⁹) When $P \in \Psi DO^{-m}(M, E)$ is elliptic with $\Re(m) >$
cardona-qft-methods_ F00194	21	■ 1.000	$\Re(m) >$	$\Re(m) >$	Theorem 2.2. (see ⁴⁹) When $P \in \Psi DO^{-m}(M, E)$ is elliptic with $\Re(m) >$
cardona-qft-methods_ F00195	21	■ 1.000	r	r	bundle of rank r over M . If $P \in \Psi DO^{-m}(M, E)$, then for any couple of
cardona-qft-methods_ F00196	21	■ 1.000	$s \in \Gamma^\infty(M, E), t^* \in \Gamma^\infty(M, E^*)$	$s \in \Gamma^\infty(M, E), t^* \in \Gamma^\infty(M, E^*)$	sections $s \in \Gamma^\infty(M, E)$, $t^* \in \Gamma^\infty(M, E^*)$, the operator $f \in C^\infty(M) \rightarrow$
cardona-qft-methods_ F00197	21	■ 1.000	$f \in C^\infty(M) \rightarrow$	$f \in C^\infty(M) \rightarrow$	sections $s \in \Gamma^\infty(M, E)$, $t^* \in \Gamma^\infty(M, E^*)$, the operator $f \in C^\infty(M) \rightarrow$
cardona-qft-methods_ F00198	21	■ 1.000	$\langle t^*, P(fs) \rangle \in C^\infty(M)$	$\langle t^*, P(fs) \rangle \in C^\infty(M)$	$\langle t^*, P(fs) \rangle \in C^\infty(M)$ is in $\Psi DO^m(M)$. This means that in a local chart
cardona-qft-methods_ F00199	21	■ 1.000	(U, ϕ)	(U, ϕ)	(U, ϕ) , these operators are $r \times r$ matrices of pseudodifferential operators
cardona-qft-methods_ F00200	21	■ 1.000	$r \times r$	$r \times r$	(U, ϕ) , these operators are $r \times r$ matrices of pseudodifferential operators
cardona-qft-methods_ F00201	21	■ 1.000	$-m$	$-m$	of order $-m$. The total symbol is in $C^\infty(T^*U) \otimes End(E)$ with $End(E) \simeq$
cardona-qft-methods_ F00202	21	■ 1.000	$C^\infty(T^*U) \otimes End(E)$	$C^\infty(T^*U) \otimes End(E)$	of order $-m$. The total symbol is in $C^\infty(T^*U) \otimes End(E)$ with $End(E) \simeq$
cardona-qft-methods_ F00203	21	■ 1.000	$End(E) \simeq$	$End(E) \simeq$	of order $-m$. The total symbol is in $C^\infty(T^*U) \otimes End(E)$ with $End(E) \simeq$
cardona-qft-methods_ F00204	21	■ 1.000	$M_r(\mathbb{C})$	$M_r(\mathbb{C})$	$M_r(\mathbb{C})$. The principal symbol can be globally defined: $\sigma_{-m}^P(x, \xi) : E_x \rightarrow$

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cardona-qft-methods_ F00205	21	■ 1.000	$\backslash\mathrm{sigma}_{\{-m\}}^{\{P\}}(x,\xi):E_{\{x\}}\rightarrow$	$\sigma_{-m}^P(x,\xi):E_x\rightarrow$	$M_r(\mathbb{C})$. The principal symbol can be globally defined: $\sigma_{-m}^P(x,\xi):E_x\rightarrow$
cardona-qft-methods_ F00206	21	■ 1.000	$E_{\{x\}}$	E_x	E_x for $x\in M$ and $\xi\in T_x^*M$, can be seen as a smooth homomorphism
cardona-qft-methods_ F00207	21	■ 1.000	$\xi\in T_{\{x\}}^*M$	$\xi\in T_x^*M$	E_x for $x\in M$ and $\xi\in T_x^*M$, can be seen as a smooth homomorphism
cardona-qft-methods_ F00208	21	■ 1.000	$T^{\{*\}}M$	T^*M	homogeneous of degree $-m$ on all fibers of T^*M . We get the simple formula
cardona-qft-methods_ F00209	21	■ 1.000	$h\in C^{\{\infty\}}(M)$	$h\in C^\infty(M)$	where $h\in C^\infty(M)$ is such that $d_xh=\xi$.
cardona-qft-methods_ F00210	21	■ 1.000	$d_{\{x\}}h=\xi$	$d_xh=\xi$	where $h\in C^\infty(M)$ is such that $d_xh=\xi$.
cardona-qft-methods_ F00211	21	■ 1.000	$\backslash\mathrm{mathbb{R}}^{\{d\}}\backslash\{0\}$	$\mathbb{R}^d\backslash\{0\}$	an extension on $\mathbb{R}^d$ of a given homogeneous symbol on $\mathbb{R}^d\backslash\{0\}$. Such ex-
cardona-qft-methods_ F00212	21	■ 1.000	$\backslash\mathrm{mathcal{S}}$	$\mathcal{S}$	The Schwartz space on $\mathbb{R}^d$ is denoted by $\mathcal{S}$ and the space of tempered
cardona-qft-methods_ F00213	21	■ 1.000	$\backslash\mathrm{mathcal{S}}^{\{\prime\}}$	$\mathcal{S}'$	distributions by $\mathcal{S}'$.
cardona-qft-methods_ F00214	21	■ 1.000	$f_{\{\lambda\}}(\xi)=f(\lambda\xi),\lambda\in\backslash\mathrm{mathbb{R}}_{\{+\}}^{\{*\}}$	$f_\lambda(\xi)=f(\lambda\xi),\lambda\in\mathbb{R}_+^*$	Definition 2.5. For $f_\lambda(\xi)=f(\lambda\xi)$, $\lambda\in\mathbb{R}_+^*$, define $\tau\in\mathcal{S}'\rightarrow\tau_\lambda$ by
cardona-qft-methods_ F00215	21	■ 1.000	$\tau\in\backslash\mathrm{mathcal{S}}^{\{\prime\}}\rightarrow\tau_{\{\lambda\}}$	$\tau\in\mathcal{S}'\rightarrow\tau_\lambda$	Definition 2.5. For $f_\lambda(\xi)=f(\lambda\xi)$, $\lambda\in\mathbb{R}_+^*$, define $\tau\in\mathcal{S}'\rightarrow\tau_\lambda$ by
cardona-qft-methods_ F00216	21	■ 0.974	$\backslash\mathrm{left}\backslash\mathrm{angle}\tau_{\{\lambda\}},f\backslash\mathrm{right}\backslash\mathrm{rangle}=\lambda^{\{-d\}}\backslash\mathrm{left}\backslash\mathrm{angle}\tau,f_{\{\lambda\}^{-1}}\backslash\mathrm{right}\backslash\mathrm{rangle}$	$\langle\tau_\lambda,f\rangle=\lambda^{-d}\langle\tau,f_{\lambda^{-1}}\rangle$	$\langle\tau_\lambda,f\rangle=\lambda^{-d}\langle\tau,f_{\lambda^{-1}}\rangle$ for all $f\in\mathcal{S}$. A distribution $\tau\in\mathcal{S}'$ is homogeneous
cardona-qft-methods_ F00217	21	■ 0.974	$f\in\backslash\mathrm{mathcal{S}}$	$f\in\mathcal{S}$	$\langle\tau_\lambda,f\rangle=\lambda^{-d}\langle\tau,f_{\lambda^{-1}}\rangle$ for all $f\in\mathcal{S}$. A distribution $\tau\in\mathcal{S}'$ is homogeneous
cardona-qft-methods_ F00218	21	■ 0.974	$\tau\in\backslash\mathrm{mathcal{S}}^{\{\prime\}}$	$\tau\in\mathcal{S}'$	$\langle\tau_\lambda,f\rangle=\lambda^{-d}\langle\tau,f_{\lambda^{-1}}\rangle$ for all $f\in\mathcal{S}$. A distribution $\tau\in\mathcal{S}'$ is homogeneous
cardona-qft-methods_ F00219	21	■ 1.000	$\tau_{\{\lambda\}}=\lambda^{\{m\}}\tau$	$\tau_\lambda=\lambda^m\tau$	of order $m\in\mathbb{C}$ when $\tau_\lambda=\lambda^m\tau$.
cardona-qft-methods_ F00220	22	■ 1.000	$\backslash\mathrm{sigma}\in C^{\{\infty\}}\backslash\mathrm{left}(\backslash\mathrm{mathbb{R}}^{\{d\}}\backslash\{0\}\backslash\mathrm{right})$	$\sigma\in C^\infty(\mathbb{R}^d\backslash\{0\})$	Prop 2.1. Let $\sigma\in C^\infty(\mathbb{R}^d\backslash\{0\})$ be a homogeneous symbol of order $k\in\mathbb{Z}$.
cardona-qft-methods_ F00221	22	■ 1.000	$k\in\backslash\mathrm{mathbb{Z}}$	$k\in\mathbb{Z}$	Prop 2.1. Let $\sigma\in C^\infty(\mathbb{R}^d\backslash\{0\})$ be a homogeneous symbol of order $k\in\mathbb{Z}$.
cardona-qft-methods_ F00222	22	■ 1.000	$k>-d$	$k>-d$	(i) If $k>-d$, then σ defines a homogeneous distribution.
cardona-qft-methods_ F00223	22	■ 1.000	$k=-d$	$k=-d$	(ii) If $k=-d$, there is a unique obstruction to an extension of σ given by
cardona-qft-methods_ F00224	22	■ 1.000	$c_{\{\mathrm{sigma}\}}=\int\backslash\mathrm{mathbb{S}}^{\{-d-1\}}\backslash\mathrm{sigma}(\xi)d\xi$	$c_\sigma=\int_{\mathbb{S}^{d-1}}\sigma(\xi)d\xi$	$c_\sigma=\int_{\mathbb{S}^{d-1}}\sigma(\xi)d\xi$, namely, one can at best extend σ in $\tau\in\mathcal{S}'$ such that
cardona-qft-methods_ F00225	22	■ 1.000	$\tau_{\{\lambda\}}=\lambda^{\{-d\}}\backslash\mathrm{left}(\tau+c_{\{\mathrm{sigma}\}}\log(\lambda)\delta_0\backslash\mathrm{right})$	$\tau_\lambda=\lambda^{-d}(\tau+c_\sigma\log(\lambda)\delta_0)$	$\tau_\lambda=\lambda^{-d}(\tau+c_\sigma\log(\lambda)\delta_0)$.
cardona-qft-methods_ F00226	22	■ 0.807	$\backslash\mathrm{operatorname{kernel}}k^{\{P\}}$	kernel k^P	of the kernel k^P for $P\in\Psi DO$. For any $\tau\in\mathcal{S}'$, we choose the decomposition
cardona-qft-methods_ F00227	22	■ 0.807	$P\in\backslash\mathrm{Psi}DO$	$P\in\Psi DO$	of the kernel k^P for $P\in\Psi DO$. For any $\tau\in\mathcal{S}'$, we choose the decomposition
cardona-qft-methods_ F00228	22	■ 0.954	$\tau=\phi\circ\tau+(1-\phi)\circ\tau$	$\tau=\phi\circ\tau+(1-\phi)\circ\tau$	as $\tau=\phi\circ\tau+(1-\phi)\circ\tau$ where $\phi\in C_c^\infty(\mathbb{R}^d)$ and $\phi=1$ near 0. We can look at
cardona-qft-methods_ F00229	22	■ 0.954	$\phi=1$	$\phi=1$	as $\tau=\phi\circ\tau+(1-\phi)\circ\tau$ where $\phi\in C_c^\infty(\mathbb{R}^d)$ and $\phi=1$ near 0. We can look at
cardona-qft-methods_ F00230	22	■ 1.000	τ	τ	the infrared behavior of τ near the origin and its ultraviolet behavior near
cardona-qft-methods_ F00231	22	■ 0.919	$\phi\circ\tau$	$\phi\circ\tau$	infinity. Remark first that, since $\phi\circ\tau$ has a compact support, $(\phi\circ\tau)^\sim\in\mathcal{S}'$,

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cardona-qft-methods_ F00232	22	■ 0.919	$(\phi \circ \tau)^{\vee} \in \mathcal{S}'$	$(\phi \circ \tau)^{\vee} \in \mathcal{S}'$	infinity. Remark first that, since $\phi \circ \tau$ has a compact support, $(\phi \circ \tau)^{\vee} \in \mathcal{S}'$,
cardona-qft-methods_ F00233	22	■ 0.529	$\tau^{\vee}$	$\tau^{\vee}$	so the regularity of $\tau^{\vee}$ depends only of its ultraviolet part $((1 - \phi) \circ \tau)^{\vee}$.
cardona-qft-methods_ F00234	22	■ 0.529	$((1 - \phi) \circ \tau)^{\vee}$	$((1 - \phi) \circ \tau)^{\vee}$	so the regularity of $\tau^{\vee}$ depends only of its ultraviolet part $((1 - \phi) \circ \tau)^{\vee}$.
cardona-qft-methods_ F00235	22	■ 0.979	$P \in \Psi D^0_m(U), m \in \mathbb{Z}$	$P \in \Psi DO^m(U), m \in \mathbb{Z}$	Prop 2.2. Let $P \in \Psi DO^m(U)$, $m \in \mathbb{Z}$. Then, in local form near the
cardona-qft-methods_ F00236	22	■ 1.000	$a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$	$a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$	where $a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$ is homogeneous of order j in y and
cardona-qft-methods_ F00237	22	■ 1.000	j	j	where $a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$ is homogeneous of order j in y and
cardona-qft-methods_ F00238	22	■ 1.000	y	y	where $a_j(x, y) \in C^\infty(U \times U \setminus \{x\})$ is homogeneous of order j in y and
cardona-qft-methods_ F00239	22	■ 1.000	$c_P(x) \in C^\infty(U)$	$c_P(x) \in C^\infty(U)$	$c_P(x) \in C^\infty(U)$ with
cardona-qft-methods_ F00240	22	■ 1.000	$P \in \Psi D^0_m(M, E), m \in \mathbb{Z}$	$P \in \Psi DO^m(M, E), m \in \mathbb{Z}$	Theorem 2.3. Let $P \in \Psi DO^m(M, E)$, $m \in \mathbb{Z}$. Then, for any trivializing
cardona-qft-methods_ F00241	22	■ 1.000	a_j	a_j	where a_j is homogeneous of degree j in y , c_P is intrinsically locally defined
cardona-qft-methods_ F00242	22	■ 1.000	y, c_P	y, c_P	where a_j is homogeneous of degree j in y , c_P is intrinsically locally defined
cardona-qft-methods_ F00243	22	■ 0.990	$c_P(x) dx $	$c_P(x) dx $	Moreover, $c_P(x) dx $ is a 1-density over M which is functorial for the dif-
cardona-qft-methods_ F00244	22	■ 1.000	c_P	c_P	The main difficulty is of course to prove that c_P is well defined.
cardona-qft-methods_ F00245	23	■ 1.000	$\int_M c_P(x) dx $	$\int_M c_P(x) dx $	The claim is that $\int_M c_P(x) dx $ is a residue. For this, we embed everything
cardona-qft-methods_ F00246	23	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	in $\mathbb{C}$. In the same spirit as in Proposition 2.1, one obtains the following
cardona-qft-methods_ F00247	23	■ 1.000	$m \in \mathbb{C} \setminus \mathbb{Z}$	$m \in \mathbb{C} \setminus \mathbb{Z}$	$m \in \mathbb{C} \setminus \mathbb{Z}$ can be uniquely extended to a homogeneous distribution.
cardona-qft-methods_ F00248	23	■ 1.000	Ω	Ω	Definition 2.6. Let U be an open set in $\mathbb{R}^d$ and Ω be a domain in $\mathbb{C}$.
cardona-qft-methods_ F00249	23	■ 1.000	$\sigma: \Omega \rightarrow S^m \left(U \times \mathbb{R}^d \right)$	$\sigma: \Omega \rightarrow S^m \left(U \times \mathbb{R}^d \right)$	A map $\sigma: \Omega \rightarrow S^m(U \times \mathbb{R}^d)$ is said to be holomorphic when
cardona-qft-methods_ F00250	23	■ 1.000	$z \in \Omega \rightarrow \sigma(z)(x, \xi)$	$z \in \Omega \rightarrow \sigma(z)(x, \xi)$	the map $z \in \Omega \rightarrow \sigma(z)(x, \xi)$ is analytic for all $x \in U$, $\xi \in \mathbb{R}^d$,
cardona-qft-methods_ F00251	23	■ 1.000	$x \in U, \xi \in \mathbb{R}^d$	$x \in U, \xi \in \mathbb{R}^d$	the map $z \in \Omega \rightarrow \sigma(z)(x, \xi)$ is analytic for all $x \in U$, $\xi \in \mathbb{R}^d$,
cardona-qft-methods_ F00252	23	■ 1.000	$m(z)$	$m(z)$	the order $m(z)$ of $\sigma(z)$ is analytic on Ω ,
cardona-qft-methods_ F00253	23	■ 1.000	$\sigma(z)$	$\sigma(z)$	the order $m(z)$ of $\sigma(z)$ is analytic on Ω ,
cardona-qft-methods_ F00254	23	■ 1.000	$\sigma(z) \simeq \sum_j \sigma_{m(z)-j}(z)$	$\sigma(z) \simeq \sum_j \sigma_{m(z)-j}(z)$	$\sigma(z) \simeq \sum_j \sigma_{m(z)-j}(z)$ are locally uniform in z .
cardona-qft-methods_ F00255	23	■ 1.000	z	z	$\sigma(z) \simeq \sum_j \sigma_{m(z)-j}(z)$ are locally uniform in z .
cardona-qft-methods_ F00256	23	■ 1.000	$z \rightarrow \sigma_{m(z)-j}(z)$	$z \rightarrow \sigma_{m(z)-j}(z)$	This hypothesis is sufficient to get that the map $z \rightarrow \sigma_{m(z)-j}(z)$ is holomor-
cardona-qft-methods_ F00257	23	■ 1.000	$C^\infty \left(U \times \mathbb{R}^d \setminus \{0\} \right)$	$C^\infty \left(U \times \mathbb{R}^d \setminus \{0\} \right)$	phic from Ω to $C^\infty(U \times \mathbb{R}^d \setminus \{0\})$ and the map $\partial_z \sigma(z)(x, \xi)$ is a classical

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cardona-qft-methods_ F00258	23	■ 1.000	$\partial_{\mathbf{z}}\sigma(\mathbf{z})(\mathbf{x}, \mathbf{\xi})$	$\partial_z\sigma(z)(x, \xi)$	phic from Ω to $C^\infty(U \times \mathbb{R}^d \setminus \{0\})$ and the map $\partial_z\sigma(z)(x, \xi)$ is a classical
cardona-qft-methods_ F00259	23	■ 1.000	$U \times \mathbb{R}^d$	$U \times \mathbb{R}^d$	symbol on $U \times \mathbb{R}^d$ such that $\partial_z\sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z\sigma_{m(z)-j}(z)(x, \xi)$.
cardona-qft-methods_ F00260	23	■ 1.000	$\partial_{\mathbf{z}}\sigma(\mathbf{z})(\mathbf{x}, \mathbf{\xi}) \simeq \sum_{j \geq 0} \partial_{\mathbf{z}}\sigma_{m(\mathbf{z})-j}(\mathbf{z})(\mathbf{x}, \mathbf{\xi})$	$\partial_z\sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z\sigma_{m(z)-j}(z)(x, \xi)$	symbol on $U \times \mathbb{R}^d$ such that $\partial_z\sigma(z)(x, \xi) \simeq \sum_{j \geq 0} \partial_z\sigma_{m(z)-j}(z)(x, \xi)$.
cardona-qft-methods_ F00261	23	■ 1.000	$P: \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$	$P: \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$	Definition 2.7. The map $P: \Omega \subset \mathbb{C} \rightarrow \Psi DO(U)$ is said to be holomorphic
cardona-qft-methods_ F00262	23	■ 1.000	$P(\mathbf{z})=\sigma(\mathbf{z})(\cdot, D)+R(\mathbf{z})$	$P(z) = \sigma(z)(\cdot, D) + R(z)$	if it has the decomposition $P(z) = \sigma(z)(\cdot, D) + R(z)$ (see definition (1))
cardona-qft-methods_ F00263	23	■ 1.000	$\sigma: \Omega \rightarrow S(U \times \mathbb{R}^d)$	$\sigma: \Omega \rightarrow S(U \times \mathbb{R}^d)$	where $\sigma: \Omega \rightarrow S(U \times \mathbb{R}^d)$ and $R: \Omega \rightarrow C^\infty(U \times U)$ are holomorphic.
cardona-qft-methods_ F00264	23	■ 1.000	$R: \Omega \rightarrow C^\infty(U \times U)$	$R: \Omega \rightarrow C^\infty(U \times U)$	where $\sigma: \Omega \rightarrow S(U \times \mathbb{R}^d)$ and $R: \Omega \rightarrow C^\infty(U \times U)$ are holomorphic.
cardona-qft-methods_ F00265	23	■ 1.000	$\Psi DO(M, E)$	$\Psi DO(M, E)$	$\Psi DO(M, E)$ with a holomorphic product (when M is compact).
cardona-qft-methods_ F00266	23	■ 1.000	$P \in \Psi DO^m(U), m \in \mathbb{C}$	$P \in \Psi DO^m(U), m \in \mathbb{C}$	<i>Elliptic operators:</i> Recall that $P \in \Psi DO^m(U)$, $m \in \mathbb{C}$, is elliptic essen-
cardona-qft-methods_ F00267	23	■ 1.000	$Q \in \Psi DO^m(M, E)$	$Q \in \Psi DO^m(M, E)$	Let $Q \in \Psi DO^m(M, E)$ with $\Re(m) > 0$. We assume that M is compact
cardona-qft-methods_ F00268	23	■ 1.000	$\Re(m)>0$	$\Re(m) > 0$	Let $Q \in \Psi DO^m(M, E)$ with $\Re(m) > 0$. We assume that M is compact
cardona-qft-methods_ F00269	23	■ 0.994	Q	Q	and Q is elliptic. Thus Q has a discrete spectrum. We assume $\text{Spectrum}(Q) \cap$
cardona-qft-methods_ F00270	23	■ 0.994	$\operatorname{Spectrum}(Q) \cap$	$\text{Spectrum}(Q) \cap$	and Q is elliptic. Thus Q has a discrete spectrum. We assume $\text{Spectrum}(Q) \cap$
cardona-qft-methods_ F00271	23	■ 1.000	$\mathbb{R}^- = \emptyset$	$\mathbb{R}^- = \emptyset$	$\mathbb{R}^- = \emptyset$ and the existence of a curve Γ whose interior contains the spectrum
cardona-qft-methods_ F00272	23	■ 1.000	Γ	Γ	$\mathbb{R}^- = \emptyset$ and the existence of a curve Γ whose interior contains the spectrum
cardona-qft-methods_ F00273	23	■ 1.000	λ^z	λ^z	and avoid branch points of λ^z at $z = 0$:
cardona-qft-methods_ F00274	23	■ 1.000	$z=0$	$z = 0$	and avoid branch points of λ^z at $z = 0$:
cardona-qft-methods_ F00275	24	■ 1.000	$\Re(s)<0, Q^s=\frac{1}{i2\pi} \int_{\Gamma} \lambda^s(\lambda-Q)^{-1} d\lambda$	$\Re(s) < 0, Q^s = \frac{1}{i2\pi} \int_{\Gamma} \lambda^s(\lambda - Q)^{-1} d\lambda$	When $\Re(s) < 0$, $Q^s = \frac{1}{i2\pi} \int_{\Gamma} \lambda^s(\lambda - Q)^{-1} d\lambda$ makes sense as operator on
cardona-qft-methods_ F00276	24	■ 1.000	$s \rightarrow Q^s$	$s \rightarrow Q^s$	The map $s \rightarrow Q^s$ is a one-parameter group containing Q^0 and Q^1 which
cardona-qft-methods_ F00277	24	■ 1.000	$Q^{\{0\}}$	Q^0	The map $s \rightarrow Q^s$ is a one-parameter group containing Q^0 and Q^1 which
cardona-qft-methods_ F00278	24	■ 1.000	$Q^{\{1\}}$	Q^1	The map $s \rightarrow Q^s$ is a one-parameter group containing Q^0 and Q^1 which
cardona-qft-methods_ F00279	24	■ 1.000	$\Re(s) \leq 0$	$\Re(s) \leq 0$	is holomorphic on $\Re(s) \leq 0$. We want to integrate symbols, so we will need
cardona-qft-methods_ F00280	24	■ 0.997	S_{int}	S_{int}	the set S_{int} of integrable symbols. Using same type of arguments as in
cardona-qft-methods_ F00281	24	■ 1.000	$L: \sigma \in S_{\text{int}}^{\mathbb{Z}}(\mathbb{R}^d) \rightarrow L(\sigma) = \check{\sigma}(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$	$L: \sigma \in S_{\text{int}}^{\mathbb{Z}}(\mathbb{R}^d) \rightarrow L(\sigma) = \check{\sigma}(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$	Prop 2.3. Let $L: \sigma \in S_{\text{int}}^{\mathbb{Z}}(\mathbb{R}^d) \rightarrow L(\sigma) = \check{\sigma}(0) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \sigma(\xi) d\xi$.
cardona-qft-methods_ F00282	24	■ 0.223	L	L	Then L has a unique holomorphic extension $\widetilde{L}$ on $S^{\mathbb{C} \setminus \mathbb{Z}}(\mathbb{R}^d)$. Moreover,
cardona-qft-methods_ F00283	24	■ 0.223	$\widetilde{L}$	$\widetilde{L}$	Then L has a unique holomorphic extension $\widetilde{L}$ on $S^{\mathbb{C} \setminus \mathbb{Z}}(\mathbb{R}^d)$. Moreover,

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cardona-qft-methods_ F00284	24	■ 0.223	$\left.S^{\mathbb{C}}(\mathbb{R}^d)\right ^{\mathbb{Z}}$	$S^{\mathbb{C}}(\mathbb{R}^d)$	Then L has a unique holomorphic extension $\tilde{L}$ on $S^{\mathbb{C}\backslash\mathbb{Z}}(\mathbb{R}^d)$. Moreover,
cardona-qft-methods_ F00285	24	■ 0.748	$\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \backslash \mathbb{Z}, \tilde{L}(\sigma) = \left(\sigma - \sum_{j \leq N} \tau_{m-j}\right)^{\sim}(0) =$	$\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \backslash \mathbb{Z}, \tilde{L}(\sigma) = \left(\sigma - \sum_{j \leq N} \tau_{m-j}\right)^{\sim}(0) =$	when $\sigma(\xi) \simeq \sum_j \sigma_{m-j}(\xi), m \in \mathbb{C} \backslash \mathbb{Z}, \tilde{L}(\sigma) = (\sigma - \sum_{j \leq N} \tau_{m-j})^{\sim}(0) =$
cardona-qft-methods_ F00286	24	■ 1.000	$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left(\sigma - \sum_{j \leq N} \tau_{m-j}\right)(\xi) d\xi$	$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left(\sigma - \sum_{j \leq N} \tau_{m-j}\right)(\xi) d\xi$	$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (\sigma - \sum_{j \leq N} \tau_{m-j})(\xi) d\xi$ where m is the order of σ , N is an integer
cardona-qft-methods_ F00287	24	■ 1.000	σ, N	σ, N	$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (\sigma - \sum_{j \leq N} \tau_{m-j})(\xi) d\xi$ where m is the order of σ , N is an integer
cardona-qft-methods_ F00288	24	■ 1.000	$N > \Re(m) + d$	$N > \Re(m) + d$	with $N > \Re(m) + d$ and τ_{m-j} is the extension of σ_{m-j} of Lemma 2.2.
cardona-qft-methods_ F00289	24	■ 1.000	τ_{m-j}	τ_{m-j}	with $N > \Re(m) + d$ and τ_{m-j} is the extension of σ_{m-j} of Lemma 2.2.
cardona-qft-methods_ F00290	24	■ 1.000	σ_{m-j}	σ_{m-j}	with $N > \Re(m) + d$ and τ_{m-j} is the extension of σ_{m-j} of Lemma 2.2.
cardona-qft-methods_ F00291	24	■ 0.976	$\sigma : \mathbb{C} \rightarrow S(\mathbb{R}^d)$	$\sigma : \mathbb{C} \rightarrow S(\mathbb{R}^d)$	Corollary 2.1. If $\sigma : \mathbb{C} \rightarrow S(\mathbb{R}^d)$ is holomorphic and $\text{order}(\sigma(s)) = s$,
cardona-qft-methods_ F00292	24	■ 0.976	$(\sigma(s)) = s$	$(\sigma(s)) = s$	Corollary 2.1. If $\sigma : \mathbb{C} \rightarrow S(\mathbb{R}^d)$ is holomorphic and $\text{order}(\sigma(s)) = s$,
cardona-qft-methods_ F00293	24	■ 1.000	$\tilde{L}(\sigma(s))$	$\tilde{L}(\sigma(s))$	then $\tilde{L}(\sigma(s))$ is meromorphic with simple poles on $\mathbb{Z}$ and for $p \in \mathbb{Z}$,
cardona-qft-methods_ F00294	24	■ 1.000	$\mathbb{Z}$	$\mathbb{Z}$	then $\tilde{L}(\sigma(s))$ is meromorphic with simple poles on $\mathbb{Z}$ and for $p \in \mathbb{Z}$,
cardona-qft-methods_ F00295	24	■ 1.000	$p \in \mathbb{Z}$	$p \in \mathbb{Z}$	then $\tilde{L}(\sigma(s))$ is meromorphic with simple poles on $\mathbb{Z}$ and for $p \in \mathbb{Z}$,
cardona-qft-methods_ F00296	24	■ 0.997	$\text{Res}_{s=p} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \sigma_{-d}(p)(\xi) d\xi$	$\text{Res}_{s=p} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \sigma_{-d}(p)(\xi) d\xi$	$\text{Res}_{s=p} \tilde{L}(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \sigma_{-d}(p)(\xi) d\xi$.
cardona-qft-methods_ F00297	24	■ 1.000	$\{ \}^{109,110}$	109,110	Wodzicki. ^{109,110}
cardona-qft-methods_ F00298	24	■ 1.000	$\mathcal{D} \in \Psi D O(M, E)$	$\mathcal{D} \in \Psi D O(M, E)$	Definition 2.8. Let $\mathcal{D} \in \Psi D O(M, E)$ be an elliptic pseudodifferential
cardona-qft-methods_ F00299	24	■ 0.935	$\Psi D O_{\text{int}}(M, E) = \{Q \in \Psi D O^{\mathbb{C}}(M, E) \mid \Re(\text{order}(Q)) < -d\}$	$\Psi D O_{\text{int}}(M, E) = \{Q \in \Psi D O^{\mathbb{C}}(M, E) \mid \Re(\text{order}(Q)) < -d\}$	Let $\Psi D O_{\text{int}}(M, E) = \{Q \in \Psi D O^{\mathbb{C}}(M, E) \mid \Re(\text{order}(Q)) < -d\}$ be
cardona-qft-methods_ F00300	24	■ 1.000	$P \in \Psi D O_{\text{int}}(M, E)$	$P \in \Psi D O_{\text{int}}(M, E)$	In particular, if $P \in \Psi D O_{\text{int}}(M, E)$, then its kernel $k^P(x, x)$ is a smooth
cardona-qft-methods_ F00301	24	■ 1.000	$k^P(x, x)$	$k^P(x, x)$	In particular, if $P \in \Psi D O_{\text{int}}(M, E)$, then its kernel $k^P(x, x)$ is a smooth
cardona-qft-methods_ F00302	24	■ 1.000	$M \times M$	$M \times M$	density on the diagonal of $M \times M$ with values in $\text{End}(E)$.
cardona-qft-methods_ F00303	24	■ 1.000	$\text{End}(E)$	$\text{End}(E)$	density on the diagonal of $M \times M$ with values in $\text{End}(E)$.
cardona-qft-methods_ F00304	24	■ 1.000	$P \in \Psi D O^{\mathbb{Z}}(M, E)$	$P \in \Psi D O^{\mathbb{Z}}(M, E)$	For $P \in \Psi D O^{\mathbb{Z}}(M, E)$, define
cardona-qft-methods_ F00305	24	■ 1.000	$P \in \Psi D O_{\text{int}}(M, E) \rightarrow \text{Tr}(P) \in \mathbb{C}$	$P \in \Psi D O_{\text{int}}(M, E) \rightarrow \text{Tr}(P) \in \mathbb{C}$	Theorem 2.4. (i) The map $P \in \Psi D O_{\text{int}}(M, E) \rightarrow \text{Tr}(P) \in \mathbb{C}$ has a
cardona-qft-methods_ F00306	24	■ 1.000	$\Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$	$\Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$	unique analytic extension on $\Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$.
cardona-qft-methods_ F00307	24	■ 1.000	$s \in \mathbb{C} \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$	$s \in \mathbb{C} \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$	(ii) If $P \in \Psi D O^{\mathbb{Z}}(M, E)$, the map: $s \in \mathbb{C} \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$ has at most

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cardona-qft-methods_F00308	25	■ 1.000	$c_{\{P\}}(x)=$	$c_P(x) =$	<i>is independent of $\mathcal{D}$. Recall (see Theorem 2.3) that $c_P(x) =$</i>
cardona-qft-methods_F00309	25	■ 1.000	$\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \text{tr}(\sigma_{-d}^P(x, \xi)) d\xi$	$\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \text{tr}(\sigma_{-d}^P(x, \xi)) d\xi$	$\frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} \text{tr}(\sigma_{-d}^P(x, \xi)) d\xi.$
cardona-qft-methods_F00310	25	■ 1.000	$\Psi D O^{\mathbb{Z}}(M, E)$	$\Psi D O^{\mathbb{Z}}(M, E)$	(iii) $W\text{Res}$ is a trace on the algebra $\Psi D O^{\mathbb{Z}}(M, E)$.
cardona-qft-methods_F00311	25	■ 1.000	$s \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$	$s \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$	Proof. (i) The map $s \rightarrow \text{Tr}(P \mathcal{D} ^{-s})$ is holomorphic on $\mathbb{C}$ and connect
cardona-qft-methods_F00312	25	■ 1.000	$P \in \Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$	$P \in \Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$	$P \in \Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$ to the set $\Psi D O_{int}(M, E)$ within $\Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$, so
cardona-qft-methods_F00313	25	■ 1.000	$\Psi D O_{\text{int}}(M, E)$	$\Psi D O_{int}(M, E)$	$P \in \Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$ to the set $\Psi D O_{int}(M, E)$ within $\Psi D O^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$, so
cardona-qft-methods_F00314	25	■ 1.000	$s=0$	$s = 0$	(2) First, fix $s = 0$. We are interested in the function $L_\phi(\sigma) =$
cardona-qft-methods_F00315	25	■ 1.000	$L_\phi(\sigma)=$	$L_\phi(\sigma) =$	(2) First, fix $s = 0$. We are interested in the function $L_\phi(\sigma) =$
cardona-qft-methods_F00316	25	■ 0.999	$\text{Tr}(\phi \sigma(x, D))$	$\text{Tr}(\phi \sigma(x, D))$	$\text{Tr}(\phi \sigma(x, D))$ with $\sigma \in S_{int}(U \times \mathbb{R}^d)$ and $\phi \in C^\infty(U)$. For instance, if
cardona-qft-methods_F00317	25	■ 0.999	$\sigma \in S_{\text{int}}(U \times \mathbb{R}^d)$	$\sigma \in S_{\text{int}}(U \times \mathbb{R}^d)$	$\text{Tr}(\phi \sigma(x, D))$ with $\sigma \in S_{int}(U \times \mathbb{R}^d)$ and $\phi \in C^\infty(U)$. For instance, if
cardona-qft-methods_F00318	25	■ 0.999	$\phi \in C^\infty(U)$	$\phi \in C^\infty(U)$	$\text{Tr}(\phi \sigma(x, D))$ with $\sigma \in S_{int}(U \times \mathbb{R}^d)$ and $\phi \in C^\infty(U)$. For instance, if
cardona-qft-methods_F00319	25	■ 1.000	$P=\sigma(\cdot, D)$	$P = \sigma(\cdot, D)$	$P = \sigma(\cdot, D),$
cardona-qft-methods_F00320	25	■ 0.999	L_ϕ	L_ϕ	so one extends L_ϕ to $S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$ with Proposition 2.3 via $\tilde{L}_\phi(\sigma) =$
cardona-qft-methods_F00321	25	■ 0.999	$S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$	$S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$	so one extends L_ϕ to $S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$ with Proposition 2.3 via $\tilde{L}_\phi(\sigma) =$
cardona-qft-methods_F00322	25	■ 0.999	$\tilde{L}_\phi(\sigma)=$	$\tilde{L}_\phi(\sigma) =$	so one extends L_ϕ to $S^{\mathbb{C}\backslash\mathbb{Z}}(U \times \mathbb{R}^d)$ with Proposition 2.3 via $\tilde{L}_\phi(\sigma) =$
cardona-qft-methods_F00323	25	■ 0.998	$\int_U \phi(x) \tilde{L}_\phi(\sigma(x, \cdot)) dx $	$\int_U \phi(x) \tilde{L}_\phi(\sigma(x, \cdot)) dx $	$\int_U \phi(x) \tilde{L}_\phi(\sigma(x, \cdot)) dx .$
cardona-qft-methods_F00324	25	■ 1.000	$\sigma(x, \xi)=\sigma(s)(x, \xi)$	$\sigma(x, \xi) = \sigma(s)(x, \xi)$	(3) If now $\sigma(x, \xi) = \sigma(s)(x, \xi)$ depends holomorphically on s , we get
cardona-qft-methods_F00325	25	■ 1.000	s	s	(3) If now $\sigma(x, \xi) = \sigma(s)(x, \xi)$ depends holomorphically on s , we get
cardona-qft-methods_F00326	25	■ 1.000	x	x	uniform bounds in x , thus we get, via Lemma 2.2 applied to $\tilde{L}_\phi(\sigma(s)(x, \cdot))$
cardona-qft-methods_F00327	25	■ 1.000	$\tilde{L}_\phi(\sigma(s)(x, \cdot))$	$\tilde{L}_\phi(\sigma(s)(x, \cdot))$	uniform bounds in x , thus we get, via Lemma 2.2 applied to $\tilde{L}_\phi(\sigma(s)(x, \cdot))$
cardona-qft-methods_F00328	25	■ 1.000	$\tilde{L}_\phi(\sigma(s))$	$\tilde{L}_\phi(\sigma(s))$	uniformly in x , yielding a natural extension to $\tilde{L}_\phi(\sigma(s))$ which is holomor-
cardona-qft-methods_F00329	25	■ 1.000	$\mathbb{C}\backslash\mathbb{Z}$	$\mathbb{C}\backslash\mathbb{Z}$	phic on $\mathbb{C}\backslash\mathbb{Z}$.
cardona-qft-methods_F00330	25	■ 1.000	$\text{order}(\sigma(s))=s$	$\text{order}(\sigma(s)) = s$	When $\text{order}(\sigma(s)) = s$, the map $\tilde{L}_\phi(\sigma(s))$ has at most simple poles on $\mathbb{Z}$ and
cardona-qft-methods_F00331	25	■ 1.000	$p \in \mathbb{Z}, \text{Res}_{s=p} \tilde{L}_\phi(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(p)(x, \xi) d\xi dx =$	$p \in \mathbb{Z}, \text{Res}_{s=p} \tilde{L}_\phi(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(p)(x, \xi) d\xi dx =$	for each $p \in \mathbb{Z}$, $\text{Res}_{s=p} \tilde{L}_\phi(\sigma(s)) = -\frac{1}{(2\pi)^d} \int_U \int_{\mathbb{S}^{d-1}} \phi(x) \sigma_{-d}(p)(x, \xi) d\xi dx =$

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cardona-qft-methods_ F08332	25	■ 1.000	$\int_U \phi(x) c_{Pp}(x) dx $	$\int_U \phi(x) c_{Pp}(x) dx $	$-\int_U \phi(x) c_{Pp}(x) dx $ where we used (5) with $P = Op(\sigma_p(x, \xi))$.
cardona-qft-methods_ F08333	25	■ 1.000	$P=0 \ p\left(\sigma_p(x, \xi)\right)$	$P = Op(\sigma_p(x, \xi))$	$-\int_U \phi(x) c_{Pp}(x) dx $ where we used (5) with $P = Op(\sigma_p(x, \xi))$.
cardona-qft-methods_ F08334	25	■ 1.000	Tr	Tr	usual trace Tr on $\Psi DO^{\mathbb{Z}}(M, E)$ that we still denoted by Tr .
cardona-qft-methods_ F08335	25	■ 1.000	$P: \mathbb{C} \rightarrow \Psi D 0^{\mathbb{Z}}(M, E)$	$P: \mathbb{C} \rightarrow \Psi DO^{\mathbb{Z}}(M, E)$	When $P: \mathbb{C} \rightarrow \Psi DO^{\mathbb{Z}}(M, E)$ is meromorphic with $\operatorname{order}(P(s)) =$
cardona-qft-methods_ F08336	25	■ 1.000	$\operatorname{order}(P(s))=$	$\operatorname{order}(P(s)) =$	When $P: \mathbb{C} \rightarrow \Psi DO^{\mathbb{Z}}(M, E)$ is meromorphic with $\operatorname{order}(P(s)) =$
cardona-qft-methods_ F08337	25	■ 0.814	$\operatorname{Tr}(P(s))$	$\operatorname{Tr}(P(s))$	s , then $\operatorname{Tr}(P(s))$ has at most poles on $\mathbb{Z}$ and $\operatorname{Res}_{s=p} \operatorname{Tr}(P(s)) =$
cardona-qft-methods_ F08338	25	■ 0.814	$\operatorname{Res}_{s=p}^{\operatorname{Tr}}(P(s))=$	$\operatorname{Res}_{s=p}^{\operatorname{Tr}}(P(s)) =$	s , then $\operatorname{Tr}(P(s))$ has at most poles on $\mathbb{Z}$ and $\operatorname{Res}_{s=p} \operatorname{Tr}(P(s)) =$
cardona-qft-methods_ F08339	25	■ 0.987	$-\int_M c_{P(p)}(x) dx $	$-\int_M c_{P(p)}(x) dx $	$-\int_M c_{P(p)}(x) dx $ for $p \in \mathbb{Z}$. So we get the claim for the family $P(s) =$
cardona-qft-methods_ F08340	25	■ 0.987	$P(s)=$	$P(s) =$	$-\int_M c_{P(p)}(x) dx $ for $p \in \mathbb{Z}$. So we get the claim for the family $P(s) =$
cardona-qft-methods_ F08341	25	■ 1.000	$P \mathcal{D} ^{-s}$	$P \mathcal{D} ^{-s}$	$P \mathcal{D} ^{-s}$.
cardona-qft-methods_ F08342	25	■ 0.997	$P_1, P_2 \in \Psi D 0^{\mathbb{Z}}(M, E)$	$P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$	(iii) Let $P_1, P_2 \in \Psi DO^{\mathbb{Z}}(M, E)$. Since Tr is a trace on $\Psi DO^{\mathbb{C}\backslash\mathbb{Z}}(M, E)$,
cardona-qft-methods_ F08343	25	■ 0.956	(i), $\operatorname{Tr}(P_1 P_2 \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2 \mathcal{D} ^{-s} P_1)$	(i), $\operatorname{Tr}(P_1 P_2 \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2 \mathcal{D} ^{-s} P_1)$	we get by (i), $\operatorname{Tr}(P_1 P_2 \mathcal{D} ^{-s}) = \operatorname{Tr}(P_2 \mathcal{D} ^{-s} P_1)$. Moreover
cardona-qft-methods_ F08344	25	■ 1.000	$P(s)$	$P(s)$	the value of $P(s)$ at $s = 0$.
cardona-qft-methods_ F08345	25	—	$\square$	$\square$	—
cardona-qft-methods_ F08346	26	■ 0.852	$\Psi D 0^{-m}(M, E), m \in \mathbb{N}$	$\Psi DO^{-m}(M, E), m \in \mathbb{N}$	tiplication by a constant) on the algebra $\Psi DO^{-m}(M, E)$, $m \in \mathbb{N}$, when M
cardona-qft-methods_ F08347	26	■ 1.000	$d \geq 2$	$d \geq 2$	is connected and $d \geq 2$.
cardona-qft-methods_ F08348	26	■ 1.000	$d=1, T^*M$	$d = 1, T^*M$	$d = 1, T^*M$ is disconnected and they are two residues.
cardona-qft-methods_ F08349	26	■ 1.000	$\sigma_{-d}^P = 0$	$\sigma_{-d}^P = 0$	2) If $\sigma_{-d}^P = 0$, then $\sigma^P(x, \xi)$ is a finite sum of commutators of symbols:
cardona-qft-methods_ F08350	26	■ 1.000	$\sigma^P(x, \xi)$	$\sigma^P(x, \xi)$	2) If $\sigma_{-d}^P = 0$, then $\sigma^P(x, \xi)$ is a finite sum of commutators of symbols:
cardona-qft-methods_ F08351	26	■ 0.998	$\sigma^P \simeq \sum_j \sigma_{m-j}^P$	$\sigma^P \simeq \sum_j \sigma_{m-j}^P$	If $\sigma^P \simeq \sum_j \sigma_{m-j}^P$ with $m = \operatorname{order}(P)$, by Euler's theorem,
cardona-qft-methods_ F08352	26	■ 0.998	$m=\operatorname{order}(P)$	$m = \operatorname{order}(P)$	If $\sigma^P \simeq \sum_j \sigma_{m-j}^P$ with $m = \operatorname{order}(P)$, by Euler's theorem,
cardona-qft-methods_ F08353	26	■ 1.000	$\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$	$\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$	$\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$ (this is false for $m = j!$) and
cardona-qft-methods_ F08354	26	■ 1.000	$m=j!$	$m = j!$	$\sum_{k=1}^d \xi_k \partial_{\xi_k} \sigma_{m-j}^P = (m-j) \sigma_{m-j}^P$ (this is false for $m = j!$) and
cardona-qft-methods_ F08355	26	■ 1.000	$\sum_{k=1}^d [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$	$\sum_{k=1}^d [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$	$\sum_{k=1}^d [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$. So $\sigma^P =$

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cardona-qft-methods_ F00356	26	■ 1.000	$\sigma^P =$	$\sigma^P =$	$\sum_{k=1}^d [x^k, \xi_k \sigma_{m-j}^P] = i \sum_{k=1}^d \partial_{\xi_k} \xi_k \sigma_{m-j}^P = i(m-j+d) \sigma_{m-j}^P$. So $\sigma^P =$
cardona-qft-methods_ F00357	26	■ 1.000	$\sum_{k=1}^d \left[\xi_k \tau, x^k \right]$	$\sum_{k=1}^d [\xi_k \tau, x^k]$	$\sum_{k=1}^d [\xi_k \tau, x^k]$.
cardona-qft-methods_ F00358	26	■ 1.000	T	T	Let T be another trace on $\Psi DO^{\mathbb{Z}}(M, E)$. Then $T(P)$ depends only on
cardona-qft-methods_ F00359	26	■ 1.000	$T(P)$	$T(P)$	Let T be another trace on $\Psi DO^{\mathbb{Z}}(M, E)$. Then $T(P)$ depends only on
cardona-qft-methods_ F00360	26	■ 1.000	σ_{-d}^P	σ_{-d}^P	σ_{-d}^P because $T([\cdot, \cdot]) = 0$.
cardona-qft-methods_ F00361	26	■ 1.000	$T([\cdot, \cdot])=0$	$T([\cdot, \cdot]) = 0$	σ_{-d}^P because $T([\cdot, \cdot]) = 0$.
cardona-qft-methods_ F00362	26	■ 1.000	$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi = 0$	$\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi = 0$	3) We have $\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x, \xi) d \xi = 0$ if and only if σ_{-d}^P is sum of deriva-
cardona-qft-methods_ F00363	26	■ 1.000	$\Delta_{\mathbb{S}} f = 0 \iff$	$\Delta_{\mathbb{S}} f = 0 \iff$	on the sphere $\mathbb{S}^{d-1}$ and these are kernels of the Laplacian: $\Delta_{\mathbb{S}} f = 0 \iff$
cardona-qft-methods_ F00364	26	■ 0.987	$df = 0 \iff f = cst$	$df = 0 \iff f = cst$	$df = 0 \iff f = cst$. Thus $\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$ has a solution h on $\mathbb{S}^{d-1}$. If
cardona-qft-methods_ F00365	26	■ 0.987	$\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$	$\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$	$df = 0 \iff f = cst$. Thus $\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$ has a solution h on $\mathbb{S}^{d-1}$. If
cardona-qft-methods_ F00366	26	■ 0.987	h	h	$df = 0 \iff f = cst$. Thus $\Delta_{\mathbb{S}^{d-1}} h = \sigma_{-d \mathbb{S}^{d-1}}^P$ has a solution h on $\mathbb{S}^{d-1}$. If
cardona-qft-methods_ F00367	26	■ 1.000	$\tilde{h}(\xi) = \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$	$\tilde{h}(\xi) = \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$	$\tilde{h}(\xi) = \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$ is its extension to $\mathbb{R}^d \setminus \{0\}$, then we get $\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$
cardona-qft-methods_ F00368	26	■ 1.000	$\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$	$\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$	$\tilde{h}(\xi) = \xi ^{-d+2} h\left(\frac{\xi}{ \xi }\right)$ is its extension to $\mathbb{R}^d \setminus \{0\}$, then we get $\Delta_{\mathbb{R}^d} \tilde{h}(\xi) =$
cardona-qft-methods_ F00369	26	■ 1.000	$ \xi \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$	$ \xi \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$	$ \xi \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$ because $\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$. This
cardona-qft-methods_ F00370	26	■ 1.000	$\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$	$\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$	$ \xi \sigma_{-d}^P\left(\frac{\xi}{ \xi }\right) = \sigma_{-d}^P(\xi)$ because $\Delta_{\mathbb{R}^d} = r^{1-d} \partial_r (r^{d-1} \partial_r) + r^{-2} \Delta_{\mathbb{S}^{d-1}}$. This
cardona-qft-methods_ F00371	26	■ 1.000	$\tilde{h}$	$\tilde{h}$	means that $\tilde{h}$ is a symbol of order $d-2$ and $\partial_{\xi} \tilde{h}$ is a symbol of order $d-1$.
cardona-qft-methods_ F00372	26	■ 1.000	$d-2$	$d-2$	means that $\tilde{h}$ is a symbol of order $d-2$ and $\partial_{\xi} \tilde{h}$ is a symbol of order $d-1$.
cardona-qft-methods_ F00373	26	■ 1.000	$\partial_{\xi} \tilde{h}$	$\partial_{\xi} \tilde{h}$	means that $\tilde{h}$ is a symbol of order $d-2$ and $\partial_{\xi} \tilde{h}$ is a symbol of order $d-1$.
cardona-qft-methods_ F00374	26	■ 1.000	$d-1$	$d-1$	means that $\tilde{h}$ is a symbol of order $d-2$ and $\partial_{\xi} \tilde{h}$ is a symbol of order $d-1$.
cardona-qft-methods_ F00375	26	■ 1.000	$\sigma_{-d}^P(x, \xi) - \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$	$\sigma_{-d}^P(x, \xi) - \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$	4) End of proof: the symbol $\sigma_{-d}^P(x, \xi) - \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$ is of order $-d$
cardona-qft-methods_ F00376	26	■ 1.000	$-d$	$-d$	4) End of proof: the symbol $\sigma_{-d}^P(x, \xi) - \frac{ \xi ^{-d}}{\text{Vol}(\mathbb{S}^{d-1})} c_P(x)$ is of order $-d$
cardona-qft-methods_ F00377	26	■ 1.000	$T(P) =$	$T(P) =$	with zero integral, thus is a sum of commutators by 3) and $T(P) =$
cardona-qft-methods_ F00378	26	■ 1.000	$T\left(Op(\xi ^{-d} c_p(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)\right)$	$T\left(Op(\xi ^{-d} c_p(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)\right)$	$T\left(Op(\xi ^{-d} c_p(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)\right)$. So the map $\mu : f \in C_c^{\infty}(U) \rightarrow$
cardona-qft-methods_ F00379	26	■ 1.000	$\mu : f \in C_c^{\infty}(U) \rightarrow$	$\mu : f \in C_c^{\infty}(U) \rightarrow$	$T\left(Op(\xi ^{-d} c_p(x)), \forall T \in \Psi DO^{\mathbb{Z}}(M, E)\right)$. So the map $\mu : f \in C_c^{\infty}(U) \rightarrow$

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cardona-qft-methods_ F00380	26	■ 1.000	$\mathrm{T}\left(0\ p\left(f\left \xi\right ^{-d}\right)\right)\right)$	$T\left(Op\left(f\left \xi\right ^{-d}\right)\right)$	$T\left(Op(f \xi ^{-d})\right)$ is linear, continuous and satisfies $\mu(\partial_{x^k}f) = 0$ because
cardona-qft-methods_ F00381	26	■ 1.000	$\mu\left(\partial_{x^k}f\right)=0$	$\mu\left(\partial_{x^k}f\right)=0$	$T\left(Op(f \xi ^{-d})\right)$ is linear, continuous and satisfies $\mu(\partial_{x^k}f) = 0$ because
cardona-qft-methods_ F00382	27	■ 1.000	$\partial_{x^k}(f)\left \xi\right ^{-d}$	$\partial_{x^k}(f) \xi ^{-d}$	$\partial_{x^k}(f) \xi ^{-d}$ is a commutator if f has a compact support and U is homeo-
cardona-qft-methods_ F00383	27	■ 1.000	μ	μ	morphic to $\mathbb{R}^d$. As a consequence, μ is a multiple of the Lebesgue integral
cardona-qft-methods_ F00384	27	■ 1.000	$(1+\Delta)^{-d/2}\in\Psi D^0(M)$	$(1+\Delta)^{-d/2}\in\Psi DO(M)$	Proof. $(1+\Delta)^{-d/2}\in\Psi DO(M)$ has order $-d$ and its principal symbol
cardona-qft-methods_ F00385	27	■ 1.000	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}$	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}$	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}$ satisfies $\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\left\ \xi\right\ _x^{-d}$. So
cardona-qft-methods_ F00386	27	■ 1.000	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\left\ \xi\right\ _x^{-d}$	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\left\ \xi\right\ _x^{-d}$	$\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}$ satisfies $\sigma_{-d}^{\left(1+\Delta\right)^{-d/2}}(x,\xi)=-\left(g_x^{ij}\xi_i\xi_j\right)^{-d/2}=-\left\ \xi\right\ _x^{-d}$. So
cardona-qft-methods_ F00387	27	■ 0.675	$\{ \ }^{\{33,49,68,87,104,105\}}$	33,49,68,87,104,105	References for this section: ^{33,49,68,87,104,105}
cardona-qft-methods_ F00388	27	■ 1.000	$\mathcal{B}(\mathcal{H})$	$\mathcal{B}(\mathcal{H})$	property, it is <i>normal</i> . Recall that Tr acting on $\mathcal{B}(\mathcal{H})$ is a particular case
cardona-qft-methods_ F00389	27	■ 1.000	ω	ω	of a weight ω acting on a von Neumann algebra $\mathcal{M}$: it is a homogeneous
cardona-qft-methods_ F00390	27	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	of a weight ω acting on a von Neumann algebra $\mathcal{M}$: it is a homogeneous
cardona-qft-methods_ F00391	27	■ 1.000	$\mathcal{M}^+=\left\{aa^*\mid a\in\mathcal{M}\right\}$	$\mathcal{M}^+=\{aa^*\mid a\in\mathcal{M}\}$	additive map from $\mathcal{M}^+=\{aa^*\mid a\in\mathcal{M}\}$ to $[0,\infty]$.
cardona-qft-methods_ F00392	27	■ 1.000	$[0,\infty]$	$[0,\infty]$	additive map from $\mathcal{M}^+=\{aa^*\mid a\in\mathcal{M}\}$ to $[0,\infty]$.
cardona-qft-methods_ F00393	27	■ 0.788	$\omega\in\mathcal{M}^*$	$\omega\in\mathcal{M}^*$	A state is a weight $\omega\in\mathcal{M}^*$ (so $\omega(a)<\infty$, $\forall a\in\mathcal{M}$) such that $\omega(1)=1$.
cardona-qft-methods_ F00394	27	■ 0.788	$\omega(a)<\infty,\forall a\in\mathcal{M}$	$\omega(a)<\infty,\forall a\in\mathcal{M}$	A state is a weight $\omega\in\mathcal{M}^*$ (so $\omega(a)<\infty$, $\forall a\in\mathcal{M}$) such that $\omega(1)=1$.
cardona-qft-methods_ F00395	27	■ 0.788	$\omega(1)=1$	$\omega(1)=1$	A state is a weight $\omega\in\mathcal{M}^*$ (so $\omega(a)<\infty$, $\forall a\in\mathcal{M}$) such that $\omega(1)=1$.
cardona-qft-methods_ F00396	27	■ 1.000	$\omega\left(aa^*\right)=\omega\left(a^*a\right)$	$\omega\left(aa^*\right)=\omega\left(a^*a\right)$	A trace is a weight such that $\omega(aa^*)=\omega(a^*a)$ for all $a\in\mathcal{M}$.
cardona-qft-methods_ F00397	27	■ 1.000	$a\in\mathcal{M}$	$a\in\mathcal{M}$	A trace is a weight such that $\omega(aa^*)=\omega(a^*a)$ for all $a\in\mathcal{M}$.
cardona-qft-methods_ F00398	27	■ 0.993	$\omega\left(\sup_{\alpha}a_{\alpha}\right)=\sup_{\alpha}\omega\left(a_{\alpha}\right)$	$\omega\left(\sup_{\alpha}a_{\alpha}\right)=\sup_{\alpha}\omega\left(a_{\alpha}\right)$	A weight ω is normal if $\omega(\sup_{\alpha}a_{\alpha})=\sup_{\alpha}\omega(a_{\alpha})$ whenever $(a_{\alpha})\subset\mathcal{M}^+$ is
cardona-qft-methods_ F00399	27	■ 0.993	$\left(a_{\alpha}\right)\subset\mathcal{M}^+$	$(a_{\alpha})\subset\mathcal{M}^+$	A weight ω is normal if $\omega(\sup_{\alpha}a_{\alpha})=\sup_{\alpha}\omega(a_{\alpha})$ whenever $(a_{\alpha})\subset\mathcal{M}^+$ is
cardona-qft-methods_ F00400	27	■ 1.000	$\left(a_{\alpha}\right)_{\alpha}$	$(a_{\alpha})_{\alpha}$	net $(a_{\alpha})_{\alpha}$ converges in $\mathcal{B}(\mathcal{H})$ and this property looks innocent since a trace
cardona-qft-methods_ F00401	28	■ 0.628	$\operatorname{Tr}_{\operatorname{Dix}}(T)''=\lim_{N\rightarrow\infty}\frac{1}{\log N}\sum_{n=0}^N\mu_n(T)$	$\mathrm{Tr}_{\mathrm{Dix}}(T)''=\lim_{N\rightarrow\infty}\frac{1}{\log N}\sum_{n=0}^N\mu_n(T)$	tially means $\mathrm{Tr}_{\mathrm{Dix}}(T)''=\lim_{N\rightarrow\infty}\frac{1}{\log N}\sum_{n=0}^N\mu_n(T)$, where the $\mu_n(T)$
cardona-qft-methods_ F00402	28	■ 0.628	$\mu_n(T)$	$\mu_n(T)$	tially means $\mathrm{Tr}_{\mathrm{Dix}}(T)''=\lim_{N\rightarrow\infty}\frac{1}{\log N}\sum_{n=0}^N\mu_n(T)$, where the $\mu_n(T)$
cardona-qft-methods_ F00403	28	■ 0.988	$P\in\Psi D^{-d}(M)$	$P\in\Psi DO^{-d}(M)$	trace via Proposition 2.2. We will see that if $P\in\Psi DO^{-d}(M)$ where M is
cardona-qft-methods_ F00404	28	■ 1.000	S^*M	S^*M	where S^*M is the cosphere bundle on M .

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cardona-qft-methods_ F00405	28	■ 1.000	$\sum_{n \in \mathbb{N}} \frac{1}{n}$	$\sum_{n \in \mathbb{N}^*} \frac{1}{n}$	The physical motivation is quite essential: we know how $\sum_{n \in \mathbb{N}^*} \frac{1}{n}$ di-
cardona-qft-methods_ F00406	28	■ 1.000	$\frac{1}{r}$	$\frac{1}{r}$	tional potentials are in $\frac{1}{r}$ which has the same singularity (in one-dimension
cardona-qft-methods_ F00407	28	■ 1.000	$\zeta: s \in \mathbb{C} \rightarrow \sum_{n \in \mathbb{N}^*} \frac{1}{n^s}$	$\zeta: s \in \mathbb{C} \rightarrow \sum_{n \in \mathbb{N}^*} \frac{1}{n^s}$	tion $\zeta: s \in \mathbb{C} \rightarrow \sum_{n \in \mathbb{N}^*} \frac{1}{n^s}$. This is also why we have already seen a
cardona-qft-methods_ F00408	28	■ 1.000	$\zeta_P(s) = \operatorname{Tr}(\left(P \mathcal{D} ^{-s}\right))$	$\zeta_P(s) = \operatorname{Tr}(P \mathcal{D} ^{-s})$	ated to a pseudodifferential operator P by $\zeta_P(s) = \operatorname{Tr}(P \mathcal{D} ^{-s})$ in (7).
cardona-qft-methods_ F00409	28	■ 1.000	$T \in \mathcal{K}(\mathcal{H})$	$T \in \mathcal{K}(\mathcal{H})$	operators: for $T \in \mathcal{K}(\mathcal{H})$ (compact operators), let $\mu_n(T) =$
cardona-qft-methods_ F00410	28	■ 1.000	$\mu_n(T) =$	$\mu_n(T) =$	operators: for $T \in \mathcal{K}(\mathcal{H})$ (compact operators), let $\mu_n(T) =$
cardona-qft-methods_ F00411	28	■ 0.707	$\inf \left\{ \left\ T_{\uparrow E^\perp} \right\ \mid E \right\}$	$\inf \left\{ \left\ T_{\uparrow E^\perp} \right\ \mid E \right\}$	$\inf \left\{ \left\ T_{\uparrow E^\perp} \right\ \mid E \text{ subspace of } \mathcal{H} \text{ with } \dim(E) = n \right\}, n \in \mathbb{N}$. This could looks
cardona-qft-methods_ F00412	28	■ 0.707	$\left\{ \dim(E) = n \right\}, n \in \mathbb{N}$	$\dim(E) = n, n \in \mathbb{N}$	$\inf \left\{ \left\ T_{\uparrow E^\perp} \right\ \mid E \text{ subspace of } \mathcal{H} \text{ with } \dim(E) = n \right\}, n \in \mathbb{N}$. This could looks
cardona-qft-methods_ F00413	28	■ 1.000	$(n+1)$	$(n+1)$	the $(n+1)$ th of eigenvalues of $ T $ sorted in decreasing order. Since
cardona-qft-methods_ F00414	28	■ 1.000	$ T $	$ T $	the $(n+1)$ th of eigenvalues of $ T $ sorted in decreasing order. Since
cardona-qft-methods_ F00415	28	■ 1.000	$\lim_{n \rightarrow \infty} \mu_n(T) = 0$	$\lim_{n \rightarrow \infty} \mu_n(T) = 0$	$\lim_{n \rightarrow \infty} \mu_n(T) = 0$, for any $\epsilon > 0$, there exists a finite-dimensional sub-
cardona-qft-methods_ F00416	28	■ 1.000	$\epsilon > 0$	$\epsilon > 0$	$\lim_{n \rightarrow \infty} \mu_n(T) = 0$, for any $\epsilon > 0$, there exists a finite-dimensional sub-
cardona-qft-methods_ F00417	28	■ 0.933	E_ϵ	E_ϵ	space E_ϵ such that $\left\ T_{\uparrow E_\epsilon^\perp} \right\ < \epsilon$ and this property being equivalent to T
cardona-qft-methods_ F00418	28	■ 0.933	$\left\ T_{\uparrow E_\epsilon^\perp} \right\ < \epsilon$	$\left\ T_{\uparrow E_\epsilon^\perp} \right\ < \epsilon$	space E_ϵ such that $\left\ T_{\uparrow E_\epsilon^\perp} \right\ < \epsilon$ and this property being equivalent to T
cardona-qft-methods_ F00419	28	■ 1.000	$N \in \mathbb{N}$	$N \in \mathbb{N}$	Definition 3.1. For $T \in \mathcal{K}(\mathcal{H})$, the partial trace of order $N \in \mathbb{N}$ is
cardona-qft-methods_ F00420	28	■ 1.000	$\sigma_N(T) = \sum_{n=0}^N \mu_n(T)$	$\sigma_N(T) = \sum_{n=0}^N \mu_n(T)$	$\sigma_N(T) = \sum_{n=0}^N \mu_n(T)$.
cardona-qft-methods_ F00421	29	■ 0.999	$\ T\ \leq \sigma_N(T) \leq N\ T\ $	$\ T\ \leq \sigma_N(T) \leq N\ T\ $	Remark that $\ T\ \leq \sigma_N(T) \leq N\ T\ $ which implies $\sigma_n \simeq \ \cdot\ $ on $\mathcal{K}(\mathcal{H})$. Then
cardona-qft-methods_ F00422	29	■ 0.999	$\sigma_n \simeq \ \cdot\ $	$\sigma_n \simeq \ \cdot\ $	Remark that $\ T\ \leq \sigma_N(T) \leq N\ T\ $ which implies $\sigma_n \simeq \ \cdot\ $ on $\mathcal{K}(\mathcal{H})$. Then
cardona-qft-methods_ F00423	29	■ 0.999	$\mathcal{K}(\mathcal{H})$	$\mathcal{K}(\mathcal{H})$	Remark that $\ T\ \leq \sigma_N(T) \leq N\ T\ $ which implies $\sigma_n \simeq \ \cdot\ $ on $\mathcal{K}(\mathcal{H})$. Then
cardona-qft-methods_ F00424	29	■ 1.000	σ_N	σ_N	This norm σ_N splits:
cardona-qft-methods_ F00425	29	■ 1.000	$\lambda \in \mathbb{R}^+$	$\lambda \in \mathbb{R}^+$	Definition 3.2. The partial trace of T of order $\lambda \in \mathbb{R}^+$ is
cardona-qft-methods_ F00426	29	■ 1.000	$\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$	$\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$	As before, $\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$, for $\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$
cardona-qft-methods_ F00427	29	■ 1.000	$\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$	$\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$	As before, $\sigma_{\lambda_1}(T_1) + \sigma_{\lambda_2}(T_2) = \sigma_{\lambda_1 + \lambda_2}(T_1 + T_2)$, for $\lambda_1, \lambda_2 \in \mathbb{R}^+, 0 \leq$
cardona-qft-methods_ F00428	29	■ 0.996	$T_1, T_2 \in \mathcal{K}(\mathcal{H})$	$T_1, T_2 \in \mathcal{K}(\mathcal{H})$	$T_1, T_2 \in \mathcal{K}(\mathcal{H})$. We define a real interpolate space between $\mathcal{L}^1(\mathcal{H})$ and $\mathcal{K}(\mathcal{H})$
cardona-qft-methods_ F00429	29	■ 1.000	$\mathcal{L}^p(\mathcal{H})$	$\mathcal{L}^p(\mathcal{H})$	If $\mathcal{L}^p(\mathcal{H})$ is the ideal of operators T such that $\operatorname{Tr}(T ^p) < \infty$, so $\sigma_\lambda(T) =$
cardona-qft-methods_ F00430	29	■ 1.000	$\operatorname{Tr}(T ^p) < \infty$	$\operatorname{Tr}(T ^p) < \infty$	If $\mathcal{L}^p(\mathcal{H})$ is the ideal of operators T such that $\operatorname{Tr}(T ^p) < \infty$, so $\sigma_\lambda(T) =$

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cardona-qft-methods_ F00431	29	■ 1.000	$\sigma_{\lambda}(\lambda)(T)=$	$\sigma_{\lambda}(T) =$	If $\mathcal{L}^p(\mathcal{H})$ is the ideal of operators T such that $\text{Tr}(T ^p) < \infty$, so $\sigma_{\lambda}(T) =$
cardona-qft-methods_ F00432	29	■ 1.000	$\mathcal{O}\left(\lambda^{1-1/p}\right)$	$\mathcal{O}\left(\lambda^{1-1/p}\right)$	$\mathcal{O}(\lambda^{1-1/p})$, we have
cardona-qft-methods_ F00433	29	■ 1.000	$\mathcal{L}^1, \infty$	$\mathcal{L}^{1,\infty}$	Lemma 3.1. $\mathcal{L}^{1,\infty}$ is a C^* -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$. Moreover,
cardona-qft-methods_ F00434	29	■ 1.000	C^*	C^*	Lemma 3.1. $\mathcal{L}^{1,\infty}$ is a C^* -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$. Moreover,
cardona-qft-methods_ F00435	29	■ 1.000	$\ \cdot\ _{1, \infty}$	$\ \cdot\ _{1,\infty}$	Lemma 3.1. $\mathcal{L}^{1,\infty}$ is a C^* -ideal of $\mathcal{B}(\mathcal{H})$ for the norm $\ \cdot\ _{1,\infty}$. Moreover,
cardona-qft-methods_ F00436	29	■ 1.000	$\frac{\sigma_{\rho}(T)}{\log \rho}$	$\frac{\sigma_{\rho}(T)}{\log \rho}$	We begin with a Cesàro mean of $\frac{\sigma_{\rho}(T)}{\log \rho}$ with respect of the Haar measure of
cardona-qft-methods_ F00437	29	■ 1.000	$\mathbb{R}_+^*$	$\mathbb{R}_+^*$	the group $\mathbb{R}_+^*$:
cardona-qft-methods_ F00438	29	■ 0.997	$\lambda \geq e$	$\lambda \geq e$	Definition 3.3. For $\lambda \geq e$ and $T \in \mathcal{K}(\mathcal{H})$, let $\tau_{\lambda}(T) = \frac{1}{\log \lambda} \int_e^{\lambda} \frac{\sigma_{\rho}(T)}{\log \rho} \frac{d\rho}{\rho}$.
cardona-qft-methods_ F00439	29	■ 0.997	$\tau_{\lambda}(\lambda)(T)=\frac{1}{\log \lambda} \int_e^{\lambda} \frac{\sigma_{\rho}(T)}{\log \rho} \frac{d\rho}{\rho}$	$\tau_{\lambda}(T) = \frac{1}{\log \lambda} \int_e^{\lambda} \frac{\sigma_{\rho}(T)}{\log \rho} \frac{d\rho}{\rho}$	Definition 3.3. For $\lambda \geq e$ and $T \in \mathcal{K}(\mathcal{H})$, let $\tau_{\lambda}(T) = \frac{1}{\log \lambda} \int_e^{\lambda} \frac{\sigma_{\rho}(T)}{\log \rho} \frac{d\rho}{\rho}$.
cardona-qft-methods_ F00440	29	■ 1.000	$\sigma_{\rho}(T) \leq \log \rho \ T\ _{1, \infty}$	$\sigma_{\rho}(T) \leq \log \rho \ T\ _{1,\infty}$	Clearly, $\sigma_{\rho}(T) \leq \log \rho \ T\ _{1,\infty}$ and $\tau_{\lambda}(T) \leq \ T\ _{1,\infty}$, thus the map: $\lambda \rightarrow$
cardona-qft-methods_ F00441	29	■ 1.000	$\tau_{\lambda}(T) \leq \ T\ _{1, \infty}$	$\tau_{\lambda}(T) \leq \ T\ _{1,\infty}$	Clearly, $\sigma_{\rho}(T) \leq \log \rho \ T\ _{1,\infty}$ and $\tau_{\lambda}(T) \leq \ T\ _{1,\infty}$, thus the map: $\lambda \rightarrow$
cardona-qft-methods_ F00442	29	■ 1.000	$\lambda \rightarrow$	$\lambda \rightarrow$	Clearly, $\sigma_{\rho}(T) \leq \log \rho \ T\ _{1,\infty}$ and $\tau_{\lambda}(T) \leq \ T\ _{1,\infty}$, thus the map: $\lambda \rightarrow$
cardona-qft-methods_ F00443	29	■ 1.000	$\tau_{\lambda}(T)$	$\tau_{\lambda}(T)$	$\tau_{\lambda}(T)$ is in $C_b([e, \infty])$. It is not additive on $\mathcal{L}^{1,\infty}$ but this defect is under
cardona-qft-methods_ F00444	29	■ 1.000	$C_b([e, \infty])$	$C_b([e, \infty])$	$\tau_{\lambda}(T)$ is in $C_b([e, \infty])$. It is not additive on $\mathcal{L}^{1,\infty}$ but this defect is under
cardona-qft-methods_ F00445	29	■ 0.905	$T_1, T_2 \in \mathcal{L}_+^{1, \infty}$	$T_1, T_2 \in \mathcal{L}_+^{1,\infty}$	when $T_1, T_2 \in \mathcal{L}_+^{1,\infty}$.
cardona-qft-methods_ F00446	30	■ 1.000	$\lambda \rightarrow \tau_{\lambda}(T)$	$\lambda \rightarrow \tau_{\lambda}(T)$	The Dixmier's idea was to force additivity: since the map $\lambda \rightarrow \tau_{\lambda}(T)$
cardona-qft-methods_ F00447	30	■ 1.000	$\lambda \rightarrow \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$	$\lambda \rightarrow \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$	is in $C_b([e, \infty])$ and $\lambda \rightarrow \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$ is in $C_0([e, \infty])$, consider the
cardona-qft-methods_ F00448	30	■ 1.000	$C_0([e, \infty])$	$C_0([e, \infty])$	is in $C_b([e, \infty])$ and $\lambda \rightarrow \left(\frac{\log 2(2 + \log \log \lambda)}{\log \lambda} \right)$ is in $C_0([e, \infty])$, consider the
cardona-qft-methods_ F00449	30	■ 1.000	$\mathcal{A} = C_b([e, \infty]) / C_0([e, \infty])$	$\mathcal{A} = C_b([e, \infty]) / C_0([e, \infty])$	C^* -algebra $\mathcal{A} = C_b([e, \infty]) / C_0([e, \infty])$.
cardona-qft-methods_ F00450	30	■ 1.000	$[\tau(T)]$	$[\tau(T)]$	If $[\tau(T)]$ is the class of the map $\lambda \rightarrow \tau_{\lambda}(T)$ in $\mathcal{A}$, previous lemma shows
cardona-qft-methods_ F00451	30	■ 1.000	$[\tau]: T \rightarrow [\tau(T)]$	$[\tau]: T \rightarrow [\tau(T)]$	that $[\tau]: T \rightarrow [\tau(T)]$ is additive and positive homogeneous from $\mathcal{L}_+^{1,\infty}$ into
cardona-qft-methods_ F00452	30	■ 1.000	$\mathcal{L}_+^{1, \infty}$	$\mathcal{L}_+^{1,\infty}$	that $[\tau]: T \rightarrow [\tau(T)]$ is additive and positive homogeneous from $\mathcal{L}_+^{1,\infty}$ into
cardona-qft-methods_ F00453	30	■ 1.000	$[\tau(UTU^*)] = [\tau(T)]$	$[\tau(UTU^*)] = [\tau(T)]$	$\mathcal{A}$ satisfying $[\tau(UTU^*)] = [\tau(T)]$ for any unitary U .
cardona-qft-methods_ F00454	30	■ 0.902	$\omega \circ [\tau(\cdot)]$	$\omega \circ [\tau(\cdot)]$	$\omega(1) = 1$. Then, $\omega \circ [\tau(\cdot)]$ is a tracial weight on $\mathcal{L}_+^{1,\infty}$. Since $\mathcal{L}^{1,\infty}$ is a C^* -
cardona-qft-methods_ F00455	30	■ 0.902	C^* -	C^* -	$\omega(1) = 1$. Then, $\omega \circ [\tau(\cdot)]$ is a tracial weight on $\mathcal{L}_+^{1,\infty}$. Since $\mathcal{L}^{1,\infty}$ is a C^* -

Identifier	Page	Conf.	LaTeX source	Rendered	Image
cardona-qft-methods_F00456	30	■ 0.915	$\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^1, \infty \rightarrow \mathbb{C}$	$\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^1, \infty \rightarrow$	elements, and this map can be extended to a map $\omega \circ [\tau(\cdot)]: T \in \mathcal{L}^1, \infty \rightarrow$
cardona-qft-methods_F00457	30	■ 0.976	$\omega([\tau(T)]) \in \mathbb{C}$	$\omega([\tau(T)]) \in \mathbb{C}$	$\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1 T_2)]) = \omega([\tau(T_2 T_1)])$ for $T_1, T_2 \in \mathcal{L}^1, \infty$.
cardona-qft-methods_F00458	30	■ 0.976	$\omega([\tau(T_1 T_2)]) = \omega([\tau(T_2 T_1)])$	$\omega([\tau(T_1 T_2)]) = \omega([\tau(T_2 T_1)])$	$\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1 T_2)]) = \omega([\tau(T_2 T_1)])$ for $T_1, T_2 \in \mathcal{L}^1, \infty$.
cardona-qft-methods_F00459	30	■ 0.976	$T_1, T_2 \in \mathcal{L}^1, \infty$	$T_1, T_2 \in \mathcal{L}^1, \infty$	$\omega([\tau(T)]) \in \mathbb{C}$ such that $\omega([\tau(T_1 T_2)]) = \omega([\tau(T_2 T_1)])$ for $T_1, T_2 \in \mathcal{L}^1, \infty$.
cardona-qft-methods_F00460	30	■ 1.000	Tr_ω	Tr_ω	Definition 3.4. The Dixmier trace Tr_ω associated to a state ω on $\mathcal{A}$ is
cardona-qft-methods_F00461	30	■ 1.000	$\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$	$\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$	$\text{Tr}_\omega(\cdot) = \omega \circ [\tau(\cdot)]$.
cardona-qft-methods_F00462	30	■ 1.000	$\text{Tr}_\omega(T) = 0$	$\text{Tr}_\omega(T) = 0$	1) Note that $\text{Tr}_\omega(T) = 0$ if $T \in \mathcal{L}^1(\mathcal{H})$ and all Dixmier traces vanish on the
cardona-qft-methods_F00463	30	■ 1.000	$T \in \mathcal{L}^1(\mathcal{H})$	$T \in \mathcal{L}^1(\mathcal{H})$	1) Note that $\text{Tr}_\omega(T) = 0$ if $T \in \mathcal{L}^1(\mathcal{H})$ and all Dixmier traces vanish on the
cardona-qft-methods_F00464	30	■ 1.000	$\ \cdot\ _{1, \infty}$	$\ \cdot\ _{1, \infty}$	closure for the norm $\ \cdot\ _{1, \infty}$ of the ideal of finite rank operators. So Dixmier
cardona-qft-methods_F00465	30	■ 1.000	$p \geq 1, \mathcal{L}^1, \infty$	$p \geq 1, \mathcal{L}^1, \infty$	$p \geq 1, \mathcal{L}^1, \infty$ is not a separable.
cardona-qft-methods_F00466	30	■ 1.000	$f \in C_b([e, \infty])$	$f \in C_b([e, \infty])$	For instance, a function $f \in C_b([e, \infty])$ has a limit $\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$ if and
cardona-qft-methods_F00467	30	■ 1.000	$\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$	$\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$	For instance, a function $f \in C_b([e, \infty])$ has a limit $\ell = \lim_{\lambda \rightarrow \infty} f(\lambda)$ if and
cardona-qft-methods_F00468	30	■ 1.000	$\ell = \omega(f)$	$\ell = \omega(f)$	only if $\ell = \omega(f)$ for all state ω .
cardona-qft-methods_F00469	30	■ 1.000	$T \in \mathcal{L}^1, \infty$	$T \in \mathcal{L}^1, \infty$	Definition 3.5. The operator $T \in \mathcal{L}^1, \infty$ is said to be measurable if $\text{Tr}_\omega(T)$
cardona-qft-methods_F00470	30	■ 1.000	$\text{Tr}_\omega(T)$	$\text{Tr}_\omega(T)$	Definition 3.5. The operator $T \in \mathcal{L}^1, \infty$ is said to be measurable if $\text{Tr}_\omega(T)$
cardona-qft-methods_F00471	30	■ 1.000	Tr_{Dix}	Tr_{Dix}	is independent of ω . In this case, Tr_ω is denoted Tr_{Dix} .
cardona-qft-methods_F00472	30	■ 1.000	$\text{Tr}_\omega(T) = \ell$	$\text{Tr}_\omega(T) = \ell$	Lemma 3.3. The operator $T \in \mathcal{L}^1, \infty$ is measurable and $\text{Tr}_\omega(T) = \ell$ if and
cardona-qft-methods_F00473	30	■ 1.000	$\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$	$\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$	only if the map $\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$ converges at infinity to ℓ .
cardona-qft-methods_F00474	30	■ 1.000	ℓ	ℓ	only if the map $\lambda \in \mathbb{R}^+ \rightarrow \tau_\lambda(T) \in \mathcal{A}$ converges at infinity to ℓ .
cardona-qft-methods_F00475	30	■ 1.000	$\{ \}^{72, 74, 75}$	$^{72, 74, 75}$	tigated, see for instance, ^{72,74,75} but we quote only the following character-
cardona-qft-methods_F00476	30	■ 1.000	$T \in \mathcal{K}_+(\mathcal{H})$	$T \in \mathcal{K}_+(\mathcal{H})$	If $T \in \mathcal{K}_+(\mathcal{H})$, then T is measurable if and only if
cardona-qft-methods_F00477	30	■ 1.000	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$	$\lim_{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^N \mu_n(T)$ exists.
cardona-qft-methods_F00478	31	■ 1.000	$\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$	$\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$	Let $\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$ be the d-dimensional torus and $\Delta = -\sum_{i=1}^d \partial_{x_i}^2$ be the
cardona-qft-methods_F00479	31	■ 1.000	$\Delta = -\sum_{i=1}^d \partial_{x_i}^2$	$\Delta = -\sum_{i=1}^d \partial_{x_i}^2$	Let $\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$ be the d-dimensional torus and $\Delta = -\sum_{i=1}^d \partial_{x_i}^2$ be the
cardona-qft-methods_F00480	31	■ 0.997	$\mathcal{H} = L^2(\mathbb{T}^d)$	$\mathcal{H} = L^2(\mathbb{T}^d)$	scalar Laplacian seen as unbounded operator on $\mathcal{H} = L^2(\mathbb{T}^d)$. We want to

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cardona-qft-methods_ F00481	31	■ 1.000	$\operatorname{Tr}_{\omega}\left((1+\Delta)^{-p}\right)$	$\operatorname{Tr}_{\omega}\left((1+\Delta)^{-p}\right)$	compute $\operatorname{Tr}_{\omega}\left((1+\Delta)^{-p}\right)$ for $\frac{d}{2} \leq p \in \mathbb{N}^*$. We use $1+\Delta$ to avoid the kernel
cardona-qft-methods_ F00482	31	■ 1.000	$\frac{d}{2} \leq p \in \mathbb{N}^*$	$\frac{d}{2} \leq p \in \mathbb{N}^*$	compute $\operatorname{Tr}_{\omega}\left((1+\Delta)^{-p}\right)$ for $\frac{d}{2} \leq p \in \mathbb{N}^*$. We use $1+\Delta$ to avoid the kernel
cardona-qft-methods_ F00483	31	■ 1.000	$1+\Delta$	$1+\Delta$	compute $\operatorname{Tr}_{\omega}\left((1+\Delta)^{-p}\right)$ for $\frac{d}{2} \leq p \in \mathbb{N}^*$. We use $1+\Delta$ to avoid the kernel
cardona-qft-methods_ F00484	31	■ 1.000	ϵ	ϵ	by any $\epsilon > 0$ and the result does not depends on ϵ .
cardona-qft-methods_ F00485	31	■ 1.000	$e_{\{k\}}(x)=\frac{1}{2\pi} e^{\mathrm{i} \, k \cdot x}$	$e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$	Notice that the functions $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$ with $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$
cardona-qft-methods_ F00486	31	■ 1.000	$x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$	$x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$	Notice that the functions $e_k(x) = \frac{1}{2\pi} e^{ik \cdot x}$ with $x \in \mathbb{T}^d, k \in \mathbb{Z}^d =$
cardona-qft-methods_ F00487	31	■ 1.000	$\left(\mathbb{T}^d\right)^*$	$(\mathbb{T}^d)^*$	$(\mathbb{T}^d)^*$ form a basis of $\mathcal{H}$ of eigenvectors: $\Delta e_k = k ^2 e_k$. For $t \in$
cardona-qft-methods_ F00488	31	■ 1.000	$\Delta e_{\{k\}}= k ^2 e_{\{k\}}$	$\Delta e_k = k ^2 e_k$	$(\mathbb{T}^d)^*$ form a basis of $\mathcal{H}$ of eigenvectors: $\Delta e_k = k ^2 e_k$. For $t \in$
cardona-qft-methods_ F00489	31	■ 1.000	$t \in$	$t \in$	$(\mathbb{T}^d)^*$ form a basis of $\mathcal{H}$ of eigenvectors: $\Delta e_k = k ^2 e_k$. For $t \in$
cardona-qft-methods_ F00490	31	■ 1.000	$\mathbb{R}_{+}^*, e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right)=\sum_{k \in \mathbb{Z}^d} e^{-t k ^2} e^{-t k ^2}=\left(\sum_{k \in \mathbb{Z}} e^{-tk^2}\right)^d$	$\mathbb{R}_{+}^*, e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right)=\sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left(\sum_{k \in \mathbb{Z}} e^{-tk^2}\right)^d$	$\mathbb{R}_{+}^*, e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right)=\sum_{k \in \mathbb{Z}^d} e^{-t k ^2} = \left(\sum_{k \in \mathbb{Z}} e^{-tk^2}\right)^d$. We know that
cardona-qft-methods_ F00491	31	■ 1.000	$\left \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2}\right \leq 1$	$\left \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2}\right \leq 1$	$ \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \leq 1$, and since the first integral is $\sqrt{\frac{\pi}{t}}$, we
cardona-qft-methods_ F00492	31	■ 1.000	$\sqrt{\frac{\pi}{t}}$	$\sqrt{\frac{\pi}{t}}$	$ \int_{-\infty}^{\infty} e^{-tx^2} dx - \sum_{k \in \mathbb{Z}} e^{-tk^2} \leq 1$, and since the first integral is $\sqrt{\frac{\pi}{t}}$, we
cardona-qft-methods_ F00493	31	■ 0.953	$e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right) \underset{t \downarrow 0+}{\sim} \left(\frac{\pi}{t}\right)^{d/2} = \alpha t^{-d/2}$	$e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right) \underset{t \downarrow 0+}{\sim} \left(\frac{\pi}{t}\right)^{d/2} = \alpha t^{-d/2}$	get $e^t \operatorname{Tr}\left(e^{-t(1+\Delta)}\right) \underset{t \downarrow 0+}{\sim} \left(\frac{\pi}{t}\right)^{d/2} = \alpha t^{-d/2}$.
cardona-qft-methods_ F00494	31	■ 0.995	$\left.\mu_n\left((1+\Delta)^{-d/2}\right)\right _{n \rightarrow \infty} \simeq \left(\alpha \frac{1}{\Gamma(d/2+1)}\right) \frac{1}{n}$	$\mu_n\left((1+\Delta)^{-d/2}\right)\Big _{n \rightarrow \infty} \simeq \left(\alpha \frac{1}{\Gamma(d/2+1)}\right) \frac{1}{n}$	We will use a Tauberian theorem: $\mu_n\left((1+\Delta)^{-d/2}\right)\Big _{n \rightarrow \infty} \simeq \left(\alpha \frac{1}{\Gamma(d/2+1)}\right) \frac{1}{n}$,
cardona-qft-methods_ F00495	31	■ 1.000	$\{ \}^{49,54}$	49,54	see. ^{49,54} Thus
cardona-qft-methods_ F00496	31	■ 1.000	$(1+\Delta)^{-d/2}$	$(1+\Delta)^{-d/2}$	So $(1+\Delta)^{-d/2}$ is measurable and $\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right)=\operatorname{Tr}_{\omega}\left((1+\Delta)^{-d/2}\right)$
cardona-qft-methods_ F00497	31	■ 1.000	$\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right)=\operatorname{Tr}_{\omega}\left((1+\Delta)^{-d/2}\right)$	$\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right)=\operatorname{Tr}_{\omega}\left((1+\Delta)^{-d/2}\right)$	So $(1+\Delta)^{-d/2}$ is measurable and $\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right)=\operatorname{Tr}_{\omega}\left((1+\Delta)^{-d/2}\right)$
cardona-qft-methods_ F00498	31	■ 0.997	$\frac{\pi^{d/2}}{\Gamma(d/2+1)}$	$\frac{\pi^{d/2}}{\Gamma(d/2+1)}$	$\Delta)^{-d/2}) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$. Since $(1+\Delta)^{-p}$ is traceable for $p > \frac{d}{2}$, $\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.
cardona-qft-methods_ F00499	31	■ 0.997	$(1+\Delta)^{-p}$	$(1+\Delta)^{-p}$	$\Delta)^{-d/2}) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$. Since $(1+\Delta)^{-p}$ is traceable for $p > \frac{d}{2}$, $\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.
cardona-qft-methods_ F00500	31	■ 0.997	$p > \frac{d}{2}, \operatorname{Tr}_{Dix}\left((1+\Delta)^{-p}\right) = 0$	$p > \frac{d}{2}, \operatorname{Tr}_{Dix}\left((1+\Delta)^{-p}\right) = 0$	$\Delta)^{-d/2}) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$. Since $(1+\Delta)^{-p}$ is traceable for $p > \frac{d}{2}$, $\operatorname{Tr}_{Dix}\left((1+\Delta)^{-d/2}\right) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.
cardona-qft-methods_ F00501	31	■ 0.843	$\Delta)^{-p} = 0$	$\Delta)^{-p} = 0$	$\Delta)^{-p} = 0$.
cardona-qft-methods_ F00502	31	■ 1.000	$P \in \Psi D^{-d}(M, E)$	$P \in \Psi DO^{-d}(M, E)$	E a vector bundle over M and $P \in \Psi DO^{-d}(M, E)$. Then, $P \in \mathcal{L}^{1,\infty}$, is

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cardona-qft-methods_ F08503	31	■ 1.000	$P \in \mathcal{L}^1, \infty$	$P \in \mathcal{L}^{1,\infty}$	E a vector bundle over M and $P \in \Psi DO^{-d}(M, E)$. Then, $P \in \mathcal{L}^{1,\infty}$, is
cardona-qft-methods_ F08504	31	■ 1.000	$\operatorname{Tr}_{\text{Dix}}(P) = \frac{1}{d} W \operatorname{Res}(P)$	$\operatorname{Tr}_{\text{Dix}}(P) = \frac{1}{d} W \operatorname{Res}(P)$	measurable and $\operatorname{Tr}_{\text{Dix}}(P) = \frac{1}{d} W \operatorname{Res}(P)$.
cardona-qft-methods_ F08505	31	■ 1.000	$\{ \}^c$	c	spin^c structures on orientable manifolds M defined by Morita equivalence
cardona-qft-methods_ F08506	31	■ 1.000	$C(M)$	$C(M)$	between the C^* -algebras $C(M)$ and $\Gamma(\mathcal{C}l M)$. Then the notion of spin struc-
cardona-qft-methods_ F08507	31	■ 1.000	$\Gamma(\mathcal{C}l M)$	$\Gamma(\mathcal{C}l M)$	between the C^* -algebras $C(M)$ and $\Gamma(\mathcal{C}l M)$. Then the notion of spin struc-
cardona-qft-methods_ F08508	32	■ 1.000	$E = \mathcal{C}l T^* M$	$E = \mathcal{C}l T^* M$	$E = \mathcal{C}l T^* M$ where the fiber $\mathcal{C}l T_x^* M$ is the Clifford algebra of the real
cardona-qft-methods_ F08509	32	■ 1.000	$T_x^* M$	$T_x^* M$	vector space $T_x^* M$ for $x \in M$ endowed with the nondegenerate quadratic
cardona-qft-methods_ F08510	32	■ 0.611	$m \in \mathbb{N}$	$m \in \mathbb{N}$	$m \in \mathbb{N}$ on E is an element of $\operatorname{Diff}^m(M, E) = \Gamma(M, \operatorname{End}(E))$.
cardona-qft-methods_ F08511	32	■ 0.611	$\operatorname{Diff}^m(M, E) = \Gamma(M, \operatorname{End}(E))$	$\operatorname{Diff}^m(M, E) = \Gamma(M, \operatorname{End}(E))$	$m \in \mathbb{N}$ on E is an element of $\operatorname{Diff}^m(M, E) = \Gamma(M, \operatorname{End}(E))$.
cardona-qft-methods_ F08512	32	■ 0.989	$\left\{ \nabla_{X_1} \cdots \nabla_{X_j} \mid X_j \in \Gamma(M, TM), j \leq m \right\}$	$\{ \nabla_{X_1} \cdots \nabla_{X_j} \mid X_j \in \Gamma(M, TM), j \leq m \}$	$\operatorname{Vect} \{ \nabla_{X_1} \cdots \nabla_{X_j} \mid X_j \in \Gamma(M, TM), j \leq m \}$.
cardona-qft-methods_ F08513	32	■ 1.000	$\operatorname{Diff}^m(M, E)$	$\operatorname{Diff}^m(M, E)$	In particular, $\operatorname{Diff}^m(M, E)$ is a subalgebra of $\operatorname{End}(\Gamma(M, E))$ and the
cardona-qft-methods_ F08514	32	■ 1.000	$\operatorname{End}(\Gamma(M, E))$	$\operatorname{End}(\Gamma(M, E))$	In particular, $\operatorname{Diff}^m(M, E)$ is a subalgebra of $\operatorname{End}(\Gamma(M, E))$ and the
cardona-qft-methods_ F08515	32	■ 0.980	$\Gamma(T^* M, \pi^* \operatorname{End}(E))$	$\Gamma(T^* M, \pi^* \operatorname{End}(E))$	operator P has a principal symbol σ_m^P in $\Gamma(T^* M, \pi^* \operatorname{End}(E))$ where $\pi :$
cardona-qft-methods_ F08516	32	■ 0.980	π	π	operator P has a principal symbol σ_m^P in $\Gamma(T^* M, \pi^* \operatorname{End}(E))$ where $\pi :$
cardona-qft-methods_ F08517	32	■ 1.000	$E = \wedge T^* M$	$E = \wedge T^* M$	<i>Example:</i> Let $E = \wedge T^* M$. The exterior product and the contraction given
cardona-qft-methods_ F08518	32	■ 1.000	$\omega, \omega_j \in E$	$\omega, \omega_j \in E$	on $\omega, \omega_j \in E$ by
cardona-qft-methods_ F08519	32	■ 1.000	$c(\omega) = \epsilon(\omega) + \iota(\omega)$	$c(\omega) = \epsilon(\omega) + \iota(\omega)$	suggest the definition $c(\omega) = \epsilon(\omega) + \iota(\omega)$, and one checks that
cardona-qft-methods_ F08520	32	■ 1.000	$e_1, \dots, e_d$	$e_1, \dots, e_d$	E has a natural scalar product: if $e_1, \dots, e_d$ is an orthonormal basis of $T_x^* M$,
cardona-qft-methods_ F08521	32	■ 1.000	$e_{i_1} \wedge \cdots \wedge e_{i_p}$	$e_{i_1} \wedge \cdots \wedge e_{i_p}$	then the scalar product is chosen such that $e_{i_1} \wedge \cdots \wedge e_{i_p}$ for $i_1 < \cdots < i_p$
cardona-qft-methods_ F08522	32	■ 1.000	$i_1 < \cdots < i_p$	$i_1 < \cdots < i_p$	then the scalar product is chosen such that $e_{i_1} \wedge \cdots \wedge e_{i_p}$ for $i_1 < \cdots < i_p$
cardona-qft-methods_ F08523	32	■ 1.000	$d \in$	$d \in$	If $d \in \operatorname{Diff}^1$ is the exterior derivative and d^* is its adjoint for the deduced
cardona-qft-methods_ F08524	32	■ 1.000	1	1	If $d \in \operatorname{Diff}^1$ is the exterior derivative and d^* is its adjoint for the deduced
cardona-qft-methods_ F08525	32	■ 1.000	d^*	d^*	If $d \in \operatorname{Diff}^1$ is the exterior derivative and d^* is its adjoint for the deduced
cardona-qft-methods_ F08526	32	■ 1.000	$\Gamma(M, E)$	$\Gamma(M, E)$	scalar product on $\Gamma(M, E)$, then their principal symbols are
cardona-qft-methods_ F08527	32	■ 1.000	$\sigma_1^d(x, \xi) = i\xi$	$\sigma_1^d(x, \xi) = i\xi$	where h is such that $d_x h = \xi$, so $\sigma_1^d(x, \xi) = i\xi$, and similarly for $\sigma_1^{d^*}$.
cardona-qft-methods_ F08528	32	■ 1.000	$\sigma_1^{d^*}$	$\sigma_1^{d^*}$	where h is such that $d_x h = \xi$, so $\sigma_1^{d^*}(x, \xi) = i\xi$, and similarly for $\sigma_1^{d^*}$.

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cardona-qft-methods_F08529	32	■ 1.000	$P \in \operatorname{Diff}^m(M)$	$P \in \operatorname{Diff}^m(M)$	More generally, if $P \in \operatorname{Diff}^m(M)$, then $\sigma_m^P(dh) = \frac{1}{i^m m!} (adh)^m(P)$ with
cardona-qft-methods_F08538	32	■ 1.000	$\sigma_m^P(dh) = \frac{1}{i^m m!} (adh)^m(P)$	$\sigma_m^P(dh) = \frac{1}{i^m m!} (adh)^m(P)$	More generally, if $P \in \operatorname{Diff}^m(M)$, then $\sigma_m^P(dh) = \frac{1}{i^m m!} (adh)^m(P)$ with
cardona-qft-methods_F08531	32	■ 0.901	$adh = [\cdot, h]$	$adh = [\cdot, h]$	$adh = [\cdot, h]$ and $\sigma_m^{P^*}(\omega) = \sigma_m^P(\omega)^*$ where the adjoint $\bar{P}^*$ is for the scalar
cardona-qft-methods_F08532	32	■ 0.901	$\sigma_m^{\bar{P}^*}(\omega) = \sigma_m^P(\omega)^*$	$\sigma_m^{P^*}(\omega) = \sigma_m^P(\omega)^*$	$adh = [\cdot, h]$ and $\sigma_m^{P^*}(\omega) = \sigma_m^P(\omega)^*$ where the adjoint $\bar{P}^*$ is for the scalar
cardona-qft-methods_F08533	32	■ 0.901	P^*	P^*	$adh = [\cdot, h]$ and $\sigma_m^{P^*}(\omega) = \sigma_m^P(\omega)^*$ where the adjoint $\bar{P}^*$ is for the scalar
cardona-qft-methods_F08534	32	■ 1.000	$E: \langle \psi, \psi' \rangle =$	$E: \langle \psi, \psi' \rangle =$	product on $\Gamma(M, E)$ associated to an hermitean metric on E : $\langle \psi, \psi' \rangle =$
cardona-qft-methods_F08535	32	■ 1.000	$\int_M \langle \psi(x), \psi'(x) \rangle_x dx $	$\int_M \langle \psi(x), \psi'(x) \rangle_x dx $	$\int_M \langle \psi(x), \psi'(x) \rangle_x dx $ is a scalar product on $\Gamma(M, E)$.
cardona-qft-methods_F08536	32	■ 1.000	$P \in \operatorname{Diff}^2(M, E)$	$P \in \operatorname{Diff}^2(M, E)$	Definition 4.1. The operator $P \in \operatorname{Diff}^2(M, E)$ is called a generalized
cardona-qft-methods_F08537	32	■ 1.000	$\sigma_2^P(x, \xi) = \xi _x^2 id_{E_x}$	$\sigma_2^P(x, \xi) = \xi _x^2 id_{E_x}$	Laplacian when its symbol satisfies $\sigma_2^P(x, \xi) = \xi _x^2 id_{E_x}$ for $x \in M, \xi \in$
cardona-qft-methods_F08538	32	■ 1.000	$x \in M, \xi \in$	$x \in M, \xi \in$	Laplacian when its symbol satisfies $\sigma_2^P(x, \xi) = \xi _x^2 id_{E_x}$ for $x \in M, \xi \in$
cardona-qft-methods_F08539	33	■ 1.000	$P =$	$P =$	This is equivalent to say that, in local coordinates, $P =$
cardona-qft-methods_F08548	33	■ 1.000	$-\sum_{i,j} g^{ij}(x) \partial_{x^i} \partial_{x^j} + b^j(x) \partial_{x^j} + c(x)$	$-\sum_{i,j} g^{ij}(x) \partial_{x^i} \partial_{x^j} + b^j(x) \partial_{x^j} + c(x)$	$-\sum_{i,j} g^{ij}(x) \partial_{x^i} \partial_{x^j} + b^j(x) \partial_{x^j} + c(x)$, where the b^j are smooth and c is
cardona-qft-methods_F08541	33	■ 1.000	b^j	b^j	$-\sum_{i,j} g^{ij}(x) \partial_{x^i} \partial_{x^j} + b^j(x) \partial_{x^j} + c(x)$, where the b^j are smooth and c is
cardona-qft-methods_F08542	33	■ 0.981	$\Gamma(M, \operatorname{End}(E))$	$\Gamma(M, \operatorname{End}(E))$	in $\Gamma(\bar{M}, \operatorname{End}(E))$.
cardona-qft-methods_F08543	33	■ 1.000	$E = E^+ \oplus E^-$	$E = E^+ \oplus E^-$	Definition 4.2. Assume that $E = E^+ \oplus E^-$ is a $\mathbb{Z}_2$ -graded vector bundle.
cardona-qft-methods_F08544	33	■ 1.000	$\mathbb{Z}_2$	$\mathbb{Z}_2$	Definition 4.2. Assume that $E = E^+ \oplus E^-$ is a $\mathbb{Z}_2$ -graded vector bundle.
cardona-qft-methods_F08545	33	■ 1.000	$D \in \operatorname{Diff}^1(M, E)$	$D \in \operatorname{Diff}^1(M, E)$	When $D \in \operatorname{Diff}^1(M, E)$ and $D = \begin{pmatrix} 0 & D^+ \\ D^- & 0 \end{pmatrix}$ (i.e. D is odd) where
cardona-qft-methods_F08546	33	■ 1.000	$D = \begin{pmatrix} 0 & D^+ \\ D^- & 0 \end{pmatrix}$	$D = \begin{pmatrix} 0 & D^+ \\ D^- & 0 \end{pmatrix}$	When $D \in \operatorname{Diff}^1(M, E)$ and $D = \begin{pmatrix} 0 & D^+ \\ D^- & 0 \end{pmatrix}$ (i.e. D is odd) where
cardona-qft-methods_F08547	33	■ 1.000	D	D	When $D \in \operatorname{Diff}^1(M, E)$ and $D = \begin{pmatrix} 0 & D^+ \\ D^- & 0 \end{pmatrix}$ (i.e. D is odd) where
cardona-qft-methods_F08548	33	■ 0.992	$D^\pm: \Gamma(M, E^\mp) \rightarrow \Gamma(M, E^\pm), D$	$D^\pm: \Gamma(M, E^\mp) \rightarrow \Gamma(M, E^\pm), D$	$D^\pm: \Gamma(M, E^\mp) \rightarrow \Gamma(M, E^\pm)$, D is called a Dirac operator if $D^2 =$
cardona-qft-methods_F08549	33	■ 0.992	$D^2 =$	$D^2 =$	$D^\pm: \Gamma(M, E^\mp) \rightarrow \Gamma(M, E^\pm)$, D is called a Dirac operator if $D^2 =$
cardona-qft-methods_F08550	33	■ 1.000	$\begin{pmatrix} D^- D^+ & 0 \\ 0 & D^+ D^- \end{pmatrix}$	$\begin{pmatrix} D^- D^+ & 0 \\ 0 & D^+ D^- \end{pmatrix}$	$\begin{pmatrix} D^- D^+ & 0 \\ 0 & D^+ D^- \end{pmatrix}$ is a generalized Laplacian.
cardona-qft-methods_F08551	33	■ 0.370	$E = \bigwedge T^* M = \bigwedge^{\text{even}} T^* M \oplus \bigwedge^{\text{odd}} T^* M$	$E = \bigwedge T^* M = \bigwedge^{\text{even}} T^* M \oplus \bigwedge^{\text{odd}} T^* M$	A good example is given by $E = \bigwedge T^* M = \bigwedge^{\text{even}} T^* M \oplus \bigwedge^{\text{odd}} T^* M$ and the
cardona-qft-methods_F08552	33	■ 1.000	$D = d + d^*$	$D = d + d^*$	de Rham operator $D = d + d^*$. It is a Dirac operator since $D^2 = dd^* + d^*d$
cardona-qft-methods_F08553	33	■ 1.000	$D^2 = dd^* + d^*d$	$D^2 = dd^* + d^*d$	de Rham operator $D = d + d^*$. It is a Dirac operator since $D^2 = dd^* + d^*d$

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cardona-qft-methods_F08554	33	■ 1.000	$D^{\wedge\{2\}}$	D^2	is a generalized Laplacian according to (12). D^2 is also called the Laplace–
cardona-qft-methods_F08555	33	■ 1.000	$\mathcal{C}\ell M$	$\mathcal{C}\ell M$	Definition 4.3. Define $\mathcal{C}\ell M$ as the vector bundle over M whose fiber
cardona-qft-methods_F08556	33	■ 1.000	$\mathcal{C}\ell T_x M$	$\mathcal{C}\ell T_x M$	in $x \in M$ is the Clifford algebra $\mathcal{C}\ell T_x^* M$ (or $\mathcal{C}\ell T_x M$ using the musical
cardona-qft-methods_F08557	33	■ 1.000	$X \in TM \leftrightarrow X^{\flat} \in T^*M$	$X \in TM \leftrightarrow X^{\flat} \in T^*M$	isomorphism $X \in TM \leftrightarrow X^{\flat} \in T^*M$).
cardona-qft-methods_F08558	33	■ 1.000	$c: \Gamma(M, \mathcal{C}\ell M) \rightarrow \text{End}(\Gamma(M, E))$	$c: \Gamma(M, \mathcal{C}\ell M) \rightarrow \text{End}(\Gamma(M, E))$	$\mathbb{Z}_2$ -graduate action $c: \Gamma(M, \mathcal{C}\ell M) \rightarrow \text{End}(\Gamma(M, E))$.
cardona-qft-methods_F08559	33	■ 0.997	$D \in \text{Diff}^1$	$D \in \text{Diff}^1$	Prop 4.1. If E is a Clifford module, every odd $D \in \text{Diff}^1$ such that
cardona-qft-methods_F08560	33	■ 1.000	$[D, f] = i c(df)$	$[D, f] = i c(df)$	$[D, f] = i c(df)$ for $f \in C^\infty(M)$ is a Dirac operator.
cardona-qft-methods_F08561	33	■ 1.000	$f \in C^\infty(M)$	$f \in C^\infty(M)$	$[D, f] = i c(df)$ for $f \in C^\infty(M)$ is a Dirac operator.
cardona-qft-methods_F08562	33	■ 1.000	$c(df) = -i[D, f]$	$c(df) = -i[D, f]$	$c(df) = -i[D, f]$.
cardona-qft-methods_F08563	33	■ 1.000	$c = i(\epsilon + \iota)$	$c = i(\epsilon + \iota)$	Clifford module for $c = i(\epsilon + \iota)$ coming from the Dirac operator $D = d + d^*$:
cardona-qft-methods_F08564	33	■ 1.000	$a \in \Gamma(M, \mathcal{C}\ell M)$	$a \in \Gamma(M, \mathcal{C}\ell M)$	E is a Clifford connection if for $a \in \Gamma(M, \mathcal{C}\ell M)$ and $X \in \Gamma(M, TM)$,
cardona-qft-methods_F08565	33	■ 1.000	$X \in \Gamma(M, TM)$	$X \in \Gamma(M, TM)$	E is a Clifford connection if for $a \in \Gamma(M, \mathcal{C}\ell M)$ and $X \in \Gamma(M, TM)$,
cardona-qft-methods_F08566	33	■ 1.000	$[\nabla_X, c(a)] = c(\nabla_X^{LC} a)$	$[\nabla_X, c(a)] = c(\nabla_X^{LC} a)$	$[\nabla_X, c(a)] = c(\nabla_X^{LC} a)$ where ∇_X^{LC} is the Levi-Civita connection after its
cardona-qft-methods_F08567	33	■ 1.000	∇_X^{LC}	∇_X^{LC}	$[\nabla_X, c(a)] = c(\nabla_X^{LC} a)$ where ∇_X^{LC} is the Levi-Civita connection after its
cardona-qft-methods_F08568	33	■ 1.000	D_∇	D_∇	A Dirac operator D_∇ is associated to a Clifford connection ∇ :
cardona-qft-methods_F08569	33	■ 1.000	$c \otimes 1$	$c \otimes 1$	where we use c for $c \otimes 1$.
cardona-qft-methods_F08570	34	■ 1.000	$\nabla = \sum_{j=1}^d dx^j \otimes \nabla_{\partial_j}$	$\nabla = \sum_{j=1}^d dx^j \otimes \nabla_{\partial_j}$	Thus if in local coordinates, $\nabla = \sum_{j=1}^d dx^j \otimes \nabla_{\partial_j}$, the associated Dirac
cardona-qft-methods_F08571	34	■ 1.000	$D_\nabla = -i \sum_j c(dx^j) \nabla_{\partial_j}$	$D_\nabla = -i \sum_j c(dx^j) \nabla_{\partial_j}$	operator is given by $D_\nabla = -i \sum_j c(dx^j) \nabla_{\partial_j}$. In particular, for $f \in C^\infty(M)$,
cardona-qft-methods_F08572	34	■ 0.529	$[D_\nabla, f id_E] = -i \sum_{i=1}^d c(dx^i) [\nabla_{\partial_j}, f] = \sum_{j=1}^d -i c(dx^j) \partial_j f = -i c(df)$	$[D_\nabla, f id_E] = -i \sum_{i=1}^d c(dx^i) [\nabla_{\partial_j}, f] = \sum_{j=1}^d -i c(dx^j) \partial_j f = -i c(df)$	$[D_\nabla, f id_E] = -i \sum_{i=1}^d c(dx^i) [\nabla_{\partial_j}, f] = \sum_{j=1}^d -i c(dx^j) \partial_j f = -i c(df)$.
cardona-qft-methods_F08573	34	■ 1.000	Spin_d	Spin_d	2) <i>The spinor bundle:</i> Recall that the spin group Spin_d is the non-trivial
cardona-qft-methods_F08574	34	■ 1.000	SO_d	SO_d	two-fold covering of SO_d , so we have $0 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}_d \xrightarrow{\xi} SO_d \rightarrow 1$.
cardona-qft-methods_F08575	34	■ 1.000	$0 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}_d \xrightarrow{\xi} SO_d \rightarrow 1$	$0 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}_d \xrightarrow{\xi} SO_d \rightarrow 1$	two-fold covering of SO_d , so we have $0 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}_d \xrightarrow{\xi} SO_d \rightarrow 1$.
cardona-qft-methods_F08576	34	■ 0.999	$\text{SO}(TM) \rightarrow M$	$\text{SO}(TM) \rightarrow M$	Let $\text{SO}(TM) \rightarrow M$ be the SO_d -principal bundle of positively oriented or-
cardona-qft-methods_F08577	34	■ 0.999	SO_d	SO_d	Let $\text{SO}(TM) \rightarrow M$ be the SO_d -principal bundle of positively oriented or-
cardona-qft-methods_F08578	34	■ 1.000	TM	TM	thonormal frames on TM of an oriented Riemannian manifold M of dimen-

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cardona-qft-methods_F00579	34	■ 0.901	$\operatorname{Spin}(T M) \xrightarrow{\pi} M$	$\operatorname{Spin}(TM) \xrightarrow{\pi} M$	(M, g) is a Spin_d -principal bundle $\operatorname{Spin}(TM) \xrightarrow{\pi} M$ with a two-fold cov-
cardona-qft-methods_F00580	34	■ 0.608	$\operatorname{Spin}(T M) \xrightarrow{\eta} \operatorname{Spin}(T M)$	$\operatorname{Spin}(TM) \xrightarrow{\eta} \operatorname{Spin}(TM)$	ering map $\operatorname{Spin}(TM) \xrightarrow{\eta} \operatorname{Spin}(TM)$ such that the following diagram com-
cardona-qft-methods_F00581	34	■ 0.999	$\operatorname{Spin}(T M)$	$\operatorname{Spin}(TM)$	fiber bundles $\operatorname{Spin}(TM)$ and $\operatorname{Spin}(TM)$.
cardona-qft-methods_F00582	34	■ 0.999	$\operatorname{Spin}(T M)$	$\operatorname{Spin}(TM)$	fiber bundles $\operatorname{Spin}(TM)$ and $\operatorname{Spin}(TM)$.
cardona-qft-methods_F00583	34	■ 1.000	Spin_d^c	Spin_d^c	In above definition, one can replace Spin_d by the group Spin_d^c which is a
cardona-qft-methods_F00584	34	■ 0.999	$\mathbb{T}: 0 \longrightarrow \mathbb{T} \longrightarrow \operatorname{Spin}_d^c \xrightarrow{\xi} \operatorname{Spin}_d^c \longrightarrow 1$	$\mathbb{T}: 0 \longrightarrow \mathbb{T} \longrightarrow \operatorname{Spin}_d^c \xrightarrow{\xi} \operatorname{Spin}_d^c \longrightarrow 1$	central extension of Spin_d by $\mathbb{T}: 0 \longrightarrow \mathbb{T} \longrightarrow \operatorname{Spin}_d^c \xrightarrow{\xi} \operatorname{Spin}_d^c \longrightarrow 1$.
cardona-qft-methods_F00585	34	■ 0.972	M, g	M, g	An oriented Riemannian manifold (M, g) is spin if and only if the sec-
cardona-qft-methods_F00586	34	■ 1.000	$H^1(M, \mathbb{Z}_2)$	$H^1(M, \mathbb{Z}_2)$	spondence with $H^1(M, \mathbb{Z}_2)$. In particular the existence of a spin structure
cardona-qft-methods_F00587	34	■ 1.000	ρ	ρ	Let ρ be an irreducible representation of $\mathcal{C}\ell\mathbb{C}^d \rightarrow \operatorname{End}_{\mathbb{C}}(\Sigma_d)$ with $\Sigma_d \simeq$
cardona-qft-methods_F00588	34	■ 1.000	$\mathcal{C}\ell\mathbb{C}^d \rightarrow \operatorname{End}_{\mathbb{C}}(\Sigma_d)$	$\mathcal{C}\ell\mathbb{C}^d \rightarrow \operatorname{End}_{\mathbb{C}}(\Sigma_d)$	Let ρ be an irreducible representation of $\mathcal{C}\ell\mathbb{C}^d \rightarrow \operatorname{End}_{\mathbb{C}}(\Sigma_d)$ with $\Sigma_d \simeq$
cardona-qft-methods_F00589	34	■ 1.000	$\Sigma_d \simeq$	$\Sigma_d \simeq$	Let ρ be an irreducible representation of $\mathcal{C}\ell\mathbb{C}^d \rightarrow \operatorname{End}_{\mathbb{C}}(\Sigma_d)$ with $\Sigma_d \simeq$
cardona-qft-methods_F00590	34	■ 0.999	$\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$	$\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$	$\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$ as set of complex spinors. Of course, $\mathcal{C}\ell\mathbb{C}^d$ is endowed with its
cardona-qft-methods_F00591	34	■ 0.999	$\mathcal{C}\ell\mathbb{C}^d$	$\mathcal{C}\ell\mathbb{C}^d$	$\mathbb{C}^{2^{\lfloor d/2 \rfloor}}$ as set of complex spinors. Of course, $\mathcal{C}\ell\mathbb{C}^d$ is endowed with its
cardona-qft-methods_F00592	34	■ 1.000	S	S	The <i>spinor bundle</i> S of M is the complex vector bundle associated
cardona-qft-methods_F00593	35	■ 1.000	$S = \operatorname{Spin}(T M) \times_{\rho_d} \Sigma_d$	$S = \operatorname{Spin}(TM) \times_{\rho_d} \Sigma_d$	$S = \operatorname{Spin}(TM) \times_{\rho_d} \Sigma_d$. Here ρ_d is a representation of Spin_d on $\operatorname{Aut}(\Sigma_d)$
cardona-qft-methods_F00594	35	■ 1.000	ρ_d	ρ_d	$S = \operatorname{Spin}(TM) \times_{\rho_d} \Sigma_d$. Here ρ_d is a representation of Spin_d on $\operatorname{Aut}(\Sigma_d)$
cardona-qft-methods_F00595	35	■ 1.000	$\operatorname{Aut}(\Sigma_d)$	$\operatorname{Aut}(\Sigma_d)$	$S = \operatorname{Spin}(TM) \times_{\rho_d} \Sigma_d$. Here ρ_d is a representation of Spin_d on $\operatorname{Aut}(\Sigma_d)$
cardona-qft-methods_F00596	35	■ 1.000	$d=2 m$	$d = 2m$	More precisely, if $d = 2m$ is even, $\rho_d = \rho^+ + \rho^-$ where $\rho^{\pm}$ are two
cardona-qft-methods_F00597	35	■ 1.000	$\rho_d = \rho^+ + \rho^-$	$\rho_d = \rho^+ + \rho^-$	More precisely, if $d = 2m$ is even, $\rho_d = \rho^+ + \rho^-$ where $\rho^{\pm}$ are two
cardona-qft-methods_F00598	35	■ 1.000	$\rho^{\pm}$	$\rho^{\pm}$	More precisely, if $d = 2m$ is even, $\rho_d = \rho^+ + \rho^-$ where $\rho^{\pm}$ are two
cardona-qft-methods_F00599	35	■ 1.000	Spin_{2m}	Spin_{2m}	nonequivalent irreducible complex representations of Spin_{2m} and $\Sigma_{2m} =$
cardona-qft-methods_F00600	35	■ 1.000	$\Sigma_{2m} =$	$\Sigma_{2m} =$	nonequivalent irreducible complex representations of Spin_{2m} and $\Sigma_{2m} =$
cardona-qft-methods_F00601	35	■ 1.000	$\Sigma_{2m}^+ \oplus \Sigma_{2m}^-$	$\Sigma_{2m}^+ \oplus \Sigma_{2m}^-$	$\Sigma_{2m}^+ \oplus \Sigma_{2m}^-$, while for $d = 2m + 1$ odd, the spinor representation ρ_d is
cardona-qft-methods_F00602	35	■ 1.000	$d=2 m+1$	$d = 2m + 1$	$\Sigma_{2m}^+ \oplus \Sigma_{2m}^-$, while for $d = 2m + 1$ odd, the spinor representation ρ_d is
cardona-qft-methods_F00603	35	■ 0.798	$S = S^+ \oplus S^-$	$S = S^+ \oplus S^-$	$S = S^+ \oplus S^-$ such that $S \simeq \bigwedge T^*M$. Due to the dimension of M , the
cardona-qft-methods_F00604	35	■ 0.798	$S \simeq \bigwedge T^*M$	$S \simeq \bigwedge T^*M$	$S = S^+ \oplus S^-$ such that $S \simeq \bigwedge T^*M$. Due to the dimension of M , the

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cardona-qft-methods_F00605	35	■ 1.000	$S \simeq M \times \mathbb{C}^{d/2}$	$S \simeq M \times \mathbb{C}^{d/2}$	Locally, the spinor bundle satisfies $S \simeq M \times \mathbb{C}^{d/2}$.
cardona-qft-methods_F00606	35	■ 0.974	$\nabla^S: \Gamma^\infty(M, S) \rightarrow \Gamma^\infty(M, S) \otimes \Gamma^\infty(M, T^*M)$	$\nabla^S: \Gamma^\infty(M, S) \rightarrow \Gamma^\infty(M, S) \otimes \Gamma^\infty(M, T^*M)$	A spin connection $\nabla^S: \Gamma^\infty(M, S) \rightarrow \Gamma^\infty(M, S) \otimes \Gamma^\infty(M, T^*M)$ is any
cardona-qft-methods_F00607	35	■ 0.977	$\left[\nabla^S, c(\cdot)\right] = c(\nabla^{LC} \cdot)$	$[\nabla^S, c(\cdot)] = c(\nabla^{LC} \cdot)$	connection which is compatible with Clifford action: $[\nabla^S, c(\cdot)] = c(\nabla^{LC} \cdot)$.
cardona-qft-methods_F00608	35	■ 1.000	$\not{D} = -i c \left(dx^j \right) \left(\partial_j - \omega_j(x) \right)$	$\not{D} = -ic(dx^j)(\partial_j - \omega_j(x))$	In coordinates, $\not{D} = -ic(dx^j)(\partial_j - \omega_j(x))$ where ω_j is the spin connec-
cardona-qft-methods_F00609	35	■ 1.000	ω_j	ω_j	In coordinates, $\not{D} = -ic(dx^j)(\partial_j - \omega_j(x))$ where ω_j is the spin connec-
cardona-qft-methods_F00610	35	■ 1.000	$\omega_j = \frac{1}{4} \left(\Gamma_{ji}^k g_{kl} - \right.$	$\omega_j = \frac{1}{4} \left(\Gamma_{ji}^k g_{kl} - \right.$	tion part which can be computed in the coordinate basis $\omega_j = \frac{1}{4} \left(\Gamma_{ji}^k g_{kl} - \right.$
cardona-qft-methods_F00611	35	■ 1.000	$\left. \partial_i (h_j^\alpha) \delta_{\alpha\beta} h_l^\beta \right) c(dx^i) c(dx^l)$	$\partial_i (h_j^\alpha) \delta_{\alpha\beta} h_l^\beta \Big) c(dx^i) c(dx^l)$	$\partial_i (h_j^\alpha) \delta_{\alpha\beta} h_l^\beta \Big) c(dx^i) c(dx^l)$ and the matrix $H = [h_j^\alpha]$ is such that $H^t H =$
cardona-qft-methods_F00612	35	■ 1.000	$H = \left[h_j^\alpha \right]$	$H = [h_j^\alpha]$	$\partial_i (h_j^\alpha) \delta_{\alpha\beta} h_l^\beta \Big) c(dx^i) c(dx^l)$ and the matrix $H = [h_j^\alpha]$ is such that $H^t H =$
cardona-qft-methods_F00613	35	■ 1.000	$H^t H =$	$H^t H =$	$\partial_i (h_j^\alpha) \delta_{\alpha\beta} h_l^\beta \Big) c(dx^i) c(dx^l)$ and the matrix $H = [h_j^\alpha]$ is such that $H^t H =$
cardona-qft-methods_F00614	35	■ 1.000	$\left[g_{ij} \right]$	$[g_{ij}]$	$[g_{ij}]$ (we use Latin letters for coordinate basis indices and Greek letters for
cardona-qft-methods_F00615	35	■ 1.000	$\sigma_1^D(x, \xi) = c(\xi) + i c \left(dx^j \right) \omega_j(x)$	$\sigma_1^D(x, \xi) = c(\xi) + ic(dx^j)\omega_j(x)$	This gives $\sigma_1^D(x, \xi) = c(\xi) + ic(dx^j)\omega_j(x)$. Thus in <i>normal coordinates</i>
cardona-qft-methods_F00616	35	■ 1.000	$x_{\{0\}}$	x_0	around x_0 ,
cardona-qft-methods_F00617	35	■ 1.000	γ	γ	where the γ 's are constant hermitean matrices.
cardona-qft-methods_F00618	35	■ 0.999	$C^\infty(M)$	$C^\infty(M)$	where we have a scalar product which is $C^\infty(M)$ -valued. On its domain
cardona-qft-methods_F00619	35	■ 1.000	$\Gamma^\infty(M, S), \not{D}$	$\Gamma^\infty(M, S), \not{D}$	$\Gamma^\infty(M, S), \not{D}$ is symmetric: $\langle \psi, \not{D}\phi \rangle = \langle \not{D}\psi, \phi \rangle$. Moreover, it has a selfad-
cardona-qft-methods_F00620	35	■ 1.000	$\langle \psi, \not{D}\phi \rangle = \langle \not{D}\psi, \phi \rangle$	$\langle \psi, \not{D}\phi \rangle = \langle \not{D}\psi, \phi \rangle$	$\Gamma^\infty(M, S), \not{D}$ is symmetric: $\langle \psi, \not{D}\phi \rangle = \langle \not{D}\psi, \phi \rangle$. Moreover, it has a selfad-
cardona-qft-methods_F00621	35	■ 0.719	$\not{D}^{**}$	$\not{D}^{**}$	joint closure (which is $\not{D}^{**}$):
cardona-qft-methods_F00622	36	■ 0.919	$\mathcal{H}, \not{D}$	$\mathcal{H}, \not{D}$	<i>mannian spin manifold without boundary. By extension to $\mathcal{H}$, $\not{D}$ is es-</i>
cardona-qft-methods_F00623	36	■ 1.000	$\Gamma^\infty(M, S)$	$\Gamma^\infty(M, S)$	<i>entially selfadjoint on its original domain $\Gamma^\infty(M, S)$. It is a differential</i>
cardona-qft-methods_F00624	36	■ 1.000	$\not{D}$	$\not{D}$	There is a nice formula which relates the Dirac operator $\not{D}$ to the spinor
cardona-qft-methods_F00625	36	■ 1.000	$R \in \Gamma^\infty \left(M, \bigwedge^2 T^*M \otimes \right.$	$R \in \Gamma^\infty \left(M, \bigwedge^2 T^*M \otimes \right.$	Before giving it, we need to fix a few notations: let $R \in \Gamma^\infty \left(M, \bigwedge^2 T^*M \otimes \right.$
cardona-qft-methods_F00626	36	■ 0.785	$\left. \operatorname{End}(TM) \right)$	$\operatorname{End}(TM))$	$\operatorname{End}(TM))$ be the <i>Riemann curvature tensor</i> with components $R_{ijkl} =$
cardona-qft-methods_F00627	36	■ 0.785	$R_{ijkl} =$	$R_{ijkl} =$	$\operatorname{End}(TM))$ be the <i>Riemann curvature tensor</i> with components $R_{ijkl} =$
cardona-qft-methods_F00628	36	■ 1.000	$g(\partial_i, R(\partial_k, \partial_l) \partial_j)$	$g(\partial_i, R(\partial_k, \partial_l) \partial_j)$	$g(\partial_i, R(\partial_k, \partial_l) \partial_j)$, the <i>Ricci tensor</i> components are $R_{jl} = g^{ik} T_{ijkl}$ and the
cardona-qft-methods_F00629	36	■ 1.000	$R_{jl} = g^{ik} T_{ijkl}$	$R_{jl} = g^{ik} T_{ijkl}$	$g(\partial_i, R(\partial_k, \partial_l) \partial_j)$, the <i>Ricci tensor</i> components are $R_{jl} = g^{ik} T_{ijkl}$ and the

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cardona-qft-methods_ F00630	36	■ 1.000	$s=g^{\{j\ l\}} R_{jl}$	$s = g^{jl} R_{jl}$	<i>scalar curvature</i> is $s = g^{jl} R_{jl}$.
cardona-qft-methods_ F00631	36	■ 1.000	$M, \not{D}^2=\Delta^S+\frac{1}{4} s$	$M, \not{D}^2 = \Delta^S + \frac{1}{4} s$	of M , $\not{D}^2 = \Delta^S + \frac{1}{4} s$.
cardona-qft-methods_ F00632	36	■ 1.000	$\not{D}^{-1}$	$\not{D}^{-1}$	We already know via Theorems 2.2 and 4.1 that $\not{D}^{-1}$ is compact so has a
cardona-qft-methods_ F00633	36	■ 1.000	$\left\{\lambda_n(T)\right\}_{n\in\mathbb{N}}$	$\{\lambda_n(T)\}_{n\in\mathbb{N}}$	For $T \in \mathcal{K}_+(\mathcal{H})$, we denote by $\{\lambda_n(T)\}_{n\in\mathbb{N}}$ its spectrum sorted in de-
cardona-qft-methods_ F00634	36	■ 1.000	T^{-1}	T^{-1}	bounded positive operator T such that T^{-1} is compact) and by $N_T(\lambda) =$
cardona-qft-methods_ F00635	36	■ 1.000	$N_{\{T\}(\lambda)}=$	$N_T(\lambda) =$	bounded positive operator T such that T^{-1} is compact) and by $N_T(\lambda) =$
cardona-qft-methods_ F00636	36	■ 0.996	$\#\left\{\lambda_n(T)\mid \lambda_n\leq\lambda\right\}$	$\#\{\lambda_n(T) \mid \lambda_n \leq \lambda\}$	$\#\{\lambda_n(T) \mid \lambda_n \leq \lambda\}$ its counting function.
cardona-qft-methods_ F00637	36	■ 0.999	$N_{ \not{D} }(\lambda)\underset{\lambda\rightarrow\infty}{\sim}\frac{2^d\mathrm{Vol}\left(\mathbb{S}^{d-1}\right)}{d(2\pi)^d}\mathrm{Vol}(M)\lambda^d$	$N_{ \not{D} }(\lambda) \underset{\lambda \rightarrow \infty}{\sim} \frac{2^d \mathrm{Vol}(\mathbb{S}^{d-1})}{d(2\pi)^d} \mathrm{Vol}(M) \lambda^d$	<i>erator counting function</i> is $N_{ \not{D} }(\lambda) \underset{\lambda \rightarrow \infty}{\sim} \frac{2^d \mathrm{Vol}(\mathbb{S}^{d-1})}{d(2\pi)^d} \mathrm{Vol}(M) \lambda^d$ where
cardona-qft-methods_ F00638	36	■ 0.600	$\operatorname{Vol}(\mathrm{M})=\int_{\mathrm{M}}\mathrm{dvol}$	$\mathrm{Vol}(\mathrm{M}) = \int_{\mathrm{M}} \mathrm{dvol}$	$\mathrm{Vol}(\mathrm{M}) = \int_{\mathrm{M}} \mathrm{dvol}$.
cardona-qft-methods_ F00639	36	■ 1.000	$S_{\{g\}}$	S_g	Recall that the spinor bundle S_g and square integrable spinors $\mathcal{H}_g$ defined
cardona-qft-methods_ F00640	36	■ 1.000	$\mathcal{H}_{\{g\}}$	$\mathcal{H}_g$	Recall that the spinor bundle S_g and square integrable spinors $\mathcal{H}_g$ defined
cardona-qft-methods_ F00641	36	■ 1.000	$M_{\{g\}}$	M_g	in (14) depends on the chosen metric g , so we note M_g instead of M and
cardona-qft-methods_ F00642	36	■ 1.000	$\mathcal{H}_{\{g\}}:=L^2\left(M_{\{g\}},S_{\{g\}}\right)$	$\mathcal{H}_g := L^2(M_g, S_g)$	$\mathcal{H}_g := L^2(M_g, S_g)$ and a natural question is: what happens to a Dirac
cardona-qft-methods_ F00643	36	■ 1.000	$g^{\{\prime\}}$	g'	Let g' be another Riemannian metric on M . Since the space of d -forms
cardona-qft-methods_ F00644	36	■ 1.000	$f_{\{g, g^{\{\prime\}}\}}: M \rightarrow \mathbb{R}^{+}$	$f_{g,g'} : M \rightarrow \mathbb{R}^+$	is one-dimensional, there exists a positive function $f_{g,g'} : M \rightarrow \mathbb{R}^+$ such
cardona-qft-methods_ F00645	36	■ 0.945	$d\,v\,o\,l_{\{g^{\{\prime\}}\}}=f_{\{g^{\{\prime\}}\}},g\}$	$dvol_{g'} = f_{g',g}$	that $dvol_{g'} = f_{g',g} dvol_g$.
cardona-qft-methods_ F00646	36	■ 1.000	$I_{\{g, g^{\{\prime\}}\}}(x): S_{\{g\}} \rightarrow S_{\{g^{\{\prime\}}\}}$	$I_{g,g'}(x) : S_g \rightarrow S_{g'}$	Let $I_{g,g'}(x) : S_g \rightarrow S_{g'}$ the natural injection on the spinors spaces
cardona-qft-methods_ F00647	36	■ 1.000	$\left I_{\{g, g^{\{\prime\}}\}}(x)\,\psi(x)\right _{\{g^{\{\prime\}}\}}=$	$ I_{g,g'}(x)\psi(x) _{g'} =$	above point $x \in M$ which is a pointwise linear isometry: $ I_{g,g'}(x)\psi(x) _{g'} =$
cardona-qft-methods_ F00648	37	■ 1.000	$ \psi(x) _{\{g\}}$	$ \psi(x) _g$	$ \psi(x) _g$. Let us first see its construction: there always exists a g -symmetric
cardona-qft-methods_ F00649	37	■ 1.000	$H_{\{g, g^{\{\prime\}}\}}$	$H_{g,g'}$	automorphism $H_{g,g'}$ of the $2^{\lfloor d/2 \rfloor}$ - dimensional vector space TM such that
cardona-qft-methods_ F00650	37	■ 1.000	$2^{\lfloor d/2 \rfloor}$	$2^{\lfloor d/2 \rfloor}$	automorphism $H_{g,g'}$ of the $2^{\lfloor d/2 \rfloor}$ - dimensional vector space TM such that
cardona-qft-methods_ F00651	37	■ 1.000	$g^{\{\prime\}}(X, Y)=g\left(H_{\{g, g^{\{\prime\}}\}} X, Y\right)$	$g'(X, Y) = g(H_{g,g'} X, Y)$	$g'(X, Y) = g(H_{g,g'} X, Y)$ for $X, Y \in TM$ so define $\iota_{g,g'} X = H_{g,g'}^{-1/2} X$.
cardona-qft-methods_ F00652	37	■ 1.000	$X, Y \in T M$	$X, Y \in TM$	$g'(X, Y) = g(H_{g,g'} X, Y)$ for $X, Y \in TM$ so define $\iota_{g,g'} X = H_{g,g'}^{-1/2} X$.
cardona-qft-methods_ F00653	37	■ 1.000	$\iota_{\{g, g^{\{\prime\}}\}} X=H_{\{g, g^{\{\prime\}}\}}^{-1 / 2} X$	$\iota_{g,g'} X = H_{g,g'}^{-1/2} X$	$g'(X, Y) = g(H_{g,g'} X, Y)$ for $X, Y \in TM$ so define $\iota_{g,g'} X = H_{g,g'}^{-1/2} X$.
cardona-qft-methods_ F00654	37	■ 1.000	$\iota_{\{g, g^{\{\prime\}}\}}$	$\iota_{g,g'}$	Note that $\iota_{g,g'}$ commutes with right action of the orthogonal group O_d

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cardona-qft-methods_ F00655	37	■ 1.000	$0_{\{d\}}$	O_d	Note that $\iota_{g,g'}$ commutes with right action of the orthogonal group O_d
cardona-qft-methods_ F00656	37	■ 0.637	$_{\{d\}}$	d	and can be lifted up to a diffeomorphism Pin_d -equivariant on the spin
cardona-qft-methods_ F00657	37	■ 0.963	$I_{\{g, g^{\prime}\}}\left(\right.$	$I_{g,g'} ($	structures associated to g and g' and this lift is denoted by $I_{g,g'}$ (see ⁶). This
cardona-qft-methods_ F00658	37	■ 0.963	$\left.^{\{6\}}\right)$	$\left.^{\{6\}}\right)$	structures associated to g and g' and this lift is denoted by $I_{g,g'}$ (see ⁶). This
cardona-qft-methods_ F00659	37	■ 1.000	$I_{\{g, g^{\prime}\}}: \mathcal{H}_{\{g\}} \rightarrow \mathcal{H}_{\{g^{\prime}\}}$	$I_{g,g'} : \mathcal{H}_g \rightarrow \mathcal{H}_{g'}$	isometry is extended as operator on the Hilbert spaces $I_{g,g'} : \mathcal{H}_g \rightarrow \mathcal{H}_{g'}$
cardona-qft-methods_ F00660	37	■ 0.999	$\left(I_{\{g, g^{\prime}\}} \psi\right)(x):=I_{\{g, g^{\prime}\}}(x) \psi(x)$	$(I_{g,g'}\psi)(x) := I_{g,g'}(x)\psi(x)$	with $(I_{g,g'}\psi)(x) := I_{g,g'}(x)\psi(x)$.
cardona-qft-methods_ F00661	37	■ 1.000	$U_{\{g^{\prime}, g\}}$	$U_{g',g}$	Then by construction, $U_{g',g}$ is a unitary operator from $\mathcal{H}_{g'}$ onto $\mathcal{H}_g$: for
cardona-qft-methods_ F00662	37	■ 1.000	$\mathcal{H}_{\{g^{\prime}\}}$	$\mathcal{H}_{g'}$	Then by construction, $U_{g',g}$ is a unitary operator from $\mathcal{H}_{g'}$ onto $\mathcal{H}_g$: for
cardona-qft-methods_ F00663	37	■ 1.000	$\psi^{\prime} \in \mathcal{H}_{\{g^{\prime}\}}$	$\psi' \in \mathcal{H}_{g'}$	$\psi' \in \mathcal{H}_{g'}$,
cardona-qft-methods_ F00664	37	■ 1.000	$\not D_{\{g^{\prime}\}}$	$\not D_{g'}$	So we can realize $\not D_{g'}$ as an operator $D_{g'}$ acting on $\mathcal{H}_g$ with
cardona-qft-methods_ F00665	37	■ 1.000	$D_{\{g^{\prime}\}}$	$D_{g'}$	So we can realize $\not D_{g'}$ as an operator $D_{g'}$ acting on $\mathcal{H}_g$ with
cardona-qft-methods_ F00666	37	■ 1.000	$H^{\{k\}}\left(M_{\{g\}}, S_{\{g\}}\right)$	$H^k(M_g, S_g)$	In the same vein, the k -th Sobolev space $H^k(M_g, S_g)$ (which is
cardona-qft-methods_ F00667	37	■ 1.000	$\Gamma^{\infty}\left(M_{\{g\}}, S_{\{g\}}\right)$	$\Gamma^{\infty}(M_g, S_g)$	the completion of the space $\Gamma^{\infty}(M_g, S_g)$ under the norm $\ \psi\ _k^2 =$
cardona-qft-methods_ F00668	37	■ 1.000	$\ \psi\ _{\{k\}}^2=$	$\ \psi\ _k^2 =$	the completion of the space $\Gamma^{\infty}(M_g, S_g)$ under the norm $\ \psi\ _k^2 =$
cardona-qft-methods_ F00669	37	■ 0.922	$\sum_{\{j=0\}}^{\{k\}} \int_{\{M\}}\left \nabla^{\{j\}} \psi(x)\right ^2 d x$	$\sum_{j=0}^k \int_M \nabla^j \psi(x) ^2 dx$	$\sum_{j=0}^k \int_M \nabla^j \psi(x) ^2 dx$; be careful, ∇ applied to $\nabla \psi$ is the tensor prod-
cardona-qft-methods_ F00670	37	■ 0.922	$\nabla \psi$	$\nabla \psi$	$\sum_{j=0}^k \int_M \nabla^j \psi(x) ^2 dx$; be careful, ∇ applied to $\nabla \psi$ is the tensor prod-
cardona-qft-methods_ F00671	37	■ 1.000	$T^{\{*\}} M_{\{g\}} \otimes S_{\{g\}}$	$T^*M_g \otimes S_g$	uct connection on $T^*M_g \otimes S_g$ etc, see Theorem 2.2) can be transported:
cardona-qft-methods_ F00672	37	■ 1.000	$U_{\{g, g^{\prime}\}}: H^{\{k\}}\left(M_{\{g\}}, S_{\{g\}}\right) \rightarrow H^{\{k\}}\left(M_{\{g^{\prime}\}}, S_{\{g^{\prime}\}}\right)$	$U_{g,g'} : H^k(M_g, S_g) \rightarrow H^k(M_{g'}, S_{g'})$	the map $U_{g,g'} : H^k(M_g, S_g) \rightarrow H^k(M_{g'}, S_{g'})$ is an isomorphism, see. ⁹⁶ In
cardona-qft-methods_ F00673	37	■ 0.801	$\{ \}^{\{2,6,46,56\}}$	2,6,46,56	g where we can compute explicitly $\not D_{g'}$ and $D_{g'}^{2,6,46,56}$
cardona-qft-methods_ F00674	37	■ 1.000	$g^{\prime}=e^{\{2 h\}} g$	$g' = e^{2h} g$	Theorem 4.3. Let $g' = e^{2h} g$ with $h \in C^{\infty}(M, \mathbb{R})$. Then there exists an
cardona-qft-methods_ F00675	37	■ 1.000	$h \in C^{\infty}\left(M, \mathbb{R}\right)$	$h \in C^{\infty}(M, \mathbb{R})$	Theorem 4.3. Let $g' = e^{2h} g$ with $h \in C^{\infty}(M, \mathbb{R})$. Then there exists an
cardona-qft-methods_ F00676	37	■ 1.000	$I_{\{g, g^{\prime}\}}$	$I_{g,g'}$	isometry $I_{g,g'}$ between the spinor bundle S_g and $S_{g'}$ such that for $\psi \in$
cardona-qft-methods_ F00677	37	■ 1.000	$S_{\{g^{\prime}\}}$	$S_{g'}$	isometry $I_{g,g'}$ between the spinor bundle S_g and $S_{g'}$ such that for $\psi \in$
cardona-qft-methods_ F00678	37	■ 1.000	$\psi \in$	$\psi \in$	isometry $I_{g,g'}$ between the spinor bundle S_g and $S_{g'}$ such that for $\psi \in$
cardona-qft-methods_ F00679	37	■ 0.987	$\Gamma^{\infty}\left(M, S_{\{g\}}\right)$	$\Gamma^{\infty}(M, S_g)$	$\Gamma^{\infty}(M, S_g)$,

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cardona-qft-methods_F00680	38	■ 1.000	$\backslash\mathrm{sigma}^{\mathrm{D}_{\{g^{\backslash\mathrm{prime}}\}}}(x,\backslash\mathrm{xi})=\mathrm{e}^{\{-h(x)\}}\mathrm{c}_{\{g\}}(\backslash\mathrm{xi})$	$\sigma^{D_{g'}}(x,\xi)=e^{-h(x)}c_g(\xi)$	symbol has an x -dependence: $\sigma^{D_{g'}}(x,\xi)=e^{-h(x)}c_g(\xi)$. The principal sym-
cardona-qft-methods_F00681	38	■ 0.965	$\backslash\mathrm{not}\mathrm{D}_{\{g\}}$	$\not{D}_g$	bols of $\not{D}_{g'}$ and $\not{D}_g$ are related by
cardona-qft-methods_F00682	38	■ 1.000	$\mathrm{c}_{\{g^{\backslash\mathrm{prime}}\}}(\backslash\mathrm{xi})=\mathrm{e}^{\{-h(x)\}}\mathrm{U}_{\{g^{\backslash\mathrm{prime}}\},\mathrm{g}\wedge\{-1\}}(x)\mathrm{c}_{\{g\}}(\backslash\mathrm{xi})\mathrm{U}_{\{g^{\backslash\mathrm{prime}}\},\mathrm{g}\}}(x),\backslash\mathrm{xi}\in\mathrm{T}_{\{x\}}^{\wedge\{*\}}\mathrm{M}$	$c_{g'}(\xi)=e^{-h(x)}U_{g',g}^{-1}(x)c_g(\xi)U_{g',g}(x),\xi\in T_x^*M$	Thus $c_{g'}(\xi)=e^{-h(x)}U_{g',g}^{-1}(x)c_g(\xi)U_{g',g}(x)$, $\xi\in T_x^*M$. This formula gives
cardona-qft-methods_F00683	38	■ 1.000	$\mathrm{g}^{\backslash\mathrm{prime}}(\backslash\mathrm{xi},\backslash\mathrm{eta})=\mathrm{e}^{\{-2\,h\}}\mathrm{g}(\backslash\mathrm{xi},\backslash\mathrm{eta})$	$g'(\xi,\eta)=e^{-2h}g(\xi,\eta)$	a verification of $g'(\xi,\eta)=e^{-2h}g(\xi,\eta)$ using $c_g(\xi)c_g(\eta)+c_g(\eta)c_g(\xi)=$
cardona-qft-methods_F00684	38	■ 1.000	$\mathrm{c}_{\{g\}}(\backslash\mathrm{xi})\mathrm{c}_{\{g\}}(\backslash\mathrm{eta})+\mathrm{c}_{\{g\}}(\backslash\mathrm{eta})\mathrm{c}_{\{g\}}(\backslash\mathrm{xi})=$	$c_g(\xi)c_g(\eta)+c_g(\eta)c_g(\xi)=$	a verification of $g'(\xi,\eta)=e^{-2h}g(\xi,\eta)$ using $c_g(\xi)c_g(\eta)+c_g(\eta)c_g(\xi)=$
cardona-qft-methods_F00685	38	■ 0.947	$2\,\mathrm{g}(\backslash\mathrm{xi},\backslash\mathrm{eta})\,\backslash\mathrm{mathrm{id}}_{\{S_{\{g\}}\}}$	$2g(\xi,\eta)\mathrm{id}_{S_g}$	$2g(\xi,\eta)\mathrm{id}_{S_g}$.
cardona-qft-methods_F00686	38	■ 1.000	$\backslash\mathrm{tilde}\{\backslash\mathrm{alpha}\}=\mathrm{H}_{\{\backslash\mathrm{alpha}^{\{*\}}\,\mathrm{g},\,\mathrm{g}\wedge\{-1\,/\,2\}}\,\mathrm{T}\,\backslash\mathrm{alpha}}$	$\tilde{\alpha}=H_{\alpha^*g,g}^{-1/2}T\alpha$	$\tilde{\alpha}=H_{\alpha^*g,g}^{-1/2}T\alpha$, and this lift also exists on S_g when α preserves both the
cardona-qft-methods_F00687	38	■ 0.999	$\mathrm{g}^{\backslash\mathrm{prime}}=\backslash\mathrm{alpha}^{\{*\}}\,\mathrm{g}$	$g'=\alpha^*g$	The pull-back $g'=\alpha^*g$ of the metric g is defined by $(\alpha^*g)_x(\xi,\eta)=$
cardona-qft-methods_F00688	38	■ 0.999	$\backslash\mathrm{left}(\backslash\mathrm{alpha}^{\{*\}}\,\mathrm{g}\backslash\mathrm{right})_{\{x\}}(\backslash\mathrm{xi},\backslash\mathrm{eta})=$	$(\alpha^*g)_x(\xi,\eta)=$	The pull-back $g'=\alpha^*g$ of the metric g is defined by $(\alpha^*g)_x(\xi,\eta)=$
cardona-qft-methods_F00689	38	■ 0.997	$\mathrm{g}_{\{\backslash\mathrm{alpha}(x)\}}\backslash\mathrm{left}(\backslash\mathrm{alpha}_{\{*\}}(\backslash\mathrm{xi}),\backslash\mathrm{alpha}_{\{*\}}\backslash\mathrm{eta}\backslash\mathrm{right}),\,x\in\mathrm{M}$	$g_{\alpha(x)}(\alpha_*(\xi),\alpha_*\eta),x\in M$	$g_{\alpha(x)}(\alpha_*(\xi),\alpha_*\eta)$, $x\in M$, where α_* is the push-forward map : $T_xM\rightarrow$
cardona-qft-methods_F00690	38	■ 0.997	$\backslash\mathrm{alpha}_{\{*\}}$	α_*	$g_{\alpha(x)}(\alpha_*(\xi),\alpha_*\eta)$, $x\in M$, where α_* is the push-forward map : $T_xM\rightarrow$
cardona-qft-methods_F00691	38	■ 0.997	$\mathrm{T}_{\{x\}}\,\mathrm{M}\,\backslash\mathrm{rightarrow}$	$T_xM\rightarrow$	$g_{\alpha(x)}(\alpha_*(\xi),\alpha_*\eta)$, $x\in M$, where α_* is the push-forward map : $T_xM\rightarrow$
cardona-qft-methods_F00692	38	■ 0.998	$\mathrm{T}_{\{\backslash\mathrm{alpha}(x)\}}\,\mathrm{M}$	$T_{\alpha(x)}M$	$T_{\alpha(x)}M$. Of course, the metric g' and g are different but the geodesic dis-
cardona-qft-methods_F00693	38	■ 1.000	$\mathrm{d}_{\{g^{\backslash\mathrm{prime}}\}}=\backslash\mathrm{alpha}^{\{*\}}\,\mathrm{d}_{\{g\}}$	$d_{g'}=\alpha^*d_g$	tances are the same and one checks that $d_{g'}=\alpha^*d_g$.
cardona-qft-methods_F00694	38	■ 1.000	$\backslash\mathrm{sigma}_{\{d\}}^{\mathrm{D}}(x,\backslash\mathrm{xi})=\mathrm{c}_{\{g\}}(\backslash\mathrm{xi})$	$\sigma_d^D(x,\xi)=c_g(\xi)$	The principal symbol of a Dirac operator D is $\sigma_d^D(x,\xi)=c_g(\xi)$ so gives
cardona-qft-methods_F00695	38	■ 1.000	$\mathrm{d}_{\{\backslash\mathrm{alpha}^{\{*\}}\,\mathrm{g}\}}$	d_{α^*g}	ric distance; see the remark after (24). The link between d_{α^*g} and d_g is
cardona-qft-methods_F00696	38	■ 1.000	$\mathrm{d}_{\{g\}}$	d_g	ric distance; see the remark after (24). The link between d_{α^*g} and d_g is
cardona-qft-methods_F00697	38	■ 1.000	$\{ \}^{\wedge\{4,43,44\}}$	$4,43,44$	References for this section: ^{4,43,44} and especially. ¹⁰⁷
cardona-qft-methods_F00698	38	■ 1.000	$\mathrm{e}^{\{t\,\backslash\mathrm{Delta}\}}$	$e^{t\Delta}$	The heat kernel is a Green function of the heat operator $e^{t\Delta}$ (recall that
cardona-qft-methods_F00699	38	■ 1.000	$-\backslash\mathrm{Delta}$	$-\Delta$	$-\Delta$ is a positive operator) which measures the temperature evolution in a
cardona-qft-methods_F00700	38	■ 1.000	$\backslash\mathrm{lim}_{\{t\,\backslash\mathrm{downarrowarrow}\,0\}}\mathrm{k}_{\{t\}}(x,\mathrm{y})=\backslash\mathrm{delta}(x-\mathrm{y})$	$\lim_{t\downarrow 0}k_t(x,y)=\delta(x-y)$	with initial condition $\lim_{t\downarrow 0}k_t(x,y)=\delta(x-y)$. Actually, $k_t(x,y)=$
cardona-qft-methods_F00701	38	■ 1.000	$\mathrm{k}_{\{t\}}(x,\mathrm{y})=$	$k_t(x,y)=$	with initial condition $\lim_{t\downarrow 0}k_t(x,y)=\delta(x-y)$. Actually, $k_t(x,y)=$
cardona-qft-methods_F00702	38	■ 1.000	$\backslash\mathrm{frac}\{1\}{2\,\backslash\mathrm{pi}}\,\backslash\mathrm{int}_{\{-\backslash\mathrm{infty}\}}^{\backslash\mathrm{infty}}\mathrm{e}^{\{-t\,\mathrm{s}^{\wedge\{2\}}\}}\mathrm{e}^{\{i\,\mathrm{s}(x-\mathrm{y})\}}\,\mathrm{d}\,\mathrm{s}$	$\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-ts^2}e^{is(x-y)}ds$	$\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-ts^2}e^{is(x-y)}ds$ when $d=1$.
cardona-qft-methods_F00703	38	■ 1.000	$\mathrm{d}=1$	$d=1$	$\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-ts^2}e^{is(x-y)}ds$ when $d=1$.
cardona-qft-methods_F00704	39	■ 1.000	$\mathrm{f}\in\backslash\mathrm{mathcal{D}}\backslash\mathrm{left}(\backslash\mathrm{mathbb{R}}^{\wedge\{d\}}\backslash\mathrm{right})$	$f\in\mathcal{D}(\mathbb{R}^d)$	Note that for $f\in\mathcal{D}(\mathbb{R}^d)$, we have $\lim_{t\downarrow 0}\int_{\mathbf{R}^d}k_t(x,y)f(y)dy=f(x)$.
cardona-qft-methods_F00705	39	■ 1.000	$\backslash\mathrm{lim}_{\{t\,\backslash\mathrm{downarrowarrow}\,0\}}\backslash\mathrm{int}_{\{\backslash\mathrm{mathbf{R}}^{\wedge\{d\}}\}}\mathrm{k}_{\{t\}}(x,\mathrm{y})\,\mathrm{f}(\mathrm{y})\,\mathrm{d}\,\mathrm{y}=\mathrm{f}(x)$	$\lim_{t\downarrow 0}\int_{\mathbf{R}^d}k_t(x,y)f(y)dy=f(x)$	Note that for $f\in\mathcal{D}(\mathbb{R}^d)$, we have $\lim_{t\downarrow 0}\int_{\mathbf{R}^d}k_t(x,y)f(y)dy=f(x)$.

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cardona-qft-methods_F00706	39	■ 1.000	λ_n	λ_n	U , let λ_n be the eigenvalues for the Dirichlet problem of minus the Lapla-
cardona-qft-methods_F00707	39	■ 1.000	$\psi_n \in L^2(U)$	$\psi_n \in L^2(U)$	If $\psi_n \in L^2(U)$ are the normalized eigenfunctions, the inverse Dirichlet
cardona-qft-methods_F00708	39	■ 1.000	Δ^{-1}	Δ^{-1}	Laplacian Δ^{-1} is a selfadjoint compact operator, $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$
cardona-qft-methods_F00709	39	■ 1.000	$0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$	$0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$	Laplacian Δ^{-1} is a selfadjoint compact operator, $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq$
cardona-qft-methods_F00710	39	■ 1.000	$\cdots, \lambda_n \rightarrow \infty$	$\cdots, \lambda_n \rightarrow \infty$	$\cdots, \lambda_n \rightarrow \infty$.
cardona-qft-methods_F00711	39	■ 1.000	$f(x) = \int_0^\infty dt e^{-tx} \phi(x)$	$f(x) = \int_0^\infty dt e^{-tx} \phi(x)$	The interest for the heat kernel is that, if $f(x) = \int_0^\infty dt e^{-tx} \phi(x)$ is the
cardona-qft-methods_F00712	39	■ 1.000	$\operatorname{Tr}(f(-\Delta)) = \int_0^\infty dt \operatorname{Tr}(e^{t\Delta})$	$\operatorname{Tr}(f(-\Delta)) = \int_0^\infty dt \phi(t) \operatorname{Tr}(e^{t\Delta})$	Laplace transform of ϕ , then $\operatorname{Tr}(f(-\Delta)) = \int_0^\infty dt \phi(t) \operatorname{Tr}(e^{t\Delta})$ (if every-
cardona-qft-methods_F00713	39	■ 0.986	$\operatorname{Tr}(e^{t\Delta}) = \int_M d\operatorname{vol}(x) \operatorname{tr}_{V_x} k_t(x, x)$	$\operatorname{Tr}(e^{t\Delta}) = \int_M d\operatorname{vol}(x) \operatorname{tr}_{V_x} k_t(x, x)$	thing makes sense) is controlled by $\operatorname{Tr}(e^{t\Delta}) = \int_M d\operatorname{vol}(x) \operatorname{tr}_{V_x} k_t(x, x)$ since
cardona-qft-methods_F00714	39	■ 1.000	$\operatorname{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$	$\operatorname{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$	$\operatorname{Tr}(e^{t\Delta}) = \sum_{n=1}^\infty e^{t\lambda_n}$ and
cardona-qft-methods_F00715	39	■ 1.000	k_t	k_t	So it is useful to know the asymptotics of the heat kernel k_t on the diagonal
cardona-qft-methods_F00716	39	■ 1.000	$t=0$	$t=0$	of $M \times M$ especially near $t=0$.
cardona-qft-methods_F00717	39	■ 1.000	$P \in \Psi DO^m(M, V)$	$P \in \Psi DO^m(M, V)$	V be a vector bundle over M and $P \in \Psi DO^m(M, V)$ be a positive elliptic
cardona-qft-methods_F00718	39	■ 1.000	$m > 0$	$m > 0$	operator of order $m > 0$. If $k_t(x, y)$ is the kernel of the heat operator e^{-tP} ,
cardona-qft-methods_F00719	39	■ 1.000	$k_t(x, y)$	$k_t(x, y)$	operator of order $m > 0$. If $k_t(x, y)$ is the kernel of the heat operator e^{-tP} ,
cardona-qft-methods_F00720	39	■ 1.000	e^{-tP}	e^{-tP}	operator of order $m > 0$. If $k_t(x, y)$ is the kernel of the heat operator e^{-tP} ,
cardona-qft-methods_F00721	39	■ 1.000	$\left k_t(x, x) - \sum_{k \leq k(n)} a_k(x) t^{(-d+k)/m} \right _{\infty, n} < c_n t^n$	$\left k_t(x, x) - \sum_{k \leq k(n)} a_k(x) t^{(-d+k)/m} \right _{\infty, n} < c_n t^n$	which means that $\left k_t(x, x) - \sum_{k \leq k(n)} a_k(x) t^{(-d+k)/m} \right _{\infty, n} < c_n t^n$ for $0 <$
cardona-qft-methods_F00722	39	■ 1.000	$0 <$	$0 <$	which means that $\left k_t(x, x) - \sum_{k \leq k(n)} a_k(x) t^{(-d+k)/m} \right _{\infty, n} < c_n t^n$ for $0 <$
cardona-qft-methods_F00723	39	■ 1.000	$t < 1$	$t < 1$	$t < 1$ where we used $ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha \leq n} \partial_x^\alpha f $ (since P is elliptic,
cardona-qft-methods_F00724	39	■ 1.000	$ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha \leq n} \partial_x^\alpha f $	$ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha \leq n} \partial_x^\alpha f $	$t < 1$ where we used $ f _{\infty, n} := \sup_{x \in M} \sum_{ \alpha \leq n} \partial_x^\alpha f $ (since P is elliptic,
cardona-qft-methods_F00725	39	■ 0.665	(t, x, y)	(t, x, y)	$k_t(x, y)$ is a smooth function of (t, x, y) for $t > 0$, see [43, section 1.6, 1.7]).
cardona-qft-methods_F00726	39	■ 1.000	$k(t, f, P) = \operatorname{Tr}(f e^{-tP})$	$k(t, f, P) = \operatorname{Tr}(f e^{-tP})$	More generally, we will use $k(t, f, P) = \operatorname{Tr}(f e^{-tP})$ where f is a smooth
cardona-qft-methods_F00727	40	■ 1.000	a_k	a_k	—
cardona-qft-methods_F00728	40	■ 1.000	$\{ \}^{43,44}$	43,44	1) The existence of this asymptotics is non-trivial. ^{43,44}
cardona-qft-methods_F00729	40	■ 1.000	$a_{2k}(f, P)$	$a_{2k}(f, P)$	2) The coefficients $a_{2k}(f, P)$ can be computed locally as integral of
cardona-qft-methods_F00730	40	■ 1.000	$a_{2k+1}(f, P) = 0$	$a_{2k+1}(f, P) = 0$	3) The odd coefficients are zero: $a_{2k+1}(f, P) = 0$.
cardona-qft-methods_F00731	40	■ 1.000	$(g^{\mu\nu})_{1 \leq \mu, \nu \leq d}$	$(g^{\mu\nu})_{1 \leq \mu, \nu \leq d}$	where $(g^{\mu\nu})_{1 \leq \mu, \nu \leq d}$ is the inverse matrix associated to the metric g on M ,

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cardona-qft-methods_ F00732	40	■ 1.000	$\mathbb{A}^\mu$	$\mathbb{A}^\mu$	and $\mathbb{A}^\mu$ and $\mathbb{B}$ are smooth $L(V)$ -sections on M (endomorphisms) (see also
cardona-qft-methods_ F00733	40	■ 1.000	$\mathbb{B}$	$\mathbb{B}$	and $\mathbb{A}^\mu$ and $\mathbb{B}$ are smooth $L(V)$ -sections on M (endomorphisms) (see also
cardona-qft-methods_ F00734	40	■ 1.000	$L(V)$	$L(V)$	and $\mathbb{A}^\mu$ and $\mathbb{B}$ are smooth $L(V)$ -sections on M (endomorphisms) (see also
cardona-qft-methods_ F00735	40	■ 1.000	X, Y	X, Y	X, Y are vector fields on M and ∇^{LC} is the Levi-Civita connection on M .
cardona-qft-methods_ F00736	40	■ 1.000	∇^{LC}	∇^{LC}	X, Y are vector fields on M and ∇^{LC} is the Levi-Civita connection on M .
cardona-qft-methods_ F00737	40	■ 1.000	$\Gamma_{\mu\nu}^\rho$	$\Gamma_{\mu\nu}^\rho$	where $\Gamma_{\mu\nu}^\rho$ are the Christoffel coefficients of ∇^{LC} . Moreover (with local
cardona-qft-methods_ F00738	40	■ 0.993	$\nabla = dx^\mu \otimes (\partial_\mu + \omega_\mu)$	$\nabla = dx^\mu \otimes (\partial_\mu + \omega_\mu)$	frames of T^*M and V), $\nabla = dx^\mu \otimes (\partial_\mu + \omega_\mu)$ and E are related to $g^{\mu\nu}$, $\mathbb{A}^\mu$
cardona-qft-methods_ F00739	40	■ 0.993	$g^{\mu\nu}, \mathbb{A}^\mu$	$g^{\mu\nu}, \mathbb{A}^\mu$	frames of T^*M and V), $\nabla = dx^\mu \otimes (\partial_\mu + \omega_\mu)$ and E are related to $g^{\mu\nu}$, $\mathbb{A}^\mu$
cardona-qft-methods_ F00740	40	■ 0.969	$a_k(f, P) = \int_M$	$a_k(f, P) = \int_M$	In this case, the coefficients $a_k(f, P) = \int_M dvol_g \operatorname{tr}_E(f(x) a_k(P)(x))$ and
cardona-qft-methods_ F00741	40	■ 0.969	$\operatorname{tr}_E(f(x) a_k(P)(x))$	$\operatorname{tr}_E(f(x) a_k(P)(x))$	In this case, the coefficients $a_k(f, P) = \int_M dvol_g \operatorname{tr}_E(f(x) a_k(P)(x))$ and
cardona-qft-methods_ F00742	40	■ 1.000	$a_k(P) = c_i \alpha_k^i(P)$	$a_k(P) = c_i \alpha_k^i(P)$	the $a_k(P) = c_i \alpha_k^i(P)$ are linear combination with constants c_i of all possible
cardona-qft-methods_ F00743	40	■ 1.000	c_i	c_i	the $a_k(P) = c_i \alpha_k^i(P)$ are linear combination with constants c_i of all possible
cardona-qft-methods_ F00744	40	■ 1.000	$\alpha_k^i(P)$	$\alpha_k^i(P)$	independent invariants $\alpha_k^i(P)$ of dimension k constructed from E, Ω, R and
cardona-qft-methods_ F00745	40	■ 1.000	E, Ω, R	E, Ω, R	independent invariants $\alpha_k^i(P)$ of dimension k constructed from E, Ω, R and
cardona-qft-methods_ F00746	40	■ 1.000	$k=2, E$	$k=2, E$	Riemann curvature tensor). As an example, for $k=2, E$ and s are the only
cardona-qft-methods_ F00747	41	■ 1.000	a_6	a_6	The coefficient a_6 was computed by Gilkey, a_8 by Amsterdamski, Berkin
cardona-qft-methods_ F00748	41	■ 1.000	a_8	a_8	The coefficient a_6 was computed by Gilkey, a_8 by Amsterdamski, Berkin
cardona-qft-methods_ F00749	41	■ 1.000	a_{10}	a_{10}	and O'Connor and a_{10} in 1998 by van de Ven. ¹⁰⁸ Some higher coefficients
cardona-qft-methods_ F00750	41	■ 1.000	$a_k(P)(x) = \frac{1}{m} c_{P(k-d)/m}(x)$	$a_k(P)(x) = \frac{1}{m} c_{P(k-d)/m}(x)$	Wodzicki has proved that, in (17), $a_k(P)(x) = \frac{1}{m} c_{P(k-d)/m}(x)$ is true not
cardona-qft-methods_ F00751	41	■ 1.000	$k=0$	$k=0$	only for $k=0$ as seen in Theorem 3.2 (where $P \leftrightarrow P^{-1}$), but for all $k \in \mathbb{N}$.
cardona-qft-methods_ F00752	41	■ 1.000	$P \leftrightarrow P^{-1}$	$P \leftrightarrow P^{-1}$	only for $k=0$ as seen in Theorem 3.2 (where $P \leftrightarrow P^{-1}$), but for all $k \in \mathbb{N}$.
cardona-qft-methods_ F00753	41	■ 1.000	$k \in \mathbb{N}$	$k \in \mathbb{N}$	only for $k=0$ as seen in Theorem 3.2 (where $P \leftrightarrow P^{-1}$), but for all $k \in \mathbb{N}$.
cardona-qft-methods_ F00754	41	■ 1.000	$\mathcal{H} = L^2(M, S)$	$\mathcal{H} = L^2(M, S)$	operator on $\mathcal{H} = L^2(M, S)$.
cardona-qft-methods_ F00755	41	■ 1.000	e^{-tD^2}	e^{-tD^2}	We are going to use the heat operator e^{-tD^2} since D^2 is related to the
cardona-qft-methods_ F00756	41	■ 0.994	$t>0, e^{-tD^2} \in \mathcal{L}^1$	$t>0, e^{-tD^2} \in \mathcal{L}^1$	For $t>0$, $e^{-tD^2} \in \mathcal{L}^1$: follows from $e^{-tD^2} = (1 + D^2)^{(d+1)/2} e^{-tD^2} (1 +$
cardona-qft-methods_ F00757	41	■ 0.994	$e^{-tD^2} = (1 + D^2)^{(d+1)/2} e^{-tD^2} (1 +$	$e^{-tD^2} = (1 + D^2)^{(d+1)/2} e^{-tD^2} (1 +$	For $t>0$, $e^{-tD^2} \in \mathcal{L}^1$: follows from $e^{-tD^2} = (1 + D^2)^{(d+1)/2} e^{-tD^2} (1 +$

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cardona-qft-methods_F00758	41	■ 1.000	$\left.D^{\wedge 2}\right)^{\{-\left(d+1\right) / 2\}}$	$D^2)^{-(d+1)/2}$	$D^2)^{-(d+1)/2}$, since $(1+D^2)^{-(d+1)/2} \in \mathcal{L}^1$ and $\lambda \rightarrow (1+\lambda^2)^{(d+1)/2} e^{-t\lambda^2}$ is
cardona-qft-methods_F00759	41	■ 1.000	$\left(1+D^{\wedge 2}\right)^{\{-\left(d+1\right) / 2\}} \in \mathcal{L}^1$	$(1+D^2)^{-(d+1)/2} \in \mathcal{L}^1$	$D^2)^{-(d+1)/2}$, since $(1+D^2)^{-(d+1)/2} \in \mathcal{L}^1$ and $\lambda \rightarrow (1+\lambda^2)^{(d+1)/2} e^{-t\lambda^2}$ is
cardona-qft-methods_F00760	41	■ 1.000	$\lambda \rightarrow\left(1+\lambda^2\right)^{\left(d+1\right) / 2} e^{-t \lambda^2}$	$\lambda \rightarrow (1+\lambda^2)^{(d+1)/2} e^{-t\lambda^2}$	$D^2)^{-(d+1)/2}$, since $(1+D^2)^{-(d+1)/2} \in \mathcal{L}^1$ and $\lambda \rightarrow (1+\lambda^2)^{(d+1)/2} e^{-t\lambda^2}$ is
cardona-qft-methods_F00761	41	■ 1.000	$\operatorname{Tr}\left(e^{-t D^{\wedge 2}}\right)=\sum _n e^{-t \lambda _n^2}<\infty$	$\operatorname{Tr}\left(e^{-t D^2}\right)=\sum _n e^{-t \lambda _n^2}<\infty$	bounded. So $\operatorname{Tr}\left(e^{-t D^2}\right)=\sum _n e^{-t \lambda _n^2}<\infty$.
cardona-qft-methods_F00762	41	■ 0.989	$\left(\left\{\right\}^{\wedge 70}\right)$	$(^{70})$	larizing $(^{70})$ and the asymptotics of its kernel is, see (16): $k_t(x, y) \underset{t \downarrow 0^+}{\sim}$
cardona-qft-methods_F00763	41	■ 0.989	$k_{\{t\}}(x, y) \underset{t \downarrow 0^+}{\sim}$	$k_t(x, y) \underset{t \downarrow 0^+}{\sim}$	larizing $(^{70})$ and the asymptotics of its kernel is, see (16): $k_t(x, y) \underset{t \downarrow 0^+}{\sim}$
cardona-qft-methods_F00764	41	■ 1.000	$\frac{1}{\left(4 \pi t\right)^{d / 2}} \sqrt{\operatorname{det} g_x} \sum _{j \geq 0} k_j(x, y) t^j e^{-d_g(x, y)^2 / 4 t}$	$\frac{1}{(4 \pi t)^{d/2}} \sqrt{\det g_x} \sum_{j \geq 0} k_j(x, y) t^j e^{-d_g(x, y)^2/4t}$	$\frac{1}{(4 \pi t)^{d/2}} \sqrt{\det g_x} \sum_{j \geq 0} k_j(x, y) t^j e^{-d_g(x, y)^2/4t}$ where k_j is a smooth section
cardona-qft-methods_F00765	41	■ 1.000	$k_{\{j\}}$	k_j	$\frac{1}{(4 \pi t)^{d/2}} \sqrt{\det g_x} \sum_{j \geq 0} k_j(x, y) t^j e^{-d_g(x, y)^2/4t}$ where k_j is a smooth section
cardona-qft-methods_F00766	41	■ 1.000	$E^{\ast} \otimes E$	$E^* \otimes E$	on $\dot{E}^* \otimes E$. Thus
cardona-qft-methods_F00767	41	■ 1.000	$a_{\{2 j\}}\left(D^{\wedge 2}\right)=\frac{1}{\left(4 \pi i\right)^{d / 2}} \operatorname{tr}\left(k_{\{j\}}(x, x)\right) \sqrt{\operatorname{det} g_x} d x , \quad a_{2 j+1}\left(D^2\right)=0$	$a_{2j}\left(D^2\right)=\frac{1}{(4 \pi)^{d / 2}} \int_M \operatorname{tr}\left(k_j(x, x)\right) \sqrt{\det g_x} d x , \quad a_{2j+1}\left(D^2\right)=0$	with $a_{2j}\left(D^2\right)=\frac{1}{(4 \pi)^{d / 2}} \int_M \operatorname{tr}\left(k_j(x, x)\right) \sqrt{\det g_x} d x , \quad a_{2j+1}\left(D^2\right)=0$, for
cardona-qft-methods_F00768	41	■ 1.000	$j \in \mathbb{N}$	$j \in \mathbb{N}$	$j \in \mathbb{N}$.
cardona-qft-methods_F00769	42	■ 1.000	$p, \theta \leq p \leq d, D^{\wedge\{-p\}} \in \Psi D 0^{\wedge\{-p\}}(M, E)$	$p, 0 \leq p \leq d, D^{-p} \in \Psi DO^{-p}(M, E)$	Theorem 5.1. For any integer $p, 0 \leq p \leq d, D^{-p} \in \Psi DO^{-p}(M, E)$ and
cardona-qft-methods_F00770	42	■ 0.953	$p=d, W \operatorname{Res}\left(D^{-d}\right)=\frac{2}{\Gamma(p / 2)} a_0\left(D^2\right)=\frac{2}{\Gamma(p / 2)} \frac{\operatorname{Rank}(E)}{(4 \pi)^{d / 2}} \operatorname{Vol}(M)$	$p=d, W \operatorname{Res}\left(D^{-d}\right)=\frac{2}{\Gamma(p / 2)} a_0\left(D^2\right)=\frac{2}{\Gamma(p / 2)} \frac{\operatorname{Rank}(E)}{(4 \pi)^{d / 2}} \operatorname{Vol}(M)$	1) If $p=d, W \operatorname{Res}\left(D^{-d}\right)=\frac{2}{\Gamma(p / 2)} a_0\left(D^2\right)=\frac{2}{\Gamma(p / 2)} \frac{\operatorname{Rank}(E)}{(4 \pi)^{d / 2}} \operatorname{Vol}(M)$.
cardona-qft-methods_F00771	42	■ 1.000	$\operatorname{Tr}\left(e^{-t D^{\wedge 2}}\right) \underset{t \downarrow 0^+}{\sim} a_{\{0\}}\left(D^{\wedge 2}\right) t^{-d / 2}$	$\operatorname{Tr}\left(e^{-t D^2}\right) \underset{t \downarrow 0^+}{\sim} a_0\left(D^2\right) t^{-d / 2}$	Since $\operatorname{Tr}\left(e^{-t D^2}\right) \underset{t \downarrow 0^+}{\sim} a_0\left(D^2\right) t^{-d / 2}$, the Tauberian theorem used in Example
cardona-qft-methods_F00772	42	■ 1.000	$D^{\wedge\{-d\}}=\left(D^{\wedge\{-2\}}\right)^{d / 2}$	$D^{-d}=\left(D^{-2}\right)^{d / 2}$	3.1 implies that $D^{-d}=\left(D^{-2}\right)^{d / 2}$ is measurable and we obtain Connes' trace
cardona-qft-methods_F00773	42	■ 1.000	$D=\not{D}$	$D=\not{D}$	2) When $D=\not{D}$ and E is the spinor bundle, the Seeley-deWit coefficient
cardona-qft-methods_F00774	42	■ 0.648	$a_{\{2\}}\left(\mathcal{D}^2\right)$	$a_2\left(\mathcal{D}^2\right)$	$a_2(\mathcal{D}^2)$ (see (19) with $f=1$) can be easily computed (see ^{43,49}): if s is the
cardona-qft-methods_F00775	42	■ 0.648	(19)	(19)	$a_2(\mathcal{D}^2)$ (see (19) with $f=1$) can be easily computed (see ^{43,49}): if s is the
cardona-qft-methods_F00776	42	■ 0.648	$f=1$	$f=1$	$a_2(\mathcal{D}^2)$ (see (19) with $f=1$) can be easily computed (see ^{43,49}): if s is the
cardona-qft-methods_F00777	42	■ 0.648	$\left\{\right\}^{\wedge 43,49}$	^{43,49}	$a_2(\mathcal{D}^2)$ (see (19) with $f=1$) can be easily computed (see ^{43,49}): if s is the
cardona-qft-methods_F00778	42	■ 0.480	$W \operatorname{Res}\left(\not{D}^{-d+2}\right)=\frac{2}{\Gamma(d / 2-1)} a_2\left(\not{D}^2\right)=c \int_M s(x) d \operatorname{vol}_g(x)$	$W \operatorname{Res}\left(\not{D}^{-d+2}\right)=\frac{2}{\Gamma(d / 2-1)} a_2\left(\not{D}^2\right)=c \int_M s(x) d \operatorname{vol}_g(x)$	So $W \operatorname{Res}\left(\not{D}^{-d+2}\right)=\frac{2}{\Gamma(d / 2-1)} a_2\left(\not{D}^2\right)=c \int_M s(x) d \operatorname{vol}_g(x)$. This is a quite
cardona-qft-methods_F00779	42	■ 1.000	$D \in \Psi D 0^{\wedge\{1\}}(M, E)$	$D \in \Psi DO^1(M, E)$	with notations of Section 2.4, let $P \in \Psi DO^{\mathbb{Z}}(M, E)$ and $D \in \Psi DO^1(M, E)$

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cardona-qft-methods_ F00780	42	■ 1.000	$\zeta_{D}^P(s)=\operatorname{Tr}\left(P D ^{\{-s\}}\right)$	$\zeta_D^P(s) = \operatorname{Tr} \left(P D ^{-s}\right)$	which is elliptic. The definition of zeta function $\zeta_D^P(s) = \operatorname{Tr} \left(P D ^{-s}\right)$ has																																				
cardona-qft-methods_ F00781	42	■ 0.992	$WRes P=\operatorname{Res}_{s=0} \zeta_{D}^P(s)=-\int_M c_P(x) d x $	$WRes P = \operatorname{Res}_{s=0} \zeta_D^P(s) = - \int_M c_P(x) dx $	been useful to prove that $WRes P = \operatorname{Res}_{s=0} \zeta_D^P(s) = - \int_M c_P(x) dx $.																																				
cardona-qft-methods_ F00782	42	■ 1.000	$\{ \}^{\{24,30,33,36,49\}}$	24,30,33,36,49	in the notion of spectral triple. References: ^{24,30,33,36,49}																																				
cardona-qft-methods_ F00783	42	■ 0.982	$\mathcal{A}=C^{\infty}(M), \mathcal{H}=L^2(M, S), \not{D}$	$\mathcal{A} = C^\infty(M), \mathcal{H} = L^2(M, S), \not{D}$	recaptured using the following triple $(\mathcal{A} = C^\infty(M), \mathcal{H} = L^2(M, S), \not{D})$.																																				
cardona-qft-methods_ F00784	42	■ 1.000	$x=\left(x^{\{1\}}, \cdots, x^{\{d\}}\right)$	$x = \left(x^1, \cdots, x^d\right)$	The coordinates $x = (x^1, \cdots, x^d)$ are exchanged with the algebra $C^\infty(M)$,																																				
cardona-qft-methods_ F00785	43	■ 0.985	$K\ 0$	KO	pseudodifferential operators, measure and integration theory, KO -theory,																																				
cardona-qft-methods_ F00786	43	■ 0.997	$\mathcal{A}, \mathcal{H}, \mathcal{D}$	$\mathcal{A}, \mathcal{H}, \mathcal{D}$	Definition 6.1. A spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ is the data of an involutive																																				
cardona-qft-methods_ F00787	43	■ 1.000	$[\mathcal{D}, \pi(a)]$	$[\mathcal{D}, \pi(a)]$	spectrum) such that $[\mathcal{D}, \pi(a)]$ is bounded for any $a \in \mathcal{A}$.																																				
cardona-qft-methods_ F00788	43	■ 1.000	$a \in \mathcal{A}$	$a \in \mathcal{A}$	spectrum) such that $[\mathcal{D}, \pi(a)]$ is bounded for any $a \in \mathcal{A}$.																																				
cardona-qft-methods_ F00789	43	■ 1.000	$\mathcal{A}:=$	$\mathcal{A} :=$	We could impose the existence of a C^* -algebra A where $\mathcal{A} :=$																																				
cardona-qft-methods_ F00790	43	■ 0.980	$\{a \in A \mid [\mathcal{D}, \pi(a)]$	$\{a \in A \mid [\mathcal{D}, \pi(a)]$	$\{a \in A \mid [\mathcal{D}, \pi(a)]$ is bounded $\}$ is norm dense in A so $\mathcal{A}$ is a pre- C^* -algebra																																				
cardona-qft-methods_ F00791	43	■ 0.980	$\}$	$\}$	$\{a \in A \mid [\mathcal{D}, \pi(a)]$ is bounded $\}$ is norm dense in A so $\mathcal{A}$ is a pre- C^* -algebra																																				
cardona-qft-methods_ F00792	43	■ 1.000	$\pi(a)$	$\pi(a)$	When there is no confusion, we will write a instead of $\pi(a)$.																																				
cardona-qft-methods_ F00793	43	■ 0.994	χ	χ	It is even if there is a grading operator χ s.t. $\chi = \chi^*, [\chi, \pi(a)] = 0, \forall a \in$																																				
cardona-qft-methods_ F00794	43	■ 0.994	$\chi=\chi^*, [\chi, \pi(a)]=0, \forall a \in$	$\chi = \chi^*, [\chi, \pi(a)] = 0, \forall a \in$	It is even if there is a grading operator χ s.t. $\chi = \chi^*, [\chi, \pi(a)] = 0, \forall a \in$																																				
cardona-qft-methods_ F00795	43	■ 1.000	$\mathcal{A}, \mathcal{D} \chi = -\chi \mathcal{D}$	$\mathcal{A}, \mathcal{D} \chi = -\chi \mathcal{D}$	$\mathcal{A}, \mathcal{D} \chi = -\chi \mathcal{D}$.																																				
cardona-qft-methods_ F00796	43	■ 1.000	$d \in \mathbb{Z} / 8$	$d \in \mathbb{Z} / 8$	It is real of KO-dimension $d \in \mathbb{Z} / 8$ if there is an antilinear isometry																																				
cardona-qft-methods_ F00797	43	■ 1.000	$J: \mathcal{H} \rightarrow \mathcal{H}$	$J : \mathcal{H} \rightarrow \mathcal{H}$	$J : \mathcal{H} \rightarrow \mathcal{H}$ such that $J\mathcal{D} = \epsilon \mathcal{D}J, J^2 = \epsilon', J\chi = \epsilon'' \chi J$ with the following																																				
cardona-qft-methods_ F00798	43	■ 1.000	$J \mathcal{D} = \epsilon \mathcal{D}J, J^2 = \epsilon', J\chi = \epsilon'' \chi J$	$J\mathcal{D} = \epsilon \mathcal{D}J, J^2 = \epsilon', J\chi = \epsilon'' \chi J$	$J : \mathcal{H} \rightarrow \mathcal{H}$ such that $J\mathcal{D} = \epsilon \mathcal{D}J, J^2 = \epsilon', J\chi = \epsilon'' \chi J$ with the following																																				
cardona-qft-methods_ F00799	43	■ 1.000	$\epsilon, \epsilon', \epsilon''$	$\epsilon, \epsilon', \epsilon''$	table for the signs $\epsilon, \epsilon', \epsilon''$																																				
cardona-qft-methods_ F00800	43	■ 1.000	ϵ'	ϵ'	<table><tr><td>d</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ϵ</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ'</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	d	0	1	2	3	4	5	6	7	ϵ	1	-1	1	1	1	-1	1	1	ϵ'	1	1	-1	-1	-1	-1	1	1	ϵ''	1		-1		1		-1	
d	0	1	2	3	4	5	6	7																																	
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ϵ'	1	1	-1	-1	-1	-1	1	1																																	
ϵ''	1		-1		1		-1																																		
cardona-qft-methods_ F00801	43	■ 1.000	ϵ''	ϵ''	<table><tr><td>d</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ϵ</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ'</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ϵ''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	d	0	1	2	3	4	5	6	7	ϵ	1	-1	1	1	1	-1	1	1	ϵ'	1	1	-1	-1	-1	-1	1	1	ϵ''	1		-1		1		-1	
d	0	1	2	3	4	5	6	7																																	
ϵ	1	-1	1	1	1	-1	1	1																																	
ϵ'	1	1	-1	-1	-1	-1	1	1																																	
ϵ''	1		-1		1		-1																																		
cardona-qft-methods_ F00802	43	■ 1.000	$\pi(a)^{\circ}=J \pi\left(a^{*}\right) J^{-1}$	$\pi(a)^{\circ} = J\pi \left(a^{*}\right) J^{-1}$	where $\pi(a)^{\circ} = J\pi \left(a^{*}\right) J^{-1}$ is a representation of the opposite algebra $\mathcal{A}^{\circ}$.																																				
cardona-qft-methods_ F00803	43	■ 1.000	$\mathcal{A}^{\circ}$	$\mathcal{A}^{\circ}$	where $\pi(a)^{\circ} = J\pi \left(a^{*}\right) J^{-1}$ is a representation of the opposite algebra $\mathcal{A}^{\circ}$.																																				

Identifier	Page	Conf.	LaTeX source	Rendered	Image
cardona-qft-methods_ F08884	43	■ 1.000	$\mu_n\backslash\left(\backslash\mathrm{mathcal{D}}^{\wedge\{-1\}}\backslash\mathrm{right})=\backslash\mathrm{mathcal{0}}\backslash\left(n^{\wedge\{-1\}}\backslash\mathrm{d}\backslash\mathrm{right})$	$\mu_n\left(\mathcal{D}^{-1}\right)=\mathcal{O}\left(n^{-1/d}\right)$	behave like $\mu_n(\mathcal{D}^{-1})=\mathcal{O}(n^{-1/d})$.
cardona-qft-methods_ F08895	43	■ 1.000	$[\backslash\mathrm{mathcal{D}},\backslash\mathrm{mathcal{A}}]$	$[\mathcal{D},\mathcal{A}]$	It is regular if $\mathcal{A}$ and $[\mathcal{D},\mathcal{A}]$ are in the domain of δ^n for all $n\in\mathbb{N}$ where
cardona-qft-methods_ F08896	43	■ 1.000	$\delta^{\wedge\{n\}}$	δ^n	It is regular if $\mathcal{A}$ and $[\mathcal{D},\mathcal{A}]$ are in the domain of δ^n for all $n\in\mathbb{N}$ where
cardona-qft-methods_ F08897	43	■ 1.000	$n\in\backslash\mathrm{in}\backslash\mathrm{mathbb{N}}$	$n\in\mathbb{N}$	It is regular if $\mathcal{A}$ and $[\mathcal{D},\mathcal{A}]$ are in the domain of δ^n for all $n\in\mathbb{N}$ where
cardona-qft-methods_ F08898	43	■ 1.000	$\delta(T)=[\backslash\mathrm{mathcal{D}} ,T]$	$\delta(T)=[\mathcal{D} ,T]$	$\delta(T)=[\mathcal{D} ,T]$.
cardona-qft-methods_ F08899	43	■ 1.000	$\backslash\mathrm{mathcal{H}}^{\wedge\{\infty\}}=$	$\mathcal{H}^\infty=$	It satisfies the finiteness condition if the space of smooth vectors $\mathcal{H}^\infty=$
cardona-qft-methods_ F08810	43	■ 1.000	$\bigcap_k\backslash\mathrm{mathcal{D}}^{\wedge\{k\}}$	$\bigcap_k\mathcal{D}^k$	$\bigcap_k\mathcal{D}^k$ is a finitely projective left $\mathcal{A}$ -module.
cardona-qft-methods_ F08811	43	■ 1.000	$c\in$	$c\in$	It satisfies the orientation condition if there is a Hochschild cycle $c\in$
cardona-qft-methods_ F08812	43	■ 1.000	$Z_{\{d\}}\backslash\left(\backslash\mathrm{mathcal{A}},\backslash\mathrm{mathcal{A}}\backslash\mathrm{otimes}\backslash\mathrm{mathcal{A}}^{\wedge\{\circ\}}\backslash\mathrm{right})$	$Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^\circ)$	$Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^\circ)$ such that $\pi_{\mathcal{D}}(c)=\chi$, where $\pi_{\mathcal{D}}((a\otimes b^\circ)\otimes a_1\otimes\cdots\otimes a_d)=$
cardona-qft-methods_ F08813	43	■ 1.000	$\pi_{\backslash\mathrm{mathcal{D}}}\backslash\left(\backslash\mathrm{mathcal{D}}\right)(c)=\chi$	$\pi_{\mathcal{D}}(c)=\chi$	$Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^\circ)$ such that $\pi_{\mathcal{D}}(c)=\chi$, where $\pi_{\mathcal{D}}((a\otimes b^\circ)\otimes a_1\otimes\cdots\otimes a_d)=$
cardona-qft-methods_ F08814	43	■ 1.000	$\pi_{\backslash\mathrm{mathcal{D}}}\backslash\left(\backslash\left(a\backslash\mathrm{otimes}b^{\wedge\{\circ\}}\backslash\mathrm{right})\backslash\mathrm{otimes}a_{\{1\}}\backslash\mathrm{otimes}\backslash\mathrm{cdots}\backslash\mathrm{otimes}a_{\{d\}}\backslash\mathrm{right})=$	$\pi_{\mathcal{D}}((a\otimes b^\circ)\otimes a_1\otimes\cdots\otimes a_d)=$	$Z_d(\mathcal{A},\mathcal{A}\otimes\mathcal{A}^\circ)$ such that $\pi_{\mathcal{D}}(c)=\chi$, where $\pi_{\mathcal{D}}((a\otimes b^\circ)\otimes a_1\otimes\cdots\otimes a_d)=$
cardona-qft-methods_ F08815	43	■ 1.000	$\pi(a)\pi(b)^{\wedge\{\circ\}}\backslash\left[\backslash\mathrm{mathcal{D}},\pi\backslash\left(a_{\{1\}}\backslash\mathrm{right})\backslash\mathrm{right}]\backslash\mathrm{cdots}\backslash\left[\backslash\mathrm{mathcal{D}},\pi\backslash\left(a_{\{d\}}\backslash\mathrm{right})\backslash\mathrm{right}]\right]$	$\pi(a)\pi(b)^\circ[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$	$\pi(a)\pi(b)^\circ[\mathcal{D},\pi(a_1)]\cdots[\mathcal{D},\pi(a_d)]$ and d is its metric dimension.
cardona-qft-methods_ F08816	44	■ 1.000	$\{ \}^{\wedge\{20,27,30\}}$	20,27,30	model. ^{20,27,30}
cardona-qft-methods_ F08817	44	■ 1.000	J	J	operator J is related to Tomita theory. ¹⁰¹
cardona-qft-methods_ F08818	44	■ 1.000	$\backslash\mathrm{mathcal{A}}\backslash\mathrm{simeq}C^{\wedge\{\infty\}}(M)$	$\mathcal{A}\simeq C^\infty(M)$	$\mathcal{A}\simeq C^\infty(M)$ and $\mathcal{D}$ is a Dirac operator.
cardona-qft-methods_ F08819	44	■ 1.000	$M=\operatorname{Sp}(\backslash\mathrm{mathcal{A}})=\operatorname{Sp}(A)$	$M=\operatorname{Sp}(\mathcal{A})=\operatorname{Sp}(A)$	The manifold is known as a set, $M=\operatorname{Sp}(\mathcal{A})=\operatorname{Sp}(A)$. Notice that $\mathcal{D}$
cardona-qft-methods_ F08820	44	■ 1.000	$\phi_{\{1\}},\phi_{\{2\}}$	ϕ_1,ϕ_2	defines a distance (eventually infinite) between two states ϕ_1,ϕ_2 on the
cardona-qft-methods_ F08821	44	■ 1.000	K	K	class represents the K -homology fundamental class of the noncommutative
cardona-qft-methods_ F08822	44	■ 1.000	$t\in\backslash\mathrm{in}\backslash\mathrm{mathbb{R}}$	$t\in\mathbb{R}$	For $t\in\mathbb{R}$ define the map $F_t:T\in\mathcal{B}(\mathcal{H})\rightarrow e^{it \mathcal{D} }Te^{-it \mathcal{D} }$ and for $\alpha\in\mathbb{R}$
cardona-qft-methods_ F08823	44	■ 1.000	$F_{\{t\}}:T\in\backslash\mathrm{mathcal{B}}(\backslash\mathrm{mathcal{H}})\backslash\mathrm{rightarrow}e^{\{i\}t \backslash\mathrm{mathcal{D}} }T e^{-\{i\}t \backslash\mathrm{mathcal{D}} }$	$F_t:T\in\mathcal{B}(\mathcal{H})\rightarrow e^{it \mathcal{D} }Te^{-it \mathcal{D} }$	For $t\in\mathbb{R}$ define the map $F_t:T\in\mathcal{B}(\mathcal{H})\rightarrow e^{it \mathcal{D} }Te^{-it \mathcal{D} }$ and for $\alpha\in\mathbb{R}$
cardona-qft-methods_ F08824	44	■ 1.000	$\alpha\in\backslash\mathrm{in}\backslash\mathrm{mathbb{R}}$	$\alpha\in\mathbb{R}$	For $t\in\mathbb{R}$ define the map $F_t:T\in\mathcal{B}(\mathcal{H})\rightarrow e^{it \mathcal{D} }Te^{-it \mathcal{D} }$ and for $\alpha\in\mathbb{R}$
cardona-qft-methods_ F08825	44	■ 0.777	$OP^{\wedge\{0\}}=\backslash\left\{T\backslash\mathrm{mid}t\backslash\mathrm{rightarrow}F_{\{t\}}(T)\in C^{\wedge\{\infty\}}(\backslash\mathrm{mathbb{R}},\backslash\mathrm{mathcal{B}}(\backslash\mathrm{mathcal{H}}))\backslash\mathrm{right}\backslash\right\}$	$OP^0=\{T\mid t\rightarrow F_t(T)\in C^\infty(\mathbb{R},\mathcal{B}(\mathcal{H}))\}$	$OP^0=\{T\mid t\rightarrow F_t(T)\in C^\infty(\mathbb{R},\mathcal{B}(\mathcal{H}))\}$ is the set of operators or order ≤ 0 ,
cardona-qft-methods_ F08826	44	■ 0.777	≤ 0	≤ 0	$OP^0=\{T\mid t\rightarrow F_t(T)\in C^\infty(\mathbb{R},\mathcal{B}(\mathcal{H}))\}$ is the set of operators or order ≤ 0 ,
cardona-qft-methods_ F08827	44	■ 1.000	$OP^{\wedge\{\alpha\}}=\backslash\left\{\backslash\left(T \backslash\mathrm{mathcal{D}} \right)^{-\alpha}\in OP^{\wedge\{0\}}\backslash\mathrm{right}\backslash\right\}$	$OP^\alpha=\left\{T \mathcal{D} ^{-\alpha}\in OP^0\right\}$	$OP^\alpha=\{T\mid T \mathcal{D} ^{-\alpha}\in OP^0\}$ is the set of operators of order $\leq \alpha$.
cardona-qft-methods_ F08828	44	■ 1.000	$\leq \alpha$	$\leq \alpha$	$OP^\alpha=\{T\mid T \mathcal{D} ^{-\alpha}\in OP^0\}$ is the set of operators of order $\leq \alpha$.
cardona-qft-methods_ F08829	45	■ 1.000	$\delta(T)=[\backslash\mathrm{mathcal{D}} ,T],\quad\backslash\mathrm{quad}\backslash\mathrm{nabla}(T)=\backslash\left[\backslash\mathrm{mathcal{D}}^{\wedge\{2\}},T\backslash\mathrm{right}]\right]$	$\delta(T)=[\mathcal{D} ,T],\quad\nabla(T)=[\mathcal{D}^2,T]$	Moreover, we set $\delta(T)=[\mathcal{D} ,T],\quad\nabla(T)=[\mathcal{D}^2,T]$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
cardona-qft-methods_F00830	45	■ 1.000	$C^{\infty}(M) = OP^0 \bigcap L^{\infty}(M)$	$C^{\infty}(M) = OP^0 \bigcap L^{\infty}(M)$	For instance, $C^{\infty}(M) = OP^0 \bigcap L^{\infty}(M)$ and $L^{\infty}(M)$ is the von Neu-
cardona-qft-methods_F00831	45	■ 1.000	$L^{\infty}(M)$	$L^{\infty}(M)$	For instance, $C^{\infty}(M) = OP^0 \bigcap L^{\infty}(M)$ and $L^{\infty}(M)$ is the von Neu-
cardona-qft-methods_F00832	45	■ 1.000	$\mathcal{A} = C^{\infty}(M)$	$\mathcal{A} = C^{\infty}(M)$	mann algebra generated by $\mathcal{A} = C^{\infty}(M)$. The spaces OP^{α} have the ex-
cardona-qft-methods_F00833	45	■ 1.000	OP^{α}	OP^{α}	mann algebra generated by $\mathcal{A} = C^{\infty}(M)$. The spaces OP^{α} have the ex-
cardona-qft-methods_F00834	45	■ 0.983	$\mathcal{A} \subset OP^0 = \bigcap_{k \geq 0} \delta^k \subset$ $\subset$	$\mathcal{A} \subset OP^0 = \bigcap_{k \geq 0} \delta^k \subset$	Prop 6.1. Assume that $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ is regular so $\mathcal{A} \subset OP^0 = \bigcap_{k \geq 0} \delta^k \subset$
cardona-qft-methods_F00835	45	■ 1.000	$\alpha, \beta \in \mathbb{R}$	$\alpha, \beta \in \mathbb{R}$	$\mathcal{B}(\mathcal{H})$. Then, for any $\alpha, \beta \in \mathbb{R}$,
cardona-qft-methods_F00836	45	■ 1.000	$X = a \mathcal{D} [\mathcal{D}, b]\mathcal{D}^{-3}$	$X = a \mathcal{D} [\mathcal{D}, b]\mathcal{D}^{-3}$	As an example, let us compute the order of $X = a \mathcal{D} [\mathcal{D}, b]\mathcal{D}^{-3}$: since the
cardona-qft-methods_F00837	45	■ 0.842	$[\mathcal{D}, b]$	$[\mathcal{D}, b]$	order of a is 0, of $ \mathcal{D} $ is 1, of $[\mathcal{D}, b]$ is 0 and of $\mathcal{D}^{-3}$ is -3, we get $X \in OP^{-2}$.
cardona-qft-methods_F00838	45	■ 0.842	$\mathcal{D}^{-3}$	$\mathcal{D}^{-3}$	order of a is 0, of $ \mathcal{D} $ is 1, of $[\mathcal{D}, b]$ is 0 and of $\mathcal{D}^{-3}$ is -3, we get $X \in OP^{-2}$.
cardona-qft-methods_F00839	45	■ 0.842	$X \in OP^{-2}$	$X \in OP^{-2}$	order of a is 0, of $ \mathcal{D} $ is 1, of $[\mathcal{D}, b]$ is 0 and of $\mathcal{D}^{-3}$ is -3, we get $X \in OP^{-2}$.
cardona-qft-methods_F00840	45	■ 0.838	$\mathcal{D}(\mathcal{A})$	$\mathcal{D}(\mathcal{A})$	Definition 6.4. Let $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ be a spectral triple and $\mathcal{D}(\mathcal{A})$ be the poly-
cardona-qft-methods_F00841	45	■ 1.000	$\mathcal{A}, \mathcal{A}^{\circ}, \mathcal{D}$	$\mathcal{A}, \mathcal{A}^{\circ}, \mathcal{D}$	nomial algebra generated by $\mathcal{A}, \mathcal{A}^{\circ}, \mathcal{D}$ and $ \mathcal{D} $. Define the set of pseudod-
cardona-qft-methods_F00842	45	■ 1.000	$OP^{-\infty}$	$OP^{-\infty}$	$OP^{-\infty}$ of smoothing operators. This explains the presence of the arbitrary
cardona-qft-methods_F00843	45	■ 1.000	N	N	N and R . In the commutative case where $\mathcal{D} \in \text{Diff}^1(M, E)$, we get the
cardona-qft-methods_F00844	45	■ 1.000	$\mathcal{D} \in \text{Diff}^1(M, E)$	$\mathcal{D} \in \text{Diff}^1(M, E)$	N and R . In the commutative case where $\mathcal{D} \in \text{Diff}^1(M, E)$, we get the
cardona-qft-methods_F00845	45	■ 1.000	$\Psi(C^{\infty}(M)) \subset \Psi DO(M, E)$	$\Psi(C^{\infty}(M)) \subset \Psi DO(M, E)$	natural inclusion $\Psi(C^{\infty}(M)) \subset \Psi DO(M, E)$.
cardona-qft-methods_F00846	45	■ 1.000	$\{ \}^{30,33,49}$	30,33,49	in ^{30,33,49} since it pays attention to the reality operator J when it is present.
cardona-qft-methods_F00847	45	■ 1.000	$P \in \Psi^*(\mathcal{A})$	$P \in \Psi^*(\mathcal{A})$	Definition 6.5. For $P \in \Psi^*(\mathcal{A})$, we define the zeta-function associated to
cardona-qft-methods_F00848	45	■ 1.000	$P($	$P($	P (and $\mathcal{D}$) by
cardona-qft-methods_F00849	45	■ 1.000	$\mathcal{D}))$	$\mathcal{D}))$	P (and $\mathcal{D}$) by
cardona-qft-methods_F00850	45	■ 1.000	$\Re(s) \gg 1, P \mathcal{D} ^{-s} \in \mathcal{L}^1(\mathcal{H})$	$\Re(s) \gg 1, P \mathcal{D} ^{-s} \in \mathcal{L}^1(\mathcal{H})$	which makes sense since for $\Re(s) \gg 1, P \mathcal{D} ^{-s} \in \mathcal{L}^1(\mathcal{H})$.
cardona-qft-methods_F00851	45	■ 0.743	Sd	Sd	The dimension spectrum Sd of $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ is the set $\{\text{poles of } \zeta_{\mathcal{D}}^P(s) \mid P \in$
cardona-qft-methods_F00852	45	■ 0.743	$\zeta_{\mathcal{D}}^P(s) \mid P \in$	$\zeta_{\mathcal{D}}^P(s) \mid P \in$	The dimension spectrum Sd of $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ is the set $\{\text{poles of } \zeta_{\mathcal{D}}^P(s) \mid P \in$
cardona-qft-methods_F00853	45	■ 1.000	$\left. \Psi(\mathcal{A}) \cap OP^0 \right\}$	$\Psi(\mathcal{A}) \cap OP^0\}$	$\Psi(\mathcal{A}) \cap OP^0\}$. It is said simple if it contains poles of order at most one.
cardona-qft-methods_F00854	45	■ 1.000	$fP = \text{Res}_{s=0} \zeta_{\mathcal{D}}^P(s)$	$fP = \text{Res}_{s=0} \zeta_{\mathcal{D}}^P(s)$	The noncommutative integral of P is defined by $fP = \text{Res}_{s=0} \zeta_{\mathcal{D}}^P(s)$.
cardona-qft-methods_F00855	45	■ 1.000	$\mathcal{D} + P, P$	$\mathcal{D} + P, P$	invertible operator $\mathcal{D} + P$, P being the projection on $\mathcal{D}$. This change does

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cardona-qft-methods_ F08856	45	■ 0.928	f X=0	$fX = 0$	not modify the computation of the integrals f which follow since $fX = 0$
cardona-qft-methods_ F08857	45	■ 1.000	X	X	when X is a trace-class operator.
cardona-qft-methods_ F08858	46	■ 1.000	$\mathrm{U}(\mathrm{A})$	$\mathcal{U}(\mathcal{A})$	The unitary group $\mathcal{U}(\mathcal{A})$ of $\mathcal{A}$ gives rise to the automorphism α_u :
cardona-qft-methods_ F08859	46	■ 1.000	α_u	α_u	The unitary group $\mathcal{U}(\mathcal{A})$ of $\mathcal{A}$ gives rise to the automorphism α_u :
cardona-qft-methods_ F08860	46	■ 1.000	$a \in \mathrm{A} \rightarrow uau^* \in \mathrm{A}$	$a \in \mathcal{A} \rightarrow uau^* \in \mathcal{A}$	$a \in \mathcal{A} \rightarrow uau^* \in \mathcal{A}$. This defines the inner automorphisms group
cardona-qft-methods_ F08861	46	■ 0.999	$\mathrm{Inn}(\mathrm{A})$	$\mathrm{Inn}(\mathcal{A})$	$\mathrm{Inn}(\mathcal{A})$ which is a normal subgroup of the automorphisms $\mathrm{Aut}(\mathcal{A}) =$
cardona-qft-methods_ F08862	46	■ 0.999	$\mathrm{Aut}(\mathrm{A})=$	$\mathrm{Aut}(\mathcal{A}) =$	$\mathrm{Inn}(\mathcal{A})$ which is a normal subgroup of the automorphisms $\mathrm{Aut}(\mathcal{A}) =$
cardona-qft-methods_ F08863	46	■ 1.000	$\{\alpha \in \mathrm{Aut}(\mathrm{A}) \mid \alpha(\mathrm{A}) \subset \mathrm{A}\}$	$\{\alpha \in \mathrm{Aut}(\mathcal{A}) \mid \alpha(\mathcal{A}) \subset \mathcal{A}\}$	$\{\alpha \in \mathrm{Aut}(\mathcal{A}) \mid \alpha(\mathcal{A}) \subset \mathcal{A}\}$. For instance, in case of a gauge theory, the
cardona-qft-methods_ F08864	46	■ 1.000	$\mathrm{A}=C^\infty(M, M_n(\mathbb{C})) \simeq C^\infty(M) \otimes M_n(\mathbb{C})$	$\mathcal{A} = C^\infty(M, M_n(\mathbb{C})) \simeq C^\infty(M) \otimes M_n(\mathbb{C})$	algebra $\mathcal{A} = C^\infty(M, M_n(\mathbb{C})) \simeq C^\infty(M) \otimes M_n(\mathbb{C})$ is typically used.
cardona-qft-methods_ F08865	46	■ 0.997	$\mathrm{G}=C^\infty(M, PSU(n))$	$\mathcal{G} = C^\infty(M, PSU(n))$	Then, $\mathrm{Inn}(\mathcal{A})$ is locally isomorphic to $\mathcal{G} = C^\infty(M, PSU(n))$. Since
cardona-qft-methods_ F08866	46	■ 1.000	$\mathrm{Aut}(C^\infty(M)) \simeq \mathrm{Diff}(M)$	$\mathrm{Aut}(C^\infty(M)) \simeq \mathrm{Diff}(M)$	$\mathrm{Aut}(C^\infty(M)) \simeq \mathrm{Diff}(M)$, we get a complete parallel analogy between fol-
cardona-qft-methods_ F08867	46	■ 0.998	$\{ \}^{24,49}$	^{24,49}	$(\mathcal{A}, \mathcal{H}, \mathcal{D})$ is Morita equivalence, see. ^{24,49}
cardona-qft-methods_ F08868	46	■ 0.852	$\Omega_{\mathcal{D}}^1(\mathrm{A}) = \mathrm{span}\{adb \mid a, b \in \mathrm{A}\}$	$\Omega_{\mathcal{D}}^1(\mathcal{A}) = \mathrm{span}\{adb \mid a, b \in \mathcal{A}\}$	One-forms are defined as $\Omega_{\mathcal{D}}^1(\mathcal{A}) = \mathrm{span}\{adb \mid a, b \in \mathcal{A}\}$, $db = [\mathcal{D}, b]$ which
cardona-qft-methods_ F08869	46	■ 0.852	$db = [\mathrm{D}, b]$	$db = [\mathcal{D}, b]$	One-forms are defined as $\Omega_{\mathcal{D}}^1(\mathcal{A}) = \mathrm{span}\{adb \mid a, b \in \mathcal{A}\}$, $db = [\mathcal{D}, b]$ which
cardona-qft-methods_ F08870	46	■ 0.998	$\mathrm{D}: \mathrm{D} \rightarrow \mathrm{D}_A =$	$\mathcal{D} : \mathcal{D} \rightarrow \mathcal{D}_A =$	the Hilbert space $\mathcal{H}$, gives a natural hermitean fluctuation of $\mathcal{D}$: $\mathcal{D} \rightarrow \mathcal{D}_A =$
cardona-qft-methods_ F08871	46	■ 1.000	$\mathrm{D}+A$	$\mathcal{D} + A$	$\mathcal{D} + A$ with $A = A^* \in \Omega_{\mathcal{D}}^1(\mathcal{A})$. For instance, in commutative geometries,
cardona-qft-methods_ F08872	46	■ 1.000	$A=A^* \in \Omega_{\mathrm{D}}^1(\mathrm{A})$	$A = A^* \in \Omega_{\mathcal{D}}^1(\mathcal{A})$	$\mathcal{D} + A$ with $A = A^* \in \Omega_{\mathcal{D}}^1(\mathcal{A})$. For instance, in commutative geometries,
cardona-qft-methods_ F08873	46	■ 0.552	$\Omega_{\mathcal{D}}^1(C^\infty(M)) = \{c(da) \mid a \in C^\infty(M)\}$	$\Omega_{\mathcal{D}}^1(C^\infty(M)) = \{c(da) \mid a \in C^\infty(M)\}$	$\Omega_{\mathcal{D}}^1(C^\infty(M)) = \{c(da) \mid a \in C^\infty(M)\}$.
cardona-qft-methods_ F08874	46	■ 1.000	$\mathcal{D}_A J = \epsilon J \mathcal{D}_A$	$\mathcal{D}_A J = \epsilon J \mathcal{D}_A$	When a reality operator J exists, we also want $\mathcal{D}_A J = \epsilon J \mathcal{D}_A$, so we
cardona-qft-methods_ F08875	46	■ 0.957	$(\mathrm{A}, \mathrm{H}, \mathrm{D}_A)$	$(\mathcal{A}, \mathcal{H}, \mathcal{D}_A)$	space $\mathcal{H}$, a fluctuation of $\mathcal{D}$ still give rise to a spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D}_A)$ or
cardona-qft-methods_ F08876	46	■ 0.990	$(\mathrm{A}, \mathrm{H}, \widetilde{\mathrm{D}}_A)$	$(\mathcal{A}, \mathcal{H}, \mathcal{D}_{\tilde{A}})$	$(\mathcal{A}, \mathcal{H}, \mathcal{D}_{\tilde{A}})$.
cardona-qft-methods_ F08877	46	■ 0.967	$A \in \Omega_{\mathrm{D}}^1$	$A \in \Omega_{\mathcal{D}}^1$	and chirality χ . If $A \in \Omega_{\mathcal{D}}^1$, the fluctuated Dirac operator $\mathcal{D}_A$ or $\mathcal{D}_{\tilde{A}}$ is
cardona-qft-methods_ F08878	46	■ 0.967	D_A	$\mathcal{D}_A$	and chirality χ . If $A \in \Omega_{\mathcal{D}}^1$, the fluctuated Dirac operator $\mathcal{D}_A$ or $\mathcal{D}_{\tilde{A}}$ is
cardona-qft-methods_ F08879	46	■ 0.967	$\widetilde{\mathrm{D}}_A$	$\mathcal{D}_{\tilde{A}}$	and chirality χ . If $A \in \Omega_{\mathcal{D}}^1$, the fluctuated Dirac operator $\mathcal{D}_A$ or $\mathcal{D}_{\tilde{A}}$ is
cardona-qft-methods_ F08880	46	■ 1.000	$\mathrm{D} \rightarrow \mathrm{D}_u = u \mathrm{D} u^*$	$\mathcal{D} \rightarrow \mathcal{D}_u = u \mathcal{D} u^*$	Note that $\mathcal{U}(\mathcal{A})$ acts on $\mathcal{D}$ by $\mathcal{D} \rightarrow \mathcal{D}_u = u \mathcal{D} u^*$ leaving invariant the
cardona-qft-methods_ F08881	46	■ 1.000	$\mathrm{D}_u = \mathrm{D} + u [\mathrm{D}, u^*]$	$\mathcal{D}_u = \mathcal{D} + u [\mathcal{D}, u^*]$	spectrum of $\mathcal{D}$. Since $\mathcal{D}_u = \mathcal{D} + u [\mathcal{D}, u^*]$ and in a C^* -algebra, any element
cardona-qft-methods_ F08882	47	■ 1.000	$\mathrm{D}_A = 0$	$\mathcal{D}_A = 0$	append that $\mathcal{D}_A = 0$ with $\mathcal{D} \neq 0$.

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cardona-qft-methods_ F00883	47	■ 1.000	$\mathcal{D} \neq 0$	$\mathcal{D} \neq 0$	append that $\mathcal{D}_A = 0$ with $\mathcal{D} \neq 0$.
cardona-qft-methods_ F00884	47	■ 0.728	$(\mathcal{A}, \mathcal{D}, \mathcal{H})$	$(\mathcal{A}, \mathcal{D}, \mathcal{H})$	Lemma 6.2. Let $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ be a spectral triple and $X \in \Psi(\mathcal{A})$. Then
cardona-qft-methods_ F00885	47	■ 0.728	$X \in \Psi(\mathcal{A})$	$X \in \Psi(\mathcal{A})$	Lemma 6.2. Let $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ be a spectral triple and $X \in \Psi(\mathcal{A})$. Then
cardona-qft-methods_ F00886	47	■ 0.592	$f X^* = \overline{f X}$	$f X^* = \overline{f X}$	$f X^* = \overline{f X}$. If the spectral triple is real, then, for $X \in \Psi(\mathcal{A})$, $JXJ^{-1} \in$
cardona-qft-methods_ F00887	47	■ 0.592	$X \in \Psi(\mathcal{A}), J X J^{-1} \in$	$X \in \Psi(\mathcal{A}), JXJ^{-1} \in$	$f X^* = \overline{f X}$. If the spectral triple is real, then, for $X \in \Psi(\mathcal{A})$, $JXJ^{-1} \in$
cardona-qft-methods_ F00888	47	■ 0.997	$\Psi(\mathcal{A})$	$\Psi(\mathcal{A})$	$\Psi(\mathcal{A})$ and $f JXJ^{-1} = f X^* = \overline{f X}$.
cardona-qft-methods_ F00889	47	■ 0.997	$f J X J^{-1} = f X^* = \overline{f X}$	$f JXJ^{-1} = f X^* = \overline{f X}$	$\Psi(\mathcal{A})$ and $f JXJ^{-1} = f X^* = \overline{f X}$.
cardona-qft-methods_ F00890	47	■ 0.999	$A = A^*$	$A = A^*$	If $A = A^*$ then, for $k, l \in \mathbb{N}$, the integrals $f A^l \mathcal{D}^{-k}$, $f (A \mathcal{D}^{-1})^k$,
cardona-qft-methods_ F00891	47	■ 0.999	$k, l \in \mathbb{N}$	$k, l \in \mathbb{N}$	If $A = A^*$ then, for $k, l \in \mathbb{N}$, the integrals $f A^l \mathcal{D}^{-k}$, $f (A \mathcal{D}^{-1})^k$,
cardona-qft-methods_ F00892	47	■ 0.999	$f A^{\{l\}} \mathcal{D}^{-k}, f \left(A \mathcal{D}^{-1} \right)^k$	$f A^l \mathcal{D}^{-k}, f (A \mathcal{D}^{-1})^k$	If $A = A^*$ then, for $k, l \in \mathbb{N}$, the integrals $f A^l \mathcal{D}^{-k}$, $f (A \mathcal{D}^{-1})^k$,
cardona-qft-methods_ F00893	47	■ 0.919	$f A^{\{l\}} \mathcal{D} ^{-k}, f \chi A^{\{l\}} \mathcal{D} ^{-k}$	$f A^l \mathcal{D} ^{-k}, f \chi A^l \mathcal{D} ^{-k}$	$f A^l \mathcal{D} ^{-k}$, $f \chi A^l \mathcal{D} ^{-k}$ and $f A^l \mathcal{D} \mathcal{D} ^{-k}$ are real valued.
cardona-qft-methods_ F00894	47	■ 0.919	$f A^{\{l\}} \mathcal{D} \mathcal{D} ^{-k}$	$f A^l \mathcal{D} \mathcal{D} ^{-k}$	$f A^l \mathcal{D} ^{-k}$, $f \chi A^l \mathcal{D} ^{-k}$ and $f A^l \mathcal{D} \mathcal{D} ^{-k}$ are real valued.
cardona-qft-methods_ F00895	47	■ 1.000	$A + \epsilon J A J^{-1}$	$A + \epsilon J A J^{-1}$	$A + \epsilon J A J^{-1}$ is always zero in this commutative case for any selfadjoint
cardona-qft-methods_ F00896	47	■ 1.000	$A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$	$A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$	and of dimension d . Let $A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$ be a selfadjoint gauge potential. Then,
cardona-qft-methods_ F00897	47	■ 0.982	$\zeta_{D_{\tilde{A}}}(0) = \zeta_D(0) + \sum_{q=1}^d \frac{(-1)^q}{q} f \left(\tilde{A} \mathcal{D}^{-1} \right)^q$	$\zeta_{D_{\tilde{A}}}(0) = \zeta_D(0) + \sum_{q=1}^d \frac{(-1)^q}{q} f \left(\tilde{A} \mathcal{D}^{-1} \right)^q$	$\zeta_{D_{\tilde{A}}}(0) = \zeta_D(0) + \sum_{q=1}^d \frac{(-1)^q}{q} f(\tilde{A} \mathcal{D}^{-1})^q$.
cardona-qft-methods_ F00898	47	■ 0.942	$\nabla(T) = [\mathcal{D}^2, T]$	$\nabla(T) = [\mathcal{D}^2, T]$	(recall that $\nabla(T) = [\mathcal{D}^2, T]$).
cardona-qft-methods_ F00899	47	■ 1.000	σ_z	σ_z	The expansion of the one-parameter group σ_z gives for $T \in OP^q$
cardona-qft-methods_ F00900	47	■ 1.000	$T \in OP^q$	$T \in OP^q$	The expansion of the one-parameter group σ_z gives for $T \in OP^q$
cardona-qft-methods_ F00901	47	■ 1.000	$g(z, r) = \frac{1}{r!} \left(\frac{z}{2} \right) \cdots \left(\frac{z}{2} - (r-1) \right) = \binom{z/2}{r}$	$g(z, r) = \frac{1}{r!} \left(\frac{z}{2} \right) \cdots \left(\frac{z}{2} - (r-1) \right) = \binom{z/2}{r}$	where $g(z, r) = \frac{1}{r!} \left(\frac{z}{2} \right) \cdots \left(\frac{z}{2} - (r-1) \right) = \binom{z/2}{r}$ with the convention $g(z, 0) =$
cardona-qft-methods_ F00902	47	■ 1.000	$g(z, 0) =$	$g(z, 0) =$	where $g(z, r) = \frac{1}{r!} \left(\frac{z}{2} \right) \cdots \left(\frac{z}{2} - (r-1) \right) = \binom{z/2}{r}$ with the convention $g(z, 0) =$
cardona-qft-methods_ F00903	48	■ 1.000	P_A	P_A	where P_A is the projection on $\mathcal{D}_A$. Remark that $\tilde{A} \in \mathcal{D}(\mathcal{A}) \cap OP^0$ and
cardona-qft-methods_ F00904	48	■ 1.000	$\tilde{A} \in \mathcal{D}(\mathcal{A}) \cap OP^0$	$\tilde{A} \in \mathcal{D}(\mathcal{A}) \cap OP^0$	where P_A is the projection on $\mathcal{D}_A$. Remark that $\tilde{A} \in \mathcal{D}(\mathcal{A}) \cap OP^0$ and
cardona-qft-methods_ F00905	48	■ 1.000	$\mathcal{D}_A \in \mathcal{D}(\mathcal{A}) \cap OP^1$	$\mathcal{D}_A \in \mathcal{D}(\mathcal{A}) \cap OP^1$	$\mathcal{D}_A \in \mathcal{D}(\mathcal{A}) \cap OP^1$.
cardona-qft-methods_ F00906	48	■ 1.000	$V_A = P_A - P_0$	$V_A = P_A - P_0$	We denote $V_A = P_A - P_0$ and, as the following lemma shows, V_A is a
cardona-qft-methods_ F00907	48	■ 1.000	V_A	V_A	We denote $V_A = P_A - P_0$ and, as the following lemma shows, V_A is a

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cardona-qft-methods_ F09908	48	■ 1.000	$\bigcap_{k \geq 1} (\mathcal{D}_A)^k \subseteq \bigcap_{k \geq 1} D ^k$ $\subseteq \bigcap_{k \geq 1} D ^k$	$\bigcap_{k \geq 1} (\mathcal{D}_A)^k \subseteq \bigcap_{k \geq 1} D ^k$	Lemma 6.3. $(i) \bigcap_{k \geq 1} (\mathcal{D}_A)^k \subseteq \bigcap_{k \geq 1} D ^k$.
cardona-qft-methods_ F09909	48	■ 1.000	$\mathcal{D}_A \subseteq \bigcap_{k \geq 1} \bar{D} ^k$	$\mathcal{D}_A \subseteq \bigcap_{k \geq 1} \bar{D} ^k$	$(ii) \mathcal{D}_A \subseteq \bigcap_{k \geq 1} \bar{D} ^k$.
cardona-qft-methods_ F09910	48	■ 0.808	$\alpha, \beta \in \mathbb{R}, D ^{\beta} P_A D ^{\alpha}$	$\alpha, \beta \in \mathbb{R}, D ^{\beta} P_A D ^{\alpha}$	(iii) For any $\bar{\alpha}, \beta \in \mathbb{R}$, $ D ^{\beta} P_A D ^{\alpha}$ is bounded.
cardona-qft-methods_ F09911	48	■ 1.000	$P_A \in OP^{-\infty}$	$P_A \in OP^{-\infty}$	$(iv) P_A \in OP^{-\infty}$.
cardona-qft-methods_ F09912	48	■ 1.000	$X = \mathcal{D}_A^2 - \mathcal{D}^2 = \tilde{A}\mathcal{D} + \mathcal{D}\tilde{A} + \tilde{A}^2$, $\mathcal{D} + \mathcal{D} \tilde{A} + \tilde{A}^2$, $X_V = X + V_A$	$X = \mathcal{D}_A^2 - \mathcal{D}^2 = \tilde{A}\mathcal{D} + \mathcal{D}\tilde{A} + \tilde{A}^2$, $X_V = X + V_A$	Let us define $X = \mathcal{D}_A^2 - \mathcal{D}^2 = \tilde{A}\mathcal{D} + \mathcal{D}\tilde{A} + \tilde{A}^2$, $X_V = X + V_A$, thus
cardona-qft-methods_ F09913	48	■ 1.000	$X \in \mathcal{D}_1(\mathcal{A}) \cap OP^1$	$X \in \mathcal{D}_1(\mathcal{A}) \cap OP^1$	$X \in \mathcal{D}_1(\mathcal{A}) \cap OP^1$ and by Lemma 6.3, $X_V \sim X \bmod OP^{-\infty}$. Now let
cardona-qft-methods_ F09914	48	■ 1.000	$X_V \sim X \bmod OP^{-\infty}$	$X_V \sim X \bmod OP^{-\infty}$	$X \in \mathcal{D}_1(\mathcal{A}) \cap OP^1$ and by Lemma 6.3, $X_V \sim X \bmod OP^{-\infty}$. Now let
cardona-qft-methods_ F09915	48	■ 1.000	$Y = \log(D_A^2) - \log(D^2)$	$Y = \log(D_A^2) - \log(D^2)$	$Y = \log(D_A^2) - \log(D^2)$ which makes sense since $D_A^2 = \mathcal{D}_A^2 + P_A$ is invertible
cardona-qft-methods_ F09916	48	■ 1.000	$D_A^2 = \mathcal{D}_A^2 + P_A$	$D_A^2 = \mathcal{D}_A^2 + P_A$	$Y = \log(D_A^2) - \log(D^2)$ which makes sense since $D_A^2 = \mathcal{D}_A^2 + P_A$ is invertible
cardona-qft-methods_ F09917	48	■ 1.000	X_V	X_V	By definition of X_V , we get $Y = \log(D^2 + X_V) - \log(D^2)$.
cardona-qft-methods_ F09918	48	■ 1.000	$Y = \log(D^2 + X_V) - \log(D^2)$	$Y = \log(D^2 + X_V) - \log(D^2)$	By definition of X_V , we get $Y = \log(D^2 + X_V) - \log(D^2)$.
cardona-qft-methods_ F09919	48	■ 1.000	Y	Y	Lemma 6.4. Y is a pseudodifferential operator in OP^{-1} with the expan-
cardona-qft-methods_ F09920	48	■ 1.000	OP^{-1}	OP^{-1}	Lemma 6.4. Y is a pseudodifferential operator in OP^{-1} with the expan-
cardona-qft-methods_ F09921	48	■ 0.996	$\bmod OP^{-N-1}$	$\bmod OP^{-N-1}$	$\bmod OP^{-N-1}$. For any $N \in \mathbb{N}$ and $s \in \mathbb{C}$,
cardona-qft-methods_ F09922	48	■ 0.996	$s \in \mathbb{C}$	$s \in \mathbb{C}$	$\bmod OP^{-N-1}$. For any $N \in \mathbb{N}$ and $s \in \mathbb{C}$,
cardona-qft-methods_ F09923	48	■ 1.000	$K_p(Y, s) \in OP^{-p}$	$K_p(Y, s) \in OP^{-p}$	with $K_p(Y, s) \in OP^{-p}$. For any $p \in \mathbb{N}$ and $r_1, \dots, r_p \in \mathbb{N}_0$,
cardona-qft-methods_ F09924	48	■ 1.000	$p \in \mathbb{N}$	$p \in \mathbb{N}$	with $K_p(Y, s) \in OP^{-p}$. For any $p \in \mathbb{N}$ and $r_1, \dots, r_p \in \mathbb{N}_0$,
cardona-qft-methods_ F09925	48	■ 1.000	$r_1, \dots, r_p \in \mathbb{N}_0$	$r_1, \dots, r_p \in \mathbb{N}_0$	with $K_p(Y, s) \in OP^{-p}$. For any $p \in \mathbb{N}$ and $r_1, \dots, r_p \in \mathbb{N}_0$,
cardona-qft-methods_ F09926	48	■ 0.977	$\varepsilon^{r_1}(Y) \dots \varepsilon^{r_p}(Y) \in \Psi(\mathcal{A})$	$\varepsilon^{r_1}(Y) \dots \varepsilon^{r_p}(Y) \in \Psi(\mathcal{A})$	$\varepsilon^{r_1}(Y) \dots \varepsilon^{r_p}(Y) \in \Psi(\mathcal{A})$.
cardona-qft-methods_ F09927	48	■ 1.000	$\zeta_{D_A}(0) - \zeta_D(0) = \operatorname{Tr}(K_1(Y, s) D ^{-s})_{s=0}$	$\zeta_{D_A}(0) - \zeta_D(0) = \operatorname{Tr}(K_1(Y, s) D ^{-s})_{s=0}$	(29) entails that $\zeta_{D_A}(0) - \zeta_D(0) = \operatorname{Tr}(K_1(Y, s) D ^{-s})_{s=0}$. Thus, with (27),
cardona-qft-methods_ F09928	48	■ 1.000	$\zeta_{D_A}(0) - \zeta_D(0) = -\frac{1}{2}fY$	$\zeta_{D_A}(0) - \zeta_D(0) = -\frac{1}{2}fY$	we get $\zeta_{D_A}(0) - \zeta_D(0) = -\frac{1}{2}fY$.
cardona-qft-methods_ F09929	48	■ 1.000	$f \log((1+S)(1+T)) = f \log(1+S) +$	$f \log((1+S)(1+T)) = f \log(1+S) +$	Now the conclusion follows from $f \log((1+S)(1+T)) = f \log(1+S) +$
cardona-qft-methods_ F09930	48	■ 0.932	$f \log(1+T)$	$f \log(1+T)$	$f \log(1+T)$ for $S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$ (since $\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$)
cardona-qft-methods_ F09931	48	■ 0.932	$S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$	$S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$	$f \log(1+T)$ for $S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$ (since $\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$)
cardona-qft-methods_ F09932	48	■ 0.932	$\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$	$\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$	$f \log(1+T)$ for $S, T \in \Psi(\mathcal{A}) \cap OP^{-1}$ (since $\log(1+S) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} S^n$)

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cardona-qft-methods_F00933	49	■ 0.999	$S=D^{-1} A$	$S = D^{-1} A$	with $S = D^{-1} A$ and $T = AD^{-1}$; so $f \log(1 + XD^{-2}) = 2f \log(1 + AD^{-1})$
cardona-qft-methods_F00934	49	■ 0.999	$T=A D^{-1}$	$T = AD^{-1}$	with $S = D^{-1} A$ and $T = AD^{-1}$; so $f \log(1 + XD^{-2}) = 2f \log(1 + AD^{-1})$
cardona-qft-methods_F00935	49	■ 0.999	$f \log \left(1+X D^{-2}\right)=2 f \log \left(1+A D^{-1}\right)$	$f \log \left(1+X D^{-2}\right)=2 f \log \left(1+A D^{-1}\right)$	with $S = D^{-1} A$ and $T = AD^{-1}$; so $f \log(1 + XD^{-2}) = 2f \log(1 + AD^{-1})$
cardona-qft-methods_F00936	49	■ 1.000	$-\frac{1}{2} f Y=\sum_{q=1}^d \frac{(-1)^q}{q} f\left(\tilde{A} D^{-1}\right)^q$	$-\frac{1}{2} f Y=\sum_{q=1}^d \frac{(-1)^q}{q} f\left(\tilde{A} D^{-1}\right)^q$	and $-\frac{1}{2} f Y=\sum_{q=1}^d \frac{(-1)^q}{q} f\left(\tilde{A} D^{-1}\right)^q$.
cardona-qft-methods_F00937	49	■ 1.000	$k \in \mathbb{N}_{\{0\}}$	$k \in \mathbb{N}_0$	Lemma 6.5. For any $k \in \mathbb{N}_0$,
cardona-qft-methods_F00938	49	■ 0.953	$\operatorname{Res}_{\{s=d-k\}} \zeta_{D\{A\}}(s)=\operatorname{Res}_{\{s=d-k\}} \zeta_D(s)+$	$\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)=\operatorname{Res}_{s=d-k} \zeta_D(s)+$	$\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)=\operatorname{Res}_{s=d-k} \zeta_D(s)+$
cardona-qft-methods_F00939	49	■ 1.000	$h(s, r, p)=(-s / 2)^{\{p\}} \int_{\{0 \leq t_{\{1\}} \leq \cdots \leq t_{\{p\}} \leq 1\}} g\left(-s t_{\{1\}}, r_{\{1\}}\right) \cdots g\left(-s t_{\{p\}}, r_{\{p\}}\right) d t$	$h(s, r, p)=(-s / 2)^p \int_{0 \leq t_1 \leq \cdots \leq t_p \leq 1} g\left(-s t_1, r_1\right) \cdots g\left(-s t_p, r_p\right) d t$	where $h(s, r, p)=(-s / 2)^p \int_{0 \leq t_1 \leq \cdots \leq t_p \leq 1} g\left(-s t_1, r_1\right) \cdots g\left(-s t_p, r_p\right) d t$.
cardona-qft-methods_F00940	49	■ 1.000	$\left D_{\{A\}}\right ^{\{k\}}$	$\left D_A\right ^k$	Our operators $\left D_A\right ^k$ are pseudodifferential operators: for any $k \in \mathbb{Z}$,
cardona-qft-methods_F00941	49	■ 1.000	$\left D_{\{A\}}\right ^{\{k\}} \in \Psi^{\{k\}}(\mathcal{A})$	$\left D_A\right ^k \in \Psi^k(\mathcal{A})$	$\left D_A\right ^k \in \Psi^k(\mathcal{A})$.
cardona-qft-methods_F00942	49	■ 1.000	$D_{\{A\}}$	D_A	f for D or D_A :
cardona-qft-methods_F00943	49	■ 0.718	$\operatorname{Res}_{\{s=0\}} \operatorname{Tr}\left(P\left D_{\{A\}}\right ^{-s}\right)=f P$	$\operatorname{Res}_{s=0} \operatorname{Tr}\left(P\left D_A\right ^{-s}\right)=f P$	Prop 6.2. If the spectral triple is simple, $\operatorname{Res}_{s=0} \operatorname{Tr}\left(P\left D_A\right ^{-s}\right)=f P$ for any
cardona-qft-methods_F00944	49	■ 0.972	$k \in \mathbb{N}_{\{0\}} f\left D_{\{A\}}\right ^{-\{d-k\}}=$	$k \in \mathbb{N}_0 f\left D_A\right ^{-(d-k)}=$	pseudodifferential operator P . In particular, for any $k \in \mathbb{N}_0 f\left D_A\right ^{-(d-k)}=$
cardona-qft-methods_F00945	49	■ 0.999	$\operatorname{Res}_{\{s=d-k\}} \zeta_{D\{A\}}(s)$	$\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)$	$\operatorname{Res}_{s=d-k} \zeta_{D_A}(s)$. Moreover,
cardona-qft-methods_F00946	49	■ 0.945	$\operatorname{Tad}_{\{\mathcal{D}\}+A}(k)$	$\operatorname{Tad}_{\mathcal{D}+A}(k)$	Definition 6.8. In $(\mathcal{A}, \mathcal{H}, \mathcal{D})$, the tadpole $\operatorname{Tad}_{\mathcal{D}+A}(k)$ of order k , for $k \in$
cardona-qft-methods_F00947	49	■ 0.945	$k \in$	$k \in$	Definition 6.8. In $(\mathcal{A}, \mathcal{H}, \mathcal{D})$, the tadpole $\operatorname{Tad}_{\mathcal{D}+A}(k)$ of order k , for $k \in$
cardona-qft-methods_F00948	49	■ 1.000	$\{d-l: l \in \mathbb{N}\}$	$\{d-l: l \in \mathbb{N}\}$	$\{d-l: l \in \mathbb{N}\}$ is the term linear in $A=A^* \in \Omega_{\mathcal{D}}^1$, in the Λ^k term of (33)
cardona-qft-methods_F00949	49	■ 1.000	$A=A^{\{*\}} \in \Omega_{\mathcal{D}}^1$	$A=A^* \in \Omega_{\mathcal{D}}^1$	$\{d-l: l \in \mathbb{N}\}$ is the term linear in $A=A^* \in \Omega_{\mathcal{D}}^1$, in the Λ^k term of (33)
cardona-qft-methods_F00950	49	■ 1.000	$\Lambda^{\{k\}}$	Λ^k	$\{d-l: l \in \mathbb{N}\}$ is the term linear in $A=A^* \in \Omega_{\mathcal{D}}^1$, in the Λ^k term of (33)
cardona-qft-methods_F00951	49	■ 1.000	$\mathcal{D} \rightarrow \mathcal{D}+A$	$\mathcal{D} \rightarrow \mathcal{D}+A$	where $\mathcal{D} \rightarrow \mathcal{D}+A$.
cardona-qft-methods_F00952	49	■ 0.655	$\mathcal{A}, \mathcal{H}, \mathcal{D}, J$	$\mathcal{A}, \mathcal{H}, \mathcal{D}, J$	If moreover, the triple $(\mathcal{A}, \mathcal{H}, \mathcal{D}, J)$ is real, the tadpole $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}(k)$ is
cardona-qft-methods_F00953	49	■ 0.655	$\operatorname{Tad}_{\{\mathcal{D}\}+\tilde{A}}(k)$	$\operatorname{Tad}_{\mathcal{D}+\tilde{A}}(k)$	If moreover, the triple $(\mathcal{A}, \mathcal{H}, \mathcal{D}, J)$ is real, the tadpole $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}(k)$ is
cardona-qft-methods_F00954	49	■ 1.000	$\mathcal{D} \rightarrow \mathcal{D}+\tilde{A}$	$\mathcal{D} \rightarrow \mathcal{D}+\tilde{A}$	the term linear in A , in the Λ^k term of (33) where $\mathcal{D} \rightarrow \mathcal{D}+\tilde{A}$.
cardona-qft-methods_F00955	50	■ 1.000	$\operatorname{Tad}_{\{\mathcal{D}\}+\tilde{A}}=2 \operatorname{Tad}_{\mathcal{D}+A}$	$\operatorname{Tad}_{\mathcal{D}+\tilde{A}}=2 \operatorname{Tad}_{\mathcal{D}+A}$	Moreover, if the triple is real, $\operatorname{Tad}_{\mathcal{D}+\tilde{A}}=2 \operatorname{Tad}_{\mathcal{D}+A}$.
cardona-qft-methods_F00956	50	■ 1.000	$\tilde{A}=0$	$\tilde{A}=0$	such that $\tilde{A}=0$, then $\operatorname{Tad}_{\mathcal{D}+A}(k)=0$ for any $k \in \mathbb{Z}, k \leq d$.
cardona-qft-methods_F00957	50	■ 1.000	$\operatorname{Tad}_{\{D\}+A}(k)=0$	$\operatorname{Tad}_{D+A}(k)=0$	such that $\tilde{A}=0$, then $\operatorname{Tad}_{D+A}(k)=0$ for any $k \in \mathbb{Z}, k \leq d$.

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cardona-qft-methods_ F08958	50	■ 1.000	$k \in \mathbb{Z}, k \leq d$	$k \in \mathbb{Z}, k \leq d$	such that $\tilde{A} = 0$, then $\text{Tad}_{D+A}(k) = 0$ for any $k \in \mathbb{Z}, k \leq d$.
cardona-qft-methods_ F08959	50	■ 0.968	$\left(\text{see}^{12}\right)$	$\left(\text{see}^{12}\right)$	The vanishing tadpole of order 0 has the following equivalence (see ¹²)
cardona-qft-methods_ F08960	50	■ 1.000	$\alpha(b) := \mathcal{D} b \mathcal{D}^{-1}$	$\alpha(b) := \mathcal{D} b \mathcal{D}^{-1}$	where $\alpha(b) := \mathcal{D} b \mathcal{D}^{-1}$.
cardona-qft-methods_ F08961	50	■ 1.000	$A=0$	$A = 0$	The existence of tadpoles is important since, for instance, $A = 0$ is not
cardona-qft-methods_ F08962	50	■ 0.999	$\left(\mathcal{A} := C^\infty(M), \mathcal{H} := L^2(M, S), \not{D}\right)$	$(\mathcal{A} := C^\infty(M), \mathcal{H} := L^2(M, S), \not{D})$	$(\mathcal{A} := C^\infty(M), \mathcal{H} := L^2(M, S), \not{D})$ where S is the spinor bundle over M .
cardona-qft-methods_ F08963	50	■ 1.000	$J a J^{-1} = a^*, \quad \forall a \in \mathcal{A}$	$J a J^{-1} = a^*, \quad \forall a \in \mathcal{A}$	$J a J^{-1} = a^*, \quad \forall a \in \mathcal{A}$, and when d is even, the grading is given by the
cardona-qft-methods_ F08964	50	■ 1.000	$\chi := (-i)^{d/2} \gamma^1 \gamma^2 \cdots \gamma^d$	$\chi := (-i)^{d/2} \gamma^1 \gamma^2 \cdots \gamma^d$	chirality matrix $\chi := (-i)^{d/2} \gamma^1 \gamma^2 \cdots \gamma^d$. Such triple is said to be a com-
cardona-qft-methods_ F08965	50	■ 0.951	$\mathcal{A}^\circ \simeq J \mathcal{A} J^{-1} \simeq \mathcal{A}$	$\mathcal{A}^\circ \simeq J \mathcal{A} J^{-1} \simeq \mathcal{A}$	the commutative case, $\mathcal{A}^\circ \simeq J \mathcal{A} J^{-1} \simeq \mathcal{A}$ which also gives $J \mathcal{A} J^{-1} =$
cardona-qft-methods_ F08966	50	■ 0.951	$J \mathcal{A} J^{-1} =$	$J \mathcal{A} J^{-1} =$	the commutative case, $\mathcal{A}^\circ \simeq J \mathcal{A} J^{-1} \simeq \mathcal{A}$ which also gives $J \mathcal{A} J^{-1} =$
cardona-qft-methods_ F08967	50	■ 0.979	$-\epsilon A^*, \quad \forall A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$	$-\epsilon A^*, \quad \forall A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$	$-\epsilon A^*, \quad \forall A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$ or $\tilde{A} = 0$ when $A = A^*$.
cardona-qft-methods_ F08968	50	■ 0.999	$f P$	$f P$	As noticed by Wodzicki, $f P$ is equal to -2 times the coefficient in
cardona-qft-methods_ F08969	50	■ 1.000	$\log t$	$\log t$	$\log t$ of the asymptotics of $\text{Tr}(P e^{-t \not{D}^2})$ as $t \rightarrow 0$. It is remarkable that
cardona-qft-methods_ F08970	50	■ 1.000	$\text{Tr}(P e^{-t \not{D}^2})$	$\text{Tr}(P e^{-t \not{D}^2})$	$\log t$ of the asymptotics of $\text{Tr}(P e^{-t \not{D}^2})$ as $t \rightarrow 0$. It is remarkable that
cardona-qft-methods_ F08971	50	■ 1.000	$t \rightarrow 0$	$t \rightarrow 0$	$\log t$ of the asymptotics of $\text{Tr}(P e^{-t \not{D}^2})$ as $t \rightarrow 0$. It is remarkable that
cardona-qft-methods_ F08972	50	■ 1.000	ζ	ζ	close relation between the ζ function and heat kernel expansion with $WRes$.
cardona-qft-methods_ F08973	51	■ 1.000	$f P = 2a'_0$	$f P = 2a'_0$	so $f P = 2a'_0$. Since $f, WRes$ are traces on $\Psi(C^\infty(M))$, Corollary 2.2 gives
cardona-qft-methods_ F08974	51	■ 1.000	$\Psi(C^\infty(M))$	$\Psi(C^\infty(M))$	so $f P = 2a'_0$. Since $f, WRes$ are traces on $\Psi(C^\infty(M))$, Corollary 2.2 gives
cardona-qft-methods_ F08975	51	■ 0.813	$f P = c$	$f P = c$	$f P = c WRes P$. Because $\text{Tr}(P \not{D}^{-2s}) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \text{Tr}(P e^{-t \not{D}^2}) dt$, the
cardona-qft-methods_ F08976	51	■ 0.813	$\text{Tr}(P \not{D}^{-2s}) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \text{Tr}(P e^{-t \not{D}^2}) dt$	$\text{Tr}(P \not{D}^{-2s}) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \text{Tr}(P e^{-t \not{D}^2}) dt$	$f P = c WRes P$. Because $\text{Tr}(P \not{D}^{-2s}) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \text{Tr}(P e^{-t \not{D}^2}) dt$, the
cardona-qft-methods_ F08977	51	■ 1.000	$a'_k, k \neq 0$	$a'_k, k \neq 0$	non-zero coefficient $a'_k, k \neq 0$ creates a pole of $\text{Tr}(P \not{D}^{-2s})$ of order $k+2$
cardona-qft-methods_ F08978	51	■ 1.000	$\text{Tr}(P \not{D}^{-2s})$	$\text{Tr}(P \not{D}^{-2s})$	non-zero coefficient $a'_k, k \neq 0$ creates a pole of $\text{Tr}(P \not{D}^{-2s})$ of order $k+2$
cardona-qft-methods_ F08979	51	■ 1.000	$k+2$	$k+2$	non-zero coefficient $a'_k, k \neq 0$ creates a pole of $\text{Tr}(P \not{D}^{-2s})$ of order $k+2$
cardona-qft-methods_ F08980	51	■ 0.998	$\int_0^1 t^{s-1} \log(t)^k dt = \frac{(-1)^k k!}{s^{k+1}}$	$\int_0^1 t^{s-1} \log(t)^k dt = \frac{(-1)^k k!}{s^{k+1}}$	as we get $\int_0^1 t^{s-1} \log(t)^k dt = \frac{(-1)^k k!}{s^{k+1}}$ and $\Gamma(s) = \frac{1}{s} + \gamma + s g(s)$ where γ is
cardona-qft-methods_ F08981	51	■ 0.998	$\Gamma(s) = \frac{1}{s} + \gamma + s g(s)$	$\Gamma(s) = \frac{1}{s} + \gamma + s g(s)$	as we get $\int_0^1 t^{s-1} \log(t)^k dt = \frac{(-1)^k k!}{s^{k+1}}$ and $\Gamma(s) = \frac{1}{s} + \gamma + s g(s)$ where γ is
cardona-qft-methods_ F08982	51	■ 0.982	$\text{Sp}(M)$	$\text{Sp}(M)$	Prop 6.4. Let $Sp(M)$ be the dimension spectrum of a commutative geom-

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cardona-qft-methods_ F09983	51	■ 0.988	$\operatorname{Sp}(M)=\{d-k \mid k \in \mathbb{N}\}$	$\operatorname{Sp}(M)=\{d-k \mid k \in \mathbb{N}\}$	etry of dimension d . Then $Sp(M)$ is simple and $Sp(M)=\{d-k \mid k \in \mathbb{N}\}$.
cardona-qft-methods_ F09984	51	■ 1.000	$\sigma_d^{\left \mathcal{D}_A\right ^{-d}}=\sigma_d^{\left \mathcal{D}\right ^{-d}}$	$\sigma_d^{\left \mathcal{D}_A\right ^{-d}}=\sigma_d^{\left \mathcal{D}\right ^{-d}}$	The equation (30) can be obtained via (8) as $\sigma_d^{\left \mathcal{D}_A\right ^{-d}}=\sigma_d^{\left \mathcal{D}\right ^{-d}}$.
cardona-qft-methods_ F09985	51	■ 1.000	$d=4$	$d=4$	In dimension $d=4$, the computation in (19) of coefficient $a_4(1, \mathcal{D}_A^2)$
cardona-qft-methods_ F09986	51	■ 1.000	$a_4\left(1, \mathcal{D}_A^2\right)$	$a_4\left(1, \mathcal{D}_A^2\right)$	In dimension $d=4$, the computation in (19) of coefficient $a_4(1, \mathcal{D}_A^2)$
cardona-qft-methods_ F09987	51	■ 1.000	$a_k\left(1, \mathcal{D}_A^2\right)$	$a_k\left(1, \mathcal{D}_A^2\right)$	see Corollary 7.1 to see precise correspondence between $a_k(1, \mathcal{D}_A^2)$ and
cardona-qft-methods_ F09988	51	■ 1.000	$\zeta_{\mathcal{D}_A}(0)$	$\zeta_{\mathcal{D}_A}(0)$	$\zeta_{\mathcal{D}_A}(0)$. One recognizes the Yang–Mills action which will be generalized
cardona-qft-methods_ F09989	51	■ 1.000	$f f(x) \not \mathcal{D}^{-d+2}=$	$f f(x) \mathcal{D}^{-d+2}=$	if s is the scalar curvature and $f \in C^\infty(M)$, we know that $f f(x) \mathcal{D}^{-d+2}=$
cardona-qft-methods_ F09990	51	■ 1.000	$\int_M f(x) s(x) d v o l(x)$	$\int_M f(x) s(x) d v o l(x)$	$\int_M f(x) s(x) d v o l(x)$. This suggests the
cardona-qft-methods_ F09991	51	■ 1.000	$\mathcal{R}: a \in \mathcal{A} \rightarrow \mathbb{C}$	$\mathcal{R}: a \in \mathcal{A} \rightarrow \mathbb{C}$	scalar curvature is the map $\mathcal{R}: a \in \mathcal{A} \rightarrow \mathbb{C}$ defined by $\mathcal{R}(a)=f a \mathcal{D}^{-d+2}$.
cardona-qft-methods_ F09992	51	■ 1.000	$\mathcal{R}(a)=f a \mathcal{D}^{-d+2}$	$\mathcal{R}(a)=f a \mathcal{D}^{-d+2}$	scalar curvature is the map $\mathcal{R}: a \in \mathcal{A} \rightarrow \mathbb{C}$ defined by $\mathcal{R}(a)=f a \mathcal{D}^{-d+2}$.
cardona-qft-methods_ F09993	51	■ 1.000	$\mathcal{R}$	$\mathcal{R}$	In the commutative case, $\mathcal{R}$ is a trace on the algebra. More generally
cardona-qft-methods_ F09994	51	■ 0.997	$f A \mathcal{D}^{-d+1}$	$f A \mathcal{D}^{-d+1}$	$f A \mathcal{D}^{-d+1}$ are zero for all $A \in \Omega_{\mathcal{D}}^1$, $\mathcal{R}$ is invariant by inner fluctuations
cardona-qft-methods_ F09995	51	■ 0.997	$A \in \Omega_{\mathcal{D}}^1, \mathcal{R}$	$A \in \Omega_{\mathcal{D}}^1, \mathcal{R}$	$f A \mathcal{D}^{-d+1}$ are zero for all $A \in \Omega_{\mathcal{D}}^1$, $\mathcal{R}$ is invariant by inner fluctuations
cardona-qft-methods_ F09996	51	■ 0.932	$\left(\mathcal{A}_i, \mathcal{D}_i, \mathcal{H}_i\right)$	$\left(\mathcal{A}_i, \mathcal{D}_i, \mathcal{H}_i\right)$	to direct product of manifolds in the commutative case. Let $\left(\mathcal{A}_i, \mathcal{D}_i, \mathcal{H}_i\right)$,
cardona-qft-methods_ F09997	52	■ 1.000	$i=1,2$	$i=1,2$	$i=1,2$, be two spectral triples of dimension d_i with simple dimension
cardona-qft-methods_ F09998	52	■ 1.000	d_i	d_i	$i=1,2$, be two spectral triples of dimension d_i with simple dimension
cardona-qft-methods_ F09999	52	■ 1.000	χ_1	χ_1	spectrum. Assume the first to be of even dimension, with grading χ_1 . The
cardona-qft-methods_ F01000	52	■ 1.000	$\mathcal{D}^2=\mathcal{D}_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2$	$\mathcal{D}^2=\mathcal{D}_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2$	The interest of χ_1 is to guarantee additivity: $\mathcal{D}^2=\mathcal{D}_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2$.
cardona-qft-methods_ F01001	52	■ 1.000	$\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \sim_{t \rightarrow 0} a_1 t^{-d_1 / 2}$	$\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \sim_{t \rightarrow 0} a_1 t^{-d_1 / 2}$	Lemma 6.6. Assume that $\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \sim_{t \rightarrow 0} a_1 t^{-d_1 / 2}$ and $\operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0}$
cardona-qft-methods_ F01002	52	■ 1.000	$\operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0}$	$\operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0}$	Lemma 6.6. Assume that $\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \sim_{t \rightarrow 0} a_1 t^{-d_1 / 2}$ and $\operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0}$
cardona-qft-methods_ F01003	52	■ 1.000	$a_2 t^{-d_2 / 2}$	$a_2 t^{-d_2 / 2}$	$a_2 t^{-d_2 / 2}$.
cardona-qft-methods_ F01004	52	■ 0.523	$d=d_1+d_2$	$d=d_1+d_2$	The triple $(\mathcal{A}, \mathcal{D}, \mathcal{H})$ has dimension $d=d_1+d_2$.
cardona-qft-methods_ F01005	52	■ 1.000	$\zeta_{\mathcal{D}}(s)=\operatorname{Tr}\left(D ^{-s}\right)$	$\zeta_{\mathcal{D}}(s)=\operatorname{Tr}\left(D ^{-s}\right)$	Moreover, the function $\zeta_{\mathcal{D}}(s)=\operatorname{Tr}\left(D ^{-s}\right)$ has a simple pole at $s=d_1+d_2$
cardona-qft-methods_ F01006	52	■ 1.000	$s=d_1+d_2$	$s=d_1+d_2$	Moreover, the function $\zeta_{\mathcal{D}}(s)=\operatorname{Tr}\left(D ^{-s}\right)$ has a simple pole at $s=d_1+d_2$
cardona-qft-methods_ F01007	52	■ 1.000	$\zeta_{\mathcal{D}}(2 s)=\sum_{n=0}^{\infty} \mu_n\left(D_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2\right)^{-s}=\sum_{n, m=0}^{\infty}\left(\mu_n\left(\mathcal{D}_1^2\right)+\right.$	$\zeta_{\mathcal{D}}(2 s)=\sum_{n=0}^{\infty} \mu_n\left(D_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2\right)^{-s}=\sum_{n, m=0}^{\infty}\left(\mu_n\left(\mathcal{D}_1^2\right)+\right.$	Proof. We get $\zeta_{\mathcal{D}}(2 s)=\sum_{n=0}^{\infty} \mu_n\left(D_1^2 \otimes 1+1 \otimes \mathcal{D}_2^2\right)^{-s}=\sum_{n, m=0}^{\infty}\left(\mu_n\left(\mathcal{D}_1^2\right)+\right.$

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cardona-qft-methods_ F01008	52	■ 1.000	$\left.\mu_m\right(\left(\mathcal{D}_{\{2\}}^2\right)\right)\right)^{-s}$	$\mu_m\left(\mathcal{D}_2^2\right)\right)^{-s}$	$\mu_m\left(\mathcal{D}_2^2\right)\right)^{-s}$. Since $\left(\mu_n\left(\mathcal{D}_1\right)^2+\mu_m\left(\mathcal{D}_2\right)^2\right)^{-(c_1+c_2)/2} \leq \mu_n\left(\mathcal{D}_1\right)^{-c_1} \mu_m\left(\mathcal{D}_2\right)^{-c_2}$,
cardona-qft-methods_ F01009	52	■ 1.000	$\left.\mu_n\right(\left(\mathcal{D}_{\{1\}}\right)\right)^{2+}\left.\mu_m\right(\left(\mathcal{D}_{\{2\}}\right)\right)^{2}\right)\right)^{-\left(c_1+c_2\right) / 2} \leq \mu_n\left(\mathcal{D}_1\right)^{-c_1} \mu_m\left(\mathcal{D}_2\right)^{-c_2}$	$\left(\mu_n\left(\mathcal{D}_1\right)^2+\mu_m\left(\mathcal{D}_2\right)^2\right)^{-(c_1+c_2)/2} \leq \mu_n\left(\mathcal{D}_1\right)^{-c_1} \mu_m\left(\mathcal{D}_2\right)^{-c_2}$	$\mu_m\left(\mathcal{D}_2^2\right)\right)^{-s}$. Since $\left(\mu_n\left(\mathcal{D}_1\right)^2+\mu_m\left(\mathcal{D}_2\right)^2\right)^{-(c_1+c_2)/2} \leq \mu_n\left(\mathcal{D}_1\right)^{-c_1} \mu_m\left(\mathcal{D}_2\right)^{-c_2}$,
cardona-qft-methods_ F01010	52	■ 0.979	$\zeta_{\left\{\mathcal{D}\right\}}\left(c_1+c_2\right) \leq \zeta_{\mathcal{D}_1}\left(c_1\right) \zeta_{\mathcal{D}_2}\left(c_2\right)$	$\zeta_{\mathcal{D}}\left(c_1+c_2\right) \leq \zeta_{\mathcal{D}_1}\left(c_1\right) \zeta_{\mathcal{D}_2}\left(c_2\right)$	$\zeta_{\mathcal{D}}\left(c_1+c_2\right) \leq \zeta_{\mathcal{D}_1}\left(c_1\right) \zeta_{\mathcal{D}_2}\left(c_2\right)$ if $c_i > d_i$, so $d:=\inf \left\{c \in \mathbb{R}^+:\zeta_{\mathcal{D}}(c)<\infty\right\} \leq$
cardona-qft-methods_ F01011	52	■ 0.979	$c_{\{i\}} > d_{\{i\}}$	$c_i > d_i$	$\zeta_{\mathcal{D}}\left(c_1+c_2\right) \leq \zeta_{\mathcal{D}_1}\left(c_1\right) \zeta_{\mathcal{D}_2}\left(c_2\right)$ if $c_i > d_i$, so $d:=\inf \left\{c \in \mathbb{R}^+:\zeta_{\mathcal{D}}(c)<\infty\right\} \leq$
cardona-qft-methods_ F01012	52	■ 0.979	$d:=\inf \left\{c \in \mathbb{R}^+:\zeta_{\left\{\mathcal{D}\right\}}(c)<\infty\right\} \leq$	$d:=\inf \left\{c \in \mathbb{R}^+:\zeta_{\mathcal{D}}(c)<\infty\right\} \leq$	$\zeta_{\mathcal{D}}\left(c_1+c_2\right) \leq \zeta_{\mathcal{D}_1}\left(c_1\right) \zeta_{\mathcal{D}_2}\left(c_2\right)$ if $c_i > d_i$, so $d:=\inf \left\{c \in \mathbb{R}^+:\zeta_{\mathcal{D}}(c)<\infty\right\} \leq$
cardona-qft-methods_ F01013	52	■ 1.000	$d_{\{1\}}+d_{\{2\}}$	d_1+d_2	d_1+d_2 . We claim that $d=d_1+d_2$: recall first that $a_i:=$
cardona-qft-methods_ F01014	52	■ 1.000	$a_{\{i\}}:=$	$a_i:=$	d_1+d_2 . We claim that $d=d_1+d_2$: recall first that $a_i:=$
cardona-qft-methods_ F01015	52	■ 0.965	$\operatorname{Res}_{s=d_{\{i\}} / 2}\left(\Gamma(s) \zeta_{\left\{\mathcal{D}_{\{i\}}\right\}}(2 s)\right)=\Gamma\left(d_{\{i\}} / 2\right) \operatorname{Res}_{s=d_{\{i\}} / 2}\left(\zeta_{\left\{\mathcal{D}_{\{i\}}\right\}}(2 s)\right)=\frac{1}{2} \Gamma\left(d_{\{i\}} / 2\right) \operatorname{Res}_{s=d_{\{i\}}}\left(\zeta_{\left\{\mathcal{D}_{\{i\}}\right\}}(s)\right)$	$\operatorname{Res}_{s=d_i / 2}\left(\Gamma(s) \zeta_{\mathcal{D}_i}(2 s)\right)=\Gamma\left(d_i / 2\right) \operatorname{Res}_{s=d_i / 2}\left(\zeta_{\mathcal{D}_i}(2 s)\right)=\frac{1}{2} \Gamma\left(d_i / 2\right) \operatorname{Res}_{s=d_i}\left(\zeta_{\mathcal{D}_i}(s)\right)$	$\operatorname{Res}_{s=d_i / 2}\left(\Gamma(s) \zeta_{\mathcal{D}_i}(2 s)\right)=\Gamma\left(d_i / 2\right) \operatorname{Res}_{s=d_i / 2}\left(\zeta_{\mathcal{D}_i}(2 s)\right)=\frac{1}{2} \Gamma\left(d_i / 2\right) \operatorname{Res}_{s=d_i}\left(\zeta_{\mathcal{D}_i}(s)\right)$. If
cardona-qft-methods_ F01016	52	■ 0.946	$f(s):=\Gamma(s) \zeta_{\mathcal{D}}(2 s), \quad \operatorname{Tr}\left(e^{-t \mathcal{D}}\right) t^{s-1} d t+g(s)=$	$f(s):=\Gamma(s) \zeta_D(2 s), \quad f(s)=\int_0^1 \operatorname{Tr}\left(e^{-t \mathcal{D}^2}\right) t^{s-1} d t+g(s)=$	$f(s):=\Gamma(s) \zeta_D(2 s), \quad f(s)=\int_0^1 \operatorname{Tr}\left(e^{-t \mathcal{D}^2}\right) t^{s-1} d t+g(s)=$
cardona-qft-methods_ F01017	52	■ 0.999	$\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) t^{s-1} d t+g(s)$	$\int_0^1 \operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) t^{s-1} d t+g(s)$	$\int_0^1 \operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) t^{s-1} d t+g(s)$ where g is holomorphic since the map
cardona-qft-methods_ F01018	52	■ 1.000	$x \in \mathbb{R} \rightarrow \int_1^\infty e^{-t x^2} t^{x-1} d t$	$x \in \mathbb{R} \rightarrow \int_1^\infty e^{-t x^2} t^{x-1} d t$	$x \in \mathbb{R} \rightarrow \int_1^\infty e^{-t x^2} t^{x-1} d t$ is in Schwartz space.
cardona-qft-methods_ F01019	52	■ 1.000	$\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2) / 2}$	$\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2) / 2}$	Since $\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2) / 2}$, the function $f(s)$ has
cardona-qft-methods_ F01020	52	■ 1.000	$f(s)$	$f(s)$	Since $\operatorname{Tr}\left(e^{-t \mathcal{D}_1^2}\right) \operatorname{Tr}\left(e^{-t \mathcal{D}_2^2}\right) \sim_{t \rightarrow 0} a_1 a_2 t^{-(d_1+d_2) / 2}$, the function $f(s)$ has
cardona-qft-methods_ F01021	52	■ 1.000	$s=\left(d_{\{1\}}+d_{\{2\}}\right) / 2$	$s=\left(d_1+d_2\right) / 2$	a simple pole at $s=\left(d_1+d_2\right) / 2$. We conclude that $\zeta_{\mathcal{D}}(s)$ has a simple
cardona-qft-methods_ F01022	52	■ 1.000	$\zeta_{\left\{\mathcal{D}\right\}}(s)$	$\zeta_{\mathcal{D}}(s)$	a simple pole at $s=\left(d_1+d_2\right) / 2$. We conclude that $\zeta_{\mathcal{D}}(s)$ has a simple
cardona-qft-methods_ F01023	52	■ 0.996	$a_{\{i\}}, \frac{1}{2} \Gamma\left(\left(d_1+d_2\right) / 2\right) \operatorname{Res}_{s=d}\left(\zeta_{\left\{\mathcal{D}\right\}}(s)\right)=$	$a_i, \frac{1}{2} \Gamma\left(\left(d_1+d_2\right) / 2\right) \operatorname{Res}_{s=d}\left(\zeta_{\mathcal{D}}(s)\right)=$	pole at $s=d_1+d_2$. Moreover, using $a_i, \frac{1}{2} \Gamma\left(\left(d_1+d_2\right) / 2\right) \operatorname{Res}_{s=d}\left(\zeta_{\mathcal{D}}(s)\right)=$
cardona-qft-methods_ F01024	52	■ 0.998	$\frac{1}{2} \Gamma\left(d_1 / 2\right) \operatorname{Res}_{s=d_1}\left(\zeta_{\mathcal{D}_1}(s)\right) \frac{1}{2} \Gamma\left(d_2 / 2\right) \operatorname{Res}_{s=d_2}\left(\zeta_{\mathcal{D}_2}(s)\right)$	$\frac{1}{2} \Gamma\left(d_1 / 2\right) \operatorname{Res}_{s=d_1}\left(\zeta_{\mathcal{D}_1}(s)\right) \frac{1}{2} \Gamma\left(d_2 / 2\right) \operatorname{Res}_{s=d_2}\left(\zeta_{\mathcal{D}_2}(s)\right)$	$\frac{1}{2} \Gamma\left(d_1 / 2\right) \operatorname{Res}_{s=d_1}\left(\zeta_{\mathcal{D}_1}(s)\right) \frac{1}{2} \Gamma\left(d_2 / 2\right) \operatorname{Res}_{s=d_2}\left(\zeta_{\mathcal{D}_2}(s)\right)$.
cardona-qft-methods_ F01025	53	■ 0.936	$S_{E H}(g)=-\int_M s_g(x) d v o l_g(x)$	$S_{E H}(g)=-\int_M s_g(x) d v o l_g(x)$	This action is $S_{E H}(g)=-\int_M s_g(x) d v o l_g(x)$ where s is the scalar curvature

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cardona-qft-methods_ F01026	53	■ 1.000	$f \not{-}^2$	$f \not{-}^2$	(chosen positive for the sphere). Up to a constant, this is $f \mathcal{D}^{-2}$ in dimension
cardona-qft-methods_ F01027	53	■ 1.000	$\mathcal{M}_1$	$\mathcal{M}_1$	This action is interesting for the following reason: Let $\mathcal{M}_1$ be the set
cardona-qft-methods_ F01028	53	■ 0.992	$\int_M d\text{vol}_g = 1$	$\int_M d\text{vol}_g = 1$	of Riemannian metrics g on M such that $\int_M d\text{vol}_g = 1$. By a theorem of
cardona-qft-methods_ F01029	53	■ 1.000	$\{ \}^5 g \in \mathcal{M}_1$	${}^5 g \in \mathcal{M}_1$	Hilbert, ⁵ $g \in \mathcal{M}_1$ is a critical point of $S_{EH}(g)$ restricted to $\mathcal{M}_1$ if and only if
cardona-qft-methods_ F01030	53	■ 1.000	$S_{EH}(g)$	$S_{EH}(g)$	Hilbert, ⁵ $g \in \mathcal{M}_1$ is a critical point of $S_{EH}(g)$ restricted to $\mathcal{M}_1$ if and only if
cardona-qft-methods_ F01031	53	■ 1.000	$g: R = c g$	$g: R = cg$	a constant to $g: R = cg$). Taking the trace, this means that $s_g = c \dim(M)$
cardona-qft-methods_ F01032	53	■ 1.000	$s_g = c \operatorname{dim}(M)$	$s_g = c \dim(M)$	a constant to $g: R = cg$). Taking the trace, this means that $s_g = c \dim(M)$
cardona-qft-methods_ F01033	53	■ 0.945	$\int_M f(s_g(x)) d\text{vol}_g(x)$	$\int_M f(s_g(x)) d\text{vol}_g(x)$	ing $\int_M f(s_g(x)) d\text{vol}_g(x)$ and they are others. ⁴² In this desire to implement
cardona-qft-methods_ F01034	53	■ 1.000	$\mathcal{D}^{-1}$	$\mathcal{D}^{-1}$	In a way, a spectral triple fits quantum field theory since $\mathcal{D}^{-1}$ can be seen
cardona-qft-methods_ F01035	53	■ 1.000	ds	ds	as the propagator (or line element ds) for (Euclidean) fermions and we
cardona-qft-methods_ F01036	53	■ 1.000	$\Omega_{\mathcal{D}}^1(\mathcal{A})$	$\Omega_{\mathcal{D}}^1(\mathcal{A})$	is associated to a one-form in $\Omega_{\mathcal{D}}^1(\mathcal{A})$. Thus, the guiding principle followed
cardona-qft-methods_ F01037	53	■ 0.627	$\mathcal{D}.^{10}$	$\mathcal{D}.^{10}$	the spectrum of $\mathcal{D}$. ¹⁰ They proposed the
cardona-qft-methods_ F01038	53	■ 0.995	$\mathcal{S}(\mathcal{D}, f, \Lambda) := \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$	$\mathcal{S}(\mathcal{D}, f, \Lambda) := \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$	$\mathcal{S}(\mathcal{D}, f, \Lambda) := \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$ where $\Lambda \in \mathbb{R}^+$ plays the role of a cut-off and
cardona-qft-methods_ F01039	53	■ 1.000	$f (\mathcal{D}^2 / \Lambda^2)$	$f (\mathcal{D}^2 / \Lambda^2)$	f is any positive function (such that $f (\mathcal{D}^2 / \Lambda^2)$ is a trace-class operator).
cardona-qft-methods_ F01040	53	■ 1.000	$\mathcal{S}(\mathcal{D}, f, \Lambda) = \operatorname{Tr} (f (\mathcal{D} / \Lambda))$	$\mathcal{S}(\mathcal{D}, f, \Lambda) = \operatorname{Tr} (f (\mathcal{D} / \Lambda))$	Remark 7.1. We can also define $\mathcal{S}(\mathcal{D}, f, \Lambda) = \operatorname{Tr} (f (\mathcal{D} / \Lambda))$ when f is
cardona-qft-methods_ F01041	53	■ 1.000	$S(\mathcal{D}, g, \Lambda) = \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$	$S(\mathcal{D}, g, \Lambda) = \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$	positive and even. With this second definition, $S(\mathcal{D}, g, \Lambda) = \operatorname{Tr} (f (\mathcal{D}^2 / \Lambda^2))$
cardona-qft-methods_ F01042	53	■ 1.000	$g(x) = f(x^2)$	$g(x) = f(x^2)$	with $g(x) = f(x^2)$.
cardona-qft-methods_ F01043	53	■ 0.939	$f(\mathcal{D} / \Lambda)$	$f(\mathcal{D} / \Lambda)$	For f , one can think of the characteristic function of $[-1, 1]$, thus $f(\mathcal{D} / \Lambda)$
cardona-qft-methods_ F01044	53	■ 1.000	$[-\Lambda, \Lambda]$	$[-\Lambda, \Lambda]$	is nothing else but the number of eigenvalues of $\mathcal{D}$ within $[-\Lambda, \Lambda]$.
cardona-qft-methods_ F01045	54	■ 1.000	$\frac{1}{2} \langle J\psi, \mathcal{D}\psi \rangle$	$\frac{1}{2} \langle J\psi, \mathcal{D}\psi \rangle$	$\frac{1}{2} \langle J\psi, \mathcal{D}\psi \rangle$ for $\psi \in \mathcal{H}$ to get a full action. In the standard model of particle
cardona-qft-methods_ F01046	54	■ 1.000	$\psi \in \mathcal{H}$	$\psi \in \mathcal{H}$	$\frac{1}{2} \langle J\psi, \mathcal{D}\psi \rangle$ for $\psi \in \mathcal{H}$ to get a full action. In the standard model of particle
cardona-qft-methods_ F01047	54	■ 1.000	$\langle \psi, \mathcal{D}\psi \rangle$	$\langle \psi, \mathcal{D}\psi \rangle$	is of KO -dimension 6, thus $\langle \psi, \mathcal{D}\psi \rangle$ has to be replaced by $\frac{1}{2} \langle J\psi, \mathcal{D}\psi \rangle$ for
cardona-qft-methods_ F01048	54	■ 0.999	$\psi = \chi \psi \in \mathcal{H}$	$\psi = \chi \psi \in \mathcal{H}$	$\psi = \chi \psi \in \mathcal{H}$, see. ³⁰
cardona-qft-methods_ F01049	54	■ 0.999	$\mathfrak{g}$	$\mathfrak{g}$	situation: let G be a compact Lie group with its Lie algebra $\mathfrak{g}$ and let $A \in$
cardona-qft-methods_ F01050	54	■ 0.999	$A \in$	$A \in$	situation: let G be a compact Lie group with its Lie algebra $\mathfrak{g}$ and let $A \in$
cardona-qft-methods_ F01051	54	■ 1.000	$\Omega^1(M, \mathfrak{g})$	$\Omega^1(M, \mathfrak{g})$	$\Omega^1(M, \mathfrak{g})$ be a connection. If $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$ is the curvature

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cardona-qft-methods_ F01052	54	■ 1.000	$F=d\, a+\frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$	$F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$	$\Omega^1(M, \mathfrak{g})$ be a connection. If $F = da + \frac{1}{2}[A, A] \in \Omega^2(M, \mathfrak{g})$ is the curvature
cardona-qft-methods_ F01053	54	■ 1.000	$S_{YM}(A)=\int_M \operatorname{tr}(F \wedge$	$S_{YM}(A) = \int_M \operatorname{tr}(F \wedge$	(or field strength) of A , then the Yang-Mills action is $S_{YM}(A) = \int_M \operatorname{tr}(F \wedge$
cardona-qft-methods_ F01054	54	■ 0.706	$\star F) \, d\, \operatorname{vol}_g$	$\star F) d\operatorname{vol}_g$	$\star F) d\operatorname{vol}_g$. In the abelian case $G = U(1)$, it is the Maxwell action and
cardona-qft-methods_ F01055	54	■ 0.706	$G=U(1)$	$G = U(1)$	$\star F) d\operatorname{vol}_g$. In the abelian case $G = U(1)$, it is the Maxwell action and
cardona-qft-methods_ F01056	54	■ 1.000	$U(1)$	$U(1)$	gauged $U(1)$ of electric charge conservation can be gauged and its gauging
cardona-qft-methods_ F01057	54	■ 1.000	$\operatorname{dim}(M)=4$	$\dim(M) = 4$	produces electromagnetism. ⁹⁷ It is conformally invariant when $\dim(M) = 4$.
cardona-qft-methods_ F01058	54	■ 0.843	$d: ^{23,24}$	$d: ^{23,24}$	spectral triple $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ of dimension d . ^{23,24} let $A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$ and curvature
cardona-qft-methods_ F01059	54	■ 1.000	$\theta=d\, A+A^2$	$\theta = dA + A^2$	$\theta = dA + A^2$; then it is natural to consider $I(A) = \operatorname{Tr}_{Dix}(\theta^2 \mathcal{D} ^{-d})$ since
cardona-qft-methods_ F01060	54	■ 1.000	$I(A)=\operatorname{Tr}_{\operatorname{Dix}}\left(\theta^2 \mathcal{D} ^{-d}\right)$	$I(A) = \operatorname{Tr}_{\operatorname{Dix}}(\theta^2 \mathcal{D} ^{-d})$	$\theta = dA + A^2$; then it is natural to consider $I(A) = \operatorname{Tr}_{Dix}(\theta^2 \mathcal{D} ^{-d})$ since
cardona-qft-methods_ F01061	54	■ 1.000	$P=\theta^2 \mathcal{D} ^{-d}$	$P = \theta^2 \mathcal{D} ^{-d}$	commutative case: if $P = \theta^2 \mathcal{D} ^{-d}$, then Theorems 2.4 and 3.2 give the
cardona-qft-methods_ F01062	54	■ 1.000	$\operatorname{tr}\left(\sigma^P(x, \xi)\right)=c \operatorname{tr}(F \wedge \star F)(x)$	$\operatorname{tr}\left(\sigma^P(x, \xi)\right) = c \operatorname{tr}(F \wedge \star F)(x)$	claim since for the principal symbol, $\operatorname{tr}\left(\sigma^P(x, \xi)\right) = c \operatorname{tr}(F \wedge \star F)(x)$.
cardona-qft-methods_ F01063	54	■ 1.000	$d\, A$	dA	There is nevertheless a problem with the definition of dA : if $A =$
cardona-qft-methods_ F01064	54	■ 1.000	$A=$	$A =$	There is nevertheless a problem with the definition of dA : if $A =$
cardona-qft-methods_ F01065	54	■ 1.000	$\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$	$\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$	$\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$, then $dA = \sum_j\left[\mathcal{D}, \pi\left(a_j\right)\right]\left[\mathcal{D}, \pi\left(b_j\right)\right]$ can be non-zero
cardona-qft-methods_ F01066	54	■ 1.000	$d\, A=\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$	$dA = \sum_j\left[\mathcal{D}, \pi\left(a_j\right)\right]\left[\mathcal{D}, \pi\left(b_j\right)\right]$	$\sum_j \pi\left(a_j\right)\left[\mathcal{D}, \pi\left(b_j\right)\right]$, then $dA = \sum_j\left[\mathcal{D}, \pi\left(a_j\right)\right]\left[\mathcal{D}, \pi\left(b_j\right)\right]$ can be non-zero
cardona-qft-methods_ F01067	54	■ 1.000	$\Omega_{\mathcal{D}}^*\left(\mathcal{A}\right)$	$\Omega_{\mathcal{D}}^*(\mathcal{A})$	tial algebra $\Omega_{\mathcal{D}}^*(\mathcal{A})$, one must divide by a junk, for instance $\Omega_{\mathcal{D}}^2 \simeq$
cardona-qft-methods_ F01068	54	■ 1.000	$\Omega_{\mathcal{D}}^2 \simeq$	$\Omega_{\mathcal{D}}^2 \simeq$	tial algebra $\Omega_{\mathcal{D}}^*(\mathcal{A})$, one must divide by a junk, for instance $\Omega_{\mathcal{D}}^2 \simeq$
cardona-qft-methods_ F01069	54	■ 1.000	$\pi\left(\Omega^2 / \pi\left(\delta\left(\operatorname{Ker}(\pi) \cap \Omega^1\right)\right)\right)$	$\pi\left(\Omega^2 / \pi\left(\delta\left(\operatorname{Ker}(\pi) \cap \Omega^1\right)\right)\right)$	$\pi(\Omega^2 / \pi(\delta(\operatorname{Ker}(\pi) \cap \Omega^1)))$ where $\Omega^k(\mathcal{A})$ is the set of universal k -forms
cardona-qft-methods_ F01070	54	■ 1.000	$\Omega^k\left(\mathcal{A}\right)$	$\Omega^k(\mathcal{A})$	$\pi(\Omega^2 / \pi(\delta(\operatorname{Ker}(\pi) \cap \Omega^1)))$ where $\Omega^k(\mathcal{A})$ is the set of universal k -forms
cardona-qft-methods_ F01071	54	■ 1.000	$a_0 \delta a_1 \cdots \delta a_k$	$a_0 \delta a_1 \cdots \delta a_k$	over $\mathcal{A}$ given by the set of $a_0 \delta a_1 \cdots \delta a_k$ (before representation on $\mathcal{H}$:
cardona-qft-methods_ F01072	54	■ 1.000	$\pi\left(a_0 \delta a_1 \cdots \delta a_k\right)=a_0\left[\mathcal{D}, a_1\right] \cdots\left[\mathcal{D}, a_k\right]$	$\pi\left(a_0 \delta a_1 \cdots \delta a_k\right)=a_0\left[\mathcal{D}, a_1\right] \cdots\left[\mathcal{D}, a_k\right]$	$\pi(a_0 \delta a_1 \cdots \delta a_k) = a_0[\mathcal{D}, a_1] \cdots [\mathcal{D}, a_k]$.
cardona-qft-methods_ F01073	54	■ 1.000	$\mathcal{H}_k$	$\mathcal{H}_k$	Let $\mathcal{H}_k$ be the Hilbert space completion of $\pi(\Omega^k(\mathcal{A}))$ with the scalar prod-
cardona-qft-methods_ F01074	54	■ 1.000	$\pi\left(\Omega^k\left(\mathcal{A}\right)\right)$	$\pi\left(\Omega^k(\mathcal{A})\right)$	Let $\mathcal{H}_k$ be the Hilbert space completion of $\pi(\Omega^k(\mathcal{A}))$ with the scalar prod-
cardona-qft-methods_ F01075	54	■ 1.000	$\left\langle A_1, A_2\right\rangle_k=\operatorname{Tr}_{\operatorname{Dix}}\left(A_2^* A_1\left \mathcal{D}\right ^{-d}\right)$	$\langle A_1, A_2 \rangle_k = \operatorname{Tr}_{\operatorname{Dix}}\left(A_2^* A_1 \mathcal{D} ^{-d}\right)$	uct defined by $\langle A_1, A_2 \rangle_k = \operatorname{Tr}_{Dix}(A_2^* A_1 \mathcal{D} ^{-d})$ for $A_j \in \pi(\Omega^k(\mathcal{A}))$.
cardona-qft-methods_ F01076	54	■ 1.000	$A_j \in \pi\left(\Omega^k\left(\mathcal{A}\right)\right)$	$A_j \in \pi\left(\Omega^k(\mathcal{A})\right)$	uct defined by $\langle A_1, A_2 \rangle_k = \operatorname{Tr}_{Dix}(A_2^* A_1 \mathcal{D} ^{-d})$ for $A_j \in \pi(\Omega^k(\mathcal{A}))$.

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cardona-qft-methods_ F01077	54	■ 1.000	$\Omega^1(\mathcal{A})$	$\Omega^1(\mathcal{A})$	The Yang–Mills action on $\Omega^1(\mathcal{A})$ is $S_{YM}(V) = \langle \delta V + V^2, \delta V + V^2 \rangle$. It is pos-
cardona-qft-methods_ F01078	54	■ 1.000	$S_{YM}(V) = \langle \left\langle \delta V + V^2, \delta V + V^2 \right\rangle, \left\langle \delta V + V^2, \delta V + V^2 \right\rangle \right\rangle$	$S_{YM}(V) = \langle \delta V + V^2, \delta V + V^2 \rangle$	The Yang–Mills action on $\Omega^1(\mathcal{A})$ is $S_{YM}(V) = \langle \delta V + V^2, \delta V + V^2 \rangle$. It is pos-
cardona-qft-methods_ F01079	54	■ 1.000	$V \rightarrow \pi(u) V \pi(u^*) + \pi(u) [\mathcal{D}, \pi(u^*)]$	$V \rightarrow \pi(u) V \pi(u^*) + \pi(u) [\mathcal{D}, \pi(u^*)]$	itive, quartic and gauge invariant under $V \rightarrow \pi(u) V \pi(u^*) + \pi(u) [\mathcal{D}, \pi(u^*)]$
cardona-qft-methods_ F01080	54	■ 1.000	$u \in \mathcal{U}(\mathcal{A})$	$u \in \mathcal{U}(\mathcal{A})$	when $u \in \mathcal{U}(\mathcal{A})$. Moreover, $S_{YM}(V) = \inf \{ I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega) = V \}$
cardona-qft-methods_ F01081	54	■ 1.000	$S_{YM}(V) = \inf \{ I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega) = V \}$	$S_{YM}(V) = \inf \{ I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega) = V \}$	when $u \in \mathcal{U}(\mathcal{A})$. Moreover, $S_{YM}(V) = \inf \{ I(\omega) \mid \omega \in \Omega^1(\mathcal{A}), \pi(\omega) = V \}$
cardona-qft-methods_ F01082	55	■ 1.000	$\mathcal{D}_A = \mathcal{D} + A$	$\mathcal{D}_A = \mathcal{D} + A$	in the artificial decomposition $\mathcal{D}_A = \mathcal{D} + A$. For instance, for the minimally
cardona-qft-methods_ F01083	55	■ 1.000	$\text{Tr}(f(\mathcal{D}/\Lambda)) \geq 0$	$\text{Tr}(f(\mathcal{D}/\Lambda)) \geq 0$	the action $\text{Tr}(f(\mathcal{D}/\Lambda)) \geq 0$ has the correct sign for a Euclidean action: the
cardona-qft-methods_ F01084	55	■ 1.000	$S(\mathcal{D}, f, \Lambda)$	$S(\mathcal{D}, f, \Lambda)$	of spectral action $S(\mathcal{D}, f, \Lambda)$ when Λ goes to infinity.
cardona-qft-methods_ F01085	55	■ 1.000	$\zeta_{\mathcal{D}}$	$\zeta_{\mathcal{D}}$	Then, for the zeta function $\zeta_{\mathcal{D}}$ defined in (25)
cardona-qft-methods_ F01086	55	■ 1.000	$\alpha < 0, \zeta_{\mathcal{D}}$	$\alpha < 0, \zeta_{\mathcal{D}}$	(i) If $\alpha < 0$, $\zeta_{\mathcal{D}}$ has a pole at -2α with $a_{\alpha} = \frac{1}{2} \Gamma(-\alpha) \text{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s)$.
cardona-qft-methods_ F01087	55	■ 1.000	-2α	-2α	(i) If $\alpha < 0$, $\zeta_{\mathcal{D}}$ has a pole at -2α with $a_{\alpha} = \frac{1}{2} \Gamma(-\alpha) \text{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s)$.
cardona-qft-methods_ F01088	55	■ 1.000	$a_{\alpha} = \frac{1}{2} \Gamma(-\alpha) \text{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s)$	$a_{\alpha} = \frac{1}{2} \Gamma(-\alpha) \text{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s)$	(i) If $\alpha < 0$, $\zeta_{\mathcal{D}}$ has a pole at -2α with $a_{\alpha} = \frac{1}{2} \Gamma(-\alpha) \text{Res}_{s=-2\alpha} \zeta_{\mathcal{D}}(s)$.
cardona-qft-methods_ F01089	55	■ 1.000	$\alpha = 0$	$\alpha = 0$	(ii) For $\alpha = 0$, we get $a_0 = \zeta_{\mathcal{D}}(0) + \dim \mathcal{D}$.
cardona-qft-methods_ F01090	55	■ 1.000	$a_0 = \zeta_{\mathcal{D}}(0) + \dim \mathcal{D}$	$a_0 = \zeta_{\mathcal{D}}(0) + \dim \mathcal{D}$	(ii) For $\alpha = 0$, we get $a_0 = \zeta_{\mathcal{D}}(0) + \dim \mathcal{D}$.
cardona-qft-methods_ F01091	55	■ 1.000	$\alpha > 0, a_{\alpha} = \zeta(-2\alpha) \text{Res}_{s=-\alpha} \Gamma(s)$	$\alpha > 0, a_{\alpha} = \zeta(-2\alpha) \text{Res}_{s=-\alpha} \Gamma(s)$	(iii) If $\alpha > 0$, $a_{\alpha} = \zeta(-2\alpha) \text{Res}_{s=-\alpha} \Gamma(s)$.
cardona-qft-methods_ F01092	55	■ 1.000	Sd^+	Sd^+	part Sd^+ of Sd :
cardona-qft-methods_ F01093	56	■ 1.000	$f_{\beta} := \int_0^{\infty} f(x) x^{\beta-1} dx$	$f_{\beta} := \int_0^{\infty} f(x) x^{\beta-1} dx$	where the dependence of the even function f is $f_{\beta} := \int_0^{\infty} f(x) x^{\beta-1} dx$ and
cardona-qft-methods_ F01094	56	■ 0.808	$\dots$	$\dots$	$\dots$ involves the full Taylor expansion of f at 0 .
cardona-qft-methods_ F01095	56	■ 0.996	$\Gamma(s/2) \mathcal{D} ^{-s} = \int_0^{\infty} e^{-t\mathcal{D}^2} t^{s/2-1} dt = \int_0^1 e^{-t\mathcal{D}^2} t^{s/2-1} dt +$	$\Gamma(s/2) \mathcal{D} ^{-s} = \int_0^{\infty} e^{-t\mathcal{D}^2} t^{s/2-1} dt = \int_0^1 e^{-t\mathcal{D}^2} t^{s/2-1} dt +$	Proof. (i): Since $\Gamma(s/2) \mathcal{D} ^{-s} = \int_0^{\infty} e^{-t\mathcal{D}^2} t^{s/2-1} dt = \int_0^1 e^{-t\mathcal{D}^2} t^{s/2-1} dt +$
cardona-qft-methods_ F01096	56	■ 1.000	$x \rightarrow$	$x \rightarrow$	$f(s)$, where the function f is holomorphic (since the map $x \rightarrow$
cardona-qft-methods_ F01097	56	■ 0.997	$\int_1^{\infty} e^{-tx^2} x^{s/2-1} dt$	$\int_1^{\infty} e^{-tx^2} x^{s/2-1} dt$	$\int_1^{\infty} e^{-tx^2} x^{s/2-1} dt$ is in the Schwartz space), the swap of $\text{Tr}(e^{-t\mathcal{D}^2})$ with a
cardona-qft-methods_ F01098	56	■ 0.997	$\text{Tr}(e^{-t\mathcal{D}^2})$	$\text{Tr}(e^{-t\mathcal{D}^2})$	$\int_1^{\infty} e^{-tx^2} x^{s/2-1} dt$ is in the Schwartz space), the swap of $\text{Tr}(e^{-t\mathcal{D}^2})$ with a
cardona-qft-methods_ F01099	56	■ 1.000	$a_{\alpha} t^{\alpha}$	$a_{\alpha} t^{\alpha}$	sum of $a_{\alpha} t^{\alpha}$ and $a_{\alpha} \int_0^1 t^{\alpha+s/2-1} dt = \frac{2\alpha_a}{s+2\alpha}$ yields (32).
cardona-qft-methods_ F01100	56	■ 1.000	$a_{\alpha} \int_0^1 t^{\alpha+s/2-1} dt = \frac{2\alpha_a}{s+2\alpha}$	$a_{\alpha} \int_0^1 t^{\alpha+s/2-1} dt = \frac{2\alpha_a}{s+2\alpha}$	sum of $a_{\alpha} t^{\alpha}$ and $a_{\alpha} \int_0^1 t^{\alpha+s/2-1} dt = \frac{2\alpha_a}{s+2\alpha}$ yields (32).
cardona-qft-methods_ F01101	56	■ 1.000	$\Gamma(s/2)^{-1} \underset{s \rightarrow 0}{\sim} s/2$	$\Gamma(s/2)^{-1} \underset{s \rightarrow 0}{\sim} s/2$	(ii): The regularity of $\Gamma(s/2)^{-1} \underset{s \rightarrow 0}{\sim} s/2$ around zero implies that only

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cardona-qft-methods_F01102	56	■ 1.000	$\int_0^\infty \operatorname{Tr}\left(e^{-t\mathcal{D}^2}\right) t^{s/2-1} dt$	$\int_0^\infty \operatorname{Tr}\left(e^{-t\mathcal{D}^2}\right) t^{s/2-1} dt$	the pole part at $s = 0$ of $\int_0^\infty \operatorname{Tr}\left(e^{-t\mathcal{D}^2}\right) t^{s/2-1} dt$ contributes to $\zeta_{\mathcal{D}}(0)$. This
cardona-qft-methods_F01103	56	■ 1.000	$\zeta_{\mathcal{D}}(0)$	$\zeta_{\mathcal{D}}(0)$	the pole part at $s = 0$ of $\int_0^\infty \operatorname{Tr}\left(e^{-t\mathcal{D}^2}\right) t^{s/2-1} dt$ contributes to $\zeta_{\mathcal{D}}(0)$. This
cardona-qft-methods_F01104	56	■ 1.000	$a_0 \int_0^1 t^{s/2-1} dt = \frac{2a_0}{s}$	$a_0 \int_0^1 t^{s/2-1} dt = \frac{2a_0}{s}$	contribution is $a_0 \int_0^1 t^{s/2-1} dt = \frac{2a_0}{s}$.
cardona-qft-methods_F01105	56	■ 1.000	$f(x)=g\left(x^2\right)$	$f(x)=g\left(x^2\right)$	(iv): Assume $f(x)=g(x^2)$ where g is a Laplace transform: $g(x):=$
cardona-qft-methods_F01106	56	■ 1.000	$g(x):=$	$g(x):=$	(iv): Assume $f(x)=g(x^2)$ where g is a Laplace transform: $g(x):=$
cardona-qft-methods_F01107	56	■ 1.000	$\int_0^\infty e^{-sx} \phi(s) ds$	$\int_0^\infty e^{-sx} \phi(s) ds$	$\int_0^\infty e^{-sx} \phi(s) ds$. We will see in Section 7.3 how to relax this hypothesis.
cardona-qft-methods_F01108	56	■ 1.000	$\alpha < 0, s^\alpha = \Gamma(-\alpha)^{-1} \int_0^\infty e^{-sy} y^{-\alpha-1} dy$	$\alpha < 0, s^\alpha = \Gamma(-\alpha)^{-1} \int_0^\infty e^{-sy} y^{-\alpha-1} dy$	When $\alpha < 0$, $s^\alpha = \Gamma(-\alpha)^{-1} \int_0^\infty e^{-sy} y^{-\alpha-1} dy$ and $\int_0^\infty s^\alpha \phi(s) ds =$
cardona-qft-methods_F01109	56	■ 1.000	$\int_0^\infty s^\alpha \phi(s) ds =$	$\int_0^\infty s^\alpha \phi(s) ds =$	When $\alpha < 0$, $s^\alpha = \Gamma(-\alpha)^{-1} \int_0^\infty e^{-sy} y^{-\alpha-1} dy$ and $\int_0^\infty s^\alpha \phi(s) ds =$
cardona-qft-methods_F01110	56	■ 1.000	$\Gamma(-\alpha)^{-1} \int_0^\infty g(y) y^{-\alpha-1} dy$	$\Gamma(-\alpha)^{-1} \int_0^\infty g(y) y^{-\alpha-1} dy$	$\Gamma(-\alpha)^{-1} \int_0^\infty g(y) y^{-\alpha-1} dy$. Thus
cardona-qft-methods_F01111	56	■ 0.997	$\frac{1}{2} \int_0^\infty g(y) y^{\beta/2-1} dy = \int_0^\infty f(x) x^{\beta-1} dx$	$\frac{1}{2} \int_0^\infty g(y) y^{\beta/2-1} dy = \int_0^\infty f(x) x^{\beta-1} dx$	Thus (33) follows from (i), (ii) and $\frac{1}{2} \int_0^\infty g(y) y^{\beta/2-1} dy = \int_0^\infty f(x) x^{\beta-1} dx$.
cardona-qft-methods_F01112	56	■ 1.000	$\mathcal{S}(\mathcal{D}, f, \Lambda) \underset{t \downarrow 0}{\sim} \sum_{k \in \{1, \dots, d\}} f_k \Lambda^k a_{d-k}(\mathcal{D}^2) + f(0) a_d(\mathcal{D}^2) + \dots$	$\mathcal{S}(\mathcal{D}, f, \Lambda) \underset{t \downarrow 0}{\sim} \sum_{k \in \{1, \dots, d\}} f_k \Lambda^k a_{d-k}(\mathcal{D}^2) + f(0) a_d(\mathcal{D}^2) + \dots$	then $\mathcal{S}(\mathcal{D}, f, \Lambda) \underset{t \downarrow 0}{\sim} \sum_{k \in \{1, \dots, d\}} f_k \Lambda^k a_{d-k}(\mathcal{D}^2) + f(0) a_d(\mathcal{D}^2) + \dots$ with
cardona-qft-methods_F01113	56	■ 0.840	$f_k := \frac{1}{\Gamma(k/2)} \int_0^\infty f(s) s^{k/2-1} ds$	$f_k := \frac{1}{\Gamma(k/2)} \int_0^\infty f(s) s^{k/2-1} ds$	$f_k := \frac{1}{\Gamma(k/2)} \int_0^\infty f(s) s^{k/2-1} ds$. Moreover,
cardona-qft-methods_F01114	56	■ 1.000	Λ^0	Λ^0	The asymptotics (33) use the value of $\zeta_{\mathcal{D}}(0)$ in the constant term Λ^0 ,
cardona-qft-methods_F01115	56	■ 1.000	$\mathcal{D} \rightarrow$	$\mathcal{D} \rightarrow$	so it is fundamental to look at its variation under a gauge fluctuation $\mathcal{D} \rightarrow$
cardona-qft-methods_F01116	57	■ 0.813	$S(\mathcal{D}, f, \Lambda) \underset{\Lambda \rightarrow +\infty}{\sim}$	$S(\mathcal{D}, f, \Lambda) \underset{\Lambda \rightarrow +\infty}{\sim}$	The spectral action asymptotic behavior $S(\mathcal{D}, f, \Lambda) \underset{\Lambda \rightarrow +\infty}{\sim}$
cardona-qft-methods_F01117	57	■ 1.000	$\sum_{n=0}^\infty c_n \Lambda^{d-n} a_n(\mathcal{D}^2)$	$\sum_{n=0}^\infty c_n \Lambda^{d-n} a_n(\mathcal{D}^2)$	$\sum_{n=0}^\infty c_n \Lambda^{d-n} a_n(\mathcal{D}^2)$ has been proved for a smooth function f which is
cardona-qft-methods_F01118	57	■ 1.000	$f(x)=e^{-x}$	$f(x)=e^{-x}$	it does not cover the heat kernel case where $f(x)=e^{-x}$.
cardona-qft-methods_F01119	57	■ 1.000	$(-1)^m b_{2m+4}(\mathcal{D}^2) = \frac{(4\pi)^2}{m!} \mu_m(\mathcal{D}^2)$	$(-1)^m b_{2m+4}(\mathcal{D}^2) = \frac{(4\pi)^2}{m!} \mu_m(\mathcal{D}^2)$	where $(-1)^m b_{2m+4}(\mathcal{D}^2) = \frac{(4\pi)^2}{m!} \mu_m(\mathcal{D}^2)$ are suitably normalized, integrated
cardona-qft-methods_F01120	57	■ 1.000	$\mathcal{K}'(\mathbb{R})$	$\mathcal{K}'(\mathbb{R})$	(see ³⁷ for definition) for f in $\mathcal{K}'(\mathbb{R})$, which is the dual of $\mathcal{K}(\mathbb{R})$. This latter
cardona-qft-methods_F01121	57	■ 1.000	$\mathcal{K}(\mathbb{R})$	$\mathcal{K}(\mathbb{R})$	(see ³⁷ for definition) for f in $\mathcal{K}'(\mathbb{R})$, which is the dual of $\mathcal{K}(\mathbb{R})$. This latter
cardona-qft-methods_F01122	57	■ 0.981	$a \in \mathbb{R}, \phi^{(k)}(x) =$	$a \in \mathbb{R}, \phi^{(k)}(x) =$	is the space of smooth functions ϕ such that for some $a \in \mathbb{R}$, $\phi^{(k)}(x) =$
cardona-qft-methods_F01123	57	■ 1.000	$\mathcal{O}(x ^{a-k})$	$\mathcal{O}(x ^{a-k})$	$\mathcal{O}(x ^{a-k})$ as $ x \rightarrow \infty$, for each $k \in \mathbb{N}$. In particular, the Schwartz functions
cardona-qft-methods_F01124	57	■ 1.000	$ x \rightarrow \infty$	$ x \rightarrow \infty$	$\mathcal{O}(x ^{a-k})$ as $ x \rightarrow \infty$, for each $k \in \mathbb{N}$. In particular, the Schwartz functions
cardona-qft-methods_F01125	57	■ 1.000	$g(\Lambda^{-1})$	$g(\Lambda^{-1})$	not converge. It is known that each function $g(\Lambda^{-1})$ defines at most a

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cardona-qft-methods_ F01126	57	■ 1.000	$M=\mathbb{T}^d$	$M = \mathbb{T}^d$	When $M = \mathbb{T}^d$ as in Example 3.1 with $\Delta = \delta^{\mu\nu} \partial_\mu \partial_\nu$, $\mathrm{Tr}(e^{t\Delta}) =$
cardona-qft-methods_ F01127	57	■ 1.000	$\Delta=\delta^{\mu\nu}\partial_\mu\partial_\nu$, $\operatorname{Tr}\left(e^{t\Delta}\right)=$	$\Delta = \delta^{\mu\nu} \partial_\mu \partial_\nu, \mathrm{Tr}(e^{t\Delta}) =$	When $M = \mathbb{T}^d$ as in Example 3.1 with $\Delta = \delta^{\mu\nu} \partial_\mu \partial_\nu$, $\mathrm{Tr}(e^{t\Delta}) =$
cardona-qft-methods_ F01128	57	■ 1.000	$\frac{(4\pi)^{-d/2}\operatorname{Vol}\left(\mathbb{T}^d\right)}{t^{d/2}}+\mathcal{O}\left(t^{-d/2}e^{-1/4t}\right)$	$\frac{(4\pi)^{-d/2}\mathrm{Vol}\left(\mathbb{T}^d\right)}{t^{d/2}} + \mathcal{O}\left(t^{-d/2}e^{-1/4t}\right)$	$\frac{(4\pi)^{-d/2}\mathrm{Vol}(\mathbb{T}^d)}{t^{d/2}} + \mathcal{O}(t^{-d/2}e^{-1/4t})$, so the asymptotic series $\mathrm{Tr}(e^{t\Delta}) \simeq$
cardona-qft-methods_ F01129	57	■ 1.000	$\operatorname{Tr}\left(e^{t\Delta}\right)\simeq$	$\mathrm{Tr}\left(e^{t\Delta}\right) \simeq$	$\frac{(4\pi)^{-d/2}\mathrm{Vol}(\mathbb{T}^d)}{t^{d/2}} + \mathcal{O}(t^{-d/2}e^{-1/4t})$, so the asymptotic series $\mathrm{Tr}(e^{t\Delta}) \simeq$
cardona-qft-methods_ F01130	57	■ 0.999	$\frac{(4\pi)^{-d/2}\operatorname{Vol}\left(\mathbb{T}^d\right)}{t^{d/2}},t\rightarrow 0$ $\rightarrow 0$	$\frac{(4\pi)^{-d/2}\mathrm{Vol}\left(\mathbb{T}^d\right)}{t^{d/2}}, t \rightarrow 0$	$\frac{(4\pi)^{-d/2}\mathrm{Vol}(\mathbb{T}^d)}{t^{d/2}}, t \rightarrow 0$, has only one term.
cardona-qft-methods_ F01131	57	■ 1.000	$\mathbb{S}^4$	$\mathbb{S}^4$	In the opposite direction, let now M be the unit four-sphere $\mathbb{S}^4$ and
cardona-qft-methods_ F01132	58	■ 0.530	$\left(\mathrm{see}^{19}\right)$	$\left(\mathrm{see}^{19}\right)$	(see^{19}) :
cardona-qft-methods_ F01133	58	■ 1.000	B_{2k}	B_{2k}	with Bernoulli numbers B_{2k} . Thus $t^2\mathrm{Tr}(e^{-t\varrho^2}) \simeq \frac{2}{3} + \frac{2}{3}t + \sum_{k=0}^\infty a_k t^{k+2}$
cardona-qft-methods_ F01134	58	■ 1.000	$t^2\operatorname{Tr}\left(e^{-t\varrho^2}\right)\simeq\frac{2}{3}+\frac{2}{3}t+\sum_{k=0}^\infty a_k t^{k+2}$	$t^2\mathrm{Tr}\left(e^{-t\varrho^2}\right) \simeq \frac{2}{3} + \frac{2}{3}t + \sum_{k=0}^\infty a_k t^{k+2}$	with Bernoulli numbers B_{2k} . Thus $t^2\mathrm{Tr}(e^{-t\varrho^2}) \simeq \frac{2}{3} + \frac{2}{3}t + \sum_{k=0}^\infty a_k t^{k+2}$
cardona-qft-methods_ F01135	58	■ 1.000	$a_k>\frac{4}{3k!}\frac{ B_{2k+4} }{2k+4}>0$	$a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$	$a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$ and $ B_{2k+4} = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq$
cardona-qft-methods_ F01136	58	■ 1.000	$ B_{2k+4} =2\frac{(2k+4)!}{(2\pi)^{2k+4}}\zeta(2k+4)\simeq$	$ B_{2k+4} = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq$	$a_k > \frac{4}{3k!} \frac{ B_{2k+4} }{2k+4} > 0$ and $ B_{2k+4} = 2 \frac{(2k+4)!}{(2\pi)^{2k+4}} \zeta(2k+4) \simeq$
cardona-qft-methods_ F01137	58	■ 1.000	$4\sqrt{\pi(k+2)}\left(\frac{k+2}{\pi e}\right)^{2k+4}\rightarrow\infty$	$4\sqrt{\pi(k+2)}\left(\frac{k+2}{\pi e}\right)^{2k+4} \rightarrow \infty$	$4\sqrt{\pi(k+2)}\left(\frac{k+2}{\pi e}\right)^{2k+4} \rightarrow \infty$ if $k \rightarrow \infty$.
cardona-qft-methods_ F01138	58	■ 1.000	$k\rightarrow\infty$	$k \rightarrow \infty$	$4\sqrt{\pi(k+2)}\left(\frac{k+2}{\pi e}\right)^{2k+4} \rightarrow \infty$ if $k \rightarrow \infty$.
cardona-qft-methods_ F01139	58	■ 1.000	$2k>d$	$2k > d$	appear in (34) when $2k > d$. Nevertheless, all coefficients are local for
cardona-qft-methods_ F01140	58	■ 1.000	$2k\leq d$	$2k \leq d$	$2k \leq d$.
cardona-qft-methods_ F01141	58	■ 1.000	$e^{-t\sqrt{-\Delta}}$	$e^{-t\sqrt{-\Delta}}$	looks at the (integral) kernel of $e^{-t\sqrt{-\Delta}}$: in $\mathbb{T}^1$, with $\mathrm{Vol}(\mathbb{T}^1) = 2\pi$, we get ³⁸
cardona-qft-methods_ F01142	58	■ 1.000	$\mathbb{T}^1$	$\mathbb{T}^1$	looks at the (integral) kernel of $e^{-t\sqrt{-\Delta}}$: in $\mathbb{T}^1$, with $\mathrm{Vol}(\mathbb{T}^1) = 2\pi$, we get ³⁸
cardona-qft-methods_ F01143	58	■ 1.000	$\operatorname{Vol}\left(\mathbb{T}^1\right)=2\pi$	$\mathrm{Vol}\left(\mathbb{T}^1\right) = 2\pi$	looks at the (integral) kernel of $e^{-t\sqrt{-\Delta}}$: in $\mathbb{T}^1$, with $\mathrm{Vol}(\mathbb{T}^1) = 2\pi$, we get ³⁸
cardona-qft-methods_ F01144	58	■ 1.000	$\operatorname{Tr}\left(e^{-t\sqrt{-\Delta}}\right)=\frac{\sinh(t)}{\cosh(t)-1}=\coth\left(\frac{t}{2}\right)=\frac{2}{t}\sum_{k=0}^\infty\frac{B_{2k}}{(2k)!}t^{2k}=\frac{2}{t}\left[1+\frac{t^2}{12}-\frac{t^4}{720}+\mathcal{O}(t^6)\right]$	$\mathrm{Tr}\left(e^{-t\sqrt{-\Delta}}\right) = \frac{\sinh(t)}{\cosh(t)-1} = \coth\left(\frac{t}{2}\right) = \frac{2}{t} \sum_{k=0}^\infty \frac{B_{2k}}{(2k)!} t^{2k} = \frac{2}{t} \left[1 + \frac{t^2}{12} - \frac{t^4}{720} + \mathcal{O}(t^6)\right]$	$\mathrm{Tr}(e^{-t\sqrt{-\Delta}}) = \frac{\sinh(t)}{\cosh(t)-1} = \coth(\frac{t}{2}) = \frac{2}{t} \sum_{k=0}^\infty \frac{B_{2k}}{(2k)!} t^{2k} = \frac{2}{t} [1 + \frac{t^2}{12} - \frac{t^4}{720} + \mathcal{O}(t^6)]$
cardona-qft-methods_ F01145	58	■ 0.998	$t<2\pi$	$t < 2\pi$	so the series converges when $t < 2\pi$ as $\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$, thus $\frac{ B_{2k} }{(2k)!} \simeq$

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cardona-qft-methods_ F01146	58	■ 0.998	$\frac{B_{2k}}{(2k)!} \{(-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}\}$	$\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$	so the series converges when $t < 2\pi$ as $\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$, thus $\frac{ B_{2k} }{(2k)!} \simeq$
cardona-qft-methods_ F01147	58	■ 0.998	$\frac{ \left B_{2k}\right }{(2k)!} \simeq$	$\frac{ B_{2k} }{(2k)!} \simeq$	so the series converges when $t < 2\pi$ as $\frac{B_{2k}}{(2k)!} = (-1)^{k+1} \frac{2\zeta(2k)}{(2\pi)^{2k}}$, thus $\frac{ B_{2k} }{(2k)!} \simeq$
cardona-qft-methods_ F01148	58	■ 1.000	$\frac{2}{(2\pi)^{2k}}$	$\frac{2}{(2\pi)^{2k}}$	$\frac{2}{(2\pi)^{2k}}$ when $k \rightarrow \infty$.
cardona-qft-methods_ F01149	58	■ 1.000	$t \rightarrow \infty$	$t \rightarrow \infty$	Thus we have an example where $t \rightarrow \infty$ cannot be used with the asymp-
cardona-qft-methods_ F01150	59	■ 1.000	$\mathbb{T}^d$	$\mathbb{T}^d$	For the flat torus $\mathbb{T}^d$, the difference between $\text{Tr}(e^{t\Delta})$ and its asymp-
cardona-qft-methods_ F01151	59	■ 1.000	$\text{Tr}(e^{t\Delta})$	$\text{Tr}(e^{t\Delta})$	For the flat torus $\mathbb{T}^d$, the difference between $\text{Tr}(e^{t\Delta})$ and its asymp-
cardona-qft-methods_ F01152	59	■ 1.000	$N(\lambda)$	$N(\lambda)$	flow on $\mathbb{T}^d$. Similarly, the counting function $N(\lambda)$ (number of eigenval-
cardona-qft-methods_ F01153	59	■ 0.999	λ	λ	ues including multiplicities of Δ less than λ) obeys Weyl's law: $N(\lambda) =$
cardona-qft-methods_ F01154	59	■ 0.999	$N(\lambda) =$	$N(\lambda) =$	ues including multiplicities of Δ less than λ) obeys Weyl's law: $N(\lambda) =$
cardona-qft-methods_ F01155	59	■ 0.892	$\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{\Gamma(d/2+1)} \lambda^{d/2} + o(\lambda^{d/2})$	$\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{\Gamma(d/2+1)} \lambda^{d/2} + o(\lambda^{d/2})$	$\frac{(4\pi)^{-d/2} \text{Vol}(\mathbb{T}^d)}{\Gamma(d/2+1)} \lambda^{d/2} + o(\lambda^{d/2})$ — see ¹ for a nice historical review on these
cardona-qft-methods_ F01156	59	■ 1.000	t	t	the small- t asymptotics of heat kernel is determined by the large- λ asymp-
cardona-qft-methods_ F01157	59	■ 0.892	λ^{38}	λ^{38}	also gives informations on intermediate values of λ . ³⁸
cardona-qft-methods_ F01158	59	■ 1.000	$\{16, 76\}$	$16, 76$	$\text{In}^{16,76}$ are examples of spectral actions on commutative geometries of
cardona-qft-methods_ F01159	59	■ 1.000	$SU_q(2)$	$SU_q(2)$	$SU_q(2)$, the spectral action itself has only 4 terms, independently of the
cardona-qft-methods_ F01160	59	■ 0.958	$A \in \Gamma^\infty(M, \text{End}(S))$	$A \in \Gamma^\infty(M, \text{End}(S))$	dimension 4. One adds a gauge connection $A \in \Gamma^\infty(M, \text{End}(S))$ to $\mathcal{D}$ such
cardona-qft-methods_ F01161	59	■ 1.000	$\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$	$\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$	that $\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$, thus with a field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$
cardona-qft-methods_ F01162	59	■ 1.000	$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$	$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$	that $\mathcal{D} = i\gamma^\mu (\partial_\mu + A_\mu)$, thus with a field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu +$
cardona-qft-methods_ F01163	59	■ 1.000	$[A_\mu, A_\nu]$	$[A_\mu, A_\nu]$	$[A_\mu, A_\nu]$. We apply (18) with $P = \mathcal{D}^2$ and find the coefficients $a_i(1, P)$ of
cardona-qft-methods_ F01164	59	■ 1.000	$P = \mathcal{D}^2$	$P = \mathcal{D}^2$	$[A_\mu, A_\nu]$. We apply (18) with $P = \mathcal{D}^2$ and find the coefficients $a_i(1, P)$ of
cardona-qft-methods_ F01165	59	■ 1.000	$a_i(1, P)$	$a_i(1, P)$	$[A_\mu, A_\nu]$. We apply (18) with $P = \mathcal{D}^2$ and find the coefficients $a_i(1, P)$ of
cardona-qft-methods_ F01166	59	■ 1.000	$i = 0, 2, 4$	$i = 0, 2, 4$	(19) with $i = 0, 2, 4$. The asymptotic expansion corresponds to a weak field
cardona-qft-methods_ F01167	59	■ 1.000	$\{20, 30\}$	$20, 30$	physics, see. ^{20,30} This approach offers a lot of interesting perspectives, for
cardona-qft-methods_ F01168	59	■ 1.000	$\{10, 15, 17, 57, 63, 64, 71, 79\}$	$10, 15, 17, 57, 63, 64, 71, 79$	each noncommutative model) is particularly instructive. ^{10,15,17,57,63,64,71,79}
cardona-qft-methods_ F01169	60	■ 1.000	$SU(2)$	$SU(2)$	$SU_q(2)$ which is not a deformation of $SU(2)$ of the type considered on the
cardona-qft-methods_ F01170	60	■ 1.000	$\mathbb{S}^3 \times \mathbb{S}^1$	$\mathbb{S}^3 \times \mathbb{S}^1$	spheres $\mathbb{S}^4$ or $\mathbb{S}^3 \times \mathbb{S}^1$ has been investigated in ^{16,19} for instance in the case

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cardona-qft-methods_F01171	60	■ 1.000	$\{ \}^{\{16,19\}}$	16,19	spheres $\mathbb{S}^4$ or $\mathbb{S}^3 \times \mathbb{S}^1$ has been investigated in ^{16,19} for instance in the case
cardona-qft-methods_F01172	60	■ 1.000	$\{ \}^{\{66,76-78,81,95\}}$	66,76–78,81,95	to inflation, see. ^{66,76–78,81,95}
cardona-qft-methods_F01173	60	■ 1.000	$\{ \}^{\{14,18,18,58,59,61\}}$	14,18,18,58,59,61	ered in. ^{14,18,18,58,59,61} The main difficulty is precisely to put nice boundary
cardona-qft-methods_F01174	60	■ 1.000	$\{ \}^{\{53,84,85\}}$	53,84,85	The case of manifolds with torsion has also been studied in, ^{53,84,85} and
cardona-qft-methods_F01175	60	■ 1.000	$C^{\infty}(\text{\texttt{\textbackslash left(\mathbb{T}}_{\text{\texttt{\textbackslash Theta}}^{\text{\texttt{\textbackslash n}}}}\text{\texttt{\textbackslash right}})}$	$C^{\infty}(\mathbb{T}_{\Theta}^n)$	Let $C^{\infty}(\mathbb{T}_{\Theta}^n)$ be the smooth noncommutative n -torus associated to a non-
cardona-qft-methods_F01176	60	■ 1.000	n	n	Let $C^{\infty}(\mathbb{T}_{\Theta}^n)$ be the smooth noncommutative n -torus associated to a non-
cardona-qft-methods_F01177	60	■ 1.000	$\text{\texttt{\textbackslash Theta}} \in M_{\text{\texttt{\textbackslash n}}}(\text{\texttt{\textbackslash mathbb{R}}})$	$\Theta \in M_n(\mathbb{R})$	zero skew-symmetric deformation matrix $\Theta \in M_n(\mathbb{R})$. It was introduced
cardona-qft-methods_F01178	60	■ 1.000	$\text{\texttt{\textbackslash mathbb{T}}}^{\text{\texttt{\textbackslash n}}}$	$\mathbb{T}^n$	by Rieffel ⁹³ and Connes ²¹ to generalize the n -torus $\mathbb{T}^n$. This means that
cardona-qft-methods_F01179	60	■ 1.000	$u_{\text{\texttt{\textbackslash i}}}, \text{\texttt{\textbackslash i}}=1, \text{\texttt{\textbackslash ldots}}, n$	$u_i, i = 1, \dots, n$	$C^{\infty}(\mathbb{T}_{\Theta}^n)$ is the algebra generated by n unitaries $u_i, i = 1, \dots, n$ subject to
cardona-qft-methods_F01180	60	■ 0.993	$u_{\text{\texttt{\textbackslash l}}} u_{\text{\texttt{\textbackslash j}}}=e^{\text{\texttt{\textbackslash i}} \text{\texttt{\textbackslash Theta}}_{\text{\texttt{\textbackslash l}} \text{\texttt{\textbackslash j}}}} u_{\text{\texttt{\textbackslash j}}} u_{\text{\texttt{\textbackslash l}}}$	$u_l u_j = e^{i\Theta_{lj}} u_j u_l$	the relations $u_l u_j = e^{i\Theta_{lj}} u_j u_l$, and with Schwartz coefficients: an element
cardona-qft-methods_F01181	60	■ 0.999	$a \in C^{\infty}(\text{\texttt{\textbackslash left(\mathbb{T}}_{\text{\texttt{\textbackslash Theta}}^{\text{\texttt{\textbackslash n}}}}\text{\texttt{\textbackslash right}})}$	$a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$	$a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$ can be written as $a = \sum_{k \in \mathbb{Z}^n} a_k U_k$, where $\{a_k\} \in \mathcal{S}(\mathbb{Z}^n)$
cardona-qft-methods_F01182	60	■ 0.999	$a=\sum_{\text{\texttt{\textbackslash k}} \in \text{\texttt{\textbackslash mathbb{Z}}}^{\text{\texttt{\textbackslash n}}}} a_{\text{\texttt{\textbackslash k}}} U_{\text{\texttt{\textbackslash k}}}$	$a = \sum_{k \in \mathbb{Z}^n} a_k U_k$	$a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$ can be written as $a = \sum_{k \in \mathbb{Z}^n} a_k U_k$, where $\{a_k\} \in \mathcal{S}(\mathbb{Z}^n)$
cardona-qft-methods_F01183	60	■ 0.999	$\text{\texttt{\textbackslash left\{a_{\text{\texttt{\textbackslash k}}}\right\}} \in \text{\texttt{\textbackslash mathcal{S}}}\text{\texttt{\textbackslash left(\mathbb{Z}}^{\text{\texttt{\textbackslash n}}}}\text{\texttt{\textbackslash right}})}$	$\{a_k\} \in \mathcal{S}(\mathbb{Z}^n)$	$a \in C^{\infty}(\mathbb{T}_{\Theta}^n)$ can be written as $a = \sum_{k \in \mathbb{Z}^n} a_k U_k$, where $\{a_k\} \in \mathcal{S}(\mathbb{Z}^n)$
cardona-qft-methods_F01184	60	■ 0.998	$U_{\text{\texttt{\textbackslash k}}}=e^{\text{\texttt{\textbackslash -}}\frac{\text{\texttt{\textbackslash i}}}{2} \text{\texttt{\textbackslash k}} \cdot \text{\texttt{\textbackslash chi}} \text{\texttt{\textbackslash k}}} u_{\text{\texttt{\textbackslash 1}}}^{\text{\texttt{\textbackslash k}}_{\text{\texttt{\textbackslash 1}}}} \text{\texttt{\textbackslash cdots}} u_{\text{\texttt{\textbackslash n}}}^{\text{\texttt{\textbackslash k}}_{\text{\texttt{\textbackslash n}}}}, \text{\texttt{\textbackslash k}} \in \text{\texttt{\textbackslash mathbb{Z}}}^{\text{\texttt{\textbackslash n}}}$	$U_k = e^{-\frac{i}{2} k \cdot \chi k} u_1^{k_1} \dots u_n^{k_n}, k \in \mathbb{Z}^n$	with the Weyl elements defined by $U_k = e^{-\frac{i}{2} k \cdot \chi k} u_1^{k_1} \dots u_n^{k_n}, k \in \mathbb{Z}^n$, the
cardona-qft-methods_F01185	61	■ 1.000	$\text{\texttt{\textbackslash Theta}}$	Θ	where χ is the matrix restriction of Θ to its upper triangular part. Thus
cardona-qft-methods_F01186	61	■ 1.000	$U_{\text{\texttt{\textbackslash k}}}$	U_k	unitary operators U_k satisfy $U_k^* = U_{-k}$ and $[U_k, U_l] = -2i \sin(\frac{1}{2} k \cdot \Theta l) U_{k+l}$.
cardona-qft-methods_F01187	61	■ 1.000	$U_{\text{\texttt{\textbackslash k}}}^* = U_{\text{\texttt{\textbackslash -k}}}$	$U_k^* = U_{-k}$	unitary operators U_k satisfy $U_k^* = U_{-k}$ and $[U_k, U_l] = -2i \sin(\frac{1}{2} k \cdot \Theta l) U_{k+l}$.
cardona-qft-methods_F01188	61	■ 1.000	$\text{\texttt{\textbackslash left}[U_{\text{\texttt{\textbackslash k}}}, U_{\text{\texttt{\textbackslash l}}}\text{\texttt{\textbackslash right}}]=-2 \text{\texttt{\textbackslash i}} \text{\texttt{\textbackslash sin}} \text{\texttt{\textbackslash left(\frac{1}{2} \text{\texttt{\textbackslash k}} \cdot \text{\texttt{\textbackslash Theta}} \text{\texttt{\textbackslash right}})} U_{\text{\texttt{\textbackslash k+l}}}}$	$[U_k, U_l] = -2i \sin\left(\frac{1}{2} k \cdot \Theta l\right) U_{k+l}$	unitary operators U_k satisfy $U_k^* = U_{-k}$ and $[U_k, U_l] = -2i \sin(\frac{1}{2} k \cdot \Theta l) U_{k+l}$.
cardona-qft-methods_F01189	61	■ 1.000	$\text{\texttt{\textbackslash tau}}\text{\texttt{\textbackslash left(\sum_{\text{\texttt{\textbackslash k}} \in \text{\texttt{\textbackslash mathbb{Z}}}^{\text{\texttt{\textbackslash n}}}} a_{\text{\texttt{\textbackslash k}}} U_{\text{\texttt{\textbackslash k}}}\text{\texttt{\textbackslash right}})}=a_{\text{\texttt{\textbackslash 0}}}$	$\tau\left(\sum_{k \in \mathbb{Z}^n} a_k U_k\right) = a_0$	Let τ be the trace on $C^{\infty}(\mathbb{T}_{\Theta}^n)$ defined by $\tau(\sum_{k \in \mathbb{Z}^n} a_k U_k) = a_0$ and $\mathcal{H}_{\tau}$
cardona-qft-methods_F01190	61	■ 1.000	$\text{\texttt{\textbackslash mathcal{H}}}_{\text{\texttt{\textbackslash tau}}}$	$\mathcal{H}_{\tau}$	Let τ be the trace on $C^{\infty}(\mathbb{T}_{\Theta}^n)$ defined by $\tau(\sum_{k \in \mathbb{Z}^n} a_k U_k) = a_0$ and $\mathcal{H}_{\tau}$
cardona-qft-methods_F01191	61	■ 1.000	$\text{\texttt{\textbackslash langle}} a, b \text{\texttt{\textbackslash rangle}}=\text{\texttt{\textbackslash tau}}\text{\texttt{\textbackslash left}(a^{\text{\texttt{\textbackslash *}}} b \text{\texttt{\textbackslash right}})}$	$\langle a, b \rangle = \tau(a^* b)$	of the norm induced by the scalar product $\langle a, b \rangle = \tau(a^* b)$.
cardona-qft-methods_F01192	61	■ 0.992	$\text{\texttt{\textbackslash mathcal{H}}}_{\text{\texttt{\textbackslash tau}}}=\text{\texttt{\textbackslash left\{ \sum_{\text{\texttt{\textbackslash k}} \in \text{\texttt{\textbackslash mathbb{Z}}}^{\text{\texttt{\textbackslash n}}}} a_{\text{\texttt{\textbackslash k}}} U_{\text{\texttt{\textbackslash k}}} \text{\texttt{\textbackslash mid}} \text{\texttt{\textbackslash left\{a_{\text{\texttt{\textbackslash k}}}\right\}}_{\text{\texttt{\textbackslash k}}} \in \text{\texttt{\textbackslash l}}^2\text{\texttt{\textbackslash left(\mathbb{Z}}^{\text{\texttt{\textbackslash n}}}}\text{\texttt{\textbackslash right}}\text{\texttt{\textbackslash right\}}}$	$\mathcal{H}_{\tau} = \left\{ \sum_{k \in \mathbb{Z}^n} a_k U_k \mid \{a_k\}_k \in l^2(\mathbb{Z}^n) \right\}$	On $\mathcal{H}_{\tau} = \{ \sum_{k \in \mathbb{Z}^n} a_k U_k \mid \{a_k\}_k \in l^2(\mathbb{Z}^n) \}$, we consider the left and
cardona-qft-methods_F01193	61	■ 1.000	$L($	$L($	denote respectively by $L(\cdot)$ and $R(\cdot)$.
cardona-qft-methods_F01194	61	■ 1.000	$)$ and $R($	$)andR($	denote respectively by $L(\cdot)$ and $R(\cdot)$.
cardona-qft-methods_F01195	61	■ 1.000	$)$.	$).$	denote respectively by $L(\cdot)$ and $R(\cdot)$.

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cardona-qft-methods_ F01196	61	■ 1.000	$\delta_{\mu}, \mu \in \{1, \dots, n\}$	$\delta_{\mu}, \mu \in \{1, \dots, n\}$	Let also $\delta_{\mu}, \mu \in \{1, \dots, n\}$, be the n (pairwise commuting) canonical
cardona-qft-methods_ F01197	61	■ 1.000	$\delta_{\mu} \left(U_k \right) = i k_{\mu} U_k$	$\delta_{\mu} (U_k) = i k_{\mu} U_k$	derivations, defined by $\delta_{\mu}(U_k) = i k_{\mu} U_k$.
cardona-qft-methods_ F01198	61	■ 1.000	$\mathcal{A}_{\Theta} = C^{\infty}(\mathbb{T}_{\Theta}^n)$	$\mathcal{A}_{\Theta} = C^{\infty}(\mathbb{T}_{\Theta}^n)$	We need to fix notations: let $\mathcal{A}_{\Theta} = C^{\infty}(\mathbb{T}_{\Theta}^n)$ acting on $\mathcal{H} = \mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$
cardona-qft-methods_ F01199	61	■ 1.000	$\mathcal{H} = \mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$	$\mathcal{H} = \mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$	We need to fix notations: let $\mathcal{A}_{\Theta} = C^{\infty}(\mathbb{T}_{\Theta}^n)$ acting on $\mathcal{H} = \mathcal{H}_{\tau} \otimes \mathbb{C}^{2^m}$
cardona-qft-methods_ F01200	61	■ 1.000	$n = 2m$	$n = 2m$	with $n = 2m$ or $n = 2m + 1$ (i.e., $m = \lfloor \frac{n}{2} \rfloor$ is the integer part of $\frac{n}{2}$),
cardona-qft-methods_ F01201	61	■ 1.000	$n = 2m + 1$	$n = 2m + 1$	with $n = 2m$ or $n = 2m + 1$ (i.e., $m = \lfloor \frac{n}{2} \rfloor$ is the integer part of $\frac{n}{2}$),
cardona-qft-methods_ F01202	61	■ 1.000	$m = \lfloor \frac{n}{2} \rfloor$	$m = \lfloor \frac{n}{2} \rfloor$	with $n = 2m$ or $n = 2m + 1$ (i.e., $m = \lfloor \frac{n}{2} \rfloor$ is the integer part of $\frac{n}{2}$),
cardona-qft-methods_ F01203	61	■ 1.000	$\frac{n}{2}$	$\frac{n}{2}$	with $n = 2m$ or $n = 2m + 1$ (i.e., $m = \lfloor \frac{n}{2} \rfloor$ is the integer part of $\frac{n}{2}$),
cardona-qft-methods_ F01204	61	■ 1.000	$\mathcal{A}_{\Theta}$	$\mathcal{A}_{\Theta}$	element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$. The Tomita conjugation
cardona-qft-methods_ F01205	61	■ 1.000	$L(a) \otimes 1_{2^m}$	$L(a) \otimes 1_{2^m}$	element of $\mathcal{A}_{\Theta}$ is represented on $\mathcal{H}$ as $L(a) \otimes 1_{2^m}$. The Tomita conjugation
cardona-qft-methods_ F01206	61	■ 1.000	$J_0(a) = a^*$	$J_0(a) = a^*$	$J_0(a) = a^*$ satisfies $[J_0, \delta_{\mu}] = 0$ and we define $J = J_0 \otimes C_0$ where C_0 is an
cardona-qft-methods_ F01207	61	■ 1.000	$[J_0, \delta_{\mu}] = 0$	$[J_0, \delta_{\mu}] = 0$	$J_0(a) = a^*$ satisfies $[J_0, \delta_{\mu}] = 0$ and we define $J = J_0 \otimes C_0$ where C_0 is an
cardona-qft-methods_ F01208	61	■ 1.000	$J = J_0 \otimes C_0$	$J = J_0 \otimes C_0$	$J_0(a) = a^*$ satisfies $[J_0, \delta_{\mu}] = 0$ and we define $J = J_0 \otimes C_0$ where C_0 is an
cardona-qft-methods_ F01209	61	■ 1.000	C_0	C_0	$J_0(a) = a^*$ satisfies $[J_0, \delta_{\mu}] = 0$ and we define $J = J_0 \otimes C_0$ where C_0 is an
cardona-qft-methods_ F01210	61	■ 1.000	$\mathbb{C}^{2^m}$	$\mathbb{C}^{2^m}$	operator on $\mathbb{C}^{2^m}$. The Dirac-like operator is given by
cardona-qft-methods_ F01211	61	■ 1.000	$C^{\infty}(\mathbb{T}_{\Theta}^n) \otimes \mathbb{C}^{2^m}$	$C^{\infty}(\mathbb{T}_{\Theta}^n) \otimes \mathbb{C}^{2^m}$	the dense subset of $\mathcal{H}$ given by $C^{\infty}(\mathbb{T}_{\Theta}^n) \otimes \mathbb{C}^{2^m}$ and $\mathcal{D}$ denotes its selfadjoint
cardona-qft-methods_ F01212	61	■ 1.000	$C_0 \gamma^{\alpha} = -\varepsilon \gamma^{\alpha} C_0$	$C_0 \gamma^{\alpha} = -\varepsilon \gamma^{\alpha} C_0$	extension. Thus $C_0 \gamma^{\alpha} = -\varepsilon \gamma^{\alpha} C_0$, and $\mathcal{D} U_k \otimes e_i = k_{\mu} U_k \otimes \gamma^{\mu} e_i$, where
cardona-qft-methods_ F01213	61	■ 1.000	$\mathcal{D} U_k \otimes e_i = k_{\mu} U_k \otimes \gamma^{\mu} e_i$	$\mathcal{D} U_k \otimes e_i = k_{\mu} U_k \otimes \gamma^{\mu} e_i$	extension. Thus $C_0 \gamma^{\alpha} = -\varepsilon \gamma^{\alpha} C_0$, and $\mathcal{D} U_k \otimes e_i = k_{\mu} U_k \otimes \gamma^{\mu} e_i$, where
cardona-qft-methods_ F01214	61	■ 0.992	(e_i)	(e_i)	(e_i) is the canonical basis of $\mathbb{C}^{2^m}$. Moreover, $C_0^2 = \pm 1_{2^m}$ depending on the
cardona-qft-methods_ F01215	61	■ 0.992	$C_0^2 = \pm 1_{2^m}$	$C_0^2 = \pm 1_{2^m}$	(e_i) is the canonical basis of $\mathbb{C}^{2^m}$. Moreover, $C_0^2 = \pm 1_{2^m}$ depending on the
cardona-qft-methods_ F01216	61	■ 1.000	$\chi = id \otimes (-i)^m \gamma^1 \dots \gamma^n$	$\chi = id \otimes (-i)^m \gamma^1 \dots \gamma^n$	parity of m . Finally, the chirality in the even case is $\chi = id \otimes (-i)^m \gamma^1 \dots \gamma^n$.
cardona-qft-methods_ F01217	61	■ 0.710	$(\mathcal{A}_{\Theta}, \mathcal{H}, \mathcal{D}, J, \chi)$	$(\mathcal{A}_{\Theta}, \mathcal{H}, \mathcal{D}, J, \chi)$	Theorem 8.1. ^{24,49} The 5-tuple $(\mathcal{A}_{\Theta}, \mathcal{H}, \mathcal{D}, J, \chi)$ is a real regular spectral
cardona-qft-methods_ F01218	61	■ 1.000	$u \in \mathcal{A}, uu^* = u^*u = U_0$	$u \in \mathcal{A}, uu^* = u^*u = U_0$	For every unitary $u \in \mathcal{A}$, $uu^* = u^*u = U_0$, the perturbed operator $V_u \mathcal{D} V_u^*$
cardona-qft-methods_ F01219	61	■ 1.000	$V_u \mathcal{D} V_u^*$	$V_u \mathcal{D} V_u^*$	For every unitary $u \in \mathcal{A}$, $uu^* = u^*u = U_0$, the perturbed operator $V_u \mathcal{D} V_u^*$
cardona-qft-methods_ F01220	61	■ 1.000	$V_u = (L(u) \otimes 1_{2^m}) J (L(u) \otimes 1_{2^m}) J^{-1}$	$V_u = (L(u) \otimes 1_{2^m}) J (L(u) \otimes 1_{2^m}) J^{-1}$	by the unitary $V_u = (L(u) \otimes 1_{2^m}) J (L(u) \otimes 1_{2^m}) J^{-1}$, must satisfy condition
cardona-qft-methods_ F01221	61	■ 0.990	$J \mathcal{D} J = \epsilon \mathcal{D} J$	$J \mathcal{D} = \epsilon \mathcal{D} J$	$J \mathcal{D} = \epsilon \mathcal{D} J$. The yields the necessity of a symmetrized covariant Dirac

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cardona-qft-methods_ F01222	61	■ 0.552	$\backslash\mathrm{mathcal{D}}_{\mathrm{A}}=\mathrm{mathcal{D}}+\mathrm{A}+\epsilon\mathrm{J}\mathrm{A}\mathrm{J}^{-1}$	$\mathcal{D}_A = \mathcal{D} + A + \epsilon J A J^{-1}$	operator $\mathcal{D}_A = \mathcal{D} + A + \epsilon J A J^{-1}$ since $V_u \mathcal{D} V_u^* = \mathcal{D}_{L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]}$.
cardona-qft-methods_ F01223	61	■ 0.552	$\mathrm{V}_{\mathrm{u}}\backslash\mathrm{mathcal{D}}\mathrm{V}_{\mathrm{u}}^{\ast}=\mathrm{mathcal{D}}_{\mathrm{L}(\mathrm{u})}\otimes\mathrm{L}\backslash\mathrm{left}[\backslash\mathrm{mathcal{D}},\mathrm{L}\backslash\mathrm{left}(\mathrm{u}^{\ast}\backslash\mathrm{right})\otimes\mathrm{L}\backslash\mathrm{right}]\}$	$V_u \mathcal{D} V_u^* = \mathcal{D}_{L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]}$	operator $\mathcal{D}_A = \mathcal{D} + A + \epsilon J A J^{-1}$ since $V_u \mathcal{D} V_u^* = \mathcal{D}_{L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]}$.
cardona-qft-methods_ F01224	61	■ 1.000	$\mathrm{V}_{\mathrm{u}}\backslash\mathrm{mathcal{D}}_{\mathrm{A}}\mathrm{V}_{\mathrm{u}}^{\ast}=\mathrm{mathcal{D}}_{\backslash\mathrm{gamma}_{\mathrm{u}}(\mathrm{A})}$	$V_u \mathcal{D}_A V_u^* = \mathcal{D}_{\gamma_u(A)}$	Moreover, we get the gauge transformation $V_u \mathcal{D}_A V_u^* = \mathcal{D}_{\gamma_u(A)}$ where the
cardona-qft-methods_ F01225	61	■ 1.000	$\backslash\mathrm{gamma}_{\mathrm{u}}(\mathrm{A})=\mathrm{u}\backslash\mathrm{left}[\backslash\mathrm{mathcal{D}},\mathrm{u}^{\ast}\backslash\mathrm{right}]+\mathrm{u}\mathrm{A}\mathrm{u}^{\ast}$	$\gamma_u(A) = u [\mathcal{D}, u^*] + u A u^*$	gauged transform one-form of A is $\gamma_u(A) = u [\mathcal{D}, u^*] + u A u^*$, with the
cardona-qft-methods_ F01226	61	■ 1.000	$\mathrm{L}(\mathrm{u})\otimes\mathrm{L}_{2^{\mathrm{m}}}\longrightarrow\mathrm{u}$	$L(u) \otimes 1_{2^m} \longrightarrow u$	shorthand $L(u) \otimes 1_{2^m} \longrightarrow u$. So the spectral action is gauge invariant:
cardona-qft-methods_ F01227	61	■ 1.000	$\backslash\mathrm{mathcal{S}}\backslash\mathrm{left}(\backslash\mathrm{mathcal{D}}_{\mathrm{A}},\mathrm{f},\backslash\mathrm{Lambda}\backslash\mathrm{right})=\backslash\mathrm{mathcal{S}}\backslash\mathrm{left}(\backslash\mathrm{mathcal{D}}_{\backslash\mathrm{gamma}_{\mathrm{u}}(\mathrm{A})},\mathrm{f},\backslash\mathrm{Lambda}\backslash\mathrm{right})$	$\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \mathcal{S}(\mathcal{D}_{\gamma_u(A)}, f, \Lambda)$	$\mathcal{S}(\mathcal{D}_A, f, \Lambda) = \mathcal{S}(\mathcal{D}_{\gamma_u(A)}, f, \Lambda)$.
cardona-qft-methods_ F01228	61	■ 1.000	$\mathrm{A}=\mathrm{L}\backslash\mathrm{left}(-\mathrm{i}\mathrm{A}_{\backslash\mathrm{alpha}}\backslash\mathrm{right})\otimes$	$A = L(-i A_\alpha) \otimes$	Any selfadjoint one-form $A \in \Omega_{\mathcal{D}}^1(\mathcal{A})$, is written as $A = L(-i A_\alpha) \otimes$
cardona-qft-methods_ F01229	61	■ 0.981	$\backslash\mathrm{gamma}^{\backslash\mathrm{alpha}},\mathrm{A}_{\backslash\mathrm{alpha}}=-\mathrm{A}_{\backslash\mathrm{alpha}}^{\ast}\in\backslash\mathrm{in}\backslash\mathrm{mathcal{A}}_{\backslash\mathrm{Theta}}$	$\gamma^\alpha, A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$	$\gamma^\alpha, A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$, thus $\mathcal{D}_A = -i(\delta_\alpha + L(A_\alpha) - R(A_\alpha)) \otimes \gamma^\alpha$. Defining
cardona-qft-methods_ F01230	61	■ 0.981	$\backslash\mathrm{mathcal{D}}_{\mathrm{A}}=-\mathrm{i}\backslash\mathrm{left}(\backslash\mathrm{delta}_{\backslash\mathrm{alpha}}+\mathrm{L}\backslash\mathrm{left}(\mathrm{A}_{\backslash\mathrm{alpha}}\backslash\mathrm{right})-\mathrm{R}\backslash\mathrm{left}(\mathrm{A}_{\backslash\mathrm{alpha}}\backslash\mathrm{right})\backslash\mathrm{right})\otimes\backslash\mathrm{gamma}^{\backslash\mathrm{alpha}}$	$\mathcal{D}_A = -i(\delta_\alpha + L(A_\alpha) - R(A_\alpha)) \otimes \gamma^\alpha$	$\gamma^\alpha, A_\alpha = -A_\alpha^* \in \mathcal{A}_\Theta$, thus $\mathcal{D}_A = -i(\delta_\alpha + L(A_\alpha) - R(A_\alpha)) \otimes \gamma^\alpha$. Defining
cardona-qft-methods_ F01231	62	■ 1.000	$\backslash\mathrm{tilde}{\mathrm{A}}_{\backslash\mathrm{alpha}}=\mathrm{L}\backslash\mathrm{left}(\mathrm{A}_{\backslash\mathrm{alpha}}\backslash\mathrm{right})-\mathrm{R}\backslash\mathrm{left}(\mathrm{A}_{\backslash\mathrm{alpha}}\backslash\mathrm{right})$	$\tilde{A}_\alpha = L(A_\alpha) - R(A_\alpha)$	$\tilde{A}_\alpha = L(A_\alpha) - R(A_\alpha)$, we get
cardona-qft-methods_ F01232	62	■ 1.000	$\backslash\mathrm{mathcal{D}}_{\mathrm{A}}^{\wedge\{2\}}=-g^{\alpha_1\alpha_2}\backslash\mathrm{alpha}_{\{1}\}\backslash\mathrm{alpha}_{\{2\}}\backslash\mathrm{left}(\backslash\mathrm{delta}_{\backslash\mathrm{alpha}_{\{1\}}}+\backslash\mathrm{tilde}{\mathrm{A}}_{\backslash\mathrm{alpha}_{\{1\}}}\backslash\mathrm{right})\backslash\mathrm{left}(\backslash\mathrm{delta}_{\backslash\mathrm{alpha}_{\{2\}}}+\backslash\mathrm{tilde}{\mathrm{A}}_{\backslash\mathrm{alpha}_{\{2\}}}\backslash\mathrm{right})\otimes\mathrm{L}_{2^{\mathrm{m}}}-\frac{1}{2}\Omega_{\alpha_1\alpha_2}\otimes\gamma^{\alpha_1\alpha_2}$	$\mathcal{D}_A^2 = -g^{\alpha_1\alpha_2}(\delta_{\alpha_1} + \tilde{A}_{\alpha_1})(\delta_{\alpha_2} + \tilde{A}_{\alpha_2}) \otimes 1_{2^m} - \frac{1}{2} \Omega_{\alpha_1\alpha_2} \otimes \gamma^{\alpha_1\alpha_2}$	$\mathcal{D}_A^2 = -g^{\alpha_1\alpha_2}(\delta_{\alpha_1} + \tilde{A}_{\alpha_1})(\delta_{\alpha_2} + \tilde{A}_{\alpha_2}) \otimes 1_{2^m} - \frac{1}{2} \Omega_{\alpha_1\alpha_2} \otimes \gamma^{\alpha_1\alpha_2}$ where
cardona-qft-methods_ F01233	62	■ 1.000	$\backslash\mathrm{mathcal{D}}=\mathrm{U}_{\{0\}}\otimes\backslash\mathrm{mathbb{C}}^{\wedge\{2^{\mathrm{m}}\}}$	$\mathcal{D} = U_0 \otimes \mathbb{C}^{2^m}$	Prop 8.1. $\mathcal{D} = U_0 \otimes \mathbb{C}^{2^m}$, so $\dim \mathcal{D} = 2^m$. For any selfadjoint one-form A ,
cardona-qft-methods_ F01234	62	■ 1.000	$\backslash\mathrm{operatorname{dim}}\backslash\mathrm{mathcal{D}}=2^{\mathrm{m}}$	$\dim \mathcal{D} = 2^m$	Prop 8.1. $\mathcal{D} = U_0 \otimes \mathbb{C}^{2^m}$, so $\dim \mathcal{D} = 2^m$. For any selfadjoint one-form A ,
cardona-qft-methods_ F01235	62	■ 1.000	$\backslash\mathrm{mathcal{D}}\backslash\mathrm{subsepeq}\backslash\mathrm{mathcal{D}}_{\mathrm{A}}$	$\mathcal{D} \subseteq \mathcal{D}_A$	$\mathcal{D} \subseteq \mathcal{D}_A$ and or any unitary $u \in \mathcal{A}$, $\mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$.
cardona-qft-methods_ F01236	62	■ 1.000	$\mathrm{u}\in\backslash\mathrm{in}\backslash\mathrm{mathcal{A}},\backslash\mathrm{mathcal{D}}_{\backslash\mathrm{gamma}_{\mathrm{u}}(\mathrm{A})}=\mathrm{V}_{\mathrm{u}}\backslash\mathrm{mathcal{D}}_{\mathrm{A}}$	$u \in \mathcal{A}, \mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$	$\mathcal{D} \subseteq \mathcal{D}_A$ and or any unitary $u \in \mathcal{A}$, $\mathcal{D}_{\gamma_u(A)} = V_u \mathcal{D}_A$.
cardona-qft-methods_ F01237	62	■ 1.000	$\backslash\mathrm{mathcal{D}}_{\mathrm{A}}=\backslash\mathrm{mathcal{D}}$	$\mathcal{D}_A = \mathcal{D}$	One shows that $\mathcal{D}_A = \mathcal{D}$ in the following cases:
cardona-qft-methods_ F01238	62	■ 1.000	$\mathrm{A}=\mathrm{A}_{\mathrm{u}}=\mathrm{L}(\mathrm{u})\otimes\mathrm{L}_{2^{\mathrm{m}}}\backslash\mathrm{left}[\backslash\mathrm{mathcal{D}},\mathrm{L}\backslash\mathrm{left}(\mathrm{u}^{\ast}\backslash\mathrm{right})\otimes\mathrm{L}_{2^{\mathrm{m}}}\backslash\mathrm{right}]$	$A = A_u = L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]$	(i) $A = A_u = L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]$ when u is a unitary in $\mathcal{A}$.
cardona-qft-methods_ F01239	62	■ 1.000	u	u	(i) $A = A_u = L(u) \otimes 1_{2^m} [\mathcal{D}, L(u^*) \otimes 1_{2^m}]$ when u is a unitary in $\mathcal{A}$.
cardona-qft-methods_ F01240	62	■ 0.995	$\backslash\mathrm{A}\backslash\mathrm{<}\frac{1}{2}$	$\ A\ < \frac{1}{2}$	(ii) $\ A\ < \frac{1}{2}$.
cardona-qft-methods_ F01241	62	■ 1.000	$\frac{1}{2\pi}\backslash\mathrm{Theta}$	$\frac{1}{2\pi} \Theta$	(iii) The matrix $\frac{1}{2\pi} \Theta$ has only integral coefficients.
cardona-qft-methods_ F01242	62	■ 1.000	$\backslash\mathrm{mathcal{D}}=\backslash\mathrm{mathcal{D}}_{\mathrm{A}}$	$\mathcal{D} = \mathcal{D}_A$	<i>Conjecture:</i> $\mathcal{D} = \mathcal{D}_A$ at least for generic Θ 's.
cardona-qft-methods_ F01243	62	■ 1.000	$\backslash\mathrm{delta}>0$	$\delta > 0$	Definition 8.1. Let $\delta > 0$. A vector $a \in \mathbb{R}^n$ is said to be δ -badly approx-
cardona-qft-methods_ F01244	62	■ 1.000	$\mathrm{a}\in\backslash\mathrm{in}\backslash\mathrm{mathbb{R}}^{\mathrm{n}}$	$a \in \mathbb{R}^n$	Definition 8.1. Let $\delta > 0$. A vector $a \in \mathbb{R}^n$ is said to be δ -badly approx-
cardona-qft-methods_ F01245	62	■ 1.000	$\backslash\mathrm{delta}$	δ	Definition 8.1. Let $\delta > 0$. A vector $a \in \mathbb{R}^n$ is said to be δ -badly approx-

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cardona-qft-methods_ F01246	62	■ 0.947	$c>0$	$c > 0$	inable if there exists $c > 0$ such that $ q \cdot a - m \geq c q ^{-\delta}$, $\forall q \in \mathbb{Z}^n \setminus \{0\}$
cardona-qft-methods_ F01247	62	■ 0.947	$ q \cdot a - m \geq c q ^{-\delta}, \forall q \in \mathbb{Z}^n \setminus \{0\}$	$ q \cdot a - m \geq c q ^{-\delta}, \forall q \in \mathbb{Z}^n \setminus \{0\}$	inable if there exists $c > 0$ such that $ q \cdot a - m \geq c q ^{-\delta}$, $\forall q \in \mathbb{Z}^n \setminus \{0\}$
cardona-qft-methods_ F01248	62	■ 1.000	$\forall m \in \mathbb{Z}$	$\forall m \in \mathbb{Z}$	and $\forall m \in \mathbb{Z}$.
cardona-qft-methods_ F01249	62	■ 0.996	$\mathcal{B}(\delta)$	$\mathcal{BV}(\delta)$	We note $\mathcal{BV}(\delta)$ the set of δ -badly approximable vectors and $\mathcal{BV} :=$
cardona-qft-methods_ F01250	62	■ 0.996	$\mathcal{B} \setminus \mathcal{V} :=$	$\mathcal{BV} :=$	We note $\mathcal{BV}(\delta)$ the set of δ -badly approximable vectors and $\mathcal{BV} :=$
cardona-qft-methods_ F01251	62	■ 1.000	$\cup_{\delta>0} \mathcal{B}(\delta)$	$\cup_{\delta>0} \mathcal{BV}(\delta)$	$\cup_{\delta>0} \mathcal{BV}(\delta)$ the set of badly approximable vectors.
cardona-qft-methods_ F01252	62	■ 1.000	$\Theta \in \mathcal{M}_n(\mathbb{R})$	$\Theta \in \mathcal{M}_n(\mathbb{R})$	A matrix $\Theta \in \mathcal{M}_n(\mathbb{R})$ (real $n \times n$ matrices) will be said to be badly
cardona-qft-methods_ F01253	62	■ 1.000	$n \times n$	$n \times n$	A matrix $\Theta \in \mathcal{M}_n(\mathbb{R})$ (real $n \times n$ matrices) will be said to be badly
cardona-qft-methods_ F01254	62	■ 1.000	$u \in \mathbb{Z}^n$	$u \in \mathbb{Z}^n$	approximable if there exists $u \in \mathbb{Z}^n$ such that ${}^t\Theta(u)$ is a badly approximable
cardona-qft-methods_ F01255	62	■ 1.000	$\{ {}^t\Theta(u) \}$	${}^t\Theta(u)$	approximable if there exists $u \in \mathbb{Z}^n$ such that ${}^t\Theta(u)$ is a badly approximable
cardona-qft-methods_ F01256	62	■ 1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	vector of $\mathbb{R}^n$.
cardona-qft-methods_ F01257	62	■ 1.000	$\delta > n$	$\delta > n$	Remark. A result from Diophantine approximation asserts that for $\delta > n$,
cardona-qft-methods_ F01258	62	■ 0.975	$\mathbb{R}^n \setminus \mathcal{BV}(\delta)$	$\mathbb{R}^n \setminus \mathcal{BV}(\delta)$	the Lebesgue measure of $\mathbb{R}^n \setminus \mathcal{BV}(\delta)$ is zero (i.e almost any element of $\mathbb{R}^n$
cardona-qft-methods_ F01259	62	■ 1.000	i	i	Let $\Theta \in \mathcal{M}_n(\mathbb{R})$. If its row of index i is a badly approximable vector
cardona-qft-methods_ F01260	62	■ 0.666	$L_i \in \mathcal{B}$	$L_i \in \mathcal{BV}$	of $\mathbb{R}^n$ (i.e. if $L_i \in \mathcal{BV}$) then ${}^t\Theta(e_i) \in \mathcal{BV}$ and thus Θ is a badly approx-
cardona-qft-methods_ F01261	62	■ 0.666	$\{ {}^t\Theta(e_i) \} \in \mathcal{B} \setminus \mathcal{V}$	${}^t\Theta(e_i) \in \mathcal{BV}$	of $\mathbb{R}^n$ (i.e. if $L_i \in \mathcal{BV}$) then ${}^t\Theta(e_i) \in \mathcal{BV}$ and thus Θ is a badly approx-
cardona-qft-methods_ F01262	62	■ 1.000	$\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$	$\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$	inable matrix. It follows that almost any matrix of $\mathcal{M}_n(\mathbb{R}) \approx \mathbb{R}^{n^2}$ is badly
cardona-qft-methods_ F01263	63	■ 1.000	$(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$	$(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$	the spectral triple $(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$ is equal to the set $\{n - k : k \in \mathbb{N}_0\}$
cardona-qft-methods_ F01264	63	■ 1.000	$\{n - k : k \in \mathbb{N}_0\}$	$\{n - k : k \in \mathbb{N}_0\}$	the spectral triple $(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$ is equal to the set $\{n - k : k \in \mathbb{N}_0\}$
cardona-qft-methods_ F01265	63	■ 1.000	$\zeta_D(0) = 0$	$\zeta_D(0) = 0$	and all these poles are simple. Moreover $\zeta_D(0) = 0$.
cardona-qft-methods_ F01266	63	■ 1.000	$\zeta_D(0)$	$\zeta_D(0)$	We can compute $\zeta_D(0)$ easily but the main difficulty is precisely to
cardona-qft-methods_ F01267	63	■ 1.000	$\zeta_{D_A}(0)$	$\zeta_{D_A}(0)$	calculate $\zeta_{D_A}(0)$.
cardona-qft-methods_ F01268	63	■ 0.813	$C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D}$	$C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D}$	Theorem 8.2. Consider the noncommutative torus $(C^\infty(\mathbb{T}_\Theta^n), \mathcal{H}, \mathcal{D})$ of di-
cardona-qft-methods_ F01269	63	■ 1.000	$A = L(-iA_\alpha) \otimes \gamma^\alpha$	$A = L(-iA_\alpha) \otimes \gamma^\alpha$	proximable matrix, and a selfadjoint one-form $A = L(-iA_\alpha) \otimes \gamma^\alpha$. Then,
cardona-qft-methods_ F01270	63	■ 1.000	$n=2, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 4\pi f_2 \Lambda^2 + \mathcal{O}(\Lambda^{-2})$	$n = 2, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 4\pi f_2 \Lambda^2 + \mathcal{O}(\Lambda^{-2})$	(i) for $n = 2$, $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = 4\pi f_2 \Lambda^2 + \mathcal{O}(\Lambda^{-2})$,
cardona-qft-methods_ F01271	63	■ 1.000	$n=4, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 8\pi^2 f_4 \Lambda^4 - \frac{4\pi^2}{3} f(0) \tau(F_{\mu\nu} F^{\mu\nu}) + \mathcal{O}(\Lambda^{-2})$	$n = 4, \mathcal{S}(\mathcal{D}_A, f, \Lambda) = 8\pi^2 f_4 \Lambda^4 - \frac{4\pi^2}{3} f(0) \tau(F_{\mu\nu} F^{\mu\nu}) + \mathcal{O}(\Lambda^{-2})$	(ii) for $n = 4$, $\mathcal{S}(\mathcal{D}_A, f, \Lambda) = 8\pi^2 f_4 \Lambda^4 - \frac{4\pi^2}{3} f(0) \tau(F_{\mu\nu} F^{\mu\nu}) + \mathcal{O}(\Lambda^{-2})$,

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cardona-qft-methods_ F01272	63	■ 1.000	$\mathcal{S}\left(\mathcal{D}_A, f, \Lambda\right)=\sum_{k=0}^n f_{n-k} c_{n-k}(A) \Lambda^{n-k}+$	$\mathcal{S}\left(\mathcal{D}_A, f, \Lambda\right)=\sum_{k=0}^n f_{n-k} c_{n-k}(A) \Lambda^{n-k}+$	(iii) More generally, in $\mathcal{S}\left(\mathcal{D}_A, f, \Lambda\right)=\sum_{k=0}^n f_{n-k} c_{n-k}(A) \Lambda^{n-k}+$
cardona-qft-methods_ F01273	63	■ 1.000	$\mathcal{O}\left(\Lambda^{-1}\right), c_{n-2}(A)=0, c_{n-k}(A)=0$	$\mathcal{O}\left(\Lambda^{-1}\right), c_{n-2}(A)=0, c_{n-k}(A)=0$	$\mathcal{O}\left(\Lambda^{-1}\right), c_{n-2}(A)=0, c_{n-k}(A)=0$ for k odd. In particular, $c_0(A)=0$
cardona-qft-methods_ F01274	63	■ 1.000	$c_{\{0\}}(A)=0$	$c_0(A)=0$	$\mathcal{O}\left(\Lambda^{-1}\right), c_{n-2}(A)=0, c_{n-k}(A)=0$ for k odd. In particular, $c_0(A)=0$
cardona-qft-methods_ F01275	63	■ 1.000	$n=4$	$n=4$	This result (for $n=4$) has also been obtained in ⁴¹ using the heat kernel
cardona-qft-methods_ F01276	63	■ 1.000	$A_{\{\alpha\}} \rightarrow \gamma_u\left(A_{\{\alpha\}}\right)=u A_{\alpha} u^{*}+u \delta_{\alpha}\left(u^{*}\right)$	$A_{\alpha} \rightarrow \gamma_u\left(A_{\alpha}\right)=u A_{\alpha} u^{*}+u \delta_{\alpha}\left(u^{*}\right)$	variant by $A_{\alpha} \rightarrow \gamma_u\left(A_{\alpha}\right)=u A_{\alpha} u^{*}+u \delta_{\alpha}\left(u^{*}\right)$. A particular case is $u=U_k$
cardona-qft-methods_ F01277	63	■ 1.000	$u=U_{\{k\}}$	$u=U_k$	variant by $A_{\alpha} \rightarrow \gamma_u\left(A_{\alpha}\right)=u A_{\alpha} u^{*}+u \delta_{\alpha}\left(u^{*}\right)$. A particular case is $u=U_k$
cardona-qft-methods_ F01278	63	■ 1.000	$U_{\{k\}} \delta_{\alpha}\left(U_{\{k\}}^{*}\right)=-i k_{\alpha} U_0$	$U_k \delta_{\alpha}\left(U_k^{*}\right)=-i k_{\alpha} U_0$	where $U_k \delta_{\alpha}\left(U_k^{*}\right)=-i k_{\alpha} U_0$.
cardona-qft-methods_ F01279	64	■ 1.000	$A, \mathcal{D}_A=\mathcal{D}$	$A, \mathcal{D}_A=\mathcal{D}$	tive case where, for any selfadjoint one-form A , $\mathcal{D}_A=\mathcal{D}$ (so A is equivalent
cardona-qft-methods_ F01280	64	■ 1.000	$\Theta=\theta\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$	$\Theta=\theta\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$	$\Theta=\theta\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$ with $\theta \in \mathbb{R}$. In fact, the Hochschild cohomology $H\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)$
cardona-qft-methods_ F01281	64	■ 1.000	$\theta \in \mathbb{R}$	$\theta \in \mathbb{R}$	$\Theta=\theta\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$ with $\theta \in \mathbb{R}$. In fact, the Hochschild cohomology $H\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)$
cardona-qft-methods_ F01282	64	■ 1.000	$H\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)$	$H\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)$	$\Theta=\theta\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$ with $\theta \in \mathbb{R}$. In fact, the Hochschild cohomology $H\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)$
cardona-qft-methods_ F01283	64	■ 0.996	$\dim H^j\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)=2$	$\dim H^j\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)=2$	satisfies $\dim H^j\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)=2$ (or 1) for $j=1$ (or $j=2$) if and only if the
cardona-qft-methods_ F01284	64	■ 0.996	$j=1$	$j=1$	satisfies $\dim H^j\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)=2$ (or 1) for $j=1$ (or $j=2$) if and only if the
cardona-qft-methods_ F01285	64	■ 0.996	$j=2$	$j=2$	satisfies $\dim H^j\left(\mathcal{A}_{\Theta}, \mathcal{A}_{\Theta}^{*}\right)=2$ (or 1) for $j=1$ (or $j=2$) if and only if the
cardona-qft-methods_ F01286	64	■ 1.000	θ	θ	irrational number θ satisfies a Diophantine condition like $\left 1-e^{i 2 \pi n \theta}\right ^{-1} =$
cardona-qft-methods_ F01287	64	■ 1.000	$\left 1-e^{i 2 \pi n \theta}\right ^{-1} =$	$\left 1-e^{i 2 \pi n \theta}\right ^{-1} =$	irrational number θ satisfies a Diophantine condition like $\left 1-e^{i 2 \pi n \theta}\right ^{-1} =$
cardona-qft-methods_ F01288	64	■ 1.000	$\mathcal{O}\left(n^k\right)$	$\mathcal{O}\left(n^k\right)$	$\mathcal{O}\left(n^k\right)$ for some k .
cardona-qft-methods_ F01289	64	■ 1.000	$\mathbb{Z}^n$	$\mathbb{Z}^n$	by its columns is dense after translation by $\mathbb{Z}^n$, see [49, Def. 12.8]), then
cardona-qft-methods_ F01290	71	■ 1.000	$\boldsymbol{G}$	$\boldsymbol{G}$	INDEX THEORY FOR NON-COMPACT G-MANIFOLDS
cardona-qft-methods_ F01291	71	■ 0.998	$\{ \}^{3-5,13,17}$	$3-5,13,17$	numerous generalizations have been made, see for example ^{3-5,13,17} to mention
cardona-qft-methods_ F01292	71	■ 0.997	$\{ \}^{11,25}$	$11,25$	non-compact manifolds, ^{11,25}
cardona-qft-methods_ F01293	72	■ 1.000	$A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$	$A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$	Consider a linear operator $A: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ between two <i>finite dimensional</i>
cardona-qft-methods_ F01294	72	■ 0.997	$\mathcal{H}_1$	$\mathcal{H}_1$	vector spaces $\mathcal{H}_1$ and $\mathcal{H}_2$. Then the <i>index</i> of A is defined to be
cardona-qft-methods_ F01295	72	■ 0.997	$\mathcal{H}_2$	$\mathcal{H}_2$	vector spaces $\mathcal{H}_1$ and $\mathcal{H}_2$. Then the <i>index</i> of A is defined to be
cardona-qft-methods_ F01296	72	■ 0.840	$\operatorname{Coker}(A):=\mathcal{H}_2 / \operatorname{Im}(A)$	$\operatorname{Coker}(A):=\mathcal{H}_2 / \operatorname{Im}(A)$	where $\operatorname{Coker}(A):=\mathcal{H}_2 / \operatorname{Im}(A)$. Notice that, although $\dim \operatorname{Ker}(A)$ and
cardona-qft-methods_ F01297	72	■ 0.840	$\dim \operatorname{Ker}(A)$	$\dim \operatorname{Ker}(A)$	where $\operatorname{Coker}(A):=\mathcal{H}_2 / \operatorname{Im}(A)$. Notice that, although $\dim \operatorname{Ker}(A)$ and

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cardona-qft-methods_ F01298	72	■ 0.974	$\operatorname{dim} \operatorname{Coker}(A)$	$\dim \operatorname{Coker}(A)$	$\dim \operatorname{Coker}(A)$ depend on A , the index
cardona-qft-methods_ F01299	72	■ 0.999	$\operatorname{Ker}(A)$	$\operatorname{Ker}(A)$	Then, in general, both $\operatorname{Ker}(A)$ and $\operatorname{Coker}(A)$ may be infinite dimensional
cardona-qft-methods_ F01300	72	■ 0.999	$\operatorname{Coker}(A)$	$\operatorname{Coker}(A)$	Then, in general, both $\operatorname{Ker}(A)$ and $\operatorname{Coker}(A)$ may be infinite dimensional
cardona-qft-methods_ F01301	72	■ 0.994	$\operatorname{Im}(A)$	$\operatorname{Im}(A)$	Exercise 1.3. Show that if the dimension of $\operatorname{Coker}(A)$ is finite, then $\operatorname{Im}(A)$
cardona-qft-methods_ F01302	72	■ 0.985	$\mathcal{H}_1, \mathcal{H}_2$	$\mathcal{H}_1, \mathcal{H}_2$	Definition 1.4. Let $\mathcal{H}_1, \mathcal{H}_2$ be two Hilbert spaces and let $A : \mathcal{H}_1 \rightarrow \mathcal{H}_2$
cardona-qft-methods_ F01303	72	■ 0.506	$\operatorname{dim} \operatorname{Ker}(A) < \infty, \operatorname{dim} \operatorname{Coker}(A) < \infty$	$\dim \operatorname{Ker}(A) < \infty, \dim \operatorname{Coker}(A) < \infty$	$\dim \operatorname{Ker}(A) < \infty, \dim \operatorname{Coker}(A) < \infty$.
cardona-qft-methods_ F01304	72	■ 1.000	$\operatorname{Ind}(A)$	$\operatorname{Ind}(A)$	If $A : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ is a Fredholm operator, then its index $\operatorname{Ind}(A)$ can be
cardona-qft-methods_ F01305	72	■ 1.000	$\{ \}^{\{20 \operatorname{a}\}}$	^{20a}	by Fritz Noether ^{20a} , who also established the following “stability” property
cardona-qft-methods_ F01306	72	■ 1.000	$A(t)$	$A(t)$	Theorem 1.5. a. If $A(t)$ is a Fredholm operator which depends continu-
cardona-qft-methods_ F01307	72	■ 1.000	$\operatorname{Ind}(A(t))$	$\operatorname{Ind}(A(t))$	ously on the parameter t , then $\operatorname{Ind}(A(t))$ is constant.
cardona-qft-methods_ F01308	72	■ 1.000	$\operatorname{Ind}(A+K)=\operatorname{Ind}(A)$	$\operatorname{Ind}(A+K) = \operatorname{Ind}(A)$	b. If K is compact and A Fredholm, then $\operatorname{Ind}(A+K) = \operatorname{Ind}(A)$.
cardona-qft-methods_ F01309	72	■ 0.992	$\{ \}^{\{\operatorname{a}\}}$	^a	^a Fritz Noether was a fine mathematician with a very interesting and tragic biography.
cardona-qft-methods_ F01310	72	■ 1.000	$\{ \}^{\{21,22\}}$	^{21,22}	reader to ^{21,22} for a short description of his biography.
cardona-qft-methods_ F01311	73	■ 1.000	$T: \mathcal{H} \rightarrow \mathcal{H}$	$T : \mathcal{H} \rightarrow \mathcal{H}$	and consider the linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ defined by
cardona-qft-methods_ F01312	73	■ 1.000	$\operatorname{dim} \operatorname{Ker}(T)=2$	$\dim \operatorname{Ker}(T) = 2$	Thus $\dim \operatorname{Ker}(T) = 2$. Since the image of A is the whole space $\mathcal{H}$, we
cardona-qft-methods_ F01313	73	■ 1.000	$\operatorname{Coker}(T)=\{0\}$	$\operatorname{Coker}(T) = \{0\}$	conclude that $\operatorname{Coker}(T) = \{0\}$. Thus $\operatorname{Ind}(T) = 2$.
cardona-qft-methods_ F01314	73	■ 1.000	$\operatorname{Ind}(T)=2$	$\operatorname{Ind}(T) = 2$	conclude that $\operatorname{Coker}(T) = \{0\}$. Thus $\operatorname{Ind}(T) = 2$.
cardona-qft-methods_ F01315	73	■ 0.999	$\operatorname{Ind}(A)=0$	$\operatorname{Ind}(A) = 0$	(2) that $\operatorname{Ind}(A) = 0$ for every linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$. The example
cardona-qft-methods_ F01316	73	■ 0.999	$A: \mathcal{H} \rightarrow \mathcal{H}$	$A : \mathcal{H} \rightarrow \mathcal{H}$	(2) that $\operatorname{Ind}(A) = 0$ for every linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$. The example
cardona-qft-methods_ F01317	73	■ 1.000	$\operatorname{dim} \mathcal{H}=\infty$	$\dim \mathcal{H} = \infty$	above shows that this is not true if $\dim \mathcal{H} = \infty$.
cardona-qft-methods_ F01318	73	■ 1.000	$B: \mathcal{H}_2 \rightarrow \mathcal{H}_3$	$B : \mathcal{H}_2 \rightarrow \mathcal{H}_3$	Exercise 1.8. Suppose $A : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ and $B : \mathcal{H}_2 \rightarrow \mathcal{H}_3$ are Fredholm
cardona-qft-methods_ F01319	73	■ 1.000	$B A: \mathcal{H}_1 \rightarrow \mathcal{H}_3$	$BA : \mathcal{H}_1 \rightarrow \mathcal{H}_3$	operators. Show that the operator $BA : \mathcal{H}_1 \rightarrow \mathcal{H}_3$ is Fredholm and that
cardona-qft-methods_ F01320	73	■ 0.983	$A^{\{*\}}: \mathcal{H}_2 \rightarrow \mathcal{H}_1$	$A^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$	adjoint operator $A^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ is Fredholm and that
cardona-qft-methods_ F01321	73	■ 0.991	$\operatorname{Ind}(A)>0$	$\operatorname{Ind}(A) > 0$	Suppose for example that $\operatorname{Ind}(A) > 0$. This means that $\operatorname{Ker}(A) \neq \{0\}$, i.e.,
cardona-qft-methods_ F01322	73	■ 0.991	$\operatorname{Ker}(A) \neq \{0\}$	$\operatorname{Ker}(A) \neq \{0\}$	Suppose for example that $\operatorname{Ind}(A) > 0$. This means that $\operatorname{Ker}(A) \neq \{0\}$, i.e.,
cardona-qft-methods_ F01323	73	■ 1.000	$A x=0$	$Ax = 0$	the equation $Ax = 0$ has a non-trivial solution. Moreover, for any compact
cardona-qft-methods_ F01324	73	■ 1.000	$\operatorname{Ind}(A+K)=\operatorname{Ind}(K)>0$	$\operatorname{Ind}(A+K) = \operatorname{Ind}(K) > 0$	operator K we have $\operatorname{Ind}(A+K) = \operatorname{Ind}(K) > 0$. Hence, the equation

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cardona-qft-methods_F01325	73	■ 1.000	B	B	an operator B we often compute the index of a simpler operator A , for
cardona-qft-methods_F01326	73	■ 1.000	$B=A+K$	$B = A + K$	that $B = A + K$ for K a compact operator. If the index of A is positive,
cardona-qft-methods_F01327	74	■ 1.000	$\operatorname{Fred}\left(\mathcal{H}_1, \mathcal{H}_2\right)$	$\operatorname{Fred}(\mathcal{H}_1, \mathcal{H}_2)$	Let $\operatorname{Fred}(\mathcal{H}_1, \mathcal{H}_2)$ denote the set of Fredholm operators $A : \mathcal{H}_1 \rightarrow \mathcal{H}_2$. It
cardona-qft-methods_F01328	74	■ 1.000	$d(A, B):=\ A-B\ $	$d(A, B) := \ A - B\ $	is a metric space with the distance defined by $d(A, B) := \ A - B\ $. The-
cardona-qft-methods_F01329	74	■ 1.000	$\operatorname{Ind}(A)=\operatorname{Ind}(B)$	$\operatorname{Ind}(A) = \operatorname{Ind}(B)$	component of $\operatorname{Fred}(\mathcal{H}_1, \mathcal{H}_2)$, then $\operatorname{Ind}(A) = \operatorname{Ind}(B)$. In other words, $\operatorname{Ind}(A)$
cardona-qft-methods_F01330	74	■ 1.000	$\operatorname{Irr}(G)$	$\operatorname{Irr}(G)$	trivial G -invariant subspaces. We denote by $\operatorname{Irr}(G)$ the set of irreducible
cardona-qft-methods_F01331	74	■ 1.000	$m_V \ V$	$m_V V$	Here $m_V V$ stands for the direct sum $V \oplus V \oplus \cdots \oplus V$ of m_V copies of V .
cardona-qft-methods_F01332	74	■ 1.000	$V \oplus V \oplus \cdots \oplus V$	$V \oplus V \oplus \cdots \oplus V$	Here $m_V V$ stands for the direct sum $V \oplus V \oplus \cdots \oplus V$ of m_V copies of V .
cardona-qft-methods_F01333	74	■ 1.000	m_V	m_V	Here $m_V V$ stands for the direct sum $V \oplus V \oplus \cdots \oplus V$ of m_V copies of V .
cardona-qft-methods_F01334	74	■ 1.000	$m_V \in \mathbb{N}$	$m_V \in \mathbb{N}$	The numbers $m_V \in \mathbb{N}$ are called the <i>multiplicities of the irreducible repre-</i>
cardona-qft-methods_F01335	74	■ 1.000	$x \in \operatorname{Ker}(A), g \in G$	$x \in \operatorname{Ker}(A), g \in G$	$x \in \operatorname{Ker}(A)$, $g \in G$ we have $gx \in \operatorname{Ker}(A)$. Thus the restriction of the action
cardona-qft-methods_F01336	74	■ 1.000	$g x \in \operatorname{Ker}(A)$	$gx \in \operatorname{Ker}(A)$	$x \in \operatorname{Ker}(A)$, $g \in G$ we have $gx \in \operatorname{Ker}(A)$. Thus the restriction of the action
cardona-qft-methods_F01337	75	■ 1.000	m_+, m_-	m_+, m_-	are actually finite, since only finitely many numbers m_+, m_- can be non-
cardona-qft-methods_F01338	75	■ 0.886	$R(G)$	$R(G)$	to it the <i>Grothendieck group</i> $R(G)$ by considering formal differences $V - U$
cardona-qft-methods_F01339	75	■ 0.886	$V-U$	$V - U$	to it the <i>Grothendieck group</i> $R(G)$ by considering formal differences $V - U$
cardona-qft-methods_F01340	75	■ 1.000	V_1-V_2	$V_1 - V_2$	Definition 1.18. Consider the set of formal differences, $V_1 - V_2$, of finite
cardona-qft-methods_F01341	75	■ 1.000	V_1	V_1	dimensional representations V_1 and V_2 of G . The quotient of this set by the
cardona-qft-methods_F01342	75	■ 1.000	V_2	V_2	dimensional representations V_1 and V_2 of G . The quotient of this set by the
cardona-qft-methods_F01343	75	■ 1.000	$\{ \ }^{\text{b}} \}$	$^{\text{b}}$	equivalence relation ^b
cardona-qft-methods_F01344	75	■ 0.997	$R(G):=\{(V, U) \mid V, U$	$R(G) := \{(V, U) \mid V, U$	$R(G) := \{(V, U) \mid V, U \text{ are finite dimensional representations of } G\} / \sim$.
cardona-qft-methods_F01345	75	■ 0.997	$G\} / \sim$	$G\} / \sim$	$R(G) := \{(V, U) \mid V, U \text{ are finite dimensional representations of } G\} / \sim$.
cardona-qft-methods_F01346	75	■ 1.000	$\{ \ }^{\mathrm{b}} \}$	$^{\text{b}}$	^b In general, the definition of the equivalence relation in the Grothendieck group is slightly
cardona-qft-methods_F01347	76	■ 1.000	$\operatorname{Ind}_G(A)$	$\operatorname{Ind}_G(A)$	Definition 1.23. The <i>equivariant index</i> $\operatorname{Ind}_G(A)$ of A is defined to be the
cardona-qft-methods_F01348	76	■ 1.000	$V_{\{0\}}$	V_0	Example 1.24. Let V_0 denote the trivial representation. If the inequality
cardona-qft-methods_F01349	76	■ 1.000	$m_{V_{\{0\}}}^+-m_{V_{\{0\}}}^->0$	$m_{V_0}^+ - m_{V_0}^- > 0$	$m_{V_0}^+ - m_{V_0}^- > 0$ holds, that means that there exist a nontrivial solution of
cardona-qft-methods_F01350	76	■ 1.000	q	q	To avoid confusion we denote by q the non-trivial element of G (thus $q =$
cardona-qft-methods_F01351	76	■ 1.000	q=	$q =$	To avoid confusion we denote by q the non-trivial element of G (thus $q =$

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cardona-qft-methods_ F01352	76	■ 0.582	$l^{\wedge\{2\}}$	l^2	$-1)$. We define an action of G on l^2 by
cardona-qft-methods_ F01353	76	■ 1.000	$V_{\{0\}}=\mathbb{C}$	$V_0 = \mathbb{C}$	Observe that G has two irreducible representations, namely: $V_0 = \mathbb{C}$ with
cardona-qft-methods_ F01354	76	■ 1.000	$V_{\{1\}}=\mathbb{C}$	$V_1 = \mathbb{C}$	the trivial action, and $V_1 = \mathbb{C}$ with the action $qz = -z$.
cardona-qft-methods_ F01355	76	■ 1.000	$q\ z=-z$	$qz = -z$	the trivial action, and $V_1 = \mathbb{C}$ with the action $qz = -z$.
cardona-qft-methods_ F01356	76	■ 0.977	$\operatorname{Ker}(T)=V_{\{0\}} \oplus V_{\{1\}}$	$\operatorname{Ker}(T) = V_0 \oplus V_1$	Show that the operator (4) is G -equivariant and that $\operatorname{Ker}(T) = V_0 \oplus V_1$.
cardona-qft-methods_ F01357	76	■ 1.000	$\operatorname{Ind}_G(T)$	$\operatorname{Ind}_G(T)$	Use this result to compute $\operatorname{Ind}_G(T)$.
cardona-qft-methods_ F01358	77	■ 1.000	$D_{\{j\}}$	D_j	We recall the definition of linear differential operators in $\mathbb{R}^n$. Let D_j :
cardona-qft-methods_ F01359	77	■ 1.000	$C^{\infty}\left(\mathbb{R}^n\right) \rightarrowtail C^{\infty}\left(\mathbb{R}^n\right)$	$C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$	$C^\infty(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n)$ denote the operator
cardona-qft-methods_ F01360	77	■ 0.999	$\alpha=\left(\alpha_{\{1\}}, \ldots, \alpha_{\{n\}}\right) \in$	$\alpha = (\alpha_1, \dots, \alpha_n) \in$	A <i>multi-index</i> is an n -tuple of non-negative integers $\alpha = (\alpha_1, \dots, \alpha_n) \in$
cardona-qft-methods_ F01361	77	■ 1.000	$\mathbb{Z}_{\geq 0}^n$	$\mathbb{Z}_{\geq 0}^n$	$\mathbb{Z}_{\geq 0}^n$. For a multiindex α we set
cardona-qft-methods_ F01362	77	■ 1.000	$a_{\{\alpha\}}(x) \in C^{\infty}\left(\mathbb{R}^n\right)$	$a_\alpha(x) \in C^\infty(\mathbb{R}^n)$	where $a_\alpha(x) \in C^\infty(\mathbb{R}^n)$.
cardona-qft-methods_ F01363	77	■ 1.000	$C_c^{\infty}\left(\mathbb{R}^n\right)$	$C_c^\infty(\mathbb{R}^n)$	Let $C_c^\infty(\mathbb{R}^n)$ denote the set of smooth functions with compact support
cardona-qft-methods_ F01364	77	■ 1.000	$\langle \cdot, \cdot \rangle$	$\langle \cdot, \cdot \rangle$	on $\mathbb{R}^n$ and let $\langle \cdot, \cdot \rangle$ denote the scalar product on $C_c^\infty(\mathbb{R}^n)$ defined by
cardona-qft-methods_ F01365	78	■ 1.000	$H^k\left(\mathbb{R}^n\right) \subset L^2\left(\mathbb{R}^n\right)$	$H^k(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$	The <i>Sobolev space</i> $H^k(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$ is the completion of the space
cardona-qft-methods_ F01366	78	■ 1.000	$H^k\left(\mathbb{R}^n\right)$	$H^k(\mathbb{R}^n)$	of the Sobolev space $H^k(\mathbb{R}^n)$:
cardona-qft-methods_ F01367	78	■ 1.000	$L^2\left(\mathbb{R}^n\right)$	$L^2(\mathbb{R}^n)$	$L^2(\mathbb{R}^n)$. <i>More explicitly</i>
cardona-qft-methods_ F01368	78	■ 1.000	$\Omega \subset \mathbb{R}^n$	$\Omega \subset \mathbb{R}^n$	Remark 2.5. If $\Omega \subset \mathbb{R}^n$ is an open set, one can define the Sobolev space
cardona-qft-methods_ F01369	78	■ 1.000	$H^k(\Omega)$	$H^k(\Omega)$	$H^k(\Omega)$ of functions on Ω . An analogue of Theorem 2.4 still holds if the
cardona-qft-methods_ F01370	78	■ 1.000	$K \subset M$	$K \subset M$	Exercise 2.6. Let $K \subset M$ be a compact set. Denote by
cardona-qft-methods_ F01371	78	■ 0.997	$m \geq k$	$m \geq k$	For every $m \geq k$ a differential operator $\mathcal{D} : C_c^\infty(\mathbb{R}^n) \rightarrow C_c^\infty(\mathbb{R}^n)$ of order
cardona-qft-methods_ F01372	78	■ 0.997	$\mathcal{D}: C_c^{\infty}\left(\mathbb{R}^n\right) \rightarrow C_c^{\infty}\left(\mathbb{R}^n\right)$	$\mathcal{D} : C_c^\infty(\mathbb{R}^n) \rightarrow C_c^\infty(\mathbb{R}^n)$	For every $m \geq k$ a differential operator $\mathcal{D} : C_c^\infty(\mathbb{R}^n) \rightarrow C_c^\infty(\mathbb{R}^n)$ of order
cardona-qft-methods_ F01373	78	■ 1.000	$\mathcal{D}: H_K^m\left(\mathbb{R}^n\right) \rightarrow$	$\mathcal{D} : H_K^m(\mathbb{R}^n) \rightarrow$	k can be extended continuously to a bounded operator $\mathcal{D} : H_K^m(\mathbb{R}^n) \rightarrow$
cardona-qft-methods_ F01374	78	■ 1.000	$K \subset \mathbb{R}^n$	$K \subset \mathbb{R}^n$	Theorem 2.7. (Rellich's lemma) <i>If $K \subset \mathbb{R}^n$ is a compact set, then the</i>
cardona-qft-methods_ F01375	78	■ 1.000	$H_K^m\left(\mathbb{R}^n\right) \hookrightarrow H^k\left(\mathbb{R}^n\right)$	$H_K^m(\mathbb{R}^n) \hookrightarrow H^k(\mathbb{R}^n)$	<i>embedding $H_K^m(\mathbb{R}^n) \hookrightarrow H^k(\mathbb{R}^n)$ is a compact operator for any $m > k$.</i>
cardona-qft-methods_ F01376	78	■ 1.000	$m>k$	$m > k$	<i>embedding $H_K^m(\mathbb{R}^n) \hookrightarrow H^k(\mathbb{R}^n)$ is a compact operator for any $m > k$.</i>
cardona-qft-methods_ F01377	78	■ 1.000	$k<m$	$k < m$	ential operator $\mathcal{D}$ of order $k < m$ to $H_K^m(\mathbb{R}^n)$ defines a compact operator
cardona-qft-methods_ F01378	78	■ 1.000	$H_K^m\left(\mathbb{R}^n\right)$	$H_K^m(\mathbb{R}^n)$	ential operator $\mathcal{D}$ of order $k < m$ to $H_K^m(\mathbb{R}^n)$ defines a compact operator

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cardona-qft-methods_ F01379	78	■ 1.000	$H_{\mathbb{K}}^m \left(\mathbb{R}^n \right) \rightarrow L^2 \left(\mathbb{R}^n \right)$	$H_K^m(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$	$H_K^m(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$.
cardona-qft-methods_ F01380	78	■ 1.000	$C^\infty \left(\mathbb{R}^n, \mathbb{R}^N \right)$	$C^\infty(\mathbb{R}^n, \mathbb{R}^N)$	$\mathbb{R}^n$. Let $C^\infty(\mathbb{R}^n, \mathbb{R}^N)$ denote the set of smooth vector valued functions
cardona-qft-methods_ F01381	78	■ 1.000	$\mathbb{R}^N$	$\mathbb{R}^N$	Using the scalar product of $\mathbb{R}^N$ we can generalize the constructions of
cardona-qft-methods_ F01382	78	■ 1.000	$L^2 \left(\mathbb{R}^n, \mathbb{R}^N \right), H^k \left(\mathbb{R}^n, \mathbb{R}^N \right)$	$L^2(\mathbb{R}^n, \mathbb{R}^N), H^k(\mathbb{R}^n, \mathbb{R}^N)$	the previous section to define the spaces $L^2(\mathbb{R}^n, \mathbb{R}^N), H^k(\mathbb{R}^n, \mathbb{R}^N)$ and
cardona-qft-methods_ F01383	78	■ 1.000	$H_K^k \left(\mathbb{R}^n, \mathbb{R}^N \right)$	$H_K^k(\mathbb{R}^n, \mathbb{R}^N)$	$H_K^k(\mathbb{R}^n, \mathbb{R}^N)$, etc.
cardona-qft-methods_ F01384	79	■ 0.998	$\mathcal{D}: C^\infty \left(\mathbb{R}^n, \mathbb{R}^{N_1} \right) \rightarrow C^\infty \left(\mathbb{R}^n, \mathbb{R}^{N_2} \right)$	$\mathcal{D}: C^\infty(\mathbb{R}^n, \mathbb{R}^{N_1}) \rightarrow C^\infty(\mathbb{R}^n, \mathbb{R}^{N_2})$	An operator $\mathcal{D}: C^\infty(\mathbb{R}^n, \mathbb{R}^{N_1}) \rightarrow C^\infty(\mathbb{R}^n, \mathbb{R}^{N_2})$ is a (matrix-valued)
cardona-qft-methods_ F01385	79	■ 1.000	$\pi: E \rightarrow M$	$\pi: E \rightarrow M$	a smooth manifold which is equipped with a <i>projection map</i> $\pi: E \rightarrow M$.
cardona-qft-methods_ F01386	79	■ 1.000	$\pi^{-1}(x)$	$\pi^{-1}(x)$	For each $x \in M$ the preimage $\pi^{-1}(x)$ is called the <i>fiber of E over x</i> and is
cardona-qft-methods_ F01387	79	■ 1.000	$E = M \times \mathcal{C}^N$	$E = M \times \mathcal{C}^N$	Example 2.11. a. The <i>trivial bundle</i> $E = M \times \mathcal{C}^N$. Here $\pi: E \rightarrow M$ is
cardona-qft-methods_ F01388	79	■ 1.000	$\mathcal{C}^N$	$\mathcal{C}^N$	isomorphic with $\mathcal{C}^N$.
cardona-qft-methods_ F01389	79	■ 1.000	$s: M \rightarrow E$	$s: M \rightarrow E$	A <i>smooth section</i> of a vector bundle E is a smooth map $s: M \rightarrow E$
cardona-qft-methods_ F01390	79	■ 1.000	$s(x) \in E_x$	$s(x) \in E_x$	In other words, for each $x \in M$ we have an element $s(x) \in E_x$. Notice that
cardona-qft-methods_ F01391	79	■ 1.000	$M \rightarrow \mathcal{C}^N$	$M \rightarrow \mathcal{C}^N$	same as defining a vector valued function $M \rightarrow \mathcal{C}^N$. Thus smooth sections
cardona-qft-methods_ F01392	79	■ 1.000	$C^\infty(M, E)$	$C^\infty(M, E)$	We denote the set of smooth sections of E by $C^\infty(M, E)$.
cardona-qft-methods_ F01393	80	■ 1.000	$f \in C^\infty(M, E)$	$f \in C^\infty(M, E)$	is <i>local</i> if for every section $f \in C^\infty(M, E)$, we have $\text{supp}(\mathcal{D}f) \subset \text{supp}(f)$.
cardona-qft-methods_ F01394	80	■ 1.000	$\text{supp}(\mathcal{D}f) \subset \text{supp}(f)$	$\text{supp}(\mathcal{D}f) \subset \text{supp}(f)$	is <i>local</i> if for every section $f \in C^\infty(M, E)$, we have $\text{supp}(\mathcal{D}f) \subset \text{supp}(f)$.
cardona-qft-methods_ F01395	80	■ 1.000	$\{U_i\}_{i=1}^m$	$\{U_i\}_{i=1}^m$	In this case, for any open cover $\{U_i\}_{i=1}^m$ of M the operator $\mathcal{D}$ is completely
cardona-qft-methods_ F01396	80	■ 1.000	U_i	U_i	determined by its restriction to functions with supports in one U_i . If the
cardona-qft-methods_ F01397	80	■ 0.881	$C^\infty(M, E) \rightarrow C^\infty(M, F)$	$C^\infty(M, E) \rightarrow C^\infty(M, F)$	Definition 2.13. A local linear operator $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ is
cardona-qft-methods_ F01398	80	■ 1.000	$i=1, \dots, m$	$i = 1, \dots, m$	of M and coordinate systems on U_i such that for every $i = 1, \dots, m$, the
cardona-qft-methods_ F01399	80	■ 1.000	$\mathcal{D}_i$	$\mathcal{D}_i$	operator $\mathcal{D}_i$ has the form (1).
cardona-qft-methods_ F01400	80	■ 1.000	$H^k(M, E)$	$H^k(M, E)$	One can define the spaces $H^k(M, E)$ of Sobolev sections of E in a way
cardona-qft-methods_ F01401	80	■ 1.000	$\langle \cdot, \cdot \rangle_k$	$\langle \cdot, \cdot \rangle_k$	a scalar product $\langle \cdot, \cdot \rangle_k$ on $C_k^\infty(M, E)$ as a combination of scalar products
cardona-qft-methods_ F01402	80	■ 1.000	$C_k^\infty(M, E)$	$C_k^\infty(M, E)$	a scalar product $\langle \cdot, \cdot \rangle_k$ on $C_k^\infty(M, E)$ as a combination of scalar products
cardona-qft-methods_ F01403	80	■ 1.000	$C_c^\infty(M, E)$	$C_c^\infty(M, E)$	is the completion of $C_c^\infty(M, E)$ with respect to the norm defined by this
cardona-qft-methods_ F01404	80	■ 1.000	$M = \mathbb{R}^n$	$M = \mathbb{R}^n$	scalar product. We refer to ²⁷ for details. As in the case $M = \mathbb{R}^n$ we have

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cardona-qft-methods_ F01485	80	■ 1.000	$\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$	$\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$	Theorem 2.15. <i>Let $\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$ be a differential operator</i>
cardona-qft-methods_ F01486	81	■ 1.000	$H_K^m(M, E) \rightarrow L^2(M, F)$	$H_K^m(M, E) \rightarrow L^2(M, F)$	$H_K^m(M, E) \rightarrow L^2(M, F)$ is compact.
cardona-qft-methods_ F01497	81	■ 1.000	$x_0 \in M$	$x_0 \in M$	of order k . Fix $x_0 \in M$, a cotangent vector $\xi \in T_{x_0}^*M$ and a vector $e \in$
cardona-qft-methods_ F01488	81	■ 1.000	$\xi \in T_{x_0}^*M$	$\xi \in T_{x_0}^*M$	of order k . Fix $x_0 \in M$, a cotangent vector $\xi \in T_{x_0}^*M$ and a vector $e \in$
cardona-qft-methods_ F01489	81	■ 1.000	$e \in$	$e \in$	of order k . Fix $x_0 \in M$, a cotangent vector $\xi \in T_{x_0}^*M$ and a vector $e \in$
cardona-qft-methods_ F01410	81	■ 1.000	E_{x_0}	E_{x_0}	E_{x_0} . Let $f \in C^{\infty}(M)$ be a smooth function, such that $df_{x_0} = \xi$ and let
cardona-qft-methods_ F01411	81	■ 1.000	$df_{x_0} = \xi$	$df_{x_0} = \xi$	E_{x_0} . Let $f \in C^{\infty}(M)$ be a smooth function, such that $df_{x_0} = \xi$ and let
cardona-qft-methods_ F01412	81	■ 1.000	$s \in C^{\infty}(M, E)$	$s \in C^{\infty}(M, E)$	$s \in C^{\infty}(M, E)$ be a section, such that $s(x_0) = e$. Set
cardona-qft-methods_ F01413	81	■ 1.000	$s(x_0) = e$	$s(x_0) = e$	$s \in C^{\infty}(M, E)$ be a section, such that $s(x_0) = e$. Set
cardona-qft-methods_ F01414	81	■ 1.000	$\sigma_L(\mathcal{D})(x_0, \xi) e$	$\sigma_L(\mathcal{D})(x_0, \xi) e$	Show that (5) is independent of the choice of f and s and that $\sigma_L(\mathcal{D})(x_0, \xi) e$
cardona-qft-methods_ F01415	81	■ 1.000	e	e	is linear in e . Thus we can view $\sigma_L(\mathcal{D})(x_0, \xi)$ as a linear map $E_{x_0} \rightarrow F_{x_0}$.
cardona-qft-methods_ F01416	81	■ 1.000	$\sigma_L(\mathcal{D})(x_0, \xi)$	$\sigma_L(\mathcal{D})(x_0, \xi)$	is linear in e . Thus we can view $\sigma_L(\mathcal{D})(x_0, \xi)$ as a linear map $E_{x_0} \rightarrow F_{x_0}$.
cardona-qft-methods_ F01417	81	■ 1.000	$E_{x_0} \rightarrow F_{x_0}$	$E_{x_0} \rightarrow F_{x_0}$	is linear in e . Thus we can view $\sigma_L(\mathcal{D})(x_0, \xi)$ as a linear map $E_{x_0} \rightarrow F_{x_0}$.
cardona-qft-methods_ F01418	81	■ 1.000	$T_{x_0}M$	$T_{x_0}M$	local coordinates we can identify $T_{x_0}M$ with $\mathbb{R}^n$. Then
cardona-qft-methods_ F01419	81	■ 0.999	$\sigma_L(\mathcal{D})(x_0, \xi) (x_0 \in M, \xi \in T_{x_0}^*M)$	$\sigma_L(\mathcal{D})(x_0, \xi) (x_0 \in M, \xi \in T_{x_0}^*M)$	ator of order k . The <i>leading symbol</i> $\sigma_L(\mathcal{D})(x_0, \xi) (x_0 \in M, \xi \in T_{x_0}^*M)$ of $\mathcal{D}$
cardona-qft-methods_ F01420	81	■ 1.000	$\operatorname{Hom}(E_x, F_x)$	$\operatorname{Hom}(E_x, F_x)$	is the element of $\operatorname{Hom}(E_x, F_x)$ defined by (5).
cardona-qft-methods_ F01421	82	■ 1.000	$\pi: T^*M \rightarrow M$	$\pi: T^*M \rightarrow M$	Let $\pi: T^*M \rightarrow M$ denote the projection. For a vector bundle E we denote
cardona-qft-methods_ F01422	82	■ 1.000	π^*E	π^*E	by π^*E the <i>pullback</i> bundle over T^*M . This is a vector bundle over T^*M
cardona-qft-methods_ F01423	82	■ 1.000	$(x, \xi) \in T^*M$	$(x, \xi) \in T^*M$	whose fiber over $(x, \xi) \in T^*M$ is the fiber E_x of E , see [27, §5.1] for a
cardona-qft-methods_ F01424	82	■ 1.000	$\sigma_L(\mathcal{D})(x, \xi)$	$\sigma_L(\mathcal{D})(x, \xi)$	$(x, \xi) \in T^*M$, the leading symbol $\sigma_L(\mathcal{D})(x, \xi)$ is a linear map
cardona-qft-methods_ F01425	82	■ 1.000	$\sigma_L(\mathcal{D})$	$\sigma_L(\mathcal{D})$	Hence, we can view $\sigma_L(\mathcal{D})$ as a section of the vector bundle $\operatorname{Hom}(\pi^*E, \pi^*F)$.
cardona-qft-methods_ F01426	82	■ 1.000	$\operatorname{Hom}(\pi^*E, \pi^*F)$	$\operatorname{Hom}(\pi^*E, \pi^*F)$	Hence, we can view $\sigma_L(\mathcal{D})$ as a section of the vector bundle $\operatorname{Hom}(\pi^*E, \pi^*F)$.
cardona-qft-methods_ F01427	82	■ 1.000	$\xi \neq 0$	$\xi \neq 0$	called <i>elliptic</i> if $\sigma_L(\mathcal{D})(x, \xi)$ is invertible for all $\xi \neq 0$.
cardona-qft-methods_ F01428	82	■ 0.999	$\sigma_L(\Delta)$	$\sigma_L(\Delta)$	on $\mathbb{R}^n$. Its leading symbol $\sigma_L(\Delta)$, given by the formula
cardona-qft-methods_ F01429	82	■ 0.999	$\xi = (\xi_1, \dots, \xi_n) \neq 0$	$\xi = (\xi_1, \dots, \xi_n) \neq 0$	is invertible for $\xi = (\xi_1, \dots, \xi_n) \neq 0$. Hence, Δ is elliptic.
cardona-qft-methods_ F01430	82	■ 0.999	$\mathcal{D}f = u$	$\mathcal{D}f = u$	a. (<i>Elliptic regularity</i>) If $\mathcal{D}f = u$ and $u \in C^{\infty}(M, F)$ then $f \in C^{\infty}(M, E)$.
cardona-qft-methods_ F01431	82	■ 0.999	$u \in C^{\infty}(M, F)$	$u \in C^{\infty}(M, F)$	a. (<i>Elliptic regularity</i>) If $\mathcal{D}f = u$ and $u \in C^{\infty}(M, F)$ then $f \in C^{\infty}(M, E)$.

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cardona-qft-methods_ F01432	82	■ 0.999	$\operatorname{Ind}(\mathcal{D})$	$\operatorname{Ind}(\mathcal{D})$	Part b. of this theorem implies that the index $\operatorname{Ind}(\mathcal{D})$ of an elliptic
cardona-qft-methods_ F01433	82	■ 1.000	$\sigma_L(\mathcal{D}_1) = \sigma_L(\mathcal{D}_2)$	$\sigma_L(\mathcal{D}_1) = \sigma_L(\mathcal{D}_2)$	leading symbol of $\mathcal{D}$, i.e. if $\sigma_L(\mathcal{D}_1) = \sigma_L(\mathcal{D}_2)$ then $\operatorname{Ind}(\mathcal{D}_1) = \operatorname{Ind}(\mathcal{D}_2)$.
cardona-qft-methods_ F01434	82	■ 1.000	$\operatorname{Ind}(\mathcal{D}_1) = \operatorname{Ind}(\mathcal{D}_2)$	$\operatorname{Ind}(\mathcal{D}_1) = \operatorname{Ind}(\mathcal{D}_2)$	leading symbol of $\mathcal{D}$, i.e. if $\sigma_L(\mathcal{D}_1) = \sigma_L(\mathcal{D}_2)$ then $\operatorname{Ind}(\mathcal{D}_1) = \operatorname{Ind}(\mathcal{D}_2)$.
cardona-qft-methods_ F01435	83	■ 1.000	$M = S^1$	$M = S^1$	Exercise 2.26. Let $M = S^1$ be the circle and consider the operator
cardona-qft-methods_ F01436	83	■ 1.000	$\operatorname{SEll}(E, F)$	$\operatorname{SEll}(E, F)$	space $\operatorname{SEll}(E, F)$ of smooth sections of $\sigma(x, \xi) \in \operatorname{Hom}(\pi^*E, \pi^*F)$, which are
cardona-qft-methods_ F01437	83	■ 1.000	$\sigma(x, \xi) \in \operatorname{Hom}(\pi^*E, \pi^*F)$	$\sigma(x, \xi) \in \operatorname{Hom}(\pi^*E, \pi^*F)$	space $\operatorname{SEll}(E, F)$ of smooth sections of $\sigma(x, \xi) \in \operatorname{Hom}(\pi^*E, \pi^*F)$, which are
cardona-qft-methods_ F01438	83	■ 1.000	$\{ \}^{14,15}$	14,15	ago, Israel Gel'fand ^{14,15} formulated a problem: how to compute the index of
cardona-qft-methods_ F01439	83	■ 1.000	$\{ \}^{6,7}$	6,7	brilliantly by Atiyah and Singer ^{6,7} (see also, for example, ^{9,23}). They did
cardona-qft-methods_ F01440	83	■ 1.000	$\{ \}^{9,23}$	9,23	brilliantly by Atiyah and Singer ^{6,7} (see also, for example, ^{9,23}). They did
cardona-qft-methods_ F01441	83	■ 0.942	$\operatorname{t-Ind}(\mathcal{D})$	$\operatorname{t-Ind}(\mathcal{D})$	associated a number – <i>the topological index</i> $\operatorname{t-Ind}(\mathcal{D})$ – to each element of
cardona-qft-methods_ F01442	84	■ 1.000	$\operatorname{Ind}(\mathcal{D}) = \operatorname{t-Ind}(\sigma_L(\mathcal{D}))$	$\operatorname{Ind}(\mathcal{D}) = \operatorname{t-Ind}(\sigma_L(\mathcal{D}))$	Theorem 3.3. (Atiyah-Singer) $\operatorname{Ind}(\mathcal{D}) = \operatorname{t-Ind}(\sigma_L(\mathcal{D}))$.
cardona-qft-methods_ F01443	84	■ 1.000	$\mathcal{D}f = 0$	$\mathcal{D}f = 0$	$\mathcal{D}f = 0$ and $\mathcal{D}^*u = 0$. The index theorem allows one to obtain at least
cardona-qft-methods_ F01444	84	■ 1.000	$\mathcal{D}^*u = 0$	$\mathcal{D}^*u = 0$	$\mathcal{D}f = 0$ and $\mathcal{D}^*u = 0$. The index theorem allows one to obtain at least
cardona-qft-methods_ F01445	84	■ 1.000	$\sigma(x) : E_x \rightarrow F_x$	$\sigma(x) : E_x \rightarrow F_x$	such that the induced map of fibers $\sigma(x) : E_x \rightarrow F_x$ is invertible for all
cardona-qft-methods_ F01446	84	■ 1.000	$K \subset X$	$K \subset X$	x outside of a compact set $K \subset X$. Two such maps $\sigma_1 : E_1 \rightarrow F_1$ and
cardona-qft-methods_ F01447	84	■ 1.000	$\sigma_1 : E_1 \rightarrow F_1$	$\sigma_1 : E_1 \rightarrow F_1$	x outside of a compact set $K \subset X$. Two such maps $\sigma_1 : E_1 \rightarrow F_1$ and
cardona-qft-methods_ F01448	84	■ 0.999	$\sigma_2 : E_2 \rightarrow F_2$	$\sigma_2 : E_2 \rightarrow F_2$	$\sigma_2 : E_2 \rightarrow F_2$ are said to be equivalent if there exist integers $k_1, k_2 \geq 0$ such
cardona-qft-methods_ F01449	84	■ 0.999	$k_1, k_2 \geq 0$	$k_1, k_2 \geq 0$	$\sigma_2 : E_2 \rightarrow F_2$ are said to be equivalent if there exist integers $k_1, k_2 \geq 0$ such
cardona-qft-methods_ F01450	85	■ 1.000	$\sigma_1 \sim \sigma_2$	$\sigma_1 \sim \sigma_2$	this case we write $\sigma_1 \sim \sigma_2$.
cardona-qft-methods_ F01451	85	■ 1.000	$K(X)$	$K(X)$	Definition 3.6. Let X be a topological space. The <i>K-theory group</i> $K(X)$
cardona-qft-methods_ F01452	85	■ 1.000	$\sigma_1, \sigma_2 : E \rightarrow F$	$\sigma_1, \sigma_2 : E \rightarrow F$	Remark 3.7. If X is compact then any two maps $\sigma_1, \sigma_2 : E \rightarrow F$ between
cardona-qft-methods_ F01453	85	■ 1.000	(E, F)	(E, F)	vector bundles (E, F) subject to an appropriate equivalence relation. We
cardona-qft-methods_ F01454	85	■ 1.000	$E - F$	$E - F$	and we, usually denote this pair by $E - F$. Thus, as mentioned above,
cardona-qft-methods_ F01455	85	■ 1.000	$X = \{pt\}$	$X = \{pt\}$	Example 3.8. If $X = \{pt\}$, then a vector bundle over M is just a vector
cardona-qft-methods_ F01456	85	■ 1.000	$E_x \rightarrow F_x$	$E_x \rightarrow F_x$	$E_x \rightarrow F_x$. In other words, $\sigma(\mathcal{D})$ defines a bundle map
cardona-qft-methods_ F01457	85	■ 1.000	$\sigma(\mathcal{D})$	$\sigma(\mathcal{D})$	$E_x \rightarrow F_x$. In other words, $\sigma(\mathcal{D})$ defines a bundle map
cardona-qft-methods_ F01458	86	■ 1.000	$X = T^*M$	$X = T^*M$	map (3) is invertible outside of a compact subset of $X = T^*M$. Hence, (3)

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cardona-qft-methods_ F01459	86	■ 1.000	$K\left(T^*M\right)$	$K\left(T^*M\right)$	defines an element of $K(T^*M)$.
cardona-qft-methods_ F01460	86	■ 1.000	$j: Y \hookrightarrow X$	$j: Y \hookrightarrow X$	given an inclusion $j: Y \hookrightarrow X$ one can define a ring homomorphism
cardona-qft-methods_ F01461	86	■ 1.000	$i:\{p\,t\} \hookrightarrow \mathcal{C}^N$	$i:\{pt\} \hookrightarrow \mathcal{C}^N$	Theorem 3.11. (Bott periodicity) <i>Let $i:\{pt\} \hookrightarrow \mathcal{C}^N$ be an embedding</i>
cardona-qft-methods_ F01462	86	■ 1.000	$M \hookrightarrow \mathbb{R}^N$	$M \hookrightarrow \mathbb{R}^N$	Consider an embedding $M \hookrightarrow \mathbb{R}^N$ (such an embedding always exists for
cardona-qft-methods_ F01463	87	■ 0.971	$i_!$	$i_!$	By Theorem 3.11 we can invert $i_!$ and define the <i>topological index</i> as the
cardona-qft-methods_ F01464	87	■ 1.000	$j: M \hookrightarrow \mathbb{R}^N$	$j: M \hookrightarrow \mathbb{R}^N$	of the number N and the embedding $j: M \hookrightarrow \mathbb{R}^N$.
cardona-qft-methods_ F01465	87	■ 1.000	$g \in G$	$g \in G$	Suppose a compact group G acts on M . That means that to each $g \in G$ is
cardona-qft-methods_ F01466	87	■ 1.000	$\phi(g): M \rightarrow M$	$\phi(g): M \rightarrow M$	assigned a diffeomorphism $\phi(g): M \rightarrow M$ such that $\phi(g_1) \circ \phi(g_2) = \phi(g_1 g_2)$.
cardona-qft-methods_ F01467	87	■ 1.000	$\phi\left(g_1\right) \circ \phi\left(g_2\right)=\phi\left(g_1 g_2\right)$	$\phi\left(g_1\right) \circ \phi\left(g_2\right)=\phi\left(g_1 g_2\right)$	assigned a diffeomorphism $\phi(g): M \rightarrow M$ such that $\phi(g_1) \circ \phi(g_2) = \phi(g_1 g_2)$.
cardona-qft-methods_ F01468	87	■ 1.000	$g \cdot x$	$g \cdot x$	When it does not lead to confusion, we often write $g \cdot x$ and $g \cdot e$ for
cardona-qft-methods_ F01469	87	■ 1.000	$g \cdot e$	$g \cdot e$	When it does not lead to confusion, we often write $g \cdot x$ and $g \cdot e$ for
cardona-qft-methods_ F01470	87	■ 1.000	$\phi(g)(x)$	$\phi(g)(x)$	$\phi(g)(x)$ and $\psi(g)(e)$ respectively.
cardona-qft-methods_ F01471	87	■ 1.000	$\psi(g)(e)$	$\psi(g)(e)$	$\phi(g)(x)$ and $\psi(g)(e)$ respectively.
cardona-qft-methods_ F01472	87	■ 1.000	$l_g: L^2(M, E) \rightarrow L^2(M, E)$	$l_g: L^2(M, E) \rightarrow L^2(M, E)$	Similarly, we define the action $l_g: L^2(M, E) \rightarrow L^2(M, E)$ on the space of
cardona-qft-methods_ F01473	87	■ 1.000	$L^2(G)$	$L^2(G)$	square integrable sections of E . In this way $L^2(G)$ becomes a representation
cardona-qft-methods_ F01474	87	■ 1.000	$g \cdot \mathcal{D}=\mathcal{D} \cdot g$	$g \cdot \mathcal{D}=\mathcal{D} \cdot g$	is called <i>G-invariant</i> if $g \cdot \mathcal{D}=\mathcal{D} \cdot g$.
cardona-qft-methods_ F01475	88	■ 1.000	$K_G(X)$	$K_G(X)$	$K_G(X)$ as the set of the equivalence classes of G -equivariant maps between
cardona-qft-methods_ F01476	88	■ 1.000	$M \subset T^* M$	$M \subset T^* M$	which is invertible outside of the compact set $M \subset T^* M$. Thus $\sigma_L \in$
cardona-qft-methods_ F01477	88	■ 1.000	$\sigma_L \in$	$\sigma_L \in$	which is invertible outside of the compact set $M \subset T^* M$. Thus $\sigma_L \in$
cardona-qft-methods_ F01478	88	■ 0.612	$K_G\left(T^* M\right)$	$K_G\left(T^* M\right)$	$K_G(T^* M)$ and its topological index $\mathrm{t}\text{-Ind}\left(\sigma_L(\mathcal{D})\right)$ is defined. The index
cardona-qft-methods_ F01479	88	■ 0.612	$\mathrm{t}\text{-Ind}\left(\sigma_L(\mathcal{D})\right)$	$\mathrm{t}\text{-Ind}\left(\sigma_L(\mathcal{D})\right)$	$K_G(T^* M)$ and its topological index $\mathrm{t}\text{-Ind}\left(\sigma_L(\mathcal{D})\right)$ is defined. The index
cardona-qft-methods_ F01480	88	■ 1.000	$\mathrm{Ind}_G(\mathcal{D})=\mathrm{t}\text{-Ind}_G\left(\sigma_L(\mathcal{D})\right)$	$\mathrm{Ind}_G(\mathcal{D})=\mathrm{t}\text{-Ind}_G\left(\sigma_L(\mathcal{D})\right)$	Theorem 3.19. (Atiyah-Singer) $\mathrm{Ind}_G(\mathcal{D})=\mathrm{t}\text{-Ind}_G\left(\sigma_L(\mathcal{D})\right)$.
cardona-qft-methods_ F01481	89	■ 1.000	$\sigma: \pi^* E \rightarrow \pi^* F$	$\sigma: \pi^* E \rightarrow \pi^* F$	F be G -equivariant vector bundles over M and let $\sigma: \pi^* E \rightarrow \pi^* F$ be an
cardona-qft-methods_ F01482	89	■ 0.920	$C^{\infty}(N) \rightarrow C^{\infty}(N)$	$C^{\infty}(N) \rightarrow C^{\infty}(N)$	Suppose N is a closed manifold and let $\mathcal{D}: C^{\infty}(N) \rightarrow C^{\infty}(N)$ be an elliptic
cardona-qft-methods_ F01483	89	■ 1.000	$\left(x_1,\ldots, x_n\right)$	$\left(x_1,\ldots, x_n\right)$	defined as follows: Suppose that in a coordinate chart $\left(x_1,\ldots, x_n\right)$ on N
cardona-qft-methods_ F01484	89	■ 0.558	$\left(y_1,\cdots, y_r\right)$	$\left(y_1,\cdots, y_r\right)$	Let $\left(y_1,\cdots, y_r\right)$ be a coordinate system on G . Then, $\left(x_1,\ldots, x_n, y_1,\ldots, y_r\right)$
cardona-qft-methods_ F01485	89	■ 0.558	$\left(x_1,\ldots, x_n, y_1,\ldots, y_r\right)$	$\left(x_1,\ldots, x_n, y_1,\ldots, y_r\right)$	Let $\left(y_1,\cdots, y_r\right)$ be a coordinate system on G . Then, $\left(x_1,\ldots, x_n, y_1,\ldots, y_r\right)$

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cardona-qft-methods_ F01486	89	■ 1.000	$\tilde{\mathcal{D}}$	$\tilde{\mathcal{D}}$	is a coordinate system on M . By definition the operator $\tilde{\mathcal{D}}$ takes the form
cardona-qft-methods_ F01487	90	■ 1.000	$\operatorname{Ker}(\tilde{\mathcal{D}})$	$\operatorname{Ker}(\tilde{\mathcal{D}})$	Thus despite the fact that $\operatorname{Ker}(\tilde{\mathcal{D}})$ and $\operatorname{Coker}(\tilde{\mathcal{D}})$ are infinite dimensional
cardona-qft-methods_ F01488	90	■ 1.000	$\operatorname{Coker}(\tilde{\mathcal{D}})$	$\operatorname{Coker}(\tilde{\mathcal{D}})$	Thus despite the fact that $\operatorname{Ker}(\tilde{\mathcal{D}})$ and $\operatorname{Coker}(\tilde{\mathcal{D}})$ are infinite dimensional
cardona-qft-methods_ F01489	90	■ 1.000	$\operatorname{dim} V$	$\dim V$	tiplicity equal to $\dim V$.
cardona-qft-methods_ F01490	90	■ 1.000	$\operatorname{Ind}(\tilde{\mathcal{D}})$	$\operatorname{Ind}(\tilde{\mathcal{D}})$	infinite. Thus $\operatorname{Ind}(\tilde{\mathcal{D}})$ lies in the completed ring of characters $\widehat{R}(G)$, cf.
cardona-qft-methods_ F01491	90	■ 1.000	$\widehat{R}(G)$	$\widehat{R}(G)$	infinite. Thus $\operatorname{Ind}(\tilde{\mathcal{D}})$ lies in the completed ring of characters $\widehat{R}(G)$, cf.
cardona-qft-methods_ F01492	90	■ 1.000	$N=M / G$	$N = M/G$	quotient $N = M/G$. The purpose of this section is to define the analogue
cardona-qft-methods_ F01493	90	■ 1.000	$\xi \in T^*M$	$\xi \in T^*M$	We say that a cotangent vector $\xi \in T^*M$ is <i>perpendicular to the orbits</i>
cardona-qft-methods_ F01494	90	■ 0.994	$v \in T \mathcal{O}(x)$	$v \in T\mathcal{O}(x)$	<i>of G</i> if for $x \in M$ and any tangent vector $v \in T\mathcal{O}(x)$, we have $\xi(v) = 0$.
cardona-qft-methods_ F01495	90	■ 0.994	$\xi(v)=0$	$\xi(v) = 0$	<i>of G</i> if for $x \in M$ and any tangent vector $v \in T\mathcal{O}(x)$, we have $\xi(v) = 0$.
cardona-qft-methods_ F01496	90	■ 1.000	$T_{\mathcal{G}}^*M$	T_G^*M	Definition 4.4. The <i>transversal cotangent bundle</i> T_G^*M is
cardona-qft-methods_ F01497	90	■ 1.000	$T_{\mathcal{G}, x}^*M := T_{\mathcal{G}}^*M \cap T_x^*M$	$T_{G,x}^*M := T_G^*M \cap T_x^*M$	We set $T_{G,x}^*M := T_G^*M \cap T_x^*M$.
cardona-qft-methods_ F01498	91	■ 1.000	$0 \neq \xi \in T_{\mathcal{G}}^*M$	$0 \neq \xi \in T_G^*M$	for every non-zero $0 \neq \xi \in T_G^*M$.
cardona-qft-methods_ F01499	91	■ 0.999	$\operatorname{Ker}(\mathcal{D})$	$\operatorname{Ker}(\mathcal{D})$	<i>the spaces $\operatorname{Ker}(\mathcal{D})$ and $\operatorname{Coker}(\mathcal{D})$ can be decomposed into a direct sum of</i>
cardona-qft-methods_ F01500	91	■ 0.999	$\operatorname{Coker}(\mathcal{D})$	$\operatorname{Coker}(\mathcal{D})$	<i>the spaces $\operatorname{Ker}(\mathcal{D})$ and $\operatorname{Coker}(\mathcal{D})$ can be decomposed into a direct sum of</i>
cardona-qft-methods_ F01501	91	■ 1.000	$m_V^{\pm} \in$	$m_V^{\pm} \in$	Notice that the sums in (5) are, in general, infinite, but the numbers $m_V^{\pm} \in$
cardona-qft-methods_ F01502	91	■ 1.000	$\mathbb{Z}_{\geq 0}$	$\mathbb{Z}_{\geq 0}$	$\mathbb{Z}_{\geq 0}$ are finite.
cardona-qft-methods_ F01503	91	■ 1.000	$m_V^{\pm} \in \mathbb{Z}_{\geq 0}$	$m_V^{\pm} \in \mathbb{Z}_{\geq 0}$	where the numbers $m_V^{\pm} \in \mathbb{Z}_{\geq 0}$ are defined in (5) and $\widehat{R}(G)$ stands for the
cardona-qft-methods_ F01504	91	■ 1.000	$T_{\mathcal{G}, x}^*M$	$T_{G,x}^*M$	fibers $T_{G,x}^*M$ might depend on x . But it is a topological space, and one can
cardona-qft-methods_ F01505	92	■ 1.000	$K_{\mathcal{G}}(\operatorname{Ker}(\mathcal{D}))$	$K_G(T_G^*M)$	define the K-theory $K_G(T_G^*M)$ as the set of equivalence classes of pairs of
cardona-qft-methods_ F01506	92	■ 1.000	$\sigma \in K_{\mathcal{G}}(\operatorname{Ker}(\mathcal{D}))$	$\sigma \in K_G(T_G^*M)$	inition is more involved. In fact, for compact manifold M , for $\sigma \in K_G(T_G^*M)$
cardona-qft-methods_ F01507	92	■ 1.000	$\sigma_L(D)=\sigma$	$\sigma_L(D) = \sigma$	one just chooses a G -invariant operator $\mathcal{D}$ with $\sigma_L(D) = \sigma$ and sets
cardona-qft-methods_ F01508	93	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	Let V be a finite dimensional vector space over $\mathbb{R}$ endowed with a scalar
cardona-qft-methods_ F01509	93	■ 1.000	W	W	Definition 5.2. A <i>Clifford action</i> of V on a complex vector space W is a
cardona-qft-methods_ F01510	93	■ 1.000	$v \in V$	$v \in V$	such that for any $v \in V$ we have
cardona-qft-methods_ F01511	93	■ 0.994	$c: V \rightarrow \operatorname{End}(W)$	$c: V \rightarrow \operatorname{End}(W)$	Exercise 5.3. Show that a linear map $c: V \rightarrow \operatorname{End}(W)$ is a Clifford action
cardona-qft-methods_ F01512	93	■ 1.000	$\mathbb{R}^3$	$\mathbb{R}^3$	defines a Clifford action of $\mathbb{R}^3$ on $\mathcal{C}^2$.

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cardona-qft-methods_ F01513	93	■ 1.000	$\mathcal{C}^2$	$\mathcal{C}^2$	defines a Clifford action of $\mathbb{R}^3$ on $\mathcal{C}^2$.
cardona-qft-methods_ F01514	93	■ 1.000	$V^{\mathcal{C}} = V \otimes_{\mathbb{R}} \mathcal{C}$	$V^{\mathcal{C}} = V \otimes_{\mathbb{R}} \mathcal{C}$	and let $V^{\mathcal{C}} = V \otimes_{\mathbb{R}} \mathcal{C}$ be its complexification. Let
cardona-qft-methods_ F01515	93	■ 1.000	$V^{\mathcal{C}}$	$V^{\mathcal{C}}$	be the exterior algebra of $V^{\mathcal{C}}$. For $v \in V$ we denote $\varepsilon(v) : W \rightarrow W$ the
cardona-qft-methods_ F01516	93	■ 1.000	$\varepsilon(v) : W \rightarrow W$	$\varepsilon(v) : W \rightarrow W$	be the exterior algebra of $V^{\mathcal{C}}$. For $v \in V$ we denote $\varepsilon(v) : W \rightarrow W$ the
cardona-qft-methods_ F01517	93	■ 1.000	v	v	exterior multiplication by v
cardona-qft-methods_ F01518	93	■ 1.000	V^*	V^*	space V^* and denote by $\iota_v : W \rightarrow W$ the interior multiplication by v . Set
cardona-qft-methods_ F01519	93	■ 1.000	$\iota_v : W \rightarrow W$	$\iota_v : W \rightarrow W$	space V^* and denote by $\iota_v : W \rightarrow W$ the interior multiplication by v . Set
cardona-qft-methods_ F01520	93	■ 1.000	$c : v \mapsto c(v)$	$c : v \mapsto c(v)$	Show that the map $c : v \mapsto c(v)$ defines a Clifford action of V on W .
cardona-qft-methods_ F01521	94	■ 1.000	$E \rightarrow$	$E \rightarrow$	Definition 5.7. A <i>Clifford bundle</i> over M is a complex vector bundle $E \rightarrow$
cardona-qft-methods_ F01522	94	■ 0.999	$v \in T^*M$	$v \in T^*M$	such that for any $v \in T^*M$ we have $c(v)^2 = - v ^2 \text{Id}$.
cardona-qft-methods_ F01523	94	■ 0.999	$c(v)^2 = - v ^2 \text{Id}$	$c(v)^2 = - v ^2 \text{Id}$	such that for any $v \in T^*M$ we have $c(v)^2 = - v ^2 \text{Id}$.
cardona-qft-methods_ F01524	94	■ 0.999	$M = \mathbb{R}^3$	$M = \mathbb{R}^3$	Example 5.8. Set $M = \mathbb{R}^3$ and $E = \mathbb{R}^3 \times \mathcal{C}^2$. Then the projection on
cardona-qft-methods_ F01525	94	■ 0.999	$E = \mathbb{R}^3 \times \mathcal{C}^2$	$E = \mathbb{R}^3 \times \mathcal{C}^2$	Example 5.8. Set $M = \mathbb{R}^3$ and $E = \mathbb{R}^3 \times \mathcal{C}^2$. Then the projection on
cardona-qft-methods_ F01526	94	■ 1.000	$E = \Lambda^*(T^*M \otimes \mathcal{C})$	$E = \Lambda^*(T^*M \otimes \mathcal{C})$	Example 5.9. Let M be a Riemannian manifold and let $E = \Lambda^*(T^*M \otimes \mathcal{C})$.
cardona-qft-methods_ F01527	94	■ 1.000	$\Omega^*(M)$	$\Omega^*(M)$	Note that the space of sections of E is just the space $\Omega^*(M)$ of complex
cardona-qft-methods_ F01528	94	■ 1.000	$\omega \in \Omega^*(M)$	$\omega \in \Omega^*(M)$	structure on E , such that for $v \in T^*M$ and $\omega \in \Omega^*(M)$ we have
cardona-qft-methods_ F01529	94	■ 0.993	$\nabla : C^\infty(M, E) \rightarrow \Omega^1(M, E)$	$\nabla : C^\infty(M, E) \rightarrow \Omega^1(M, E)$	Definition 5.11. A connection $\nabla : C^\infty(M, E) \rightarrow \Omega^1(M, E)$ on a Clifford
cardona-qft-methods_ F01530	94	■ 1.000	$u \in TM, v \in T^*M, s \in C^\infty(M, E)$	$u \in TM, v \in T^*M, s \in C^\infty(M, E)$	for all $u \in TM, v \in T^*M, s \in C^\infty(M, E)$.
cardona-qft-methods_ F01531	95	■ 0.949	$e_1, \dots, e_n$	$e_1, \dots, e_n$	where $e_1, \dots, e_n$ is an orthonormal basis of TM . ^c
cardona-qft-methods_ F01532	95	■ 0.949	TM^c	TM^c	where $e_1, \dots, e_n$ is an orthonormal basis of TM . ^c
cardona-qft-methods_ F01533	95	■ 1.000	$d : \Omega^*(M, E) \rightarrow$	$d : \Omega^*(M, E) \rightarrow$	Example 5.18. Let M be a Riemannian manifold and let $d : \Omega^*(M, E) \rightarrow$
cardona-qft-methods_ F01534	95	■ 1.000	$\Omega^{*+1}(M, E)$	$\Omega^{*+1}(M, E)$	$\Omega^{*+1}(M, E)$ denote the de Rham differential. The Riemannian metric in-
cardona-qft-methods_ F01535	95	■ 0.997	$\Omega^*(M, E)$	$\Omega^*(M, E)$	duces a scalar product $\langle \cdot, \cdot \rangle$ on the space $\Omega^*(M, E)$ of differential forms.
cardona-qft-methods_ F01536	95	■ 1.000	$\{ \}^{\mathfrak{c}}$	^c	^c As above, we use the Riemannian metric to identify TM and T^*M .
cardona-qft-methods_ F01537	97	■ 1.000	$C^\infty(M, E^+)$	$C^\infty(M, E^+)$	We denote the restriction of $\mathcal{D}$ to $C^\infty(M, E^+)$ (respectively,
cardona-qft-methods_ F01538	97	■ 0.950	$C^\infty(M, E^-)$	$C^\infty(M, E^-)$	$C^\infty(M, E^-)$) by $\mathcal{D}^+$ (respectively, $\mathcal{D}^-$). It follows from the exercise above
cardona-qft-methods_ F01539	97	■ 0.950	$\mathcal{D}^+$	$\mathcal{D}^+$	$C^\infty(M, E^-)$) by $\mathcal{D}^+$ (respectively, $\mathcal{D}^-$). It follows from the exercise above

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cardona-qft-methods_ F01540	97	■ 0.950	$\mathcal{D}^{\{-}}$	$\mathcal{D}^-$	$C^\infty(M, E^-)$ by $\mathcal{D}^+$ (respectively, $\mathcal{D}^-$). It follows from the exercise above
cardona-qft-methods_ F01541	97	■ 1.000	$\mathcal{D} = \mathcal{D}^+ \oplus \mathcal{D}^{\{-}}$	$\mathcal{D} = \mathcal{D}^+ \oplus \mathcal{D}^-$	To save space we often write (13) as $\mathcal{D} = \mathcal{D}^+ \oplus \mathcal{D}^-$. When a Dirac operator
cardona-qft-methods_ F01542	97	■ 1.000	$\mathcal{D}^+ : C^\infty(\infty)\left(M, E^+\right) \rightarrow$ $\rightarrow$	$\mathcal{D}^+ : C^\infty(M, E^+) \rightarrow$	theory one usually considers the index of the operator $\mathcal{D}^+ : C^\infty(M, E^+) \rightarrow$
cardona-qft-methods_ F01543	97	■ 0.827	$C^\infty(\infty)\left(M, E^{\{-}}\right)$	$C^\infty(M, E^-)$	$C^\infty(M, E^-)$.
cardona-qft-methods_ F01544	97	■ 1.000	$E = \Lambda^*(T^*M \otimes$	$E = \Lambda^*(T^*M \otimes$	Example 5.22. Let M be a Riemannian manifold and let $E = \Lambda^*(T^*M \otimes$
cardona-qft-methods_ F01545	97	■ 0.997	$\mathcal{C}$	$\mathcal{C}$	
cardona-qft-methods_ F01546	97	■ 0.998	$\operatorname{Ind}\left(\mathcal{D}^+\right)$	$\operatorname{Ind}(\mathcal{D}^+)$	index $\operatorname{Ind}(\mathcal{D}^+)$ of the de Rham-Dirac operator is equal to the Euler char-
cardona-qft-methods_ F01547	97	■ 1.000	$\operatorname{dim} M = n = 2l$	$\dim M = n = 2l$	Example 5.24. Suppose the dimension of M is even, $\dim M = n = 2l$.
cardona-qft-methods_ F01548	97	■ 1.000	$* : E \rightarrow E$	$* : E \rightarrow E$	Let $* : E \rightarrow E$ denote the Hodge star operator [27, §19.1]. Define the
cardona-qft-methods_ F01549	97	■ 1.000	$\Gamma : E \rightarrow E$	$\Gamma : E \rightarrow E$	<i>chirality operator</i> $\Gamma : E \rightarrow E$ by
cardona-qft-methods_ F01550	97	■ 0.998	$\Gamma^2 = 1$	$\Gamma^2 = 1$	One can show [27, §19.1] that $\Gamma^2 = 1$. It follows that the spectrum of Γ is
cardona-qft-methods_ F01551	97	■ 1.000	$\{1, -1\}$	$\{1, -1\}$	$\{1, -1\}$. Let E^+ (respectively, E^-) denote the eigenspace of Γ correspond-
cardona-qft-methods_ F01552	97	■ 1.000	$E^{\{+}}$	E^+	$\{1, -1\}$. Let E^+ (respectively, E^-) denote the eigenspace of Γ correspond-
cardona-qft-methods_ F01553	97	■ 1.000	$E^{\{-}}$	E^-	$\{1, -1\}$. Let E^+ (respectively, E^-) denote the eigenspace of Γ correspond-
cardona-qft-methods_ F01554	98	■ 0.925	$\mathcal{D}^{\pm}$	$\mathcal{D}^\pm$	the connection ∇ , and the grading on E . Then the operators $\mathcal{D}^\pm$ are G -
cardona-qft-methods_ F01555	99	■ 0.963	$\mathfrak{g} = \operatorname{Lie} S^1$	$\mathfrak{g} = \operatorname{Lie} S^1$	be the circle group. Then the Lie algebra $\mathfrak{g} = \operatorname{Lie} S^1$ is naturally isomorphic
cardona-qft-methods_ F01556	99	■ 1.000	$\mathbf{v} : M \rightarrow$	$\mathbf{v} : M \rightarrow$	to $\mathbb{R}$. Suppose G acts on a complete Riemannian manifold M . Let $\mathbf{v} : M \rightarrow$
cardona-qft-methods_ F01557	99	■ 0.998	$\mathbf{v}(x) \equiv 1$	$\mathbf{v}(x) \equiv 1$	$\mathfrak{g}$ be the constant map $\mathbf{v}(x) \equiv 1$. The corresponding vector field v is called
cardona-qft-methods_ F01558	99	■ 1.000	$G = S^1$	$G = S^1$	the <i>generating vector field</i> for the action of $G = S^1$, since it completely
cardona-qft-methods_ F01559	99	■ 1.000	$M = \mathcal{C}$	$M = \mathcal{C}$	Exercise 6.3. Let $G = S^1$ acts on $M = \mathcal{C}$ by multiplications: $(e^{it}, z) \mapsto e^{it}z$.
cardona-qft-methods_ F01560	99	■ 1.000	$\left(e^{it}, z\right) \mapsto e^{it} z$	$(e^{it}, z) \mapsto e^{it}z$	Exercise 6.3. Let $G = S^1$ acts on $M = \mathcal{C}$ by multiplications: $(e^{it}, z) \mapsto e^{it}z$.
cardona-qft-methods_ F01561	99	■ 1.000	$v(x)$	$v(x)$	Compute the generating vector field $v(x)$.
cardona-qft-methods_ F01562	99	■ 0.885	$(M, \mathbf{v})$	$(M, \mathbf{v})$	$(M, \mathbf{v})$, where M is a complete Riemannian manifold and $\mathbf{v}$ is a taming
cardona-qft-methods_ F01563	99	■ 0.885	$\mathbf{v}$	$\mathbf{v}$	$(M, \mathbf{v})$, where M is a complete Riemannian manifold and $\mathbf{v}$ is a taming
cardona-qft-methods_ F01564	99	■ 1.000	$\pi : T_G^*M \rightarrow M$	$\pi : T_G^*M \rightarrow M$	transversal cotangent bundle to M , cf. Definition 4.4, and let $\pi : T_G^*M \rightarrow M$
cardona-qft-methods_ F01565	99	■ 1.000	$\pi^*E = \pi^*E^+ \oplus \pi^*E^{\{-}}$	$\pi^*E = \pi^*E^+ \oplus \pi^*E^-$	denote the projection. Consider the pullback bundle $\pi^*E = \pi^*E^+ \oplus \pi^*E^-$
cardona-qft-methods_ F01566	99	■ 1.000	$x \in M, \xi \in T^*M$	$x \in M, \xi \in T^*M$	(3) as an element of T^*M . Then for $x \in M, \xi \in T^*M$, we can consider the

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cardona-qft-methods_ F01567	100	■ 1.000	$(x, \xi) \in T_G^* M$	$(x, \xi) \in T_G^* M$	The collection of these maps for all $(x, \xi) \in T_G^* M$ defines a bundle map
cardona-qft-methods_ F01568	100	■ 1.000	$c(\xi + v(x))$	$c(\xi + v(x))$	Exercise 6.7. Show that the map $c(\xi + v(x))$ is invertible for all $(x, \xi) \in$
cardona-qft-methods_ F01569	100	■ 1.000	$(x, \xi) \in$	$(x, \xi) \in$	Exercise 6.7. Show that the map $c(\xi + v(x))$ is invertible for all $(x, \xi) \in$
cardona-qft-methods_ F01570	100	■ 1.000	$v(x) \neq 0, \xi \neq 0$	$v(x) \neq 0, \xi \neq 0$	$T_G^* M$ such that $v(x) \neq 0, \xi \neq 0$. Conclude that the bundle map (4) defines
cardona-qft-methods_ F01571	100	■ 0.572	$(E, \mathbf{v})$	$(E, \mathbf{v})$	over a tamed G -manifold $(M, \mathbf{v})$. We refer to the pair $(E, \mathbf{v})$ as a <i>tamed</i>
cardona-qft-methods_ F01572	100	■ 1.000	$\mathrm{t}\text{-}\operatorname{Ind}_G(E, \mathbf{v})$	$\mathbf{t} - \operatorname{Ind}_G(E, \mathbf{v})$	<i>Clifford bundle over M</i> . The topological index $\mathbf{t}\text{-}\operatorname{Ind}_G(E, \mathbf{v})$ of a tamed
cardona-qft-methods_ F01573	100	■ 1.000	$\{ \}^{24,25}$	24,25	E. Paradan. ^{24,25} Our purpose is to construct a Fredholm operator, whose
cardona-qft-methods_ F01574	100	■ 1.000	$c(\xi)$	$c(\xi)$	$c(\xi)$. However, if M is not compact, the operator $\mathcal{D}$ does not have to be
cardona-qft-methods_ F01575	100	■ 0.991	$\boldsymbol{v}$	$\boldsymbol{v}$	6.10. A rescaling of $\boldsymbol{v}$
cardona-qft-methods_ F01576	100	■ 1.000	$f(x) v(x)$	$f(x)v(x)$	which we mean the product $f(x)v(x)$, where $f : M \rightarrow [0, \infty)$ is a smooth
cardona-qft-methods_ F01577	100	■ 1.000	$f: M \rightarrow [0, \infty)$	$f : M \rightarrow [0, \infty)$	which we mean the product $f(x)v(x)$, where $f : M \rightarrow [0, \infty)$ is a smooth
cardona-qft-methods_ F01578	100	■ 1.000	M, E	M, E	the geometry of M, E and ∇ . If a function f satisfies these conditions we
cardona-qft-methods_ F01579	100	■ 1.000	∇	∇	the geometry of M, E and ∇ . If a function f satisfies these conditions we
cardona-qft-methods_ F01580	100	■ 0.817	$M, E, \nabla, \mathbf{v}$	$M, E, \nabla, \mathbf{v}$	call it <i>admissible for the quadruple</i> $(M, E, \nabla, \mathbf{v})$. The index which we are
cardona-qft-methods_ F01581	101	■ 0.994	$(M, E, \nabla, \mathbf{v})$	$(M, E, \nabla, \mathbf{v})$	Let f be an admissible function for $(M, E, \nabla, \mathbf{v})$. Consider the <i>deformed</i>
cardona-qft-methods_ F01582	101	■ 1.000	$\mathcal{D}_{f v}^+$	$\mathcal{D}_{f v}^+$	and let $\mathcal{D}_{f v}^+$ denote the restriction of $\mathcal{D}_{f v}$ to $C^\infty(M, E^+)$. This operator is
cardona-qft-methods_ F01583	101	■ 1.000	$\mathcal{D}_{f v}$	$\mathcal{D}_{f v}$	and let $\mathcal{D}_{f v}^+$ denote the restriction of $\mathcal{D}_{f v}$ to $C^\infty(M, E^+)$. This operator is
cardona-qft-methods_ F01584	101	■ 1.000	$\operatorname{Ker}(\mathcal{D}_{f v}^+)$	$\operatorname{Ker}(\mathcal{D}_{f v}^+)$	G acts on $\operatorname{Ker}(\mathcal{D}_{f v}^+)$ and $\operatorname{Coker}(\mathcal{D}_{f v}^+)$.
cardona-qft-methods_ F01585	101	■ 1.000	$\operatorname{Coker}(\mathcal{D}_{f v}^+)$	$\operatorname{Coker}(\mathcal{D}_{f v}^+)$	G acts on $\operatorname{Ker}(\mathcal{D}_{f v}^+)$ and $\operatorname{Coker}(\mathcal{D}_{f v}^+)$.
cardona-qft-methods_ F01586	101	■ 1.000	$D_{f v}^+$	$D_{f v}^+$	<i>the cokernel of the deformed Dirac operator $D_{f v}^+$ decompose into an infinite</i>
cardona-qft-methods_ F01587	101	■ 1.000	$m_V^\pm$	$m_V^\pm$	<i>where $m_V^\pm$ are non-negative integers. In particular, each irreducible repre-</i>
cardona-qft-methods_ F01588	101	■ 1.000	$\operatorname{Ker} \mathcal{D}_{f v}^\pm$	$\operatorname{Ker} \mathcal{D}_{f v}^\pm$	<i>sentation of G appears in $\operatorname{Ker} \mathcal{D}_{f v}^\pm$ with finite multiplicity.</i>
cardona-qft-methods_ F01589	101	■ 1.000	$V \in \operatorname{Irr}(G)$	$V \in \operatorname{Irr}(G)$	Theorem 6.13. For an irreducible representation $V \in \operatorname{Irr}(G)$, let $m_V^\pm$ be
cardona-qft-methods_ F01590	101	■ 1.000	$m_V^+ - m_V^-(V \in \operatorname{Irr}(G))$	$m_V^+ - m_V^-(V \in \operatorname{Irr}(G))$	<i>defined by (7). Then the differences $m_V^+ - m_V^-$ ($V \in \operatorname{Irr}(G)$) are independent</i>
cardona-qft-methods_ F01591	101	■ 0.611	$(\mathcal{D}, \mathbf{v})$	$(\mathcal{D}, \mathbf{v})$	Definition 6.14. We refer to the pair $(\mathcal{D}, \mathbf{v})$ as a <i>tamed Dirac operator</i> .
cardona-qft-methods_ F01592	102	■ 0.666	$M, \mathbf{v}$	$M, \mathbf{v}$	Exercise 6.15. Let $(M, \mathbf{v})$ be the tamed $G = S^1$ -manifold defined in Exam-

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cardona-qft-methods_ F01593	102	■ 1.000	$\sigma_{\{1\}}$	σ_1	where σ_1 and σ_2 are the first two Pauli matrices, cf. (2).
cardona-qft-methods_ F01594	102	■ 1.000	$\sigma_{\{2\}}$	σ_2	where σ_1 and σ_2 are the first two Pauli matrices, cf. (2).
cardona-qft-methods_ F01595	102	■ 1.000	$E^{\pm}$	$E^{\pm}$	The G action on M lifts to $E^{\pm}$ so that
cardona-qft-methods_ F01596	102	■ 1.000	$f \equiv 1$	$f \equiv 1$	b. One can show that $f \equiv 1$ is an admissible function for E . Compute
cardona-qft-methods_ F01597	102	■ 1.000	$\mathcal{D}_v$	$\mathcal{D}_v$	the operators $\mathcal{D}_v$ and $\mathcal{D}_v^2$.
cardona-qft-methods_ F01598	102	■ 1.000	$\mathcal{D}_v^2$	$\mathcal{D}_v^2$	the operators $\mathcal{D}_v$ and $\mathcal{D}_v^2$.
cardona-qft-methods_ F01599	102	■ 1.000	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v})$	$\operatorname{Ind}_G(\mathcal{D}, \mathbf{v})$	d. Compute $\operatorname{Ind}_G(\mathcal{D}, \mathbf{v})$.
cardona-qft-methods_ F01600	103	■ 0.854	$\Sigma \subset M$	$\Sigma \subset M$	Let $(M, \mathbf{v})$ be a tamed G -manifold. Suppose $\Sigma \subset M$ is a smooth G -invariant
cardona-qft-methods_ F01601	103	■ 1.000	$M \setminus \Sigma$	$M \setminus \Sigma$	hypersurface in M such that $M \setminus \Sigma$ is a disjoint union of two open manifolds
cardona-qft-methods_ F01602	103	■ 1.000	$M_{\{1\}}$	M_1	M_1 and M_2 :
cardona-qft-methods_ F01603	103	■ 1.000	$M_{\{2\}}$	M_2	M_1 and M_2 :
cardona-qft-methods_ F01604	103	■ 1.000	Σ	Σ	For simplicity, we assume that Σ is compact. Assume also that the vector
cardona-qft-methods_ F01605	103	■ 1.000	$M_{\{j\}} (j=1,2)$	$M_j (j=1,2)$	complete Riemannian metric on $M_j (j=1,2)$. Let $\mathbf{v}_j$ denote the restriction
cardona-qft-methods_ F01606	103	■ 1.000	$\mathbf{v}_{\{j\}}$	$\mathbf{v}_j$	complete Riemannian metric on $M_j (j=1,2)$. Let $\mathbf{v}_j$ denote the restriction
cardona-qft-methods_ F01607	103	■ 0.783	$M_{\{j\}}$	M_j	of $\mathbf{v}$ to M_j . Then $(M_j, \mathbf{v}_j) (j=1,2)$ are tamed G -manifolds.
cardona-qft-methods_ F01608	103	■ 0.783	$\left(M_{\{j\}}, \mathbf{v}_{\{j\}}\right) (j=1,2)$	$(M_j, \mathbf{v}_j) (j=1,2)$	of $\mathbf{v}$ to M_j . Then $(M_j, \mathbf{v}_j) (j=1,2)$ are tamed G -manifolds.
cardona-qft-methods_ F01609	103	■ 1.000	$E_{\{j\}}$	E_j	over M . Denote by E_j the restriction of E to $M_j (j=1,2)$. Let $\mathcal{D}_j (j=1,2)$
cardona-qft-methods_ F01610	103	■ 1.000	$\mathcal{D}_{\{j\}} (j=1,2)$	$\mathcal{D}_j (j=1,2)$	over M . Denote by E_j the restriction of E to $M_j (j=1,2)$. Let $\mathcal{D}_j (j=1,2)$
cardona-qft-methods_ F01611	104	■ 0.663	$K \ K$	KK	17. G. G. Kasparov, <i>Equivariant KK-theory and the Novikov conjecture</i> , Invent.
cardona-qft-methods_ F01612	107	■ 0.374	$\{ \}^{\text{AM11a}}, \{ \}^{\text{AM11b}}, \{ \}^{\text{AM11c}}$	AM11a , AM11b , AM11c	with Matilde Marcolli (especially ^{AM11a,AM11b,AM11c}). We define a notion
cardona-qft-methods_ F01613	107	■ 1.000	$C_G(t)$	$C_G(t)$	man rules, including a polynomial invariant $C_G(t)$ defined by means of the
cardona-qft-methods_ F01614	107	■ 0.987	$\{ \}^{\text{Mar10}}$	Mar10	may be found in. ^{Mar10}
cardona-qft-methods_ F01615	108	■ 1.000	$\mathbb{C}[\mathbf{x}] = \mathbb{C}[x_1, \dots, x_n]$	$\mathbb{C}[x_1, \dots, x_n]$	in $\mathbb{C}[x_1, \dots, x_n]$, viewed as a subset of affine space $\mathbb{A}_{\mathbb{C}}^n (= \mathbb{C}^n)$.
cardona-qft-methods_ F01616	108	■ 1.000	$\mathbb{A}_{\mathbb{C}}^n (= \mathbb{C}^n)$	$\mathbb{A}_{\mathbb{C}}^n (= \mathbb{C}^n)$	in $\mathbb{C}[x_1, \dots, x_n]$, viewed as a subset of affine space $\mathbb{A}_{\mathbb{C}}^n (= \mathbb{C}^n)$.
cardona-qft-methods_ F01617	108	■ 1.000	$\mathbb{C}[\mathbf{z}] = \mathbb{C}[z_0, \dots, z_n]$	$\mathbb{C}[z_0, \dots, z_n]$	homogeneous polynomials in $\mathbb{C}[z_0, \dots, z_n]$, viewed as a subset of projective
cardona-qft-methods_ F01618	108	■ 1.000	$\mathbb{P}^n = \mathbb{P}_{\mathbb{C}}^n$	$\mathbb{P}^n = \mathbb{P}_{\mathbb{C}}^n$	space $\mathbb{P}^n = \mathbb{P}_{\mathbb{C}}^n$.
cardona-qft-methods_ F01619	108	■ 1.000	$Y \setminus Z$	$Y \setminus Z$	riety, i.e., a set which may be written as $Y \setminus Z$ where both Y and Z are

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cardona-qft-methods_ F01620	108	■ 1.000	Z	Z	riety, i.e., a set which may be written as $Y \setminus Z$ where both Y and Z are
cardona-qft-methods_ F01621	108	■ 0.947	$X \rightarrowtail Y$	$X \rightarrow Y$	$X \dashrightarrow Y$ is the datum of an isomorphism between dense open subsets of
cardona-qft-methods_ F01622	108	■ 1.000	$Y \dashv X$	$Y; X$	X and Y ; X and Y are then said to be ‘birationally isomorphic’ or ‘bi-
cardona-qft-methods_ F01623	108	■ 1.000	$\mathbb{A}^n$	$\mathbb{A}^n$	equations defining the variety. For example, a hypersurface X of $\mathbb{A}^n$ given
cardona-qft-methods_ F01624	108	■ 1.000	$f(x_1, \dots, x_n) = 0$	$f(x_1, \dots, x_n) = 0$	by $f(x_1, \dots, x_n) = 0$ is nonsingular at a point p if the gradient $\nabla_p f$ is not
cardona-qft-methods_ F01625	108	■ 1.000	$\nabla_p f$	$\nabla_p f$	by $f(x_1, \dots, x_n) = 0$ is nonsingular at a point p if the gradient $\nabla_p f$ is not
cardona-qft-methods_ F01626	108	■ 1.000	TX	TX	spaces forms the <i>tangent bundle</i> TX to the variety. If Z is a nonsingu-
cardona-qft-methods_ F01627	108	■ 1.000	$N_Z X$	$N_Z X$	$N_Z X$, that is, the quotient $TX _Z/TZ$. The <i>blow-up</i> of a variety X along a
cardona-qft-methods_ F01628	108	■ 1.000	$TX _Z/TZ$	$TX _Z/TZ$	$N_Z X$, that is, the quotient $TX _Z/TZ$. The <i>blow-up</i> of a variety X along a
cardona-qft-methods_ F01629	108	■ 1.000	$X \setminus Z$	$X \setminus Z$	over the complement $X \setminus Z$ (so that blow-ups are birational), and replaces
cardona-qft-methods_ F01630	109	■ 1.000	Hir64	Hir64	<i>resolution of singularities</i> (Hir64), every variety is birationally isomorphic
cardona-qft-methods_ F01631	109	■ 0.988	$\sum \chi(X) := \sum_i (-1)^i \dim H_c^i(X, \mathbb{Q})$	$\sum \chi(X) := \sum_i (-1)^i \dim H_c^i(X, \mathbb{Q})$	$\sum \chi(X) := \sum_i (-1)^i \dim H_c^i(X, \mathbb{Q})$.
cardona-qft-methods_ F01632	109	■ 1.000	H_c	H_c	Here H_c denotes cohomology with compact support. As it happens, ev-
cardona-qft-methods_ F01633	109	■ 1.000	Ful93	Ful93	the ordinary Euler characteristic. (See e.g., ^{Ful93} p. 95 and 141-2.)
cardona-qft-methods_ F01634	109	■ 1.000	Hir75	Hir75	As every projective variety X may be triangulated (Hir75), we can also
cardona-qft-methods_ F01635	109	■ 1.000	s_i	s_i	where s_i is the number of real-dimension i simplices in a triangulation. If
cardona-qft-methods_ F01636	109	■ 1.000	$X = Y \setminus Z$	$X = Y \setminus Z$	X is quasi-projective, say realized as $X = Y \setminus Z$ with Y and Z projective,
cardona-qft-methods_ F01637	109	■ 0.999	$\chi(X) = \chi(Y) - \chi(Z)$	$\chi(X) = \chi(Y) - \chi(Z)$	then $\chi(X) = \chi(Y) - \chi(Z)$ independently of the realization.
cardona-qft-methods_ F01638	109	■ 0.999	X'	X'	(i) If X and X' are isomorphic, then $\chi(X) = \chi(X')$;
cardona-qft-methods_ F01639	109	■ 0.999	$\chi(X) = \chi(X')$	$\chi(X) = \chi(X')$	(i) If X and X' are isomorphic, then $\chi(X) = \chi(X')$;
cardona-qft-methods_ F01640	109	■ 1.000	$Z \subseteq X$	$Z \subseteq X$	(ii) If $Z \subseteq X$ is a closed subvariety, then $\chi(X) = \chi(Z) + \chi(X \setminus Z)$;
cardona-qft-methods_ F01641	109	■ 1.000	$\chi(X) = \chi(Z) + \chi(X \setminus Z)$	$\chi(X) = \chi(Z) + \chi(X \setminus Z)$	(ii) If $Z \subseteq X$ is a closed subvariety, then $\chi(X) = \chi(Z) + \chi(X \setminus Z)$;
cardona-qft-methods_ F01642	109	■ 1.000	$\chi(X \times Y) = \chi(X) \cdot \chi(Y)$	$\chi(X \times Y) = \chi(X) \cdot \chi(Y)$	(iii) $\chi(X \times Y) = \chi(X) \cdot \chi(Y)$.
cardona-qft-methods_ F01643	109	■ 1.000	$\chi(X) = \chi(Y) \cdot \chi(F)$	$\chi(X) = \chi(Y) \cdot \chi(F)$	that if $X \rightarrow Y$ is locally trivial, with fiber F , then $\chi(X) = \chi(Y) \cdot \chi(F)$.
cardona-qft-methods_ F01644	109	■ 0.744	K_0	K_0	2.2. $K_0(\text{Var})$
cardona-qft-methods_ F01645	110	■ 1.000	$\mathbb{Q}, \mathbb{F}_q$	$\mathbb{Q}, \mathbb{F}_q$	to consider other possibilities, such as $\mathbb{Q}, \mathbb{F}_q$ (the finite field with a prime
cardona-qft-methods_ F01646	110	■ 0.792	K_0	K_0	We are going to define a group $K_0(\text{Var})$, called the ‘Grothendieck group

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cardona-qft-methods_ F01647	110	■ 0.792	)	)	We are going to define a group $K_0(\text{Var})$, called the ‘Grothendieck group
cardona-qft-methods_ F01648	110	■ 1.000	$[M]=\left[M^{\{\text{prime}\}}\right]+\left[M^{\{\text{prime}\}\text{prime}}\right]$	$[M] = [M'] + [M'']$	of R -modules by a relation $[M] = [M'] + [M'']$ for every exact sequence
cardona-qft-methods_ F01649	110	■ 1.000	$0 \rightarrow M^{\{\text{prime}\}} \rightarrow M \rightarrow M^{\{\text{prime}\}\text{prime}} \rightarrow 0$	$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$	$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$. More generally, the same concept may be
cardona-qft-methods_ F01650	110	■ 1.000	$\{ \}^{\{\text{Lev08}\}}$	Lev08	group arise from triangles. The reader may consult ^{Lev08} for a general re-
cardona-qft-methods_ F01651	110	■ 0.590	$2 \text{ in}^{\{\text{Mar10}\}}$	$2\text{in}^{\text{Mar10}}$	view, and Chapter 2 in ^{Mar10} for a treatment very close to the context of
cardona-qft-methods_ F01652	110	■ 1.000	Z, X	Z, X	gle with vertices Z, X , and $X \setminus Z$ for every variety X and every closed
cardona-qft-methods_ F01653	110	■ 1.000	$K_{\{0\}}\left(\operatorname{Var}_{\{k\}}\right):=\frac{\text{Free abelian group on isom. classes of quasi-proj. } k \text{ varieties }}{\langle [X]=[Z]+[X \backslash Z] \text{ for every closed embedding } Z \subseteq X \rangle}$	$K_0(\text{Var}_k) := \frac{\text{Free abelian group on isom. classes of quasi-proj. } k\text{-varieties}}{\langle [X] = [Z] + [X \setminus Z] \text{ for every closed embedding } Z \subseteq X \rangle}$	$K_0(\text{Var}_k) := \frac{\text{Free abelian group on isom. classes of quasi-proj. } k\text{-varieties}}{\langle [X] = [Z] + [X \setminus Z] \text{ for every closed embedding } Z \subseteq X \rangle}$
cardona-qft-methods_ F01654	110	■ 1.000	$[X] \cdot [Y]=\left[X \times_k Y\right]$	$[X] \cdot [Y] = [X \times_k Y]$	operation defined on generators by $[X] \cdot [Y] = [X \times_k Y]$.
cardona-qft-methods_ F01655	110	■ 0.995	$K_{\{0\}}\left(\operatorname{Var}_{\{k\}}\right)$	$K_0(\text{Var}_k)$	$K_0(\text{Var}_k)$.
cardona-qft-methods_ F01656	110	■ 1.000	$k=\mathbb{C}$	$k = \mathbb{C}$	I will usually (but not always) take $k = \mathbb{C}$, and omit the subscript k in
cardona-qft-methods_ F01657	111	■ 1.000	$[X]$	$[X]$	ring of pure Chow motives, assigning to the Grothendieck class $[X]$ the class
cardona-qft-methods_ F01658	111	■ 0.916	$K_{\{0\}}\left(\operatorname{Var}\right)\left(\{ \}^{\{\text{Bit04}\}}\right)$	$K_0(\text{Var}) \left(\text{Bit04}\right)$	alternative description of $K_0(\text{Var})$ (^{Bit04}) which will be recalled below.
cardona-qft-methods_ F01659	111	■ 1.000	$e(p)=1$	$e(p) = 1$	• If p is a point, then $e(p) = 1$;
cardona-qft-methods_ F01660	111	■ 1.000	$e(X)=e\left(X^{\{\text{prime}\}}\right)$	$e(X) = e(X')$	• If X and X' are isomorphic, then $e(X) = e(X')$;
cardona-qft-methods_ F01661	111	■ 1.000	$e(X)=e(Z)+e(X \backslash Z)$	$e(X) = e(Z) + e(X \setminus Z)$	• If $Z \subseteq X$ is a closed subvariety, then $e(X) = e(Z) + e(X \setminus Z)$;
cardona-qft-methods_ F01662	111	■ 1.000	$e(X \times Y)=e(X) e(Y)$	$e(X \times Y) = e(X)e(Y)$	• $e(X \times Y) = e(X)e(Y)$.
cardona-qft-methods_ F01663	111	■ 0.968	$K_{\{0\}}\left(\operatorname{Var}\right) \rightarrow R$	$K_0(\text{Var}) \rightarrow R$	Then e is induced by a unique ring homomorphism $K_0(\text{Var}) \rightarrow R$.
cardona-qft-methods_ F01664	111	■ 0.970	$\} \rightarrow \mathbb{Z}$	$\} \rightarrow \mathbb{Z}$	defined in §2.1 is induced by a unique ring homomorphism $K_0(\text{Var}) \rightarrow \mathbb{Z}$,
cardona-qft-methods_ F01665	111	■ 0.621	$[X] \in K_{\{0\}}\left(\operatorname{Var}\right)$	$[X] \in K_0(\text{Var})$	to its Grothendieck class $[X] \in K_0(\text{Var})$ is a ‘universal Euler characteristic’
cardona-qft-methods_ F01666	111	■ 0.623	$K_{\{0\}}\left(\operatorname{Var}\right)$	$K_0(\text{Var})$	$K_0(\text{Var})$: as mentioned above, there is a homomorphism from $K_0(\text{Var})$ to
cardona-qft-methods_ F01667	111	■ 1.000	$\mathbb{Z}[u, v]$	$\mathbb{Z}[u, v]$	takes values in a polynomial ring $\mathbb{Z}[u, v]$. Also: for varieties defined over $\mathbb{Z}$
cardona-qft-methods_ F01668	111	■ 1.000	$K\left(\operatorname{Var}_{\mathbb{Z}}\right) \rightarrow \mathbb{Z}$	$K(\text{Var}_{\mathbb{Z}}) \rightarrow \mathbb{Z}$	and every prime power q we can define homomorphisms $K(\text{Var}_{\mathbb{Z}}) \rightarrow \mathbb{Z}$
cardona-qft-methods_ F01669	111	■ 1.000	$\mathbb{F}_q$	$\mathbb{F}_q$	by sending $[X]$ to the number of points of X over the finite field $\mathbb{F}_q$ (i.e.,
cardona-qft-methods_ F01670	112	■ 1.000	$\{ \}^{\{\text{Bit04}\}}$	Bit04	The reader is addressed to ^{Bit04} for an excellent treatment of the uni-
cardona-qft-methods_ F01671	112	■ 1.000	$Z \subseteq X$	$Z \subset X$	lar and $Z \subset X$ is a nonsingular subvariety, then there exists a ‘blow-up’

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cardona-qft-methods_ F01672	112	■ 1.000	$\widetilde{X} \rightarrow X$	$\widetilde{X} \rightarrow X$	morphism $\widetilde{X} \rightarrow X$ which is an isomorphism over $X \setminus Z$; the inverse image
cardona-qft-methods_ F01673	112	■ 1.000	$[X]=[Z]+[X \backslash Z]$	$[X] = [Z] + [X \setminus Z]$	type relations $[X] = [Z] + [X \setminus Z]$ used in Definition 2.
cardona-qft-methods_ F01674	112	■ 1.000	$\mathbb{L} :=$	$\mathbb{L} :=$	Definition 3. The <i>Lefschetz (naive) motive</i> is the Grothendieck class $\mathbb{L} :=$
cardona-qft-methods_ F01675	112	■ 0.994	$\left[\mathbb{A}^1\right]$	$[\mathbb{A}^1]$	$[\mathbb{A}^1]$ of the affine line. Elements of the subring $\mathbb{Z}[\mathbb{L}] \subseteq K_0(\text{Var})$ are <i>mixed-</i>
cardona-qft-methods_ F01676	112	■ 0.994	$\mathbb{Z}[\mathbb{L}] \subseteq K_0(\mathbb{A}^1)$	$\mathbb{Z}[\mathbb{L}] \subseteq K_0(\mathbb{A}^1)$	$[\mathbb{A}^1]$ of the affine line. Elements of the subring $\mathbb{Z}[\mathbb{L}] \subseteq K_0(\text{Var})$ are <i>mixed-</i>
cardona-qft-methods_ F01677	112	■ 1.000	$\mathbb{Z}[\mathbb{L}]$	$\mathbb{Z}[\mathbb{L}]$	that classes in $\mathbb{Z}[\mathbb{L}]$ correspond to <i>mixed-Tate motives</i> in the category of
cardona-qft-methods_ F01678	112	■ 1.000	$\widetilde{X}$	$\widetilde{X}$	variety, and $\widetilde{X}$ is the blow-up of X along Z , then $[\widetilde{X}]$ is in $\mathbb{Z}[\mathbb{L}]$ if $[X]$
cardona-qft-methods_ F01679	112	■ 1.000	$[\widetilde{X}]$	$[\widetilde{X}]$	variety, and $\widetilde{X}$ is the blow-up of X along Z , then $[\widetilde{X}]$ is in $\mathbb{Z}[\mathbb{L}]$ if $[X]$
cardona-qft-methods_ F01680	112	■ 1.000	$[Z]$	$[Z]$	and $[Z]$ are in $\mathbb{Z}[\mathbb{L}]$. As we will see in §3, the question of whether the
cardona-qft-methods_ F01681	113	■ 1.000	$[X] \in \mathbb{Z}[\mathbb{L}] \subseteq K_0(\text{Var}_{\mathbb{Z}})$	$[X] \in \mathbb{Z}[\mathbb{L}] \subseteq K_0(\text{Var}_{\mathbb{Z}})$	variety to be mixed-Tate, note that if $[X] \in \mathbb{Z}[\mathbb{L}] \subseteq K_0(\text{Var}_{\mathbb{Z}})$, say
cardona-qft-methods_ F01682	113	■ 0.644	$\chi(X) = \sum_i a_i$	$\chi(X) = \sum_i a_i$	is 1, if the class of X is given by (2), then necessarily $\chi(X) = \sum_i a_i$: the
cardona-qft-methods_ F01683	113	■ 0.984	$\mathbb{F}_1$	$\mathbb{F}_1$	of a mixed-Tate motive is $[X]$ is the ‘number of points of X over $\mathbb{F}_1$ ’.
cardona-qft-methods_ F01684	113	■ 0.999	c_{SM}	c_{SM}	2.3. c_{SM} classes
cardona-qft-methods_ F01685	113	■ 1.000	$c_{\text{SM}}(X)$	$c_{\text{SM}}(X)$	denote by $c_{\text{SM}}(X)$. It takes values in the homology of X , or more properly
cardona-qft-methods_ F01686	113	■ 1.000	$A_* X$	$A_* X$	in the <i>Chow group</i> $A_* X$ of X . This is not the place to discuss rational
cardona-qft-methods_ F01687	113	■ 1.000	Ful84b	Ful84b	from consulting. ^{Ful84b}) In the case of projective space, $A_* \mathbb{P}^n$ agrees with
cardona-qft-methods_ F01688	113	■ 1.000	$A_* \mathbb{P}^n$	$A_* \mathbb{P}^n$	from consulting. ^{Ful84b}) In the case of projective space, $A_* \mathbb{P}^n$ agrees with
cardona-qft-methods_ F01689	113	■ 1.000	$[\mathbb{P}^0], \dots, [\mathbb{P}^n]$	$[\mathbb{P}^0], \dots, [\mathbb{P}^n]$	homology, and equals the free abelian group on generators $[\mathbb{P}^0], \dots, [\mathbb{P}^n]$. It
cardona-qft-methods_ F01690	113	■ 0.950	$r = \dim X$	$r = \dim X$	mentioned in §2.1; the rank of this bundle is $r = \dim X$. Now, every vector
cardona-qft-methods_ F01691	114	■ 1.000	$c_i(E) \cap [X]$	$c_i(E) \cap [X]$	E , denoted $c_i(E) \cap [X]$, is a class in codimension i recording information
cardona-qft-methods_ F01692	114	■ 0.994	$\text{rk } E - i + 1$	$\text{rk } E - i + 1$	on the subset of X where $\text{rk } E - i + 1$ general sections of E are linearly
cardona-qft-methods_ F01693	114	■ 1.000	Ful84a	Ful84a	capturing the gist of Examples 14.4.1 and 14.4.2 in. ^{Ful84a} According to this
cardona-qft-methods_ F01694	114	■ 0.988	$c_{\dim X}(TX) \cap [X]$	$c_{\dim X}(TX) \cap [X]$	rough interpretation, the ‘top’ Chern class $c_{\dim X}(TX) \cap [X]$ of the tangent
cardona-qft-methods_ F01695	114	■ 0.981	$\dim X$	$\dim X$	bundle of X is a class in codimension $\dim X$ (i.e., in dimension 0), recording
cardona-qft-methods_ F01696	114	■ 1.000	$\int$	$\int$	Here, the integral sign $\int$ denotes the degree of the dimension 0 part
cardona-qft-methods_ F01697	114	■ 1.000	$\chi(X) < 0$	$\chi(X) < 0$	$\chi(X) < 0$?) A similar statement may be given relating all Chern classes of
cardona-qft-methods_ F01698	114	■ 0.885	Mac74	Mac74	framework, later confirmed by R. MacPherson (^{Mac74}), of which Theo-

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cardona-qft-methods_ F01699	115	■ 0.594	$\left(,^{\mathrm{BS}}_{81\mathrm{AB}}{}_{08}\right)$	$\left(,^{\mathrm{BS81AB08}}\right)$	MacPherson were found to agree ($,^{\mathrm{BS81AB08}}$); they are commonly called
cardona-qft-methods_ F01700	115	■ 0.987	c_{SM}	c_{SM}	<i>Chern-Schwartz-MacPherson classes</i> , c_{SM} classes for short.
cardona-qft-methods_ F01701	115	■ 0.987	$c_{\mathrm{SM}}(\varphi)$	$c_{\mathrm{SM}}(\varphi)$	In MacPherson’s approach, one defines a class $c_{\mathrm{SM}}(\varphi)$ in A_*X
cardona-qft-methods_ F01702	115	■ 1.000	$\varphi=\sum m_i\mathbb{1}_{Z_i}$	$\varphi=\sum m_i\mathbb{1}_{Z_i}$	functions of subvarieties: $\varphi=\sum m_i\mathbb{1}_{Z_i}$. Constructible functions on X form
cardona-qft-methods_ F01703	115	■ 1.000	$C(X)$	$C(X)$	an abelian group $C(X)$, and in fact C is a <i>covariant functor</i> : if $f:X\rightarrow Y$
cardona-qft-methods_ F01704	115	■ 1.000	C	C	an abelian group $C(X)$, and in fact C is a <i>covariant functor</i> : if $f:X\rightarrow Y$
cardona-qft-methods_ F01705	115	■ 1.000	$f:X\rightarrowtail Y$	$f:X\rightarrow Y$	an abelian group $C(X)$, and in fact C is a <i>covariant functor</i> : if $f:X\rightarrow Y$
cardona-qft-methods_ F01706	115	■ 1.000	$C(X)\rightarrowtail C(Y)$	$C(X)\rightarrow C(Y)$	is a regular map, we can define a homomorphism $C(X)\rightarrow C(Y)$ by setting
cardona-qft-methods_ F01707	115	■ 1.000	$p\in Y$	$p\in Y$	for any $p\in Y$ and any subvariety Z , and extending this prescription by
cardona-qft-methods_ F01708	115	■ 1.000	$X, c_{\mathrm{SM}}:C(X)\rightarrowtail A_*X$	$X, c_{\mathrm{SM}}:C(X)\rightarrow A_*X$	• For any X , $c_{\mathrm{SM}}:C(X)\rightarrow A_*X$ is a homomorphism;
cardona-qft-methods_ F01709	115	■ 1.000	$c_{\mathrm{SM}}(\mathbb{1}_X)=c(TX)\cap[X]$	$c_{\mathrm{SM}}(\mathbb{1}_X)=c(TX)\cap[X]$	• If X is nonsingular and compact, then $c_{\mathrm{SM}}(\mathbb{1}_X)=c(TX)\cap[X]$;
cardona-qft-methods_ F01710	115	■ 1.000	$\varphi\in C(X)$	$\varphi\in C(X)$	• If $f:X\rightarrow Y$ is a proper morphism and $\varphi\in C(X)$, then
cardona-qft-methods_ F01711	115	■ 1.000	$c_{\mathrm{SM}}(f_*\varphi)=f_*c_{\mathrm{SM}}(\varphi)$	$c_{\mathrm{SM}}(f_*\varphi)=f_*c_{\mathrm{SM}}(\varphi)$	$c_{\mathrm{SM}}(f_*\varphi)=f_*c_{\mathrm{SM}}(\varphi)$.
cardona-qft-methods_ F01712	115	■ 0.549	$\{ \}^{\mathrm{Ken90Alu06}}$	Ken90Alu06	been considered ($,^{\mathrm{Ken90Alu06}}$). Resolution of singularities and the covariance
cardona-qft-methods_ F01713	115	■ 1.000	$c_{\mathrm{SM}}(\mathbb{1}_X)$	$c_{\mathrm{SM}}(\mathbb{1}_X)$	The class $c_{\mathrm{SM}}(\mathbb{1}_X)$ is usually denoted $c_{\mathrm{SM}}(X)$, and called the <i>Chern-</i>
cardona-qft-methods_ F01714	115	■ 1.000	$c_{\mathrm{SM}}(Y)$	$c_{\mathrm{SM}}(Y)$	<i>Schwartz-MacPherson</i> class of X . Abusing language, we denote by $c_{\mathrm{SM}}(Y)$
cardona-qft-methods_ F01715	115	■ 0.998	$c_{\mathrm{SM}}(\mathbb{1}_Y)$	$c_{\mathrm{SM}}(\mathbb{1}_Y)$	the class $c_{\mathrm{SM}}(\mathbb{1}_Y)$ in A_*X for any subvariety (closed or otherwise) of X .
cardona-qft-methods_ F01716	115	■ 0.821	$\kappa:X\rightarrowtail\{\mathrm{pt}\}$	$\kappa:X\rightarrow\{\mathrm{pt}\}$	extremely special case of the constant function $\kappa:X\rightarrow\{\mathrm{pt}\}$ (where X is
cardona-qft-methods_ F01717	116	■ 0.736	$\{ \}^{\mathrm{Kwi92}}$	Kwi92	describe here, and for which the reader may consult ^{Kwi92} or ^{.Alu06} There is a
cardona-qft-methods_ F01718	116	■ 0.736	$\{ \}^{\mathrm{Alu06}}$	Alu06	describe here, and for which the reader may consult ^{Kwi92} or ^{.Alu06} There is a
cardona-qft-methods_ F01719	116	■ 1.000	$c_{\mathrm{SM}}(X')$	$c_{\mathrm{SM}}(X')$	of course $c_{\mathrm{SM}}(X)$ and $c_{\mathrm{SM}}(X')$ agree up to the identification $A_*X\cong A_*X'$.
cardona-qft-methods_ F01720	116	■ 1.000	$A_*X\cong A_*X'$	$A_*X\cong A_*X'$	of course $c_{\mathrm{SM}}(X)$ and $c_{\mathrm{SM}}(X')$ agree up to the identification $A_*X\cong A_*X'$.
cardona-qft-methods_ F01721	116	■ 1.000	$c_{\mathrm{SM}}(\mathbb{1}_{X'})$	$c_{\mathrm{SM}}(\mathbb{1}_{X'})$	space, then $c_{\mathrm{SM}}(\mathbb{1}_X)$ and $c_{\mathrm{SM}}(\mathbb{1}_{X'})$ will depend on the embeddings, and
cardona-qft-methods_ F01722	116	■ 0.999	$\mathbb{P}^1$	$\mathbb{P}^1$	Example 2.2. —Viewing $\mathbb{P}^1$ as a subvariety of itself, $c_{\mathrm{SM}}(\mathbb{P}^1)=[\mathbb{P}^1]+$
cardona-qft-methods_ F01723	116	■ 0.999	$c_{\mathrm{SM}}(\mathbb{P}^1)=[\mathbb{P}^1]+$	$c_{\mathrm{SM}}(\mathbb{P}^1)=[\mathbb{P}^1]+$	Example 2.2. —Viewing $\mathbb{P}^1$ as a subvariety of itself, $c_{\mathrm{SM}}(\mathbb{P}^1)=[\mathbb{P}^1]+$
cardona-qft-methods_ F01724	116	■ 1.000	$2[\mathbb{P}^0]$	$2[\mathbb{P}^0]$	$2[\mathbb{P}^0]$: indeed, since $\mathbb{P}^1$ is nonsingular we have
cardona-qft-methods_ F01725	116	■ 0.974	$T\mathbb{P}^n$	$T\mathbb{P}^n$	eties, a standard exact sequence for $T\mathbb{P}^n$ ($,^{\mathrm{Ful84a}}\mathrm{B}.5.8$), and the <i>Whit-</i>

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cardona-qft-methods_ F01726	116	■ 1.000	$\int c_{\mathrm{SM}}(\mathbb{P}^n) = \binom{n+1}{n} = n+1 = \chi(\mathbb{P}^n)$	$\int c_{\mathrm{SM}}(\mathbb{P}^n) = \binom{n+1}{n} = n+1 = \chi(\mathbb{P}^n)$	$\int c_{\mathrm{SM}}(\mathbb{P}^n) = \binom{n+1}{n} = n+1 = \chi(\mathbb{P}^n)$, as it should. A shorthand for (6)
cardona-qft-methods_ F01727	116	■ 1.000	H	H	where H is the ‘hyperplane class’, i.e., H^i stands for $[\mathbb{P}^{n-i}]$. (And in par-
cardona-qft-methods_ F01728	116	■ 1.000	$H^{\{i\}}$	H^i	where H is the ‘hyperplane class’, i.e., H^i stands for $[\mathbb{P}^{n-i}]$. (And in par-
cardona-qft-methods_ F01729	116	■ 1.000	$[\mathbb{P}^{n-i}]$	$[\mathbb{P}^{n-i}]$	where H is the ‘hyperplane class’, i.e., H^i stands for $[\mathbb{P}^{n-i}]$. (And in par-
cardona-qft-methods_ F01730	116	■ 1.000	$H^{n+1}=0$	$H^{n+1}=0$	ticular $H^{n+1}=0$.)
cardona-qft-methods_ F01731	117	■ 1.000	$\mathbb{P}^2$	$\mathbb{P}^2$	—On the other hand, a nonsingular <i>conic</i> C in $\mathbb{P}^2$ is easily seen to be
cardona-qft-methods_ F01732	117	■ 1.000	$c_{\mathrm{SM}}(\mathbb{P}^1)$	$c_{\mathrm{SM}}(\mathbb{P}^1)$	This agrees with $c_{\mathrm{SM}}(\mathbb{P}^1)$ via the isomorphism, but the result of the push-
cardona-qft-methods_ F01733	117	■ 0.999	L_1, L_2	L_1, L_2	—A union X of two distinct lines L_1, L_2 in $\mathbb{P}^2$ is a <i>singular</i> conic. By
cardona-qft-methods_ F01734	117	■ 1.000	$c_{\mathrm{SM}}(X) = c_{\mathrm{SM}}(L_1 \cup L_2)$	$c_{\mathrm{SM}}(X) = c_{\mathrm{SM}}(L_1 \cup L_2)$	inclusion-exclusion, $c_{\mathrm{SM}}(X) = c_{\mathrm{SM}}(L_1 \cup L_2)$ is given by
cardona-qft-methods_ F01735	117	■ 0.988	c_{SM}	c_{SM}	singular curves of the same degree. In general, c_{SM} classes may be used
cardona-qft-methods_ F01736	117	■ 1.000	$\mathrm{PP01}$	$\mathrm{PP01}$	<i>classes</i> , see e.g., ^{PP01} Indeed, one of the main motivations in the study of
cardona-qft-methods_ F01737	117	■ 1.000	$\mathrm{AM09}$	$\mathrm{AM09}$	Grothendieck class. The precise statement may be found in, ^{AM09} Propo-
cardona-qft-methods_ F01738	117	■ 0.693	$\mathrm{Alu12}$	$\mathrm{Alu12}$	<i>polynomial</i> of the arrangement (cf. ^{Alu12}). For instance, for graphical ar-
cardona-qft-methods_ F01739	117	■ 0.998	$X \subseteq \mathbb{P}^n$	$X \subseteq \mathbb{P}^n$	For $X \subseteq \mathbb{P}^n$, the information carried by $c_{\mathrm{SM}}(\mathbb{I}_X)$ is precisely equivalent
cardona-qft-methods_ F01740	117	■ 0.996	Alub	Alub	of X (Alub).
cardona-qft-methods_ F01741	117	■ 0.980	$\mathrm{BSY10}$	$\mathrm{BSY10}$	terms of a relative Grothendieck group ($\mathrm{BSY10}$). For projective sets, there
cardona-qft-methods_ F01742	117	■ 1.000	$\mathbb{P}^\infty$	$\mathbb{P}^\infty$	is a group homomorphism from a Grothendieck group associated with $\mathbb{P}^\infty$
cardona-qft-methods_ F01743	117	■ 1.000	$\mathbb{Z}[T]$	$\mathbb{Z}[T]$	to the corresponding Chow group, a polynomial ring $\mathbb{Z}[T]$, which yields
cardona-qft-methods_ F01744	117	■ 0.753	Alub	Alub	the c_{SM} class (Alub). The multiplicativity properties of this morphism are
cardona-qft-methods_ F01745	118	■ 1.000	$L(\phi)$	$L(\phi)$	in dimension D , where $L(\phi)$ is a ‘Lagrangian density’. These integrals may
cardona-qft-methods_ F01746	118	■ 1.000	$\langle G \rangle$	$\langle G \rangle$	Lagrangian, and the contribution $\langle G \rangle$ of a graph is of the form $\frac{G(\phi)}{\#\mathrm{Aut}(G)}$;
cardona-qft-methods_ F01747	118	■ 1.000	$\frac{G(\phi)}{\#\mathrm{Aut}(G)}$	$\frac{G(\phi)}{\#\mathrm{Aut}(G)}$	Lagrangian, and the contribution $\langle G \rangle$ of a graph is of the form $\frac{G(\phi)}{\#\mathrm{Aut}(G)}$;
cardona-qft-methods_ F01748	118	■ 1.000	$G(\phi)$	$G(\phi)$	we will say more about $G(\phi)$ below. Further manipulations may be used
cardona-qft-methods_ F01749	118	■ 0.970	$\mathrm{Zee10}$	$\mathrm{Zee10}$	The reader is addressed to any text in quantum field theory (such as ^{Zee10})
cardona-qft-methods_ F01750	118	■ 1.000	p_i	p_i	Here N is the number of external edges attached to G , and p_i denotes a
cardona-qft-methods_ F01751	118	■ 1.000	$\sum_i p_i = 0$	$\sum_i p_i = 0$	$\sum_i p_i = 0$ to enforce momentum conservation; $\hat{\phi}$ denotes Fourier transform,

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cardona-qft-methods_ F01752	118	■ 1.000	$\hat{\phi}$	$\hat{\phi}$	$\sum_i p_i = 0$ to enforce momentum conservation; $\hat{\phi}$ denotes Fourier transform,
cardona-qft-methods_ F01753	118	■ 0.874	$\text{\text{CM}}$	CM	been understood very recently, see. ^{CM} In the above integral,
cardona-qft-methods_ F01754	119	■ 1.000	I_G	I_G	where ℓ is the number of loops in G , and the core contribution I_G may
cardona-qft-methods_ F01755	119	■ 1.000	$U(G)$	$U(G)$	‘Feynman trick’) yields an expression for $U(G)$ of the following form:
cardona-qft-methods_ F01756	119	■ 1.000	$\sigma_n = \left\{ \sum_i t_i = 1 \right\} ; \omega_n$	$\sigma_n = \left\{ \sum_i t_i = 1 \right\} ; \omega_n$	(real) simplex $\sigma_n = \{\sum_i t_i = 1\}$; ω_n is the volume form; and P_G, Ψ_G are
cardona-qft-methods_ F01757	119	■ 1.000	P_G, Ψ_G	P_G, Ψ_G	(real) simplex $\sigma_n = \{\sum_i t_i = 1\}$; ω_n is the volume form; and P_G, Ψ_G are
cardona-qft-methods_ F01758	119	■ 1.000	Ψ_G	Ψ_G	this polynomial Ψ_G (see §3.2) is <i>the only one</i> that will be really relevant
cardona-qft-methods_ F01759	119	■ 0.957	$P_G(\underline{t}, p)$	$P_G(\underline{t}, p)$	in. ^{Mar10} The polynomial $P_G(\underline{t}, p)$ is given by
cardona-qft-methods_ F01760	119	■ 1.000	s_C	s_C	where the sum ranges over the <i>cut sets</i> C of G , and the coefficient s_C is a
cardona-qft-methods_ F01761	119	■ 1.000	$D \ell / 2 - n \geq 0$	$D\ell/2 - n \geq 0$	that D is large enough that $D\ell/2 - n \geq 0$, so that P_G is really at numerator
cardona-qft-methods_ F01762	119	■ 1.000	P_G	P_G	that D is large enough that $D\ell/2 - n \geq 0$, so that P_G is really at numerator
cardona-qft-methods_ F01763	120	■ 1.000	$\Gamma(n - D \ell / 2)$	$\Gamma(n - D\ell/2)$	and Ψ_G is really at denominator in (2). The Gamma factor $\Gamma(n - D\ell/2)$
cardona-qft-methods_ F01764	120	■ 1.000	$M \geq 0$	$M \geq 0$	with $M \geq 0$, an integral of an algebraic differential form over the <i>comple-</i>
cardona-qft-methods_ F01765	120	■ 1.000	$\hat{X}_G$	$\hat{X}_G$	<i>ment of the hypersurface $\hat{X}_G$ defined by $\Psi_G = 0$.</i> This hypersurface com-
cardona-qft-methods_ F01766	120	■ 1.000	$\Psi_G = 0$	$\Psi_G = 0$	<i>ment of the hypersurface $\hat{X}_G$ defined by $\Psi_G = 0$.</i> This hypersurface com-
cardona-qft-methods_ F01767	120	■ 1.000	$\overline{\mathbb{Q}}$	$\overline{\mathbb{Q}}$	Integrals of algebraic forms over algebraic domains (all defined over $\overline{\mathbb{Q}}$)
cardona-qft-methods_ F01768	120	■ 1.000	$\text{\text{KZ01}}$	KZ01	view article by M. Kontsevich and D. Zagier, ^{KZ01} The set of periods forms
cardona-qft-methods_ F01769	120	■ 1.000	$\frac{1}{2\pi i}$	$\frac{1}{2\pi i}$	a ring (often enlarged by including $\frac{1}{2\pi i}$, for technical reasons), and is <i>count-</i>
cardona-qft-methods_ F01770	120	■ 1.000	$\text{\text{Bro12}}$	Bro12	long-standing conjecture, and has recently been proved by F. Brown, ^{Bro12}
cardona-qft-methods_ F01771	120	■ 1.000	$n_i \geq 1, n_r \geq 2$	$n_i \geq 1, n_r \geq 2$	with $n_i \geq 1, n_r \geq 2$. Brown was able to confirm that all periods of mixed
cardona-qft-methods_ F01772	120	■ 1.000	$\mathbb{Q} \left[\frac{1}{2\pi i} \right]$	$\mathbb{Q} \left[\frac{1}{2\pi i} \right]$	Tate motives over $\mathbb{Z}$ are $\mathbb{Q}[\frac{1}{2\pi i}]$ -linear combinations of multiple zeta values.
cardona-qft-methods_ F01773	120	■ 1.000	$\hat{Y}_G$	$\hat{Y}_G$	for the (motive corresponding to) the complement $\hat{Y}_G$ of the <i>graph hyper-</i>
cardona-qft-methods_ F01774	120	■ 1.000	$\hat{X}_G \subseteq \mathbb{A}^n$	$\hat{X}_G \subseteq \mathbb{A}^n$	<i>surface $\hat{X}_G \subseteq \mathbb{A}^n$ defined by $\Psi_G(\underline{t}) = 0$.</i> Again, let me point out that there
cardona-qft-methods_ F01775	120	■ 1.000	$\Psi_G(\underline{t}) = 0$	$\Psi_G(\underline{t}) = 0$	<i>surface $\hat{X}_G \subseteq \mathbb{A}^n$ defined by $\Psi_G(\underline{t}) = 0$.</i> Again, let me point out that there
cardona-qft-methods_ F01776	121	■ 1.000	$\text{\text{BK97}}$	BK97	these computations is, ^{BK97} the amazing observation stemming from these
cardona-qft-methods_ F01777	121	■ 1.000	t_e	t_e	of spanning trees of the connected components of G), and t_e is a variable

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cardona-qft-methods_ F01778	122	■ 1.000	$n>0$	$n > 0$	for all $n > 0$.
cardona-qft-methods_ F01779	123	■ 1.000	$\Psi_{\{G\}}(\underline{t})$	$\Psi_G(\underline{t})$	for G is the same for all such forests (exercise!), therefore $\Psi_G(\underline{t})$ is homoge-
cardona-qft-methods_ F01780	123	■ 0.943	$X_{\{G\}}$	X_G	neous. Thus, its vanishing defines a hypersurface X_G in $\mathbb{P}^{n-1}$ (n =number
cardona-qft-methods_ F01781	123	■ 0.943	$\mathbb{P}^{n-1}$	$\mathbb{P}^{n-1}$	neous. Thus, its vanishing defines a hypersurface X_G in $\mathbb{P}^{n-1}$ (n =number
cardona-qft-methods_ F01782	123	■ 0.943	$n=$	$n =$	neous. Thus, its vanishing defines a hypersurface X_G in $\mathbb{P}^{n-1}$ (n =number
cardona-qft-methods_ F01783	123	■ 1.000	$Y_{\{G\}}$	Y_G	corresponding to the points of X_G). We will denote by Y_G the complement
cardona-qft-methods_ F01784	123	■ 1.000	$b_{\{1\}}(G)$	$b_1(G)$	definition that the degree of Ψ_G equals the number of loops $b_1(G)$ of G .
cardona-qft-methods_ F01785	123	■ 1.000	$G \setminus e$	$G \setminus e$	smaller graphs $G \setminus e$. The study of singularities of X_G was initiated by
cardona-qft-methods_ F01786	123	■ 0.980	$\{ \}^{\text{Pat10}}$	Pat10	E. Patterson (^{Pat10}); some of our own work reviewed in §4 is aimed at the
cardona-qft-methods_ F01787	123	■ 0.974	$\{ \}^{\text{SMJO } 77}$	SMJO 77	irreducible; and it is known (see for example, ^{SMJO77} Proposition 5.2) that
cardona-qft-methods_ F01788	123	■ 0.998	$\{ \}^{\text{MA}}$	MA	subgraphs, or graphs joined at a vertex. Also see, ^{MA} §5, for a discussion of
cardona-qft-methods_ F01789	123	■ 1.000	$\left[\hat{Y}_{\{G\}} \right]$	$\left[\hat{Y}_G \right]$	the Grothendieck class $[\hat{Y}_G]$ should belong to the subring $\mathbb{Z}[\mathbb{L}]$ of the
cardona-qft-methods_ F01790	123	■ 1.000	$K_{\{0\}}(\operatorname{Var}_{\{\mathbb{Z}\}})$	$K_0(\operatorname{Var}_{\mathbb{Z}})$	Grothendieck ring $K_0(\operatorname{Var}_{\mathbb{Z}})$; and it would follow that $\hat{Y}_G$ is <i>polynomially</i>
cardona-qft-methods_ F01791	123	■ 1.000	$\mathbb{F}_q[t_{\{1\}}, \ldots, t_{\{n\}}]$	$\mathbb{F}_q[t_1, \ldots, t_n]$	Ψ_G defines a polynomial in $\mathbb{F}_q[t_1, \ldots, t_n]$ for all finite fields $\mathbb{F}_q$ (here q is
cardona-qft-methods_ F01792	123	■ 1.000	$N_{\{q\}}(G)$	$N_q(G)$	of course a prime power); one can then let $N_q(G)$ be the number of tuples
cardona-qft-methods_ F01793	123	■ 1.000	$\left(t_{\{1\}}, \ldots, t_{\{n\}} \right)$	$(t_1, \ldots, t_n)$	$(t_1, \ldots, t_n)$ in $\mathbb{F}_q^n$ such that $\Psi_G(\bar{t}) \neq 0$ in $\mathbb{F}_q$. We have the implications
cardona-qft-methods_ F01794	123	■ 1.000	$\mathbb{F}_q^{\{n\}}$	$\mathbb{F}_q^n$	$(t_1, \ldots, t_n)$ in $\mathbb{F}_q^n$ such that $\Psi_G(\bar{t}) \neq 0$ in $\mathbb{F}_q$. We have the implications
cardona-qft-methods_ F01795	123	■ 1.000	$\Psi_{\{G\}}(\bar{t}) \neq 0$	$\Psi_G(\bar{t}) \neq 0$	$(t_1, \ldots, t_n)$ in $\mathbb{F}_q^n$ such that $\Psi_G(\bar{t}) \neq 0$ in $\mathbb{F}_q$. We have the implications
cardona-qft-methods_ F01796	124	■ 0.977	$\{ \}^{\text{Sta98}}$	Sta98	rials R. Stanley (^{Sta98}), J. R. Stembridge (^{Ste98}), and others appeared
cardona-qft-methods_ F01797	124	■ 0.977	$\{ \}^{\text{Ste98}}$	Ste98	rials R. Stanley (^{Sta98}), J. R. Stembridge (^{Ste98}), and others appeared
cardona-qft-methods_ F01798	124	■ 0.997	$\{ \}^{\text{BB03}}$	BB03	<i>proved</i> in a remarkable paper by P. Belkale and P. Brosnan (^{BB03}), showing
cardona-qft-methods_ F01799	124	■ 0.999	$\{ \}^{\text{Dor11Sch11}}$	Dor11Sch11	O. Schnetz (^{Dor11Sch11}), it has been shown that all graphs with 13 or fewer
cardona-qft-methods_ F01800	124	■ 1.000	$\mathbb{P}^{\{3\}}$	$\mathbb{P}^3$	related to the number of points on a K3 in $\mathbb{P}^3$. The appearance of K3s in
cardona-qft-methods_ F01801	124	■ 1.000	$\{ \}^{\text{BS12}}$	BS12	this context is further studied in the more recent. ^{BS12}
cardona-qft-methods_ F01802	124	■ 1.000	$H^{\{1\}}(G)$	$H^1(G)$	for $H^1(G)$. Thus, graph hypersurfaces may be viewed as specializations of a
cardona-qft-methods_ F01803	124	■ 1.000	$\{ \}^{\text{AM10}}$	AM10	out in, ^{AM10} but hits against a wall: even though determinant hypersurfaces

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cardona-qft-methods_ F01804	124	■ 1.000	$\{ \}^{\text{\texttt{Vak06}}}$	Vak06	is one expression of R. Vakil’s “Murphy’s law”, ^{Vak06}).
cardona-qft-methods_ F01805	124	■ 0.948	$K_{\{0\}}(\operatorname{Var})$	$K_0(\operatorname{Var})$	class $[\hat{Y}_G]$ in $K_0(\operatorname{Var})$ with the combinatorics of G . We will also have some-
cardona-qft-methods_ F01806	125	■ 0.990	$c_{\{\mathrm{SM}\}}\left(X_G\right)$	$c_{\mathrm{SM}}(X_G)$	any case we view $c_{\mathrm{SM}}(X_G)$ as a means to get global information on the sin-
cardona-qft-methods_ F01807	125	■ 0.749	$\sum_{\text{\texttt{graphs}}\{ \}_G}\angle G\angle$	$\sum_{\text{graphs } G} \langle G \rangle$	(1): $\sum_{\text{graphs } G} \langle G \rangle$, where $\langle G \rangle$ is a contribution due to G and accounting for
cardona-qft-methods_ F01808	125	■ 0.999	$G_{\{i\}}$	G_i	a symmetry factor. If G_i are the connected components of G , then
cardona-qft-methods_ F01809	125	■ 1.000	$G_{\{1\}}$	G_1	if G is obtained by joining G_1 and G_2 by an edge joining a vertex of G_1 to
cardona-qft-methods_ F01810	125	■ 1.000	$G_{\{2\}}$	G_2	if G is obtained by joining G_1 and G_2 by an edge joining a vertex of G_1 to
cardona-qft-methods_ F01811	126	■ 1.000	$\{ \}^{\text{\texttt{AM11a}}}$	AM11a	This material is explored in more detail in. ^{AM11a} The reader should be
cardona-qft-methods_ F01812	126	■ 1.000	$G_{\{1\}}, G_{\{2\}}$	G_1, G_2	Lemma 4.1. <i>Let G be the disjoint union of two graphs G_1, G_2. Then</i>
cardona-qft-methods_ F01813	126	■ 1.000	$\Psi_{\{G\}}=\left(\Psi_{\{G_{\{1\}}\}}\right)\left(\Psi_{\{G_{\{2\}}\}}\right)$	$\Psi_G = (\Psi_{G_1})(\Psi_{G_2})$	$\Psi_G = (\Psi_{G_1})(\Psi_{G_2})$.
cardona-qft-methods_ F01814	127	■ 1.000	$t_{\{4\}}$	t_4	with a spanning tree of the triangle on the right by the bridge t_4 . It follows
cardona-qft-methods_ F01815	127	■ 0.988	$\left(t_{\{1\}}+t_{\{2\}}+t_{\{3\}}\right)\left(t_{\{5\}}+t_{\{6\}}+t_{\{7\}}\right)$	$(t_1 + t_2 + t_3)(t_5 + t_6 + t_7)$	easily that the graph polynomial equals $(t_1 + t_2 + t_3)(t_5 + t_6 + t_7)$.
cardona-qft-methods_ F01816	127	■ 1.000	$\Psi_{\{G\}}\left(t_{\{1\}}, \ldots, t_{\{n\}}\right)=0$	$\Psi_G(t_1, \dots, t_n) = 0$	surface $\Psi_G(t_1, \dots, t_n) = 0$ in $\mathbb{P}^{n-1}$, by $\hat{X}_G$ the affine cone in $\mathbb{A}^n$, and by
cardona-qft-methods_ F01817	127	■ 1.000	$\mathbb{A}^n \backslash \hat{X}_{\{G\}}$	$\mathbb{A}^n \setminus \hat{X}_G$	$\hat{Y}_G$ the complement $\mathbb{A}^n \setminus \hat{X}_G$.
cardona-qft-methods_ F01818	127	■ 1.000	$G=G_{\{1\}} \amalg G_{\{2\}}$	$G = G_1 \amalg G_2$	• If $G = G_1 \amalg G_2$, then $\hat{Y}_G \cong \hat{Y}_{G_1} \times \hat{Y}_{G_2}$.
cardona-qft-methods_ F01819	127	■ 1.000	$\hat{Y}_{\{G\}} \cong \hat{Y}_{\{G_{\{1\}}\}} \times \hat{Y}_{\{G_{\{2\}}\}}$	$\hat{Y}_G \cong \hat{Y}_{G_1} \times \hat{Y}_{G_2}$	• If $G = G_1 \amalg G_2$, then $\hat{Y}_G \cong \hat{Y}_{G_1} \times \hat{Y}_{G_2}$.
cardona-qft-methods_ F01820	127	■ 1.000	$\hat{Y}_{\{G\}} \cong$	$\hat{Y}_G \cong$	• If G is obtained by joining G_1 and G_2 by a bridge, then $\hat{Y}_G \cong$
cardona-qft-methods_ F01821	127	■ 1.000	$\hat{Y}_{\{G_{\{1\}}\}} \times \hat{Y}_{\{G_{\{2\}}\}} \times \mathbb{A}^1$	$\hat{Y}_{G_1} \times \hat{Y}_{G_2} \times \mathbb{A}^1$	$\hat{Y}_{G_1} \times \hat{Y}_{G_2} \times \mathbb{A}^1$.
cardona-qft-methods_ F01822	127	■ 1.000	$t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}}$	$t_1, \dots, t_{n_1}$	Proof. Let $t_1, \dots, t_{n_1}$ be the variables corresponding to the edges of
cardona-qft-methods_ F01823	127	■ 1.000	$u_{\{1\}}, \ldots, u_{\{n_{\{2\}}\}}$	$u_1, \dots, u_{n_2}$	G_1 , and let $u_1, \dots, u_{n_2}$ be the variables corresponding to the edges of
cardona-qft-methods_ F01824	127	■ 1.000	$\Psi_{\{G\}}=\Psi_{\{G_{\{1\}}\}} \Psi_{\{G_{\{2\}}\}}$	$\Psi_G = \Psi_{G_1} \Psi_{G_2}$	G_2 . If $G = G_1 \amalg G_2$, then $\Psi_G = \Psi_{G_1} \Psi_{G_2}$ as observed above. A point
cardona-qft-methods_ F01825	127	■ 1.000	$\left(t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}}, u_{\{1\}}, \ldots, u_{\{n_{\{2\}}\}}\right)$	$(t_1, \dots, t_{n_1}, u_1, \dots, u_{n_2})$	$(t_1, \dots, t_{n_1}, u_1, \dots, u_{n_2})$ of $\mathbb{A}^{n_1+n_2}$ is in $\hat{Y}_G$ if and only if
cardona-qft-methods_ F01826	127	■ 1.000	$\mathbb{A}^{n_{\{1\}}+n_{\{2\}}}$	$\mathbb{A}^{n_1+n_2}$	$(t_1, \dots, t_{n_1}, u_1, \dots, u_{n_2})$ of $\mathbb{A}^{n_1+n_2}$ is in $\hat{Y}_G$ if and only if
cardona-qft-methods_ F01827	127	■ 1.000	$\left(t_{\{1\}}, \ldots, t_{\{n_{\{1\}}\}}\right) \in \hat{Y}_{\{G_{\{1\}}\}}$	$(t_1, \dots, t_{n_1}) \in \hat{Y}_{G_1}$	if and only if $(t_1, \dots, t_{n_1}) \in \hat{Y}_{G_1}$ and $(u_1 \dots, u_{n_2}) \in \hat{Y}_2$. The first assertion
cardona-qft-methods_ F01828	127	■ 1.000	$\left(u_{\{1\}} \ldots, u_{\{n_{\{2\}}\}}\right) \in \hat{Y}_{\{G_{\{2\}}\}}$	$(u_1 \dots, u_{n_2}) \in \hat{Y}_2$	if and only if $(t_1, \dots, t_{n_1}) \in \hat{Y}_{G_1}$ and $(u_1 \dots, u_{n_2}) \in \hat{Y}_2$. The first assertion
cardona-qft-methods_ F01829	128	■ 1.000	$\mathbb{A}^1$	$\mathbb{A}^1$	extra factor of $\mathbb{A}^1$, giving the second assertion. (In fact, the second assertion

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cardona-qft-methods_ F01836	128	■ 1.000	$G \mapsto \mathbb{U}(G)$	$G \mapsto U(G)$	assignment from graphs to rings, $G \mapsto U(G)$, such that
cardona-qft-methods_ F01831	128	■ 1.000	$\mathbb{U}(G)=\mathbb{U}(\text{left}(G_1)\text{right}) \mathbb{U}(\text{left}(G_2)\text{right})$	$U(G) = U(G_1)U(G_2)$	vertex, then $U(G) = U(G_1)U(G_2)$.
cardona-qft-methods_ F01832	128	■ 1.000	$\mathbb{U}(G)=\mathbb{U}(\text{left}(G_1)\text{right}) \mathbb{U}(\text{left}(G_2)\text{right}) \mathbb{U}(e)$	$U(G) = U(G_1)U(G_2)U(e)$	In particular, we have that $U(G) = U(G_1)U(G_2)U(e)$ if G is obtained
cardona-qft-methods_ F01833	128	■ 1.000	$\mathbb{U}(e)$	$U(e)$	by joining G_1 and G_2 by a bridge e . This identifies $U(e)$ as the (inverse)
cardona-qft-methods_ F01834	128	■ 1.000	$\text{cal}\{S_B\}(G) \in \mathbb{Z}$	$\mathcal{SB}(G) \in \mathbb{Z}$	Example 4.2. Let $\mathcal{SB}(G) \in \mathbb{Z}$ be 0 if G has edges that are not looping
cardona-qft-methods_ F01835	128	■ 1.000	$(-1)^n$	$(-1)^n$	edges, and $(-1)^n$ if G has n looping edges and no other edge.
cardona-qft-methods_ F01836	128	■ 1.000	$F \rightarrow R$	$F \rightarrow R$	correspond precisely to ring homomorphisms $F \rightarrow R$. The hypersurface
cardona-qft-methods_ F01837	129	■ 1.000	$\text{cal}\{S_B\}(G)$	$\mathcal{SB}(G)$	instance, is the assignment $\mathcal{SB}(G)$ of Example 4.2 algebro-geometric?
cardona-qft-methods_ F01838	129	■ 1.000	$\chi(\text{left}(Y_G)\text{right})$	$\chi(Y_G)$	characteristic $\chi(Y_G)$ of the complement of the graph hypersurface in pro-
cardona-qft-methods_ F01839	129	■ 0.960	$\text{left}[Y_G\text{right}]$	$[Y_G]$	G_2 is a forest, then the class $[Y_G]$ is a multiple of $(\mathbb{L} - 1)$ (exercise!), so
cardona-qft-methods_ F01840	129	■ 0.960	$(\mathbb{L}-1)$	$(\mathbb{L} - 1)$	G_2 is a forest, then the class $[Y_G]$ is a multiple of $(\mathbb{L} - 1)$ (exercise!), so
cardona-qft-methods_ F01841	129	■ 1.000	$\chi(\text{left}(Y_G)\text{right})=0$	$\chi(Y_G) = 0$	$\chi(Y_G) = 0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-
cardona-qft-methods_ F01842	129	■ 1.000	$\chi(\text{left}(Y_{G_1})\text{right})$	$\chi(Y_{G_1})$	$\chi(Y_G) = 0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-
cardona-qft-methods_ F01843	129	■ 1.000	$\chi(\text{left}(Y_{G_2})\text{right})$	$\chi(Y_{G_2})$	$\chi(Y_G) = 0$ no matter what $\chi(Y_{G_1})$ and $\chi(Y_{G_2})$ may be. By itself, this in-
cardona-qft-methods_ F01844	129	■ 1.000	$\mathbb{U}(G):=\text{left}[\hat{Y}_G\text{right}]$	$\mathbb{U}(G) := [\hat{Y}_G]$	Simply send the class of $\hat{Y}_G$ in F to the class $\mathbb{U}(G) := [\hat{Y}_G]$ in the ordinary
cardona-qft-methods_ F01845	129	■ 0.999	$\mathbb{L}=(\mathbb{T}+1)$	$\mathbb{L} = (\mathbb{T} + 1)$	The ‘motivic propagator’ is the class $\mathbb{L} = (\mathbb{T} + 1)$ of $\mathbb{A}^1$. It is a good
cardona-qft-methods_ F01846	129	■ 1.000	$\mathbb{U}(G)$	$\mathbb{U}(G)$	exercise to establish the effect of some simple graph operations on $\mathbb{U}(G)$; for
cardona-qft-methods_ F01847	129	■ 0.516	$(\mathbb{T}+1)$	$(\mathbb{T} + 1)$	example, splitting an edge also multiplies $\mathbb{U}(G)$ by the propagator $(\mathbb{T} + 1)$,
cardona-qft-methods_ F01848	129	■ 1.000	$\mathbb{T}$	$\mathbb{T}$	while adding a looping edge multiplies $\mathbb{U}(G)$ by $\mathbb{T}$.
cardona-qft-methods_ F01849	129	■ 1.000	$\text{left}[\hat{X}_G\text{right}],\text{left}[X_G\text{right}]$	$[\hat{X}_G], [X_G]$	variations on this theme—such as $[\hat{X}_G]$, $[X_G]$, or the class $[Y_G]$ of the hy-
cardona-qft-methods_ F01850	129	■ 1.000	$\hat{X}$	$\hat{X}$	Consider any affine cone $\hat{X}$ in $\mathbb{A}^n$ as a locally closed subset of the pro-
cardona-qft-methods_ F01851	129	■ 0.996	$\mathbb{P}^n$	$\mathbb{P}^n$	jective space $\mathbb{P}^n$ obtained by adding the hyperplane at infinity. The c_{SM}
cardona-qft-methods_ F01852	129	■ 1.000	$F_{\hat{X}}(T):=a_0+a_1T+\cdots+a_nT^n$	$F_{\hat{X}}(T) := a_0 + a_1T + \cdots + a_nT^n$	Associate with $\hat{X}$ the polynomial $F_{\hat{X}}(T) := a_0 + a_1T + \cdots + a_nT^n$.
cardona-qft-methods_ F01853	129	■ 1.000	$F_{\mathbb{A}^n}(T)$	$F_{\mathbb{A}^n}(T)$	Example 4.3. What is $F_{\mathbb{A}^n}(T)$? Denoting by H the hyperplane class in
cardona-qft-methods_ F01854	130	■ 1.000	$F_{\mathbb{A}^n}(T)=(1+T)^n$	$F_{\mathbb{A}^n}(T) = (1 + T)^n$	Therefore, $F_{\mathbb{A}^n}(T) = (1 + T)^n$.
cardona-qft-methods_ F01855	130	■ 1.000	$F \rightarrow \mathbb{Z}[T]$	$F \rightarrow \mathbb{Z}[T]$	Prop 4.2 (Easy). F defines a group homomorphism $F \rightarrow \mathbb{Z}[T]$.
cardona-qft-methods_ F01856	130	■ 1.000	$F \rightarrow$	$F \rightarrow$	Theorem 4.1 (Not so easy). F defines a ring homomorphism $F \rightarrow$

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cardona-qft-methods_ F01857	130	■ 1.000	$\left[\mathbb{P}^i\right]$	$[\mathbb{P}^i]$	<i>priori</i> , there does not seem to be any reasonable sense in which $[\mathbb{P}^i]$ and
cardona-qft-methods_ F01858	130	■ 1.000	$\left[\mathbb{P}^j\right]$	$[\mathbb{P}^j]$	$[\mathbb{P}^j]$ should ‘multiply’ to $[\mathbb{P}^{i+j}]$, especially if $i + j$ exceeds the dimension of
cardona-qft-methods_ F01859	130	■ 1.000	$\left[\mathbb{P}^{i+j}\right]$	$[\mathbb{P}^{i+j}]$	$[\mathbb{P}^j]$ should ‘multiply’ to $[\mathbb{P}^{i+j}]$, especially if $i + j$ exceeds the dimension of
cardona-qft-methods_ F01860	130	■ 1.000	$i+j$	$i + j$	$[\mathbb{P}^j]$ should ‘multiply’ to $[\mathbb{P}^{i+j}]$, especially if $i + j$ exceeds the dimension of
cardona-qft-methods_ F01861	130	■ 1.000	$C_{\{G\}}(T) :=$	$C_G(T) :=$	Feynman rules by associating with a graph G the polynomial $C_G(T) :=$
cardona-qft-methods_ F01862	130	■ 0.997	$F_{\{\hat{Y}_{\{G\}}(T) \in \mathbb{Z}[T]$	$F_{\hat{Y}_G}(T) \in \mathbb{Z}[T]$	$F_{\hat{Y}_G}(T) \in \mathbb{Z}[T]$. The c_{SM} propagator is the polynomial $C_{\mathbb{A}^1}(T) = 1 + T$.
cardona-qft-methods_ F01863	130	■ 0.997	$C_{\{\mathbb{A}^1\}}(T)=1+T$	$C_{\mathbb{A}^1}(T) = 1 + T$	$F_{\hat{Y}_G}(T) \in \mathbb{Z}[T]$. The c_{SM} propagator is the polynomial $C_{\mathbb{A}^1}(T) = 1 + T$.
cardona-qft-methods_ F01864	130	■ 1.000	$C_{\{G\}}(T)=T(1+T)^{n-1}$	$C_G(T) = T(1 + T)^{n-1}$	Example 4.4. If G is an n -sided polygon, then $C_G(T) = T(1 + T)^{n-1}$.
cardona-qft-methods_ F01865	130	■ 1.000	$t_1+\cdots+t_n$	$t_1 + \cdots + t_n$	G is simply $t_1 + \cdots + t_n$. Therefore, $\hat{Y}_G$ consists of the complement of a
cardona-qft-methods_ F01866	130	■ 1.000	$\mathbb{A}^{n-1}$	$\mathbb{A}^{n-1}$	hyperplane $\mathbb{A}^{n-1}$ in $\mathbb{A}^n$. As verified in Example 4.3, $F_{\mathbb{A}^r}(T) = (1 + T)^r$,
cardona-qft-methods_ F01867	130	■ 1.000	$F_{\{\mathbb{A}^r\}}(T)=(1+T)^r$	$F_{\mathbb{A}^r}(T) = (1 + T)^r$	hyperplane $\mathbb{A}^{n-1}$ in $\mathbb{A}^n$. As verified in Example 4.3, $F_{\mathbb{A}^r}(T) = (1 + T)^r$,
cardona-qft-methods_ F01868	131	■ 1.000	$C_{\{G\}}(T)$	$C_G(T)$	A number of general properties of $C_G(T)$ may be established easily; the
cardona-qft-methods_ F01869	131	■ 1.000	T^{n-1}	T^{n-1}	(ii) The coefficient of T^{n-1} in $C_G(T)$ equals $n - b_1(G)$;
cardona-qft-methods_ F01870	131	■ 1.000	$n-b_1(G)$	$n - b_1(G)$	(ii) The coefficient of T^{n-1} in $C_G(T)$ equals $n - b_1(G)$;
cardona-qft-methods_ F01871	131	■ 0.975	$C_{\{G\}}(T)=(1+T)^n$	$C_G(T) = (1 + T)^n$	(iii) If G is a forest with n edges, then $C_G(T) = (1 + T)^n$;
cardona-qft-methods_ F01872	131	■ 1.000	$C_{\{G\}}(0)=0$	$C_G(0) = 0$	(iv) If G is not a forest, then $C_G(0) = 0$;
cardona-qft-methods_ F01873	131	■ 1.000	$C_{\{G^{\prime}\}}(0)$	$C'_G(0)$	(v) $C'_G(0)$ equals the Euler characteristic of $Y_G = \mathbb{P}^{n-1} \setminus X_G$;
cardona-qft-methods_ F01874	131	■ 1.000	$Y_{\{G\}}=\mathbb{P}^{n-1} \backslash X_{\{G\}}$	$Y_G = \mathbb{P}^{n-1} \setminus X_G$	(v) $C'_G(0)$ equals the Euler characteristic of $Y_G = \mathbb{P}^{n-1} \setminus X_G$;
cardona-qft-methods_ F01875	131	■ 1.000	$G^{\{\prime\}}$	G'	(vi) If G' is obtained from G by attaching an edge or splitting an edge, then
cardona-qft-methods_ F01876	131	■ 0.829	$C_{\{G^{\{\prime\}}\}}(T)=(1+T) C_{\{G\}}(T)$	$C_{G'}(T) = (1 + T)C_G(T)$	$C_{G'}(T) = (1 + T)C_G(T)$;
cardona-qft-methods_ F01877	131	■ 1.000	$C_{\{G^{\{\prime\}}\}}(T)=$	$C_{G'}(T) =$	(vii) If G' is obtained from G by attaching a looping edge, then $C_{G'}(T) =$
cardona-qft-methods_ F01878	131	■ 1.000	$T C_{\{G\}}(T)$	$TC_G(T)$	$TC_G(T)$;
cardona-qft-methods_ F01879	131	■ 0.999	$C_{\{G\}}(-1)=0$	$C_G(-1) = 0$	(viii) If G is not 1-particle irreducible, then $C_G(-1) = 0$.
cardona-qft-methods_ F01880	131	■ 1.000	$n-1$	$n - 1$	single looping edge is T ; and then applying (6) $n - 1$ times we obtain again
cardona-qft-methods_ F01881	131	■ 1.000	$T(1+T)^{n-1}$	$T(1 + T)^{n-1}$	that the invariant for an n -sided polygon is $T(1 + T)^{n-1}$.
cardona-qft-methods_ F01882	131	■ 1.000	$C_{\{G_1\}}(T)=\chi\left(Y_{\{G_1\}}\right) T+$	$C_{G_1}(T) = \chi(Y_{G_1})T +$	G_2 are not forests, then $C_{G_1}(T) = \chi(Y_{G_1})T +$ higher order terms and
cardona-qft-methods_ F01883	131	■ 1.000	$C_{\{G_2\}}(T)=\chi\left(Y_{\{G_2\}}\right) T+$	$C_{G_2}(T) = \chi(Y_{G_2})T +$	$C_{G_2}(T) = \chi(Y_{G_2})T +$ h.o.t. By Theorem 4.1,

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cardona-qft-methods_ F01884	131	■ 1.000	$\chi\left(Y_{G_1}\amalg G_2\right)=0$	$\chi(Y_{G_1\amalg G_2}) = 0$	this confirms that $\chi(Y_{G_1\amalg G_2}) = 0$, but tells us that the product
cardona-qft-methods_ F01885	131	■ 1.000	$\chi\left(Y_{G_1}\right)\chi\left(Y_{G_2}\right)$	$\chi(Y_{G_1})\chi(Y_{G_2})$	$\chi(Y_{G_1})\chi(Y_{G_2})$ is not lost in the process: it just appears as the coefficient
cardona-qft-methods_ F01886	131	■ 1.000	$T^{\{2\}}$	T^2	of T^2 in the corresponding c_{SM} Feynman rules.
cardona-qft-methods_ F01887	131	■ 0.803	± 1	± 1	Incidentally: <i>In all examples I know</i> , $\chi(Y_G)$ is always 0 or ± 1 . (A large
cardona-qft-methods_ F01888	131	■ 1.000	$\text{\texttt{Str11}}$	Str11	number of computations for small graphs are collected in. ^{Str11}) I have not
cardona-qft-methods_ F01889	131	■ 1.000	$c_{\text{SM}}\left(\mathbb{P}^{n-1}\backslash X_G\right)$	$c_{\text{SM}}\left(\mathbb{P}^{n-1}\backslash X_G\right)$	obtained from $c_{\text{SM}}(\mathbb{P}^{n-1}\backslash X_G)$ by replacing $[\mathbb{P}^k]$ with T^{k+1} . Naive varia-
cardona-qft-methods_ F01890	131	■ 1.000	$\left[\mathbb{P}^k\right]$	$[\mathbb{P}^k]$	obtained from $c_{\text{SM}}(\mathbb{P}^{n-1}\backslash X_G)$ by replacing $[\mathbb{P}^k]$ with T^{k+1} . Naive varia-
cardona-qft-methods_ F01891	131	■ 1.000	$T^{\{k+1\}}$	T^{k+1}	obtained from $c_{\text{SM}}(\mathbb{P}^{n-1}\backslash X_G)$ by replacing $[\mathbb{P}^k]$ with T^{k+1} . Naive varia-
cardona-qft-methods_ F01892	131	■ 0.997	$c_{\text{SM}}\left(\mathbb{P}^1\backslash X_G\right)$	$c_{\text{SM}}(\mathbb{P}^1\backslash X_G)$	tions on this recipe, e.g., using $c_{\text{SM}}(\mathbb{P}^1\backslash X_G)$ instead, simply do not work. The
cardona-qft-methods_ F01893	132	■ 1.000	$t_1t_2+t_1t_3+t_2t_3$	$t_1t_2+t_1t_3+t_2t_3$	The corresponding graph polynomial is $t_1t_2+t_1t_3+t_2t_3$; therefore, the
cardona-qft-methods_ F01894	132	■ 1.000	$c_{\text{SM}}\left(\mathbb{P}^1\backslash X_G\right)=$	$c_{\text{SM}}(\mathbb{P}^1\backslash X_G) =$	graph hypersurface X_G is a nonsingular conic in $\mathbb{P}^2$, hence $c_{\text{SM}}(\mathbb{P}^1\backslash X_G) =$
cardona-qft-methods_ F01895	132	■ 1.000	$2\left[\mathbb{P}^0\right]+2\left[\mathbb{P}^1\right]$	$2[\mathbb{P}^0]+2[\mathbb{P}^1]$	$2[\mathbb{P}^0]+2[\mathbb{P}^1]$. It follows that $c_{\text{SM}}(\mathbb{P}^2\backslash X_G) = [\mathbb{P}^0]+[\mathbb{P}^1]+[\mathbb{P}^2]$, and therefore
cardona-qft-methods_ F01896	132	■ 1.000	$c_{\text{SM}}\left(\mathbb{P}^2\backslash X_G\right)=\left[\mathbb{P}^0\right]+\left[\mathbb{P}^1\right]+\left[\mathbb{P}^2\right]$	$c_{\text{SM}}(\mathbb{P}^2\backslash X_G) = [\mathbb{P}^0]+[\mathbb{P}^1]+[\mathbb{P}^2]$	$2[\mathbb{P}^0]+2[\mathbb{P}^1]$. It follows that $c_{\text{SM}}(\mathbb{P}^2\backslash X_G) = [\mathbb{P}^0]+[\mathbb{P}^1]+[\mathbb{P}^2]$, and therefore
cardona-qft-methods_ F01897	132	■ 0.651	$C_G(T)=T\left(1+T+T^2\right)$	$C_G(T) = T(1+T+T^2)$	$C_G(T) = T(1+T+T^2)$.
cardona-qft-methods_ F01898	132	■ 1.000	$\text{\texttt{Web12}}$	Web12	natural proof) in an <i>equivariant</i> setting; see, ^{Web12} Proposition 3.
cardona-qft-methods_ F01899	132	■ 1.000	$\text{\texttt{Bol98}}$	Bol98	cializations of the ‘Tutte polynomial’ (see e.g., ^{Bol98} Chapter X). This poly-
cardona-qft-methods_ F01900	132	■ 1.000	$T_G(x,y)=x^iy^j$	$T_G(x,y) = x^iy^j$	polynomial is $T_G(x,y) = x^iy^j$ if G contains i bridges, j looping edges, and
cardona-qft-methods_ F01901	132	■ 0.999	$G/e, G\backslash e$	$G/e, G\backslash e$	$G/e, G\backslash e$ denote respectively the operations of contracting an edge e and
cardona-qft-methods_ F01902	132	■ 1.000	$R[\alpha, \beta, \gamma, x, y]$	$R[\alpha, \beta, \gamma, x, y]$	of graphs to $R[\alpha, \beta, \gamma, x, y]$, where R is a ring (usually taken to be $\mathbb{C}$) such
cardona-qft-methods_ F01903	132	■ 1.000	$\tau(G)=\gamma^{\#\text{ vertices }}$	$\tau(G) = \gamma^{\#\text{ vertices }}$	• $\tau(G) = \gamma^{\#\text{ vertices }}$ if G has no edges;
cardona-qft-methods_ F01904	132	■ 1.000	$\tau(G)=x\tau(G\backslash e)$	$\tau(G) = x\tau(G\backslash e)$	• $\tau(G) = x\tau(G\backslash e)$ if e is a bridge;
cardona-qft-methods_ F01905	132	■ 1.000	$\tau(G)=y\tau(G/e)$	$\tau(G) = y\tau(G/e)$	• $\tau(G) = y\tau(G/e)$ if e is a looping edge;
cardona-qft-methods_ F01906	132	■ 0.999	$\tau(G)=\alpha\tau(G/e)+\beta\tau(G\backslash e)$	$\tau(G) = \alpha\tau(G/e) + \beta\tau(G\backslash e)$	• $\tau(G) = \alpha\tau(G/e) + \beta\tau(G\backslash e)$ if e is neither a bridge nor a looping edge.
cardona-qft-methods_ F01907	133	■ 1.000	$\alpha=\beta=\gamma=1$	$\alpha = \beta = \gamma = 1$	The Tutte polynomial is the special case $\alpha = \beta = \gamma = 1$.
cardona-qft-methods_ F01908	133	■ 1.000	$b_0(G)$	$b_0(G)$	satisfies the recursion spelled out above, where $b_0(G)$ is the number of
cardona-qft-methods_ F01909	134	■ 1.000	$x=1-\lambda, y=0$	$x = 1 - \lambda, y = 0$	covered from the Tutte polynomial by setting $x = 1 - \lambda, y = 0$ (and taking

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cardona-qft-methods_ F01910	134	■ 1.000	$(1-\lambda)^2+(1-\lambda)=(1-\lambda)(2-\lambda)$	$(1-\lambda)^2+(1-\lambda)=(1-\lambda)(2-\lambda)$	this gives $(1-\lambda)^2+(1-\lambda)=(1-\lambda)(2-\lambda)$, which equals the chromatic
cardona-qft-methods_ F01911	134	■ 1.000	$T_{\{G\}}(x, y)$	$T_G(x, y)$	factor is x . Also, the recursive definition of $T_G(x, y)$ implies a formula for
cardona-qft-methods_ F01912	134	■ 1.000	$G_{\{2\ e\}}$	G_{2e}	doubling a single edge in a graph: if e is an edge of G , denote by G_{2e} the
cardona-qft-methods_ F01913	134	■ 0.998	x^2+x+y	x^2+x+y	polynomial x^2+x+y for the triangle and the polynomial $x+y$ for a 2-
cardona-qft-methods_ F01914	134	■ 0.998	$x+y$	$x+y$	polynomial x^2+x+y for the triangle and the polynomial $x+y$ for a 2-
cardona-qft-methods_ F01915	135	■ 0.997	$\{ \}^{\text{\texttt{AM11b}}}$	AM11b	found in ^{AM11b} and. ^{Alua}
cardona-qft-methods_ F01916	135	■ 0.997	$\{ \}^{\text{\texttt{Alua}}}$	Alua	found in ^{AM11b} and. ^{Alua}
cardona-qft-methods_ F01917	135	■ 0.453	$\mathbb{U}(G)=\mathbb{L}\cdot\left[\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}\right]-\mathbb{U}(G\setminus e)$	$\mathbb{U}(G)=\mathbb{L}\cdot\left[\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}\right]-\mathbb{U}(G\setminus e)$	• $\mathbb{U}(G)=\mathbb{L}\cdot\left[\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}\right]-\mathbb{U}(G\setminus e)$,
cardona-qft-methods_ F01918	135	■ 1.000	$\mathbb{U}(\left(G_{\{2\ e\}}\right))=(\mathbb{T}-1)\cdot\mathbb{U}(G)+\mathbb{T}\cdot\mathbb{U}(G\setminus e)+(\mathbb{T}+1)\cdot\mathbb{U}(G/e)$	$\mathbb{U}(G_{2e})=(\mathbb{T}-1)\cdot\mathbb{U}(G)+\mathbb{T}\cdot\mathbb{U}(G\setminus e)+(\mathbb{T}+1)\cdot\mathbb{U}(G/e)$	• $\mathbb{U}(G_{2e})=(\mathbb{T}-1)\cdot\mathbb{U}(G)+\mathbb{T}\cdot\mathbb{U}(G\setminus e)+(\mathbb{T}+1)\cdot\mathbb{U}(G/e)$.
cardona-qft-methods_ F01919	135	■ 1.000	$\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}$	$\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}$	combinatorial: the term in the middle involves a locus, $\hat{Y}_{G\setminus e}\cup\hat{Y}_{G/e}$, which
cardona-qft-methods_ F01920	135	■ 1.000	$\mathbb{L}^3-\mathbb{L}^2=\mathbb{T}(\mathbb{T}+1)^2, (\mathbb{T}+1)^2, \mathbb{T}(\mathbb{T}+1)$	$\mathbb{L}^3-\mathbb{L}^2=\mathbb{T}(\mathbb{T}+1)^2, (\mathbb{T}+1)^2, \mathbb{T}(\mathbb{T}+1)$	in Figure 1, to wit $\mathbb{L}^3-\mathbb{L}^2=\mathbb{T}(\mathbb{T}+1)^2, (\mathbb{T}+1)^2, \mathbb{T}(\mathbb{T}+1)$ (exercise!), we
cardona-qft-methods_ F01921	135	■ 0.997	$G, G\setminus e, G/e$	$G, G\setminus e, G/e$	Fig. 1. $G, G\setminus e, G/e$.
cardona-qft-methods_ F01922	135	■ 0.999	$\mathbb{T}(\mathbb{T}+1)^2$	$\mathbb{T}(\mathbb{T}+1)^2$	edge in a 3-banana, and the class of a 3-banana is $\mathbb{T}(\mathbb{T}+1)^2$. A complete
cardona-qft-methods_ F01923	136	■ 1.000	$G\setminus e, G/e$	$G\setminus e, G/e$	$G\setminus e, G/e$ are polynomially countable, then so are all G_{me} .
cardona-qft-methods_ F01924	136	■ 1.000	$G_{\{m\ e\}}$	G_{me}	$G\setminus e, G/e$ are polynomially countable, then so are all G_{me} .
cardona-qft-methods_ F01925	136	■ 1.000	$T_{\{y\}}$	T_y	are functions used in defining Hirzebruch's T_y genus, cf. §11 of Chapter III
cardona-qft-methods_ F01926	136	■ 1.000	$\{ \}^{\text{\texttt{Hir95}}}$	Hir95	of. ^{Hir95}
cardona-qft-methods_ F01927	136	■ 1.000	$\mathbb{U}(G_{\{m\ e\}})$	$\mathbb{U}(G_{me})$	(iii) The generating function for $\mathbb{U}(G_{me})$ implies that
cardona-qft-methods_ F01928	136	■ 1.000	$m\geq 2$	$m\geq 2$	for $m\geq 2$, still under the assumption that e is neither a bridge nor a looping
cardona-qft-methods_ F01929	136	■ 1.000	$\chi(Y_{G\setminus e})$	$\chi(Y_{G/e})$	necessarily the case, then this formula would imply that $\chi(Y_G)$ and $\chi(Y_{G/e})$
cardona-qft-methods_ F01930	136	■ 0.590	$G=\mathrm{a}^2$	$G=\mathbf{a}2$	(iv) For $G=\mathbf{a}2$ -banana (so that $\mathbb{U}(G)=\mathbb{T}(\mathbb{T}+1)$, $\mathbb{U}(G\setminus e)=\mathbb{T}+1$,
cardona-qft-methods_ F01931	136	■ 0.590	$\mathbb{U}(G)=\mathbb{T}(\mathbb{T}+1), \mathbb{U}(G\setminus e)=\mathbb{T}+1$	$\mathbb{U}(G)=\mathbb{T}(\mathbb{T}+1), \mathbb{U}(G\setminus e)=\mathbb{T}+1$	(iv) For $G=\mathbf{a}2$ -banana (so that $\mathbb{U}(G)=\mathbb{T}(\mathbb{T}+1)$, $\mathbb{U}(G\setminus e)=\mathbb{T}+1$,
cardona-qft-methods_ F01932	136	■ 0.998	$\mathbb{U}(G/e)=\mathbb{T}$	$\mathbb{U}(G/e)=\mathbb{T}$	and $\mathbb{U}(G/e)=\mathbb{T}$), we obtain a generating function for the class of $G_{me} =$
cardona-qft-methods_ F01933	136	■ 0.998	$G_{\{m\ e\}}=$	$G_{me} =$	and $\mathbb{U}(G/e)=\mathbb{T}$), we obtain a generating function for the class of $G_{me} =$
cardona-qft-methods_ F01934	136	■ 1.000	$(m+1)$	$(m+1)$	the $(m+1)$ -th banana. This recovers the result of the direct computations
cardona-qft-methods_ F01935	137	■ 1.000	G/e	G/e	maximal forests of G/e . Denoting by t_e the variable corresponding to e ,

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cardona-qft-methods_ F01936	137	■ 0.998	$\underline{t} \in \mathbb{A}^n, n =$	$\underline{t} \in \mathbb{A}^n, n =$	Now let $\underline{t} \in \mathbb{A}^n$, n =number of edges of G . We have
cardona-qft-methods_ F01937	137	■ 1.000	$\underline{t}^{\prime}$	$\underline{t}'$	where $\underline{t}'$ denotes omission of t_e . Now a simple case-by-case analysis proves
cardona-qft-methods_ F01938	137	■ 1.000	$\underline{t}^{\prime} \notin Y_{G \setminus e}$	$\underline{t}' \notin Y_{G \setminus e}$	the first formula in Theorem 4.2. For example, the case in which $\underline{t}' \notin Y_{G \setminus e}$
cardona-qft-methods_ F01939	137	■ 1.000	$\underline{t}^{\prime} \in Y_{G/e}$	$\underline{t}' \in Y_{G/e}$	and $\underline{t}' \in Y_{G/e}$ accounts for a term
cardona-qft-methods_ F01940	137	■ 1.000	$\mathbb{L}$	$\mathbb{L}$	with the $\mathbb{L}$ factor due to the freely varying t_e . Adding the terms obtained
cardona-qft-methods_ F01941	137	■ 1.000	$e^{\prime}$	e'	For the second part, assume that e and e' are parallel edges in G_{2e} .
cardona-qft-methods_ F01942	137	■ 0.999	$\left[Y_{G_{2e} \setminus e} \cup \right.$	$\left[Y_{G_{2e}/e} \cup \right.$	This formula can be used to obtain a relation between the classes $[Y_{G_{2e}/e} \cup$
cardona-qft-methods_ F01943	137	■ 0.950	$\left. Y_{G_{2e} \setminus e} \right]$	$Y_{G_{2e} \setminus e}]$	$Y_{G_{2e} \setminus e}]$ and $[Y_{G/e} \cup Y_{G \setminus e}]$. A serendipitous cancellation gets rid of these
cardona-qft-methods_ F01944	137	■ 0.950	$\left[Y_{G/e} \cup Y_{G \setminus e} \right]$	$[Y_{G/e} \cup Y_{G \setminus e}]$	$Y_{G_{2e} \setminus e}]$ and $[Y_{G/e} \cup Y_{G \setminus e}]$. A serendipitous cancellation gets rid of these
cardona-qft-methods_ F01945	138	■ 1.000	$C_{G \setminus e, G/e}(T)$	$C_{G \setminus e, G/e}(T)$	Here, $C_{G \setminus e, G/e}(T)$ is a ‘non-combinatorial term’ obtained from the
cardona-qft-methods_ F01946	138	■ 1.000	$\Psi_{G/e}(T)$	$\Psi_{G/e}(T)$	$\Psi_{G/e}(T)$ is required to belong to the Jacobian ideal of $\Psi_{G \setminus e}$. It is a little
cardona-qft-methods_ F01947	138	■ 1.000	$\Psi_{G \setminus e}$	$\Psi_{G \setminus e}$	$\Psi_{G/e}(T)$ is required to belong to the Jacobian ideal of $\Psi_{G \setminus e}$. It is a little
cardona-qft-methods_ F01948	138	■ 0.573	$C_{3\text{-}\text{banana}}(T) =$	$C_{3\text{-}\text{banana}}(T) =$	may be obtained by splitting an edge of a 3-banana, and $C_{3\text{-}\text{banana}}(T) =$
cardona-qft-methods_ F01949	138	■ 1.000	$T(1 + T + T^2)$	$T(1 + T + T^2)$	$T(1 + T + T^2)$ (Example 4.5).
cardona-qft-methods_ F01950	139	■ 1.000	$m \geq 1$	$m \geq 1$	for $m \geq 1$. Note the differentiation relations among the coefficients; I was
cardona-qft-methods_ F01951	139	■ 1.000	$p \in \mathbb{P}^{n-1}$	$p \in \mathbb{P}^{n-1}$	G has n edges and (for simplicity) that X_G is irreducible. Let $p \in \mathbb{P}^{n-1}$ be
cardona-qft-methods_ F01952	139	■ 1.000	$B_{\ell_p} X_G$	$B_{\ell_p} X_G$	(1) to verify that the blow-up $B_{\ell_p} X_G$ of X_G at p is isomorphic to the blow-
cardona-qft-methods_ F01953	139	■ 1.000	$\mathbb{P}^{n-2}$	$\mathbb{P}^{n-2}$	up of $\mathbb{P}^{n-2}$ along $X_{G \setminus e} \cap X_{G/e}$. (Note that it follows that irreducible graph
cardona-qft-methods_ F01954	139	■ 1.000	$X_{G \setminus e} \cap X_{G/e}$	$X_{G \setminus e} \cap X_{G/e}$	up of $\mathbb{P}^{n-2}$ along $X_{G \setminus e} \cap X_{G/e}$. (Note that it follows that irreducible graph
cardona-qft-methods_ F01955	139	■ 0.998	$X \times \mathbb{P}^{\ell}$	$X \times \mathbb{P}^{\ell}$	varieties X, Y are <i>stably birationally equivalent</i> if $X \times \mathbb{P}^{\ell}$ and $Y \times \mathbb{P}^m$ are
cardona-qft-methods_ F01956	139	■ 0.998	$Y \times \mathbb{P}^m$	$Y \times \mathbb{P}^m$	varieties X, Y are <i>stably birationally equivalent</i> if $X \times \mathbb{P}^{\ell}$ and $Y \times \mathbb{P}^m$ are
cardona-qft-methods_ F01957	139	■ 1.000	ℓ, m	ℓ, m	birationally equivalent for some ℓ, m .
cardona-qft-methods_ F01958	140	■ 1.000	X_1	X_1	birational equivalence is a weakening of this condition. If X_1 is (stably)
cardona-qft-methods_ F01959	140	■ 1.000	X_2	X_2	birationally equivalent to X_2 and Y_1 is (stably) birationally equivalent to
cardona-qft-methods_ F01960	140	■ 1.000	Y_1	Y_1	birationally equivalent to X_2 and Y_1 is (stably) birationally equivalent to
cardona-qft-methods_ F01961	140	■ 1.000	Y_2	Y_2	Y_2 , then $X_1 \times Y_1$ is (stably) birationally equivalent to $X_2 \times Y_2$.
cardona-qft-methods_ F01962	140	■ 1.000	$X_1 \times Y_1$	$X_1 \times Y_1$	Y_2 , then $X_1 \times Y_1$ is (stably) birationally equivalent to $X_2 \times Y_2$.

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cardona-qft-methods_ F01963	140	■ 1.000	$X_2 \times Y_2$	$X_2 \times Y_2$	Y_2 , then $X_1 \times Y_1$ is (stably) birationally equivalent to $X_2 \times Y_2$.
cardona-qft-methods_ F01964	140	■ 1.000	≥ 2	≥ 2	dimension ≥ 2 , every variety is birationally equivalent to infinitely many
cardona-qft-methods_ F01965	140	■ 1.000	$\{ \}^{\text{AKMW02}}$	AKMW02	‘weak factorization theorem’ of, ^{AKMW02} a fundamental tool in the study of
cardona-qft-methods_ F01966	140	■ 0.873	$\left(\{ \}^{\text{LL03}} \right)$	$\left(\text{LL03} \right)$	Lunts define (LL03) a <i>ring</i> of stable birational equivalence classes as follows:
cardona-qft-methods_ F01967	140	■ 1.000	$S \ B$	SB	• Let SB be the monoid of stable birational equivalence classes of smooth
cardona-qft-methods_ F01968	140	■ 1.000	$X \times Y$	$X \times Y$	irreducible compact varieties, defining the class of $X \times Y$ as the product
cardona-qft-methods_ F01969	140	■ 1.000	$\mathbb{Z}[S \ B]$	$\mathbb{Z}[SB]$	• Then let $\mathbb{Z}[SB]$ be the corresponding monoid ring.
cardona-qft-methods_ F01970	140	■ 1.000	$[X]_{S \ B}$	$[X]_{SB}$	Denote by $[X]_{SB}$ the class of X in $\mathbb{Z}[SB]$. Thus, elements of $\mathbb{Z}[SB]$ may
cardona-qft-methods_ F01971	140	■ 1.000	$[X]_{S \ B}=1$	$[X]_{SB} = 1$	smooth irreducible compact <i>and</i> rational, then $[X]_{SB} = 1$.
cardona-qft-methods_ F01972	141	■ 1.000	$\mathbb{P}^1 \backslash \mathbb{P}^0$	$\mathbb{P}^1 \setminus \mathbb{P}^0$	$\mathbb{P}^1 \setminus \mathbb{P}^0$, hence map it to $[\mathbb{P}^1]_{SB} - [\mathbb{P}^0]_{SB} = 1 - 1 = 0$.
cardona-qft-methods_ F01973	141	■ 1.000	$\left[\mathbb{P}^1 \right]_{S \ B} - \left[\mathbb{P}^0 \right]_{S \ B} = 1 - 1 = 0$	$[\mathbb{P}^1]_{SB} - [\mathbb{P}^0]_{SB} = 1 - 1 = 0$	$\mathbb{P}^1 \setminus \mathbb{P}^0$, hence map it to $[\mathbb{P}^1]_{SB} - [\mathbb{P}^0]_{SB} = 1 - 1 = 0$.
cardona-qft-methods_ F01974	141	■ 0.922	$\left[\mathbb{P}^1 \right]_{S \ B} -$	$[\mathbb{P}^1]_{SB} -$	may be realized as the complement of 2 points in $\mathbb{P}^1$, so it has class $[\mathbb{P}^1]_{SB} -$
cardona-qft-methods_ F01975	141	■ 1.000	$2 \left[\mathbb{P}^0 \right]_{S \ B} = -1$	$2 [\mathbb{P}^0]_{SB} = -1$	$2[\mathbb{P}^0]_{SB} = -1$; adding the node back gives 0.
cardona-qft-methods_ F01976	141	■ 0.999	$\{ \}^{\text{AM11c}}$	AM11c	This is Corollary 3.3 in, ^{AM11c} its proof is very easy modulo, ^{LL03} and is
cardona-qft-methods_ F01977	141	■ 0.999	$\{ \}^{\text{LL03}}$	LL03	This is Corollary 3.3 in, ^{AM11c} its proof is very easy modulo, ^{LL03} and is
cardona-qft-methods_ F01978	141	■ 0.998	$X \mapsto [X]_{S \ B}$	$X \mapsto [X]_{SB}$	<i>The assignment $X \mapsto [X]_{SB}$ for X smooth, irreducible, complete descends</i>
cardona-qft-methods_ F01979	141	■ 0.996	$K_0(\text{Var}) \rightarrow \mathbb{Z}[S \ B]$	$K_0(\text{Var}) \rightarrow \mathbb{Z}[SB]$	<i>to an onto ring homomorphism $K_0(\text{Var}) \rightarrow \mathbb{Z}[SB]$, whose kernel is the ideal</i>
cardona-qft-methods_ F01980	141	■ 1.000	$(\mathbb{L})$	$(\mathbb{L})$	$(\mathbb{L})$.
cardona-qft-methods_ F01981	142	■ 1.000	$B \times \mathbb{P}^r$	$B \times \mathbb{P}^r$	indeed, E is birationally equivalent to $B \times \mathbb{P}^r$ for some r . It follows that if X
cardona-qft-methods_ F01982	142	■ 1.000	$B \subset X$	$B \subset X$	is nonsingular and compact, and $B \subset X$ is a nonsingular closed subvariety,
cardona-qft-methods_ F01983	142	■ 1.000	$B \ell_B X$	$B\ell_B X$	where $B\ell_B X$ is the blow-up of X along B , and E is the exceptional divisor.
cardona-qft-methods_ F01984	142	■ 0.903	$) \rightarrow \mathbb{Z}[S \ B]$	$) \rightarrow \mathbb{Z}[SB]$	This shows that there is a homomorphism $K_0(\text{Var}) \rightarrow \mathbb{Z}[SB]$ as claimed,
cardona-qft-methods_ F01985	142	■ 0.998	$\left[B \ell_B X \right] \equiv [X] \text{ modulo } (\mathbb{L})$	$[B\ell_B X] \equiv [X] \text{ modulo } (\mathbb{L})$	ters, the key point is that $[B\ell_B X] \equiv [X] \text{ modulo } (\mathbb{L})$ in $K_0(\text{Var})$.
cardona-qft-methods_ F01986	142	■ 1.000	$F \rightarrow \mathbb{Z}[S \ B]$	$F \rightarrow \mathbb{Z}[SB]$	$\mathbb{U}(G)$ of §4.1 modulo $\mathbb{L}$: this defines a homomorphism $F \rightarrow \mathbb{Z}[SB]$ from
cardona-qft-methods_ F01987	142	■ 1.000	$\left[\hat{Y}_G \right] = \mathbb{U}(G)$	$[\hat{Y}_G] = \mathbb{U}(G)$	of X_G modulo $\mathbb{L}$. Equivalently, we can compute the class of $[\hat{Y}_G] = \mathbb{U}(G)$
cardona-qft-methods_ F01988	143	■ 1.000	$\mathbb{U}(G) \bmod$	$\mathbb{U}(G) \bmod$	(as we saw in §4.1). These recursions reduce the computation of $\mathbb{U}(G) \bmod$
cardona-qft-methods_ F01989	143	■ 1.000	$\mathbb{Z} \rightarrow \mathbb{Z}[S \ B]$	$\mathbb{Z} \rightarrow \mathbb{Z}[SB]$	by the inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[SB]$.

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cardona-qft-methods_ F01990	143	■ 0.939	15 \mathrm{in}^{\text{BS12}}	$15\text{in}^{\text{BS12}}$	looping edge may be strengthened: Lemma 15 in ^{BS12} shows that $[\hat{Y}_G]$ is a
cardona-qft-methods_ F01991	143	■ 1.000	$\mathbb{L}^{\wedge 2}$	$\mathbb{L}^2$	multiple of $\mathbb{L}^2$ for all ‘physically significant’ graphs.
cardona-qft-methods_ F01992	143	■ 0.475	$c+\mathbb{L}\cdot M^{\{\prime\}}$	$c+\mathbb{L}\cdot M'$	$c+\mathbb{L}\cdot M'$ for some $c\in\mathbb{Z}$ and $M'\in K_0(\text{Var})$. This is very special—for
cardona-qft-methods_ F01993	143	■ 0.475	$c\in\mathbb{Z}$	$c\in\mathbb{Z}$	$c+\mathbb{L}\cdot M'$ for some $c\in\mathbb{Z}$ and $M'\in K_0(\text{Var})$. This is very special—for
cardona-qft-methods_ F01994	143	■ 0.475	$M^{\{\prime\}}\in K_{\{0\}}(\mathrm{Var})$	$M'\in K_0(\text{Var})$	$c+\mathbb{L}\cdot M'$ for some $c\in\mathbb{Z}$ and $M'\in K_0(\text{Var})$. This is very special—for
cardona-qft-methods_ F01995	143	■ 1.000	$h^{\{p,\,q\}}$	$h^{p,q}$	example, it implies that the Hodge numbers $h^{p,q}$ of M must be zero if
cardona-qft-methods_ F01996	143	■ 1.000	$p=0,\,q>0$	$p=0,\,q>0$	$p=0,\,q>0$ or $p>0,\,q=0$.
cardona-qft-methods_ F01997	143	■ 1.000	$p>0,\,q=0$	$p>0,\,q=0$	$p=0,\,q>0$ or $p>0,\,q=0$.
cardona-qft-methods_ F01998	144	■ 1.000	$\mathbb{L}^{\wedge n}-1\,(n>0)$	$\mathbb{L}^n-1\,(n>0)$	Brosnan work in a <i>localization</i> of $K_0(\text{Var}_{\mathbb{Z}})$, in which both $\mathbb{L}^n-1$ ($n>0$)
cardona-qft-methods_ F01999	144	■ 1.000	$\mathbb{L}^{\wedge n}-1$	$\mathbb{L}^n-1$	if only $\mathbb{L}^n-1$ are inverted. (Indeed, these elements are already invertible
cardona-qft-methods_ F02000	144	■ 1.000	$K\left(\operatorname{Var}_{\mathbb{Z}}[\mathbb{L}]\right)_{\mathbb{L}}$	$K(\text{Var}_{\mathbb{Z}})_{\mathbb{L}}$	<i>Question:</i> Do graph hypersurfaces generate $K(\text{Var}_{\mathbb{Z}})_{\mathbb{L}}$ over $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$?
cardona-qft-methods_ F02001	144	■ 1.000	$\mathbb{Z}[\mathbb{L}][\mathbb{L}]_{\mathbb{L}}$	$\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$	<i>Question:</i> Do graph hypersurfaces generate $K(\text{Var}_{\mathbb{Z}})_{\mathbb{L}}$ over $\mathbb{Z}[\mathbb{L}]_{\mathbb{L}}$?
cardona-qft-methods_ F02002	145	■ 1.000	ϕ^4	ϕ^4	BS12. Francis Brown and Oliver Schnetz. A K3 in ϕ^4 . <i>Duke Math. J.</i> ,
cardona-qft-methods_ F02003	149	■ 1.000	$\{ \}^{\{2,3\}}$	2,3	applications in physics have been written, ^{2,3} for which these notes are no
cardona-qft-methods_ F02004	149	■ 1.000	$\{ \}^{\{4-6\}}$	$^{4-6}$	applications of CS gravities, like. ⁴⁻⁶ Here, the emphasis is on the construc-
cardona-qft-methods_ F02005	149	■ 1.000	$S_{\{0(D,\,1)\}}$	$SO(D,1)$	theories: those with $SO(D,1)$ (de Sitter), $SO(D-1,2)$ (anti-de Sitter), or
cardona-qft-methods_ F02006	149	■ 1.000	$S_{\{0(D-1,2)\}}$	$SO(D-1,2)$	theories: those with $SO(D,1)$ (de Sitter), $SO(D-1,2)$ (anti-de Sitter), or
cardona-qft-methods_ F02007	149	■ 0.926	$ISO_{\{0(D-1,1)\}}$	$ISO(D-1,1)$	$ISO(D-1,1)$ (Poincaré), symmetry groups. Section 5 shows how CS forms
cardona-qft-methods_ F02008	149	■ 1.000	$2p$	$2p$	larities, or zero-branes), or extended localized sources such as $2p$ -branes.
cardona-qft-methods_ F02009	149	■ 1.000	$t_{\{1\}<0}$	$t_1<0$	<i>write There is a $t_1<0$ such that the image of t_1 under the natural mapping</i>
cardona-qft-methods_ F02010	149	■ 1.000	$t_{\{1\}}$	t_1	<i>write There is a $t_1<0$ such that the image of t_1 under the natural mapping</i>
cardona-qft-methods_ F02011	149	■ 0.985	$t_{\{1\}}\rightarrow\operatorname{Petya}\left(t_{\{1\}}\right)$	$t_1\rightarrow\text{Petya}(t_1)$	$t_1\rightarrow\text{Petya}(t_1)$ belongs to the set of dirty hands, and a t_2 , $t_1<t_2\leq 0$,
cardona-qft-methods_ F02012	149	■ 0.985	$t_{\{2\}},\,t_{\{1\}<t_{\{2\}}\}\leq 0$	$t_2,\,t_1<t_2\leq 0$	$t_1\rightarrow\text{Petya}(t_1)$ belongs to the set of dirty hands, and a t_2 , $t_1<t_2\leq 0$,
cardona-qft-methods_ F02013	149	■ 1.000	$t_{\{2\}}$	t_2	<i>such that the image of t_2 under the above-mentioned mappings belongs to</i>
cardona-qft-methods_ F02014	150	■ 1.000	$\mathbb{G}$	$\mathbb{G}$	theory i.e., the structure group $\mathbb{G}$ of the principal fibre bundle over the
cardona-qft-methods_ F02015	150	■ 1.000	$\forall x\in M,\,g(x)\in\mathbb{G}$	$\forall x\in M,\,g(x)\in\mathbb{G}$	where $\forall x\in M,\,g(x)\in\mathbb{G}$ defines a gauge transformation that can be
cardona-qft-methods_ F02016	150	■ 0.999	$\mathcal{C}(\mathbf{A})$	$\mathcal{C}(\mathbf{A})$	form, but in general $\mathcal{C}(\mathbf{A})$ are $(2n+1)$ -forms with integer $n\geq 0$.

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cardona-qft-methods_F02017	150	■ 0.999	(2 n+1)	$(2n + 1)$	form, but in general $\mathcal{C}(\mathbf{A})$ are $(2n + 1)$ -forms with integer $n \geq 0$.
cardona-qft-methods_F02018	150	■ 0.999	$n \geq 0$	$n \geq 0$	form, but in general $\mathcal{C}(\mathbf{A})$ are $(2n + 1)$ -forms with integer $n \geq 0$.
cardona-qft-methods_F02019	150	■ 1.000	A_{μ}	A_{μ}	• The coupling between the electromagnetic potential A_{μ} and an external
cardona-qft-methods_F02020	151	■ 1.000	$\partial_{\mu} j^{\mu} = 0$	$\partial_{\mu} j^{\mu} = 0$	law $\partial_{\mu} j^{\mu} = 0$ does not require a metric, since ∂ is the ordinary derivative,
cardona-qft-methods_F02021	151	■ 1.000	∂	∂	law $\partial_{\mu} j^{\mu} = 0$ does not require a metric, since ∂ is the ordinary derivative,
cardona-qft-methods_F02022	151	■ 0.686	$(\mathbb{G})$	$(\mathbb{G})$	under a group of transformations $(\mathbb{G})$ by a total derivative $d\Omega$, where Ω is
cardona-qft-methods_F02023	151	■ 0.686	$d \Omega$	$d\Omega$	under a group of transformations $(\mathbb{G})$ by a total derivative $d\Omega$, where Ω is
cardona-qft-methods_F02024	151	■ 1.000	$(2 n)$	$(2n)$	an arbitrary $(2n)$ -form, then $\mathcal{C}$ makes a good candidate for the Lagrangian of
cardona-qft-methods_F02025	151	■ 1.000	j^{μ}	j^{μ}	defined by a conserved current, analogous to j^{μ} . As we will show by the
cardona-qft-methods_F02026	151	■ 1.000	∂M	∂M	be turned into a a surface integral over ∂M , which can be dropped for
cardona-qft-methods_F02027	151	■ 1.000	$d \star j = 0$	$d \star j = 0$	only if $d \star j = 0$. Hence, the “minimal coupling” (5) is $\mathbb{G}$ -invariant provided
cardona-qft-methods_F02028	151	■ 0.989	$\{ \}^{\text{a}}$	a	^a For a p -form $\alpha = (p!)^{-1} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p}$, its $\star$ -dual is the $(D - p)$ -
cardona-qft-methods_F02029	151	■ 0.989	$\alpha = (p!)^{-1} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p}$	$\alpha = (p!)^{-1} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p}$	^a For a p -form $\alpha = (p!)^{-1} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p}$, its $\star$ -dual is the $(D - p)$ -
cardona-qft-methods_F02030	151	■ 0.989	$(D - p)$	$(D - p)$	^a For a p -form $\alpha = (p!)^{-1} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p}$, its $\star$ -dual is the $(D - p)$ -
cardona-qft-methods_F02031	151	■ 0.694	$\star \alpha = [p!(D - p)!]^{-1} \epsilon_{\nu_1 \nu_2 \dots \nu_{D-p}}^{\mu_1 \mu_2 \dots \mu_p} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\nu_1} \wedge dx^{\nu_2} \wedge \dots \wedge dx^{\nu_{D-p}}$	$\star \alpha = [p!(D - p)!]^{-1} \epsilon_{\nu_1 \nu_2 \dots \nu_{D-p}}^{\mu_1 \mu_2 \dots \mu_p} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\nu_1} \wedge dx^{\nu_2} \wedge \dots \wedge dx^{\nu_{D-p}}$	form $\star \alpha = [p!(D - p)!]^{-1} \epsilon_{\nu_1 \nu_2 \dots \nu_{D-p}}^{\mu_1 \mu_2 \dots \mu_p} \alpha_{\mu_1 \mu_2 \dots \mu_p} dx^{\nu_1} \wedge dx^{\nu_2} \wedge \dots \wedge dx^{\nu_{D-p}}$.
cardona-qft-methods_F02032	152	■ 1.000	$\star j$	$\star j$	dimensional, a $(2n + 1)$ -form $\mathcal{C}$ couples to a source j , whose dual $\star j$ is a
cardona-qft-methods_F02033	152	■ 0.998	$(D - 2n - 1)$	$(D - 2n - 1)$	$(D - 2n - 1)$ form.
cardona-qft-methods_F02034	152	■ 1.000	$j =$	$j =$	In electromagnetism, the current 1-form that enters in (4) is $j =$
cardona-qft-methods_F02035	152	■ 1.000	$q \delta(\Gamma) dz$	$q \delta(\Gamma) dz$	$q \delta(\Gamma) dz$, where $\delta(\Gamma)$ is the Dirac delta with support on the history traced
cardona-qft-methods_F02036	152	■ 1.000	$\delta(\Gamma)$	$\delta(\Gamma)$	$q \delta(\Gamma) dz$, where $\delta(\Gamma)$ is the Dirac delta with support on the history traced
cardona-qft-methods_F02037	152	■ 1.000	$2 n$	$2n$	history of a $2n$ -dimensional membrane in the D -dimensional spacetime.
cardona-qft-methods_F02038	152	■ 1.000	$\mathbf{K}_a (a = 1, 2, \dots, N)$	$\mathbf{K}_a (a = 1, 2, \dots, N)$	where $\mathbf{K}_a$ ($a = 1, 2, \dots, N$) provide a matrix representation for the genera-
cardona-qft-methods_F02039	152	■ 1.000	$\mathbf{K}_a$	$\mathbf{K}_a$	tors of the gauge group $\mathbb{G}$. That is, the $\mathbf{K}_a$ ’s form a basis of the Lie algebra
cardona-qft-methods_F02040	152	■ 1.000	$g(x) = \exp [\alpha^a(x) \mathbf{K}_a]$	$g(x) = \exp [\alpha^a(x) \mathbf{K}_a]$	representation of the gauge group. If $g(x) = \exp [\alpha^a(x) \mathbf{K}_a]$ is an element of
cardona-qft-methods_F02041	152	■ 1.000	$D = d - \mathbf{A}$	$D = d - \mathbf{A}$	derivative, $D = d - \mathbf{A}$, that generalizes $\partial_{\mu} - i A_{\mu}$ in electrodynamics, or
cardona-qft-methods_F02042	152	■ 1.000	$\partial_{\mu} - i A_{\mu}$	$\partial_{\mu} - i A_{\mu}$	derivative, $D = d - \mathbf{A}$, that generalizes $\partial_{\mu} - i A_{\mu}$ in electrodynamics, or

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cardona-qft-methods_ F02043	152	■ 1.000	$\nabla_{\mu}=\partial_{\mu}+\boldsymbol{\Gamma}_{\mu}$	$\nabla_{\mu} = \partial_{\mu} + \boldsymbol{\Gamma}_{\mu}$	$\nabla_{\mu} = \partial_{\mu} + \boldsymbol{\Gamma}_{\mu}$ in Riemannian geometry.
cardona-qft-methods_ F02044	152	■ 1.000	$\mathbf{F}=d\mathbf{A}+\mathbf{A}\wedge\mathbf{A}$	$\mathbf{F} = d\mathbf{A} + \mathbf{A} \wedge \mathbf{A}$	(8), the curvature two form (field strength) $\mathbf{F} = d\mathbf{A} + \mathbf{A} \wedge \mathbf{A}$, transforms
cardona-qft-methods_ F02045	152	■ 1.000	$\vec{E}$	$\vec{E}$	The curvature, like $\vec{E}$ and $\vec{B}$, has directly observable local features. For
cardona-qft-methods_ F02046	152	■ 1.000	$\vec{B}$	$\vec{B}$	The curvature, like $\vec{E}$ and $\vec{B}$, has directly observable local features. For
cardona-qft-methods_ F02047	152	■ 1.000	$2\ k$	$2k$	example, the $2k$ -form
cardona-qft-methods_ F02048	153	■ 0.993	$\mathbf{A}: M \mapsto \mathbb{L}$	$\mathbf{A}: M \mapsto \mathbb{L}$	the Lie algebra $\mathbb{L}$ in which the connection one form $\mathbf{A}: M \mapsto \mathbb{L}$ takes its
cardona-qft-methods_ F02049	153	■ 1.000	$P_{2\ k}(\mathbf{F})=\left\langle\mathbf{F}^k\right\rangle$	$P_{2k}(\mathbf{F}) = \langle \mathbf{F}^k \rangle$	denote by $P_{2k}(\mathbf{F}) = \langle \mathbf{F}^k \rangle$ one of those invariants, where $\langle \dots \rangle$ stands for a
cardona-qft-methods_ F02050	153	■ 1.000	$\left\langle\cdots\right\rangle$	$\langle \dots \rangle$	denote by $P_{2k}(\mathbf{F}) = \langle \mathbf{F}^k \rangle$ one of those invariants, where $\langle \dots \rangle$ stands for a
cardona-qft-methods_ F02051	153	■ 0.973	Tr	Tr	define “ Tr ” (the algebra of rotations, for example, has two). Then, P_{2k}
cardona-qft-methods_ F02052	153	■ 0.973	$P_{2\ k}$	P_{2k}	define “ Tr ” (the algebra of rotations, for example, has two). Then, P_{2k}
cardona-qft-methods_ F02053	153	■ 1.000	$\delta_{\text{gauge }} P_{2\ k}=0$	$\delta_{\text{gauge }} P_{2k} = 0$	as $\delta_{\text{gauge }} P_{2k} = 0$.
cardona-qft-methods_ F02054	153	■ 1.000	$d\ P_{2\ k}=0$	$dP_{2k} = 0$	ii. It is closed, $dP_{2k} = 0$.
cardona-qft-methods_ F02055	153	■ 1.000	$(2\ k-1)$	$(2k - 1)$	iii. It can be locally expressed as the derivative of a $(2k - 1)$ -form, $P_{2k} =$
cardona-qft-methods_ F02056	153	■ 1.000	$P_{2\ k}=$	$P_{2k} =$	iii. It can be locally expressed as the derivative of a $(2k - 1)$ -form, $P_{2k} =$
cardona-qft-methods_ F02057	153	■ 1.000	$d\ \mathcal{C}_{2\ k-1}$	$d\mathcal{C}_{2k-1}$	$d\mathcal{C}_{2k-1}$.
cardona-qft-methods_ F02058	153	■ 1.000	$\int_M P_{2\ k}=c_{2\ k}(M)\in\mathbf{Z}$	$\int_M P_{2k} = c_{2k}(M) \in \mathbf{Z}$	boundary, is a topological invariant, $\int_M P_{2k} = c_{2k}(M) \in \mathbf{Z}$.
cardona-qft-methods_ F02059	153	■ 0.988	$\mathbf{F}$	$\mathbf{F}$	curvature $\mathbf{F}$ vanishes identically, $D\mathbf{F} = d\mathbf{F} + [\mathbf{A}, \mathbf{F}] \equiv 0$. Condition (iii)
cardona-qft-methods_ F02060	153	■ 0.988	$D\ \mathbf{F}=d\ \mathbf{F}+[\mathbf{A},\ \mathbf{F}]\equiv 0$	$D\mathbf{F} = d\mathbf{F} + [\mathbf{A}, \mathbf{F}] \equiv 0$	curvature $\mathbf{F}$ vanishes identically, $D\mathbf{F} = d\mathbf{F} + [\mathbf{A}, \mathbf{F}] \equiv 0$. Condition (iii)
cardona-qft-methods_ F02061	153	■ 1.000	$d\ \phi=0$	$d\phi = 0$	follows from (ii). By Poincaré lemma: <i>If $d\phi = 0$ then, locally ϕ can be</i>
cardona-qft-methods_ F02062	153	■ 0.996	$\phi=d$	$\phi = d$	<i>written as $\phi = d(\text{something})$.</i>
cardona-qft-methods_ F02063	153	■ 1.000	$\mathcal{C}_{2\ n-1}$	$\mathcal{C}_{2n-1}$	The CS forms $\mathcal{C}_{2n-1}$ identified in ¹ are given by the trace of some polyno-
cardona-qft-methods_ F02064	153	■ 0.991	$d\ \mathbf{A}$	$d\mathbf{A}$	mial in $\mathbf{A}$ and $d\mathbf{A}$ that <i>cannot</i> be written as a local function involving only
cardona-qft-methods_ F02065	153	■ 0.993	$\text{Tr}\left[\mathbf{F}^k\right]$	$\text{Tr}[\mathbf{F}^k]$	$\text{Tr}[\mathbf{F}^k]$, due to the cyclic property of the trace). Performing a gauge trans-
cardona-qft-methods_ F02066	154	■ 1.000	$\mathcal{C}_{2\ k-1}$	$\mathcal{C}_{2k-1}$	$\mathcal{C}_{2k-1}$ can be written locally as an exact form,
cardona-qft-methods_ F02067	154	■ 1.000	$(2\ n-1)$	$(2n - 1)$	as an abelian connection. This is sufficient to ensure that a CS $(2n - 1)$ -form
cardona-qft-methods_ F02068	154	■ 1.000	$M^{2\ n-1}$	M^{2n-1}	manifold M^{2n-1} ; a metric structure may not even be defined on it. This is a
cardona-qft-methods_ F02069	155	■ 1.000	ξ^{μ}	ξ^{μ}	These are gauge-like transformations in the sense that ξ^{μ} is an arbitrary

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cardona-qft-methods_ F02676	155	■ 0.999	$v^{\{\prime \mu\}}$	v'^{μ}	as the original field, whereas the argument of v'^{μ} in (16) is x' and not x ,
cardona-qft-methods_ F02671	155	■ 0.999	$x^{\{\prime\}}$	x'	as the original field, whereas the argument of v'^{μ} in (16) is x' and not x ,
cardona-qft-methods_ F02672	156	■ 1.000	$\mathcal{H}_{\mu}$	$\mathcal{H}_{\mu}$	The generators of diffeomorphisms $\mathcal{H}_{\mu}$ form an algebra whose Poisson
cardona-qft-methods_ F02673	156	■ 1.000	$g^{\{i\ j\}}(x)$	$g^{ij}(x)$	where $g^{ij}(x)$ is the inverse of the spatial metric at x , $\mathcal{H}_{\perp}$ generates diffeo-
cardona-qft-methods_ F02674	156	■ 1.000	$x, \mathcal{H}_{\perp}$	$x, \mathcal{H}_{\perp}$	where $g^{ij}(x)$ is the inverse of the spatial metric at x , $\mathcal{H}_{\perp}$ generates diffeo-
cardona-qft-methods_ F02675	156	■ 1.000	$x^{\{\perp\}}, \mathcal{H}_{\{i\}}$	$x^{\perp}, \mathcal{H}_i$	morphisms along the normal direction $x^{\perp}$, $\mathcal{H}_i$ generates diffeomorphisms
cardona-qft-methods_ F02676	156	■ 1.000	$x^{\{i\}} ; \delta(x, y)$	$x^i; \delta(x, y)$	along the spatial direction x^i ; $\delta(x, y)$ is the Dirac delta with support on
cardona-qft-methods_ F02677	156	■ 0.965	$x=y$	$x = y$	$x=y$, and $\delta(x, y)_{,i} = (\partial/\partial y^i)\delta(x, y)$.
cardona-qft-methods_ F02678	156	■ 0.965	$\delta(x, y)_{, \{ \}_{i}} = \left(\partial / \partial y^{\{i\}} \right) \delta(x, y)$	$\delta(x, y)_{, i} = (\partial/\partial y^i) \delta(x, y)$	$x=y$, and $\delta(x, y)_{,i} = (\partial/\partial y^i)\delta(x, y)$.
cardona-qft-methods_ F02679	157	■ 0.999	$S\ 0(3,1)$	$SO(3,1)$	gravitation a gauge theory for the group $SO(3,1)$, the first nonabelian gauge
cardona-qft-methods_ F02680	158	■ 0.833	$\mathbb{G} \supseteq S\ 0(3,1)$	$\mathbb{G} \supseteq SO(3,1)$	$\mathbb{G} \supseteq SO(3,1)$. Natural options are the de Sitter ($SO(4,1)$), anti-de Sitter
cardona-qft-methods_ F02681	158	■ 0.833	$(S\ 0(4,1))$	$(SO(4,1))$	$\mathbb{G} \supseteq SO(3,1)$. Natural options are the de Sitter ($SO(4,1)$), anti-de Sitter
cardona-qft-methods_ F02682	158	■ 0.855	$(S\ 0(3,2))$	$(SO(3,2))$	$(SO(3,2))$, conformal ($SO(4,2)$) and Poincaré $ISO(3,1)$ groups, or some
cardona-qft-methods_ F02683	158	■ 0.855	$(S\ 0(4,2))$	$(SO(4,2))$	$(SO(3,2))$, conformal ($SO(4,2)$) and Poincaré $ISO(3,1)$ groups, or some
cardona-qft-methods_ F02684	158	■ 0.855	$I\ S\ 0(3,1))$	$ISO(3,1))$	$(SO(3,2))$, conformal ($SO(4,2)$) and Poincaré $ISO(3,1)$ groups, or some
cardona-qft-methods_ F02685	158	■ 1.000	$S\ 0(D-1,1)$	$SO(D-1,1)$	that correspond to some nontrivial representation of $SO(D-1,1)$, where
cardona-qft-methods_ F02686	158	■ 1.000	$d^{\{2\}} \equiv 0$	$d^2 \equiv 0$	and since in exterior calculus, $d^2 \equiv 0$, the classical fields satisfy at most
cardona-qft-methods_ F02687	158	■ 1.000	$e_{\mu}^{\{a\}}(x)$	$e_{\mu}^a(x)$	vielbein $e_{\mu}^a(x)$ characterizing the metric structure, an the Lorentz connec-
cardona-qft-methods_ F02688	158	■ 0.615	$\omega^{\{a\}\{ \}_{b\ \mu}}(x)$	$\omega^a_{b\mu}(x)$	tion, $\omega^a_{b\mu}(x)$, that codifies the affine features. Until further notice, these
cardona-qft-methods_ F02689	159	■ 1.000	$(-1,1,1, \cdots, 1)$	$(-1,1,1, \cdots, 1)$	$(-1,1,1, \cdots, 1)$. At every point on $x \in M$ there is a D -dimensional tan-
cardona-qft-methods_ F02690	159	■ 1.000	$T_{\{x\}}$	T_x	gent space T_x , which is a good approximation of the manifold M in the
cardona-qft-methods_ F02691	159	■ 1.000	$\left\{ T_{\{x\}}, x \in M \right\}$	$\{T_x, x \in M\}$	<i>manifold</i> , $\{T_x, x \in M\}$. Either way, the essential point is that the manifold
cardona-qft-methods_ F02692	159	■ 1.000	$d\ x^{\{\mu\}}$	dx^{μ}	coordinate separation dx^{μ} between two infinitesimally close points on M .
cardona-qft-methods_ F02693	159	■ 1.000	$z^{\{a\}}$	z^a	where z^a represent an orthonormal coordinate basis in the tangent space.
cardona-qft-methods_ F02694	159	■ 1.000	$\eta_{\{a\ b\}}$	η_{ab}	Since T_x is a standard Minkowski space, it has a natural metric, η_{ab} , which
cardona-qft-methods_ F02695	159	■ 1.000	$e_{\mu}^{\{a\}}$	e_{μ}^a	defines a metric on M through the isomorphism e_{μ}^a . In fact,
cardona-qft-methods_ F02696	160	■ 1.000	$T_{\{x\}}, S\ 0(D-1,1)$	$T_x, SO(D-1,1)$	rotations of T_x , $SO(D-1,1)$, as

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cardona-qft-methods_ F02097	160	■ 1.000	$\boldsymbol{\Lambda}(x)$	$\Lambda(x)$	where the matrix $\Lambda(x)$ leaves the metric in the tangent space unchanged,
cardona-qft-methods_ F02098	160	■ 1.000	$D(D+1) / 2$	$D(D+1)/2$	nents, whereas the metric has only $D(D+1)/2$. The mismatch is exactly
cardona-qft-methods_ F02099	160	■ 1.000	$D(D-1) / 2$	$D(D-1)/2$	$D(D-1)/2$, the number of independent rotations in D dimensions.
cardona-qft-methods_ F02100	160	■ 1.000	$e^{\{a\}}(x)=e_{\{\mu\}}^{\{a\}}(x) \, d x^{\{\mu\}}$	$e^a(x) = e_{\mu}^a(x)dx^{\mu}$	In this way, the vielbein one-forms $e^a(x) = e_{\mu}^a(x)dx^{\mu}$, define the
cardona-qft-methods_ F02101	160	■ 1.000	$\Lambda(x)$	$\Lambda(x)$	formations $\Lambda(x)$, even if they act independently at each spacetime point.
cardona-qft-methods_ F02102	160	■ 0.955	$\{ \}^{\{\mathrm{d}\}}$	d	This is achieved by introducing the Lorentz connection, ^d which measures
cardona-qft-methods_ F02103	160	■ 1.000	$T_{\{x+d \, x\}}$	T_{x+dx}	T_x and T_{x+dx} .
cardona-qft-methods_ F02104	160	■ 1.000	$\phi^{\{a\}}(x)$	$\phi^a(x)$	Consider a field $\phi^a(x)$ that transforms as a vector under Lorentz rota-
cardona-qft-methods_ F02105	160	■ 1.000	$x+d \, x$	$x + dx$	tions defined on the tangent T_x . The parallel-transported field from $x + dx$
cardona-qft-methods_ F02106	160	■ 1.000	$\{ \}^{\{2,7,16\}}$	$2,7,16$	several texts such as those of Refs. ^{2,7,16} and ¹⁷
cardona-qft-methods_ F02107	161	■ 0.979	$\phi^{\{a\}}$	ϕ^a	In this case, both ϕ^a and $\phi_{ }^a$ are vectors defined at the same point, therefore
cardona-qft-methods_ F02108	161	■ 0.979	$\phi_{\{ \} }^{\{a\}}$	$\phi_{ }^a$	In this case, both ϕ^a and $\phi_{ }^a$ are vectors defined at the same point, therefore
cardona-qft-methods_ F02109	161	■ 0.993	$\omega_{\{b \, \mu\}}^{\{a\}}$	$\omega_{b\mu}^a$	spaces, but it is always possible to express $\omega_{b\mu}^a$ as a function of e_{μ}^a and $\Gamma_{\lambda\rho}^{\mu}$.
cardona-qft-methods_ F02110	161	■ 0.993	$\Gamma_{\{\lambda \rho\}}^{\{\mu\}}$	$\Gamma_{\lambda\rho}^{\mu}$	spaces, but it is always possible to express $\omega_{b\mu}^a$ as a function of e_{μ}^a and $\Gamma_{\lambda\rho}^{\mu}$.
cardona-qft-methods_ F02111	161	■ 1.000	$\omega^{\{a\}}_{\{ \}_b}=\omega^{\{a\}}_{\{ \}_b \, \mu} \, d x^{\{\mu\}}$	$\omega^a_b = \omega^a_{b\mu}dx^{\mu}$	where $\omega^a_b = \omega^a_{b\mu}dx^{\mu}$ is the connection one-form .
cardona-qft-methods_ F02112	161	■ 1.000	$\epsilon_{\{a_1 \, a_2 \, \cdots \, a_D\}}$	$\epsilon_{a_1 a_2 \cdots a_D}$	tensor, $\epsilon_{a_1 a_2 \cdots a_D}$. Because they are the same in every tangent space, they
cardona-qft-methods_ F02113	161	■ 1.000	$\left(\partial_{\{\mu\}}\eta_{\{a \, b\}}=0, \, \partial_{\{\mu\}}\epsilon_{\{a_1 \, a_2 \, \cdots \, a_D\}}=0\right)$	$(\partial_{\mu}\eta_{ab} = 0, \partial_{\mu}\epsilon_{a_1 a_2 \cdots a_D} = 0)$	are constant ($\partial_{\mu}\eta_{ab} = 0, \partial_{\mu}\epsilon_{a_1 a_2 \cdots a_D} = 0$) and since they are also invariant,
cardona-qft-methods_ F02114	162	■ 1.000	$\omega^{\{a \, b\}}=-\omega^{\{b \, a\}}$	$\omega^{ab} = -\omega^{ba}$	symmetry of the connection $\omega^{ab} = -\omega^{ba}$. The second relation (13) does
cardona-qft-methods_ F02115	162	■ 0.977	$D^{\{2\}}(D-1) / 2$	$D^2(D-1)/2$	tion. Then, the number of independent components of $\omega_{b\mu}^a$ is $D^2(D-1)/2$,
cardona-qft-methods_ F02116	162	■ 0.968	$\left(D^{\{2\}}(D+1) / 2\right)$	$(D^2(D+1)/2)$	symbol $(D^2(D+1)/2)$.
cardona-qft-methods_ F02117	162	■ 1.000	$d x^{\{\mu\}} \, \partial_{\{\mu\}} \, \wedge$	$dx^{\mu}\partial_{\mu}\wedge$	The 1-form exterior derivative operator, $dx^{\mu}\partial_{\mu}\wedge$ is such that acting on a
cardona-qft-methods_ F02118	162	■ 0.468	$\alpha_{\{p\}}$	α_p	p-form, α_p , it yields a (p+1)-form, $d\alpha_p$. One of the fundamental properties
cardona-qft-methods_ F02119	162	■ 0.468	$d \, \alpha_{\{p\}}$	$d\alpha_p$	p-form, α_p , it yields a (p+1)-form, $d\alpha_p$. One of the fundamental properties
cardona-qft-methods_ F02120	162	■ 1.000	$\{ \}^{\{16,17\}}$	$16,17$	Lorentz tensor known as the curvature two-form (see, e.g., ^{16,17} for a
cardona-qft-methods_ F02121	162	■ 0.924	$R^{\{a\}}_{\{ \}_b}$	R^a_b	formal definition of R^a_b),
cardona-qft-methods_ F02122	162	■ 0.788	$\omega^{\{a\}}_{\{ \}_b}(x)$	$\omega^a_b(x)$	The fact that $\omega^a_b(x)$ and the gauge potential in Yang-Mills theory, $A^a_b =$

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cardona-qft-methods_ F02123	162	■ 0.788	$A^{\{a\}\}_{b}=$	$A^a{}_b =$	The fact that $\omega^a_b(x)$ and the gauge potential in Yang-Mills theory, $A^a{}_b =$
cardona-qft-methods_ F02124	162	■ 0.955	$A^{\{a\}\}_{b\ \mu}\ d\ x^{\{\mu\}}$	$A^a{}_{b\mu}dx^\mu$	$A^a{}_{b\mu}dx^\mu$, are both 1-forms and have similar properties is not an accident
cardona-qft-methods_ F02125	162	■ 0.953	$R_{\{b\}^{\{a\}}$	R^a_b	have the same form, and the curvature R^a_b is completely analogous to the
cardona-qft-methods_ F02126	162	■ 1.000	$R^{\{\alpha\ \beta\}\}_{\{\mu\ \nu\}}$	$R^{\alpha\beta}{}_{\mu\nu}$	gent space, related to the Riemann tensor $R^{\alpha\beta}{}_{\mu\nu}$ through
cardona-qft-methods_ F02127	163	■ 1.000	$e^{\{a\}}$	e^a	to construct a tensor two-form solely out of e^a and its exterior derivatives,
cardona-qft-methods_ F02128	163	■ 1.000	$T^{\{a\}}$	T^a	which involves both the vielbein and the connection. In contrast with T^a ,
cardona-qft-methods_ F02129	163	■ 0.999	$\bar{\omega}\{\omega\}$	$\bar{\omega}$	into a torsion-free part $\bar{\omega}$, and the so-called contorsion κ ,
cardona-qft-methods_ F02130	163	■ 0.999	κ	κ	into a torsion-free part $\bar{\omega}$, and the so-called contorsion κ ,
cardona-qft-methods_ F02131	163	■ 1.000	$\bar{R}_{\{b\}^{\{a\}}$	$\bar{R}^a_b$	where $\bar{R}^a_b$ and $\bar{D}$ are the curvature and the covariant derivative constructed
cardona-qft-methods_ F02132	163	■ 1.000	$\bar{D}\}$	$\bar{D}$	where $\bar{R}^a_b$ and $\bar{D}$ are the curvature and the covariant derivative constructed
cardona-qft-methods_ F02133	164	■ 0.896	$\omega^{\{a\}\}_{b\ \mu}\}$	$\omega^a{}_{b\mu}$	restrict the form of the field $\omega^a_{b\mu}$ in any way. This can be checked explicitly
cardona-qft-methods_ F02134	164	■ 1.000	$R^{\{a\ b\}}$	R^{ab}	The Bianchi identity implies that the curvature R^{ab} is “transparent” for
cardona-qft-methods_ F02135	164	■ 1.000	$e^{\{a\}},\ \omega^{\{a\ b\}}$	e^a,ω^{ab}	cessive exterior derivatives of e^a , ω^{ab} and T^a one does not generate new
cardona-qft-methods_ F02136	164	■ 0.793	$\omega_{\{b\}^{\{a\}}$	ω^a_b	ω^a_b , and the two-forms R^a_b, T^a . In order to produce Lorentz invariants with
cardona-qft-methods_ F02137	164	■ 0.793	$R_{\{b\}^{\{a\}},\ T^{\{a\}}$	R^a_b, T^a	ω^a_b , and the two-forms R^a_b, T^a . In order to produce Lorentz invariants with
cardona-qft-methods_ F02138	164	■ 0.508	$I[e,\ \omega]$	$I[e,\omega]$	pressed by an action functional of the form $I[e,\omega]$. Since e^a and ω^a_b describe
cardona-qft-methods_ F02139	164	■ 0.508	$\omega^{\{a\}\}_{b\}}$	$\omega^a{}_b$	pressed by an action functional of the form $I[e,\omega]$. Since e^a and ω^a_b describe
cardona-qft-methods_ F02140	164	■ 0.852	$\mu,\ \nu$	μ,ν	nate indices (μ, ν , etc.), these 1-forms, like all exterior forms, are invariant
cardona-qft-methods_ F02141	165	■ 0.666	$e^{\{a\}},\ \omega^{\{a\}\}_{b\}}$	$e^a,\omega^a{}_b$	above, we expect it to be a local functional of the one-forms e^a , ω^a_b and their
cardona-qft-methods_ F02142	165	■ 1.000	$\epsilon_{a_1\cdots a_D}\}$	$\epsilon_{a_1\cdots a_D}$	group, η_{ab} , and $\epsilon_{a_1\cdots a_D}$ can be used to raise, lower and contract indices. One
cardona-qft-methods_ F02143	165	■ 1.000	$\{ \}^{\{\mathrm{e}\}}\ R_{\{\mu\ \nu\}}=E_{\{a\}^{\{\lambda\}}\ \eta_{\{b\ c\}}\ e_{\{\mu\}^{\{c\}}\ R_{\{\lambda\beta\}\ \nu\}^{\{a\ b\}}$	${}^eR_{\mu\nu} = E_a^\lambda\eta_{bc}e_\mu^cR_{\lambda\nu}^{ab}$	cludes tensors like the Ricci tensor ^e $R_{\mu\nu} = E_a^\lambda\eta_{bc}e_\mu^cR_{\lambda\nu}^{ab}$, or $R_{\alpha\beta}R_{\mu\nu}R^{\alpha\mu\beta\nu}$,
cardona-qft-methods_ F02144	165	■ 1.000	$R_{\{\alpha\ \beta\}\ R_{\{\mu\ \nu\}}\ R^{\{\alpha\ \mu\ \beta\}\ \nu\}}$	$R_{\alpha\beta}R_{\mu\nu}R^{\alpha\mu\beta\nu}$	cludes tensors like the Ricci tensor ^e $R_{\mu\nu} = E_a^\lambda\eta_{bc}e_\mu^cR_{\lambda\nu}^{ab}$, or $R_{\alpha\beta}R_{\mu\nu}R^{\alpha\mu\beta\nu}$,
cardona-qft-methods_ F02145	165	■ 0.988	$e^{\{a\}},\ R_{\{b\}^{\{a\}},\ T^{\{a\}}$	e^a, R^a_b, T^a	products of e^a , R^a_b , T^a , contracted with η_{ab} and $\epsilon_{a_1\cdots a_D}$, and no ω . Such
cardona-qft-methods_ F02146	165	■ 1.000	$\mathbf{P}_{\{2\ k\}}$	$\mathbf{P}_{2k}$	Among these local Lorentz invariants, the Pontryagin forms $\mathbf{P}_{2k}$ and the
cardona-qft-methods_ F02147	165	■ 1.000	$\mathbf{E}_{\{2\ n\}}$	$\mathbf{E}_{2n}$	Euler forms $\mathbf{E}_{2n}$ define topological invariants in $4k$ and $2n$ dimensions,
cardona-qft-methods_ F02148	165	■ 1.000	$4\ k$	$4k$	Euler forms $\mathbf{E}_{2n}$ define topological invariants in $4k$ and $2n$ dimensions,
cardona-qft-methods_ F02149	165	■ 0.774	$\{ \}^{\{\mathrm{e}\}}$	^e	^e Here E_a^λ is the inverse vielbein, $E_a^\lambda e^b{}_\lambda = \delta^a_b$.

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cardona-qft-methods_ F02177	168	■ 1.000	$\omega^{a\ b}$	ω^{ab}	Extremizing the action (9) with respect to e^a and ω^{ab} , yields
cardona-qft-methods_ F02178	168	■ 1.000	$I_{\{D\}}$	I_D	modulo surface terms. The condition for I_D to have an extremum to first
cardona-qft-methods_ F02179	168	■ 1.000	$\mathcal{E}_{\{a\}}$	$\mathcal{E}_a$	order under arbitrary infinitesimal variations is that $\mathcal{E}_a$ and $\mathcal{E}_{ab}$ vanish:
cardona-qft-methods_ F02180	168	■ 1.000	$\mathcal{E}_{\{a\ b\}}$	$\mathcal{E}_{ab}$	order under arbitrary infinitesimal variations is that $\mathcal{E}_a$ and $\mathcal{E}_{ab}$ vanish:
cardona-qft-methods_ F02181	168	■ 1.000	$d^2=0$	$d^2 = 0$	simply because $d^2 = 0$. Indeed, R^{ab} and T^a only involve first derivatives
cardona-qft-methods_ F02182	168	■ 0.826	$\left(E_{\{a\}}^{\{\mu\}}\right)$	(E_a^μ)	tion can be solved for ω as a function of the inverse vielbein (E_a^μ) and its
cardona-qft-methods_ F02183	168	■ 1.000	$\Gamma_{\mu\nu}^{\lambda}$	$\Gamma_{\mu\nu}^\lambda$	where $\Gamma_{\mu\nu}^\lambda$ is symmetric in $\mu\nu$ and can be identified as the Christoffel symbol
cardona-qft-methods_ F02184	168	■ 1.000	$\mu\ \nu$	$\mu\nu$	where $\Gamma_{\mu\nu}^\lambda$ is symmetric in $\mu\nu$ and can be identified as the Christoffel symbol
cardona-qft-methods_ F02185	169	■ 1.000	$g_{\{\mu\ \nu\}}$	$g_{\mu\nu}$	the fact that the Lagrangian involves the second derivatives of $g_{\mu\nu}$. This
cardona-qft-methods_ F02186	169	■ 1.000	$f(R)$	$f(R)$	to vanish identically. In $f(R)$ theories, for example, this first condition is not
cardona-qft-methods_ F02187	169	■ 1.000	$D\leq 4$	$D \leq 4$	very different for $D \leq 4$ and for $D > 4$ is that in the former case the field
cardona-qft-methods_ F02188	169	■ 1.000	$D>4$	$D > 4$	very different for $D \leq 4$ and for $D > 4$ is that in the former case the field
cardona-qft-methods_ F02189	169	■ 1.000	$(15,16)$	$(15,16)$	equations (15, 16) are linear in the curvature tensor, while in the latter case
cardona-qft-methods_ F02190	170	■ 1.000	$I[\omega, e]=I[\omega(e, \partial e), e]$	$I[\omega, e] = I[\omega(e, \partial e), e]$	$I[\omega, e] = I[\omega(e, \partial e), e]$.
cardona-qft-methods_ F02191	170	■ 1.000	$\omega(e, \partial e)$	$\omega(e, \partial e)$	braic in ω , it is impossible to solve for $\omega(e, \partial e)$ globally, and therefore the
cardona-qft-methods_ F02192	170	■ 0.857	$T^{\{a\}=0\ .\}^{\{18\}}$	$T^a = 0$. ¹⁸	possible Lorentz invariants involving torsion that would vanish if $T^a = 0$. ¹⁸
cardona-qft-methods_ F02193	170	■ 1.000	$\mathbf{P}_{\{4\}}\ v_{\{1\}}\ \tau_{\{0\}}\ \zeta_{\{0\}}$	$\mathbf{P}_4 v_1 \tau_0 \zeta_0$	in fifteen dimensions could be $\mathbf{P}_4 v_1 \tau_0 \zeta_0$. Let us examine other examples:
cardona-qft-methods_ F02194	170	■ 0.964	$S\ 0(5)$	$SO(5)$	to the difference between the Pontryagin classes for $SO(5)$ and $SO(4)$ (or
cardona-qft-methods_ F02195	170	■ 0.964	$S\ 0(4)$	$SO(4)$	to the difference between the Pontryagin classes for $SO(5)$ and $SO(4)$ (or
cardona-qft-methods_ F02196	170	■ 0.948	$S\ 0(n,\ 5-n)$	$SO(n, 5 - n$	their related groups $SO(n, 5 - n$ and $SO(m, 4 - m)$). ²⁸
cardona-qft-methods_ F02197	170	■ 0.948	$S\ 0(m,\ 4-m)\)\ .^{\{28\}}$	$SO(m, 4 - m)$. ²⁸	their related groups $SO(n, 5 - n$ and $SO(m, 4 - m)$. ²⁸
cardona-qft-methods_ F02198	171	■ 1.000	$\mathbf{P}_{\{4\}}$	$\mathbf{P}_4$	and the Nieh-Yan forms in four dimensions, $\mathbf{P}_4$ and $\mathbf{N}_4$, respectively. The
cardona-qft-methods_ F02199	171	■ 1.000	$\mathbf{N}_{\{4\}}$	$\mathbf{N}_4$	and the Nieh-Yan forms in four dimensions, $\mathbf{P}_4$ and $\mathbf{N}_4$, respectively. The
cardona-qft-methods_ F02200	171	■ 0.982	$S\ 0(2,1)$	$SO(2,1)$	Both these terms are invariant under $SO(2,1)$ (Lorentz invariant in three
cardona-qft-methods_ F02201	171	■ 1.000	$4\ k-1$	$4k - 1$	there is a CS form which provides a sensible action for gravity in $4k - 1$
cardona-qft-methods_ F02202	171	■ 1.000	$D=7$	$D = 7$	dimensions. For example, in $D = 7$, the Lorentz CS form is
cardona-qft-methods_ F02203	172	■ 1.000	$v,\ \tau$	v, τ	the families v, τ and ζ , grows wildly with the dimension. This proliferation

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cardona-qft-methods_ F02284	172	■ 0.999	$\{ \}^{-2}$	-2	and the cosmological constant have dimensions of $[\text{length}]^2$ and $[\text{length}]^{-2}$
cardona-qft-methods_ F02285	173	■ 1.000	(κ)	(κ)	seen to depend on a unique (dimensionless) coefficient (κ) , analogous to
cardona-qft-methods_ F02286	173	■ 1.000	l^{D-2p}	l^{D-2p}	l^{D-2p} . This is because the canonical dimension of the vielbein is $[e^a] = l$,
cardona-qft-methods_ F02287	173	■ 1.000	$\left[e^a\right]=l$	$[e^a] = l$	l^{D-2p} . This is because the canonical dimension of the vielbein is $[e^a] = l$,
cardona-qft-methods_ F02288	173	■ 1.000	$\left[\omega^{ab}\right]=l^0$	$[\omega^{ab}] = l^0$	while the Lorentz connection has dimensions $[\omega^{ab}] = l^0$, as a true gauge
cardona-qft-methods_ F02289	173	■ 1.000	$\hat{\omega}^{ab}=\omega^{ab}+\xi\epsilon^{abc}e_c$	$\hat{\omega}^{ab} = \omega^{ab} + \xi\epsilon^{abc}e_c$	$\hat{\omega}^{ab} = \omega^{ab} + \xi\epsilon^{abc}e_c$ define proper connections for the Lorentz group.
cardona-qft-methods_ F02218	173	■ 1.000	$\lambda^a{}_b$	$\lambda^a{}_b$	Under an infinitesimal Lorentz transformation with parameter $\lambda^a{}_b$, the
cardona-qft-methods_ F02211	174	■ 1.000	e^c, R^{ab}	e^c, R^{ab}	while e^c , R^{ab} and ϵ_{abc} transform as tensors,
cardona-qft-methods_ F02212	174	■ 1.000	ϵ_{abc}	ϵ_{abc}	while e^c , R^{ab} and ϵ_{abc} transform as tensors,
cardona-qft-methods_ F02213	174	■ 1.000	L_3	L_3	Combining these relations, the Lorentz invariance of L_3 can be explicitly
cardona-qft-methods_ F02214	174	■ 0.531	$\{ \}^{\mathrm{h}}$	h	for the translation group. ^h In fact, if the vielbein transforms under “local
cardona-qft-methods_ F02215	174	■ 1.000	λ^a	λ^a	translations” in tangent space, parametrized by λ^a as
cardona-qft-methods_ F02216	174	■ 1.000	δe	δe	using the infinitesimal transformations δe and $\delta\omega$ to compute the commu-
cardona-qft-methods_ F02217	174	■ 1.000	$\delta\omega$	$\delta\omega$	using the infinitesimal transformations δe and $\delta\omega$ to compute the commu-
cardona-qft-methods_ F02218	174	■ 0.439	$2+1$	$2+1$	5.1.2. <i>(Anti-)de Sitter group in 2+1 dimensions</i>
cardona-qft-methods_ F02219	174	■ 1.000	$\Lambda=\mp l^{-2}$	$\Lambda = \mp l^{-2}$	In the presence of a cosmological constant $\Lambda = \mp l^{-2}$ it is also possible to
cardona-qft-methods_ F02220	175	■ 1.000	$\Lambda\neq 0$	$\Lambda \neq 0$	maximally symmetric solution of the Einstein equations. If $\Lambda \neq 0$, it is no
cardona-qft-methods_ F02221	175	■ 1.000	$\Lambda>0$	$\Lambda > 0$	the Einstein equations for $\Lambda > 0$ or $\Lambda < 0$, respectively.
cardona-qft-methods_ F02222	175	■ 1.000	$\Lambda<0$	$\Lambda < 0$	the Einstein equations for $\Lambda > 0$ or $\Lambda < 0$, respectively.
cardona-qft-methods_ F02223	175	■ 0.995	W^{AB}	W^{AB}	where the 1-form W^{AB} and the 0-form Λ^{AB} stand for the combinations
cardona-qft-methods_ F02224	175	■ 0.995	Λ^{AB}	Λ^{AB}	where the 1-form W^{AB} and the 0-form Λ^{AB} stand for the combinations
cardona-qft-methods_ F02225	175	■ 0.999	$a, b, \dots=1,2, \dots D$	$a, b, .. = 1, 2, ..D$	(here $a, b, .. = 1, 2, ..D$, while $A, B, ... = 1, 2, ..., D+1$). Clearly, W^{AB} trans-
cardona-qft-methods_ F02226	175	■ 0.999	$A, B, \ldots=1,2, \dots, D+1$	$A, B, \dots = 1, 2, ..., D+1$	(here $a, b, .. = 1, 2, ..D$, while $A, B, ... = 1, 2, ..., D+1$). Clearly, W^{AB} trans-
cardona-qft-methods_ F02227	175	■ 1.000	$\Lambda^{AB}=-\Lambda^{BA}$	$\Lambda^{AB} = -\Lambda^{BA}$	formation parameters, but for which group? Since $\Lambda^{AB} = -\Lambda^{BA}$, this indi-
cardona-qft-methods_ F02228	175	■ 0.957	$SO(r, s)$	$SO(r, s)$	form, so it must be a group in the $SO(r, s)$ family. The signs $(\pm)$ in the
cardona-qft-methods_ F02229	176	■ 1.000	Υ^{AB}	Υ^{AB}	and lowered with the metric Υ^{AB} given by
cardona-qft-methods_ F02230	176	■ 1.000	$W^{AB}=\Upsilon^{BC}W^A_C$	$W^{AB} = \Upsilon^{BC}W^A_C$	so that, for example, $W^{AB} = \Upsilon^{BC}W^A_C$. Then, the covariant derivative in

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cardona-qft-methods_F02231	176	■ 1.000	$W^{\{A\ B\}}+W^{\{B\ A\}}=\emptyset$	$W^{AB} + W^{BA} = 0$	Since Υ^{AB} is constant, this last expression implies $W^{AB} + W^{BA} = 0$, in
cardona-qft-methods_F02232	176	■ 1.000	$\omega^{\{a\ b\}}+\omega^{\{b\ a\}}=$	$\omega^{ab} + \omega^{ba} =$	exact analogy with what happens with the Lorentz connection, $\omega^{ab} + \omega^{ba} =$
cardona-qft-methods_F02233	176	■ 0.957	$\omega^{\{a\ b\}} \equiv \eta^{\{b\ c\}} \omega^{\{a\} \}_{c}}$	$\omega^{ab} \equiv \eta^{bc} \omega^a{}_c$	0, where $\omega^{ab} \equiv \eta^{bc} \omega^a{}_c$. Indeed, this is a way to discover in hindsight that the
cardona-qft-methods_F02234	176	■ 0.940	$l \rightarrow \infty (\lambda \rightarrow 0)$	$l \rightarrow \infty (\lambda \rightarrow 0)$	$l \rightarrow \infty (\lambda \rightarrow 0)$. In that case, instead of (7, 8) one has
cardona-qft-methods_F02235	176	■ 0.940	(7,8)	(7,8)	$l \rightarrow \infty (\lambda \rightarrow 0)$. In that case, instead of (7, 8) one has
cardona-qft-methods_F02236	176	■ 0.516	$(c \rightarrow$	$(c \rightarrow$	from the Poincaré symmetry in the limit of infinite speed of light ($c \rightarrow$
cardona-qft-methods_F02237	176	■ 0.788	∞	∞	∞). These deformations are examples of what is known as a İnönü-Wigner
cardona-qft-methods_F02238	176	■ 1.000	$\{ \}^{\{31,32\}}$	31,32	contraction. ^{31,32} The procedure starts from a semisimple Lie algebra and
cardona-qft-methods_F02239	176	■ 1.000	l	l	some generators are rescaled by a parameter (l or λ in the above example).
cardona-qft-methods_F02240	176	■ 1.000	$l \rightarrow \infty$	$l \rightarrow \infty$	Lagrangian (6) in the limit $l \rightarrow \infty$ takes the usual Einstein Hilbert form
cardona-qft-methods_F02241	177	■ 1.000	≥ 3	≥ 3	≥ 3 which are also invariant, not just under the Lorentz group, but under
cardona-qft-methods_F02242	177	■ 0.999	$S O(D, 1), S O(D-1,2), I S O(D-1,1)$	$SO(D,1), SO(D-1,2), ISO(D-1,1)$	some of its extensions, $SO(D,1), SO(D-1,2), ISO(D-1,1)$? As we
cardona-qft-methods_F02243	177	■ 1.000	$D=2\ n-1$	$D = 2n - 1$	There exist gravity actions for every $D = 2n - 1$, invariant under local
cardona-qft-methods_F02244	177	■ 0.995	$S O(2\ n-2,2), S O(2\ n-1,1)$	$SO(2n-2,2), SO(2n-1,1)$	$SO(2n-2,2), SO(2n-1,1)$ or $ISO(2n-2,1)$ transformations, where the
cardona-qft-methods_F02245	177	■ 0.995	$I S O(2\ n-2,1)$	$ISO(2n-2,1)$	$SO(2n-2,2), SO(2n-1,1)$ or $ISO(2n-2,1)$ transformations, where the
cardona-qft-methods_F02246	177	■ 0.994	$S O(3,1), S O(2,2)$	$SO(3,1), SO(2,2)$	local $SO(2,1)$ to local $SO(3,1), SO(2,2)$ and $ISO(2,1)$? What happens if
cardona-qft-methods_F02247	177	■ 0.994	$I S O(2,1)$	$ISO(2,1)$	local $SO(2,1)$ to local $SO(3,1), SO(2,2)$ and $ISO(2,1)$? What happens if
cardona-qft-methods_F02248	178	■ 1.000	$T^{\{a\}}=\emptyset$	$T^a = 0$	unless one imposes the field equation $T^a = 0$. But this means that the
cardona-qft-methods_F02249	178	■ 0.991	$(l \rightarrow \infty)$	$(l \rightarrow \infty)$	transformations, whose limit for vanishing cosmological constant ($l \rightarrow \infty$)
cardona-qft-methods_F02250	178	■ 0.757	a, b	a, b	where a, b run from 1 to 3 (or 0 to 2) and A, B from 1 to 4 (or 0 to 3).
cardona-qft-methods_F02251	178	■ 0.757	A, B	A, B	where a, b run from 1 to 3 (or 0 to 2) and A, B from 1 to 4 (or 0 to 3).
cardona-qft-methods_F02252	178	■ 0.998	$\epsilon_{\{A\ B\ C\ D\}}$	ϵ_{ABCD}	Since the (A)dS group has an invariant tensor ϵ_{ABCD} , one can construct
cardona-qft-methods_F02253	179	■ 1.000	$\{ \}^{\{\mathrm{i}\}}$	$^{\mathrm{i}}$	multiplicative factor, as the Euler form ⁱ for a four-dimensional manifold
cardona-qft-methods_F02254	179	■ 1.000	$\Upsilon^{\{A\ B\}}=\operatorname{diag}$	$\Upsilon^{AB} = \operatorname{diag}$	whose tangent space is not Minkowskian, but has the metric $\Upsilon^{AB} = \operatorname{diag}$
cardona-qft-methods_F02255	179	■ 0.998	$\left(\eta^{\{a\ b\}}, \mp 1\right)$	$(\eta^{ab}, \mp 1)$	$(\eta^{ab}, \mp 1)$. The form $\mathbf{E}_4$ can also be written explicitly in terms of R^{ab}, T^a ,
cardona-qft-methods_F02256	179	■ 0.998	$R^{\{a\ b\}}, T^{\{a\}}$	R^{ab}, T^a	$(\eta^{ab}, \mp 1)$. The form $\mathbf{E}_4$ can also be written explicitly in terms of R^{ab}, T^a ,
cardona-qft-methods_F02257	179	■ 0.997	$(\mathrm{A})\ \mathrm{dS}_3$	$(A)\mathrm{dS}_3$	This is the key point: the l.h.s. of (26) is invariant under local (A)dS ₃

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cardona-qft-methods_F02258	179	■0.936	$\delta L_{\{3\}^{\{A\}d\,S}}=d\,\Omega$	$\delta L_3^{AdS}=d\Omega$	which in turn means, by Poincaré’s Lemma ³⁴ that, locally, $\delta L_3^{AdS}=d\Omega$,																		
cardona-qft-methods_F02259	179	■1.000	$2\,n-1$	$2n-1$	The steps to generalize the (A)dS Lagrangian from 3 to $2n-1$ dimen-																		
cardona-qft-methods_F02260	179	■1.000	$\{ \}^{\mathrm{j}}$	j	sions are now clear ^j :																		
cardona-qft-methods_F02261	179	■1.000	$L_{\{2\,n-1\}}$	L_{2n-1}	• Write this as the exterior derivative of a $(2n-1)$ -form L_{2n-1} .																		
cardona-qft-methods_F02262	179	■1.000	$\{ \}^{\{36,37\}}$	^{36,37}	^j The construction we outline here was discussed in, ^{36,37} and also in. ^{38,39}																		
cardona-qft-methods_F02263	179	■1.000	$\{ \}^{\{38,39\}}$	^{38,39}	^j The construction we outline here was discussed in, ^{36,37} and also in. ^{38,39}																		
cardona-qft-methods_F02264	180	■1.000	$\bar{\alpha}_{\{p\}}$	$\bar{\alpha}_p$	Lagrangian in which all the coefficients $\bar{\alpha}_p$ have been fixed to take the values																		
cardona-qft-methods_F02265	180	■1.000	$R_{\{t\}^{\{a\}b\}}:=R^{\{a\}b\}+\left(t^2/l^2\right)\,e^{\{a\}}e^{\{b\}}$	$R_t^{ab}:=R^{ab}+\left(t^2/l^2\right)e^ae^b$	where we have defined $R_t^{ab}:=R^{ab}+\left(t^2/l^2\right)e^ae^b$.																		
cardona-qft-methods_F02266	180	■1.000	$a_{\{3\}}$	a_3	where all indices are contracted appropriately and the coefficients a_3 ,																		
cardona-qft-methods_F02267	180	■1.000	$\cdots a_{\{2\,n-1\}}$	$\cdots a_{2n-1}$	$\cdots a_{2n-1}$ are rational numbers fixed by the condition $dL_{2n-1}^{(A)dS}=\mathbf{E}_{2n}$.																		
cardona-qft-methods_F02268	180	■1.000	$d\,L_{\{2\,n-1\}^{\{(A)\,d\,S\}}}=\mathbf{E}_{\{2\,n\}}$	$dL_{2n-1}^{(A)dS}=\mathbf{E}_{2n}$	$\cdots a_{2n-1}$ are rational numbers fixed by the condition $dL_{2n-1}^{(A)dS}=\mathbf{E}_{2n}$.																		
cardona-qft-methods_F02269	181	■1.000	$\{ \}^{\{2,41\}}$	^{2,41}	theories (see, e. g., ^{2,41}). Tables 1 and 2 give examples of CS forms which																		
cardona-qft-methods_F02270	181	—	$L_{\{3\}^{\{(A)\,d\,S\}}}=\epsilon_{\{a\}b\,c}\left(R^{\{a\}b\}\pm\frac{e^{\{a\}}e^{\{b\}}}{3\,l^2}\right)\,e^{\{c\}}$	$L_3^{(A)dS}=\epsilon_{abc}\left(R^{ab}\pm\frac{e^ae^b}{3l^2}\right)e^c$	—																		
cardona-qft-methods_F02271	181	■0.546	$\mathbf{E}_{\{4\}}=\epsilon_{\{a\}b\,c}\left(R^{\{a\}b\}\pm\frac{e^{\{a\}}e^{\{b\}}}{l^2}\right)\,T^{\{c\}}$	$\mathbf{E}_4=\epsilon_{abc}\left(R^{ab}\pm\frac{e^ae^b}{l^2}\right)T^c$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$</td><td>$\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$	$\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$	$SU(N)$
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cardona-qft-methods_F02272	181	■0.546	$S\,O(2,2)$	$SO(2,2)$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$</td><td>$\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a{}_bd\omega^b{}_a+\frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a$	$\mathbf{P}_4^{Lor}=R^a{}_bR^b{}_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{A}d\mathbf{A}+\frac{2}{3}\mathbf{A}\mathbf{A}\mathbf{A}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{F}\mathbf{F}]$	$SU(N)$
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cardona-qft-methods_ F02273	181	■ 0.546	$L_3^{\text{\texttt{Lor}}}=\omega^a_{\text{\texttt{a}}}\{\text{\texttt{ }}\}_b\text{\texttt{ d }}\omega^b_{\text{\texttt{b}}}\{\text{\texttt{ }}\}_a+\frac{2}{3}\omega^a_{\text{\texttt{a}}}\{\text{\texttt{ }}\}_b\omega^b_{\text{\texttt{b}}}\{\text{\texttt{ }}\}_c\omega^c_{\text{\texttt{c}}}\{\text{\texttt{ }}\}_a$	$L_3^{\text{Lor}}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$</td><td>$\mathbf{P}_4^{Lor}=R^a_bR^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_ F02274	181	■ 0.546	$\mathbf{P}_4^{\text{\texttt{Lor}}}=\mathbf{R}^a_{\text{\texttt{a}}}\{\text{\texttt{ }}\}_b\mathbf{R}^b_{\text{\texttt{b}}}\{\text{\texttt{ }}\}_a$	$\mathbf{P}_4^{\text{Lor}}=R^a_bR^b_a$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$</td><td>$\mathbf{P}_4^{Lor}=R^a_bR^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_ F02275	181	■ 0.546	$L_3^{\text{\texttt{T o r}}}=\mathbf{e}^a_{\text{\texttt{a}}}\text{\texttt{ T }}_a$	$L_3^{Tor}=e^aT_a$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$</td><td>$\mathbf{P}_4^{Lor}=R^a_bR^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$	$SU(N)$
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					$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$																
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cardona-qft-methods_ F02276	181	■ 0.546	$\mathbf{N}_4=\mathbf{T}^a_{\text{\texttt{a}}}\text{\texttt{ T }}_a-\mathbf{e}^a_{\text{\texttt{a}}}\text{\texttt{ e }}^b_{\text{\texttt{b}}}\text{\texttt{ R }}_{a\text{\texttt{ b}}}$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$</td><td>$\mathbf{P}_4^{Lor}=R^a_bR^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_ F02277	181	■ 0.546	$L_3^{\text{\texttt{U(1)}}}=\mathbf{A}\text{\texttt{ d }}\mathbf{A}$	$L_3^{U(1)}=AdA$	<table><tr><th>$D=3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$</td><td>$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$</td><td>$\mathbf{P}_4^{Lor}=R^a_bR^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor}=e^aT_a$</td><td>$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)}=AdA$</td><td>$\mathbf{P}_4^{U(1)}=FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D=3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS}=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{3l^2})e^c$	$\mathbf{E}_4=\epsilon_{abc}(R^{ab}\pm\frac{e^ae^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor}=\omega^a_b d\omega^b_a+\frac{2}{3}\omega^a_b\omega^b_c\omega^c_a$	$\mathbf{P}_4^{Lor}=R^a_bR^b_a$	$SO(2,1)$	$L_3^{Tor}=e^aT_a$	$\mathbf{N}_4=T^aT_a-e^ae^bR_{ab}$	$SO(2,1)$	$L_3^{U(1)}=AdA$	$\mathbf{P}_4^{U(1)}=FF$	$U(1)$	$L_3^{SU(N)}=Tr[\mathbf{AdA}+\frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)}=Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_F02278	181	0.546	$\mathbf{P}_4^{\mathrm{U}(1)} = F\ F$	$\mathbf{P}_4^{U(1)} = FF$	<table><tr><th>$D = 3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$</td><td>$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$</td><td>$\mathbf{P}_4^{Lor} = R^a_b R^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$\mathbf{P}_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D = 3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_F02279	181	0.546	$L_3^{\mathrm{SU}(N)} = \operatorname{Tr}\left[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}\right]$	$L_3^{SU(N)} = \operatorname{Tr}\left[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}\right]$	<table><tr><th>$D = 3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$</td><td>$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$</td><td>$\mathbf{P}_4^{Lor} = R^a_b R^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$\mathbf{P}_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D = 3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_F02280	181	0.546	$\mathbf{P}_4^{\mathrm{SU}(4)} = \operatorname{Tr}[\mathbf{F}\ \mathbf{F}]$	$\mathbf{P}_4^{SU(4)} = \operatorname{Tr}[\mathbf{FF}]$	<table><tr><th>$D = 3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$</td><td>$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$</td><td>$\mathbf{P}_4^{Lor} = R^a_b R^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$\mathbf{P}_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D = 3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_F02281	181	0.546	$S\ U(N)$	$SU(N)$	<table><tr><th>$D = 3$ CS Lagrangians</th><th>Characteristic classes</th><th>Groups</th></tr><tr><td>$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$</td><td>$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$</td><td>$SO(4)$, $SO(3,1)$ or $SO(2,2)$</td></tr><tr><td>$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$</td><td>$\mathbf{P}_4^{Lor} = R^a_b R^b_a$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{Tor} = e^a T_a$</td><td>$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$</td><td>$SO(2,1)$</td></tr><tr><td>$L_3^{U(1)} = AdA$</td><td>$\mathbf{P}_4^{U(1)} = FF$</td><td>$U(1)$</td></tr><tr><td>$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$</td><td>$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$</td><td>$SU(N)$</td></tr></table>	$D = 3$ CS Lagrangians	Characteristic classes	Groups	$L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c$	$\mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c$	$SO(4)$, $SO(3,1)$ or $SO(2,2)$	$L_3^{Lor} = \omega^a_b d\omega^b_a + \frac{2}{3}\omega^a_b \omega^b_c \omega^c_a$	$\mathbf{P}_4^{Lor} = R^a_b R^b_a$	$SO(2,1)$	$L_3^{Tor} = e^a T_a$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab}$	$SO(2,1)$	$L_3^{U(1)} = AdA$	$\mathbf{P}_4^{U(1)} = FF$	$U(1)$	$L_3^{SU(N)} = Tr[\mathbf{AdA} + \frac{2}{3}\mathbf{AAA}]$	$\mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}]$	$SU(N)$
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cardona-qft-methods_F02282	181	0.951	$R,\ F$	R, F	Here R , F , and $\mathbf{F}$ are the curvatures for the connections ω^a_b , A and $\mathbf{A}$,																		
cardona-qft-methods_F02283	181	0.951	ω_b^a, A	ω_b^a, A	Here R , F , and $\mathbf{F}$ are the curvatures for the connections ω^a_b , A and $\mathbf{A}$,																		
cardona-qft-methods_F02284	181	0.951	$\mathbf{A}$	$\mathbf{A}$	Here R , F , and $\mathbf{F}$ are the curvatures for the connections ω^a_b , A and $\mathbf{A}$,																		
cardona-qft-methods_F02285	181	1.000	$d\ L_3^{\mathrm{AdS}}$	dL_3^{AdS}	field equations as the standard dL_3^{AdS} , but interchanged: varying dL_3^{AdS}																		
cardona-qft-methods_F02286	182	1.000	$L_3^{\mathrm{\text{Exotic}}}$	L_3^{Exotic}	with respect to e^a gives the same equation as varying L_3^{Exotic} with respect																		
cardona-qft-methods_F02287	182	1.000	$R^{\mathrm{a\ b}}\ T_{\mathrm{a}}\ e_{\mathrm{b}}$	$R^{ab}T_a e_b$	T^a is $R^{ab}T_a e_b$, which is a total derivative, and therefore there are no new																		

Identifier	Page	Conf.	LaTeX source	Rendered	Image								
cardona-qft-methods_ F02288	182	0.562	D=7 \mathrm{CS}	$D = 7\mathrm{CS}$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
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$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02289	182	0.562	$L_7^{\text{\texttt{Lor}}} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$L_7^{\text{Lor}} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
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$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02290	182	0.562	$\mathbf{P}_8^{\text{\texttt{Lor}}} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$\mathbf{P}_8^{\text{Lor}} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$												
$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02291	182	0.562	$L_7^{\text{\texttt{I}}} = \left(\omega^a{}_b d\omega^b{}_a + \frac{2}{3} \omega^a{}_b \omega^b{}_c \omega^c{}_a \right) R^a{}_b R^b{}_a$	$L_7^I = \left(\omega^a{}_b d\omega^b{}_a + \frac{2}{3} \omega^a{}_b \omega^b{}_c \omega^c{}_a \right) R^a{}_b R^b{}_a$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$												
$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02292	182	0.562	$\left(\mathbf{P}_4^{\text{\texttt{Lor}}} \right)^2 = (R^a{}_b R^b{}_a)^2$	$\left(\mathbf{P}_4^{\text{Lor}} \right)^2 = (R^a{}_b R^b{}_a)^2$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$												
$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02293	182	0.562	$L_7^{\text{\texttt{I I}}} = (e^a T_a) (R^a{}_b R^b{}_a)$	$L_7^{II} = (e^a T_a) (R^a{}_b R^b{}_a)$	<table><tr><td>$D = 7$ CS Lagrangians</td><td>Characteristic classes</td></tr><tr><td>$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$</td><td>$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$</td></tr><tr><td>$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$</td><td>$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$</td></tr><tr><td>$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$</td><td>$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$</td></tr></table>	$D = 7$ CS Lagrangians	Characteristic classes	$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$	$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$	$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$
$D = 7$ CS Lagrangians	Characteristic classes												
$L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7$	$\mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a$												
$L_7^I = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b \omega^b{}_c \omega^c{}_a) R^a{}_b R^b{}_a$	$(\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2$												
$L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a)$	$\mathbf{N}_4 \mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$												
cardona-qft-methods_ F02294	182	—	$\begin{aligned} & \& \mathbf{N}_4 \mathbf{P}_4^{\text{\texttt{Lor}}} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c \\ & \& \left(T^a T_a - e^a e^b R_{ab} \right) R^c{}_d R^d{}_c \end{aligned}$	$\mathbf{N}_4 \mathbf{P}_4^{\text{Lor}} = (T^a T_a - e^a e^b R_{ab}) R^c{}_d R^d{}_c$	—								
cardona-qft-methods_ F02295	182	1.000	$O(D-n, n)$	$O(D - n, n)$	• The Euler class, associated with the $O(D - n, n)$ groups. For $D = 2$,								
cardona-qft-methods_ F02296	182	1.000	(g)	(g)	the Euler number is related to the genus (g) of the surface, $\chi = 2 - 2g$.								
cardona-qft-methods_ F02297	182	1.000	$\chi=2-2\ g$	$\chi = 2 - 2g$	the Euler number is related to the genus (g) of the surface, $\chi = 2 - 2g$.								
cardona-qft-methods_ F02298	182	1.000	$S\ 0(D+1)$	$SO(D + 1)$	gin classes for $SO(D + 1)$ and $SO(D)$ in D dimensions. ²⁸								
cardona-qft-methods_ F02299	182	1.000	$S\ 0(D)$	$SO(D)$	gin classes for $SO(D + 1)$ and $SO(D)$ in D dimensions. ²⁸								
cardona-qft-methods_ F02300	183	1.000	$\Omega^{\prime}$	Ω'	predictions as that over a different manifold, Ω' . In order for this change to								

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cardona-qft-methods_ F02301	183	■ 1.000	$\Omega^{\prime} \rightarrow \widetilde{\Omega^{\prime}} = -\Omega^{\prime}$	$\Omega' \rightarrow \widetilde{\Omega'} = -\Omega'$	orientation of the second manifold $\Omega' \rightarrow \widetilde{\Omega'} = -\Omega'$, and $\int_{\Omega'} \rightarrow -\int_{\widetilde{\Omega'}}$, so
cardona-qft-methods_ F02302	183	■ 1.000	$\int_{\Omega^{\prime}} \rightarrow -\int_{\widetilde{\Omega^{\prime}}}$	$\int_{\Omega'} \rightarrow -\int_{\widetilde{\Omega'}}$	orientation of the second manifold $\Omega' \rightarrow \widetilde{\Omega'} = -\Omega'$, and $\int_{\Omega'} \rightarrow -\int_{\widetilde{\Omega'}}$, so
cardona-qft-methods_ F02303	183	■ 1.000	$\widetilde{\Omega^{\prime}}$	$\widetilde{\Omega'}$	Ω and $\widetilde{\Omega'}$ along M in the right way to produce an orientable manifold,
cardona-qft-methods_ F02304	183	■ 0.955	$\chi[\Omega \cup \widetilde{\Omega'}]$	$\chi[\Omega \cup \widetilde{\Omega'}]$	$\chi[\Omega \cup \widetilde{\Omega'}]$. This integral can take an arbitrary integer value and hence κ
cardona-qft-methods_ F02305	183	■ 1.000	$D=2n+1$	$D = 2n + 1$	simple form (we take $D = 2n + 1$ and the sign corresponding to anti-de
cardona-qft-methods_ F02306	184	■ 0.979	$R^{ab} = \frac{1}{l^2} e^a e^b$	$R^{ab} = \frac{1}{l^2} e^a e^b$	one has a n-fold degenerate root $R^{ab} = \frac{1}{l^2} e^a e^b$, which corresponds to a
cardona-qft-methods_ F02307	184	■ 0.840	$\{ \}^{\text{k}}$	k	AdS solution ^k is a maximally symmetric geometry, a configuration with no
cardona-qft-methods_ F02308	184	■ 1.000	AdS/K	AdS/K	topologically distinct spaces of the form AdS/K , obtained by identification
cardona-qft-methods_ F02309	184	■ 0.793	$\alpha_p S$	$\alpha_p S$	by a choice of α_p s such that the Lagrangian reads
cardona-qft-methods_ F02310	184	■ 1.000	$\{ \}^{38,39,42}$	$38,39,42$	to a CS theory in even dimensions. ^{38,39,42} The name originates from the
cardona-qft-methods_ F02311	184	■ 1.000	M^{ab}	M^{ab}	fact that the Pfaffian of an antisymmetric matrix M^{ab} is related to its
cardona-qft-methods_ F02312	184	■ 0.995	$\{ \}^{\text{k}}$	k	^k Global AdS is denoted as $SO(D,2)/SO(D-1,1)$, which means a D -dimensional space
cardona-qft-methods_ F02313	184	■ 0.995	$S^0(D,2)/S^0(D-1,1)$	$SO(D,2)/SO(D-1,1)$	^k Global AdS is denoted as $SO(D,2)/SO(D-1,1)$, which means a D -dimensional space
cardona-qft-methods_ F02314	184	■ 1.000	$S^0(D,2)$	$SO(D,2)$	of Lorentzian signature and isometry group $SO(D,2)$.
cardona-qft-methods_ F02315	185	■ 1.000	L_{2n}^{BI}	L_{2n}^{BI}	Hence, one can understand L_{2n}^{BI} as $\sqrt{\det[R^{ab} + \frac{1}{l^2} e^a e^b]}$, which is reminiscent
cardona-qft-methods_ F02316	185	■ 1.000	$\sqrt{\det[R^{ab} + \frac{1}{l^2} e^a e^b]}$	$\sqrt{\det[R^{ab} + \frac{1}{l^2} e^a e^b]}$	Hence, one can understand L_{2n}^{BI} as $\sqrt{\det[R^{ab} + \frac{1}{l^2} e^a e^b]}$, which is reminiscent
cardona-qft-methods_ F02317	185	■ 1.000	$L_{EM}^{BI} = \sqrt{\det[F_{\mu\nu} - \lambda\eta_{\mu\nu}]}$	$L_{EM}^{BI} = \sqrt{\det[F_{\mu\nu} - \lambda\eta_{\mu\nu}]}$	of the electromagnetic Born-Infeld Lagrangian, $L_{EM}^{BI} = \sqrt{\det[F_{\mu\nu} - \lambda\eta_{\mu\nu}]}$.
cardona-qft-methods_ F02318	185	■ 1.000	$\{ \}^{45,46}$	$45,46$	nontrivial critical points, ^{45,46} or more generally, systems whose symplectic
cardona-qft-methods_ F02319	185	■ 1.000	$R^{ab} + \frac{1}{l^2} e^a e^b = 0$	$R^{ab} + \frac{1}{l^2} e^a e^b = 0$	$R^{ab} + \frac{1}{l^2} e^a e^b = 0$, in contrast with the situation when all α_p 's are arbitrary.
cardona-qft-methods_ F02320	186	■ 1.000	$I[\phi^r]$	$I[\phi^r]$	To illustrate the problem consider the variation of an action $I[\phi^r]$ with
cardona-qft-methods_ F02321	186	■ 1.000	$\Theta=0$	$\Theta = 0$	consider changing the boundary conditions in order to make sure $\Theta = 0$,
cardona-qft-methods_ F02322	186	■ 1.000	$B(\phi^r _{\partial M})$	$B(\phi^r _{\partial M})$	adding a suitably defined function of the fields at the boundary, $B(\phi^r _{\partial M})$,
cardona-qft-methods_ F02323	187	■ 0.901	$\left(\epsilon_{abc\dots} \rightarrow \epsilon\right)$	$(\epsilon_{abc\dots} \rightarrow \epsilon$	where we have suppressed all Lorentz indices ($\epsilon_{abc\dots} \rightarrow \epsilon$, etc.). Here $\widetilde{R}$ and
cardona-qft-methods_ F02324	187	■ 0.901	$\widetilde{R}$	$\widetilde{R}$	where we have suppressed all Lorentz indices ($\epsilon_{abc\dots} \rightarrow \epsilon$, etc.). Here $\widetilde{R}$ and
cardona-qft-methods_ F02325	188	■ 1.000	$\bar{A}$	$\bar{A}$	where A and $\bar{A}$ are two connections in the same Lie algebra. There is an
cardona-qft-methods_ F02326	188	■ 1.000	$\mathcal{B}_{2n}$	$\mathcal{B}_{2n}$	term $\mathcal{B}_{2n}$ in the transgression form. The interpretation now presents some

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cardona-qft-methods_ F02327	188	■ 0.892	$(\bar{A})$	$(\bar{A})$	a second dynamically independent set of fields $(\bar{A})$, having exactly the same
cardona-qft-methods_ F02328	189	■ 1.000	$\mathcal{B}_{2n}(A, \bar{A})$	$\mathcal{B}_{2n}(A, \bar{A})$	$\mathcal{B}_{2n}(A, \bar{A})$, and therefore the bulk where A is defined need not be the same
cardona-qft-methods_ F02329	189	■ 1.000	$\mathcal{C}(\bar{A})$	$\mathcal{C}(\bar{A})$	The negative sign in front of $\mathcal{C}(\bar{A})$ is an indication that the orientation
cardona-qft-methods_ F02330	189	■ 1.000	$\bar{M}$	$\bar{M}$	where the orientation of $\bar{M}$ has been reversed so that at the common bound-
cardona-qft-methods_ F02331	189	■ 1.000	$\partial M = -\partial \bar{M}$	$\partial M = -\partial \bar{M}$	ary, $\partial M = -\partial \bar{M}$. In other words, the two cobordant manifolds M and $\bar{M}$,
cardona-qft-methods_ F02332	189	■ 0.998	(M)	(M)	of spacetime (M) characterized by the dynamical field A . At he boundary
cardona-qft-methods_ F02333	190	■ 1.000	$\{ \}^{9,52}$	9,52	field theory, ^{9,52} recently a different use has been identified. As noted in
cardona-qft-methods_ F02334	190	■ 1.000	$\{ \}^{53-55}$	$^{53-55}$	some gauge color. ⁵³⁻⁵⁵
cardona-qft-methods_ F02335	190	■ 1.000	$A_{\mu_1 \mu_2 \cdots \mu_p} j^{\mu_1 \mu_2 \cdots \mu_p}$	$A_{\mu_1 \mu_2 \cdots \mu_p} j^{\mu_1 \mu_2 \cdots \mu_p}$	be achieved by the “more natural” minimal couplings $A_{\mu_1 \mu_2 \cdots \mu_p} j^{\mu_1 \mu_2 \cdots \mu_p}$,
cardona-qft-methods_ F02336	190	■ 1.000	I_{Int}	I_{Int}	The invariance of I_{Int} under gauge transformations is a consequence of
cardona-qft-methods_ F02337	190	■ 1.000	$j^{\prime \mu} = j^\mu$	$j^{\prime \mu} = j^\mu$	two properties of the current: its gauge invariance, $j^{\prime \mu} = j^\mu$ –as expected
cardona-qft-methods_ F02338	191	■ 0.999	$\sqrt{ g }$	$\sqrt{ g }$	and, therefore, the integrand in (1) is coordinate invariant. No “ $\sqrt{ g }$ ” is
cardona-qft-methods_ F02339	191	■ 1.000	$j = q * \delta(\Gamma)$	$j = q * \delta(\Gamma)$	the particle, $j = q * \delta(\Gamma)$ where e is the electric charge. Hence, $A_\mu j^\mu d^4 x =$
cardona-qft-methods_ F02340	191	■ 1.000	$A_\mu j^\mu d^4 x =$	$A_\mu j^\mu d^4 x =$	the particle, $j = q * \delta(\Gamma)$ where e is the electric charge. Hence, $A_\mu j^\mu d^4 x =$
cardona-qft-methods_ F02341	191	■ 1.000	$q A \wedge \delta(\Gamma)$	$q A \wedge \delta(\Gamma)$	$q A \wedge \delta(\Gamma)$, and (1) reads
cardona-qft-methods_ F02342	191	■ 1.000	z^μ	z^μ	where the coordinate z^μ is any convenient parametrization of the worldline
cardona-qft-methods_ F02343	191	■ 0.484	$(2p+1)$	$(2p+1)$	A $(2p+1)$ -CS form describes the coupling between a connection $\mathbf{A}$ and a
cardona-qft-methods_ F02344	192	■ 0.881	$\mathfrak{C}_{2p+1}$	$\mathfrak{C}_{2p+1}$	where $\mathfrak{C}_{2p+1}$ is the algebra-valued form whose trace is the $(2p+1)$ -CS form
cardona-qft-methods_ F02345	192	■ 1.000	$\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$	$\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$	living on the brane history, $\mathcal{C}_{2p+1}(\mathbf{A}) = \langle \mathfrak{C}_{2p+1}(\mathbf{A}) \rangle$. The current generated
cardona-qft-methods_ F02346	192	■ 0.852	$D-2p-1$	$D-2p-1$	by the $2p$ -brane is represented by the $(D-2p-1)$ -form $\mathbf{j}$ supported on the
cardona-qft-methods_ F02347	192	■ 0.852	$\mathbf{j}$	$\mathbf{j}$	by the $2p$ -brane is represented by the $(D-2p-1)$ -form $\mathbf{j}$ supported on the
cardona-qft-methods_ F02348	192	■ 1.000	dx^{α_i}	dx^{α_i}	where dx^{α_i} are transverse directions to Γ . The integration over the $D-2p-1$
cardona-qft-methods_ F02349	192	■ 1.000	$I[\mathbf{A}; \mathbf{j}]$	$I[\mathbf{A}; \mathbf{j}]$	can be checked directly by noting that, $I[\mathbf{A}; \mathbf{j}]$ changes by a (locally) exact
cardona-qft-methods_ F02350	192	■ 1.000	$\mathbf{j}_{2p}$	$\mathbf{j}_{2p}$	which can be independently checked for (4). Moreover, if the current $\mathbf{j}_{2p}$
cardona-qft-methods_ F02351	192	■ 0.999	$j^{a_1 a_2 \cdots a_s} \mathbf{K}_{a_1} \mathbf{K}_{a_2} \cdots \mathbf{K}_{a_s}$	$j^{a_1 a_2 \cdots a_s} \mathbf{K}_{a_1} \mathbf{K}_{a_2} \cdots \mathbf{K}_{a_s}$	reduced to the subgroup that commutes with $j^{a_1 a_2 \cdots a_s} \mathbf{K}_{a_1} \mathbf{K}_{a_2} \cdots \mathbf{K}_{a_s}$.
cardona-qft-methods_ F02352	192	■ 0.737	AdS_3	AdS_3	identification is made in the spatial slice of AdS_3 , using a rotational Killing
cardona-qft-methods_ F02353	193	■ 1.000	$\{ \}^{62,63}$	62,63	critical temperature. ^{62,63}

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cardona-qft-methods_F02354	193	0.531	(2+1)	$(2 + 1)$	A (2+1)-dimensional CS system is naturally generated at the interface
cardona-qft-methods_F02355	193	1.000	θ_1	θ_1	Θ takes a constant value θ_1 in the region $\tilde{M} \subset M$ and θ_2 elsewhere, then
cardona-qft-methods_F02356	193	1.000	$\tilde{M} \subset M$	$\tilde{M} \subset M$	Θ takes a constant value θ_1 in the region $\tilde{M} \subset M$ and θ_2 elsewhere, then
cardona-qft-methods_F02357	193	1.000	θ_2	θ_2	Θ takes a constant value θ_1 in the region $\tilde{M} \subset M$ and θ_2 elsewhere, then
cardona-qft-methods_F02358	193	1.000	$j = d\Theta = (\theta_1 - \theta_2)\delta(\Sigma)dz$	$j = d\Theta = (\theta_1 - \theta_2)\delta(\Sigma)dz$	where the surface current is the one-form $j = d\Theta = (\theta_1 - \theta_2)\delta(\Sigma)dz$, and z
cardona-qft-methods_F02359	193	1.000	$\tilde{M}, \Sigma = \partial\tilde{M}$	$\tilde{M}, \Sigma = \partial\tilde{M}$	is the coordinate along outward normal to the surface of $\tilde{M}$, $\Sigma = \partial\tilde{M}$. Since
cardona-qft-methods_F02360	193	1.000	$\tilde{M}$	$\tilde{M}$	the θ -term is locally exact, the field equations, both inside and outside $\tilde{M}$
cardona-qft-methods_F02361	193	0.934	$\theta \neq 0$	$\theta \neq 0$	in which $\theta \neq 0$. ⁶⁵ It turns out that the so-called topological insulators
cardona-qft-methods_F02362	200	1.000	$\{12, 33, 35, 36, 46\}$	12,33,35,36,46	tions to cosmology, along the lines of the work. ^{12,33,35,36,46} This writeup
cardona-qft-methods_F02363	201	0.996	$SU(3)$	$SU(3)$	ometrically in terms of bundles (typically $SU(2)$ or $SU(3)$ -bundles) over
cardona-qft-methods_F02364	202	1.000	$S = S_{EH} + S_{SM}$	$S = S_{EH} + S_{SM}$	of the form $S = S_{EH} + S_{SM}$ with S_{SM} the action functional for a particle
cardona-qft-methods_F02365	202	1.000	S_{SM}	S_{SM}	of the form $S = S_{EH} + S_{SM}$ with S_{SM} the action functional for a particle
cardona-qft-methods_F02366	202	1.000	$f(R, R^{\mu\nu}, C_{\lambda\mu\nu\kappa})$	$f(R, R^{\mu\nu}, C_{\lambda\mu\nu\kappa})$	terms $f(R, R^{\mu\nu}, C_{\lambda\mu\nu\kappa})$ that include conformal gravity. This means that
cardona-qft-methods_F02367	203	1.000	$C^\infty(X)$	$C^\infty(X)$	subalgebras like $C^\infty(X)$ if X is a smooth manifold, leads to good algebraic
cardona-qft-methods_F02368	204	1.000	$(\mathcal{A}, \mathcal{H}, D)$	$(\mathcal{A}, \mathcal{H}, D)$	data of a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ consist of the following:
cardona-qft-methods_F02369	204	1.000	$\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})$	$\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})$	• a representation $\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})$ in the algebra $\mathcal{L}(\mathcal{H})$ of bounded linear
cardona-qft-methods_F02370	204	1.000	$\mathcal{L}(\mathcal{H})$	$\mathcal{L}(\mathcal{H})$	• a representation $\pi: \mathcal{A} \rightarrow \mathcal{L}(\mathcal{H})$ in the algebra $\mathcal{L}(\mathcal{H})$ of bounded linear
cardona-qft-methods_F02371	204	1.000	$D = D^*$	$D = D^*$	• a self adjoint operator $D = D^*$ acting on $\mathcal{H}$ with a dense domain;
cardona-qft-methods_F02372	204	1.000	$(1 + D^2)^{-1/2}$	$(1 + D^2)^{-1/2}$	• the compact resolvent condition that $(1 + D^2)^{-1/2}$ belongs to the algebra
cardona-qft-methods_F02373	204	1.000	$\mathcal{K}$	$\mathcal{K}$	$\mathcal{K}$ of compact operators;
cardona-qft-methods_F02374	204	0.999	$[a, D]$	$[a, D]$	• the condition that commutators $[a, D]$ of elements in the algebra with
cardona-qft-methods_F02375	204	1.000	$\mathbb{Z}/2$	$\mathbb{Z}/2$	• The spectral triples is <i>even</i> if there is also a $\mathbb{Z}/2$ -grading γ on $\mathcal{H}$ such
cardona-qft-methods_F02376	204	0.895	$(C^\infty(M), L^2(M, S), \neq_M)$	$(C^\infty(M), L^2(M, S), \neq_M)$	the canonical spectral triple has the form $(C^\infty(M), L^2(M, S), \not\partial_M)$, with
cardona-qft-methods_F02377	204	1.000	$L^2(M, S)$	$L^2(M, S)$	S the spinor bundle and $L^2(M, S)$ the Hilbert space of square integrable
cardona-qft-methods_F02378	204	1.000	$D = \partial_M$	$D = \partial_M$	operator $D = \not\partial_M$. In the 4-dimensional case we will be mostly interested
cardona-qft-methods_F02379	204	0.997	γ_5	γ_5	in here, the even structure is given by the chirality operator γ_5 , the product
cardona-qft-methods_F02380	204	1.000	$\{22, 31, 49\}$	22,31,49	interesting generalizations of this basic case, see. ^{22,31,49}

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cardona-qft-methods_F02381	205	■1.000	$n \in \mathbb{Z} / 8 \mathbb{Z}$	$n \in \mathbb{Z} / 8 \mathbb{Z}$	$n \in \mathbb{Z} / 8 \mathbb{Z}$, an integer modulo 8.																																				
cardona-qft-methods_F02382	205	■1.000	$\epsilon, \epsilon', \epsilon'' \in \{\pm 1\}$	$\epsilon, \epsilon', \epsilon'' \in \{\pm 1\}$	where $\epsilon, \epsilon', \epsilon'' \in \{\pm 1\}$ are signed satisfying one of the possibilities listed in																																				
cardona-qft-methods_F02383	205	■0.993	ε	ε	<table><tr><th>n</th><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1	
n	0	1	2	3	4	5	6	7																																	
ε	1	1	-1	-1	-1	-1	1	1																																	
ε'	1	-1	1	1	1	-1	1	1																																	
ε''	1		-1		1		-1																																		
cardona-qft-methods_F02384	205	■0.993	ε'	ε'	<table><tr><th>n</th><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1	
n	0	1	2	3	4	5	6	7																																	
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ε'	1	-1	1	1	1	-1	1	1																																	
ε''	1		-1		1		-1																																		
cardona-qft-methods_F02385	205	■0.993	ε''	ε''	<table><tr><th>n</th><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>ε</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε'</td><td>1</td><td>-1</td><td>1</td><td>1</td><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>ε''</td><td>1</td><td></td><td>-1</td><td></td><td>1</td><td></td><td>-1</td><td></td></tr></table>	n	0	1	2	3	4	5	6	7	ε	1	1	-1	-1	-1	-1	1	1	ε'	1	-1	1	1	1	-1	1	1	ε''	1		-1		1		-1	
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ε	1	1	-1	-1	-1	-1	1	1																																	
ε'	1	-1	1	1	1	-1	1	1																																	
ε''	1		-1		1		-1																																		
cardona-qft-methods_F02386	205	■0.999	$\left[a, b^0\right]=0$	$\left[a, b^0\right]=0$	relation $\left[a, b^0\right]=0$, for all $a, b \in \mathcal{A}$, where the notation b^0 stands for $b^0 =$																																				
cardona-qft-methods_F02387	205	■0.999	$a, b \in \mathcal{A}$	$a, b \in \mathcal{A}$	relation $\left[a, b^0\right]=0$, for all $a, b \in \mathcal{A}$, where the notation b^0 stands for $b^0 =$																																				
cardona-qft-methods_F02388	205	■0.999	b^0	b^0	relation $\left[a, b^0\right]=0$, for all $a, b \in \mathcal{A}$, where the notation b^0 stands for $b^0 =$																																				
cardona-qft-methods_F02389	205	■0.999	$b^0=$	$b^0=$	relation $\left[a, b^0\right]=0$, for all $a, b \in \mathcal{A}$, where the notation b^0 stands for $b^0 =$																																				
cardona-qft-methods_F02390	205	■1.000	$J b^* J^{-1}$	$J b^* J^{-1}$	$J b^* J^{-1}$ for $b \in \mathcal{A}$, and the <i>order one condition</i> :																																				
cardona-qft-methods_F02391	205	■1.000	$b \in \mathcal{A}$	$b \in \mathcal{A}$	$J b^* J^{-1}$ for $b \in \mathcal{A}$, and the <i>order one condition</i> :																																				
cardona-qft-methods_F02392	206	■1.000	$\{ \}^{5,9}$	5,9	They have been studied in this general form in. ^{5,9}																																				
cardona-qft-methods_F02393	206	■1.000	$F=$	$F=$	A <i>finite noncommutative space</i> means a spectral triple $F=$																																				
cardona-qft-methods_F02394	206	■0.986	$\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$	$\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$	$\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$ where A_F is a <i>finite dimensional</i> algebra. This means that,																																				
cardona-qft-methods_F02395	206	■0.986	A_F	A_F	$\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$ where A_F is a <i>finite dimensional</i> algebra. This means that,																																				
cardona-qft-methods_F02396	206	■1.000	$F=\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$	$F=\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$	The data of $F=\left(\mathcal{A}_F, \mathcal{H}_F, D_F\right)$ are given by																																				
cardona-qft-methods_F02397	206	■1.000	$\mathbb{K}_i=\mathbb{R}$	$\mathbb{K}_i=\mathbb{R}$	where $\mathbb{K}_i=\mathbb{R}$ or $\mathbb{C}$ or the quaternions $\mathbb{H}$;																																				
cardona-qft-methods_F02398	206	■1.000	$\mathbb{H}$	$\mathbb{H}$	where $\mathbb{K}_i=\mathbb{R}$ or $\mathbb{C}$ or the quaternions $\mathbb{H}$;																																				
cardona-qft-methods_F02399	206	■1.000	$\mathcal{H}_F$	$\mathcal{H}_F$	• a representation on a finite dimensional Hilbert space $\mathcal{H}_F$;																																				
cardona-qft-methods_F02400	207	■1.000	$D^*=D$	$D^*=D$	• a self adjoint liner operator $D^*=D$ satisfying the order one condition																																				
cardona-qft-methods_F02401	207	■0.957	$\left[[D, a], b^0\right]=0$	$\left[[D, a], b^0\right]=0$	$\left[[D, a], b^0\right]=0$.																																				
cardona-qft-methods_F02402	207	■0.992	$\left(\mathcal{A}_F, \mathcal{H}_F\right)$	$\left(\mathcal{A}_F, \mathcal{H}_F\right)$	tary equivalence on a given $\left(\mathcal{A}_F, \mathcal{H}_F\right)$. These moduli spaces will provide																																				

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cardona-qft-methods_ F02483	207	■ 1.000	$\left(\lambda, q_L, q_R, m\right) \mapsto \left(\bar{\lambda}, \bar{q}_L, \bar{q}_R, m^*\right)$	$(\lambda, q_L, q_R, m) \mapsto (\bar{\lambda}, \bar{q}_L, \bar{q}_R, m^*)$	This has an involution $(\lambda, q_L, q_R, m) \mapsto (\bar{\lambda}, \bar{q}_L, \bar{q}_R, m^*)$ and two subalgebras:
cardona-qft-methods_ F02484	207	■ 1.000	$\mathbb{C} \oplus M_3(\mathbb{C})$	$\mathbb{C} \oplus M_3(\mathbb{C})$	$\mathbb{C} \oplus M_3(\mathbb{C})$, regarded as the integer spin $\mathbb{C}$ -algebra, and $\mathbb{H}_L \oplus \mathbb{H}_R$, the half-
cardona-qft-methods_ F02485	207	■ 1.000	$\mathbb{H}_L \oplus \mathbb{H}_R$	$\mathbb{H}_L \oplus \mathbb{H}_R$	$\mathbb{C} \oplus M_3(\mathbb{C})$, regarded as the integer spin $\mathbb{C}$ -algebra, and $\mathbb{H}_L \oplus \mathbb{H}_R$, the half-
cardona-qft-methods_ F02486	208	■ 0.651	$s=(1,-1,-1,1)$	$s = (1, -1, -1, 1)$	is odd of $s = (1, -1, -1, 1)$ acts by $Ad(s) = -1$. These bimodules can
cardona-qft-methods_ F02487	208	■ 0.651	$Ad(s)=-1$	$Ad(s) = -1$	is odd of $s = (1, -1, -1, 1)$ acts by $Ad(s) = -1$. These bimodules can
cardona-qft-methods_ F02488	208	■ 1.000	$\mathcal{B} =$	$\mathcal{B} =$	equivalently be described as representations of the complex algebra $\mathcal{B} =$
cardona-qft-methods_ F02489	208	■ 0.990	$\left(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op}\right)_p$	$(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op})_p$	$(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op})_p$, where $p = \frac{1}{2}(1 - s \otimes s^0)$, with $\mathcal{A}^0 = \mathcal{A}^{op}$, see. ^{17,24} This
cardona-qft-methods_ F02410	208	■ 0.990	$p=\frac{1}{2}\left(1-s \otimes s^0\right)$	$p = \frac{1}{2} (1 - s \otimes s^0)$	$(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op})_p$, where $p = \frac{1}{2}(1 - s \otimes s^0)$, with $\mathcal{A}^0 = \mathcal{A}^{op}$, see. ^{17,24} This
cardona-qft-methods_ F02411	208	■ 0.990	$\mathcal{A}^0=\mathcal{A}^{op}$	$\mathcal{A}^0 = \mathcal{A}^{op}$	$(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op})_p$, where $p = \frac{1}{2}(1 - s \otimes s^0)$, with $\mathcal{A}^0 = \mathcal{A}^{op}$, see. ^{17,24} This
cardona-qft-methods_ F02412	208	■ 0.990	$\{ \}^{17,24}$	^{17,24}	$(\mathcal{A}_{LR} \otimes_{\mathbb{R}} \mathcal{A}_{LR}^{op})_p$, where $p = \frac{1}{2}(1 - s \otimes s^0)$, with $\mathcal{A}^0 = \mathcal{A}^{op}$, see. ^{17,24} This
cardona-qft-methods_ F02413	208	■ 1.000	$\mathcal{M}_F$	$\mathcal{M}_F$	and one can consider the bimodule $\mathcal{M}_F$ which is the sum of all the inequiv-
cardona-qft-methods_ F02414	208	■ 0.999	$\mathcal{A}_{LR}$	$\mathcal{A}_{LR}$	alent irreducible odd $\mathcal{A}_{LR}$ -bimodules. This has the following properties: as
cardona-qft-methods_ F02415	208	■ 1.000	$\dim_{\mathbb{C}} \mathcal{M}_F=32$	$\dim_{\mathbb{C}} \mathcal{M}_F = 32$	a vector space $\dim_{\mathbb{C}} \mathcal{M}_F = 32$; it decomposes as $\mathcal{M}_F = \mathcal{E} \oplus \mathcal{E}^0$, with
cardona-qft-methods_ F02416	208	■ 1.000	$\mathcal{M}_F=\mathcal{E} \oplus \mathcal{E}^0$	$\mathcal{M}_F = \mathcal{E} \oplus \mathcal{E}^0$	a vector space $\dim_{\mathbb{C}} \mathcal{M}_F = 32$; it decomposes as $\mathcal{M}_F = \mathcal{E} \oplus \mathcal{E}^0$, with
cardona-qft-methods_ F02417	208	■ 0.993	$\mathbb{C}, \mathbb{H}$	$\mathbb{C}, \mathbb{H}$	where 1 , 2 , 3 denote the standard representations of $\mathbb{C}$, $\mathbb{H}$ and $M_3(\mathbb{C})$ of
cardona-qft-methods_ F02418	208	■ 0.993	$M_3(\mathbb{C})$	$M_3(\mathbb{C})$	where 1 , 2 , 3 denote the standard representations of $\mathbb{C}$, $\mathbb{H}$ and $M_3(\mathbb{C})$ of
cardona-qft-methods_ F02419	208	■ 1.000	$\mathcal{B}$	$\mathcal{B}$	representations of $\mathcal{B}$,
cardona-qft-methods_ F02420	209	■ 1.000	$\mathcal{M}_F \cong \mathcal{M}_F^0$	$\mathcal{M}_F \cong \mathcal{M}_F^0$	There is an antilinear isomorphism $\mathcal{M}_F \cong \mathcal{M}_F^0$ given by $J_F(\xi, \bar{\eta}) =$
cardona-qft-methods_ F02421	209	■ 1.000	$J_F(\xi, \bar{\eta}) =$	$J_F(\xi, \bar{\eta}) =$	There is an antilinear isomorphism $\mathcal{M}_F \cong \mathcal{M}_F^0$ given by $J_F(\xi, \bar{\eta}) =$
cardona-qft-methods_ F02422	209	■ 1.000	$(\eta, \bar{\xi})$	$(\eta, \bar{\xi})$	$(\eta, \bar{\xi})$ for $\xi, \eta \in \mathcal{E}$
cardona-qft-methods_ F02423	209	■ 1.000	$\xi, \eta \in \mathcal{E}$	$\xi, \eta \in \mathcal{E}$	$(\eta, \bar{\xi})$ for $\xi, \eta \in \mathcal{E}$
cardona-qft-methods_ F02424	209	■ 1.000	$\mathbb{Z} / 2 \mathbb{Z}$	$\mathbb{Z} / 2 \mathbb{Z}$	There is a $\mathbb{Z} / 2 \mathbb{Z}$ -grading $\gamma_F = c - J_F c J_F$ with $c = (0, 1, -1, 0) \in \mathcal{A}_{LR}$,
cardona-qft-methods_ F02425	209	■ 1.000	$\gamma_F=c-J_F c J_F$	$\gamma_F = c - J_F c J_F$	There is a $\mathbb{Z} / 2 \mathbb{Z}$ -grading $\gamma_F = c - J_F c J_F$ with $c = (0, 1, -1, 0) \in \mathcal{A}_{LR}$,
cardona-qft-methods_ F02426	209	■ 1.000	$c=(0,1,-1,0) \in \mathcal{A}_{LR}$	$c = (0, 1, -1, 0) \in \mathcal{A}_{LR}$	There is a $\mathbb{Z} / 2 \mathbb{Z}$ -grading $\gamma_F = c - J_F c J_F$ with $c = (0, 1, -1, 0) \in \mathcal{A}_{LR}$,
cardona-qft-methods_ F02427	209	■ 0.500	$\bmod 8$	mod 8	mod 8, by the table of signs for real structures.
cardona-qft-methods_ F02428	209	■ 0.899	$ \uparrow\rangle$	$ \uparrow\rangle$	the particles. One fixes a notation for a basis $ \uparrow\rangle$ and $ \downarrow\rangle$ of 2 determined
cardona-qft-methods_ F02429	209	■ 0.899	$ \downarrow\rangle$	$ \downarrow\rangle$	the particles. One fixes a notation for a basis $ \uparrow\rangle$ and $ \downarrow\rangle$ of 2 determined

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cardona-qft-methods_ F02430	209	■ 0.899	$\mathbf{2}$	2	the particles. One fixes a notation for a basis $ \uparrow\rangle$ and $ \downarrow\rangle$ of 2 determined
cardona-qft-methods_ F02431	209	■ 1.000	$\mathbf{2}_L \otimes \mathbf{1}^0$	$2_L \otimes \mathbf{1}^0$	• the subspace $2_L \otimes \mathbf{1}^0$: gives the neutrinos $\nu_L \in \uparrow\rangle_L \otimes \mathbf{1}^0$ and the charged
cardona-qft-methods_ F02432	209	■ 1.000	$\nu_L \in \uparrow\rangle_L \otimes \mathbf{1}^0$	$\nu_L \in \uparrow\rangle_L \otimes \mathbf{1}^0$	• the subspace $2_L \otimes \mathbf{1}^0$: gives the neutrinos $\nu_L \in \uparrow\rangle_L \otimes \mathbf{1}^0$ and the charged
cardona-qft-methods_ F02433	209	■ 1.000	$e_L \in \downarrow\rangle_L \otimes \mathbf{1}^0$	$e_L \in \downarrow\rangle_L \otimes \mathbf{1}^0$	leptons $e_L \in \downarrow\rangle_L \otimes \mathbf{1}^0$;
cardona-qft-methods_ F02434	209	■ 1.000	$\mathbf{2}_R \otimes \mathbf{1}^0$	$2_R \otimes \mathbf{1}^0$	• the subspace $2_R \otimes \mathbf{1}^0$ gives the right-handed neutrinos $\nu_R \in \uparrow\rangle_R \otimes \mathbf{1}^0$
cardona-qft-methods_ F02435	209	■ 1.000	$\nu_R \in \uparrow\rangle_R \otimes \mathbf{1}^0$	$\nu_R \in \uparrow\rangle_R \otimes \mathbf{1}^0$	• the subspace $2_R \otimes \mathbf{1}^0$ gives the right-handed neutrinos $\nu_R \in \uparrow\rangle_R \otimes \mathbf{1}^0$
cardona-qft-methods_ F02436	209	■ 1.000	$e_R \in \downarrow\rangle_R \otimes \mathbf{1}^0$	$e_R \in \downarrow\rangle_R \otimes \mathbf{1}^0$	and the charged leptons $e_R \in \downarrow\rangle_R \otimes \mathbf{1}^0$;
cardona-qft-methods_ F02437	209	■ 0.724	$\mathbf{2}_L \otimes \mathbf{3}^0$	$2_L \otimes \mathbf{3}^0$	• the subspace $2_L \otimes \mathbf{3}^0$, where 3 represents the color indices gives the u/c/t
cardona-qft-methods_ F02438	209	■ 0.984	$u_L \in \uparrow\rangle_L \otimes \mathbf{3}^0$	$u_L \in \uparrow\rangle_L \otimes \mathbf{3}^0$	quarks $u_L \in \uparrow\rangle_L \otimes \mathbf{3}^0$ and the d/s/b quarks $d_L \in \downarrow\rangle_L \otimes \mathbf{3}^0$;
cardona-qft-methods_ F02439	209	■ 0.984	$d_L \in \downarrow\rangle_L \otimes \mathbf{3}^0$	$d_L \in \downarrow\rangle_L \otimes \mathbf{3}^0$	quarks $u_L \in \uparrow\rangle_L \otimes \mathbf{3}^0$ and the d/s/b quarks $d_L \in \downarrow\rangle_L \otimes \mathbf{3}^0$;
cardona-qft-methods_ F02440	209	■ 0.749	$\mathbf{2}_R \otimes \mathbf{3}^0$	$2_R \otimes \mathbf{3}^0$	• the subspace $2_R \otimes \mathbf{3}^0$ (with color indices) gives the u/c/t quarks $u_R \in$
cardona-qft-methods_ F02441	209	■ 0.749	$\mathrm{u} / \mathrm{c} / \mathrm{t}$	u/c/t	• the subspace $2_R \otimes \mathbf{3}^0$ (with color indices) gives the u/c/t quarks $u_R \in$
cardona-qft-methods_ F02442	209	■ 0.749	$u_R \in$	$u_R \in$	• the subspace $2_R \otimes \mathbf{3}^0$ (with color indices) gives the u/c/t quarks $u_R \in$
cardona-qft-methods_ F02443	209	■ 0.967	$ \uparrow\rangle_R \otimes \mathbf{3}^0$	$ \uparrow\rangle_R \otimes \mathbf{3}^0$	$ \uparrow\rangle_R \otimes \mathbf{3}^0$ and the d/s/b quarks $d_R \in \downarrow\rangle_R \otimes \mathbf{3}^0$;
cardona-qft-methods_ F02444	209	■ 0.967	$d_R \in \downarrow\rangle_R \otimes \mathbf{3}^0$	$d_R \in \downarrow\rangle_R \otimes \mathbf{3}^0$	$ \uparrow\rangle_R \otimes \mathbf{3}^0$ and the d/s/b quarks $d_R \in \downarrow\rangle_R \otimes \mathbf{3}^0$;
cardona-qft-methods_ F02445	209	■ 0.999	$\mathbf{1} \otimes \mathbf{2}_{L,R}^0$	$1 \otimes \mathbf{2}_{L,R}^0$	• the subspace $1 \otimes \mathbf{2}_{L,R}^0$ gives the antineutrinos $\bar{\nu}_{L,R} \in \mathbf{1} \otimes \uparrow\rangle_{L,R}^0$ and the
cardona-qft-methods_ F02446	209	■ 0.999	$\bar{\nu}_{L,R} \in \mathbf{1} \otimes \uparrow\rangle_{L,R}^0$	$\bar{\nu}_{L,R} \in \mathbf{1} \otimes \uparrow\rangle_{L,R}^0$	• the subspace $1 \otimes \mathbf{2}_{L,R}^0$ gives the antineutrinos $\bar{\nu}_{L,R} \in \mathbf{1} \otimes \uparrow\rangle_{L,R}^0$ and the
cardona-qft-methods_ F02447	209	■ 1.000	$\bar{e}_{L,R} \in \mathbf{1} \otimes \downarrow\rangle_{L,R}^0$	$\bar{e}_{L,R} \in \mathbf{1} \otimes \downarrow\rangle_{L,R}^0$	charged antileptons $\bar{e}_{L,R} \in \mathbf{1} \otimes \downarrow\rangle_{L,R}^0$;
cardona-qft-methods_ F02448	209	■ 1.000	$\mathbf{3} \otimes \mathbf{2}_{L,R}^0$	$3 \otimes \mathbf{2}_{L,R}^0$	• the subspace $3 \otimes \mathbf{2}_{L,R}^0$ (with color indices) gives the antiquarks $\bar{u}_{L,R} \in$
cardona-qft-methods_ F02449	209	■ 1.000	$\bar{u}_{L,R} \in$	$\bar{u}_{L,R} \in$	• the subspace $3 \otimes \mathbf{2}_{L,R}^0$ (with color indices) gives the antiquarks $\bar{u}_{L,R} \in$
cardona-qft-methods_ F02450	209	■ 1.000	$\mathbf{3} \otimes \uparrow\rangle_{L,R}^0$	$3 \otimes \uparrow\rangle_{L,R}^0$	$3 \otimes \uparrow\rangle_{L,R}^0$ and $\bar{d}_{L,R} \in \mathbf{3} \otimes \downarrow\rangle_{L,R}^0$.
cardona-qft-methods_ F02451	209	■ 1.000	$\bar{d}_{L,R} \in \mathbf{3} \otimes \downarrow\rangle_{L,R}^0$	$\bar{d}_{L,R} \in \mathbf{3} \otimes \downarrow\rangle_{L,R}^0$	$3 \otimes \uparrow\rangle_{L,R}^0$ and $\bar{d}_{L,R} \in \mathbf{3} \otimes \downarrow\rangle_{L,R}^0$.
cardona-qft-methods_ F02452	209	■ 1.000	$N=3$	$N = 3$	$N = 3$. This means taking as the Hilbert space of the finite geometry
cardona-qft-methods_ F02453	209	■ 1.000	$\mathcal{H}_F = \mathcal{M}_F \oplus \mathcal{M}_F \oplus \mathcal{M}_F$	$\mathcal{H}_F = \mathcal{M}_F \oplus \mathcal{M}_F \oplus \mathcal{M}_F$	$\mathcal{H}_F = \mathcal{M}_F \oplus \mathcal{M}_F \oplus \mathcal{M}_F$. The left action of $\mathcal{A}_{LR}$ is a sum of representations
cardona-qft-methods_ F02454	209	■ 1.000	$\pi _{\mathcal{H}_f} \oplus \pi' _{\mathcal{H}_{\bar{f}}}$ with $\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$ and $\mathcal{H}_{\bar{f}} = \mathcal{E}^0 \oplus \mathcal{E}^0 \oplus \mathcal{E}^0$ and with	$\pi _{\mathcal{H}_f} \oplus \pi' _{\mathcal{H}_{\bar{f}}}$	$\pi _{\mathcal{H}_f} \oplus \pi' _{\mathcal{H}_{\bar{f}}}$ with $\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$ and $\mathcal{H}_{\bar{f}} = \mathcal{E}^0 \oplus \mathcal{E}^0 \oplus \mathcal{E}^0$ and with
cardona-qft-methods_ F02455	209	■ 1.000	$\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$	$\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$	$\pi _{\mathcal{H}_f} \oplus \pi' _{\mathcal{H}_{\bar{f}}}$ with $\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$ and $\mathcal{H}_{\bar{f}} = \mathcal{E}^0 \oplus \mathcal{E}^0 \oplus \mathcal{E}^0$ and with

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cardona-qft-methods_ F02456	209	■ 1.000	$\mathcal{H}_{\bar{f}} = \mathcal{H}_{\bar{f}} = \mathcal{H}_{\bar{f}} \oplus \mathcal{H}_{\bar{f}} \oplus \mathcal{H}_{\bar{f}}$	$\mathcal{H}_{\bar{f}} = \mathcal{E}^0 \oplus \mathcal{E}^0 \oplus \mathcal{E}^0$	$\pi _{\mathcal{H}_f} \oplus \pi' _{\mathcal{H}_{\bar{f}}}$ with $\mathcal{H}_f = \mathcal{E} \oplus \mathcal{E} \oplus \mathcal{E}$ and $\mathcal{H}_{\bar{f}} = \mathcal{E}^0 \oplus \mathcal{E}^0 \oplus \mathcal{E}^0$ and with
cardona-qft-methods_ F02457	210	■ 1.000	$\mathcal{H}_f$	$\mathcal{H}_f$	mixes $\mathcal{H}_f$ and $\mathcal{H}_{\bar{f}}$ would not satisfy the order one condition for $\mathcal{A}_{LR}$. Thus,
cardona-qft-methods_ F02458	210	■ 1.000	$\mathcal{H}_{\bar{f}}$	$\mathcal{H}_{\bar{f}}$	mixes $\mathcal{H}_f$ and $\mathcal{H}_{\bar{f}}$ would not satisfy the order one condition for $\mathcal{A}_{LR}$. Thus,
cardona-qft-methods_ F02459	210	■ 1.000	$(\mathcal{A}_F, D_F)$	$(\mathcal{A}_F, D_F)$	as a coupled problem for the pair $(\mathcal{A}_F, D_F)$, for fixed $\mathcal{H}_F$ as above, where
cardona-qft-methods_ F02460	210	■ 0.973	$\mathcal{A}_F \subset \mathcal{A}_{LR}$	$\mathcal{A}_F \subset \mathcal{A}_{LR}$	$\mathcal{A}_F \subset \mathcal{A}_{LR}$ is a subalgebra and D_F has off diagonal terms. The algebra for
cardona-qft-methods_ F02461	210	■ 0.973	D_F	D_F	$\mathcal{A}_F \subset \mathcal{A}_{LR}$ is a subalgebra and D_F has off diagonal terms. The algebra for
cardona-qft-methods_ F02462	210	■ 0.996	$\mathcal{A}_F$	$\mathcal{A}_F$	the finite spectral triple will be the maximal $\mathcal{A}_F$ (up to isomorphism) that
cardona-qft-methods_ F02463	210	■ 1.000	$\text{Aut}(\mathcal{A}_{LR})$	$\text{Aut}(\mathcal{A}_{LR})$	unique up to $\text{Aut}(\mathcal{A}_{LR})$, which represents a spontaneous breaking of the
cardona-qft-methods_ F02464	210	■ 1.000	$\lambda \in \text{U}(1)$	$\lambda \in \text{U}(1)$	in powers of $\lambda \in \text{U}(1)$ gives
cardona-qft-methods_ F02465	210	■ 1.000	$(\mathcal{A}_F, \mathcal{H}_F, \gamma_F, J_F)$	$(\mathcal{A}_F, \mathcal{H}_F, \gamma_F, J_F)$	possible Dirac operators for $(\mathcal{A}_F, \mathcal{H}_F, \gamma_F, J_F)$, namely all the self-adjoint
cardona-qft-methods_ F02466	210	■ 1.000	J_F	J_F	linear maps D_F on $\mathcal{H}_F$, commuting with J_F , anticommuting with γ_F and
cardona-qft-methods_ F02467	210	■ 1.000	γ_F	γ_F	linear maps D_F on $\mathcal{H}_F$, commuting with J_F , anticommuting with γ_F and
cardona-qft-methods_ F02468	210	■ 1.000	$a, b \in \mathcal{A}_F$	$a, b \in \mathcal{A}_F$	satisfying $[[D, a], b^0] = 0$, for all $a, b \in \mathcal{A}_F$. Here one also imposes an addi-
cardona-qft-methods_ F02469	211	■ 1.000	S_3	S_3	same for S_3 , with
cardona-qft-methods_ F02470	211	■ 1.000	$Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \text{GL}_3(\mathbb{C})$	$Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \text{GL}_3(\mathbb{C})$	$Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \text{GL}_3(\mathbb{C})$ and Y_R symmetric:
cardona-qft-methods_ F02471	211	■ 1.000	Y_R	Y_R	$Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_{(\downarrow 3)}, Y_{(\uparrow 3)} \in \text{GL}_3(\mathbb{C})$ and Y_R symmetric:
cardona-qft-methods_ F02472	211	■ 1.000	$\mathcal{C}_3 \times \mathcal{C}_1$	$\mathcal{C}_3 \times \mathcal{C}_1$	moduli space $\mathcal{C}_3 \times \mathcal{C}_1$, where $\mathcal{C}_3$ is the set of pairs $(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$ modulo the
cardona-qft-methods_ F02473	211	■ 1.000	$\mathcal{C}_3$	$\mathcal{C}_3$	moduli space $\mathcal{C}_3 \times \mathcal{C}_1$, where $\mathcal{C}_3$ is the set of pairs $(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$ modulo the
cardona-qft-methods_ F02474	211	■ 1.000	$(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$	$(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$	moduli space $\mathcal{C}_3 \times \mathcal{C}_1$, where $\mathcal{C}_3$ is the set of pairs $(Y_{(\downarrow 3)}, Y_{(\uparrow 3)})$ modulo the
cardona-qft-methods_ F02475	211	■ 1.000	W_j	W_j	where the W_j are unitary matrices, so that
cardona-qft-methods_ F02476	211	■ 1.000	$G = \text{GL}_3(\mathbb{C})$	$G = \text{GL}_3(\mathbb{C})$	with $G = \text{GL}_3(\mathbb{C})$ and $K = U(3)$. The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$
cardona-qft-methods_ F02477	211	■ 1.000	$K = U(3)$	$K = U(3)$	with $G = \text{GL}_3(\mathbb{C})$ and $K = U(3)$. The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$
cardona-qft-methods_ F02478	211	■ 1.000	$\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$	$\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$	with $G = \text{GL}_3(\mathbb{C})$ and $K = U(3)$. The real dimension is $\dim_{\mathbb{R}} \mathcal{C}_3 = 10 =$
cardona-qft-methods_ F02479	211	■ 0.978	$3+3+4$	$3 + 3 + 4$	$3 + 3 + 4$. These correspond to $3 + 3$ eigenvalues, 3 angles and 1 phase.
cardona-qft-methods_ F02480	211	■ 0.978	$3+3$	$3 + 3$	$3 + 3 + 4$. These correspond to $3 + 3$ eigenvalues, 3 angles and 1 phase.
cardona-qft-methods_ F02481	211	■ 1.000	$\mathcal{C}_1$	$\mathcal{C}_1$	Similarly, $\mathcal{C}_1$ is the set of triplets $(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R)$ with Y_R a symmetric
cardona-qft-methods_ F02482	211	■ 1.000	$(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R)$	$(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R)$	Similarly, $\mathcal{C}_1$ is the set of triplets $(Y_{(\downarrow 1)}, Y_{(\uparrow 1)}, Y_R)$ with Y_R a symmetric

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cardona-qft-methods_F02483	211	■ 1.000	$\pi: \mathcal{C}_1 \rightarrow \mathcal{C}_3$	$\pi: \mathcal{C}_1 \rightarrow \mathcal{C}_3$	by unitary matrices. There is a projection $\pi: \mathcal{C}_1 \rightarrow \mathcal{C}_3$ that forgets Y_R , so
cardona-qft-methods_F02484	211	■ 1.000	$Y_{\{R\}} \mapsto \lambda^2 Y_{\{R\}}$	$Y_R \mapsto \lambda^2 Y_R$	the fiber is given by symmetric matrices modulo $Y_R \mapsto \lambda^2 Y_R$ and the real
cardona-qft-methods_F02485	211	■ 1.000	$\operatorname{dim}_{\mathbb{R}}(\mathbb{R}) \times \left(\mathcal{C}_{\{3\}} \times \mathcal{C}_{\{1\}} \right) = 31$	$\dim_{\mathbb{R}}(\mathcal{C}_3 \times \mathcal{C}_1) = 31$	dimension is therefore $\dim_{\mathbb{R}}(\mathcal{C}_3 \times \mathcal{C}_1) = 31$, with the dimension of the fiber
cardona-qft-methods_F02486	211	■ 1.000	$12-1=11$	$12 - 1 = 11$	$12 - 1 = 11$.
cardona-qft-methods_F02487	212	■ 1.000	$\delta_{\uparrow}, \delta_{\downarrow}$	$\delta_{\uparrow}, \delta_{\downarrow}$	with $\delta_{\uparrow}, \delta_{\downarrow}$ diagonal matrices specifying the Dirac masses and
cardona-qft-methods_F02488	212	■ 1.000	$c_i = \cos \theta_i, s_i = \sin \theta_i, e_{\delta} = \exp(i\delta)$	$c_i = \cos \theta_i, s_i = \sin \theta_i, e_{\delta} = \exp(i\delta)$	specifying the angles and phase, $c_i = \cos \theta_i, s_i = \sin \theta_i, e_{\delta} = \exp(i\delta)$.
cardona-qft-methods_F02489	212	■ 1.000	$U_{\{C\ K\ M\}}$	U_{CKM}	Cabibbo–Kobayashi–Maskawa mixing matrix U_{CKM} and the Pontecorvo–
cardona-qft-methods_F02490	212	■ 1.000	$U_{\{P\ M\ N\ S\}}$	U_{PMNS}	Maki–Nakagawa–Sakata mixing matrix U_{PMNS} . In particular, because of
cardona-qft-methods_F02491	212	■ 0.969	$V^* V$	$V^* V$	$V^* V$ add up to 0. There are experimental constraints also on the neutrino
cardona-qft-methods_F02492	212	■ 1.000	$(K \times K) \backslash (G \times G) / K$	$(K \times K) \backslash (G \times G) / K$	metric spaces given by double quotients of Lie groups $(K \times K) \backslash (G \times G) / K$.
cardona-qft-methods_F02493	212	■ 1.000	ν	ν	with what is known as the ν MSM: an extension of the Minimal Standard
cardona-qft-methods_F02494	213	■ 0.699	$(\mathcal{A}_i, \mathcal{H}_i, D_i, \gamma_i, J_i, \gamma_i, J_i)$	$(\mathcal{A}_i, \mathcal{H}_i, D_i, \gamma_i, J_i)$	Given two spectral triples $(\mathcal{A}_i, \mathcal{H}_i, D_i, \gamma_i, J_i)$ of KO -dim 4 and 6, their
cardona-qft-methods_F02495	214	■ 1.000	J, γ	J, γ	the commutation relations of J, γ , and D ;
cardona-qft-methods_F02496	214	■ 1.000	$\zeta_{a,D}(s) =$	$\zeta_{a,D}(s) =$	• The <i>dimension spectrum</i> is the set of poles of the zeta functions $\zeta_{a,D}(s) =$
cardona-qft-methods_F02497	214	■ 1.000	$\operatorname{Tr}(a D ^{-s})$	$\operatorname{Tr}(a D ^{-s})$	$\operatorname{Tr}(a D ^{-s})$ associated to the spectral triple.
cardona-qft-methods_F02498	214	■ 1.000	$X=M \times F$	$X = M \times F$	8, etc. In our setting $X = M \times F$ is metrically four dimensional; its KO -
cardona-qft-methods_F02499	214	■ 0.974	$10=4+6$	$10 = 4 + 6$	dimension is $10 = 4 + 6$ (equal 2 mod 8) and the dimension spectrum is
cardona-qft-methods_F02500	214	■ 0.974	$2 \bmod 8$	$2 \bmod 8$	dimension is $10 = 4 + 6$ (equal 2 mod 8) and the dimension spectrum is
cardona-qft-methods_F02501	214	■ 0.983	$\{k \in \mathbb{Z}_{\geq 0}, k \leq 4\}$	$\{k \in \mathbb{Z}_{\geq 0}, k \leq 4\}$	given by the set $\{k \in \mathbb{Z}_{\geq 0}, k \leq 4\}$.
cardona-qft-methods_F02502	214	■ 1.000	$\mathcal{E}$	$\mathcal{E}$	• M is a smooth manifold, $\mathcal{E}$ is an algebra bundle with fiber $\mathcal{E}_x$ a finite
cardona-qft-methods_F02503	214	■ 1.000	$\mathcal{E}_{\{x\}}$	$\mathcal{E}_x$	• M is a smooth manifold, $\mathcal{E}$ is an algebra bundle with fiber $\mathcal{E}_x$ a finite
cardona-qft-methods_F02504	214	■ 1.000	$C^{\infty}(M, \mathcal{E})$	$C^{\infty}(M, \mathcal{E})$	• $C^{\infty}(M, \mathcal{E})$ are the smooth sections of the algebra bundle $\mathcal{E}$
cardona-qft-methods_F02505	214	■ 1.000	$\mathcal{D}_{\mathcal{E}} = c \circ (\nabla^{\mathcal{E}} \otimes 1 + 1 \otimes \nabla^S)$	$\mathcal{D}_{\mathcal{E}} = c \circ (\nabla^{\mathcal{E}} \otimes 1 + 1 \otimes \nabla^S)$	• The Dirac operator is of the form $\mathcal{D}_{\mathcal{E}} = c \circ (\nabla^{\mathcal{E}} \otimes 1 + 1 \otimes \nabla^S)$ with a
cardona-qft-methods_F02506	215	■ 0.998	$\operatorname{U}(\mathcal{A}) = \{u \in \mathcal{A} \mid uu^* = u^*u = 1\}$	$\operatorname{U}(\mathcal{A}) = \{u \in \mathcal{A} \mid uu^* =$	Given an algebra $\mathcal{A}$, its unitary group is given by $\operatorname{U}(\mathcal{A}) = \{u \in \mathcal{A} \mid uu^* =$
cardona-qft-methods_F02507	215	■ 0.999	$u^*u = 1\}$	$u^*u = 1\}$	$u^*u = 1\}$. In the case of a finite dimensional algebra $\mathcal{A}_F$ represented on a
cardona-qft-methods_F02508	216	■ 1.000	$\operatorname{Ad}(u)$	$\operatorname{Ad}(u)$	the unimodular subgroup of $\operatorname{U}(\mathcal{A})$ in the adjoint representation $\operatorname{Ad}(u)$ on
cardona-qft-methods_F02509	216	■ 0.991	$U(1), SU(2)$	$U(1), SU(2)$	of SM: $U(1), SU(2)$ and $SU(3)$ gauge bosons. On the other hand, the in-

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cardona-qft-methods_F02510	216	■ 1.000	$A^{\{(1,0)\}}=\sum_i a_i \left[\not\partial_M \otimes \right]$	$A^{(1,0)} = \sum_i a_i [\partial_M \otimes$	More precisely, the inner fluctuations of the form $A^{(1,0)} = \sum_i a_i [\partial_M \otimes$
cardona-qft-methods_F02511	216	■ 0.749	$\left.1, a_i^{\prime}\right]$	$1, a_i']$	$1, a_i']$, with $a_i = (\lambda_i, q_i, m_i)$ and $a_i' = (\lambda_i', q_i', m_i')$ in $\mathcal{A} = C^\infty(M, \mathcal{A}_F)$ give
cardona-qft-methods_F02512	216	■ 0.749	$a_i=\left(\lambda_{i}, q_{i}, m_{i}\right)$	$a_i = (\lambda_i, q_i, m_i)$	$1, a_i']$, with $a_i = (\lambda_i, q_i, m_i)$ and $a_i' = (\lambda_i', q_i', m_i')$ in $\mathcal{A} = C^\infty(M, \mathcal{A}_F)$ give
cardona-qft-methods_F02513	216	■ 0.749	$a_i^{\prime}=\left(\lambda_{i}^{\prime}, q_{i}^{\prime}, m_{i}^{\prime}\right)$	$a_i' = (\lambda_i', q_i', m_i')$	$1, a_i']$, with $a_i = (\lambda_i, q_i, m_i)$ and $a_i' = (\lambda_i', q_i', m_i')$ in $\mathcal{A} = C^\infty(M, \mathcal{A}_F)$ give
cardona-qft-methods_F02514	216	■ 0.749	$\mathcal{A}=\mathcal{C}^{\infty}\left(M,\mathcal{A}_F\right)$	$\mathcal{A} = C^\infty(M, \mathcal{A}_F)$	$1, a_i']$, with $a_i = (\lambda_i, q_i, m_i)$ and $a_i' = (\lambda_i', q_i', m_i')$ in $\mathcal{A} = C^\infty(M, \mathcal{A}_F)$ give
cardona-qft-methods_F02515	216	■ 1.000	$\Lambda=\sum_i \lambda_{i} d \lambda_{i}^{\prime}=\sum_i \lambda_{i}\left[\not\partial_M \otimes 1, \lambda_{i}^{\prime}\right]$	$\Lambda = \sum_i \lambda_i d \lambda_i' = \sum_i \lambda_i [\partial_M \otimes 1, \lambda_i']$	• a $U(1)$ gauge field $\Lambda = \sum_i \lambda_i d \lambda_i' = \sum_i \lambda_i [\partial_M \otimes 1, \lambda_i']$;
cardona-qft-methods_F02516	216	■ 0.988	$Q=\sum_i q_i d q_i^{\prime}$	$Q = \sum_i q_i d q_i'$	• an $SU(2)$ gauge field $Q = \sum_i q_i d q_i'$, with $q = f_0 + \sum_\alpha i f_\alpha \sigma^\alpha$ and $Q =$
cardona-qft-methods_F02517	216	■ 0.988	$q=f_0+\sum_{\alpha} i f_{\alpha} \sigma^{\alpha}$	$q = f_0 + \sum_\alpha i f_\alpha \sigma^\alpha$	• an $SU(2)$ gauge field $Q = \sum_i q_i d q_i'$, with $q = f_0 + \sum_\alpha i f_\alpha \sigma^\alpha$ and $Q =$
cardona-qft-methods_F02518	216	■ 0.988	$Q=$	$Q =$	• an $SU(2)$ gauge field $Q = \sum_i q_i d q_i'$, with $q = f_0 + \sum_\alpha i f_\alpha \sigma^\alpha$ and $Q =$
cardona-qft-methods_F02519	216	■ 0.966	$\sum_{\alpha} f_{\alpha}\left[\not\partial_M \otimes 1, i f_{\alpha}^{\prime} \sigma^{\alpha}\right] ;$	$\sum_\alpha f_\alpha [\partial_M \otimes 1, i f_\alpha' \sigma^\alpha];$	$\sum_\alpha f_\alpha [\partial_M \otimes 1, i f_\alpha' \sigma^\alpha];$
cardona-qft-methods_F02520	216	■ 1.000	$U(3)$	$U(3)$	• a $U(3)$ gauge field $V' = \sum_i m_i d m_i' = \sum_i m_i [\partial_M \otimes 1, m_i']$;
cardona-qft-methods_F02521	216	■ 1.000	$V^{\prime}=\sum_i m_i d m_i^{\prime}=\sum_i m_i\left[\not\partial_M \otimes 1, m_i^{\prime}\right]$	$V' = \sum_i m_i d m_i' = \sum_i m_i [\partial_M \otimes 1, m_i']$	• a $U(3)$ gauge field $V' = \sum_i m_i d m_i' = \sum_i m_i [\partial_M \otimes 1, m_i']$;
cardona-qft-methods_F02522	216	■ 0.806	$V^{\prime}$	V'	• after reducing the gauge field V' to $SU(3)$, passing to the unimodular
cardona-qft-methods_F02523	216	■ 1.000	$\operatorname{SU}\left(\mathcal{A}_F\right)$	$SU(\mathcal{A}_F)$	subgroup $SU(\mathcal{A}_F)$ and unimodular gauge potential with $\text{Tr}(A) = 0$, one
cardona-qft-methods_F02524	216	■ 1.000	$\operatorname{Tr}(A)=0$	$\text{Tr}(A) = 0$	subgroup $SU(\mathcal{A}_F)$ and unimodular gauge potential with $\text{Tr}(A) = 0$, one
cardona-qft-methods_F02525	216	■ 1.000	$(1,0)$	$(1,0)$	bosons. This can be seen from the fact that the $(1,0)$ part of $A + JAJ^{-1}$
cardona-qft-methods_F02526	216	■ 1.000	$A+J A J^{-1}$	$A + JAJ^{-1}$	bosons. This can be seen from the fact that the $(1,0)$ part of $A + JAJ^{-1}$
cardona-qft-methods_F02527	217	■ 1.000	$A^{\{(0,1)\}}$	$A^{(0,1)}$	form $A^{(0,1)}$, in the F -space direction. These are given by
cardona-qft-methods_F02528	217	■ 1.000	$\varphi_1=\sum \lambda_{i}\left(\alpha_{i}^{\prime}-\lambda_{i}^{\prime}\right), \varphi_2=\sum \lambda_i \beta_i^{\prime} \varphi_1^{\prime}=\sum \alpha_i\left(\lambda_{i}^{\prime}-\alpha_{i}^{\prime}\right)+\beta_i \bar{\beta}_i^{\prime}$	$\varphi_1 = \sum \lambda_i (\alpha_i' - \lambda_i'), \varphi_2 = \sum \lambda_i \beta_i' \varphi_1' = \sum \alpha_i (\lambda_i' - \alpha_i') + \beta_i \bar{\beta}_i'$	with $\varphi_1 = \sum \lambda_i (\alpha_i' - \lambda_i'), \varphi_2 = \sum \lambda_i \beta_i' \varphi_1' = \sum \alpha_i (\lambda_i' - \alpha_i') + \beta_i \bar{\beta}_i'$ and
cardona-qft-methods_F02529	217	■ 1.000	$\varphi_2^{\prime}=\sum\left(-\alpha_i \beta_i^{\prime}+\beta_i\left(\bar{\lambda}_i^{\prime}-\bar{\alpha}_i^{\prime}\right)\right)$	$\varphi_2' = \sum (-\alpha_i \beta_i' + \beta_i (\bar{\lambda}_i' - \bar{\alpha}_i'))$	$\varphi_2' = \sum (-\alpha_i \beta_i' + \beta_i (\bar{\lambda}_i' - \bar{\alpha}_i'))$, for $a_i(x) = (\lambda_i, q_i, m_i)$ and $a_i'(x) = (\lambda_i', q_i', m_i')$
cardona-qft-methods_F02530	217	■ 1.000	$a_i(x)=\left(\lambda_{i}, q_{i}, m_{i}\right)$	$a_i(x) = (\lambda_i, q_i, m_i)$	$\varphi_2' = \sum (-\alpha_i \beta_i' + \beta_i (\bar{\lambda}_i' - \bar{\alpha}_i'))$, for $a_i(x) = (\lambda_i, q_i, m_i)$ and $a_i'(x) = (\lambda_i', q_i', m_i')$
cardona-qft-methods_F02531	217	■ 1.000	$a_i^{\prime}(x)=\left(\lambda_{i}^{\prime}, q_{i}^{\prime}, m_{i}^{\prime}\right)$	$a_i'(x) = (\lambda_i', q_i', m_i')$	$\varphi_2' = \sum (-\alpha_i \beta_i' + \beta_i (\bar{\lambda}_i' - \bar{\alpha}_i'))$, for $a_i(x) = (\lambda_i, q_i, m_i)$ and $a_i'(x) = (\lambda_i', q_i', m_i')$
cardona-qft-methods_F02532	217	■ 1.000	$H=\varphi_1+\varphi_2 j$	$H = \varphi_1 + \varphi_2 j$	valued function $H = \varphi_1 + \varphi_2 j$ or $\varphi = (\varphi_1, \varphi_2)$, which is the Higgs field of
cardona-qft-methods_F02533	217	■ 1.000	$\varphi=\left(\varphi_1, \varphi_2\right)$	$\varphi = (\varphi_1, \varphi_2)$	valued function $H = \varphi_1 + \varphi_2 j$ or $\varphi = (\varphi_1, \varphi_2)$, which is the Higgs field of

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cardona-qft-methods_F02534	217	■ 0.998	$D_{\{A\}^2}=\nabla^{\{*\}}\ \nabla\!-\!E$	$D_A^2=\nabla^*\nabla-E$	and this gives $D_A^2=\nabla^*\nabla-E$ where $\nabla^*\nabla$ is the Laplacian of $\nabla=\nabla^s+\mathbb{A}$
cardona-qft-methods_F02535	217	■ 0.998	$\nabla^{\{*\}}\ \nabla\!-\!$	$\nabla^*\nabla$	and this gives $D_A^2=\nabla^*\nabla-E$ where $\nabla^*\nabla$ is the Laplacian of $\nabla=\nabla^s+\mathbb{A}$
cardona-qft-methods_F02536	217	■ 0.998	$\nabla\!=\!\nabla^{\{s\}}+\!\mathbb{A}$	$\nabla=\nabla^s+\mathbb{A}$	and this gives $D_A^2=\nabla^*\nabla-E$ where $\nabla^*\nabla$ is the Laplacian of $\nabla=\nabla^s+\mathbb{A}$
cardona-qft-methods_F02537	218	■ 0.996	$s\!=\!-\!R$	$s=-R$	with $s=-R$ the scalar curvature and $\mathbb{F}_{\mu\nu}$ the curvature of $\mathbb{A}$, with
cardona-qft-methods_F02538	218	■ 0.996	$\mathbb{F}_{\mu\nu}$	$\mathbb{F}_{\mu\nu}$	with $s=-R$ the scalar curvature and $\mathbb{F}_{\mu\nu}$ the curvature of $\mathbb{A}$, with
cardona-qft-methods_F02539	218	■ 0.996	$\mathbb{A}$	$\mathbb{A}$	with $s=-R$ the scalar curvature and $\mathbb{F}_{\mu\nu}$ the curvature of $\mathbb{A}$, with
cardona-qft-methods_F02540	218	■ 1.000	$D\ /\ \Lambda$	D/Λ	makes D/Λ dimensionless, and $f>0$ is an even smooth function, which is
cardona-qft-methods_F02541	218	■ 1.000	$f\!>\!0$	$f>0$	makes D/Λ dimensionless, and $f>0$ is an even smooth function, which is
cardona-qft-methods_F02542	218	■ 1.000	$f_{\{k\}}\!=\!\int_0^{\infty}\!f(v)\ v^{\{k-1\}}\ d\ v$	$f_k=\int_0^{\infty}f(v)v^{k-1}dv$	where $f_k=\int_0^{\infty}f(v)v^{k-1}dv$ are the momenta of f and where the summa-
cardona-qft-methods_F02543	218	■ 0.999	$\operatorname{DimSp}(\mathcal{A},\ \mathcal{H},\ D)$	$\operatorname{DimSp}(\mathcal{A},\mathcal{H},D)$	tion runs over the part of the dimension spectrum $\operatorname{DimSp}(\mathcal{A},\mathcal{H},D)$, namely
cardona-qft-methods_F02544	218	■ 1.000	$\zeta_{\{b,\ D\}}(s)\!=\!\operatorname{Tr}\!\left(b D ^{\{-s\}}\right)$	$\zeta_{b,D}(s)=\operatorname{Tr}\left(b D ^{-s}\right)$	the poles of $\zeta_{b,D}(s)=\operatorname{Tr}(b D ^{-s})$, contained in the positive axis.
cardona-qft-methods_F02545	218	■ 1.000	$a_{\{\alpha\}}$	a_α	More precisely, a non-zero terms a_α with $\alpha<0$ corresponds to a <i>pole</i> of ζ_D
cardona-qft-methods_F02546	218	■ 1.000	$\alpha\!<\!0$	$\alpha<0$	More precisely, a non-zero terms a_α with $\alpha<0$ corresponds to a <i>pole</i> of ζ_D
cardona-qft-methods_F02547	218	■ 1.000	$\zeta_{\{D\}}$	ζ_D	More precisely, a non-zero terms a_α with $\alpha<0$ corresponds to a <i>pole</i> of ζ_D
cardona-qft-methods_F02548	219	■ 1.000	$\int_0^1\!t^{\{\alpha+s\ / \ 2-1\}}\ d\ t\!=\!(\alpha+s\ / \ 2)^{\{-1\}}$	$\int_0^1t^{\alpha+s/2-1}dt=(\alpha+s/2)^{-1}$	with $\int_0^1t^{\alpha+s/2-1}dt=(\alpha+s/2)^{-1}$. Moreover, the fact that no $\log t$ terms
cardona-qft-methods_F02549	219	■ 1.000	$\zeta_{\{D\}}(0)\!=\!a_{\{0\}}$	$\zeta_D(0)=a_0$	are present in the expansion implies regularity at 0 for ζ_D with $\zeta_D(0)=a_0$.
cardona-qft-methods_F02550	219	■ 1.000	$a_{\{0\}}\ \int_0^1\!t^{\{s\ / \ 2-1\}}\ d\ t\!=\!a_{\{0\}}\ \frac{2}{s}$	$a_0\int_0^1t^{s/2-1}dt=a_0\frac{2}{s}$	is given by $a_0\int_0^1t^{s/2-1}dt=a_0\frac{2}{s}$.
cardona-qft-methods_F02551	219	■ 1.000	$\mathcal{H}_{cl}^{\{+\}}$	$\mathcal{H}_{cl}^+$	real structure, and $\mathcal{H}_{cl}^+$ the space of classical spinors, seen as Grassmann
cardona-qft-methods_F02552	219	■ 0.999	$\mathfrak{A}(\tilde{\xi})$	$\mathfrak{A}(\tilde{\xi})$	is an antisymmetric form $\mathfrak{A}(\tilde{\xi})$ on
cardona-qft-methods_F02553	219	■ 1.000	$\left\langle\!\xi,\ D_{\{A\}}\ \xi\right\rangle$	$\langle\xi,D_A\xi\rangle$	symmetric $\langle\xi,D_A\xi\rangle$ for a finite noncommutative space with <i>KO</i> -dimension
cardona-qft-methods_F02554	220	■ 1.000	$f_{\{0\}},\ f_{\{2\}},\ f_{\{4\}}$	f_0,f_2,f_4	• f_0,f_2,f_4 are three free parameters, which depend only on the test func-
cardona-qft-methods_F02555	220	■ 1.000	$f_{\{0\}}\!=\!f(0)$	$f_0=f(0)$	tion f , through $f_0=f(0)$ and, for $k>0$,
cardona-qft-methods_F02556	220	■ 1.000	$k\!>\!0$	$k>0$	tion f , through $f_0=f(0)$ and, for $k>0$,
cardona-qft-methods_F02557	220	■ 0.997	$\mathfrak{a},\ \mathfrak{b},\ \mathfrak{c},\ \mathfrak{d},\ \mathfrak{e}$	$\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{d},\mathfrak{e}$	• the parameters $\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{d},\mathfrak{e}$ are functions of Yukawa parameters of νMSM
cardona-qft-methods_F02558	221	■ 1.000	$P\!=\!\left(g^{\{\mu\}\ \{\nu\}}\ I\ \partial_{\{\mu\}}\ \partial_{\{\nu\}}\!+\!A^{\{\mu\}\ \{\nu\}}\ \partial_{\{\mu\}}\!+\!B\right)$	$P=-(g^{\mu\nu}I\partial_\mu\partial_\nu+A^\mu\partial_\mu+B)$	<i>theorem</i> for a differential operator $P=-(g^{\mu\nu}I\partial_\mu\partial_\nu+A^\mu\partial_\mu+B)$ with A ,
cardona-qft-methods_F02559	221	■ 1.000	$m\!=\!\operatorname{dim}\ M$	$m=\operatorname{dim} M$	B bundle endomorphisms and $m=\operatorname{dim} M$ (see a detailed account in §1

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cardona-qft-methods_ F02560	221	■ 1.000	$P=\nabla^*\nabla-E$	$P = \nabla^* \nabla - E$	where $P = \nabla^* \nabla - E$ and $E_{;\mu}{}^\mu := \nabla_\mu \nabla^\mu E$, with
cardona-qft-methods_ F02561	221	■ 1.000	$E_{;\mu}{}^\mu := \nabla_\mu \nabla^\mu E$	$E_{;\mu}{}^\mu := \nabla_\mu \nabla^\mu E$	where $P = \nabla^* \nabla - E$ and $E_{;\mu}{}^\mu := \nabla_\mu \nabla^\mu E$, with
cardona-qft-methods_ F02562	221	■ 0.712	$H=\frac{\sqrt{a}f_0}{\pi}\varphi$	$H = \frac{\sqrt{a}f_0}{\pi}\varphi$	• Rescale Higgs field by $H = \frac{\sqrt{a}f_0}{\pi}\varphi$ to normalize the kinetic term to
cardona-qft-methods_ F02563	222	■ 1.000	$R^*R^*=\frac{1}{4}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}R_{\mu\nu}^{\alpha\beta}R_{\rho\sigma}^{\gamma\delta}$	$R^*R^* = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}R_{\mu\nu}^{\alpha\beta}R_{\rho\sigma}^{\gamma\delta}$	where $R^*R^* = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}R_{\mu\nu}^{\alpha\beta}R_{\rho\sigma}^{\gamma\delta}$ integrates to the Euler characteristic
cardona-qft-methods_ F02564	222	■ 1.000	$\chi(M)$	$\chi(M)$	$\chi(M)$ and $C^{\mu\nu\rho\sigma}$ is the Weyl curvature tensor. The coefficients are given
cardona-qft-methods_ F02565	222	■ 1.000	$C^{\mu\nu\rho\sigma}$	$C^{\mu\nu\rho\sigma}$	$\chi(M)$ and $C^{\mu\nu\rho\sigma}$ is the Weyl curvature tensor. The coefficients are given
cardona-qft-methods_ F02566	222	■ 1.000	10^{15}	10^{15}	10^{15} and 10^{17} GeV. The common value of the coupling constants at unifi-
cardona-qft-methods_ F02567	222	■ 1.000	$10^{17}\,\mathrm{GeV}$	$10^{17}\mathrm{GeV}$	10^{15} and 10^{17} GeV. The common value of the coupling constants at unifi-
cardona-qft-methods_ F02568	222	■ 1.000	f_0	f_0	cation is related to the first parameter f_0 of the model by the relation
cardona-qft-methods_ F02569	223	■ 1.000	$\{ \}^{17,29,30,33}$	17,29,30,33	models following. ^{17,29,30,33} A renormalization group analysis of NCG mod-
cardona-qft-methods_ F02570	223	■ 0.983	$\nu\,\mathrm{M\,S\,M}$	νMSM	4.1. Renormalization group equations for the νMSM
cardona-qft-methods_ F02571	223	■ 0.999	$\Lambda\frac{df}{d\Lambda}=\beta_f(\Lambda)$	$\Lambda\frac{df}{d\Lambda} = \beta_f(\Lambda)$	At one 1-loop, the RGE equations take the form $\Lambda\frac{df}{d\Lambda} = \beta_f(\Lambda)$ with
cardona-qft-methods_ F02572	224	■ 0.978	Λ_{unif}	Λ_{unif}	• Run the RGE flow from unification energy Λ_{unif} down to first see-saw
cardona-qft-methods_ F02573	224	■ 0.822	$Y_\nu^{(3)}$	$Y_\nu^{(3)}$	• Pass to an effective field theory with $Y_\nu^{(3)}$ obtained by removing the last
cardona-qft-methods_ F02574	224	■ 1.000	Y_ν	Y_ν	row of Y_ν in a basis in which M is diagonal and with $M^{(3)}$ obtained by
cardona-qft-methods_ F02575	224	■ 1.000	$M^{(3)}$	$M^{(3)}$	row of Y_ν in a basis in which M is diagonal and with $M^{(3)}$ obtained by
cardona-qft-methods_ F02576	224	■ 0.997	$Y_\nu^{(2)}$	$Y_\nu^{(2)}$	• Introduce a second effective field theory with $Y_\nu^{(2)}$ and $M^{(2)}$ again ob-
cardona-qft-methods_ F02577	224	■ 0.997	$M^{(2)}$	$M^{(2)}$	• Introduce a second effective field theory with $Y_\nu^{(2)}$ and $M^{(2)}$ again ob-
cardona-qft-methods_ F02578	224	■ 1.000	$Y_\nu^{(1)}$	$Y_\nu^{(1)}$	• Introduce a new effective field theory with $Y_\nu^{(1)}$ and $M^{(1)}$ again obtained
cardona-qft-methods_ F02579	224	■ 1.000	$M^{(1)}$	$M^{(1)}$	• Introduce a new effective field theory with $Y_\nu^{(1)}$ and $M^{(1)}$ again obtained
cardona-qft-methods_ F02580	224	■ 1.000	Λ_{ew}	Λ_{ew}	• Run the induced RGE down to the electoweak energy scale Λ_{ew} .
cardona-qft-methods_ F02581	224	■ 1.000	$Y_\nu^{(N)}$	$Y_\nu^{(N)}$	Using the effective field theories $Y_\nu^{(N)}$ and $M^{(N)}$ between see-saw scales
cardona-qft-methods_ F02582	224	■ 1.000	$M^{(N)}$	$M^{(N)}$	Using the effective field theories $Y_\nu^{(N)}$ and $M^{(N)}$ between see-saw scales
cardona-qft-methods_ F02583	225	■ 1.000	$t=\log\left(\Lambda/M_Z\right)$	$t = \log(\Lambda/M_Z)$	in the variable $t = \log(\Lambda/M_Z)$, where the beta functions for the three
cardona-qft-methods_ F02584	225	■ 1.000	$\Lambda=M_Z$	$\Lambda = M_Z$	at $\Lambda = M_Z$, determined in agreement with experiments, are

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cardona-qft-methods_F02585	225	■ 1.000	$g_{\{3\}^2}=g_{\{2\}^2}=5\ g_{\{1\}^2}\ / \ 3$	$g_3^2 = g_2^2 = 5g_1^2/3$	coupling constants do <i>not</i> meet in the expected relation $g_3^2 = g_2^2 = 5g_1^2/3$,
cardona-qft-methods_F02586	226	■ 0.994	$\mathfrak{c}$	$\mathfrak{c}$	• See-saw mechanism and $\mathfrak{c}$ constraint
cardona-qft-methods_F02587	227	■ 1.000	$g_{\{1\}},\ g_{\{2\}},\ g_{\{3\}}$	g_1, g_2, g_3	where g_1, g_2, g_3 are the solutions for the equations for the coupling constants
cardona-qft-methods_F02588	228	■ 1.000	$\tilde{\lambda}$	$\tilde{\lambda}$	and uses the relation at unification, where $\tilde{\lambda}$ is the $ \mathbf{H} ^4$ coupling,
cardona-qft-methods_F02589	228	■ 1.000	$ \mathbf{H} ^4$	$ \mathbf{H} ^4$	and uses the relation at unification, where $\tilde{\lambda}$ is the $ \mathbf{H} ^4$ coupling,
cardona-qft-methods_F02590	228	■ 0.585	$\lambda\left(M_Z\right) \sim 0.241$	$\lambda\left(M_Z\right) \sim 0.241$	With $\lambda(M_Z) \sim 0.241$ one obtains a heavy Higgs mass ~ 170 GeV, which
cardona-qft-methods_F02591	228	■ 0.585	$\sim 170\ \mathrm{GeV}$	$\sim 170\mathrm{GeV}$	With $\lambda(M_Z) \sim 0.241$ one obtains a heavy Higgs mass ~ 170 GeV, which
cardona-qft-methods_F02592	229	■ 1.000	$\kappa_{\{0\}},\ \gamma_{\{0\}},\ \alpha_{\{0\}},\ \tau_{\{0\}},\ \mu_{\{0\}},\ \xi_{\{0\}},\ \lambda_{\{0\}}$	$\kappa_0, \gamma_0, \alpha_0, \tau_0, \mu_0, \xi_0, \lambda_0$	cients $\kappa_0, \gamma_0, \alpha_0, \tau_0, \mu_0, \xi_0, \lambda_0$ in the asymptotic expansion of the spectral
cardona-qft-methods_F02593	229	■ 0.609	$\{ \ }^{\{3,18,27\}}$	3,18,27	tion group analysis of modified gravity models of, ^{3,18,27} for a gravity action
cardona-qft-methods_F02594	230	■ 1.000	10^{-12}	10^{-12}	in the cosmological timeline to only at most 10^{-12} seconds after the Big
cardona-qft-methods_F02595	230	■ 1.000	$\lambda_{\{0\}}=\lambda\ \dot{\left(\pi^2\ \mathfrak{b}\right)}\ /\left(f_0\mathfrak{a}^2\right)$	$\lambda_0 = \lambda \cdot (\pi^2 \mathfrak{b}) / (f_0 \mathfrak{a}^2)$	instead of applying the relation $\lambda_0 = \lambda \cdot (\pi^2 \mathfrak{b}) / (f_0 \mathfrak{a}^2)$ only at the initial
cardona-qft-methods_F02596	230	■ 0.992	$x_{\{j\}}$	x_j	where x_j are, as before, the running parameters with β_j^{SM} is the Standard
cardona-qft-methods_F02597	230	■ 0.992	$\beta_{\{j\}}^{\mathrm{SM}}$	β_j^{SM}	where x_j are, as before, the running parameters with β_j^{SM} is the Standard
cardona-qft-methods_F02598	230	■ 1.000	$\beta_{\{j\}}^{\mathrm{grav}}$	β_j^{grav}	Model beta function for x_j described previously, and where β_j^{grav} are the
cardona-qft-methods_F02599	230	■ 1.000	$\rho_{\{0\}} \sim 0.024$	$\rho_0 \sim 0.024$	with $\rho_0 \sim 0.024$. The anomalous dimensions are usually taken to be $a_1 =$
cardona-qft-methods_F02600	230	■ 1.000	$\mathfrak{a}_{\{1\}}=$	$a_1 =$	with $\rho_0 \sim 0.024$. The anomalous dimensions are usually taken to be $a_1 =$
cardona-qft-methods_F02601	230	■ 0.993	$\mathfrak{a}_{\{2\}}=\mathfrak{a}_{\{3\}}=\mathfrak{a}_{\{g\}}$	$a_2 = a_3 = a_g$	$a_2 = a_3 = a_g$, with $ a_g \sim 1$ and a negative sign for the coupling constants
cardona-qft-methods_F02602	230	■ 0.993	$ \mathfrak{a}_{\{g\}} \sim 1$	$ a_g \sim 1$	$a_2 = a_3 = a_g$, with $ a_g \sim 1$ and a negative sign for the coupling constants
cardona-qft-methods_F02603	230	■ 0.473	$\mathfrak{a}_{\{g\}} \leq \mathfrak{a}_{\{y\}} < 0$	$a_g \leq a_y < 0$	and $a_g \leq a_y < 0$, while a_λ is expected to be positive (see ⁴⁵).
cardona-qft-methods_F02604	230	■ 0.473	$\mathfrak{a}_{\{\lambda\}}$	a_λ	and $a_g \leq a_y < 0$, while a_λ is expected to be positive (see ⁴⁵).
cardona-qft-methods_F02605	230	■ 0.473	$\left(\right.$	$($	and $a_g \leq a_y < 0$, while a_λ is expected to be positive (see ⁴⁵).
cardona-qft-methods_F02606	230	■ 0.473	$\left.^{\{45\}}\right)$	$\left.^{\{45\}}\right)$	and $a_g \leq a_y < 0$, while a_λ is expected to be positive (see ⁴⁵).
cardona-qft-methods_F02607	231	■ 1.000	$m_{\{H\}} \sim 125.4$	$m_H \sim 125.4$	that it admits solutions that give a realistic Higgs mass $m_H \sim 125.4$, com-
cardona-qft-methods_F02608	231	■ 0.999	$m_{\{t\}} \sim 171.3\ \mathrm{GeV}$	$m_t \sim 171.3\mathrm{GeV}$	patible with a realistic top quark mass of $m_t \sim 171.3$ GeV.
cardona-qft-methods_F02609	231	■ 1.000	$\{ \ }^{\{42-44\}}$	$^{42-44}$	of Shaposhnikov and Tkachev, ⁴²⁻⁴⁴ For a recent and extensive reference
cardona-qft-methods_F02610	236	■ 1.000	10^{-32}	10^{-32}	or atomic scales, being 10^{-32} centimetres. This is 10^9 times smaller than the
cardona-qft-methods_F02611	236	■ 1.000	$10^{\{9\}}$	10^9	or atomic scales, being 10^{-32} centimetres. This is 10^9 times smaller than the

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cardona-qft-methods_ F02612	236	■ 1.000	$L \simeq 10^{-32} \mathrm{~cm}$	$L \simeq 10^{-32} \mathrm{~cm}$	scale $L \simeq 10^{-32} \mathrm{cm}$, the Compton wavelength λ_C of a probe of mass M
cardona-qft-methods_ F02613	236	■ 1.000	λ_C	λ_C	scale $L \simeq 10^{-32} \mathrm{cm}$, the Compton wavelength λ_C of a probe of mass M
cardona-qft-methods_ F02614	236	■ 1.000	$L^3 \simeq \left(10^{-32} \mathrm{~cm}\right)^3$	$L^3 \simeq (10^{-32} \mathrm{~cm})^3$	Such a huge mass in the small volume $L^3 \simeq (10^{-32} \mathrm{cm})^3$ will strongly affect
cardona-qft-methods_ F02615	236	■ 1.000	Δx	Δx	where Δx and Δp represent the uncertainties in measurements of position
cardona-qft-methods_ F02616	236	■ 1.000	Δp	Δp	where Δx and Δp represent the uncertainties in measurements of position
cardona-qft-methods_ F02617	236	■ 0.994	$x p - p x = i \hbar$	$x p - p x = i \hbar$	achieve this, namely, we can “make x and p noncommuting”: $x p - p x = i \hbar$.
cardona-qft-methods_ F02618	236	■ 1.000	Δy	Δy	where Δx and Δy stand for the intrinsic uncertainties in measurements
cardona-qft-methods_ F02619	237	■ 0.640	$G - M$	$G - M$	$(G-M)$ plane. It is defined as the algebra $\mathcal{A}_\theta$ of functions on $\mathbb{R}^{d+1}$ with a
cardona-qft-methods_ F02620	237	■ 0.640	$\mathcal{A}_\theta$	$\mathcal{A}_\theta$	$(G-M)$ plane. It is defined as the algebra $\mathcal{A}_\theta$ of functions on $\mathbb{R}^{d+1}$ with a
cardona-qft-methods_ F02621	237	■ 0.640	$\mathbb{R}^{d+1}$	$\mathbb{R}^{d+1}$	$(G-M)$ plane. It is defined as the algebra $\mathcal{A}_\theta$ of functions on $\mathbb{R}^{d+1}$ with a
cardona-qft-methods_ F02622	238	■ 1.000	$\mathbb{C}G$	$\mathbb{C}G$	principle. We then formulate them in terms of $\mathbb{C}G$, the group algebra of G .
cardona-qft-methods_ F02623	238	■ 1.000	$U(g) \Psi$	$U(g) \Psi$	$U(g) \Psi$ under the action of $g \in G$, then we can superpose Ψ and $U(g) \Psi$
cardona-qft-methods_ F02624	239	■ 1.000	$\mathcal{H}^{(1)}$	$\mathcal{H}^{(1)}$	Let us consider a general many-particle quantum system, with $\mathcal{H}^{(1)}$ denot-
cardona-qft-methods_ F02625	239	■ 1.000	$ 0\rangle$	$ 0\rangle$	of the theory. N -particle states are created from the vacuum $ 0\rangle$ and corre-
cardona-qft-methods_ F02626	239	■ 1.000	$\mathcal{H}^{(N)} = \mathcal{H}^{(1)} \otimes \dots \otimes \mathcal{H}^{(1)}$	$\mathcal{H}^{(N)} = \mathcal{H}^{(1)} \otimes \dots \otimes \mathcal{H}^{(1)}$	spond to state vectors in the N -fold tensor product $\mathcal{H}^{(N)} = \mathcal{H}^{(1)} \otimes \dots \otimes \mathcal{H}^{(1)}$.
cardona-qft-methods_ F02627	239	■ 1.000	$\mathcal{F} = \bigoplus_N \mathcal{H}^{(N)}$	$\mathcal{F} = \bigoplus_N \mathcal{H}^{(N)}$	acts on the Fock space $\mathcal{F} = \bigoplus_N \mathcal{H}^{(N)}$ through a unitary representation of
cardona-qft-methods_ F02628	239	■ 1.000	$U^{(N)}$	$U^{(N)}$	where each $U^{(N)}$ acts on the corresponding $\mathcal{H}^{(N)}$ and, therefore, does not
cardona-qft-methods_ F02629	239	■ 1.000	$\mathcal{H}^{(N)}$	$\mathcal{H}^{(N)}$	where each $U^{(N)}$ acts on the corresponding $\mathcal{H}^{(N)}$ and, therefore, does not
cardona-qft-methods_ F02630	239	■ 1.000	$U^{(1)}$	$U^{(1)}$	Usually, the representation $U^{(N)}$ is given as the N -fold product of $U^{(1)}$,
cardona-qft-methods_ F02631	240	■ 1.000	α^*	α^*	with α a function on G . The product of two elements α^* and β^* is defined
cardona-qft-methods_ F02632	240	■ 1.000	β^*	β^*	with α a function on G . The product of two elements α^* and β^* is defined
cardona-qft-methods_ F02633	240	■ 1.000	$\alpha * \beta$	$\alpha * \beta$	where $\alpha * \beta$ stands for the <i>convolution product</i>
cardona-qft-methods_ F02634	240	■ 1.000	$\{ \}^{7,8}$	7,8	Definitions (4) and (5) for the product and convolution carry over as well. ^{7,8}
cardona-qft-methods_ F02635	240	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	$\mathcal{H}^{(N)}$, as well as on $\mathcal{F}$ by means of
cardona-qft-methods_ F02636	240	■ 1.000	$U^{(N)} (N \geq 2)$	$U^{(N)} (N \geq 2)$	We now examine $U^{(N)}$. We will see that $U^{(1)}$ determines $U^{(N)}$ ($N \geq 2$)
cardona-qft-methods_ F02637	240	■ 1.000	$\mathbb{C}G \otimes \mathbb{C}G$	$\mathbb{C}G \otimes \mathbb{C}G$	only if we have a homomorphism Δ from $\mathbb{C}G$ to $\mathbb{C}G \otimes \mathbb{C}G$ (and hence from
cardona-qft-methods_ F02638	240	■ 1.000	$U^{(0)}$	$U^{(0)}$	(i) <i>The trivial representation $U^{(0)}$:</i>

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cardona-qft-methods_ F02639	240	■ 1.000	$U^{\{0\}}(g) 0\rangle= 0\rangle$	$U^{(0)}(g) 0\rangle= 0\rangle$	$U^{(0)}(g) 0\rangle= 0\rangle$.
cardona-qft-methods_ F02640	241	■ 1.000	$D(g)$	$D(g)$	where $D(g)$ is the matrix for $U^{(1)}(g)$.
cardona-qft-methods_ F02641	241	■ 1.000	$U^{\{1\}}(g)$	$U^{(1)}(g)$	where $D(g)$ is the matrix for $U^{(1)}(g)$.
cardona-qft-methods_ F02642	241	■ 1.000	$U^{\{2\}}$	$U^{(2)}$	(iii) The two-particle representation $U^{(2)}$:
cardona-qft-methods_ F02643	241	■ 1.000	$\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$	$\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$	need a homomorphism $\Delta: G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$ to fix $U^{(2)}$. Given Δ , we can
cardona-qft-methods_ F02644	241	■ 1.000	$U^{\{2\}}(g)=U^{\{1\}} \otimes U^{\{1\}} \Delta(g)$	$U^{(2)}(g)=U^{(1)} \otimes U^{(1)}\Delta(g)$	write $U^{(2)}(g)=U^{(1)} \otimes U^{(1)}\Delta(g)$. A standard choice for Δ is
cardona-qft-methods_ F02645	241	■ 1.000	$\Delta: \mathbb{C}G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$	$\Delta: \mathbb{C}G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$	$\Delta: \mathbb{C}G \rightarrow \mathbb{C}G \otimes \mathbb{C}G$ (with further properties discussed below).
cardona-qft-methods_ F02646	241	■ 1.000	$\Delta(g)=g \otimes g$	$\Delta(g)=g \otimes g$	For the choice $\Delta(g)=g \otimes g$, we get the familiar answer
cardona-qft-methods_ F02647	242	■ 1.000	$\mathcal{E}_3$	$\mathcal{E}_3$	$\mathcal{E}_3$. Here the natural choices for state vector labels are momenta and com-
cardona-qft-methods_ F02648	242	■ 1.000	$g \in \mathcal{E}_3$	$g \in \mathcal{E}_3$	ponents of spin. Define now, for $g \in \mathcal{E}_3$,
cardona-qft-methods_ F02649	242	■ 1.000	$\theta_{\mu\nu}$	$\theta_{\mu\nu}$	$\theta_{\mu\nu}$ being a real antisymmetric matrix and the P_ν being the translation
cardona-qft-methods_ F02650	242	■ 1.000	P_ν	P_ν	$\theta_{\mu\nu}$ being a real antisymmetric matrix and the P_ν being the translation
cardona-qft-methods_ F02651	242	■ 1.000	$G \otimes G$	$G \otimes G$	of $G \otimes G$. Rather it is an element of $\mathbb{C}G \otimes \mathbb{C}G$. This example can be gen-
cardona-qft-methods_ F02652	242	■ 1.000	$\mathcal{P}_+^\uparrow$	$\mathcal{P}_+^\uparrow$	eralized if we replace the Euclidean group by the Poincaré group $\mathcal{P}_+^\uparrow$, thus
cardona-qft-methods_ F02653	242	■ 1.000	Δ_θ	Δ_θ	Problem 2.1. Check coassociativity of the coproduct Δ_θ from the previous
cardona-qft-methods_ F02654	243	■ 1.000	g^{-1}	g^{-1}	2) Every element $g \in G$ has an inverse g^{-1} . We define an antipode S on
cardona-qft-methods_ F02655	243	■ 1.000	$g \rightarrow g^{-1}$	$g \rightarrow g^{-1}$	G as the map $g \rightarrow g^{-1}$. It is an antihomomorphism ($S(gh)=S(h)S(g)$)
cardona-qft-methods_ F02656	243	■ 1.000	$(S(gh)=S(h)S(g))$	$(S(gh)=S(h)S(g))$	G as the map $g \rightarrow g^{-1}$. It is an antihomomorphism ($S(gh)=S(h)S(g)$)
cardona-qft-methods_ F02657	243	■ 0.954	$U(g^{-1})=U(g)^*$	$U(g^{-1})=U(g)^*$	U of G , which means that $U(g^{-1})=U(g)^*$, $*$ denoting the adjoint
cardona-qft-methods_ F02658	243	■ 0.970	$U(\hat{\alpha}^*)=U(\hat{\alpha})^* \cdot U$	$U(\hat{\alpha}^*)=U(\hat{\alpha})^* \cdot U$	condition $U(\hat{\alpha}^*)=U(\hat{\alpha})^*$. U is thus a “*-representation” of $\mathbb{C}G$.
cardona-qft-methods_ F02659	243	■ 1.000	$(\alpha\beta)\gamma=\alpha(\beta\gamma)$	$(\alpha\beta)\gamma=\alpha(\beta\gamma)$	(i) $(\alpha\beta)\gamma=\alpha(\beta\gamma)$ for $\alpha,\beta,\gamma \in H$,
cardona-qft-methods_ F02660	243	■ 1.000	$\alpha,\beta,\gamma \in H$	$\alpha,\beta,\gamma \in H$	(i) $(\alpha\beta)\gamma=\alpha(\beta\gamma)$ for $\alpha,\beta,\gamma \in H$,
cardona-qft-methods_ F02661	244	■ 1.000	$*: \alpha \in H \rightarrow \alpha^* \in H$	$*: \alpha \in H \rightarrow \alpha^* \in H$	(ii) $*: \alpha \in H \rightarrow \alpha^* \in H$,
cardona-qft-methods_ F02662	244	■ 0.953	$(\alpha\beta)^*=\beta^*\alpha^*, \alpha,\beta \in H$	$(\alpha\beta)^*=\beta^*\alpha^*, \alpha,\beta \in H$	$(\alpha\beta)^*=\beta^*\alpha^*, \alpha,\beta \in H$,
cardona-qft-methods_ F02663	244	■ 0.992	$(\lambda\alpha)^*=\bar{\lambda}\alpha^* \quad \lambda \in \mathbb{C}, \alpha \in H$	$(\lambda\alpha)^*=\bar{\lambda}\alpha^* \quad \lambda \in \mathbb{C}, \alpha \in H$	$(\lambda\alpha)^*=\bar{\lambda}\alpha^* \quad \lambda \in \mathbb{C}, \alpha \in H$,
cardona-qft-methods_ F02664	244	■ 0.719	$e^*=e, \quad \alpha e=e\alpha=\alpha, \forall \alpha \in H$	$e^*=e, \quad \alpha e=e\alpha=\alpha, \forall \alpha \in H$	(iii) $e^*=e, \quad \alpha e=e\alpha=\alpha, \forall \alpha \in H$,

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cardona-qft-methods_ F02665	244	■ 0.672	$\backslash\epsilon: H \rightarrow \mathbb{C}$	$\epsilon : H \rightarrow \mathbb{C}$	(2) The counit $\epsilon : H \rightarrow \mathbb{C}$ is a $*$ -homomorphism subject to the condition
cardona-qft-methods_ F02666	244	■ 0.683	$S: H \rightarrow H$	$S : H \rightarrow H$	(3) The antipode $S : H \rightarrow H$ is a $*$ -antihomomorphism, $S(\alpha\beta) = S(\beta\alpha)$
cardona-qft-methods_ F02667	244	■ 0.683	$S(\backslash\alpha \backslash\beta)=S(\backslash\beta \backslash\alpha)$	$S(\alpha\beta) = S(\beta\alpha)$	(3) The antipode $S : H \rightarrow H$ is a $*$ -antihomomorphism, $S(\alpha\beta) = S(\beta\alpha)$
cardona-qft-methods_ F02668	244	■ 1.000	$m_{13} \otimes m_{24}$	$m_{13} \otimes m_{24}$	Then, as for $\mathbb{C}G$, we can also define $m_{13} \otimes m_{24}$:
cardona-qft-methods_ F02669	244	■ 1.000	$H \otimes H \otimes H \otimes H$	$H \otimes H \otimes H \otimes H$	This operation extends by linearity to all of $H \otimes H \otimes H \otimes H$. Then we
cardona-qft-methods_ F02670	245	■ 1.000	b	b	and an element b such that $b^n = 0$ and $ab = rba$, where r is a primitive
cardona-qft-methods_ F02671	245	■ 1.000	$b^{\{n\}}=0$	$b^n = 0$	and an element b such that $b^n = 0$ and $ab = rba$, where r is a primitive
cardona-qft-methods_ F02672	245	■ 1.000	$a \, b=r \, b \, a$	$ab = rba$	and an element b such that $b^n = 0$ and $ab = rba$, where r is a primitive
cardona-qft-methods_ F02673	245	■ 0.999	$\backslash\Delta(a)=a \otimes a, \backslash\Delta(b)=a \otimes b+b \otimes \backslash a^{-1}, \backslash\epsilon(a)=1, \backslash\epsilon(b)=0$	$\Delta(a) = a \otimes a, \Delta(b) = a \otimes b + b \otimes a^{-1}, \epsilon(a) = 1, \epsilon(b) = 0$	counit defined by $\Delta(a) = a \otimes a, \Delta(b) = a \otimes b + b \otimes a^{-1}, \epsilon(a) = 1, \epsilon(b) = 0$
cardona-qft-methods_ F02674	245	■ 0.998	$H \otimes H$	$H \otimes H$	Problem 2.3. Let H be a Hopf algebra. A twist F is an element of $H \otimes H$
cardona-qft-methods_ F02675	245	■ 1.000	$\backslash\Delta_{\{F\}}$	Δ_F	Show that Δ_F defined by $\Delta_F(h) = F^{-1} \Delta(h) F$ satisfies the coassociativity
cardona-qft-methods_ F02676	245	■ 1.000	$\backslash\Delta_{\{F\}}(h)=F^{\{-1\}} \backslash\Delta(h) \, F$	$\Delta_F(h) = F^{-1} \Delta(h) F$	Show that Δ_F defined by $\Delta_F(h) = F^{-1} \Delta(h) F$ satisfies the coassociativity
cardona-qft-methods_ F02677	245	■ 1.000	$\backslash\left(\backslash\Delta_{\{F\}} \otimes I \, d\right) \backslash\Delta_{\{F\}}=\backslash\left(I \, d \otimes \backslash\Delta_{\{F\}}\right) \backslash\Delta_{\{F\}}$	$(\Delta_F \otimes Id) \Delta_F = (Id \otimes \Delta_F) \Delta_F$	condition $(\Delta_F \otimes Id) \Delta_F = (Id \otimes \Delta_F) \Delta_F$.
cardona-qft-methods_ F02678	245	■ 1.000	$U_{\{q\}}(S \, L(S, \mathbb{C}))$	$U_q(SL(S, \mathbb{C}))$	physics. One such well-known example is $U_q(SL(S, \mathbb{C}))$. The Poincar alge-
cardona-qft-methods_ F02679	245	■ 1.000	$\{ \}^{\{6,7,9-12\}}$	$^{6,7,9-12}$	besides our book, ⁸ are ^{6,7,9-12} .
cardona-qft-methods_ F02680	245	■ 1.000	$V \otimes V$	$V \otimes V$	space is based on $V \otimes V$. For the conventional choice of the coproduct Δ ,
cardona-qft-methods_ F02681	245	■ 1.000	$\backslash\xi \otimes \backslash\eta \in V \otimes V$	$\xi \otimes \eta \in V \otimes V$	the symmetry group G acts on $\xi \otimes \eta \in V \otimes V$ according to
cardona-qft-methods_ F02682	245	■ 0.744	$(\backslash\rho \otimes \backslash\rho) \backslash\Delta(g)$	$(\rho \otimes \rho) \Delta(g)$	Problem 2.4. Show that τ commutes with $(\rho \otimes \rho) \Delta(g)$, for any $g \in G$.
cardona-qft-methods_ F02683	245	■ 1.000	$V \otimes_{\{S\}} V$	$V \otimes_S V$	Use this to explain how $V \otimes V$ splits as the sum of two eigenspaces $V \otimes_S V$
cardona-qft-methods_ F02684	246	■ 0.985	$V \otimes_{\{A\}} V$	$V \otimes_A V$	and $V \otimes_A V$ for τ (corresponding, respectively, to the eigenvalues 1 and -1)
cardona-qft-methods_ F02685	246	■ 1.000	$V \otimes V \otimes \cdots \otimes V$	$V \otimes V \otimes \cdots \otimes V$	states. Thus on $V \otimes V \otimes \cdots \otimes V$, we can define transposition operators
cardona-qft-methods_ F02686	246	■ 1.000	$S_{\{N\}}$	S_N	They generate the full permutation group S_N and commute with the ac-
cardona-qft-methods_ F02687	246	■ 1.000	$\backslash\tau_{\{i, \, i+1\}} \rightarrow \pm 1$	$\tau_{i,i+1} \rightarrow \pm 1$	by the representations where $\tau_{i,i+1} \rightarrow \pm 1$ respectively. The corresponding
cardona-qft-methods_ F02688	246	■ 1.000	$V \otimes_{\{S, \, A\}} V \otimes_{\{S, \, A\}} \cdots \otimes_{\{S, \, A\}} V \equiv V^{\otimes_{\{S\}}}, V^{\otimes_{\{A\}}}$	$V \otimes_{S,A} V \otimes_{S,A} \cdots \otimes_{S,A} V \equiv V^{\otimes_S}, V^{\otimes_A}$	$V \otimes_{S,A} V \otimes_{S,A} \cdots \otimes_{S,A} V \equiv V^{\otimes_S}, V^{\otimes_A}$.
cardona-qft-methods_ F02689	246	■ 0.998	$\backslash\Delta^{\{\mathrm{op}\}}$	Δ^{op}	Δ , there is another coproduct Δ^{op} (‘op’ standing for ‘opposite’):
cardona-qft-methods_ F02690	246	■ 1.000	$R \in H \otimes H$	$R \in H \otimes H$	There exists an invertible R -matrix $R \in H \otimes H$ such that $\Delta^{\mathrm{op}}(\eta) =$
cardona-qft-methods_ F02691	246	■ 1.000	$\backslash\Delta^{\{\mathrm{op}\}}(\backslash\eta)=$	$\Delta^{\mathrm{op}}(\eta) =$	There exists an invertible R -matrix $R \in H \otimes H$ such that $\Delta^{\mathrm{op}}(\eta) =$

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cardona-qft-methods_ F02692	246	■ 1.000	$R \Delta(\eta) R^{-1}$	$R\Delta(\eta)R^{-1}$	$R\Delta(\eta)R^{-1}$ and fulfills also certain further properties (Yang-Baxter rela-
cardona-qft-methods_ F02693	246	■ 0.658	ρ	ρ	triangular”. From these properties it follows that, if ρ is a representation
cardona-qft-methods_ F02694	246	■ 1.000	$(\rho \otimes \rho) \Delta(\alpha)$	$(\rho \otimes \rho) \Delta(\alpha)$	of H , then $(\rho \otimes \rho) \Delta(\alpha)$ commutes with $\tau(\rho \otimes \rho)R$. This means that, at least
cardona-qft-methods_ F02695	246	■ 1.000	$\tau(\rho \otimes \rho) R$	$\tau(\rho \otimes \rho) R$	of H , then $(\rho \otimes \rho) \Delta(\alpha)$ commutes with $\tau(\rho \otimes \rho)R$. This means that, at least
cardona-qft-methods_ F02696	246	■ 1.000	τR	τR	for two identical particles, we can use τR instead of τ to define statistics,
cardona-qft-methods_ F02697	247	■ 1.000	$\tau_{i, i+1}$	$\tau_{i, i+1}$	generalizes $\tau_{i, i+1}$ of $\mathbb{C}G$ on $V \otimes V \otimes \dots \otimes V$. The group it generates replaces
cardona-qft-methods_ F02698	247	■ 1.000	$\mathcal{B}_N$	$\mathcal{B}_N$	S_N (it is the braid group $\mathcal{B}_N$). Notice that the square of τR is not neces-
cardona-qft-methods_ F02699	247	■ 1.000	$R \Delta R^{-1}$	$R\Delta R^{-1}$	They can be derived from the requirement that Δ^{op} is $R\Delta R^{-1}$ and both
cardona-qft-methods_ F02700	247	■ 0.974	Δ_{op}	Δ_{op}	Δ and Δ_{op} satisfy the identity coming from coassociativity. This imposes a
cardona-qft-methods_ F02701	247	■ 1.000	$(\rho \otimes \rho) \Delta_0(g)$	$(\rho \otimes \rho) \Delta_0(g)$	for the Poincar group (cf. 2.1). Since τ commutes with with $(\rho \otimes \rho) \Delta_0(g)$,
cardona-qft-methods_ F02702	247	■ 1.000	$\Delta_\theta(g)$	$\Delta_\theta(g)$	one can see from the structure of $\Delta_\theta(g)$ that
cardona-qft-methods_ F02703	247	■ 1.000	$(\rho \otimes \rho) \Delta_\theta(g)$	$(\rho \otimes \rho) \Delta_\theta(g)$	commutes with $(\rho \otimes \rho) \Delta_\theta(g)$. We can also write τ_θ as
cardona-qft-methods_ F02704	247	■ 1.000	τ_θ	τ_θ	commutes with $(\rho \otimes \rho) \Delta_\theta(g)$. We can also write τ_θ as
cardona-qft-methods_ F02705	247	■ 0.666	$\Delta_\theta^{\text{op}}$	$\Delta_\theta^{\text{op}}$	We can also show this result by examining $\Delta_\theta^{\text{op}}$. It is simple to show that
cardona-qft-methods_ F02706	247	■ 1.000	$\Delta_\theta^{\text{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$	$\Delta_\theta^{\text{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$	$\Delta_\theta^{\text{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$. Clearly, $R_\theta \Delta_\theta(g) = \Delta_\theta^{\text{op}}(g) R_\theta$, as we want. A point
cardona-qft-methods_ F02707	247	■ 1.000	$R_\theta \Delta_\theta(g) = \Delta_\theta^{\text{op}}(g) R_\theta$	$R_\theta \Delta_\theta(g) = \Delta_\theta^{\text{op}}(g) R_\theta$	$\Delta_\theta^{\text{op}}(g) = F_\theta(g \otimes g) F_\theta^{-1}$. Clearly, $R_\theta \Delta_\theta(g) = \Delta_\theta^{\text{op}}(g) R_\theta$, as we want. A point
cardona-qft-methods_ F02708	247	■ 1.000	$\tau_\theta^2 = \text{Id}$	$\tau_\theta^2 = \text{Id}$	to note is that $\tau_\theta^2 = \text{Id}$. Hence the statistics group is still the permutation
cardona-qft-methods_ F02709	248	■ 1.000	$R(t)$	$R(t)$	comes from the Kramers-Kronig relation. Suppose $R(t)$ is a response func-
cardona-qft-methods_ F02710	248	■ 1.000	$t < 0$	$t < 0$	tion for a system that is undisturbed for $t < 0$. The system should not
cardona-qft-methods_ F02711	248	■ 1.000	$A > B$	$A > B$	order: If A is an event that is to the future of an event B , we say that $A > B$.
cardona-qft-methods_ F02712	248	■ 1.000	O	O	So is O stands for an observer, for an arbitrary event C in the future light
cardona-qft-methods_ F02713	248	■ 1.000	$C > O$	$C > O$	cone of O we will have $C > O$ and for any event A in the past light cone
cardona-qft-methods_ F02714	248	■ 1.000	$O > A$	$O > A$	we will have $O > A$. An event P that is spacelike relative to O is not
cardona-qft-methods_ F02715	248	■ 0.817	$>$	$>$	causally related to O . Thus, the relation $>$ is a partial order on the set of
cardona-qft-methods_ F02716	248	■ 1.000	$\rho(x), \eta(y)$	$\rho(x), \eta(y)$	Local observables $\rho(x)$, $\eta(y)$ commute if x and y are spacelike separated:
cardona-qft-methods_ F02717	249	■ 1.000	x_μ	x_μ	Here the coordinate functions x_μ take on an operator meaning: $\widehat{x}_\mu$.
cardona-qft-methods_ F02718	249	■ 1.000	$\widehat{x}_\mu$	$\widehat{x}_\mu$	Here the coordinate functions x_μ take on an operator meaning: $\widehat{x}_\mu$.

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cardona-qft-methods_ F02719	249	■ 1.000	G M	GM	field theories on the GM plane and indicate its physical consequences. The
cardona-qft-methods_ F02728	249	■ 1.000	$\mathcal{H}_{\{i\}}$	$\mathcal{H}_i$	G with elements g acting on single particle Hilbert spaces $\mathcal{H}_i$ by unitary
cardona-qft-methods_ F02721	249	■ 1.000	$g \rightarrow U_{\{i\}}(g)$	$g \rightarrow U_i(g)$	representations $g \rightarrow U_i(g)$, then conventionally they act on $\mathcal{H}_1 \otimes \mathcal{H}_2$ by the
cardona-qft-methods_ F02722	249	■ 1.000	$\mathcal{H}_{\{1\}} \otimes \mathcal{H}_{\{2\}}$	$\mathcal{H}_1 \otimes \mathcal{H}_2$	representations $g \rightarrow U_i(g)$, then conventionally they act on $\mathcal{H}_1 \otimes \mathcal{H}_2$ by the
cardona-qft-methods_ F02723	250	■ 1.000	$m_{\{0\}}$	m_0	where m_0 is the point-wise multiplication map of $\mathcal{A}_0$:
cardona-qft-methods_ F02724	250	■ 1.000	$\mathcal{A}_{\{0\}}$	$\mathcal{A}_0$	where m_0 is the point-wise multiplication map of $\mathcal{A}_0$:
cardona-qft-methods_ F02725	250	■ 1.000	$x \in \mathbb{R}^N$	$x \in \mathbb{R}^N$	Now let Λ be an element of the Poincaré group $\mathcal{P}_+^\dagger$. For $x \in \mathbb{R}^N$,
cardona-qft-methods_ F02726	250	■ 1.000	$\mathcal{A}_{\{\theta\}}(\mathbb{R}^N)$	$\mathcal{A}_\theta(\mathbb{R}^N)$	paper shows that $\mathcal{P}_+^\dagger$ acts on $\mathcal{A}_\theta(\mathbb{R}^N)$ compatibly with m_θ if its coproduct
cardona-qft-methods_ F02727	250	■ 1.000	$m_{\{\theta\}}$	m_θ	paper shows that $\mathcal{P}_+^\dagger$ acts on $\mathcal{A}_\theta(\mathbb{R}^N)$ compatibly with m_θ if its coproduct
cardona-qft-methods_ F02728	250	■ 1.000	$\theta^{\mu\nu} = 0$	$\theta^{\mu\nu} = 0$	Such a two-particle system, for $\theta^{\mu\nu} = 0$, is a function of two sets vari-
cardona-qft-methods_ F02729	250	■ 1.000	$\mathcal{A}_{\{0\}} \otimes \mathcal{A}_{\{0\}}$	$\mathcal{A}_0 \otimes \mathcal{A}_0$	ables, “lives” in $\mathcal{A}_0 \otimes \mathcal{A}_0$ and transforms according to the usual coproduct
cardona-qft-methods_ F02730	250	■ 1.000	$\Delta_{\{0\}}$	Δ_0	Δ_0 . Similarly, in the noncommutative case, the wavefunction lives in $\mathcal{A}_\theta \otimes \mathcal{A}_\theta$
cardona-qft-methods_ F02731	250	■ 1.000	$\mathcal{A}_{\{\theta\}} \otimes \mathcal{A}_{\{\theta\}}$	$\mathcal{A}_\theta \otimes \mathcal{A}_\theta$	Δ_0 . Similarly, in the noncommutative case, the wavefunction lives in $\mathcal{A}_\theta \otimes \mathcal{A}_\theta$
cardona-qft-methods_ F02732	250	■ 1.000	$\theta_{\mu\nu} = 0$	$\theta_{\mu\nu} = 0$	For $\theta_{\mu\nu} = 0$ we require that the physical wave functions describing iden-
cardona-qft-methods_ F02733	251	■ 1.000	$\tau_{\{0\}}$	τ_0	Thus let τ_0 be the statistics (flip) operator associated with exchange for
cardona-qft-methods_ F02734	251	■ 1.000	$\Delta_{\{0\}}(\Lambda) = \Lambda \otimes \Lambda$	$\Delta_0(\Lambda) = \Lambda \otimes \Lambda$	Lorentz group action, $\Delta_0(\Lambda) = \Lambda \otimes \Lambda$, must and does commute with the
cardona-qft-methods_ F02735	251	■ 1.000	$\phi \otimes \chi$	$\phi \otimes \chi$	the same eigenvalue. Given an element $\phi \otimes \chi$ of the tensor product, the
cardona-qft-methods_ F02736	252	■ 1.000	$\mathcal{P}$	$\mathcal{P}$	As we have seen, the action of the Poincaré group $\mathcal{P}$ on the Minkowski
cardona-qft-methods_ F02737	252	■ 1.000	M^{d+1}	M^{d+1}	space M^{d+1} is given by
cardona-qft-methods_ F02738	252	■ 1.000	$\mathcal{A}_{\{0\}}(M^{d+1})$	$\mathcal{A}_0(M^{d+1})$	The commutative algebra of functions on M^{d+1} is denoted by $\mathcal{A}_0(M^{d+1})$.
cardona-qft-methods_ F02739	252	■ 1.000	φ	φ	If φ is a quantum relativistic scalar field on M^{d+1} , we require that there
cardona-qft-methods_ F02740	252	■ 1.000	$\varphi \in L(\mathcal{H}) \otimes S(M^{d+1})$	$\varphi \in L(\mathcal{H}) \otimes S(M^{d+1})$	Note that for a quantum field φ we have $\varphi \in L(\mathcal{H}) \otimes S(M^{d+1})$ where
cardona-qft-methods_ F02741	252	■ 1.000	$L(\mathcal{H})$	$L(\mathcal{H})$	$L(\mathcal{H})$ are linear operators on $\mathcal{H}$ and $S(M^{d+1})$ are distributions on M^{d+1} .
cardona-qft-methods_ F02742	252	■ 1.000	$S(M^{d+1})$	$S(M^{d+1})$	$L(\mathcal{H})$ are linear operators on $\mathcal{H}$ and $S(M^{d+1})$ are distributions on M^{d+1} .
cardona-qft-methods_ F02743	253	■ 1.000	$\varphi \otimes \varphi \otimes \dots \otimes \varphi$	$\varphi \otimes \varphi \otimes \dots \otimes \varphi$	This can be understood as the element $\varphi \otimes \varphi \otimes \dots \otimes \varphi$ belonging to
cardona-qft-methods_ F02744	253	■ 1.000	$L(\mathcal{H}) \otimes (S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1}))$	$L(\mathcal{H}) \otimes (S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1}))$	$L(\mathcal{H}) \otimes (S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1}))$ evaluated at $x_1, x_2, \dots, x_N$.

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cardona-qft-methods_ F02745	253	■ 1.000	$x_{\{1\}}, x_{\{2\}}, \ldots x_{\{N\}}$	$x_1, x_2, \dots x_N$	$L(\mathcal{H}) \otimes (S(M^{d+1}) \otimes S(M^{d+1}) \otimes \dots \otimes S(M^{d+1}))$ evaluated at $x_1, x_2, \dots x_N$.
cardona-qft-methods_ F02746	253	■ 1.000	$\varphi^{\otimes N} = \varphi \otimes \varphi \otimes \dots \otimes \varphi \in L(\mathcal{H}) \otimes \left(S(M^{d+1})^{\otimes N} \right)$	$\varphi^{\otimes N} = \varphi \otimes \varphi \otimes \dots \otimes \varphi \in L(\mathcal{H}) \otimes \left(S(M^{d+1})^{\otimes N} \right)$	Thus for $\varphi^{\otimes N} = \varphi \otimes \varphi \otimes \dots \otimes \varphi \in L(\mathcal{H}) \otimes (S(M^{d+1})^{\otimes N})$, we will have:
cardona-qft-methods_ F02747	253	■ 1.000	$U(a, \Lambda)$	$U(a, \Lambda)$	Fock space and $U(a, \Lambda)$ acts by conjugation on it for all N . But that is not
cardona-qft-methods_ F02748	253	■ 1.000	$S(\left(M^{d+1}\right)^{\otimes N})$	$S(M^{d+1})^{\otimes N}$	the case for $S(M^{d+1})^{\otimes N}$. The Poincaré group acts on it by means of the
cardona-qft-methods_ F02749	253	■ 1.000	(a, Λ)	(a, Λ)	of (a, Λ) . We will denote it by Δ_0^N in what follows. Thus
cardona-qft-methods_ F02750	253	■ 1.000	Δ_0^N	Δ_0^N	of (a, Λ) . We will denote it by Δ_0^N in what follows. Thus
cardona-qft-methods_ F02751	253	■ 1.000	$(a, \Lambda) \in \mathcal{P}$	$(a, \Lambda) \in \mathcal{P}$	for all $(a, \Lambda) \in \mathcal{P}$.
cardona-qft-methods_ F02752	253	■ 1.000	$c_{\{p\}}, c_{\{p\}}^{\dagger}$	$c_p, c_p^{\dagger}$	where $c_p, c_p^{\dagger}$ are the standard annihilation and creation operators, and $\varphi^{(\pm)}$
cardona-qft-methods_ F02753	253	■ 1.000	$\varphi^{(\pm)}$	$\varphi^{(\pm)}$	where $c_p, c_p^{\dagger}$ are the standard annihilation and creation operators, and $\varphi^{(\pm)}$
cardona-qft-methods_ F02754	253	■ 1.000	$\varphi^{(-)}$	$\varphi^{(-)}$	fulfill the covariance requirement. Let us consider $\varphi^{(-)}$. We have that
cardona-qft-methods_ F02755	253	■ 1.000	P_{μ}	P_{μ}	Let us check this for translations. Let P_{μ} be the translation generators on
cardona-qft-methods_ F02756	254	■ 0.999	$\mathcal{P}_{\mu} = -i \partial_{\mu}$	$\mathcal{P}_{\mu} = -i \partial_{\mu}$	and let $\mathcal{P}_{\mu} = -i \partial_{\mu}$ be the representation of the translation generator on
cardona-qft-methods_ F02757	254	■ 1.000	$\mathcal{P}_{\mu}$	$\mathcal{P}_{\mu}$	The coproduct Δ_0 gives for the Lie algebra element $\mathcal{P}_{\mu}$,
cardona-qft-methods_ F02758	254	■ 0.999	$\underline{\nu}$	$\underline{\nu}$	If $\underline{\nu}$ is the representation of the Lie algebra of $\mathcal{P}$ on functions, and $\hat{P}_{\mu}$ is
cardona-qft-methods_ F02759	254	■ 0.999	$\widehat{P}_{\mu}$	$\hat{P}_{\mu}$	If $\underline{\nu}$ is the representation of the Lie algebra of $\mathcal{P}$ on functions, and $\hat{P}_{\mu}$ is
cardona-qft-methods_ F02760	254	■ 1.000	$\underline{\nu}(\widehat{P}_{\mu}) = \mathcal{P}_{\mu}$	$\underline{\nu}(\hat{P}_{\mu}) = \mathcal{P}_{\mu}$	the Lie algebra generator in the abstract group $\mathcal{P}$ so that $\underline{\nu}(\hat{P}_{\mu}) = \mathcal{P}_{\mu}$,
cardona-qft-methods_ F02761	254	■ 1.000	$e_{\{p\}}$	e_p	on e_p according to
cardona-qft-methods_ F02762	254	■ 1.000	$\Lambda e_p = e_{\Lambda p}$	$\Lambda e_p = e_{\Lambda p}$	or $\Lambda e_p = e_{\Lambda p}$. For Lorentz transformations Λ , covariance is thus the identity
cardona-qft-methods_ F02763	254	■ 1.000	$d \geq 3$	$d \geq 3$	tics of N-particle state vectors on commutative spacetimes for $d \geq 3$. We
cardona-qft-methods_ F02764	255	■ 0.964	(a, Δ)	(a, Δ)	(a, Δ) acts on $e_{p_1} \otimes e_{p_2} \otimes \dots \otimes e_{p_N}$ via the coproduct. This action commutes
cardona-qft-methods_ F02765	255	■ 0.964	$e_{\{p_1\}} \otimes e_{\{p_2\}} \otimes \dots \otimes e_{\{p_N\}}$	$e_{p_1} \otimes e_{p_2} \otimes \dots \otimes e_{p_N}$	(a, Δ) acts on $e_{p_1} \otimes e_{p_2} \otimes \dots \otimes e_{p_N}$ via the coproduct. This action commutes
cardona-qft-methods_ F02766	255	■ 1.000	$e_{\{p_i\}}$	e_{p_i}	fermions by totally symmetrizing or antisymmetrizing e_{p_i} . The important
cardona-qft-methods_ F02767	255	■ 1.000	$\mathbb{C} \mathcal{P}$	$\mathbb{C} \mathcal{P}$	point here is that the group algebras $\mathbb{C} \mathcal{P}$ and $\mathbb{C} S_N$ are commutants of each
cardona-qft-methods_ F02768	255	■ 1.000	$\mathbb{C} S_N$	$\mathbb{C} S_N$	point here is that the group algebras $\mathbb{C} \mathcal{P}$ and $\mathbb{C} S_N$ are commutants of each
cardona-qft-methods_ F02769	255	■ 0.833	$U(k)$	$U(k)$	symmetry group is $U(k)$. Here $U(k)$ can be the k-flavour symmetry group.
cardona-qft-methods_ F02770	255	■ 1.000	$\mathbb{C}^k$	$\mathbb{C}^k$	It acts on $\mathbb{C}^k$. Then to reduce the representation of $U(k)$ on $\mathbb{C}^{k \otimes N}$, we use

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cardona-qft-methods_ F02771	255	■ 1.000	$\mathbb{C}^k \otimes N$	$\mathbb{C}^{k \otimes N}$	It acts on $\mathbb{C}^k$. Then to reduce the representation of $U(k)$ on $\mathbb{C}^{k \otimes N}$, we use
cardona-qft-methods_ F02772	255	■ 1.000	$\mathbb{C} \cup (k)$	$\mathbb{C}U(k)$	the fact that $\mathbb{C}S_N$ commutes with $\mathbb{C}U(k)$. That lets us use Young tableaux
cardona-qft-methods_ F02773	255	■ 1.000	τ_{ij}	τ_{ij}	1 to N and let τ_{ij} denote the transposition of objects i and j . Then S_N has
cardona-qft-methods_ F02774	255	■ 0.999	$\alpha: p \rightarrow \alpha p, p \in M$	$\alpha: p \rightarrow \alpha p, p \in M$	the set of smooth functions on a manifold M . If $\alpha: p \rightarrow \alpha p, p \in M$, is a
cardona-qft-methods_ F02775	256	■ 1.000	$\mathbb{C}^{\infty}(M)$	$\mathbb{C}^{\infty}(M)$	$\mathbb{C}^{\infty}(M)$ is closed under point-wise multiplication: If $f_1, f_2 \in C^{\infty}(M)$, then:
cardona-qft-methods_ F02776	256	■ 1.000	$f_1, f_2 \in C^{\infty}(M)$	$f_1, f_2 \in C^{\infty}(M)$	$\mathbb{C}^{\infty}(M)$ is closed under point-wise multiplication: If $f_1, f_2 \in C^{\infty}(M)$, then:
cardona-qft-methods_ F02777	256	■ 1.000	$C^0(M)$	$C^0(M)$	algebra of $C^0(M)$, a commutative C^* -algebra. By the Gelfand-Naimark
cardona-qft-methods_ F02778	256	■ 1.000	$C^{\star}$	$C^{\star}$	algebra of $C^0(M)$, a commutative C^* -algebra. By the Gelfand-Naimark
cardona-qft-methods_ F02779	256	■ 1.000	$\Psi^{\dagger}$	$\Psi^{\dagger}$	we require that $\Psi^{\dagger}$ is also a covariant complex field. That is fulfilled if
cardona-qft-methods_ F02780	256	■ 0.996	$U(a, \Delta)$	$U(a, \Delta)$	$U(a, \Delta)$ is unitary. Thus $\star$ -covariance is linked to unitarity of time-evolution
cardona-qft-methods_ F02781	256	■ 1.000	$\mathcal{A}_{\theta}(M^{d+1})$	$\mathcal{A}_{\theta}(M^{d+1})$	The Moyal plane $\mathcal{A}_{\theta}(M^{d+1})$ is the algebra of smooth functions on $M^{(d+1)}$
cardona-qft-methods_ F02782	256	■ 1.000	$M^{(d+1)}$	$M^{(d+1)}$	The Moyal plane $\mathcal{A}_{\theta}(M^{d+1})$ is the algebra of smooth functions on $M^{(d+1)}$
cardona-qft-methods_ F02783	257	■ 0.978	$\mathcal{A}_{\theta}(M^{(d+1)})$	$\mathcal{A}_{\theta}(M^{(d+1)})$	This action extends to the algebra $\mathcal{A}_{\theta}(M^{(d+1)})$ compatibly with the product
cardona-qft-methods_ F02784	257	■ 1.000	F_{θ}	F_{θ}	with F_{θ} the Drinfel'd twist. Here $\hat{P}_{\mu}$ is as before the translation genera-
cardona-qft-methods_ F02785	257	■ 1.000	$\hat{P}_{\mu}$	$\hat{P}_{\mu}$	with F_{θ} the Drinfel'd twist. Here $\hat{P}_{\mu}$ is as before the translation genera-
cardona-qft-methods_ F02786	257	■ 1.000	φ_{θ}	φ_{θ}	Let φ_{θ} be the twisted analogue of the field φ and let U_{θ} be the unitary
cardona-qft-methods_ F02787	257	■ 1.000	U_{θ}	U_{θ}	Let φ_{θ} be the twisted analogue of the field φ and let U_{θ} be the unitary
cardona-qft-methods_ F02788	257	■ 1.000	$\varphi_{\theta}^{\dagger}$	$\varphi_{\theta}^{\dagger}$	requires that $\varphi_{\theta}^{\dagger}$ is also covariant. There is also one further requirement,
cardona-qft-methods_ F02789	257	■ 1.000	$U_{\theta}(a, \Lambda) 0\rangle = 0\rangle$	$U_{\theta}(a, \Lambda) 0\rangle = 0\rangle$	that $U_{\theta}(a, \Lambda) 0\rangle = 0\rangle$:
cardona-qft-methods_ F02790	258	■ 1.000	$a_p, a_p^{\dagger}$	$a_p, a_p^{\dagger}$	twisted coproduct. This affects the properties of $a_p, a_p^{\dagger}$ in a $\theta_{\mu\nu}$ -dependent
cardona-qft-methods_ F02791	258	■ 1.000	$e_{p_1} \otimes e_{p_2}$	$e_{p_1} \otimes e_{p_2}$	Since translations act in the usual way on $e_{p_1} \otimes e_{p_2}$,
cardona-qft-methods_ F02792	258	■ 1.000	$a_{p_1}^{\dagger}$	$a_{p_1}^{\dagger}$	Translational covariance requires the standard transformation of $a_{p_1}^{\dagger}$:
cardona-qft-methods_ F02793	258	■ 1.000	P_{μ}^{θ}	P_{μ}^{θ}	P_{μ}^{θ} is the possibly θ dependent translation generator. Lorentz transforma-
cardona-qft-methods_ F02794	258	■ 1.000	$a_p^{\dagger}$	$a_p^{\dagger}$	The covariance requirement is solved by writing $a_p^{\dagger}$ in terms of the $c_p^{\dagger}$ and
cardona-qft-methods_ F02795	258	■ 1.000	$c_p^{\dagger}$	$c_p^{\dagger}$	The covariance requirement is solved by writing $a_p^{\dagger}$ in terms of the $c_p^{\dagger}$ and
cardona-qft-methods_ F02796	258	■ 0.996	c_p	c_p	twist c_p on left <i>or</i> on right, we can write φ_{θ} as a twist applied to $\varphi_0 \equiv \varphi$:

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cardona-qft-methods_ F02797	258	■ 0.996	$\varphi_{\{0\}} \equiv \varphi_{\text{phi}}$	$\varphi_0 \equiv \varphi :$	twist c_p on left <i>or</i> on right, we can write φ_θ as a twist applied to $\varphi_0 \equiv \varphi :$
cardona-qft-methods_ F02798	258	■ 1.000	$\varphi_{\{0\}} \rightarrow \varphi_{\{\theta\}}$	$\varphi_0 \rightarrow \varphi_\theta$	The transformation $\varphi_0 \rightarrow \varphi_\theta$ is an example of a <i> dressing transformation</i> .
cardona-qft-methods_ F02799	258	■ 1.000	$\varphi_{\{0\}}$	φ_0	This is well-defined for a fully interacting Heisenberg field φ_0 if P_μ stands
cardona-qft-methods_ F02800	259	■ 0.999	$\langle 0 $	$\langle 0 $	with a similar equation for the vacuum $\langle 0 $ put on the left. Since the vacuum
cardona-qft-methods_ F02801	259	■ 1.000	α, β	α, β	where α, β are single particle wave functions. As defined, τ_0 and τ_θ act on
cardona-qft-methods_ F02802	259	■ 1.000	$S_{\{2\}}$	S_2	two-particle wave functions and generate S_2 since
cardona-qft-methods_ F02803	259	■ 1.000	$S_{\{\theta\}}$	S_θ	Let us focus on S_θ :
cardona-qft-methods_ F02804	260	■ 0.932	$a^{\dagger}$	$a^\dagger$	multinomials in $a^\dagger$'s and as . In the N -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld
cardona-qft-methods_ F02805	260	■ 0.932	$\mathcal{F}_\theta^{ij}$	$\mathcal{F}_\theta^{ij}$	multinomials in $a^\dagger$'s and as . In the N -particle sector, call $\mathcal{F}_\theta^{ij}$ the Drinfeld
cardona-qft-methods_ F02806	260	■ 0.939	$\partial_\mu \otimes \partial_\nu, \partial_\mu$	$\partial_\mu \otimes \partial_\nu, \partial_\mu$	twist where in $\partial_\mu \otimes \partial_\nu$, ∂_μ acts on the i^{th} and ∂_ν on the j^{th} factor in the
cardona-qft-methods_ F02807	260	■ 0.939	i^{th}	i^{th}	twist where in $\partial_\mu \otimes \partial_\nu$, ∂_μ acts on the i^{th} and ∂_ν on the j^{th} factor in the
cardona-qft-methods_ F02808	260	■ 0.939	∂_ν	∂_ν	twist where in $\partial_\mu \otimes \partial_\nu$, ∂_μ acts on the i^{th} and ∂_ν on the j^{th} factor in the
cardona-qft-methods_ F02809	260	■ 0.939	j^{th}	j^{th}	twist where in $\partial_\mu \otimes \partial_\nu$, ∂_μ acts on the i^{th} and ∂_ν on the j^{th} factor in the
cardona-qft-methods_ F02810	260	■ 0.976	τ_0^{ij}	τ_0^{ij}	where τ_0^{ij} flips the entries of an N -fold tensor product by flipping the i^{th}
cardona-qft-methods_ F02811	260	■ 0.976	i^{th}	i^{th}	where τ_0^{ij} flips the entries of an N -fold tensor product by flipping the i^{th}
cardona-qft-methods_ F02812	260	■ 0.999	$\tau_\theta^{i,i+1} \mathbf{s}$	$\tau_\theta^{i,i+1} \mathbf{s}$	which is not obvious. It follows from (23) that the $\tau_\theta^{i,i+1}$'s generate S_N in
cardona-qft-methods_ F02813	260	■ 1.000	$\varphi_{\{0\}}^* = \varphi_{\{0\}}$	$\varphi_0^* = \varphi_0$	$\varphi_0^* = \varphi_0$, we want that $\varphi_\theta^* = \varphi_\theta$ Now
cardona-qft-methods_ F02814	260	■ 1.000	$\varphi_{\{\theta\}}^* = \varphi_{\{\theta\}}$	$\varphi_\theta^* = \varphi_\theta$	$\varphi_0^* = \varphi_0$, we want that $\varphi_\theta^* = \varphi_\theta$ Now
cardona-qft-methods_ F02815	260	■ 1.000	∂_μ	∂_μ	where ∂_μ acts just on φ_0 , P_ν acts on φ_0 and all that may follow. But since
cardona-qft-methods_ F02816	260	■ 1.000	$\varphi_{\{0\}}, P_{\{\nu\}}$	φ_0, P_ν	where ∂_μ acts just on φ_0 , P_ν acts on φ_0 and all that may follow. But since
cardona-qft-methods_ F02817	260	■ 1.000	$\varphi_{\{0\}} - \partial_\nu \varphi_{\{0\}}$	$\varphi_0 - \partial_\nu \varphi_0$	P_ν acting on $\varphi_0 - \partial_\nu \varphi_0$ and $\partial \wedge \partial = 0$, we see that
cardona-qft-methods_ F02818	260	■ 1.000	$\partial \wedge \partial = 0$	$\partial \wedge \partial = 0$	P_ν acting on $\varphi_0 - \partial_\nu \varphi_0$ and $\partial \wedge \partial = 0$, we see that
cardona-qft-methods_ F02819	261	■ 1.000	$\{ \}^{14-17}$	$^{14-17}$	reader may consult our papers ^{14–17} where all details can be found.
cardona-qft-methods_ F02820	261	■ 0.997	$V_{\{S \ T \ A \ T\}}$	V_{STAT}	In an interesting paper ¹⁸ , the statistical potential V_{STAT} between fermions
cardona-qft-methods_ F02821	261	■ 1.000	$\left x_{\{1\}}, x_{\{2\}} \right\rangle$	$ x_1, x_2\rangle$	Here $ x_1, x_2\rangle$ has twisted antisymmetry:
cardona-qft-methods_ F02822	261	■ 1.000	$0^{16} \left(C^{12} \right)$	$O^{16} (C^{12})$	forbidden transitions from $O^{16}(C^{12})$ to $\tilde{O}^{16}(\tilde{C}^{12})$ where the tilde nuclei

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cardona-qft-methods_ F02823	261	■ 1.000	$\tilde{0}^{16}\left(\tilde{C}^{12}\right)$	$\tilde{O}^{16}\left(\tilde{C}^{12}\right)$	forbidden transitions from $O^{16}(C^{12})$ to $\tilde{O}^{16}(\tilde{C}^{12})$ where the tilde nuclei
cardona-qft-methods_ F02824	261	■ 1.000	$1\ S_{1 / 2}$	$1S_{1/2}$	have an extra nucleon in the filled $1S_{1/2}$ level are found to have lifetimes
cardona-qft-methods_ F02825	261	■ 1.000	$10^{\{27\}}$	10^{27}	greater than 10^{27} years.
cardona-qft-methods_ F02826	261	■ 1.000	$10^{\{-25\}}$	10^{-25}	of crystals done in Maryland which give branching ratios less than 10^{-25}
cardona-qft-methods_ F02827	261	■ 1.000	ϕ_{θ}	ϕ_{θ}	On the Moyal plane the fields ϕ_{θ} are not local, as can be seen from
cardona-qft-methods_ F02828	262	■ 1.000	$\rho_{\theta}, \eta_{\theta}$	$\rho_{\theta}, \eta_{\theta}$	a) For $\rho_{\theta}, \eta_{\theta}$ observables,
cardona-qft-methods_ F02829	262	■ 1.000	ω_{β}	ω_{β}	b) The same holds for the Gibbs state ω_{β} given by:
cardona-qft-methods_ F02830	262	■ 1.000	$H_{\{I\}}$	H_I	invariant. The reason is loss of causality. We can see this as follows. Let H_I
cardona-qft-methods_ F02831	262	■ 1.000	$x \sim y$	$x \sim y$	where $x \sim y$ means x and y are space-like separated. That is because
cardona-qft-methods_ F02832	263	■ 1.000	$S^{\{2\}}$	$S^{(2)}$	This is $S^{(2)}$ only if
cardona-qft-methods_ F02833	263	■ 0.998	$\vec{P}_{\text {inc }}$	$\vec{P}_{\text {inc }}$	ing amplitudes is striking. They depend on total incident momentum $\vec{P}_{inc}$
cardona-qft-methods_ F02834	263	■ 1.000	$\theta_{0\ i}$	θ_{0i}	but otherwise there is dependence on θ_{0i} . The violation is of order
cardona-qft-methods_ F02835	263	■ 0.866	$\left\{ \right\}_{0\ i}\left(\vec{P}_{\text {inc }}\right)_{\{i\}}$	$_{0i}\left(\vec{P}_{\text {inc }}\right)_i$	$\theta_{0i}\left(\vec{P}_{inc}\right)_i$ in cross sections.
cardona-qft-methods_ F02836	263	■ 1.000	$Z^{\{0\}} \rightarrow 2\ \gamma$	$Z^0 \rightarrow 2\gamma$	Let us remark that even with noncommutativity the decay $Z^0 \rightarrow 2\gamma$
cardona-qft-methods_ F02837	264	■ 1.000	$a_{\{l\ m\}}$	a_{lm}	The a_{lm} can be written in terms of perturbations to Newtonian potential
cardona-qft-methods_ F02838	264	■ 1.000	Φ	Φ	—
cardona-qft-methods_ F02839	264	■ 1.000	$\Delta_{\{l\}^{\{T\}}(k)$	$\Delta_l^T(k)$	where $\Delta_l^T(k)$ are called the transfer functions. We find
cardona-qft-methods_ F02840	264	■ 1.000	$\vec{\theta}^{\{0\}}=\theta(0,0,1)\ .\ P_{\{\Phi\}}$	$\vec{\theta}^0=\theta(0,0,1).P_{\Phi}$	where $\vec{\theta}^0=\theta(0,0,1)$. P_{Φ} is the power spectrum
cardona-qft-methods_ F02841	264	■ 1.000	$\left\langle a_{\{l\ m\}} a_{\{l^{\prime}\ m^{\prime}\}}^{\{*\}}\right\rangle_{\theta}$	$\left\langle a_{lm} a_{l^{\prime} m^{\prime}}^*\right\rangle_{\theta}$	correlator $\left\langle a_{lm} a_{l^{\prime} m^{\prime}}^*\right\rangle_{\theta}$ is θ dependent indicating a preferred direction. Cor-
cardona-qft-methods_ F02842	264	■ 1.000	$C\ M\ B$	CMB	It clearly breaks isotropy in the CMB radiation. On fitting data, one
cardona-qft-methods_ F02843	264	■ 1.000	$\sqrt{\theta} \leq 10^{\{-32\}}$	$\sqrt{\theta} \leq 10^{-32}$	finds, $\sqrt{\theta} \leq 10^{-32}$ meters or energy scale $\geq 10^{16}$ GeV.
cardona-qft-methods_ F02844	264	■ 1.000	$\geq 10^{\{16\}}\ \mathrm{GeV}$	$\geq 10^{16}\mathrm{GeV}$	finds, $\sqrt{\theta} \leq 10^{-32}$ meters or energy scale $\geq 10^{16}$ GeV.
cardona-qft-methods_ F02845	264	■ 1.000	$x_{\{1\}} \sim$	$x_1 \sim$	by uncertainty principle, and it is non-zero by causality violation, for $x_1 \sim$
cardona-qft-methods_ F02846	264	■ 1.000	$x_{\{2\}}$	x_2	x_2 . This implies a new uncertainty principle.
cardona-qft-methods_ F02847	267	■ 1.000	$5+5=10$	$5+5=10$	ding space is the maximally supersymmetric completion of the $5+5=10$
cardona-qft-methods_ F02848	267	■ 0.998	$A\ d\ S_{\{5\}}\ \times S^{\{5\}}$	$AdS_5 \times S^5$	dimensional product spaces $AdS_5 \times S^5$, the 5-dimensional Anti-de-Sitter
cardona-qft-methods_ F02849	267	■ 1.000	$A\ d\ S_{\{5\}}$	AdS_5	sidered to be “brackets” stapling the AdS_5 and S^5 spaces together. The

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cardona-qft-methods_ F02850	267	■ 1.000	S^5	S^5	sidered to be “brackets” stapling the AdS_5 and S^5 spaces together. The
cardona-qft-methods_ F02851	267	■ 1.000	AdS	AdS	reason is that all supercharges carry one index referring to the AdS space
cardona-qft-methods_ F02852	267	■ 1.000	CFT_4	CFT_4	partition function at the boundary of AdS_5 to the CFT_4 partition func-
cardona-qft-methods_ F02853	268	■ 1.000	XXX	XXX	chain followed by the Heisenberg spin chain, XXX model. After that we
cardona-qft-methods_ F02854	268	■ 1.000	IIB	IIB	with remarks on the symmetry structure of IIB string theory on AdS and
cardona-qft-methods_ F02855	268	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	Let us recall spin- $\frac{1}{2}$ particles from quantum mechanics like for example
cardona-qft-methods_ F02856	268	■ 1.000	$\mathbb{C}^2$	$\mathbb{C}^2$	the electron. The spin part of the Hilbert space of the system is $\mathbb{C}^2$. The
cardona-qft-methods_ F02857	268	■ 1.000	$\vec{S}$	$\vec{S}$	operator $\vec{S}$ associated to the spin can be expressed via the Pauli matrices
cardona-qft-methods_ F02858	268	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	tion of the Lie algebra $\mathfrak{su}(2)$ defined by the commutation relations
cardona-qft-methods_ F02859	269	■ 1.000	$s=\frac{1}{2}$	$s = \frac{1}{2}$	In the case $s = \frac{1}{2}$ the ladder operators take the form
cardona-qft-methods_ F02860	269	■ 1.000	σ^3	σ^3	A convenient basis for $\mathbb{C}^2$ that diagonalizes σ^3 is
cardona-qft-methods_ F02861	270	■ 1.000	S^+	S^+	From these relations the name ladder operator becomes obvious, S^+ raises
cardona-qft-methods_ F02862	270	■ 1.000	S^-	S^-	the spin and S^- lowers it. Usually $ \uparrow\rangle$ is called highest weight state and
cardona-qft-methods_ F02863	270	■ 1.000	2^L	2^L	with total dimension 2^L . An element of this space in the introduced basis
cardona-qft-methods_ F02864	270	■ 1.000	$L=2$	$L = 2$	In the case $L = 2$ the Hilbert space is just $\mathbb{C}^2 \otimes \mathbb{C}^2$. The tensor product
cardona-qft-methods_ F02865	270	■ 1.000	$\mathbb{C}^2 \otimes \mathbb{C}^2$	$\mathbb{C}^2 \otimes \mathbb{C}^2$	In the case $L = 2$ the Hilbert space is just $\mathbb{C}^2 \otimes \mathbb{C}^2$. The tensor product
cardona-qft-methods_ F02866	271	■ 1.000	$L(L \sim 10)$	$L(L \sim 10)$	trivial. For larger and larger L ($L \sim 10$) it becomes more and more in-
cardona-qft-methods_ F02867	271	■ 1.000	$2^L \times 2^L$	$2^L \times 2^L$	Hamiltonian is $2^L \times 2^L$.
cardona-qft-methods_ F02868	271	■ 0.873	l^{th}	l^{th}	is the permutation operator acting on the l^{th} and $l+1^{\text{th}}$ space. The Hamil-
cardona-qft-methods_ F02869	271	■ 0.873	$l+1^{\text{th}}$	$l+1^{\text{th}}$	is the permutation operator acting on the l^{th} and $l+1^{\text{th}}$ space. The Hamil-
cardona-qft-methods_ F02870	271	■ 0.953	$\vec{S}_{\text{tot}}$	$\vec{S}_{\text{tot}}$	where $\vec{S}_{\text{tot}}$ is the total spin operator $\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$ and $\vec{S}_l$ is the local spin
cardona-qft-methods_ F02871	271	■ 0.953	$\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$	$\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$	where $\vec{S}_{\text{tot}}$ is the total spin operator $\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$ and $\vec{S}_l$ is the local spin
cardona-qft-methods_ F02872	271	■ 0.953	$\vec{S}_l$	$\vec{S}_l$	where $\vec{S}_{\text{tot}}$ is the total spin operator $\vec{S}_{\text{tot}} = \sum_{l=1}^L \vec{S}_l$ and $\vec{S}_l$ is the local spin
cardona-qft-methods_ F02873	272	■ 0.985	$L+1$	$L+1$	is block-diagonal. In total it contains $L+1$ blocks of size $\binom{L}{M} \times \binom{L}{M}$, where

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cardona-qft-methods_ F02874	272	■ 0.985	$\binom{L}{M} \times \binom{L}{M}$	$\binom{L}{M} \times \binom{L}{M}$	is block-diagonal. In total it contains $L + 1$ blocks of size $\binom{L}{M} \times \binom{L}{M}$, where
cardona-qft-methods_ F02875	272	■ 1.000	$[\hat{H}, \hat{U}] = 0$	$[\hat{H}, \hat{U}] = 0$	case of the closed chain, the shift operator satisfies $[\hat{H}, \hat{U}] = 0$. If one acts
cardona-qft-methods_ F02876	272	■ 1.000	$\hat{U}$	$\hat{U}$	L -times with $\hat{U}$ on some state one obtains back the same state
cardona-qft-methods_ F02877	272	■ 1.000	$n=0,1, \ldots, L-1$	$n = 0, 1, \dots, L - 1$	where $n = 0, 1, \dots, L - 1$.
cardona-qft-methods_ F02878	272	■ 1.000	u_k	u_k	Here the complex variables u_k are called Bethe roots and one of them
cardona-qft-methods_ F02879	273	■ 0.883	l_i	l_i	where l_i labels the positions of the i^{th} down spins and $ 0\rangle = \uparrow \cdots \uparrow\rangle$ is the
cardona-qft-methods_ F02880	273	■ 0.883	$ 0\rangle = \uparrow \cdots \uparrow\rangle$	$ 0\rangle = \uparrow \cdots \uparrow\rangle$	where l_i labels the positions of the i^{th} down spins and $ 0\rangle = \uparrow \cdots \uparrow\rangle$ is the
cardona-qft-methods_ F02881	273	■ 1.000	S_n	S_n	where S_n is the permutation group of a set containing n elements. The
cardona-qft-methods_ F02882	273	■ 1.000	$A(\tau)$	$A(\tau)$	amplitude $A(\tau)$ only depends on the permutation and not on the positions
cardona-qft-methods_ F02883	273	■ 1.000	$\operatorname{sgn}(\tau)$	$\operatorname{sgn}(\tau)$	where $\operatorname{sgn}(\tau)$ gives the signature of the permutation τ . The corresponding
cardona-qft-methods_ F02884	274	■ 1.000	$\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$	$\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$	acting on three spaces $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$.
cardona-qft-methods_ F02885	274	■ 0.999	$L_{a,l}(u) = R_{a,l}\left(u - \frac{i}{2}\right)$	$L_{a,l}(u) = R_{a,l}\left(u - \frac{i}{2}\right)$	<i>quantum Lax operator</i> $L_{a,l}(u) = R_{a,l}(u - \frac{i}{2})$, which is just the R -matrix
cardona-qft-methods_ F02886	274	■ 0.876	8×8	8×8	^b This is an equation for 8×8 matrices and can be easily verified by hand (see also ¹⁵)!
cardona-qft-methods_ F02887	275	■ 1.000	$(\mathbb{C}^2)^{\otimes L}$	$(\mathbb{C}^2)^{\otimes L}$	This transfer matrix is an operator acting on the tensor product $(\mathbb{C}^2)^{\otimes L}$ of
cardona-qft-methods_ F02888	275	■ 1.000	$1, \ldots, L$	$1, \dots, L$	the quantum spaces $1, \dots, L$, which is the Hilbert space of the whole spin
cardona-qft-methods_ F02889	275	■ 1.000	$1, 2, \ldots, L$	$1, 2, \dots, L$	one just has to move the lines $1, 2, \dots, L$ through the intersection of lines a
cardona-qft-methods_ F02890	276	■ 1.000	$R_{a,b}(u - u')$	$R_{a,b}(u - u')$	and b . After multiplying this equation with the inverse of $R_{a,b}(u - u')$ from
cardona-qft-methods_ F02891	276	■ 1.000	$u = \frac{i}{2}$	$u = \frac{i}{2}$	that at the special point $u = \frac{i}{2}$ of the spectral parameter the Lax operator
cardona-qft-methods_ F02892	276	■ 1.000	$L_{a,l}\left(\frac{i}{2}\right) = iP_{a,l}$	$L_{a,l}\left(\frac{i}{2}\right) = iP_{a,l}$	reduces essentially to a permutation: $L_{a,l}(\frac{i}{2}) = iP_{a,l}$. Using this observation
cardona-qft-methods_ F02893	276	■ 1.000	$\hat{P}$	$\hat{P}$	total momentum $\hat{P}$. Next we compute
cardona-qft-methods_ F02894	276	■ 1.000	$[\hat{T}(u), \hat{T}(u')] = 0$	$[\hat{T}(u), \hat{T}(u')] = 0$	From $[\hat{T}(u), \hat{T}(u')] = 0$, we conclude that all charges have to commute
cardona-qft-methods_ F02895	276	■ 1.000	$\{\hat{Q}_n\}$	$\{\hat{Q}_n\}$	commuting charges. Because the Hamiltonian is among the $\{\hat{Q}_n\}$, we also
cardona-qft-methods_ F02896	277	■ 1.000	$\hat{H} \psi\rangle = E \psi\rangle$	$\hat{H} \psi\rangle = E \psi\rangle$	i.e. a method to actually diagonalize the Hamiltonian: $\hat{H} \psi\rangle = E \psi\rangle$. This
cardona-qft-methods_ F02897	277	■ 1.000	$ \psi\rangle$	$ \psi\rangle$	To construct the eigenvectors $ \psi\rangle$ we look at the structure of the monodromy
cardona-qft-methods_ F02898	277	■ 1.000	$\hat{A}(u), \hat{B}(u), \hat{C}(u)$	$\hat{A}(u), \hat{B}(u), \hat{C}(u)$	where the matrix elements $\hat{A}(u)$, $\hat{B}(u)$, $\hat{C}(u)$ and $\hat{D}(u)$ are operators acting
cardona-qft-methods_ F02899	277	■ 1.000	$\hat{D}(u)$	$\hat{D}(u)$	where the matrix elements $\hat{A}(u)$, $\hat{B}(u)$, $\hat{C}(u)$ and $\hat{D}(u)$ are operators acting

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cardona-qft-methods_F02900	277	■ 1.000	$\hat{C}(u) 0\rangle=0$	$\hat{C}(u) 0\rangle=0$	the vacuum state $ 0\rangle= \uparrow\cdots\uparrow\rangle$, which satisfies $\hat{C}(u) 0\rangle=0$. Then one can
cardona-qft-methods_F02901	277	■ 0.960	$\left\{u_k\right\}$	$\{u_k\}$	yields eigenvectors of the transfer matrix, if the <i>Bethe roots</i> $\{u_k\}$ satisfy
cardona-qft-methods_F02902	277	■ 1.000	$\hat{T}(u)$	$\hat{T}(u)$	The corresponding eigenvalue of the transfer matrix $\hat{T}(u)$ is
cardona-qft-methods_F02903	277	■ 0.894	$T(u)$	$T(u)$	By construction $\hat{T}(u)$ (and thus also its eigenvalues $T(u)$) is a poly-
cardona-qft-methods_F02904	277	■ 1.000	$u=u_k$	$u=u_k$	$u=u_k$ have to vanish. Demanding $\text{res}_{u=u_k}T(u)=0$ yields again the Bethe
cardona-qft-methods_F02905	277	■ 1.000	$\operatorname{res}_{u=u_k}T(u)=0$	$\text{res}_{u=u_k}T(u)=0$	$u=u_k$ have to vanish. Demanding $\text{res}_{u=u_k}T(u)=0$ yields again the Bethe
cardona-qft-methods_F02906	278	■ 0.706	TQ	TQ	This function satisfies the so-called <i>TQ-relation</i>
cardona-qft-methods_F02907	278	■ 1.000	$\operatorname{SU}(N)$	$\text{SU}(N)$	local $\text{SU}(N)$ gauge theory with the maximal amount of supersymmetry in
cardona-qft-methods_F02908	278	■ 1.000	β	β	language of quantum field theory, this means that its β -function vanishes
cardona-qft-methods_F02909	278	■ 1.000	$Y_{\mu}=\sum_{a=1}^{\left(N^2-1\right)}Y_{\mu}^aT_a$	$Y_{\mu}=\sum_{a=1}^{\left(N^2-1\right)}Y_{\mu}^aT_a$	be expanded as $Y_{\mu}=\sum_{a=1}^{\left(N^2-1\right)}Y_{\mu}^aT_a$, where $\mu\in\{0,1,2,3\}$ (if present) is a
cardona-qft-methods_F02910	278	■ 1.000	$\mu\in\{0,1,2,3\}$	$\mu\in\{0,1,2,3\}$	be expanded as $Y_{\mu}=\sum_{a=1}^{\left(N^2-1\right)}Y_{\mu}^aT_a$, where $\mu\in\{0,1,2,3\}$ (if present) is a
cardona-qft-methods_F02911	278	■ 1.000	T_a	T_a	generic space-time index and T_a are the generators of the symmetry algebra
cardona-qft-methods_F02912	278	■ 1.000	$N\times N$	$N\times N$	in the adjoint representation. They are $N\times N$ matrices that fulfil the usual
cardona-qft-methods_F02913	278	■ 1.000	$\left[T_a,T_b\right]=if_{abc}T_c$	$[T_a,T_b]=if_{abc}T_c$	Lie algebra relation $[T_a,T_b]=if_{abc}T_c$. Under gauge transformations U ,
cardona-qft-methods_F02914	278	■ 0.954	$Y\rightarrow UYU^{-1}$	$Y\rightarrow UYU^{-1}$	fermions and scalars transform as $Y\rightarrow UYU^{-1}$, where $U\equiv U(x)\in\text{SU}(N)$
cardona-qft-methods_F02915	278	■ 0.954	$U\equiv U(x)\in\operatorname{SU}(N)$	$U\equiv U(x)\in\text{SU}(N)$	fermions and scalars transform as $Y\rightarrow UYU^{-1}$, where $U\equiv U(x)\in\text{SU}(N)$
cardona-qft-methods_F02916	279	■ 1.000	$x\in\mathbb{R}^{1,3}$	$x\in\mathbb{R}^{1,3}$	with $x\in\mathbb{R}^{1,3}$. The gauge field, A_{μ} , transforms as $A_{\mu}\rightarrow UA_{\mu}U^{-1}-i(\partial_{\mu}U)$.
cardona-qft-methods_F02917	279	■ 1.000	$A_{\mu}\rightarrow UA_{\mu}U^{-1}-i\left(\partial_{\mu}U\right)$	$A_{\mu}\rightarrow UA_{\mu}U^{-1}-i(\partial_{\mu}U)$	with $x\in\mathbb{R}^{1,3}$. The gauge field, A_{μ} , transforms as $A_{\mu}\rightarrow UA_{\mu}U^{-1}-i(\partial_{\mu}U)$.
cardona-qft-methods_F02918	279	■ 1.000	U^{-1}	U^{-1}	U^{-1} . This transformation ensures the gauge invariance of the theory. The
cardona-qft-methods_F02919	279	■ 1.000	φ_m	φ_m	• 6 massless, real scalars, φ_m with $m\in\{1,...,6\}$ with classical mass-
cardona-qft-methods_F02920	279	■ 1.000	$m\in\{1,\ldots,6\}$	$m\in\{1,\dots,6\}$	• 6 massless, real scalars, φ_m with $m\in\{1,...,6\}$ with classical mass-
cardona-qft-methods_F02921	279	■ 1.000	$\Delta_0=1$	$\Delta_0=1$	dimension $\Delta_0=1$. Equivalently, these may be represented by 3 com-
cardona-qft-methods_F02922	279	■ 1.000	$X=\varphi_1+i\varphi_2, Y=\varphi_3+i\varphi_4, Z=\varphi_5+i\varphi_6$	$X=\varphi_1+i\varphi_2, Y=\varphi_3+i\varphi_4, Z=\varphi_5+i\varphi_6$	plex scalars: $X=\varphi_1+i\varphi_2$, $Y=\varphi_3+i\varphi_4$, $Z=\varphi_5+i\varphi_6$ and their
cardona-qft-methods_F02923	279	■ 1.000	$\bar{X},\bar{Y},\bar{Z}$	$\bar{X},\bar{Y},\bar{Z}$	complex conjugates, denoted by $\bar{X},\bar{Y},\bar{Z}$. Yet another, important, no-
cardona-qft-methods_F02924	279	■ 1.000	φ_{ab}	φ_{ab}	tation is to express the scalars as antisymmetric bi-spinors φ_{ab} , where
cardona-qft-methods_F02925	279	■ 1.000	$a,b\in\{1,\ldots,\mathcal{N}=4\}$	$a,b\in\{1,\dots,\mathcal{N}=4\}$	$a,b\in\{1,...,\mathcal{N}=4\}$.

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cardona-qft-methods_ F02926	279	■ 1.000	$\psi_{a\alpha}$	$\psi_{a\alpha}$	• 4 massless Weyl-fermions, $\psi_{a\alpha}$, and their 4 conjugates $\bar{\psi}_{\dot{\alpha}}^a$. Here $a \in$
cardona-qft-methods_ F02927	279	■ 1.000	$\bar{\psi}_{\dot{\alpha}}^a$	$\bar{\psi}_{\dot{\alpha}}^a$	• 4 massless Weyl-fermions, $\psi_{a\alpha}$, and their 4 conjugates $\bar{\psi}_{\dot{\alpha}}^a$. Here $a \in$
cardona-qft-methods_ F02928	279	■ 1.000	$a \in$	$a \in$	• 4 massless Weyl-fermions, $\psi_{a\alpha}$, and their 4 conjugates $\bar{\psi}_{\dot{\alpha}}^a$. Here $a \in$
cardona-qft-methods_ F02929	279	■ 1.000	$\{1, \ldots \mathcal{N}=4\}$	$\{1, \dots \mathcal{N} = 4\}$	$\{1, \dots \mathcal{N} = 4\}$ labels the flavour of the (anti-) fermion and α or $\dot{\alpha}$ are
cardona-qft-methods_ F02930	279	■ 1.000	$\dot{\alpha}$	$\dot{\alpha}$	$\{1, \dots \mathcal{N} = 4\}$ labels the flavour of the (anti-) fermion and α or $\dot{\alpha}$ are
cardona-qft-methods_ F02931	279	■ 0.993	$\alpha \in \{1, 2\}$	$\alpha \in \{1, 2\}$	connected to their helicities. Concretely $\alpha \in \{1, 2\}$ labels the $\mathfrak{su}_L(2)$
cardona-qft-methods_ F02932	279	■ 0.993	$\mathfrak{su}_L(2)$	$\mathfrak{su}_L(2)$	connected to their helicities. Concretely $\alpha \in \{1, 2\}$ labels the $\mathfrak{su}_L(2)$
cardona-qft-methods_ F02933	279	■ 1.000	$\dot{\alpha} \in \{\dot{1}, \dot{2}\}$	$\dot{\alpha} \in \{\dot{1}, \dot{2}\}$	algebra and $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$ the $\mathfrak{su}_R(2)$ in the Lie algebra given by the iso-
cardona-qft-methods_ F02934	279	■ 1.000	$\mathfrak{su}_R(2)$	$\mathfrak{su}_R(2)$	algebra and $\dot{\alpha} \in \{\dot{1}, \dot{2}\}$ the $\mathfrak{su}_R(2)$ in the Lie algebra given by the iso-
cardona-qft-methods_ F02935	279	■ 1.000	$\mathfrak{so}(1, 3) \simeq$	$\mathfrak{so}(1, 3) \simeq$	morphism with the complexification of the Lorentz algebra, $\mathfrak{so}(1, 3) \simeq$
cardona-qft-methods_ F02936	279	■ 1.000	$\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$	$\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$	$\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$. The fermions have classical mass-dimension $\Delta_0 = 3/2$.
cardona-qft-methods_ F02937	279	■ 1.000	$\Delta_0 = 3/2$	$\Delta_0 = 3/2$	$\mathfrak{su}_L(2) \oplus \mathfrak{su}_R(2)$. The fermions have classical mass-dimension $\Delta_0 = 3/2$.
cardona-qft-methods_ F02938	279	■ 1.000	$F_{\mu\nu} = i[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]$	$F_{\mu\nu} = i[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]$	• 1 field strength tensor, $F_{\mu\nu} = i[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu]$,
cardona-qft-methods_ F02939	279	■ 1.000	$D_\mu = \partial_\mu - iA_\mu$	$D_\mu = \partial_\mu - iA_\mu$	with $D_\mu = \partial_\mu - iA_\mu$. The field strength has classical mass-dimensions
cardona-qft-methods_ F02940	279	■ 1.000	$\Delta_0 = 2$	$\Delta_0 = 2$	$\Delta_0 = 2$.
cardona-qft-methods_ F02941	279	■ 1.000	$D_\mu \star = \partial_\mu \star - i[A_\mu, \star]$	$D_\mu \star = \partial_\mu \star - i[A_\mu, \star]$	the adjoint representation as $D_\mu \star = \partial_\mu \star - i[A_\mu, \star]$. The classical mass-
cardona-qft-methods_ F02942	279	■ 1.000	$\lambda = Ng_{YM}^2$	$\lambda = Ng_{YM}^2$	Here $\lambda = Ng_{YM}^2$ is the 't Hooft coupling and $(\Sigma_m)^{ab}$ is the 6-dimensional
cardona-qft-methods_ F02943	279	■ 1.000	$(\Sigma_m)^{ab}$	$(\Sigma_m)^{ab}$	Here $\lambda = Ng_{YM}^2$ is the 't Hooft coupling and $(\Sigma_m)^{ab}$ is the 6-dimensional
cardona-qft-methods_ F02944	279	■ 0.523	$(\sigma_\mu)^{\alpha\dot{\beta}}$	$(\sigma_\mu)^{\alpha\dot{\beta}}$	analogon to the 4-dimensional $(\sigma_\mu)^{\alpha\dot{\beta}}$ -symbol. The free parameters of this
cardona-qft-methods_ F02945	279	■ 1.000	N^{-1}	N^{-1}	action are λ and N^{-1} .
cardona-qft-methods_ F02946	280	■ 1.000	$N \rightarrow \infty, g_{YM} \rightarrow 0$	$N \rightarrow \infty, g_{YM} \rightarrow 0$	$N \rightarrow \infty, g_{YM} \rightarrow 0$ while keeping λ fixed. In contrast to the total number
cardona-qft-methods_ F02947	280	■ 1.000	$Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$	$Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$	$Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$, with $i, j \in \{1, 2, \dots, N\}$.
cardona-qft-methods_ F02948	280	■ 1.000	$i, j \in \{1, 2, \dots, N\}$	$i, j \in \{1, 2, \dots, N\}$	$Y_\mu^{ij} = Y_\mu^a (T_a)^{ij}$, with $i, j \in \{1, 2, \dots, N\}$.
cardona-qft-methods_ F02949	280	■ 0.878	$\mathfrak{sl}(2, \mathbb{C}) \simeq \mathfrak{so}(1, 3)$	$\mathfrak{sl}(2, \mathbb{C}) \simeq \mathfrak{so}(1, 3)$	• Lorentz group: 3 rotations + 3 boosts, $\mathfrak{sl}(2, \mathbb{C}) \simeq \mathfrak{so}(1, 3)$, generators:
cardona-qft-methods_ F02950	280	■ 1.000	$M_{\mu\nu}$	$M_{\mu\nu}$	$M_{\mu\nu}$.
cardona-qft-methods_ F02951	280	■ 1.000	$\partial_\mu \rightarrow D_\mu$	$\partial_\mu \rightarrow D_\mu$	• 4 (covariant) translations, $\partial_\mu \rightarrow D_\mu$, generators: $P_\mu = iD_\mu$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image									
cardona-qft-methods_ F02952	280	■1.000	$P_{\mu}=i D_{\mu}$	$P_{\mu}=iD_{\mu}$	4 (covariant) translations, $\partial_{\mu} \rightarrow D_{\mu}$, generators: $P_{\mu}=iD_{\mu}$.									
cardona-qft-methods_ F02953	280	■1.000	$\left(M_{\mu\nu}, P_{\rho}\right)$	$(M_{\mu\nu}, P_{\rho})$	The two above algebras form the Poincaré algebra, $(M_{\mu\nu}, P_{\rho})$. With									
cardona-qft-methods_ F02954	280	■1.000	$\eta_{\mu\nu}=\operatorname{diag}(1,-1,-1,-1)$	$\eta_{\mu\nu}=\mathbf{diag}(1,-1,-1,-1)$	metric $\eta_{\mu\nu}=\mathbf{diag}(1,-1,-1,-1)$ it is given by the following symmetry									
cardona-qft-methods_ F02955	280	■1.000	K_{μ}	K_{μ}	tions, K_{μ} . To find these first define the inversion, $I: x_{\mu} \rightarrow x_{\mu}/x^2$. With									
cardona-qft-methods_ F02956	280	■1.000	$I: x_{\mu} \rightarrow x_{\mu} / x^2$	$I: x_{\mu} \rightarrow x_{\mu}/x^2$	tions, K_{μ} . To find these first define the inversion, $I: x_{\mu} \rightarrow x_{\mu}/x^2$. With									
cardona-qft-methods_ F02957	280	■1.000	$K_{\mu}=I P_{\mu} I$	$K_{\mu}=IP_{\mu}I$	this one can define $K_{\mu}=IP_{\mu}I$ and finds the remaining two non-trivial									
cardona-qft-methods_ F02958	281	■1.000	$6+4+4+1=15$	$6+4+4+1=15$	The combination of the $6+4+4+1=15$ symmetry generators listed									
cardona-qft-methods_ F02959	281	■1.000	$\mathfrak{so}(2,4) \simeq \mathfrak{su}(2,2)$	$\mathfrak{so}(2,4) \simeq \mathfrak{su}(2,2)$	above forms the <i>conformal algebra</i> , $\mathfrak{so}(2,4) \simeq \mathfrak{su}(2,2)$. Under the given set									
cardona-qft-methods_ F02960	281	■1.000	$\mathfrak{so}(6)$	$\mathfrak{so}(6)$	the $\mathfrak{so}(6)$ symmetry transforming the 6 scalars into each other,									
cardona-qft-methods_ F02961	281	■0.998	$\mathfrak{su}(4)$	$\mathfrak{su}(4)$	the $\mathfrak{su}(4)$ symmetry acting on the 4 fermionic flavours.									
cardona-qft-methods_ F02962	281	■0.907	2×2	2×2	the relatively simple $\mathfrak{su}(2)$ sector. Grouping its generators into a 2×2 table									
cardona-qft-methods_ F02963	281	■0.971	$\overline{\mathfrak{su}(2)}$	$\overline{\mathfrak{su}}(2)$	Now by gluing together a $\mathfrak{su}(2)$ and a $\overline{\mathfrak{su}}(2)$ one obtains the 4×4 table									
cardona-qft-methods_ F02964	281	■0.574	$p \ p \ p \ p$	$pppp$	<table><tr><td>L</td><td>$p \ p$</td></tr><tr><td>$k \ k$</td><td>$\bar{L}$</td></tr></table> ,	L	$p \ p$	$k \ k$	$\bar{L}$					
L	$p \ p$													
$k \ k$	$\bar{L}$													
cardona-qft-methods_ F02965	281	■0.574	$k \ k$	kk	<table><tr><td>L</td><td>$p \ p$</td></tr><tr><td>$k \ k$</td><td>$\bar{L}$</td></tr></table> ,	L	$p \ p$	$k \ k$	$\bar{L}$					
L	$p \ p$													
$k \ k$	$\bar{L}$													
cardona-qft-methods_ F02966	281	■0.574	$\bar{L}$	$\bar{L}$	<table><tr><td>L</td><td>$p \ p$</td></tr><tr><td>$k \ k$</td><td>$\bar{L}$</td></tr></table> ,	L	$p \ p$	$k \ k$	$\bar{L}$					
L	$p \ p$													
$k \ k$	$\bar{L}$													
cardona-qft-methods_ F02967	281	■0.967	$\mathfrak{su}(2,2)$	$\mathfrak{su}(2,2)$	incorporates the generators of a $\mathfrak{su}(2,2)$.									
cardona-qft-methods_ F02968	281	■0.999	$\mathfrak{psu}(2,2 \mid 4)$	$\mathfrak{psu}(2,2 \mid 4)$	arrives at the $\mathfrak{psu}(2,2 4)$ algebra, expressed in the the 8×8 table									
cardona-qft-methods_ F02969	281	■0.887	$\bar{Q}$	$\bar{Q}$	<table><tr><td>L</td><td>P</td><td>Q</td></tr><tr><td>K</td><td>$\bar{L}$</td><td>$\bar{S}$</td></tr><tr><td>S</td><td>$\bar{Q}$</td><td>R</td></tr></table> .	L	P	Q	K	$\bar{L}$	$\bar{S}$	S	$\bar{Q}$	R
L	P	Q												
K	$\bar{L}$	$\bar{S}$												
S	$\bar{Q}$	R												
cardona-qft-methods_ F02970	282	■1.000	$Q_{a \ \alpha}$	$Q_{a\alpha}$	action Eq. (1) is invariant under it. In the above table $Q_{a\alpha}$ and $\bar{Q}_{\dot{\alpha}}^a$ (with									
cardona-qft-methods_ F02971	282	■1.000	$\bar{Q}_{\dot{\alpha}}^a$	$\bar{Q}_{\dot{\alpha}}^a$	action Eq. (1) is invariant under it. In the above table $Q_{a\alpha}$ and $\bar{Q}_{\dot{\alpha}}^a$ (with									

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cardona-qft-methods_ F02972	282	■ 1.000	$a \in \{1, \dots, \mathcal{N}=4\}$	$a \in \{1, \dots, \mathcal{N} = 4\}$	$a \in \{1, \dots, \mathcal{N} = 4\}$ are the Poincaré supercharges, fulfilling the (anti-)
cardona-qft-methods_ F02973	282	■ 1.000	$\{ \}^{7,18,19}$	7,18,19	For further reading consider for example. ^{7,18,19}
cardona-qft-methods_ F02974	282	■ 0.998	$\mathcal{N}=4$ S Y M	$\mathcal{N} = 4SYM$	$\mathcal{N} = 4$ <i>SYM</i>
cardona-qft-methods_ F02975	282	■ 0.999	$\mathcal{O}$	$\mathcal{O}$	At the origin the action of the dilatation operator on $\mathcal{O}$ takes the form of
cardona-qft-methods_ F02976	282	■ 1.000	$\Delta=\Delta_0+\gamma$	$\Delta = \Delta_0 + \gamma$	where $\Delta = \Delta_0 + \gamma$ now consists of the classical scaling-dimension, Δ_0 and
cardona-qft-methods_ F02977	282	■ 1.000	$\mathcal{O}(x)$	$\mathcal{O}(x)$	classical scaling-dimensions of the constituent field in $\mathcal{O}(x)$, the anomalous
cardona-qft-methods_ F02978	282	■ 1.000	$\mathcal{N} =$	$\mathcal{N} =$	With the notion of composite operators the spectral problem of $\mathcal{N} =$
cardona-qft-methods_ F02979	283	■ 1.000	$N=\infty$	$N = \infty$	within $\mathcal{O}$ correspond to the spins. In the $N = \infty$ limit the fusion and fission
cardona-qft-methods_ F02980	283	■ 1.000	$\mathcal{O} =$	$\mathcal{O} =$	The simplest example of such a correlator is given by the operator $\mathcal{O} =$
cardona-qft-methods_ F02981	283	■ 1.000	$\operatorname{Tr}\left(Z^j\right)$	$\operatorname{Tr}\left(Z^j\right)$	$\operatorname{Tr}(Z^j)$. This operator corresponds to a Heisenberg spin chain with all spins
cardona-qft-methods_ F02982	283	■ 1.000	$\left \uparrow^j\right\rangle$	$ \uparrow^j\rangle$	pointing in the same direction, namely $ \uparrow^j\rangle$. In this case $\mathcal{O}$ is an BPS state
cardona-qft-methods_ F02983	283	■ 1.000	$\Delta_0=j$	$\Delta_0 = j$	and thus one has $\Delta_0 = j$ and $\gamma = 0$.
cardona-qft-methods_ F02984	283	■ 1.000	$\gamma=0$	$\gamma = 0$	and thus one has $\Delta_0 = j$ and $\gamma = 0$.
cardona-qft-methods_ F02985	283	■ 1.000	$\mathcal{O}=\operatorname{Tr}\left(Z^{j-1} X\right)$	$\mathcal{O} = \operatorname{Tr}\left(Z^{j-1} X\right)$	For a non-trivial example take the operator $\mathcal{O} = \operatorname{Tr}(Z^{j-1} X)$. this situation
cardona-qft-methods_ F02986	283	■ 1.000	$\left \uparrow^{j-1} \downarrow\right\rangle$	$ \uparrow^{j-1}\downarrow\rangle$	one spin is flipped down, $ \uparrow^{j-1}\downarrow\rangle$. The solution to the spectral problem of
cardona-qft-methods_ F02987	283	■ 0.879	$\boldsymbol{A} \boldsymbol{S}_5 \times \boldsymbol{S}^5$	$AdS_5 \times S^5$	6. IIB Superstring Theory on $AdS_5 \times S^5$
cardona-qft-methods_ F02988	284	■ 1.000	AdS_5	AdS_5	which is constructed by the direct product of $AdS_5 \times S^5$. Where AdS_5 is the
cardona-qft-methods_ F02989	284	■ 1.000	S_5	S_5	Lorentzian hyperboloid Anti de Sitter and S_5 is a compact space. The inner
cardona-qft-methods_ F02990	284	■ 0.985	$(-,+,+,+,-)$	$(-,+,+,+,+,-)$	products in these spaces are characterized by the metrics $(-,+,+,+,+,-)$
cardona-qft-methods_ F02991	284	■ 0.535	$(+,+,+,+,+)$	$(+,+,+,+,+,+)$	and $(+,+,+,+,+,+)$, or more explicitly:
cardona-qft-methods_ F02992	284	■ 0.562	$AdS_5=\mathrm{SO}(4,2) / \mathrm{SO}(1,4)$	$AdS_5 = \mathrm{SO}(4,2)/\mathrm{SO}(1,4)$	struction the background space can be viewed as $AdS_5 = \mathrm{SO}(4,2)/\mathrm{SO}(1,4)$
cardona-qft-methods_ F02993	284	■ 0.718	$S^5=\operatorname{SO}(6) / \operatorname{SO}(5)$	$S^5 = \mathrm{SO}(6)/\mathrm{SO}(5)$	and $S^5 = \mathrm{SO}(6)/\mathrm{SO}(5)$. With this one can construct
cardona-qft-methods_ F02994	284	■ 0.935	$\operatorname{PSU}(2,2 \mid 4)$	$\operatorname{PSU}(2,2 \mid 4)$	where the tilde signifies an infinite cover. $\operatorname{PSU}(2,2 \mid 4)$ is the super-isometry
cardona-qft-methods_ F02995	284	■ 1.000	$M \in\{0,1, \ldots, 5\}, N \in\{1, \ldots, 6\}$	$M \in \{0,1, \dots, 5\}, N \in \{1, \dots, 6\}$	with $M \in \{0,1, \dots, 5\}, N \in \{1, \dots, 6\}$, the radius R of AdS_5 and S^5 and with
cardona-qft-methods_ F02996	284	■ 1.000	$1 / \alpha^{\prime}$	$1/\alpha'$	the string tension $1/\alpha'$. The partial derivatives act on the worldsheet with
cardona-qft-methods_ F02997	284	■ 0.918	$(+,+)$	$(+,+)$	metric $(+,+)$ and are identified as $\partial_0 = \partial_\tau, \partial_1 = \partial_\sigma$. The string tension
cardona-qft-methods_ F02998	284	■ 0.918	$\partial_0=\partial_\tau, \partial_1=\partial_\sigma$	$\partial_0 = \partial_\tau, \partial_1 = \partial_\sigma$	metric $(+,+)$ and are identified as $\partial_0 = \partial_\tau, \partial_1 = \partial_\sigma$. The string tension

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cardona-qft-methods_ F02999	284	■ 1.000	$\alpha^{\{\prime\}}$	α'	effectively influences the size of appearing strings. For small α' the tension						
cardona-qft-methods_ F03000	284	■ 0.999	$\alpha^{\{\prime\}} \rightarrow 0$	$\alpha' \rightarrow 0$	is large and thus strings are short and vice versa. In the limit $\alpha' \rightarrow 0$ strings						
cardona-qft-methods_ F03001	285	■ 0.848	$\mathcal{N}=4 \mathrm{SYM}$	$\mathcal{N} = 4\mathrm{SYM}$	theory on an $AdS_5 \times S^5$ background is strictly equivalent to $\mathcal{N} = 4 \mathrm{SYM}$, ²						
cardona-qft-methods_ F03002	285	■ 1.000	$\{ \ }^{\{3,4\}}$	3,4	see e.g. ^{3,4} for explanations accessible to students. Note that this means a						
cardona-qft-methods_ F03003	285	■ 0.996	$\lambda=g_{\{\mathrm{YM}\}} \ N^{\{2\}}$	$\lambda = g_{\mathrm{YM}} N^2$	is the 't Hooft coupling constant $\lambda = g_{\mathrm{YM}} N^2$, where $N - 1$ is the rank of the						
cardona-qft-methods_ F03004	285	■ 0.996	$N-1$	$N - 1$	is the 't Hooft coupling constant $\lambda = g_{\mathrm{YM}} N^2$, where $N - 1$ is the rank of the						
cardona-qft-methods_ F03005	285	■ 0.915	$g_{\{\mathrm{YM}\}}$	g_{YM}	gauge group $SU(N)$, and g_{YM} the coupling constant. On the string theory						
cardona-qft-methods_ F03006	285	■ 1.000	$g_{\{\mathrm{s}\}}$	g_s	g_s . The AdS/CFT correspondence now states that theses parameters are						
cardona-qft-methods_ F03007	285	■ 1.000	$N \rightarrow \infty$	$N \rightarrow \infty$	the planar limit the degree of the gauge group goes to infinity, $N \rightarrow \infty$,						
cardona-qft-methods_ F03008	285	■ 1.000	$g_{\{\mathrm{s}\}} \rightarrow 0$	$g_s \rightarrow 0$	which means that the string coupling goes to zero, $g_s \rightarrow 0$. So the planar						
cardona-qft-methods_ F03009	286	■ 1.000	$\phi_{\{1\}}, \ \phi_{\{2\}}, \ \phi_{\{3\}}$	ϕ_1, ϕ_2, ϕ_3	the SYM case. The three angles ϕ_1, ϕ_2, ϕ_3 , are associated with the string						
cardona-qft-methods_ F03010	286	■ 1.000	$J_{\{1\}}, \ J_{\{2\}}, \ J_{\{3\}}$	J_1, J_2, J_3	angular momenta J_1, J_2, J_3 and correspond to the 3 <i>Cartan generators</i> of						
cardona-qft-methods_ F03011	286	■ 0.852	$\frac{s}{\mathrm{o}(6)} \ \simeq \ \frac{s}{\mathrm{u}(4)}$	$\mathfrak{so}(6) \simeq \mathfrak{su}(4)$	$\mathfrak{so}(6) \simeq \mathfrak{su}(4)$ (generate rotations around three orthogonal equators). For						
cardona-qft-methods_ F03012	286	■ 1.000	$t, \ \alpha_{\{1\}}, \ \alpha_{\{2\}}$	t, α_1, α_2	This time the three angles t, α_1, α_2 give the three conserved quantities E ,						
cardona-qft-methods_ F03013	286	■ 1.000	$S_{\{1\}}$	S_1	S_1 and S_2 and they come from the three <i>Cartan generators</i> of $\mathfrak{so}(2,4) \simeq$						
cardona-qft-methods_ F03014	286	■ 1.000	$\frac{s}{\mathrm{o}(2,4)} \ \simeq$	$\mathfrak{so}(2,4) \simeq$	S_1 and S_2 and they come from the three <i>Cartan generators</i> of $\mathfrak{so}(2,4) \simeq$						
cardona-qft-methods_ F03015	286	■ 0.474	$k \ k \ k$	kkk	<table><tr><td>L</td><td>$p \ p$</td></tr><tr><td>$k \ k$</td><td>$p \ p$</td></tr><tr><td>$k \ k$</td><td>$\bar{L}$</td></tr></table> .	L	$p \ p$	$k \ k$	$p \ p$	$k \ k$	$\bar{L}$
L	$p \ p$										
$k \ k$	$p \ p$										
$k \ k$	$\bar{L}$										
cardona-qft-methods_ F03016	286	■ 0.954	$30 \ \mathrm{mid} \ 32$	$30 \mid 32$	tum integrable at non-zero λ . The AdS/CFT scattering picture has 30 32						
cardona-qft-methods_ F03017	287	■ 0.514	$\frac{s}{\mathrm{u}(2 \ \mathrm{mid} \ 2)}$	$\mathfrak{su}(2 \mid 2)$	<table><tr><td>$\mathfrak{su}(2 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy $8 + 8$ magnons</td><td>$\overline{\mathfrak{su}(2 2)}$</td></tr></table>	$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons	destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$		
$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons										
destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$										
cardona-qft-methods_ F03018	287	■ 0.514	$8_{\{\mathrm{B}\}}+8_{\{\mathrm{F}\}}$	$8_B + 8_F$	<table><tr><td>$\mathfrak{su}(2 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy $8 + 8$ magnons</td><td>$\overline{\mathfrak{su}(2 2)}$</td></tr></table>	$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons	destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$		
$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons										
destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$										

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cardona-qft-methods_ F03019	287	0.514	$\overline{\mathfrak{s} \mathfrak{u}(2 \mid 2)}$	$\overline{\mathfrak{su}(2 \mid 2)}$	<table><tr><td>$\mathfrak{su}(2 2)$</td><td>create $8_B + 8_F$ magnons</td></tr><tr><td>destroy $8 + 8$ magnons</td><td>$\overline{\mathfrak{su}(2 2)}$</td></tr></table>	$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons	destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$
$\mathfrak{su}(2 2)$	create $8_B + 8_F$ magnons								
destroy $8 + 8$ magnons	$\overline{\mathfrak{su}(2 2)}$								
cardona-qft-methods_ F03020	287	1.000	$\dot{A}$	$\dot{A}$	where A and $\dot{A}$ are fundamental super indices. Everything else are compos-				
cardona-qft-methods_ F03021	287	1.000	$\bar{Z} \sim X \bar{X} + Y \bar{Y}$	$\bar{Z} \sim X \bar{X} + Y \bar{Y}$	ite states, e.g.: $\bar{Z} \sim X \bar{X} + Y \bar{Y}$ or $F \sim DD$.				
cardona-qft-methods_ F03022	287	1.000	$F \sim D D$	$F \sim DD$	ite states, e.g.: $\bar{Z} \sim X \bar{X} + Y \bar{Y}$ or $F \sim DD$.				
cardona-qft-methods_ F03023	288	1.000	$\mathrm{AdS}_5 \times \mathrm{S}^5$	$\mathrm{AdS}_5 \times \mathrm{S}^5$	20. G. Arutyunov and S. Frolov, “Foundations of the $\mathrm{AdS}_5 \times \mathrm{S}^5$ Superstring. Part				
cardona-qft-methods_ F03024	289	0.747	$\left. \mathrm{ple}^1 - {}^4 \right)$	$\left. \mathrm{ple}^1 - {}^4 \right)$	ple^1 - ⁴). In order to extract any information about four-dimensional physics,				
cardona-qft-methods_ F03025	290	1.000	$(2, 0)$	$(2, 0)$	string on Calabi-Yau manifolds or on exact (2,0) backgrounds (some of the				
cardona-qft-methods_ F03026	290	0.991	$\{ \}^5 - \{ \}^7$	${}^5 - {}^7$	early constructions are ⁵⁻⁷), and exceptionally type I theory. With the intro-				
cardona-qft-methods_ F03027	290	0.422	$\{ \}^8 -, ^{13}$	${}^8 -, ^{13}$	references are ^{8-, 13} for a review and more references, see for example ¹⁴).				
cardona-qft-methods_ F03028	291	0.502	$\{ \}^{19-} \{ \}^{21}$	${}^{19-} {}^{21}$	a larger framework ^{19-, 21} and thereby finds a common language for the two				
cardona-qft-methods_ F03029	292	0.999	$(0 \leq \sigma \leq \pi$	$(0 \leq \sigma \leq \pi$	the “worldsheet”, labeled by the coordinates σ along the string ($0 \leq \sigma \leq \pi$				
cardona-qft-methods_ F03030	292	1.000	$0 \leq \sigma \leq 2 \pi$	$0 \leq \sigma \leq 2\pi$	for an open string and $0 \leq \sigma \leq 2\pi$ for a closed string with 0 and 2π				
cardona-qft-methods_ F03031	292	1.000	2π	2π	for an open string and $0 \leq \sigma \leq 2\pi$ for a closed string with 0 and 2π				
cardona-qft-methods_ F03032	292	1.000	$M=0, \ldots, D-1 ; \alpha=\tau, \sigma$	$M = 0, \dots, D - 1; \alpha = \tau, \sigma$	$M = 0, \dots, D - 1; \alpha = \tau, \sigma$, which just measures the area of the string				
cardona-qft-methods_ F03033	292	1.000	$X^{\{M\}}$	X^M	worldsheet inside D -dimensional spacetime. In this equation, X^M are the				
cardona-qft-methods_ F03034	292	1.000	$X^{\{\mu\}}(\tau, \sigma)$	$X^\mu(\tau, \sigma)$	Fig. 1. Embedding of an open string in spacetime by the function $X^\mu(\tau, \sigma)$.				
cardona-qft-methods_ F03035	292	1.000	$\gamma^{\alpha \beta}$	$\gamma^{\alpha \beta}$	1); $\gamma^{\alpha \beta}$ is the worldsheet metric, η_{MN} is the spacetime Minkowski metric				
cardona-qft-methods_ F03036	292	1.000	$\eta_{M N}$	η_{MN}	1); $\gamma^{\alpha \beta}$ is the worldsheet metric, η_{MN} is the spacetime Minkowski metric				
cardona-qft-methods_ F03037	293	1.000	$l_{\{s\}}$	l_s	l_s , which has units of length, is called the “string scale”. Its inverse gives				
cardona-qft-methods_ F03038	293	0.999	$M_{\{s\}}$	M_s	the “string mass” M_s (we are using units in which $c = \hbar = 1$, to obtain a				
cardona-qft-methods_ F03039	293	0.999	$c=\hbar=1$	$c = \hbar = 1$	the “string mass” M_s (we are using units in which $c = \hbar = 1$, to obtain a				
cardona-qft-methods_ F03040	293	1.000	$\hbar / c$	$\hbar/c$	mass from the inverse of the string scale we should multiply by $\hbar/c$), which				
cardona-qft-methods_ F03041	293	1.000	$\sigma^{\pm}=\tau \pm \sigma$	$\sigma^\pm = \tau \pm \sigma$	By introducing left- and right-moving worldsheet coordinates $\sigma^\pm = \tau \pm \sigma$,				
cardona-qft-methods_ F03042	293	0.999	$X^{\{M\}}(\tau, 0)=X^{\{M\}}(\tau, 2 \pi), X^{\{\prime M\}}(\tau, 0)=X^{\{\prime M\}}(\tau, 2 \pi)$	$X^M(\tau, 0) = X^M(\tau, 2\pi), X'^M(\tau, 0) = X'^M(\tau, 2\pi)$	ary conditions $X^M(\tau, 0) = X^M(\tau, 2\pi), X'^M(\tau, 0) = X'^M(\tau, 2\pi)$ (with prime				

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cardona-qft-methods_F03043	293	■ 1.000	$x^{\{M\}}$	x^M	where x^M and p^M are the centre of mass position and momentum, respec-
cardona-qft-methods_F03044	293	■ 1.000	$p^{\{M\}}$	p^M	where x^M and p^M are the centre of mass position and momentum, respec-
cardona-qft-methods_F03045	293	■ 1.000	$\alpha_{-n}^{\{M\}}=\left(\alpha_n^{\{M\}}\right)^{*}$	$\alpha_{-n}^M = (\alpha_n^M)^*$	real solution we impose $\alpha_{-n}^M = (\alpha_n^M)^*$ and $\tilde{\alpha}_{-n}^M = (\tilde{\alpha}_n^M)^*$.
cardona-qft-methods_F03046	293	■ 1.000	$\tilde{\alpha}_{-n}^{\{M\}}=\left(\tilde{\alpha}_n^{\{M\}}\right)^{*}$	$\tilde{\alpha}_{-n}^M = (\tilde{\alpha}_n^M)^*$	real solution we impose $\alpha_{-n}^M = (\alpha_n^M)^*$ and $\tilde{\alpha}_{-n}^M = (\tilde{\alpha}_n^M)^*$.
cardona-qft-methods_F03047	293	■ 1.000	$\gamma_{\alpha \beta}$	$\gamma_{\alpha\beta}$	Varying (1) with respect to the worldsheet metric $\gamma_{\alpha\beta}$ gives the extra
cardona-qft-methods_F03048	294	■ 1.000	$\alpha_n^{\{M\}}, \tilde{\alpha}_n^{\{M\}}$	$\alpha_n^M, \tilde{\alpha}_n^M$	We can therefore interpret $\alpha_n^M, \tilde{\alpha}_n^M$ as creation operators and $\alpha_{-n}^M, \tilde{\alpha}_{-n}^M$ with
cardona-qft-methods_F03049	294	■ 1.000	$\alpha_{-n}^{\{M\}}, \tilde{\alpha}_{-n}^{\{M\}}$	$\alpha_{-n}^M, \tilde{\alpha}_{-n}^M$	We can therefore interpret $\alpha_n^M, \tilde{\alpha}_n^M$ as creation operators and $\alpha_{-n}^M, \tilde{\alpha}_{-n}^M$ with
cardona-qft-methods_F03050	294	■ 0.890	$\mathrm{in}^{\{1\}}-\{ \}^{\{3\}}$	$\mathrm{in}^1 -^3$	can be found, for example, in ¹⁻³) but only mention the most important of
cardona-qft-methods_F03051	294	■ 1.000	$N=\tilde{N}$	$N = \tilde{N}$	vanishes on the physical states. Thus, we have to impose $N = \tilde{N}$, where
cardona-qft-methods_F03052	294	■ 1.000	$N, \tilde{N}$	$N, \tilde{N}$	$N, \tilde{N}$ are the total sums of oscillator levels excited on the left and on the
cardona-qft-methods_F03053	294	■ 1.000	$k^{\{M\}}, k \cdot k=0$	$k^M, k \cdot k = 0$	and a center of mass momentum k^M , $k \cdot k = 0$. It is not hard to check
cardona-qft-methods_F03054	294	■ 1.000	$\xi, \tilde{\xi}$	$\xi, \tilde{\xi}$	that the norm of the state is positive only if $\xi, \tilde{\xi}$ are space-like vectors. The
cardona-qft-methods_F03055	294	■ 1.000	$\xi \cdot k=\tilde{\xi} \cdot k=0$	$\xi \cdot k = \tilde{\xi} \cdot k = 0$	classical conditions (8) impose $\xi \cdot k = \tilde{\xi} \cdot k = 0$, i.e., the polarization vectors
cardona-qft-methods_F03056	294	■ 0.999	$\vec{k}=(k, k, 0, \ldots, 0)$	$\vec{k} = (k, k, 0, \dots, 0)$	where $\vec{k} = (k, k, 0, \dots, 0)$, we get that $\xi_M, \tilde{\xi}_M$ belong to the $D-2$ -dimensional
cardona-qft-methods_F03057	294	■ 0.999	$\xi_{\{M\}}, \tilde{\xi}_{\{M\}}$	$\xi_M, \tilde{\xi}_M$	where $\vec{k} = (k, k, 0, \dots, 0)$, we get that $\xi_M, \tilde{\xi}_M$ belong to the $D-2$ -dimensional
cardona-qft-methods_F03058	294	■ 0.999	$2, \ldots, D-1$	$2, \dots, D-1$	space parametrized by the coordinates $2, \dots, D-1$. The states are therefore
cardona-qft-methods_F03059	294	■ 0.987	$\operatorname{SO}(D-2)$	$\mathrm{SO}(D-2)$	classified by their $\mathrm{SO}(D-2)$ representations. The tensor $\xi_{MN} \equiv \xi_M \tilde{\xi}_N$
cardona-qft-methods_F03060	294	■ 0.987	$\xi_{\{M\} N} \equiv \xi_{\{M\}} \tilde{\xi}_{\{N\}}$	$\xi_{MN} \equiv \xi_M \tilde{\xi}_N$	classified by their $\mathrm{SO}(D-2)$ representations. The tensor $\xi_{MN} \equiv \xi_M \tilde{\xi}_N$
cardona-qft-methods_F03061	295	■ 1.000	$\xi_{\{M\} N}$	ξ_{MN}	The state corresponding to the polarization ξ_{MN} is a massless state of spin
cardona-qft-methods_F03062	295	■ 1.000	$\xi_{\{t\}}$	ξ_t	2: the graviton. The state corresponding to the scalar ξ_t is the dilaton, while
cardona-qft-methods_F03063	295	■ 0.968	$-\frac{4}{\alpha'}$	$-\frac{4}{\alpha'}$	is $-\frac{4}{\alpha'}$). The appearance of this tachyon means that the bosonic string is
cardona-qft-methods_F03064	295	■ 1.000	$G_{\{M\} N}(X)$	$G_{MN}(X)$	field dependent couplings given by the spacetime metric $G_{MN}(X)$. The
cardona-qft-methods_F03065	295	■ 1.000	$G_{\{M\} N}=$	$G_{MN} =$	following sense: If we consider a small deviation from flat space, $G_{MN} =$
cardona-qft-methods_F03066	295	■ 1.000	$\eta_{\{M\} N}+h_{\{M\} N}(X)$	$\eta_{MN} + h_{MN}(X)$	$\eta_{MN} + h_{MN}(X)$, with h small, and expand the path integral ²⁶
cardona-qft-methods_F03067	295	■ 1.000	$h_{\{M\} N} \propto \xi_{\{M\} N}^s$	$h_{MN} \propto \xi_{MN}^s$	sion vertex operator with $h_{MN} \propto \xi_{MN}^s$. Thus, a background metric is
cardona-qft-methods_F03068	296	■ 1.000	$\hbar$	$\hbar$	dimensions. Note that α' in this action behaves like $\hbar$: the action is large

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cardona-qft-methods_ F03096	299	■ 1.000	$z=e^{\tau-\mathrm{i}\sigma}$	$z = e^{\tau-i\sigma}$	introduced the complex coordinates $z = e^{\tau-i\sigma}$. The equations of motion for
cardona-qft-methods_ F03097	299	■ 0.514	$\tilde{\Psi}$	$\tilde{\Psi}$	Ψ and $\tilde{\Psi}$ admit two possible boundary conditions, denoted “Ramond” (R)
cardona-qft-methods_ F03098	299	■ 0.997	$\tilde{\psi}$	$\tilde{\psi}$	and similarly for $\tilde{\psi}$. When expanding in modes as in (6), the first type of
cardona-qft-methods_ F03099	299	■ 1.000	$D=10$	$D = 10$	anomaly cancellation imposes a fixed dimension for spacetime: $D = 10$.
cardona-qft-methods_ F03100	300	■ 1.000	ξ_M	ξ_M	In the same way as for the bosonic string, the polarizations ξ_M have to be
cardona-qft-methods_ F03101	300	■ 1.000	$\vec{k}$	$\vec{k}$	orthogonal to the center of mass momentum $\vec{k}$, and are therefore classified
cardona-qft-methods_ F03102	300	■ 0.850	$\mathrm{SO}(8)$	$\mathrm{SO}(8)$	by $\mathrm{SO}(8)$ representations.
cardona-qft-methods_ F03103	300	■ 1.000	Ψ_0	Ψ_0	Since Ψ_0 obeys a Clifford algebra, we can form raising and lowering oper-
cardona-qft-methods_ F03104	300	■ 1.000	$d_{\pm}^i=\frac{1}{\sqrt{2}}(\psi_0^{2i}\pm\psi_0^{2i+1})$	$d_{\pm}^i = \frac{1}{\sqrt{2}}(\psi_0^{2i} \pm \psi_0^{2i+1})$	ators $d_{\pm}^i = \frac{1}{\sqrt{2}}(\psi_0^{2i} \pm \psi_0^{2i+1})$, with $i = 1, \dots, 4$. The Ramond ground states
cardona-qft-methods_ F03105	300	■ 1.000	$i=1, \dots, 4$	$i = 1, \dots, 4$	ators $d_{\pm}^i = \frac{1}{\sqrt{2}}(\psi_0^{2i} \pm \psi_0^{2i+1})$, with $i = 1, \dots, 4$. The Ramond ground states
cardona-qft-methods_ F03106	300	■ 0.922	$\pm \frac{1}{2}$	$\pm \frac{1}{2}$	where $\pm \frac{1}{2}$ are the chiralities in the four 2-dimensional planes i . ^c The state
cardona-qft-methods_ F03107	300	■ 0.922	i^c	i^c	where $\pm \frac{1}{2}$ are the chiralities in the four 2-dimensional planes i . ^c The state
cardona-qft-methods_ F03108	300	■ 0.994	$\psi_0^M 0\rangle$	$\psi_0^M 0\rangle$	$\psi_0^M 0\rangle$ is therefore a <i>spacetime</i> fermion, in the representation 16 . The 16
cardona-qft-methods_ F03109	300	■ 1.000	$\mathbf{8}_s \oplus \mathbf{8}_c$	$\mathbf{8}_s \oplus \mathbf{8}_c$	reduces into $\mathbf{8}_s \oplus \mathbf{8}_c$. In order to get spacetime supersymmetry, we need only
cardona-qft-methods_ F03110	300	■ 1.000	$8\,X^M$	$8X^M$	8 physical fermions (to be the superpartners of the $8\,X^M$). We therefore
cardona-qft-methods_ F03111	300	■ 1.000	$\sum_{i=1}^4 s_i=0(\bmod 2)$	$\sum_{i=1}^4 s_i = 0(\bmod 2)$	Olive ³⁰), which requires $\sum_{i=1}^4 \bar{s}_i = 0 \pmod{2}$. One can view the GSO
cardona-qft-methods_ F03112	300	■ 0.996	$\xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle$	$\xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle$	^c The RR massless state $\xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle$ should therefore be understood as $ s\rangle \otimes \bar{s}\rangle$.
cardona-qft-methods_ F03113	300	■ 0.996	$ s\rangle \otimes \tilde{s}\rangle$	$ s\rangle \otimes \tilde{s}\rangle$	^c The RR massless state $\xi_{MN}\psi_0^M\tilde{\psi}_0^N 0\rangle$ should therefore be understood as $ s\rangle \otimes \bar{s}\rangle$.
cardona-qft-methods_ F03114	301	■ 1.000	$\mathbf{8}_s$	$\mathbf{8}_s$	the spinors of positive and negative chirality by $\mathbf{8}_s$ and $\mathbf{8}_c$, respectively,
cardona-qft-methods_ F03115	301	■ 1.000	$\mathbf{8}_c$	$\mathbf{8}_c$	the spinors of positive and negative chirality by $\mathbf{8}_s$ and $\mathbf{8}_c$, respectively,
cardona-qft-methods_ F03116	301	■ 0.996	$\mathbf{8}_v$	$\mathbf{8}_v$	while the vector representation is called $\mathbf{8}_v$. Actually, there is a discrete
cardona-qft-methods_ F03117	301	■ 0.998	$\mathbf{8}_s, \mathbf{8}_c, \mathbf{8}_v$	$\mathbf{8}_s, \mathbf{8}_c, \mathbf{8}_v$	trio $\mathbf{8}_s, \mathbf{8}_c, \mathbf{8}_v$), and plays a very important role in string theory.
cardona-qft-methods_ F03118	301	■ 0.993	C_n	C_n	In this equation, we have defined C_n to be an n -form, called a R-R potential.
cardona-qft-methods_ F03119	301	■ 1.000	C_4	C_4	In type IIB there are even-degree R-R potentials (with C_4 satisfying a
cardona-qft-methods_ F03120	301	■ 1.000	$d\,C_4=*d\,C_4$	$dC_4 = *dC_4$	duality condition $dC_4 = *dC_4$). In type IIA, the R-R potentials are odd.
cardona-qft-methods_ F03121	301	■ 0.988	$\lambda^A(\mathrm{A}=1,2)$	$\lambda^A(\,A = 1, 2)$	The fermions in the $\mathbf{8}$ representation λ^A ($A=1,2$) are called the dilatini,

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cardona-qft-methods_ F03122	301	■ 0.984	$\mathbf{56}, \Psi^{A, M}$	56 , $\Psi^{A, M}$	and the ones in the 56 , $\Psi^{A, M}$ are the gravitini. The name “type II” refers
cardona-qft-methods_ F03123	302	■ 1.000	S^1	S^1	S^1 of radius R , the center of mass momentum for the string along that
cardona-qft-methods_ F03124	302	■ 1.000	$1 / R$	$1/R$	direction will be quantized, in units of $1/R$. On the other hand, there are
cardona-qft-methods_ F03125	302	■ 1.000	$X^M(\tau, 2\pi) =$	$X^M(\tau, 2\pi) =$	“winding states”, namely states with boundary conditions $X^M(\tau, 2\pi) =$
cardona-qft-methods_ F03126	302	■ 1.000	$X^M(\tau, 0) + 2\pi m R$	$X^M(\tau, 0) + 2\pi m R$	$X^M(\tau, 0) + 2\pi m R$. A string with n units of momentum, m units of winding
cardona-qft-methods_ F03127	302	■ 0.943	$\alpha'^{\prime} / R!$	$\alpha' / R!$	words winding and momentum states) if we also exchange R with $\alpha' / R!$
cardona-qft-methods_ F03128	303	■ 1.000	$X_L^M + X_R^M$	$X_L^M + X_R^M$	amounts to exchanging $X_L^M + X_R^M$ with $X_L^M - X_R^M$, and in the superstring
cardona-qft-methods_ F03129	303	■ 1.000	$X_L^M - X_R^M$	$X_L^M - X_R^M$	amounts to exchanging $X_L^M + X_R^M$ with $X_L^M - X_R^M$, and in the superstring
cardona-qft-methods_ F03130	303	■ 0.978	ψ^M	ψ^M	ψ^M with $\bar{\psi}^M$. This exchanges representations $\mathbf{8}_s$ with $\mathbf{8}_c$ and therefore the
cardona-qft-methods_ F03131	303	■ 0.978	$\bar{\psi}^M$	$\bar{\psi}^M$	ψ^M with $\bar{\psi}^M$. This exchanges representations $\mathbf{8}_s$ with $\mathbf{8}_c$ and therefore the
cardona-qft-methods_ F03132	303	■ 1.000	$O(n, n, \mathbb{Z})$	$O(n, n, \mathbb{Z})$	symmetry group $O(n, n, \mathbb{Z})$, the so-called T-duality group.
cardona-qft-methods_ F03133	303	■ 1.000	$O(8, 8)$	$O(8, 8)$	ric even under <i>continuous</i> transformations in the group $O(8, 8)$. Apart
cardona-qft-methods_ F03134	303	■ 0.960	$Gl(8)$	$Gl(8)$	from Lorentz-transformations in $Gl(8)$ that act equally on left- and right-
cardona-qft-methods_ F03135	303	■ 1.000	$O(8)$	$O(8)$	algebra for $O(8)$ on their own. Together, they combine into the Clifford
cardona-qft-methods_ F03136	303	■ 1.000	$\{ \}^{\text{e}}$	e	but is also locally a symmetry of their field equations in ten dimensions. ^e
cardona-qft-methods_ F03137	303	■ 0.995	$\mathrm{D}(p \pm 1)$	$D(p \pm 1)$	Dp -brane changes its dimension and becomes a $D(p \pm 1)$ -brane.
cardona-qft-methods_ F03138	303	■ 1.000	$O(10, 10)$	$O(10, 10)$	^e The symmetry group of the supergravity theory is in fact $O(10, 10)$ and thus larger, but
cardona-qft-methods_ F03139	304	■ 0.998	$T_p = 8\pi^7 \alpha'^{7/2} (4\pi^2 \alpha')^{-p/2}$	$T_p = 8\pi^7 \alpha'^{7/2} (4\pi^2 \alpha')^{-p/2}$	(where $T_p = 8\pi^7 \alpha'^{7/2} (4\pi^2 \alpha')^{-p/2}$ supplemented by the Chern-Simons term
cardona-qft-methods_ F03140	304	■ 0.989	$SU(3) \times SU(2) \times$	$SU(3) \times SU(2) \times$	group of the Standard Model of particle physics, namely $SU(3) \times SU(2) \times$
cardona-qft-methods_ F03141	305	■ 1.000	$O(n, n)$	$O(n, n)$	in Section 4 that it is often useful to consider the group $O(n, n)$, where n is
cardona-qft-methods_ F03142	305	■ 1.000	$\{ \}^{19, 20}$	19,20	geometry was first proposed by Hitchin and his students in ^{19,20} in order to
cardona-qft-methods_ F03143	305	■ 1.000	$\{ \}^{22, 23}$	22,23	being utilized to describe flux compactifications of string theory. ^{22,23} In this
cardona-qft-methods_ F03144	305	■ 0.998	T^*	T^*	bundle T^* , whose sections are 1-forms ζ . In generalized complex geometry
cardona-qft-methods_ F03145	305	■ 1.000	$T \oplus T^*$	$T \oplus T^*$	the tangent and cotangent bundle are joined as a single bundle, $T \oplus T^*$. Its
cardona-qft-methods_ F03146	305	■ 1.000	$X + \zeta$	$X + \zeta$	sections are the sum of a vector field plus a one-form $X + \zeta$. The bundle
cardona-qft-methods_ F03147	306	■ 1.000	$n=6$	$n = 6$	of dimension $n = 6$.
cardona-qft-methods_ F03148	306	■ 1.000	$O(6, 6)$	$O(6, 6)$	Of particular interest is the nilpotent subgroup of $O(6, 6)$ defined by the

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cardona-qft-methods_ F03149	306	■ 1.000	$U_{\{\alpha\}}$	U_{α}	In going from one coordinate patch U_{α} to another U_{β} , we do not only patch
cardona-qft-methods_ F03150	306	■ 1.000	$U_{\{\beta\}}$	U_{β}	In going from one coordinate patch U_{α} to another U_{β} , we do not only patch
cardona-qft-methods_ F03151	306	■ 0.460	$\mathrm{Gl}(6)$	$\mathrm{Gl}(6)$	vectors and one-forms by a diffeomorphism-induced $\mathrm{Gl}(6)$ -matrix $a_{(\alpha\beta)}$, but
cardona-qft-methods_ F03152	306	■ 0.460	$a_{(\{\alpha\} \{\beta\})}$	$a_{(\alpha\beta)}$	vectors and one-forms by a diffeomorphism-induced $\mathrm{Gl}(6)$ -matrix $a_{(\alpha\beta)}$, but
cardona-qft-methods_ F03153	306	■ 0.998	$b_{(\{\alpha\} \{\beta\})}$	$b_{(\alpha\beta)}$	also use such two-form shifts $b_{(\alpha\beta)}$ to fiber T^* over T . In total, the patching
cardona-qft-methods_ F03154	306	■ 1.000	$a^{-T}=\left(a^{-1}\right)^T$	$a^{-T}=\left(a^{-1}\right)^T$	where $a^{-T}=\left(a^{-1}\right)^T$. In fact, for consistency the two-form shift must satisfy
cardona-qft-methods_ F03155	306	■ 1.000	$b_{(\{\alpha\} \{\beta\})}=-\mathrm{d} \Lambda_{(\{\alpha\} \{\beta\})}$	$b_{(\alpha\beta)}=-\mathrm{d} \Lambda_{(\alpha\beta)}$	$b_{(\alpha\beta)}=-\mathrm{d} \Lambda_{(\alpha\beta)}$, where $\Lambda_{(\alpha\beta)}$ are required to satisfy
cardona-qft-methods_ F03156	306	■ 1.000	$\Lambda_{(\{\alpha\} \{\beta\})}$	$\Lambda_{(\alpha\beta)}$	$b_{(\alpha\beta)}=-\mathrm{d} \Lambda_{(\alpha\beta)}$, where $\Lambda_{(\alpha\beta)}$ are required to satisfy
cardona-qft-methods_ F03157	306	■ 1.000	$U_{\{\alpha\}} \cap U_{\{\beta\}} \cap U_{\{\gamma\}}$	$U_{\alpha} \cap U_{\beta} \cap U_{\gamma}$	on any triple intersection $U_{\alpha} \cap U_{\beta} \cap U_{\gamma}$ and $g_{\alpha\beta\gamma}:=\mathrm{i} \alpha$ is a $U(1)$ element.
cardona-qft-methods_ F03158	306	■ 1.000	$g_{\{\alpha\} \{\beta\} \{\gamma\}}:=\mathrm{i} \alpha$	$g_{\alpha\beta\gamma}:=\mathrm{i} \alpha$	on any triple intersection $U_{\alpha} \cap U_{\beta} \cap U_{\gamma}$ and $g_{\alpha\beta\gamma}:=\mathrm{i} \alpha$ is a $U(1)$ element.
cardona-qft-methods_ F03159	306	■ 1.000	$\mathcal{J}$	$\mathcal{J}$	structure $\mathcal{J}$, which is a map of E to itself that squares to $-\mathbb{I}_{2d}$ (d is real
cardona-qft-methods_ F03160	306	■ 1.000	$-\mathbb{I}_{2d}$	$-\mathbb{I}_{2d}$	structure $\mathcal{J}$, which is a map of E to itself that squares to $-\mathbb{I}_{2d}$ (d is real
cardona-qft-methods_ F03161	306	■ 1.000	$I_{\{m\}}^{\{n\}}$	$I_m^{\ n}$	structure $I_m^{\ n}$, which is a bundle map from T to itself that squares to $-\mathbb{I}_d$.
cardona-qft-methods_ F03162	306	■ 1.000	$-\mathbb{I}_{\{d\}}$	$-\mathbb{I}_d$	structure $I_m^{\ n}$, which is a bundle map from T to itself that squares to $-\mathbb{I}_d$.
cardona-qft-methods_ F03163	306	■ 1.000	$\mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$	$\mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$	condition $\mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$, with respect to the natural metric $\mathcal{G}$ on $T \oplus T^*$
cardona-qft-methods_ F03164	306	■ 1.000	$\mathcal{G}$	$\mathcal{G}$	condition $\mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$, with respect to the natural metric $\mathcal{G}$ on $T \oplus T^*$
cardona-qft-methods_ F03165	307	■ 1.000	$I^2=-\mathbb{I}_{\{d\}}$	$I^2=-\mathbb{I}_d$	with $I_m^{\ n}$ a regular almost complex structure (i.e. $I^2=-\mathbb{I}_d$). This $\mathcal{J}$ sat-
cardona-qft-methods_ F03166	307	■ 1.000	$\mathcal{J}^2=-\mathbb{I}_{2d}, \mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$	$\mathcal{J}^2=-\mathbb{I}_{2d}, \mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$	isfies the desired properties, namely $\mathcal{J}^2=-\mathbb{I}_{2d}, \mathcal{J}^t \mathcal{G} \mathcal{J}=\mathcal{G}$. Another ex-
cardona-qft-methods_ F03167	307	■ 1.000	$J_{\{m\} \{n\}}$	J_{mn}	degenerate two-form J_{mn} ,
cardona-qft-methods_ F03168	307	■ 1.000	$\pi_{\{\pm\}}=\frac{1}{2}\left(\mathbb{I}_{\{d\}} \pm \mathrm{i} I\right)$	$\pi_{\pm}=\frac{1}{2}\left(\mathbb{I}_d \pm \mathrm{i} I\right)$	and antiholomorphic projectors $\pi_{\pm}=\frac{1}{2}\left(\mathbb{I}_d \pm \mathrm{i} I\right)$. Correspondingly, projec-
cardona-qft-methods_ F03169	307	■ 1.000	$\Pi_{\{\pm\}}=$	$\Pi_{\pm}=$	tors can be built out of a generalized almost complex structure, $\Pi_{\pm}=$
cardona-qft-methods_ F03170	307	■ 1.000	$\frac{1}{2}\left(\mathbb{I}_{2d} \pm \mathrm{i} \mathcal{J}\right)$	$\frac{1}{2}\left(\mathbb{I}_{2d} \pm \mathrm{i} \mathcal{J}\right)$	$\frac{1}{2}\left(\mathbb{I}_{2d} \pm \mathrm{i} \mathcal{J}\right)$. There is an integrability condition for generalized almost com-
cardona-qft-methods_ F03171	307	■ 0.999	$\pi_{\{\mp\}}\left[\pi_{\{\pm\}} X, \pi_{\{\pm\}} Y\right]=0$	$\pi_{\mp}\left[\pi_{\pm} X, \pi_{\pm} Y\right]=0$	$\pi_{\mp}\left[\pi_{\pm} X, \pi_{\pm} Y\right]=0$, i.e. the Lie bracket of two holomorphic vectors should
cardona-qft-methods_ F03172	307	■ 1.000	Π	Π	by Π and $X+\zeta$, and the Lie bracket replaced by the Courant bracket on
cardona-qft-methods_ F03173	307	■ 1.000	$T M \oplus T^{*} M$	$T M \oplus T^{*} M$	$T M \oplus T^{*} M$, which is defined as follows
cardona-qft-methods_ F03174	307	■ 1.000	$\mathcal{J}_{\{1\}}$	$\mathcal{J}_1$	above, $\mathcal{J}_1$ and $\mathcal{J}_2$, integrability condition turns into a condition on their
cardona-qft-methods_ F03175	307	■ 1.000	$\mathcal{J}_{\{2\}}$	$\mathcal{J}_2$	above, $\mathcal{J}_1$ and $\mathcal{J}_2$, integrability condition turns into a condition on their

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cardona-qft-methods_ F03176	307	■ 1.000	$d\,J=0$	$dJ = 0$	form J_{mn} , integrability imposes $dJ = 0$, thus making J into a symplectic
cardona-qft-methods_ F03177	307	■ 1.000	$\mathrm{C}^k \times \left(\mathbb{R}^{d-2k}, J\right)$	$\mathbb{C}^k \times (\mathbb{R}^{d-2k}, J)$	$\mathbb{C}^k \times (\mathbb{R}^{d-2k}, J)$, where $J = dx^{2k+1} \wedge dx^{2k+2} + \dots + dx^{d-1} \wedge dx^d$ is the
cardona-qft-methods_ F03178	307	■ 1.000	$J=\mathrm{d}\,x^{\{2\,k+1\}}\wedge\mathrm{d}\,x^{\{2\,k+2\}}+\dots+\mathrm{d}\,x^{\{d-1\}}\wedge\mathrm{d}\,x^{\{d\}}$	$J = \mathrm{d}x^{2k+1} \wedge \mathrm{d}x^{2k+2} + \dots + \mathrm{d}x^{d-1} \wedge \mathrm{d}x^d$	$\mathbb{C}^k \times (\mathbb{R}^{d-2k}, J)$, where $J = \mathrm{d}x^{2k+1} \wedge \mathrm{d}x^{2k+2} + \dots + \mathrm{d}x^{d-1} \wedge \mathrm{d}x^d$ is the
cardona-qft-methods_ F03179	307	■ 1.000	$k \leq d/2$	$k \leq d/2$	standard symplectic structure and $k \leq d/2$ is called rank, which can be
cardona-qft-methods_ F03180	308	■ 1.000	$\mathrm{J}_i, i=1,2$	$\mathcal{J}_i, i = 1, 2$	$\mathcal{J}_i, i = 1, 2$, we can define from this a generalized metric via
cardona-qft-methods_ F03181	308	■ 0.998	J_i	$\mathcal{J}_i$	Commutation and Hermiticity of the $\mathcal{J}_i$ ensures that $\mathcal{H}$ is indeed a gener-
cardona-qft-methods_ F03182	308	■ 1.000	$+i$	$+i$	It maps the $+i$ eigenspace of $\mathcal{J}$ to the annihilator space of the spinor Φ . In
cardona-qft-methods_ F03183	308	■ 1.000	$\Phi_{\pm}$	$\Phi_{\pm}$	From now on, we will use $\Phi_{\pm}$ to denote just the forms. The subindices plus
cardona-qft-methods_ F03184	309	■ 1.000	$e^{-B}\,\Phi=$	$e^{-B}\Phi =$	where on the polyform associated to the spinor, the action is $e^{-B}\Phi =$
cardona-qft-methods_ F03185	309	■ 0.995	$\left(1-B\wedge+\frac{1}{2}\,B\wedge B\wedge+\ldots\right)\Phi$	$\left(1-B\wedge+\frac{1}{2}B\wedge B\wedge+\dots\right)\Phi$	$(1-B\wedge+\frac{1}{2}B\wedge B\wedge+\dots)\Phi$. We will refer to Φ as naked pure spinor, while
cardona-qft-methods_ F03186	309	■ 1.000	Φ^D	Φ^D	Φ^D will be called dressed pure spinor. The former lives in the spinor bundle
cardona-qft-methods_ F03187	309	■ 1.000	$\eta^1=\eta^2\equiv\eta$	$\eta^1 = \eta^2 \equiv \eta$	For the particular case $\eta^1 = \eta^2 \equiv \eta$, the pure spinors in (11) reduce to
cardona-qft-methods_ F03188	309	■ 1.000	η	η	the existence of a nowhere-vanishing spinor η or of the forms just described
cardona-qft-methods_ F03189	309	■ 1.000	$SO(6)$	$SO(6)$	in the 4 of $SO(6)$, can be further decomposed in representations of $SU(3)$
cardona-qft-methods_ F03190	309	■ 0.891	$\mathbf{4} \rightarrow \mathbf{3} + \mathbf{1}$	$\mathbf{4} \rightarrow \mathbf{3} + \mathbf{1}$	as $\mathbf{4} \rightarrow \mathbf{3} + \mathbf{1}$. There is therefore an $SU(3)$ singlet in the decomposition,
cardona-qft-methods_ F03191	309	■ 0.985	$\mathbf{10} + \mathbf{10}^*$	$\mathbf{10} + \mathbf{10}^*$	two-form 15 and three-form (self-dual and anti-self-dual) $\mathbf{10} + \mathbf{10}^*$ in rep-
cardona-qft-methods_ F03192	309	■ 0.649	$SU(3)$	$SU(3)$	resentations of $SU(3)$. This yields
cardona-qft-methods_ F03193	310	■ 0.695	$(J,\,\rho)$	(J, ρ)	forms (J, ρ) where $\rho = Re\Omega$. The forms cannot be arbitrary but must be
cardona-qft-methods_ F03194	310	■ 0.695	$\rho=Re\,\Omega$	$\rho = Re\,\Omega$	forms (J, ρ) where $\rho = Re\Omega$. The forms cannot be arbitrary but must be
cardona-qft-methods_ F03195	310	■ 0.952	$GL(6,\,\mathbb{R})$	$GL(6, \mathbb{R})$	general transformations $GL(6, \mathbb{R})$ in the tangent space at each point. A sta-
cardona-qft-methods_ F03196	310	■ 0.994	$Sp(6,\,\mathbb{R})$	$Sp(6, \mathbb{R})$	ble two-form J then defines a $Sp(6, \mathbb{R})$ structure while a stable real form ρ
cardona-qft-methods_ F03197	310	■ 0.585	$SL(3,\,\mathbb{C})$	$SL(3, \mathbb{C})$	defines a $SL(3, \mathbb{C})$ structure. Together they define an $SU(3)$ structure pro-
cardona-qft-methods_ F03198	310	■ 0.997	$J\wedge\rho=0$	$J \wedge \rho = 0$	This requires $J \wedge \rho = 0$. Note that since $SU(3) \subset SO(6)$ the pair (J, ρ)
cardona-qft-methods_ F03199	310	■ 0.997	$SU(3)\subset SO(6)$	$SU(3) \subset SO(6)$	This requires $J \wedge \rho = 0$. Note that since $SU(3) \subset SO(6)$ the pair (J, ρ)
cardona-qft-methods_ F03200	310	■ 0.999	$\eta^2=\left(v+\mathrm{i}\,v'\right)_m\gamma^m\eta^1$	$\eta^2 = (v + \mathrm{i}v')_m \gamma^m \eta^1$	vector (i.e., one can write $\eta^2 = (v + \mathrm{i}v')_m \gamma^m \eta^1$, and the corresponding pure
cardona-qft-methods_ F03201	310	■ 0.632	$\left(\eta^1,\,\eta^2\right)$	(η^1, η^2)	These are given in terms of the local $SU(2)$ structure defined by (η^1, η^2) :
cardona-qft-methods_ F03202	310	■ 0.859	$(1,1)$	$(1, 1)$	j and ω are the (1,1) and (2,0)-forms on the local four dimensional space

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cardona-qft-methods_ F03203	310	■ 0.992	$v^{\prime\prime} \cdot \Phi_+$	$v' \cdot \Phi_+$	orthogonal to v and v' . Φ_+ describes therefore a generalized almost complex
cardona-qft-methods_ F03204	310	■ 1.000	Φ_-	Φ_-	rank of Φ_- is one. In the most generic case, the rank of the pure spinors
cardona-qft-methods_ F03205	311	■ 0.712	$\Leftrightarrow \exists$	$\Leftrightarrow \exists$	$\mathcal{I}$ is integrable $\Leftrightarrow \exists$ vector v and 1-form ζ such that $d\Phi = (v \lrcorner + \zeta \wedge) \Phi$
cardona-qft-methods_ F03206	311	■ 0.712	$\mathrm{d} \Phi = (v \lrcorner + \zeta \wedge) \Phi$	$d\Phi = (v \lrcorner + \zeta \wedge) \Phi$	$\mathcal{I}$ is integrable $\Leftrightarrow \exists$ vector v and 1-form ζ such that $d\Phi = (v \lrcorner + \zeta \wedge) \Phi$
cardona-qft-methods_ F03207	311	■ 0.997	$\Leftrightarrow \exists \Phi$	$\Leftrightarrow \exists \Phi$	Generalized Calabi-Yau $\Leftrightarrow \exists \Phi$ pure such that $d\Phi = 0$
cardona-qft-methods_ F03208	311	■ 0.997	$\mathrm{d} \Phi = 0$	$d\Phi = 0$	Generalized Calabi-Yau $\Leftrightarrow \exists \Phi$ pure such that $d\Phi = 0$
cardona-qft-methods_ F03209	311	■ 1.000	W_5	W_5	torsion class W_5 (i.e. if $\exists f$ such that $\Phi = e^{-f} \Omega$ is closed). More generally,
cardona-qft-methods_ F03210	311	■ 1.000	$\exists f$	$\exists f$	torsion class W_5 (i.e. if $\exists f$ such that $\Phi = e^{-f} \Omega$ is closed). More generally,
cardona-qft-methods_ F03211	311	■ 1.000	$\Phi = e^{-f} \Omega$	$\Phi = e^{-f} \Omega$	torsion class W_5 (i.e. if $\exists f$ such that $\Phi = e^{-f} \Omega$ is closed). More generally,
cardona-qft-methods_ F03212	311	■ 1.000	$d - H \wedge$	$d - H \wedge$	such that the differential becomes $d - H \wedge$. Similar, the Courant bracket is
cardona-qft-methods_ F03213	311	■ 0.980	$\mathrm{d} \Phi^D = 0$	$d\Phi^D = 0$	$d\Phi^D = 0$.
cardona-qft-methods_ F03214	312	■ 1.000	$\eta_{\mu\nu}$	$\eta_{\mu\nu}$	use (1) to be the string frame metric), and $\eta_{\mu\nu}$ is the Minkowski (or AdS
cardona-qft-methods_ F03215	312	■ 0.840	$\langle Q_\epsilon \chi \rangle = \langle \delta_\epsilon \chi \rangle = 0$	$\langle Q_\epsilon \chi \rangle = \langle \delta_\epsilon \chi \rangle = 0$	and solve the supersymmetry conditions $\langle Q_\epsilon \chi \rangle = \langle \delta_\epsilon \chi \rangle = 0$, where Q
cardona-qft-methods_ F03216	313	■ 0.703	$\langle \delta_\epsilon \chi \rangle = 0$	$\langle \delta_\epsilon \chi \rangle = 0$	that the supersymmetry conditions $\langle \delta_\epsilon \chi \rangle = 0$ are already solved if χ is a
cardona-qft-methods_ F03217	313	■ 1.000	$\psi_M^A, A = 1, 2$	$\psi_M^A, A = 1, 2$	(36), the fermionic fields in type II theories are two gravitinos $\psi_M^A, A = 1, 2$
cardona-qft-methods_ F03218	313	■ 1.000	λ^A	λ^A	and two dilatinos λ^A . Their supersymmetry transformations in the string
cardona-qft-methods_ F03219	313	■ 1.000	$M = 0, \dots, 9, \psi_M$	$M = 0, \dots, 9, \psi_M$	In these equations $M = 0, \dots, 9, \psi_M$ stands for the column vector $\psi_M =$
cardona-qft-methods_ F03220	313	■ 1.000	$\psi_M =$	$\psi_M =$	In these equations $M = 0, \dots, 9, \psi_M$ stands for the column vector $\psi_M =$
cardona-qft-methods_ F03221	313	■ 0.769	$\binom{\psi_M^1}{\psi_M^2}$	$\binom{\psi_M^1}{\psi_M^2}$	$\binom{\psi_M^1}{\psi_M^2}$ containing the two Majorana-Weyl spinors of the same chirality in
cardona-qft-methods_ F03222	313	■ 0.994	$\mathcal{P}_n$	$\mathcal{P}_n$	2×2 matrices $\mathcal{P}$ and $\mathcal{P}_n$ are different in IIA and IIB: for IIA $\mathcal{P} = \Gamma_{11}$ and
cardona-qft-methods_ F03223	313	■ 0.994	$\mathcal{P} = \Gamma_{11}$	$\mathcal{P} = \Gamma_{11}$	2×2 matrices $\mathcal{P}$ and $\mathcal{P}_n$ are different in IIA and IIB: for IIA $\mathcal{P} = \Gamma_{11}$ and
cardona-qft-methods_ F03224	313	■ 1.000	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$, while for IIB $\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$ for $\frac{n+1}{2}$ even and $\mathcal{P}_n = i\sigma^2$
cardona-qft-methods_ F03225	313	■ 1.000	$\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$	$\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$, while for IIB $\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$ for $\frac{n+1}{2}$ even and $\mathcal{P}_n = i\sigma^2$
cardona-qft-methods_ F03226	313	■ 1.000	$\frac{n+1}{2}$	$\frac{n+1}{2}$	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$, while for IIB $\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$ for $\frac{n+1}{2}$ even and $\mathcal{P}_n = i\sigma^2$
cardona-qft-methods_ F03227	313	■ 1.000	$\mathcal{P}_n = i\sigma^2$	$\mathcal{P}_n = i\sigma^2$	$\mathcal{P}_n = \Gamma_{11}^{(n/2)} \sigma^1$, while for IIB $\mathcal{P} = -\sigma^3, \mathcal{P}_n = \sigma^1$ for $\frac{n+1}{2}$ even and $\mathcal{P}_n = i\sigma^2$
cardona-qft-methods_ F03228	313	■ 0.670	$\mathcal{F}_n = \frac{1}{n!} F_{P_1 \dots P_N} \Gamma^{P_1 \dots P_N}$	$\mathcal{F}_n = \frac{1}{n!} F_{P_1 \dots P_N} \Gamma^{P_1 \dots P_N}$	$\mathcal{F}_n = \frac{1}{n!} F_{P_1 \dots P_N} \Gamma^{P_1 \dots P_N}$, and $H_M \equiv \frac{1}{2} H_{MNP} \Gamma^{NP}$. We are using here the

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cardona-qft-methods_F03229	313	0.670	$H_{\{M\}} \equiv \frac{1}{2} H_{\{M \ N \ P\}} \Gamma^{\{N \ P\}}$	$H_M \equiv \frac{1}{2} H_{MNP} \Gamma^{NP}$	$\mathcal{F}_n = \frac{1}{n!} F_{P_1 \dots P_N} \Gamma^{P_1 \dots P_N}$, and $H_M \equiv \frac{1}{2} H_{MNP} \Gamma^{NP}$. We are using here the
cardona-qft-methods_F03230	313	0.919	$C_{\{1\}} \ldots C_{\{9\}}$	$C_1 \dots C_9$	all RR potentials ($C_1 \dots C_9$ in IIA, and $C_0, C_2 \dots C_{10}$ in IIB, instead of only
cardona-qft-methods_F03231	313	0.919	$C_{\{0\}}, C_{\{2\}} \ldots C_{\{10\}}$	$C_0, C_2 \dots C_{10}$	all RR potentials ($C_1 \dots C_9$ in IIA, and $C_0, C_2 \dots C_{10}$ in IIB, instead of only
cardona-qft-methods_F03232	313	1.000	$F^{\{(10)\}}$	$F^{(10)}$	where $F^{(10)}$ is the formal sum of all even (odd) fluxes in IIA (IIB), $\hat{F} =$
cardona-qft-methods_F03233	313	1.000	$\hat{F} =$	$\hat{F} =$	where $F^{(10)}$ is the formal sum of all even (odd) fluxes in IIA (IIB), $\hat{F} =$
cardona-qft-methods_F03234	313	0.999	$\mathrm{d} C + m e^B$	$\mathrm{d} C + m e^B$	$\mathrm{d} C + m e^B$, and $m \equiv F_0^{(10)} = \hat{F}_0$ is the mass parameter of IIA. These RR
cardona-qft-methods_F03235	313	0.999	$m \equiv F_0^{(10)} = \hat{F}_0$	$m \equiv F_0^{(10)} = \hat{F}_0$	$\mathrm{d} C + m e^B$, and $m \equiv F_0^{(10)} = \hat{F}_0$ is the mass parameter of IIA. These RR
cardona-qft-methods_F03236	313	1.000	$SO(n)$	$SO(n)$	n -dimensional Riemannian manifold, the holonomy group is a subgroup of $SO(n)$.
cardona-qft-methods_F03237	314	1.000	$\nabla_M \epsilon = 0$	$\nabla_M \epsilon = 0$	ten-dimensional manifold, i.e. $\nabla_M \epsilon = 0$. The spacetime component of this
cardona-qft-methods_F03238	314	0.998	$\Gamma^\mu = \gamma^\mu \otimes \mathbf{1}, \Gamma^m = \gamma_5 \otimes \gamma^m$	$\Gamma^\mu = \gamma^\mu \otimes \mathbf{1}, \Gamma^m = \gamma_5 \otimes \gamma^m$	gamma matrices ($\Gamma^\mu = \gamma^\mu \otimes \mathbf{1}, \Gamma^m = \gamma_5 \otimes \gamma^m$ for the external and internal
cardona-qft-methods_F03239	314	1.000	$e^{2A} \tilde{g}_{\mu\nu}$	$e^{2A} \tilde{g}_{\mu\nu}$	gamma's), and written the four dimensional metric as $e^{2A} \tilde{g}_{\mu\nu}$. This yields
cardona-qft-methods_F03240	314	0.968	$\gamma_{\mu\nu}$	$\gamma_{\mu\nu}$	Since $\gamma_{\mu\nu}$ is invertible, the integrability condition reads $\nabla_m A \nabla^m A = 0$, i.e.
cardona-qft-methods_F03241	314	0.968	$\nabla_m A \nabla^m A = 0$	$\nabla_m A \nabla^m A = 0$	Since $\gamma_{\mu\nu}$ is invertible, the integrability condition reads $\nabla_m A \nabla^m A = 0$, i.e.
cardona-qft-methods_F03242	314	0.998	$\gamma_{11} \epsilon_{\text{IIA}}^1 = \epsilon_{\text{IIA}}^1$	$\gamma_{11} \epsilon_{\text{IIA}}^1 = \epsilon_{\text{IIA}}^1$	for type IIA, where $\gamma_{11} \epsilon_{\text{IIA}}^1 = \epsilon_{\text{IIA}}^1$ and $\gamma_{11} \epsilon_{\text{IIA}}^2 = -\epsilon_{\text{IIA}}^2$, and the four and
cardona-qft-methods_F03243	314	0.998	$\gamma_{11} \epsilon_{\text{IIA}}^2 = -\epsilon_{\text{IIA}}^2$	$\gamma_{11} \epsilon_{\text{IIA}}^2 = -\epsilon_{\text{IIA}}^2$	for type IIA, where $\gamma_{11} \epsilon_{\text{IIA}}^1 = \epsilon_{\text{IIA}}^1$ and $\gamma_{11} \epsilon_{\text{IIA}}^2 = -\epsilon_{\text{IIA}}^2$, and the four and
cardona-qft-methods_F03244	314	1.000	$\xi_{-}^{1,2} = \left(\xi_{+}^{1,2} \right)^*$	$\xi_{-}^{1,2} = \left(\xi_{+}^{1,2} \right)^*$	six-dimensional spinors obey $\xi_{-}^{1,2} = \left(\xi_{+}^{1,2} \right)^*$, and $\eta_{-}^A = \left(\eta_{+}^A \right)^*$. (By a slight
cardona-qft-methods_F03245	314	1.000	$\eta_{-}^A = \left(\eta_{+}^A \right)^*$	$\eta_{-}^A = \left(\eta_{+}^A \right)^*$	six-dimensional spinors obey $\xi_{-}^{1,2} = \left(\xi_{+}^{1,2} \right)^*$, and $\eta_{-}^A = \left(\eta_{+}^A \right)^*$. (By a slight
cardona-qft-methods_F03246	314	1.000	$\eta^1 = \eta^2$	$\eta^1 = \eta^2$	stant spinor (at least because $\eta^1 = \eta^2$ is a perfectly valid choice), which
cardona-qft-methods_F03247	315	0.677	$\mathrm{Hol}(\nabla) \subseteq \mathrm{SU}(3)^{\mathrm{h}}$	$\mathrm{Hol}(\nabla) \subseteq \mathrm{SU}(3)^{\mathrm{h}}$	implies that it should have reduced holonomy, $\mathrm{Hol}(\nabla) \subseteq \mathrm{SU}(3)^{\mathrm{h}}$. A su-
cardona-qft-methods_F03248	315	1.000	ξ^1	ξ^1	four-dimensional supersymmetry parameters, ξ^1 and ξ^2 . This compactifica-
cardona-qft-methods_F03249	315	1.000	ξ^2	ξ^2	four-dimensional supersymmetry parameters, ξ^1 and ξ^2 . This compactifica-
cardona-qft-methods_F03250	315	1.000	$\mathbb{N} = 2$	$\mathbb{N} = 2$	tion preserves therefore eight supercharges, i.e. $\mathbb{N} = 2$ in four dimensions.
cardona-qft-methods_F03251	315	1.000	$\mathrm{K}3 \times T^2$	$\mathrm{K}3 \times T^2$	preserved supersymmetries. For example, on $\mathrm{K}3 \times T^2$ there are two covari-
cardona-qft-methods_F03252	315	1.000	$\mathbb{N} = 4$	$\mathbb{N} = 4$	preserves $\mathbb{N} = 4$ supersymmetry. Compactifications on T^6 preserve all the
cardona-qft-methods_F03253	315	1.000	T^6	T^6	preserves $\mathbb{N} = 4$ supersymmetry. Compactifications on T^6 preserve all the
cardona-qft-methods_F03254	315	0.826	$\mathbb{N} = 8$	$\mathbb{N} = 8$	supersymmetries (32, or $\mathbb{N} = 8$ in four dimensions).

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cardona-qft-methods_ F03255	315	■ 1.000	$\mathbb{N}=1$	$\mathbb{N} = 1$	at most $\mathbb{N} = 1$ supersymmetry. Besides, compactifications on Calabi-Yau
cardona-qft-methods_ F03256	315	■ 1.000	$F_{\{n\}}, H_{\{3\}}$	F_n, H_3	today is to turn on fluxes for the field strengths F_n, H_3 . Due to the topo-
cardona-qft-methods_ F03257	316	■ 1.000	$\mathcal{N}=2$	$\mathcal{N} = 2$	while the second implies that this action has an $\mathcal{N} = 2$ Minkowski vacuum.
cardona-qft-methods_ F03258	316	■ 1.000	$S O(d)$	$SO(d)$	dimension d has automatically structure group $SO(d)$. All vector, tensor
cardona-qft-methods_ F03259	317	■ 1.000	$c_{\{1\}}=0$	$c_1 = 0$	additionally $c_1 = 0$). Let us now see how these differential conditions get
cardona-qft-methods_ F03260	317	■ 1.000	$\delta \Psi_{\{m\}}=$	$\delta \Psi_m =$	As we reviewed in the previous sections, as a result of demanding $\delta \Psi_m =$
cardona-qft-methods_ F03261	317	■ 1.000	$\delta \lambda=0$	$\delta \lambda = 0$	$\delta \lambda = 0$, supersymmetry imposes differential conditions on the internal
cardona-qft-methods_ F03262	318	■ 0.924	$\operatorname{SU}(3) \times \operatorname{SU}(3)$	$SU(3) \times SU(3)$	eral $SU(3) \times SU(3)$ case, $\mathbb{N} = 1$ vacua can be realized in hybrid complex–
cardona-qft-methods_ F03263	318	■ 1.000	$6-2\ k$	$6 - 2k$	symplectic manifolds, i.e. manifolds with k complex dimensions and $6 - 2k$
cardona-qft-methods_ F03264	318	■ 0.942	$(13,14)$	$(13,14)$	to satisfy Eqs. (13, 14) for $\tilde{\Phi}_{\pm} = \Phi_{\pm}$, $F_{\text{IIA}} = F_{\text{IIB}} = A = 0$. ³⁶
cardona-qft-methods_ F03265	318	■ 0.942	$\tilde{\Phi}_{\pm}=\Phi_{\pm}, F_{\text{IIA}}=F_{\text{IIB}}=A=0 \text{ .} \}^{\{36\}}$	$\tilde{\Phi}_{\pm} = \Phi_{\pm}, F_{\text{IIA}} = F_{\text{IIB}} = A = 0$. ³⁶	to satisfy Eqs. (13, 14) for $\tilde{\Phi}_{\pm} = \Phi_{\pm}$, $F_{\text{IIA}} = F_{\text{IIB}} = A = 0$. ³⁶
cardona-qft-methods_ F03266	318	■ 1.000	$T^{\{6\}}, K\ 3 \times T^{\{2\}}$	$T^6, K3 \times T^2$	omy, such as $T^6, K3 \times T^2$ or CY_3 . These preserve different number of
cardona-qft-methods_ F03267	318	■ 1.000	$C\ Y_{\{3\}}$	CY_3	omy, such as $T^6, K3 \times T^2$ or CY_3 . These preserve different number of
cardona-qft-methods_ F03268	318	■ 0.999	$2\ h^{\{2,1\}}$	$2h^{2,1}$	the metric, or an infinite family of CY_3 ’s parameterized by $2h^{2,1}$ and $h^{1,1}$
cardona-qft-methods_ F03269	318	■ 0.999	$h^{\{1,1\}}$	$h^{1,1}$	the metric, or an infinite family of CY_3 ’s parameterized by $2h^{2,1}$ and $h^{1,1}$
cardona-qft-methods_ F03270	318	■ 0.731	$(6,6)$	$(6,6)$	Clifford(6,6) action with opposite chirality. The one whose chirality equals
cardona-qft-methods_ F03271	318	■ 1.000	$\{ \}^{\{\mathrm{i}\}} H$	${}^i H$	${}^i H$ in Eq.(14) does not “twist” the (usual) complex structure, as $(d - H \wedge) \Omega = 0$ implies
cardona-qft-methods_ F03272	318	■ 1.000	$(d-H \wedge) \Omega=0$	$(d - H \wedge) \Omega = 0$	${}^i H$ in Eq.(14) does not “twist” the (usual) complex structure, as $(d - H \wedge) \Omega = 0$ implies
cardona-qft-methods_ F03273	318	■ 1.000	$d \Omega=0$	$d \Omega = 0$	in particular $d \Omega = 0$.
cardona-qft-methods_ F03274	322	■ 1.000	$\mathrm{N}=(2,2)$	$\mathbf{N} = (2, 2)$	geometry and manifest $\mathbf{N} = (2,2)$ supersymmetric nonlinear sigma-models,”
cardona-qft-methods_ F03275	322	■ 0.976	$\mathrm{D}=10$	$\mathbf{D} = 10$	formulations of $\mathbf{D} = 10$ supersymmetry and D8 - O8 domain walls,” Class.
cardona-qft-methods_ F03276	322	■ 0.978	$\mathrm{N}=1$	$\mathbf{N} = 1$	general $\mathbf{N} = 1$ backgrounds,” arXiv:hep-th/0507099.
cardona-qft-methods_ F03277	326	■ 1.000	$(G,\ M)$	(G, M)	• Is there a Lie groupoid (G, M) such that its infinitesimal version corre-
cardona-qft-methods_ F03278	326	■ 1.000	$(A,\ M)$	(A, M)	sponds to a given Lie algebroid (A, M) ?
cardona-qft-methods_ F03279	326	■ 1.000	$A=T^{\{*\}}\ M$	$A = T^* M$	For the case where $A = T^* M$ and M is a Poisson manifold the answer is not
cardona-qft-methods_ F03280	327	■ 1.000	$(M,\ \Pi)$	(M, Π)	symplectic realization for (M, Π) .
cardona-qft-methods_ F03281	328	■ 1.000	$\{ \}^{\{2,5\}}$	2,5	of dual pairs in the Poisson sigma model with boundary. ^{2,5}
cardona-qft-methods_ F03282	328	■ 1.000	$\Pi \in \Gamma(T\ M \wedge T\ M)$	$\Pi \in \Gamma(TM \wedge TM)$	Recall that a bivector field $\Pi \in \Gamma(TM \wedge TM)$ is called Poisson if the

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cardona-qft-methods_ F03283	328	■ 1.000	$\backslash\{ \} :, \backslash\mathrm{mathcal}\{C\}^{\backslash\infty}\{M\} \backslash\mathrm{otimes} \backslash\mathrm{mathcal}\{C\}^{\backslash\infty}\{M\} \backslash\mathrm{rightarrow} \backslash\mathrm{mathcal}\{C\}^{\backslash\infty}\{M\}$	$\{ \} :, \mathcal{C}^\infty(M) \otimes \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$	the bracket $\{, \} : \mathcal{C}^\infty(M) \otimes \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$, defined by
cardona-qft-methods_ F03284	329	■ 1.000	$T \backslash\Sigma$	$T\Sigma$	the space of vector bundle morphisms between $T\Sigma$ and T^*M . This space
cardona-qft-methods_ F03285	329	■ 0.968	$(X, \backslash\eta)$	(X, η)	can be parametrized by the pair (X, η) , where X is a $\mathcal{C}^{k+1}$ -map from Σ
cardona-qft-methods_ F03286	329	■ 0.968	$\backslash\mathrm{mathcal}\{C\}^{\backslash k+1}$	$\mathcal{C}^{k+1}$	can be parametrized by the pair (X, η) , where X is a $\mathcal{C}^{k+1}$ -map from Σ
cardona-qft-methods_ F03287	329	■ 1.000	$\backslash\eta \backslash\mathrm{in} \backslash\Gamma^{\backslash k}\backslash\mathrm{left}(\backslash\Sigma, T^{\backslash*}\backslash\Sigma \backslash\mathrm{otimes} X^{\backslash*} T^{\backslash*} M\backslash\mathrm{right})$	$\eta \in \Gamma^k(\Sigma, T^*\Sigma \otimes X^*T^*M)$	to M and $\eta \in \Gamma^k(\Sigma, T^*\Sigma \otimes X^*T^*M)$, where $k \in \{0, 1, \dots\}$ denotes the
cardona-qft-methods_ F03288	329	■ 1.000	$k \backslash\mathrm{in}\backslash\{0,1, \backslash\mathrm{cdots}\}$	$k \in \{0, 1, \dots\}$	to M and $\eta \in \Gamma^k(\Sigma, T^*\Sigma \otimes X^*T^*M)$, where $k \in \{0, 1, \dots\}$ denotes the
cardona-qft-methods_ F03289	329	■ 1.000	$\backslash\Pi^{\backslash\#}$	$\Pi^\#$	• $\Pi^\#$ is the map from $T^*M \rightarrow TM$ induced from the Poisson bivector
cardona-qft-methods_ F03290	329	■ 1.000	$T^{\backslash*}\backslash M \backslash\mathrm{rightarrow} T M$	$T^*M \rightarrow TM$	• $\Pi^\#$ is the map from $T^*M \rightarrow TM$ induced from the Poisson bivector
cardona-qft-methods_ F03291	329	■ 1.000	$d X$	dX	• dX and η are regarded as elements in $\Omega^1(\Sigma, X^*(TM))$ and
cardona-qft-methods_ F03292	329	■ 1.000	$\backslash\mathrm{omega}^{\backslash 1}\backslash\mathrm{left}(\backslash\Sigma, X^{\backslash*}(T M)\backslash\mathrm{right})$	$\Omega^1(\Sigma, X^*(TM))$	• dX and η are regarded as elements in $\Omega^1(\Sigma, X^*(TM))$ and
cardona-qft-methods_ F03293	329	■ 1.000	$\backslash\mathrm{omega}^{\backslash 1}\backslash\mathrm{left}(\backslash\Sigma, X^{\backslash*}\backslash\mathrm{left}(T^{\backslash*} M\backslash\mathrm{right})\backslash\mathrm{right})$	$\Omega^1(\Sigma, X^*(T^*M))$	$\Omega^1(\Sigma, X^*(T^*M))$, respectively.
cardona-qft-methods_ F03294	329	■ 1.000	$\backslash\langle$	$\langle$	• $\langle, \rangle$ denotes the pairing between $\Omega^1(\Sigma, X^*(TM))$ and
cardona-qft-methods_ F03295	329	■ 1.000	$\rangle\mathrm{denotes\thepairing\between}\backslash\mathrm{omega}^{\backslash 1}\backslash\mathrm{left}(\backslash\Sigma, X^{\backslash*}(T M)\backslash\mathrm{right})$	$\rangle\mathrm{denotes\thepairing\between}\Omega^1(\Sigma, X^*(TM))$	• $\langle, \rangle$ denotes the pairing between $\Omega^1(\Sigma, X^*(TM))$ and
cardona-qft-methods_ F03296	329	■ 1.000	$\backslash\mathrm{mathcal}\{L\}$	$\mathcal{L}$	The integrand, called the Lagrangian, will be denoted by $\mathcal{L}$. Associated
cardona-qft-methods_ F03297	329	■ 1.000	$\backslash\delta S=0$	$\delta S = 0$	to this action, the corresponding variational problem $\delta S = 0$ induces the
cardona-qft-methods_ F03298	330	■ 1.000	$\{ \}^{\backslash 6} \backslash\Phi_{\backslash\mathrm{partial}}$	${}^6\Phi_\partial$	Following, ⁶ Φ_∂ is endowed with a symplectic form and a surjective sub-
cardona-qft-methods_ F03299	330	■ 1.000	$p: \backslash\Phi \backslash\mathrm{rightarrow} \backslash\Phi_{\backslash\mathrm{partial}}$	$p : \Phi \rightarrow \Phi_\partial$	mersion $p : \Phi \rightarrow \Phi_\partial$. We define
cardona-qft-methods_ F03300	330	■ 1.000	$C_{\backslash\Pi}$	C_Π	Finally, we define C_Π as the set of fields in Φ_∂ which can be completed
cardona-qft-methods_ F03301	330	■ 1.000	$\backslash\Phi_{\backslash\mathrm{partial}}$	Φ_∂	Finally, we define C_Π as the set of fields in Φ_∂ which can be completed
cardona-qft-methods_ F03302	330	■ 1.000	$L_{\backslash\Sigma^{\backslash\mathrm{prime}}}$	$L_{\Sigma'}$	to a field in $L_{\Sigma'}$, with $\Sigma' := \partial\Sigma \times [0, \varepsilon]$, for some ε .
cardona-qft-methods_ F03303	330	■ 1.000	$\backslash\Sigma^{\backslash\mathrm{prime}}:=\backslash\mathrm{partial} \backslash\Sigma \backslash\mathrm{times}[0, \backslash\mathrm{varepsilon}]$	$\Sigma' := \partial\Sigma \times [0, \varepsilon]$	to a field in $L_{\Sigma'}$, with $\Sigma' := \partial\Sigma \times [0, \varepsilon]$, for some ε .
cardona-qft-methods_ F03304	330	■ 1.000	$T^{\backslash*}\backslash(P M)$	$T^*(PM)$	It turns out that Φ_∂ can be identified with $T^*(PM)$, the cotangent bundle
cardona-qft-methods_ F03305	330	■ 1.000	$S(X, \backslash\eta)$	$S(X, \eta)$	If we define $S(X, \eta)$ and C_Π in the same way as before, without assuming
cardona-qft-methods_ F03306	330	■ 0.988	$(A, \backslash\rho)$	(A, ρ)	A pair (A, ρ) , where A is a vector bundle over M and ρ (called the anchor
cardona-qft-methods_ F03307	330	■ 1.000	$[,]_{\backslash A}$	$[,]_A$	(i) There is Lie bracket $[,]_A$ on $\Gamma(A)$ such that the induced map ρ_* :
cardona-qft-methods_ F03308	330	■ 1.000	$\backslash\Gamma(A)$	$\Gamma(A)$	(i) There is Lie bracket $[,]_A$ on $\Gamma(A)$ such that the induced map ρ_* :
cardona-qft-methods_ F03309	330	■ 1.000	$\backslash\rho_{\backslash*}$	ρ_*	(i) There is Lie bracket $[,]_A$ on $\Gamma(A)$ such that the induced map ρ_* :

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cardona-qft-methods_ F03310	330	■ 1.000	$\Gamma(A) \rightarrow \mathfrak{X}(M)$	$\Gamma(A) \rightarrow \mathfrak{X}(M)$	$\Gamma(A) \rightarrow \mathfrak{X}(M)$ is a Lie algebra homomorphism.
cardona-qft-methods_ F03311	330	■ 0.516	$[,]_{T^*M}$	$[,]_{T^*M}$	Poisson manifold T^*M , where $[,]_{T^*M}$ is the Koszul bracket for 1-forms,
cardona-qft-methods_ F03312	331	■ 1.000	$\Pi^\# : T^*M \rightarrow TM$	$\Pi^\# : T^*M \rightarrow TM$	given by $\Pi^\# : T^*M \rightarrow TM$, is a central example of Lie algebroids. To
cardona-qft-methods_ F03313	331	■ 1.000	$\Lambda^\bullet A^*$	$\Lambda^\bullet A^*$	define a morphism of Lie algebroids we consider the complex $\Lambda^\bullet A^*$, where
cardona-qft-methods_ F03314	331	■ 1.000	A^*	A^*	A^* is the dual bundle and a differential δ_A is defined by the rules
cardona-qft-methods_ F03315	331	■ 1.000	δ_A	δ_A	A^* is the dual bundle and a differential δ_A is defined by the rules
cardona-qft-methods_ F03316	331	■ 1.000	$f \in \mathcal{C}^\infty(M)$	$f \in \mathcal{C}^\infty(M)$	(i) for all $f \in \mathcal{C}^\infty(M)$,
cardona-qft-methods_ F03317	331	■ 1.000	$X, Y \in \Gamma(A), \alpha \in \Gamma(A^*)$	$X, Y \in \Gamma(A), \alpha \in \Gamma(A^*)$	(ii) for all $X, Y \in \Gamma(A), \alpha \in \Gamma(A^*)$,
cardona-qft-methods_ F03318	331	■ 0.998	$\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$	$\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$	where $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$.
cardona-qft-methods_ F03319	331	■ 0.998	$\Gamma(A^*)$	$\Gamma(A^*)$	where $\langle \cdot, \cdot \rangle$ is the natural pairing between $\Gamma(A)$ and $\Gamma(A^*)$.
cardona-qft-methods_ F03320	331	■ 1.000	$\varphi : A \rightarrow B$	$\varphi : A \rightarrow B$	A vector bundle morphism $\varphi : A \rightarrow B$ is a Lie algebroid morphism if
cardona-qft-methods_ F03321	331	■ 1.000	$\gamma(A)$	$\gamma(A)$	the anchor maps and the structure functions for $\gamma(A)$ and $\Gamma(B)$ should
cardona-qft-methods_ F03322	331	■ 1.000	$\Gamma(B)$	$\Gamma(B)$	the anchor maps and the structure functions for $\gamma(A)$ and $\Gamma(B)$ should
cardona-qft-methods_ F03323	331	■ 1.000	$T(\partial\Sigma)$	$T(\partial\Sigma)$	the space of Lie algebroid morphisms between $T(\partial\Sigma)$ and T^*M where the
cardona-qft-methods_ F03324	331	■ 1.000	T^*PM	T^*PM	coisotropic submanifold of T^*PM . Its symplectic reduction, i.e. the space
cardona-qft-methods_ F03325	331	■ 0.866	M^4	M^4	over M^4 . More precisely, a groupoid is a small category with invertible
cardona-qft-methods_ F03326	331	■ 1.000	$G \rightrightarrows M$	$G \rightrightarrows M$	folds, a Lie groupoid over M , denoted by $G \rightrightarrows M$, can be rephrased as the
cardona-qft-methods_ F03327	331	■ 1.000	s, t, ι, μ	s, t, ι, μ	where s, t, ι, μ and ε denote the source, target, inverse, multiplication
cardona-qft-methods_ F03328	331	■ 0.578	$G_{(x,y)} := s^{-1}(x) \cap t^{-1}(y)$	$G_{(x,y)} := s^{-1}(x) \cap t^{-1}(y)$	ing $G_{(x,y)} := s^{-1}(x) \cap t^{-1}(y)$:
cardona-qft-methods_ F03329	331	■ 1.000	$s \circ \varepsilon = t \circ \varepsilon = id_M$	$s \circ \varepsilon = t \circ \varepsilon = id_M$	(A.1) $s \circ \varepsilon = t \circ \varepsilon = id_M$
cardona-qft-methods_ F03330	331	■ 1.000	$g \in G_{(x,y)}$	$g \in G_{(x,y)}$	(A.2) If $g \in G_{(x,y)}$ and $h \in G_{(y,z)}$ then $\mu(g, h) \in G_{(x,z)}$
cardona-qft-methods_ F03331	331	■ 1.000	$h \in G_{(y,z)}$	$h \in G_{(y,z)}$	(A.2) If $g \in G_{(x,y)}$ and $h \in G_{(y,z)}$ then $\mu(g, h) \in G_{(x,z)}$
cardona-qft-methods_ F03332	331	■ 1.000	$\mu(g, h) \in G_{(x,z)}$	$\mu(g, h) \in G_{(x,z)}$	(A.2) If $g \in G_{(x,y)}$ and $h \in G_{(y,z)}$ then $\mu(g, h) \in G_{(x,z)}$
cardona-qft-methods_ F03333	331	■ 0.647	${}^aG \times_{(s,t)} G$	${}^aG \times_{(s,t)} G$	${}^aG \times_{(s,t)} G$ is a smooth manifold whenever s (or t) is a surjective submersion.
cardona-qft-methods_ F03334	332	■ 1.000	$\mu(\varepsilon \circ s \times id_G) = \mu(id_G \times \varepsilon \circ t) = id_G$	$\mu(\varepsilon \circ s \times id_G) = \mu(id_G \times \varepsilon \circ t) = id_G$	(A.3) $\mu(\varepsilon \circ s \times id_G) = \mu(id_G \times \varepsilon \circ t) = id_G$
cardona-qft-methods_ F03335	332	■ 1.000	$\mu(id_G \times i) = \varepsilon \circ t$	$\mu(id_G \times i) = \varepsilon \circ t$	(A.4) $\mu(id_G \times i) = \varepsilon \circ t$
cardona-qft-methods_ F03336	332	■ 1.000	$\mu(i \times id_G) = \varepsilon \circ s$	$\mu(i \times id_G) = \varepsilon \circ s$	(A.5) $\mu(i \times id_G) = \varepsilon \circ s$

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cardona-qft-methods_ F03337	332	■ 1.000	$\mu\backslashleft(\mu\times i d_{\{G\}}\backslashright)=\mu\backslashleft(i d_{\{G\}}\times\mu\backslashright)$	$\mu(\mu\times id_G)=\mu(id_G\times\mu)$	(A.6) $\mu(\mu\times id_G)=\mu(id_G\times\mu)$.
cardona-qft-methods_ F03338	332	■ 1.000	$G\times G\times\bar{G}$	$G\times G\times\bar{G}$	is a lagrangian submanifold of $G\times G\times\bar{G}$, where $\bar{G}$ denotes the sign reversed
cardona-qft-methods_ F03339	332	■ 1.000	$\bar{G}$	$\bar{G}$	is a lagrangian submanifold of $G\times G\times\bar{G}$, where $\bar{G}$ denotes the sign reversed
cardona-qft-methods_ F03340	332	■ 0.984	$\underline{C_{\{\Pi\}}}$	$\underline{C_{\Pi}}$	morphic to a disc, the symplectic reduction $\underline{C_{\Pi}}$ of C_{Π} (the space of leaves
cardona-qft-methods_ F03341	332	■ 1.000	$\mathcal{A}=\mathcal{T}^*M$	$\mathcal{A}=T^*M$	called $\mathcal{A}$ –paths modulo $\mathcal{A}$ – homotopy, ¹⁰ with $\mathcal{A}=T^*M$.
cardona-qft-methods_ F03342	332	■ 0.999	$\text{\text{\text{Ext}}}$	Ext	<i>Symplectic Category</i> , denoted by Sym^{Ext} and defined as follows:
cardona-qft-methods_ F03343	333	■ 0.905	$\text{\text{\text{E x t}}}$	Ext	Definition 3.1. Sym^{Ext} is a category in which the objects are symplec-
cardona-qft-methods_ F03344	333	■ 1.000	$L:\mathcal{M}\rightarrow\mathcal{N}$	$L:\mathcal{M}\rightarrow\mathcal{N}$	that $L:\mathcal{M}\rightarrow\mathcal{N}$ is an immersed canonical relation between two symplectic
cardona-qft-methods_ F03345	333	■ 1.000	$\mathcal{N}$	$\mathcal{N}$	manifolds $\mathcal{M}$ and $\mathcal{N}$ by definition if L is an immersed Lagrangian subman-
cardona-qft-methods_ F03346	333	■ 0.755	$\overline{\mathcal{M}}\times\mathcal{N}$	$\overline{\mathcal{M}}\times\mathcal{N}$	ifold of $\bar{\mathcal{M}}\times\mathcal{N}$. ${}^c Sym^{Ext}$ carries an involution $\dagger:(Sym^{Ext})^{op}\rightarrow Sym^{Ext}$
cardona-qft-methods_ F03347	333	■ 0.755	$\text{\text{\text{c}}}$	c	ifold of $\bar{\mathcal{M}}\times\mathcal{N}$. ${}^c Sym^{Ext}$ carries an involution $\dagger:(Sym^{Ext})^{op}\rightarrow Sym^{Ext}$
cardona-qft-methods_ F03348	333	■ 0.755	$\left(\text{\text{\text{Sym}}}\text{\text{\text{Ext}}}\right)^{op}\rightarrow$	$\left(Sym^{Ext}\right)^{op}\rightarrow$	ifold of $\bar{\mathcal{M}}\times\mathcal{N}$. ${}^c Sym^{Ext}$ carries an involution $\dagger:(Sym^{Ext})^{op}\rightarrow Sym^{Ext}$
cardona-qft-methods_ F03349	333	■ 0.755	$\text{\text{\text{Ext}}}$	Ext	ifold of $\bar{\mathcal{M}}\times\mathcal{N}$. ${}^c Sym^{Ext}$ carries an involution $\dagger:(Sym^{Ext})^{op}\rightarrow Sym^{Ext}$
cardona-qft-methods_ F03350	333	■ 1.000	$f:A\rightarrow B$	$f:A\rightarrowtail B$	that is the identity in objects and in morphisms, for $f:A\rightarrowtail B$,
cardona-qft-methods_ F03351	333	■ 1.000	$f^{\dag}:=\{(b,a)\in B\times A\mid(a,b)\in f\}$	$f^{\dagger}:=\{(b,a)\in B\times A\mid(a,b)\in f\}$	$f^{\dagger}:=\{(b,a)\in B\times A\mid(a,b)\in f\}$.
cardona-qft-methods_ F03352	333	■ 0.863	$(\mathcal{G},L,I)$	$(\mathcal{G},L,I)$	Definition 3.2. A relational symplectic groupoid is a triple $(\mathcal{G},L,I)$
cardona-qft-methods_ F03353	333	■ 1.000	$\mathcal{G}^3$	$\mathcal{G}^3$	• L is an immersed Lagrangian submanifold of $\mathcal{G}^3$.
cardona-qft-methods_ F03354	333	■ 1.000	$1L$	$1L$	• A.1 L is cyclically symmetric (i.e. if $(x,y,z)\in L$, then $(y,z,x)\in L$)
cardona-qft-methods_ F03355	333	■ 1.000	$(x,y,z)\in L$	$(x,y,z)\in L$	• A.1 L is cyclically symmetric (i.e. if $(x,y,z)\in L$, then $(y,z,x)\in L$)
cardona-qft-methods_ F03356	333	■ 1.000	$(y,z,x)\in L$	$(y,z,x)\in L$	• A.1 L is cyclically symmetric (i.e. if $(x,y,z)\in L$, then $(y,z,x)\in L$)
cardona-qft-methods_ F03357	333	■ 0.995	$2I$	$2I$	• A.2 I is an involution (i.e. $I^2=Id$).
cardona-qft-methods_ F03358	333	■ 0.995	$I^2=Id$	$I^2=Id$	• A.2 I is an involution (i.e. $I^2=Id$).
cardona-qft-methods_ F03359	333	■ 1.000	$\mathcal{G}\times\mathcal{G}\rightarrow\overline{\mathcal{G}}$	$\mathcal{G}\times\mathcal{G}\rightarrowtail\bar{\mathcal{G}}$	Notation. L is a canonical relation $\mathcal{G}\times\mathcal{G}\rightarrowtail\bar{\mathcal{G}}$ and will be denoted by
cardona-qft-methods_ F03360	333	■ 1.000	$L_{\text{\text{\text{rel}}}}$	L_{rel}	L_{rel} . Since the graph of I is a Lagrangian submanifold of $\mathcal{G}\times\mathcal{G}$, I is a
cardona-qft-methods_ F03361	333	■ 1.000	$\mathcal{G}\times\mathcal{G},I$	$\mathcal{G}\times\mathcal{G},I$	L_{rel} . Since the graph of I is a Lagrangian submanifold of $\mathcal{G}\times\mathcal{G}$, I is a
cardona-qft-methods_ F03362	333	■ 0.996	$\overline{\mathcal{G}}\rightarrow\mathcal{G}$	$\bar{\mathcal{G}}\rightarrowtail\mathcal{G}$	canonical relation $\bar{\mathcal{G}}\rightarrowtail\mathcal{G}$ and will be denoted by I_{rel} .

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cardona-qft-methods_ F03363	333	■ 0.996	$I_{\text{\texttt{rel}}}$	I_{rel}	canonical relation $\bar{\mathcal{G}} \rightarrow \mathcal{G}$ and will be denoted by I_{rel} .
cardona-qft-methods_ F03364	333	■ 0.995	$\overline{L_{\text{\texttt{r e l}}}}$	$\overline{L_{\text{rel}}}$	respectively and will be denoted by $\overline{L_{\text{rel}}}$ and $\overline{I_{\text{rel}}}$. The transposition
cardona-qft-methods_ F03365	333	■ 0.995	$\overline{I_{\text{\texttt{r e l}}}}$	$\overline{I_{\text{rel}}}$	respectively and will be denoted by $\overline{L_{\text{rel}}}$ and $\overline{I_{\text{rel}}}$. The transposition
cardona-qft-methods_ F03366	333	■ 1.000	$\omega^{\#}: T^*M \rightarrow TM$	$\omega^{\#}: T^*M \rightarrow TM$	$\omega^{\#}: T^*M \rightarrow TM$ is injective. For finite dimensional manifolds, the notion of weak
cardona-qft-methods_ F03367	334	■ 0.999	$Id: \mathcal{G} \rightarrow \mathcal{G}$	$Id: \mathcal{G} \rightarrow \mathcal{G}$	The identity map $Id: \mathcal{G} \rightarrow \mathcal{G}$ as a relation will be denoted by Id_{rel} :
cardona-qft-methods_ F03368	334	■ 0.999	$Id_{\text{\texttt{rel}}}$	Id_{rel}	The identity map $Id: \mathcal{G} \rightarrow \mathcal{G}$ as a relation will be denoted by Id_{rel} :
cardona-qft-methods_ F03369	334	■ 1.000	$\mathcal{G} \rightarrow \mathcal{G}$	$\mathcal{G} \rightarrow \mathcal{G}$	$\mathcal{G} \rightarrow \mathcal{G}$ and by $\overline{Id_{\text{rel}}}: \bar{\mathcal{G}} \rightarrow \bar{\mathcal{G}}$.
cardona-qft-methods_ F03370	334	■ 1.000	$\overline{Id_{\text{\texttt{r e l}}}}: \overline{\mathcal{G}}$ $\rightarrow \overline{\mathcal{G}}$	$\overline{Id_{\text{rel}}}: \bar{\mathcal{G}} \rightarrow \bar{\mathcal{G}}$	$\mathcal{G} \rightarrow \mathcal{G}$ and by $\overline{Id_{\text{rel}}}: \bar{\mathcal{G}} \rightarrow \bar{\mathcal{G}}$.
cardona-qft-methods_ F03371	334	■ 0.927	$3 I_{\text{\texttt{rel}}} \circ L_{\text{\texttt{rel}}} = \bar{L}_{\text{\texttt{rel}}} \circ \bar{T}_{\text{\texttt{rel}}} \circ (\overline{I_{\text{\texttt{rel}}}} \circ \overline{I_{\text{\texttt{rel}}}}) : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$	$3I_{\text{rel}} \circ L_{\text{rel}} = \bar{L}_{\text{rel}} \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \circ \overline{I_{\text{rel}}}) : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$	• A.3 $I_{\text{rel}} \circ L_{\text{rel}} = \bar{L}_{\text{rel}} \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \circ \overline{I_{\text{rel}}}) : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$.
cardona-qft-methods_ F03372	334	■ 0.997	$\overline{I_{\text{\texttt{I}}\text{\texttt{rel}}}} \circ L_3 = \bar{L}_3 \circ \bar{T}_{\text{\texttt{rel}}} \circ (\overline{I_{\text{\texttt{rel}}}} \times \overline{I_{\text{\texttt{rel}}}})$	$\overline{I_{\text{rel}}} \circ L_3 = \bar{L}_3 \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \times \overline{I_{\text{rel}}})$	Corollary 3.1. $\overline{I_{\text{rel}}} \circ L_3 = \bar{L}_3 \circ \bar{T}_{\text{rel}} \circ (\overline{I_{\text{rel}}} \times \overline{I_{\text{rel}}})$.
cardona-qft-methods_ F03373	334	■ 1.000	$4 L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightarrow \mathcal{G}$	$4L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightarrow \mathcal{G}$	• A.4 $L_3 \circ (L_3 \times Id) = L_3 \circ (Id \times L_3) : \mathcal{G}^3 \rightarrow \mathcal{G}$ is an immersed Lagrangian
cardona-qft-methods_ F03374	334	■ 0.999	$(Id \times L_3)$	$(Id \times L_3)$	L_3 is Lagrangian, and so $(Id \times L_3)$ and $(L_3 \times Id)$. The graph of the
cardona-qft-methods_ F03375	334	■ 0.999	$(L_3 \times Id)$	$(L_3 \times Id)$	L_3 is Lagrangian, and so $(Id \times L_3)$ and $(L_3 \times Id)$. The graph of the
cardona-qft-methods_ F03376	334	■ 1.000	$* \rightarrow \mathcal{G} \times \mathcal{G}$	$* \rightarrow \mathcal{G} \times \mathcal{G}$	map I , as a relation $* \rightarrow \mathcal{G} \times \mathcal{G}$ will be denoted by L_I .
cardona-qft-methods_ F03377	334	■ 1.000	L_I	L_I	map I , as a relation $* \rightarrow \mathcal{G} \times \mathcal{G}$ will be denoted by L_I .
cardona-qft-methods_ F03378	334	■ 1.000	$5 L_3 \circ L_I$	$5L_3 \circ L_I$	• A.5 $L_3 \circ L_I$ is an immersed Lagrangian submanifold of $\mathcal{G}$.
cardona-qft-methods_ F03379	334	■ 1.000	$L_1 := L_3 \circ L_I : * \rightarrow \mathcal{G}$	$L_1 := L_3 \circ L_I : * \rightarrow \mathcal{G}$	Let $L_1 := L_3 \circ L_I : * \rightarrow \mathcal{G}$. From the definitions above we get the
cardona-qft-methods_ F03380	334	■ 1.000	L_1	L_1	where L_1 is regarded as an immersed Lagrangian submanifold of $\mathcal{G}$.
cardona-qft-methods_ F03381	335	■ 1.000	$6 L_3 \circ (L_1 \times Id)$	$6L_3 \circ (L_1 \times Id)$	• A.6 $L_3 \circ (L_1 \times Id)$ is an immersed Lagrangian submanifold of $\bar{\mathcal{G}} \times \mathcal{G}$.
cardona-qft-methods_ F03382	335	■ 1.000	$\overline{\mathcal{G}} \times \mathcal{G}$	$\bar{\mathcal{G}} \times \mathcal{G}$	• A.6 $L_3 \circ (L_1 \times Id)$ is an immersed Lagrangian submanifold of $\bar{\mathcal{G}} \times \mathcal{G}$.
cardona-qft-methods_ F03383	335	■ 1.000	L_2	L_2	Corollary 3.5. L_2 leaves invariant L_1 , L_2 and L_3 , i.e.
cardona-qft-methods_ F03384	335	■ 1.000	$* \rightarrow \mathcal{G}$	$* \rightarrow \mathcal{G}$	$* \rightarrow \mathcal{G}$ denoted by $\mathcal{G}_{\text{rel}}$.
cardona-qft-methods_ F03385	335	■ 1.000	$\mathcal{G}_{\text{\texttt{rel}}}$	$\mathcal{G}_{\text{rel}}$	$* \rightarrow \mathcal{G}$ denoted by $\mathcal{G}_{\text{rel}}$.
cardona-qft-methods_ F03386	335	■ 0.989	$7 L_2 \circ \mathcal{G}_{\text{\texttt{rel}}}$	$7L_2 \circ \mathcal{G}_{\text{rel}}$	• A.7 $L_2 \circ \mathcal{G}_{\text{rel}}$ is an immersed coisotropic relation.

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cardona-qft-methods_ F03387	335	■ 1.000	$C := L_{\{2\}} \circ \text{circ } \mathcal{G}_{\{\text{rel}\}}$	$C := L_2 \circ \mathcal{G}_{\text{rel}}$	Corollary 3.7. <i>Setting $C := L_2 \circ \mathcal{G}_{\text{rel}}$ the following corollary holds.</i>
cardona-qft-methods_ F03388	335	■ 0.999	$\underline{L_{\{1\}}} = L_{\{1\}} / L_{\{2\}}$	$\underline{L_1} = L_1 / L_2$	• A.8 The reduction $\underline{L_1} = L_1 / L_2$ is a finite dimensional smooth mani-
cardona-qft-methods_ F03389	335	■ 1.000	$\underline{L_{\{1\}}}$	$\underline{L_1}$	fold. We will denote $\underline{L_1}$ by M .
cardona-qft-methods_ F03390	336	■ 1.000	$9 \ S := \left\{ (c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (l, c, g) \in L_{\{3\}} \right\}$	$9S := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (l, c, g) \in L_3\}$	• A.9 $S := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (l, c, g) \in L_3\}$ is an immersed
cardona-qft-methods_ F03391	336	■ 1.000	$\mathcal{G} \times M$	$\mathcal{G} \times M$	submanifold of $\mathcal{G} \times M$.
cardona-qft-methods_ F03392	336	■ 0.655	$\mathcal{G}, L, I$	$\mathcal{G}, L, I$	Conjecture 3.1. <i>Let $(\mathcal{G}, L, I)$ be a regular relational symplectic</i>
cardona-qft-methods_ F03393	336	■ 1.000	$G := C / L_{\{2\}}$	$G := C / L_2$	Conjecture 3.2. <i>Assume $G := C / L_2$ is smooth. Then G is a sym-</i>
cardona-qft-methods_ F03394	336	■ 1.000	$s := S / L_{\{2\}}, t := T / L_{\{2\}}, \mu :=$	$s := S / L_2, t := T / L_2, \mu :=$	<i>plectic groupoid on M with structure maps $s := S / L_2, t := T / L_2, \mu :=$</i>
cardona-qft-methods_ F03395	336	■ 0.898	$L_{\{\text{rel}\}} / L_{\{2\}}, \iota = I, \varepsilon = L_{\{1\}} / L_{\{2\}}$	$L_{\text{rel}} / L_2, \iota = I, \varepsilon = L_1 / L_2$	$L_{\text{rel}} / L_2, \iota = I, \varepsilon = L_1 / L_2$.
cardona-qft-methods_ F03396	336	■ 0.998	$\left(G, L_{\{G\}}, I_{\{G\}} \right)$	(G, L_G, I_G)	groupoids (G, L_G, I_G) and (H, L_H, I_H) is a map F from G to H satis-
cardona-qft-methods_ F03397	336	■ 0.998	$\left(H, L_{\{H\}}, I_{\{H\}} \right)$	(H, L_H, I_H)	groupoids (G, L_G, I_G) and (H, L_H, I_H) is a map F from G to H satis-
cardona-qft-methods_ F03398	336	■ 1.000	$G \times \bar{H}$	$G \times \bar{H}$	(i) F is a Lagrangian subspace of $G \times \bar{H}$.
cardona-qft-methods_ F03399	336	■ 1.000	$F \circ I_{\{G\}} = I_{\{H\}} \circ F$	$F \circ I_G = I_H \circ F$	(ii) $F \circ I_G = I_H \circ F$.
cardona-qft-methods_ F03400	336	■ 1.000	$F^{\{3\}} \left(L_{\{G\}} \right) = L_{\{H\}}$	$F^3(L_G) = L_H$	(iii) $F^3(L_G) = L_H$.
cardona-qft-methods_ F03401	336	■ 1.000	$F: G \rightarrow$	$F: G \rightarrow$	Definition 3.5. A morphism of relational symplectic groupoids $F: G \rightarrow$
cardona-qft-methods_ F03402	336	■ 0.998	$F^{\dagger}$	$F^\dagger$	H is called an equivalence if the transpose canonical relation $F^\dagger$ is also a
cardona-qft-methods_ F03403	336	■ 0.965	$\mathcal{C}^k$	$\mathcal{C}^k$	(i) Different differentiability degrees (the $\mathcal{C}^k$ - type of the maps X and η)
cardona-qft-methods_ F03404	337	■ 0.909	(G, ω)	(G, ω)	Let (G, ω) be a symplectic manifold and $\mathcal{L}$ an immersed Lagrangian sub-
cardona-qft-methods_ F03405	337	■ 0.998	$G_{\{1\}} = G, G_{\{2\}}$	$G_{(1)} = G, G_{(2)}$	Let us denote $G_{(1)} = G, G_{(2)}$ the fiber product $G \times_{(s,t)} G, G_{(3)} = G \times_{(s,t)}$
cardona-qft-methods_ F03406	337	■ 0.998	$G \times_{\{s,t\}} G, G_{\{3\}} = G \times_{\{s,t\}}$	$G \times_{(s,t)} G, G_{(3)} = G \times_{(s,t)}$	Let us denote $G_{(1)} = G, G_{(2)}$ the fiber product $G \times_{(s,t)} G, G_{(3)} = G \times_{(s,t)}$
cardona-qft-methods_ F03407	337	■ 0.615	$\left(G \times_{\{s,t\}} G \right)$	$(G \times_{(s,t)} G)$	$(G \times_{(s,t)} G)$ and so on . It can be proven the following
cardona-qft-methods_ F03408	337	■ 0.998	$G_{\{n\}}$	$G_{(n)}$	(i) $G_{(n)}$ is a coisotropic submanifold of G^n .
cardona-qft-methods_ F03409	337	■ 0.998	$G^{\{n\}}$	G^n	(i) $G_{(n)}$ is a coisotropic submanifold of G^n .
cardona-qft-methods_ F03410	338	■ 1.000	$P_{\{n\}}: \underline{G_{\{n\}}} \rightarrow G^{\{n\}}$	$P_n: \underline{G_{(n)}} \rightarrow G^n$	We have natural canonical relations $P_n: \underline{G_{(n)}} \rightarrow G^n$ defined as:
cardona-qft-methods_ F03411	338	■ 1.000	$G_{\{i\}}$	$G_{(i)}$	Proposition 3.2. $G_{(i)}$ is equivalent to $G_{(j)}, \forall i, j \geq 1$ and the equivalence
cardona-qft-methods_ F03412	338	■ 1.000	$G_{\{j\}}, \forall i, j \geq 1$	$G_{(j)}, \forall i, j \geq 1$	Proposition 3.2. $G_{(i)}$ is equivalent to $G_{(j)}, \forall i, j \geq 1$ and the equivalence

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cardona-qft-methods_ F03413	338	■ 1.000	$P_{\{i\}} \circ P_{\{j\}}^{\dagger}$	$P_i \circ P_j^{\dagger}$	<i>is given by $P_i \circ P_j^{\dagger}$.</i>
cardona-qft-methods_ F03414	338	■ 0.966	$\sim$	$\sim$	an equivalence relation, called T^*M -homotopy, ¹⁰ and denoted by $\sim$. More
cardona-qft-methods_ F03415	339	■ 1.000	$(\overline{X, \eta}) = \left\{ (X, \eta) \mid X \equiv X_0, \eta \in \ker \Pi^{\#} \right\}$	$(\overline{X, \eta}) = \{ (X, \eta) \mid X \equiv X_0, \eta \in \ker \Pi^{\#} \}$	where $(\overline{X, \eta}) = \{ (X, \eta) \mid X \equiv X_0, \eta \in \ker \Pi^{\#} \}$, we can prove the following
cardona-qft-methods_ F03416	339	■ 1.000	$N(\Gamma_0(T^*PM))$	$N(\Gamma_0(T^*PM))$	$N(\Gamma_0(T^*PM))$ such that $\overline{L_1} \cap N(\Gamma_0(T^*PM))$ is an embedded submanifold
cardona-qft-methods_ F03417	339	■ 1.000	$\overline{L_1} \cap N(\Gamma_0(T^*PM))$	$\overline{L_1} \cap N(\Gamma_0(T^*PM))$	$N(\Gamma_0(T^*PM))$ such that $\overline{L_1} \cap N(\Gamma_0(T^*PM))$ is an embedded submanifold
cardona-qft-methods_ F03418	339	■ 1.000	$L_1 \cap N(\Gamma_0(T^*PM)), L_2 \cap$	$L_1 \cap N(\Gamma_0(T^*PM)), L_2 \cap$	Theorem 3.3. ³ If M is integrable, then $L_1 \cap N(\Gamma_0(T^*PM)), L_2 \cap$
cardona-qft-methods_ F03419	339	■ 1.000	$N(\Gamma_0(T^*PM))^2$	$N(\Gamma_0(T^*PM))^2$	$N(\Gamma_0(T^*PM))^2$ and $L_3 \cap N(\Gamma_0(T^*PM))^3$ are embedded Lagrangian sub-
cardona-qft-methods_ F03420	339	■ 1.000	$L_3 \cap N(\Gamma_0(T^*PM))^3$	$L_3 \cap N(\Gamma_0(T^*PM))^3$	$N(\Gamma_0(T^*PM))^2$ and $L_3 \cap N(\Gamma_0(T^*PM))^3$ are embedded Lagrangian sub-
cardona-qft-methods_ F03421	339	■ 0.971	$\rangle$ and an asso-	$\rangle$ and an asso-	• A Hilbert space H , equipped with an inner product $\langle, \rangle$ and an asso-
cardona-qft-methods_ F03422	339	■ 1.000	$\{ \}^{\sim}$	$\sim$	where $\sim$ denotes complex conjugation, the following axioms holds:
cardona-qft-methods_ F03423	339	■ 1.000	$\{e_i\}$	$\{e_i\}$	(ii) <i>Projectability</i> : Choosing an orthonormal basis $\{e_i\}$ of H and as-
cardona-qft-methods_ F03424	340	■ 0.998	$(H, m, \langle, \rangle_H)$	$(H, m, \langle, \rangle_H)$	Remark 6. Under the assumptions above, $(H, m, \langle, \rangle_H)$ is not an unital
cardona-qft-methods_ F03425	342	■ 1.000	LM	LM	smooth loops LM on a closed and oriented manifold M . In particular alge-
cardona-qft-methods_ F03426	342	■ 1.000	$-n$	$-n$	having total degree $-n$.
cardona-qft-methods_ F03427	343	■ 1.000	$[*/G]$	$[*/G]$	to the case of $[*/G]$ and give geometric descriptions of the string topology
cardona-qft-methods_ F03428	343	■ 1.000	$H^*(LBG)$	$H^*(LBG)$	product on the cohomology $H^*(LBG)$.
cardona-qft-methods_ F03429	343	■ 1.000	$H_p(M) \cong (H^p(M))^*$	$H_p(M) \cong (H^p(M))^*$	universal coefficients theorem $H_p(M) \cong (H^p(M))^*$ and we have that the
cardona-qft-methods_ F03430	343	■ 1.000	$H_{*+n}(M)$	$H_{*+n}(M)$	$H_{*+n}(M)$ (shifted by the dimension) of a closed oriented manifold has a
cardona-qft-methods_ F03431	344	■ 1.000	$\sigma \in H_p(M), \tau \in H_q(M)$	$\sigma \in H_p(M), \tau \in H_q(M)$	when two homology classes $\sigma \in H_p(M), \tau \in H_q(M)$ can be represented
cardona-qft-methods_ F03432	344	■ 0.958	${}^a P^p \subseteq M$	${}^a P^p \subseteq M$	by submanifolds ${}^a P^p \subseteq M$ and $Q^q \subseteq M$ respectively, then the product is
cardona-qft-methods_ F03433	344	■ 0.958	$Q^q \subseteq M$	$Q^q \subseteq M$	by submanifolds ${}^a P^p \subseteq M$ and $Q^q \subseteq M$ respectively, then the product is
cardona-qft-methods_ F03434	344	■ 1.000	$P \cap Q$	$P \cap Q$	represented by the tranverse intersection $P \cap Q$.
cardona-qft-methods_ F03435	344	■ 0.951	$\Delta: M \rightarrow M \times M$	$\Delta: M \rightarrow M \times M$	Pontrjagin collapse map. Take the diagonal embedding $\Delta: M \rightarrow M \times M$
cardona-qft-methods_ F03436	344	■ 1.000	ν_{Δ}	ν_{Δ}	which has a tubular neighborhood ν_{Δ} homeomorphic to the total space
cardona-qft-methods_ F03437	344	■ 1.000	$(M \times M) / \nu_{\Delta}^c$	$(M \times M) / \nu_{\Delta}^c$	$(M \times M) / \nu_{\Delta}^c$ is homeomorphic to the Thom space M^{TM} of the normal
cardona-qft-methods_ F03438	344	■ 1.000	M^{TM}	M^{TM}	$(M \times M) / \nu_{\Delta}^c$ is homeomorphic to the Thom space M^{TM} of the normal
cardona-qft-methods_ F03439	344	■ 1.000	$e: P \rightarrow Q$	$e: P \rightarrow Q$	They key fact of what we did is that if a map $e: P \rightarrow Q$ is an em-

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cardona-qft-methods_ F03440	344	■ 1.000	$Q \rightarrow P^{\nu(e)}$	$Q \rightarrow P^{\nu(e)}$	Pontrjagin collapse map $Q \rightarrow P^{\nu(e)}$ from Q to the Thom space of the
cardona-qft-methods_ F03441	344	■ 1.000	$\nu(e) \rightarrow P$	$\nu(e) \rightarrow P$	normal bundle $\nu(e) \rightarrow P$. If the normal bundle is orientable after ap-
cardona-qft-methods_ F03442	344	■ 1.000	$e_! : H_*(Q) \rightarrow H_{*-\dim \nu(e)}(P)$	$e_! : H_*(Q) \rightarrow H_{*-\dim \nu(e)}(P)$	$e_! : H_*(Q) \rightarrow H_{*-\dim \nu(e)}(P)$.
cardona-qft-methods_ F03443	344	■ 1.000	$\sigma \in H_p(M)$	$\sigma \in H_p(M)$	^a We say that a homology class $\sigma \in H_p(M)$ can be represented by submanifold if there
cardona-qft-methods_ F03444	345	■ 1.000	$\operatorname{Map}(8, M)$	$\operatorname{Map}(8, M)$	Let $\operatorname{Map}(8, M)$ be the space of piecewise smooth maps from the figure
cardona-qft-methods_ F03445	345	■ 0.991	$ev : LM \rightarrow M$	$ev : LM \rightarrow M$	Since the evaluation at zero $ev : LM \rightarrow M$ can be considered as a fiber
cardona-qft-methods_ F03446	345	■ 0.570	$\operatorname{Map}(8, M) \rightarrow LM \times LM$	$\operatorname{Map}(8, M) \rightarrow LM \times LM$	bundle of infinite dimensional manifolds, the map $\operatorname{Map}(8, M) \rightarrow LM \times LM$
cardona-qft-methods_ F03447	345	■ 1.000	$M \rightarrow M \times M$	$M \rightarrow M \times M$	inverse image of the normal bundle to $M \rightarrow M \times M$. Thus we have a
cardona-qft-methods_ F03448	345	■ 0.927	$\tau_\Delta : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$	$\tau_\Delta : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$	Thom-Pontrjagin collapse map $\tau_\Delta : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$.
cardona-qft-methods_ F03449	345	■ 0.745	$\operatorname{Map}(8, M) \rightarrow LM$	$\operatorname{Map}(8, M) \rightarrow LM$	There is a concatenation map $\operatorname{Map}(8, M) \rightarrow LM$ that extends the usual
cardona-qft-methods_ F03450	345	■ 1.000	$\Omega M \times \Omega M \rightarrow \Omega M$	$\Omega M \times \Omega M \rightarrow \Omega M$	Pontrjagin product $\Omega M \times \Omega M \rightarrow \Omega M$. Composing the Kunneth map with
cardona-qft-methods_ F03451	346	■ 1.000	LM^{-TM}	LM^{-TM}	where LM^{-TM} is the Thom spectrum of a virtual bundle over the loop
cardona-qft-methods_ F03452	346	■ 1.000	$-TM \rightarrow M$	$-TM \rightarrow M$	normal bundle $-TM \rightarrow M$ (a virtual bundle of dimension $-n$)
cardona-qft-methods_ F03453	346	■ 1.000	$M^{-TM} \simeq F(M_+, S)$	$M^{-TM} \simeq F(M_+, S)$	and the Atiyah duality $M^{-TM} \simeq F(M_+, S)$, that identifies M^{-TM}
cardona-qft-methods_ F03454	346	■ 1.000	M^{-TM}	M^{-TM}	and the Atiyah duality $M^{-TM} \simeq F(M_+, S)$, that identifies M^{-TM}
cardona-qft-methods_ F03455	346	■ 0.999	M_+^b	M_+^b	as the Spanier-Whitehead dual of M_+^b . Therefore $H_*(M^{-TM}) \cong$
cardona-qft-methods_ F03456	346	■ 0.999	$H_*(M^{-TM}) \cong$	$H_*(M^{-TM}) \cong$	as the Spanier-Whitehead dual of M_+^b . Therefore $H_*(M^{-TM}) \cong$
cardona-qft-methods_ F03457	346	■ 1.000	$H_*(F(M_+, S)) \cong H^{-*}(M)$	$H_*(F(M_+, S)) \cong H^{-*}(M)$	$H_*(F(M_+, S)) \cong H^{-*}(M)$.
cardona-qft-methods_ F03458	346	■ 1.000	$e : M \hookrightarrow \mathbb{R}^k$	$e : M \hookrightarrow \mathbb{R}^k$	given an embedding $e : M \hookrightarrow \mathbb{R}^k$, let $\nu(e) \rightarrow M$ the normal bundle of the
cardona-qft-methods_ F03459	346	■ 1.000	$\nu(e) \rightarrow M$	$\nu(e) \rightarrow M$	given an embedding $e : M \hookrightarrow \mathbb{R}^k$, let $\nu(e) \rightarrow M$ the normal bundle of the
cardona-qft-methods_ F03460	346	■ 0.999	$M^{\nu(e)}$	$M^{\nu(e)}$	embedding and denote by $M^{\nu(e)}$ its Thom space. Define the spectrum
cardona-qft-methods_ F03461	346	■ 0.894	$M \times M \rightarrow M^{TM}$	$M \times M \rightarrow M^{TM}$	collapse map $M \times M \rightarrow M^{TM}$ that can be twisted by the bundle $-2TM$
cardona-qft-methods_ F03462	346	■ 0.894	$-2TM$	$-2TM$	collapse map $M \times M \rightarrow M^{TM}$ that can be twisted by the bundle $-2TM$
cardona-qft-methods_ F03463	346	■ 0.997	$\operatorname{Map}(8, M) \rightarrow$	$\operatorname{Map}(8, M) \rightarrow$	In a similar fashion, we can twist the concatenation map $\operatorname{Map}(8, M) \rightarrow$
cardona-qft-methods_ F03464	346	■ 1.000	$ev^*(-TM)$	$ev^*(-TM)$	LM by the virtual bundle $ev^*(-TM)$ to obtain a map of spectra
cardona-qft-methods_ F03465	346	■ 0.481	$\operatorname{Map}(8, M)^{ev^*(-TM)} \rightarrow LM^{ev^*(-TM)}$	$\operatorname{Map}(8, M)^{ev^*(-TM)} \rightarrow LM^{ev^*(-TM)}$	$\operatorname{Map}(8, M)^{ev^*(-TM)} \rightarrow LM^{ev^*(-TM)}$, and the Thom-Pontrjagin collapse
cardona-qft-methods_ F03466	346	■ 0.998	$ev^*(-2TM)$	$ev^*(-2TM)$	map $\tau_\Delta : LM \times LM \rightarrow \operatorname{Map}(8, M)^{ev^*(TM)}$ by $ev^*(-2TM)$ to get a map of

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cardona-qft-methods_ F03467	346	■ 0.999	$X_{\{+\}}$	X_+	^b For a space X , the disjoint union of X with a point is denoted by X_+ . For a space X
cardona-qft-methods_ F03468	346	■ 1.000	$F\left(X_{\{+\}}, Y\right)$	$F(X_+, Y)$	the collection of unbased maps from X to a spectrum Y is denoted by $F(X_+, Y)$.
cardona-qft-methods_ F03469	347	■ 0.716	$1.2\left(\text{\texttt{^{\{7\}}}}\right)$	$1.2\left(\text{\texttt{^{\{7\}}}}\right)$	Theorem 1.2 ⁽⁷⁾ . <i>There exists a ring spectrum structure on the Thom</i>
cardona-qft-methods_ F03470	347	■ 0.811	$L M^{\mathrm{e} v^*}(-T M)$	$LM^{ev^*}(-TM)$	<i>spectrum $LM^{ev^*}(-TM)$ which induces the Chas-Sullivan product on homol-</i>
cardona-qft-methods_ F03471	347	■ 0.999	$1.3\left(\text{\texttt{\{ }^{\{\mathbf{7\}}}}\right)$	$1.3\left(\text{\texttt{^{\{7\}}}}\right)$	Theorem 1.3 ⁽⁷⁾ . <i>For M a simply connected, oriented, closed manifold,</i>
cardona-qft-methods_ F03472	348	■ 0.979	$\bar{U}, G, U$	$\bar{U}, G, U$	<i>orbifold chart</i> on X is a triple $(\bar{U}, G, U)$, where $\bar{U}$ is a connected manifold,
cardona-qft-methods_ F03473	348	■ 0.979	$\bar{U}$	$\bar{U}$	<i>orbifold chart</i> on X is a triple $(\bar{U}, G, U)$, where $\bar{U}$ is a connected manifold,
cardona-qft-methods_ F03474	348	■ 1.000	$\bar{U} / G$	$\bar{U} / G$	to $\bar{U} / G$.
cardona-qft-methods_ F03475	348	■ 0.910	$\bar{U}_{\alpha}, G_{\alpha}, U_{\alpha}$	$\bar{U}_{\alpha}, G_{\alpha}, U_{\alpha}$	$(\bar{U}_{\alpha}, G_{\alpha}, U_{\alpha})$, that is compatible and covers X . An <i>orbifold structure</i> on
cardona-qft-methods_ F03476	348	■ 1.000	$[M / G]$	$[M / G]$	to an orbifold structure that we will denoted by $[M / G]$.
cardona-qft-methods_ F03477	348	■ 0.936	$x \in X$	$x \in X$	Given a point $x \in X$, take a chart $(\bar{U}, G, U)$ around x , let $\bar{x} \in \bar{U}$ be a
cardona-qft-methods_ F03478	348	■ 0.936	$\bar{x} \in \bar{U}$	$\bar{x} \in \bar{U}$	Given a point $x \in X$, take a chart $(\bar{U}, G, U)$ around x , let $\bar{x} \in \bar{U}$ be a
cardona-qft-methods_ F03479	348	■ 1.000	G_x	G_x	if G_x is trivial, and <i>singular</i> otherwise.
cardona-qft-methods_ F03480	348	■ 1.000	$\mathbb{C}^2 - \{0\}$	$\mathbb{C}^2 - \{0\}$	Example 2.3. Consider the action of S^1 on $\mathbb{C}^2 - \{0\}$ given by $\lambda(z_1, z_2) =$
cardona-qft-methods_ F03481	348	■ 1.000	$\lambda(z_1, z_2) =$	$\lambda(z_1, z_2) =$	Example 2.3. Consider the action of S^1 on $\mathbb{C}^2 - \{0\}$ given by $\lambda(z_1, z_2) =$
cardona-qft-methods_ F03482	348	■ 0.999	$(\lambda^2 z, \lambda z)$	$(\lambda^2 z, \lambda z)$	$(\lambda^2 z, \lambda z)$, it is a an action with finite stabilizers, and therefore the under-
cardona-qft-methods_ F03483	348	■ 0.999	$\mathcal{X}$	$\mathcal{X}$	To an orbifold $\mathcal{X}$ we can associate a topological space $B\mathcal{X}$ called the
cardona-qft-methods_ F03484	348	■ 0.999	$B \mathcal{X}$	$B\mathcal{X}$	To an orbifold $\mathcal{X}$ we can associate a topological space $B\mathcal{X}$ called the
cardona-qft-methods_ F03485	348	■ 1.000	$\mathcal{X} = [M / G]$	$\mathcal{X} = [M / G]$	$\mathcal{X} = [M / G]$ with G a finite group acting on M , $B\mathcal{X}$ is homotopy equivalent
cardona-qft-methods_ F03486	348	■ 1.000	$M, B \mathcal{X}$	$M, B\mathcal{X}$	$\mathcal{X} = [M / G]$ with G a finite group acting on M , $B\mathcal{X}$ is homotopy equivalent
cardona-qft-methods_ F03487	349	■ 1.000	$E G \times_G M$	$EG \times_G M$	to the homotopy orbit space $EG \times_G M$, where $EG \rightarrow BG$ is the universal
cardona-qft-methods_ F03488	349	■ 1.000	$EG \rightarrow BG$	$EG \rightarrow BG$	to the homotopy orbit space $EG \times_G M$, where $EG \rightarrow BG$ is the universal
cardona-qft-methods_ F03489	349	■ 0.993	$ \mathcal{X} $	$ \mathcal{X} $	lying space $ \mathcal{X} $ and the classifying space $B\mathcal{X}$; there is a map $B\mathcal{X} \rightarrow \mathcal{X} $,
cardona-qft-methods_ F03490	349	■ 0.993	$B \mathcal{X} \rightarrow \mathcal{X} $	$B\mathcal{X} \rightarrow \mathcal{X} $	lying space $ \mathcal{X} $ and the classifying space $B\mathcal{X}$; there is a map $B\mathcal{X} \rightarrow \mathcal{X} $,
cardona-qft-methods_ F03491	349	■ 1.000	$[P_G M / G]$	$[P_G M / G]$	group, Lupercio and Uribe ¹³ constructed the loop groupoid $[P_G M / G]$ as
cardona-qft-methods_ F03492	349	■ 1.000	$M \times_G EG$	$M \times_G EG$	the homotopy quotient $M \times_G EG$, i.e.
cardona-qft-methods_ F03493	349	■ 1.000	$[X / G]$	$[X / G]$	Denote by $[X / G]$ the topological groupoid whose objects are X and whose

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cardona-qft-methods_ F03494	349	■ 1.000	$X \times G$	$X \times G$	morphisms are $X \times G$ with $s(x, g) = x$ and $t(x, g) = xg$.
cardona-qft-methods_ F03495	349	■ 1.000	$s(x, g)=x$	$s(x, g) = x$	morphisms are $X \times G$ with $s(x, g) = x$ and $t(x, g) = xg$.
cardona-qft-methods_ F03496	349	■ 1.000	$t(x, g)=x \ g$	$t(x, g) = xg$	morphisms are $X \times G$ with $s(x, g) = x$ and $t(x, g) = xg$.
cardona-qft-methods_ F03497	349	■ 1.000	$[\mathbf{R} / \mathbf{Z}]$	$[\mathbf{R}/\mathbf{Z}]$	are the functors from the groupoid $[\mathbf{R}/\mathbf{Z}]$ to $[X/G]$ and whose morphisms
cardona-qft-methods_ F03498	349	■ 1.000	$\left[P_{\{G\}} X / G\right]$	$[P_G X/G]$	Definition 3.1. The loop groupoid for $[X/G]$ is the groupoid $[P_G X/G]$
cardona-qft-methods_ F03499	350	■ 1.000	$e_{\{t\}}: P_{\{g\}} M \rightarrowtail M$	$e_t : P_g M \rightarrow M$	Now, for $t \in \mathbb{R}$ let $e_t : P_g M \rightarrow M$ be the evaluation of a map at the
cardona-qft-methods_ F03500	350	■ 1.000	$t: e_{\{t\}}(f):=f(t)$	$t : e_t(f) := f(t)$	time t : $e_t(f) := f(t)$. Consider the space of composable maps
cardona-qft-methods_ F03501	350	■ 1.000	$e_{\{0\}} \circ \pi_1$	$e_0 \circ \pi_1$	where the downward arrow $e_0 \circ \pi_1$ is the evaluation at zero of the first map.
cardona-qft-methods_ F03502	350	■ 1.000	$\Delta_{\{g\}}$	Δ_g	Δ_g and therefore we can construct the Thom-Pontrjagin collapse maps
cardona-qft-methods_ F03503	350	■ 0.820	$P_{\{g\}} M \times P_{\{h\}} M \rightarrowtail P_{\{g\}} M: \{ \}_{_1} \times_0 P_{\{h\}} M^{\pi_1^*} e_0^* N_g$	$P_g M \times P_h M \rightarrow P_g M: {}_1 \times_0 P_h M^{\pi_1^*} e_0^* N_g$	$P_g M \times P_h M \rightarrow P_g M: {}_1 \times_0 P_h M^{\pi_1^*} e_0^* N_g$, and obtain a map of spectra
cardona-qft-methods_ F03504	350	■ 0.806	$P_{\{g\}} M^{\{-e_{\{0\}}^*\} T M} \wedge P_{\{h\}} M^{\{-e_{\{0\}}^*\} T M} \rightarrowtail P_{\{g\}} M: \{ \}_{_1} \times_0 P_{\{h\}} M^{\{-\pi_1^*\} e_0^* T M}$	$P_g M^{-e_0^* T M} \wedge P_h M^{-e_0^* T M} \rightarrow P_g M: {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* T M$	$P_g M^{-e_0^* T M} \wedge P_h M^{-e_0^* T M} \rightarrow P_g M: {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* T M$. The concatenation
cardona-qft-methods_ F03505	350	■ 0.991	$P_{\{g\}} M: \{ \}_{_1} \times_0 P_{\{h\}} M$	$P_g M: {}_1 \times_0 P_h M$	of paths in $P_g M: {}_1 \times_0 P_h M$ produces a map $P_g M: {}_1 \times_0 P_h M \rightarrow P_{gh} M$ which
cardona-qft-methods_ F03506	350	■ 0.991	$P_{\{g\}} M: \{ \}_{_1} \times_0 P_{\{h\}} M \rightarrowtail P_{\{g h\}} M$	$P_g M: {}_1 \times_0 P_h M \rightarrow P_{gh} M$	of paths in $P_g M: {}_1 \times_0 P_h M$ produces a map $P_g M: {}_1 \times_0 P_h M \rightarrow P_{gh} M$ which
cardona-qft-methods_ F03507	350	■ 0.926	$P_{\{g\}} M: \{ \}_{_1} \times_0 P_{\{h\}} M^{\{-\pi_1^*\} e_0^* T M} \rightarrowtail P_{\{g h\}} M^{\{-e_{\{0\}}^*\} T M}$	$P_g M: {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* T M \rightarrow P_{gh} M^{-e_0^* T M}$	induces the map of spectra $P_g M: {}_1 \times_0 P_h M^{-\pi_1^*} e_0^* T M \rightarrow P_{gh} M^{-e_0^* T M}$ that
cardona-qft-methods_ F03508	350	■ 1.000	$\mu_{\{g, h\}}: P_{\{g\}} M^{\{-e_{\{0\}}^*\} T M} \wedge P_{\{h\}} M^{\{-e_{\{0\}}^*\} T M} \rightarrowtail P_{\{g h\}} M^{\{-e_{\{0\}}^*\} T M}$	$\mu_{g,h} : P_g M^{-e_0^* T M} \wedge P_h M^{-e_0^* T M} \rightarrow P_{gh} M^{-e_0^* T M}$	defines the map $\mu_{g,h} : P_g M^{-e_0^* T M} \wedge P_h M^{-e_0^* T M} \rightarrow P_{gh} M^{-e_0^* T M}$. Define
cardona-qft-methods_ F03509	350	■ 1.000	$\mu_{\{g, h\}}$	$\mu_{g,h}$	which by assembling the maps $\mu_{g,h}$ define a map that is associative up to
cardona-qft-methods_ F03510	350	■ 0.546	$3.1 \left(\{ \}^{\mathbf{14}} \right) . P_{\{G\}} M^{\{-T M\}}$	$3.1 \left(\mathbf{14} \right) . P_G M^{-T M}$	Theorem 3.1 (14). $P_G M^{-T M}$ is a ring spectrum with a G -action
cardona-qft-methods_ F03511	350	■ 1.000	$H_{\{*\}} \left(P_{\{G\}} M^{\{-T M\}} ; \mathbb{Z} \right)$	$H_* \left(P_G M^{-T M}; \mathbb{Z} \right)$	$H_* (P_G M^{-T M}; \mathbb{Z})$, the induced ring structure on the invariant set
cardona-qft-methods_ F03512	351	■ 1.000	$C^{\{*\}}(M ; \mathbb{Q}) \# G$	$C^*(M; \mathbb{Q}) \# G$	recovered as the Hochschild cohomology ring of the dga $C^*(M; \mathbb{Q}) \# G$, i.e.
cardona-qft-methods_ F03513	351	■ 1.000	$C^{\{*\}}(M) \# G$	$C^*(M) \# G$	where $C^*(M) \# G$ is the convolution differential graded algebra,
cardona-qft-methods_ F03514	351	■ 1.000	$C^{\{*\}}(X) \# G$	$C^*(X) \# G$	space X and giving an explicit cofibrant replacement for $C^*(X) \# G$ as a
cardona-qft-methods_ F03515	351	■ 1.000	$\left(C^{\{*\}}(X) \# G \right)^e$	$(C^*(X) \# G)^e$	$(C^*(X) \# G)^e$ -module, together with an explicit diagonal map. It is shown
cardona-qft-methods_ F03516	351	■ 1.000	$g \cdot (x \otimes h) \mapsto g(x) \otimes g h g^{-1}$	$g \cdot (x \otimes h) \mapsto g(x) \otimes g h g^{-1}$	$g \cdot (x \otimes h) \mapsto g(x) \otimes g h g^{-1}$, then we have that
cardona-qft-methods_ F03517	351	■ 0.951	$H H^{\{*\}} \left(C^{\{*\}}(X) \# G, C^{\{*\}}(X) \# G \right) \cong \operatorname{Ext}_{\mathbb{Z}}^* \left(\mathbb{Z}, \mathcal{R} \operatorname{Hom}_{C^{\{*\}}(X)^e} \left(C^{\{*\}}(X), C^{\{*\}}(X) \# G \right) \right)$	$H H^* (C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Ext}_{\mathbb{Z} G}^* (\mathbb{Z}, \mathcal{R} \operatorname{Hom}_{C^*(X)^e} (C^*(X), C^*(X) \# G))$	$H H^* (C^*(X) \# G, C^*(X) \# G) \cong \operatorname{Ext}_{\mathbb{Z} G}^* (\mathbb{Z}, \mathcal{R} \operatorname{Hom}_{C^*(X)^e} (C^*(X), C^*(X) \# G))$
cardona-qft-methods_ F03518	351	■ 0.984	$P_{\{G\}} X, \operatorname{in}^3$	$P_G X, \operatorname{in}^3$	Using a cosimplicial description for the space $P_G X$, in^3 it is proved that

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cardona-qft-methods_F03519	351	■ 1.000	$P_{\{G\}} M$	$P_G M$	erties of Atiyah duality and a cosimplicial model of $P_G M$ in ³ it is proved
cardona-qft-methods_F03520	352	■ 1.000	$E \ G$	EG	sort of Poincare duality for EG .
cardona-qft-methods_F03521	352	■ 1.000	$E \ G_{\{1\}} \ \subset \cdots \ \subset E \ G_{\{n\}} \ \subset E \ G_{\{n+1\}} \ \subset \cdots \ E \ G$	$EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots EG$	Let $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots EG$ be a finite dimensional
cardona-qft-methods_F03522	352	■ 1.000	$E \ G \rightarrow$	$EG \rightarrow$	closed manifold approximation of the universal G -principal bundle $EG \rightarrow$
cardona-qft-methods_F03523	352	■ 1.000	$B \ G$	BG	BG . Consider the maps
cardona-qft-methods_F03524	352	■ 1.000	$(f, \ \lambda) \ \text{in} \ P_{\{G\}} M \ \times_{\{G\}} E \ G_{\{n\}}$	$(f, \lambda) \in P_G M \times_G EG_n$	$(f, \lambda) \in P_G M \times_G EG_n$ also by e_t . Let us call $M_n = M \times_G EG_n$.
cardona-qft-methods_F03525	352	■ 1.000	$e_{\{t\}}$	e_t	$(f, \lambda) \in P_G M \times_G EG_n$ also by e_t . Let us call $M_n = M \times_G EG_n$.
cardona-qft-methods_F03526	352	■ 1.000	$M_{\{n\}} = M \ \times_{\{G\}} E \ G_{\{n\}}$	$M_n = M \times_G EG_n$	$(f, \lambda) \in P_G M \times_G EG_n$ also by e_t . Let us call $M_n = M \times_G EG_n$.
cardona-qft-methods_F03527	352	■ 1.000	$M_{\{n\}}$	M_n	gent bundle of M_n under e_0
cardona-qft-methods_F03528	352	■ 1.000	$e_{\{0\}}$	e_0	gent bundle of M_n under e_0
cardona-qft-methods_F03529	352	■ 1.000	$E \ G_{\{n\}} \ \subset E \ G_{\{n+1\}} \ \subset \cdots \ E \ G$	$EG_n \subset EG_{n+1} \subset \cdots EG$	The inclusions $EG_n \subset EG_{n+1} \subset \cdots EG$ induce an inverse system of maps
cardona-qft-methods_F03530	352	■ 1.000	$\mathbb{L}(\left(M \ \times_{\{G\}} E \ G\right)^{-T} \ \left(M \ \times_{\{G\}} E \ G\right))$	$\mathbb{L}(M \times_G EG)^{-T(M \times_G EG)}$	that defines a pro-ring spectrum called $\mathbb{L}(M \times_G EG)^{-T(M \times_G EG)}$. The ho-
cardona-qft-methods_F03531	353	■ 0.548	$3.5 \left(\left(\mathbf{3} \right) \right)$	$3.5 \left(\mathbf{3} \right)$	Theorem 3.5 (³). <i>If M is simply connected and G is finite</i>
cardona-qft-methods_F03532	353	■ 1.000	$M = *$	$M = *$	When $M = *$ the orbifold string topology of $[*/G]$ is precisely the pro-
cardona-qft-methods_F03533	353	■ 1.000	$L \ B \ G^{-T} B \ G$	LBG^{-TBG}	ring spectrum LBG^{-TBG} which has been studied in, ¹⁰¹¹ and ¹² for general
cardona-qft-methods_F03534	353	■ 0.919	$\mathrm{L}[* \ / \ G]$	$L[* / G]$	$L[* / G]$ is the groupoid $[G^{ad} / G]$ that is the translation groupoid associated
cardona-qft-methods_F03535	353	■ 0.919	$\left[G^{\wedge \{a \ d\}} \ / \ G \right]$	$[G^{ad} / G]$	$L[* / G]$ is the groupoid $[G^{ad} / G]$ that is the translation groupoid associated
cardona-qft-methods_F03536	353	■ 1.000	$C(g)$	$C(g)$	where the union runs over the conjugacy classes of elements in G and $C(g)$
cardona-qft-methods_F03537	353	■ 1.000	$H_{\{*\}} \left(P_{\{G\}} M^{-T} M \ ; \ \mathbb{Q} \right)^{\{G\}} \ \cong \ \mathbb{Q} \ G$	$H_* \left(P_G M^{-TM}; \mathbb{Q} \right)^G \cong \mathbb{Q} G$	have $H_*(P_G M^{-TM}; \mathbb{Q})^G \cong \mathbb{Q} G$ concentraded in degree zero and where the
cardona-qft-methods_F03538	353	■ 1.000	$B \ G_{\{n\}} := E \ G_{\{n\}} \ / \ G$	$BG_n := EG_n / G$	dimensional closed manifolds. By definition $BG_n := EG_n / G$ and the pro-
cardona-qft-methods_F03539	353	■ 1.000	$\left(G^{\wedge \{a \ d\}} \ \times_{\{G\}} E \ G_{\{n\}} \right)^{-T} B \ G_{\{n\}}$	$\left(G^{ad} \times_G EG_n \right)^{-TBG_n}$	spaces $\left(G^{ad} \times_G EG_n \right)^{-TBG_n}$.
cardona-qft-methods_F03540	353	■ 0.999	$G^{\wedge \{a \ d\}} \ \times_{\{G\}} E \ G_{\{n\}}$	$G^{ad} \times_G EG_n$	The space $G^{ad} \times_G EG_n$ decomposes into a disjoint union of spaces
cardona-qft-methods_F03541	353	■ 1.000	$\bigsqcup_{\{g\}} E \ G_{\{n\}} \ / \ C(g)$	$\bigsqcup_{(g)} EG_n / C(g)$	$\bigsqcup_{(g)} EG_n / C(g)$ and the tangent space $T \left(G^{ad} \times_G EG_n \right)$ is isomorphic to
cardona-qft-methods_F03542	353	■ 1.000	$T \left(G^{\wedge \{a \ d\}} \ \times_{\{G\}} E \ G_{\{n\}} \right)$	$T \left(G^{ad} \times_G EG_n \right)$	$\bigsqcup_{(g)} EG_n / C(g)$ and the tangent space $T \left(G^{ad} \times_G EG_n \right)$ is isomorphic to
cardona-qft-methods_F03543	353	■ 1.000	$\bigsqcup_{\{g\}} T \left(E \ G_{\{n\}} \ / \ C(g) \right)$	$\bigsqcup_{(g)} T \left(EG_n / C(g) \right)$	$\bigsqcup_{(g)} T \left(EG_n / C(g) \right)$. Since any model of EG is also a model of EH for any
cardona-qft-methods_F03544	353	■ 1.000	$E \ H$	EH	$\bigsqcup_{(g)} T \left(EG_n / C(g) \right)$. Since any model of EG is also a model of EH for any

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cardona-qft-methods_ F03545	353	■ 1.000	$H \subseteq G$	$H \subseteq G$	subgroup $H \subseteq G$ we can take $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots$ to
cardona-qft-methods_ F03546	353	■ 1.000	$E G_{\{1\}} \subseteq \cdots \subseteq E G_{\{n\}} \subseteq E G_{\{n+1\}} \subseteq \cdots$	$EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots$	subgroup $H \subseteq G$ we can take $EG_1 \subset \cdots \subset EG_n \subset EG_{n+1} \subset \cdots$ to
cardona-qft-methods_ F03547	353	■ 1.000	$EC(g)$	$EC(g)$	be an approximation of each $EC(g)$. Therefore we have that the pro-ring
cardona-qft-methods_ F03548	353	■ 1.000	$\bigoplus_{(g)} H_* \left(\left(BC(g)_n \right)^{-TBC(g)_n} \right)$	$\bigoplus_{(g)} H_* \left(\left(BC(g)_n \right)^{-TBC(g)_n} \right)$	$\bigoplus_{(g)} H_* \left(\left(BC(g)_n \right)^{-TBC(g)_n} \right)$, which is by Atiyah Duality the inverse sys-
cardona-qft-methods_ F03549	353	■ 1.000	$\bigoplus_{(g)} H^* \left(BC(g)_n \right)$	$\bigoplus_{(g)} H^* \left(BC(g)_n \right)$	tem of $\bigoplus_{(g)} H^* \left(BC(g)_n \right)$, that is just $\bigoplus_{(g)} H^* \left(BC(g) \right)$. To get the ring
cardona-qft-methods_ F03550	353	■ 1.000	$\bigoplus_{(g)} H^* \left(BC(g) \right)$	$\bigoplus_{(g)} H^* \left(BC(g) \right)$	tem of $\bigoplus_{(g)} H^* \left(BC(g)_n \right)$, that is just $\bigoplus_{(g)} H^* \left(BC(g) \right)$. To get the ring
cardona-qft-methods_ F03551	354	■ 1.000	$k_n = \dim \left(EG_n \right)$	$k_n = \dim \left(EG_n \right)$	with $k_n = \dim \left(EG_n \right)$, is Poincaré dual to the pull-back map in cohomology.
cardona-qft-methods_ F03552	354	■ 0.979	$H_*^{\text{pro}} \left(\mathbb{L} B G^{-T B G} \right)$	$H_*^{\text{pro}} \left(\mathbb{L} B G^{-T B G} \right)$	Therefore the ring structure in $H_*^{\text{pro}} \left(\mathbb{L} B G^{-T B G} \right)$ is obtained by taking
cardona-qft-methods_ F03553	354	■ 1.000	$H^* \left(BC(g) \right)$	$H^* \left(BC(g) \right)$	classes in $H^* \left(BC(g) \right)$ and $H^* \left(BC(h) \right)$, pulling them back to $H^* \left(B \left(C(g) \cap \right.$
cardona-qft-methods_ F03554	354	■ 1.000	$H^* \left(BC(h) \right)$	$H^* \left(BC(h) \right)$	classes in $H^* \left(BC(g) \right)$ and $H^* \left(BC(h) \right)$, pulling them back to $H^* \left(B \left(C(g) \cap \right.$
cardona-qft-methods_ F03555	354	■ 1.000	$H^* \left(B \left(C(g) \cap \right.$	$H^* \left(B \left(C(g) \cap \right.$	classes in $H^* \left(BC(g) \right)$ and $H^* \left(BC(h) \right)$, pulling them back to $H^* \left(B \left(C(g) \cap \right.$
cardona-qft-methods_ F03556	354	■ 0.835	$C(h) \left. \right)$	$C(h) \left. \right)$	$C(h) \left. \right)$ and then pushing them forward to $H^* \left(BC(gh) \right)$. This procedure is
cardona-qft-methods_ F03557	354	■ 0.835	$H^* \left(BC(gh) \right)$	$H^* \left(BC(gh) \right)$	$C(h) \left. \right)$ and then pushing them forward to $H^* \left(BC(gh) \right)$. This procedure is
cardona-qft-methods_ F03558	354	■ 1.000	$H^* \left(B G \right) \otimes \mathbb{Z} G$	$H^* \left(B G \right) \otimes \mathbb{Z} G$	is just $H^* \left(B G \right) \otimes \mathbb{Z} G$.
cardona-qft-methods_ F03559	354	■ 1.000	$L B G^{-ad}$	$L B G^{-ad}$	of Christensen and Isaksen ⁸) to a spectrum $L B G^{-ad}$. When G is finite this
cardona-qft-methods_ F03560	354	■ 1.000	$L B G$	$L B G$	is just the suspension spectrum of the space $L B G$ and therefore we have
cardona-qft-methods_ F03561	354	■ 1.000	$L B G \cong E G \times_G G$	$L B G \cong E G \times_G G$	G on itself by conjugation $L B G \cong E G \times_G G$.
cardona-qft-methods_ F03562	354	■ 1.000	$\mu : G \times G \rightarrow G$	$\mu : G \times G \rightarrow G$	Furthermore when G is finite the multiplication map $\mu : G \times G \rightarrow G$ is a
cardona-qft-methods_ F03563	354	■ 1.000	$m_! : G_+ \rightarrow G \wedge G_+$	$m_! : G_+ \rightarrow G \wedge G_+$	(trivial) covering space and defines a stable transfer map $m_! : G_+ \rightarrow G \wedge G_+$.
cardona-qft-methods_ F03564	354	■ 1.000	$E G \times_G G \rightarrow E G \times_G (G \times G)$	$E G \times_G G \rightarrow E G \times_G (G \times G)$	spectrum of the Borel construction $E G \times_G G \rightarrow E G \times_G (G \times G)$.
cardona-qft-methods_ F03565	354	■ 1.000	$E G \times_G (G \times G) \rightarrow (E G \times_G G) \times (E G \times_G G)$	$E G \times_G (G \times G) \rightarrow (E G \times_G G) \times (E G \times_G G)$	$E G \times_G (G \times G) \rightarrow (E G \times_G G) \times (E G \times_G G)$ we obtain a map $E G \times_G G \rightarrow$
cardona-qft-methods_ F03566	354	■ 1.000	$E G \times_G G \rightarrow$	$E G \times_G G \rightarrow$	$E G \times_G (G \times G) \rightarrow (E G \times_G G) \times (E G \times_G G)$ we obtain a map $E G \times_G G \rightarrow$
cardona-qft-methods_ F03567	355	■ 0.887	$(E G \times_G G) \times (E G \times_G G)$	$(E G \times_G G) \times (E G \times_G G)$	$(E G \times_G G) \times (E G \times_G G)$, that induces on cohomology a product which
cardona-qft-methods_ F03568	355	■ 0.989	$\text{Map}(\Theta, B G) \rightarrow L B G$	$\text{Map}(\Theta, B G) \rightarrow L B G$	The map $\text{Map}(\Theta, B G) \rightarrow L B G$ is a fibration with fiber $\Omega B G$, which
cardona-qft-methods_ F03569	355	■ 0.989	$\Omega B G$	$\Omega B G$	The map $\text{Map}(\Theta, B G) \rightarrow L B G$ is a fibration with fiber $\Omega B G$, which
cardona-qft-methods_ F03570	355	■ 0.927	$H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow$	$H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow$	cohomology. Using this transfer map we obtain a map $H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow$
cardona-qft-methods_ F03571	355	■ 0.751	$H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow H^* \left(L B G \right)$	$H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow H^* \left(L B G \right)$	Composing the map $H^* \left(\text{Map}(\Theta, B G) \right) \rightarrow H^* \left(L B G \right)$ with the natural

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cardona-qft-methods_ F03572	355	■ 0.915	$H^{\{*\}}(L \ B \ G) \ \otimes H^{\{*\}}(L \ B \ G) \ \rightarrow H^{\{*\}}(\operatorname{Map}(\Theta, \ B \ G))$	$H^*(LBG) \otimes H^*(LBG) \rightarrow H^*(\operatorname{Map}(\Theta, BG))$	map $H^*(LBG) \otimes H^*(LBG) \rightarrow H^*(\operatorname{Map}(\Theta, BG))$ we obtain a product
cardona-qft-methods_ F03573	357	■ 1.000	$\mathbb{P}_{\mathbb{P}}^n$	$\mathbb{P}_k^{n-1}$	We define a special type of hypersurface varieties inside $\mathbb{P}_k^{n-1}$ arising from
cardona-qft-methods_ F03574	358	■ 0.999	$\{ \}^{\{1,4\}}$	$1,4$	given by the class of the product, ^{1,4} .
cardona-qft-methods_ F03575	358	■ 1.000	$\mathcal{V}_k$	$\mathcal{V}_k$	algebraic varieties $\mathcal{V}_k$ defined over it. Then we look at the abelian group
cardona-qft-methods_ F03576	358	■ 1.000	$K_0(\mathcal{V}_k)$	$K_0(\mathcal{V}_k)$	$K_0(\mathcal{V}_k)$ of isomorphism classes $[X]$ of varieties X over the field k with the
cardona-qft-methods_ F03577	358	■ 1.000	$Y \subset X$	$Y \subset X$	where $Y \subset X$ is a closed subvariety of X . Notice that setting $X = Y$ gives
cardona-qft-methods_ F03578	358	■ 1.000	$X=Y$	$X = Y$	where $Y \subset X$ is a closed subvariety of X . Notice that setting $X = Y$ gives
cardona-qft-methods_ F03579	358	■ 1.000	$\chi: \mathcal{V}_k \rightarrow R$	$\chi: \mathcal{V}_k \rightarrow R$	for invariants. We call an <i>additive invariant</i> a map $\chi: \mathcal{V}_k \rightarrow R$, with values
cardona-qft-methods_ F03580	358	■ 1.000	$\chi(X)=\chi(Y)$	$\chi(X) = \chi(Y)$	(i) $\chi(X) = \chi(Y)$ if X and Y are isomorphic
cardona-qft-methods_ F03581	358	■ 1.000	$\chi(X \setminus Y)=\chi(X)-\chi(Y)$	$\chi(X \setminus Y) = \chi(X) - \chi(Y)$	(ii) $\chi(X \setminus Y) = \chi(X) - \chi(Y)$, for $Y \subset X$ closed
cardona-qft-methods_ F03582	358	■ 1.000	$\chi(X \times Y)=\chi(X) \chi(Y)$	$\chi(X \times Y) = \chi(X)\chi(Y)$	(iii) $\chi(X \times Y) = \chi(X)\chi(Y)$
cardona-qft-methods_ F03583	358	■ 1.000	$\mathbb{L}=\left[\mathbb{A}_k^1\right]$	$\mathbb{L} = [\mathbb{A}_k^1]$	For the following discussion, let $\mathbb{L} = [\mathbb{A}_k^1]$ be the equivalence class in the
cardona-qft-methods_ F03584	358	■ 1.000	$\mathbb{A}_k^1$	$\mathbb{A}_k^1$	Grothendieck ring of varieties of the one dimensional affine space $\mathbb{A}_k^1$. From
cardona-qft-methods_ F03585	358	■ 1.000	$\mathbb{A}_k^n$	$\mathbb{A}_k^n$	spaces $\mathbb{A}_k^n$ are given in terms of $\mathbb{L}$ as
cardona-qft-methods_ F03586	359	■ 1.000	$k^{\times}$	$k^{\times}$	Also, let $\mathbb{T}$ be the class of the multiplicative group $k^{\times}$, which is just $\mathbb{A}_k^1$
cardona-qft-methods_ F03587	359	■ 1.000	$X \setminus (X \cap Y) = (X \cup Y) \setminus Y$	$X \setminus (X \cap Y) = (X \cup Y) \setminus Y$	<i>Since we have that $X \setminus (X \cap Y) = (X \cup Y) \setminus Y$, then</i>
cardona-qft-methods_ F03588	359	■ 1.000	$\mathbb{P}_k^{n+1} \setminus \mathbb{A}_k^{n+1} \simeq \mathbb{P}_k^n$	$\mathbb{P}_k^{n+1} \setminus \mathbb{A}_k^{n+1} \simeq \mathbb{P}_k^n$	Proof. <i>Since $\mathbb{P}_k^{n+1} \setminus \mathbb{A}_k^{n+1} \simeq \mathbb{P}_k^n$ for all $n \in \mathbb{N}$, by induction and the previous</i>
cardona-qft-methods_ F03589	359	■ 0.992	$t_1, t_2, \dots, t_n$	$t_1, t_2, \dots, t_n$	them with the variables $t_1, t_2, \dots, t_n$ of the polynomial ring $Z[t_1, t_2, \dots, t_n]$.
cardona-qft-methods_ F03590	359	■ 0.992	$Z[t_1, t_2, \dots, t_n]$	$Z[t_1, t_2, \dots, t_n]$	them with the variables $t_1, t_2, \dots, t_n$ of the polynomial ring $Z[t_1, t_2, \dots, t_n]$.
cardona-qft-methods_ F03591	360	■ 1.000	$E(T)$	$E(T)$	where T runs through all spanning trees of Γ and $E(T)$ is the set of all
cardona-qft-methods_ F03592	360	■ 1.000	$v-1$	$v-1$	spanning trees have $v-1$ edges, as otherwise there would be a cycle. Hence
cardona-qft-methods_ F03593	360	■ 1.000	$\Psi_{\Gamma}(t)$	$\Psi_{\Gamma}(t)$	the polynomial $\Psi_{\Gamma}(t)$ is a homogeneous polynomial of degree $n-v+1$ and
cardona-qft-methods_ F03594	360	■ 1.000	$n-v+1$	$n-v+1$	the polynomial $\Psi_{\Gamma}(t)$ is a homogeneous polynomial of degree $n-v+1$ and
cardona-qft-methods_ F03595	360	■ 1.000	X_{Γ}	X_{Γ}	Since $\Psi_{\Gamma}(t)$ is homogeneous, X_{Γ} is well defined as a projective variety.
cardona-qft-methods_ F03596	360	■ 1.000	$\Gamma^{\vee}$	$\Gamma^{\vee}$	Given a planar connected graph Γ , define the dual graph $\Gamma^{\vee}$ by the follow-
cardona-qft-methods_ F03597	360	■ 1.000	$\mathbb{S}^2$	$\mathbb{S}^2$	(i) embed Γ in $\mathbb{S}^2$
cardona-qft-methods_ F03598	360	■ 0.997	$\Psi_{\Gamma^{\vee}}(t)$	$\Psi_{\Gamma^{\vee}}(t)$	the resulting graph polynomials $\Psi_{\Gamma^{\vee}}(t)$ are the same up to relabeling. This

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cardona-qft-methods_ F03599	360	■ 0.820	$\{ \}^{\{1,2\}}$	$1,2$	below, ^{1,2} This makes the graph hypersurface of the dual graph $X_{\Gamma^\vee}$ a well
cardona-qft-methods_ F03600	360	■ 0.820	$X_{\{\Gamma \vee\}}$	$X_{\Gamma^\vee}$	below, ^{1,2} This makes the graph hypersurface of the dual graph $X_{\Gamma^\vee}$ a well
cardona-qft-methods_ F03601	361	■ 1.000	$\mathbb{P}_{\{k\}^{n-1}} \times \mathbb{P}_{\{k\}^{n-1}}$	$\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$	transform in $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$ and define $\mathcal{G}(\mathcal{C})$ to be its closure. Then we have
cardona-qft-methods_ F03602	361	■ 1.000	$\mathcal{G}(\mathcal{C})$	$\mathcal{G}(\mathcal{C})$	transform in $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$ and define $\mathcal{G}(\mathcal{C})$ to be its closure. Then we have
cardona-qft-methods_ F03603	361	■ 1.000	$\pi_{\{1\}}$	π_1	the two projections π_1 and π_2 of $\mathcal{G}(\mathcal{C})$ into $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$ related by
cardona-qft-methods_ F03604	361	■ 1.000	$\pi_{\{2\}}$	π_2	the two projections π_1 and π_2 of $\mathcal{G}(\mathcal{C})$ into $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$ related by
cardona-qft-methods_ F03605	361	■ 0.902	$\left(t_{\{1\}} : t_{\{2\}} : \cdots : t_{\{2n\}} \right)$	$(t_1 : t_2 : \cdots : t_{2n})$	where we used the coordinates $(t_1 : t_2 : \cdots : t_{2n})$ for $\mathbb{P}_k^{n-1} \times \mathbb{P}_k^{n-1}$.
cardona-qft-methods_ F03606	361	■ 1.000	$\pi_{\{i\}}$	π_i	where π_i are the projections given on (6). Also, we have that the Cremona
cardona-qft-methods_ F03607	361	■ 1.000	$\pi_{\{2\}} : \mathcal{G}(\mathcal{C}) \rightarrow \mathbb{P}_{\{k\}^{n-1}}$	$\pi_2 : \mathcal{G}(\mathcal{C}) \rightarrow \mathbb{P}_k^{n-1}$	and the projection $\pi_2 : \mathcal{G}(\mathcal{C}) \rightarrow \mathbb{P}_k^{n-1}$ restricts to an isomorphism
cardona-qft-methods_ F03608	362	■ 1.000	$\mathcal{G}(\mathcal{C})$	$\mathcal{G}(\mathcal{C})$	ties of $\mathcal{G}(\mathcal{C})$ (see, ¹ Lemma 1.2, Lemma 1.3, Corollary 1.4).
cardona-qft-methods_ F03609	362	■ 0.998	$X_{\{\Gamma \vee\}}$	$X_{\Gamma^\vee}$	Now, for $X_{\Gamma^\vee}$, recall that from (5), we have that the complement of Σ_n
cardona-qft-methods_ F03610	362	■ 0.998	$\Sigma_{\{n\}}$	Σ_n	Now, for $X_{\Gamma^\vee}$, recall that from (5), we have that the complement of Σ_n
cardona-qft-methods_ F03611	362	■ 1.000	$\left(k^\times \right)^{n-1}$	$(k^\times)^{n-1}$	is a copy of $(k^\times)^{n-1}$ and gives the class of $\mathbb{T}^{n-1}$.
cardona-qft-methods_ F03612	362	■ 1.000	$\mathbb{T}^{\{n-1\}}$	$\mathbb{T}^{n-1}$	is a copy of $(k^\times)^{n-1}$ and gives the class of $\mathbb{T}^{n-1}$.
cardona-qft-methods_ F03613	362	■ 1.000	$X_{\{\Gamma \vee\}} = \Sigma_{\{n\}} = \mathbb{P}_{\{k\}^{n-1}} \backslash \left(k^\times \right)^{n-1}$	$X_{\Gamma^\vee} = \Sigma_n = \mathbb{P}_k^{n-1} \backslash (k^\times)^{n-1}$	Proof. Since $X_{\Gamma^\vee} = \Sigma_n = \mathbb{P}_k^{n-1} \backslash (k^\times)^{n-1}$, the result follows from the
cardona-qft-methods_ F03614	363	■ 0.996	$t_{\{1\}}, t_{\{2\}}, \ldots, t_{\{n-1\}}$	$t_1, t_2, \dots, t_{n-1}$	Notice that choosing any $n-1$ points for $t_1, t_2, \dots, t_{n-1}$ in $\mathcal{L}$ gives a unique
cardona-qft-methods_ F03615	363	■ 1.000	$t_{\{n\}}$	t_n	value for t_n , and hence we can identify $\mathcal{L}$ with $\mathbb{P}_k^{n-2}$. Thus we find the
cardona-qft-methods_ F03616	363	■ 1.000	$\mathbb{P}_{\{k\}^{n-2}}$	$\mathbb{P}_k^{n-2}$	value for t_n , and hence we can identify $\mathcal{L}$ with $\mathbb{P}_k^{n-2}$. Thus we find the
cardona-qft-methods_ F03617	363	■ 0.997	$(n-1)$	$(n-1)$	mial is then given by the $(n-1)$ -th elementary symmetrical polynomial
cardona-qft-methods_ F03618	363	■ 1.000	$t_{\{i\}}$	t_i	in t_i ,
cardona-qft-methods_ F03619	363	■ 1.000	$\hat{t}_{\{i\}}$	$\hat{t}_i$	where $\hat{t}_i$ means that the variable t_i is omitted from the product. In order to
cardona-qft-methods_ F03620	363	■ 1.000	$X_{\{\Gamma\}} \backslash \Sigma_{\{n\}}$	$X_\Gamma \backslash \Sigma_n$	find its Grothendieck class, we first use the isomorphism between $X_\Gamma \backslash \Sigma_n$
cardona-qft-methods_ F03621	363	■ 0.930	$X_{\{\Gamma \vee\}} \backslash \Sigma_{\{n\}}$	$X_{\Gamma^\vee} \backslash \Sigma_n$	and $X_{\Gamma^\vee} \backslash \Sigma_n$ to have
cardona-qft-methods_ F03622	363	■ 1.000	$X_{\{\Gamma\}} \backslash \Sigma$	$X_\Gamma \backslash \Sigma$	From this, we can find the class of $X_\Gamma \backslash \Sigma$ by studying $\mathcal{L} \cap \Sigma_n$. First, we need
cardona-qft-methods_ F03623	363	■ 1.000	$\mathcal{L} \cap \Sigma_{\{n\}}$	$\mathcal{L} \cap \Sigma_n$	From this, we can find the class of $X_\Gamma \backslash \Sigma$ by studying $\mathcal{L} \cap \Sigma_n$. First, we need
cardona-qft-methods_ F03624	363	■ 1.000	$C = g(\mathbb{T})$	$C = g(\mathbb{T})$	$C = g(\mathbb{T})$. Then the transformation

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cardona-qft-methods_F03625	364	■ 1.000	$g(\mathbb{T})=\left[\mathbb{P}^n\right]=\frac{(\mathbb{T}-1)^n-1}{\mathbb{T}}$	$g(\mathbb{T})=[\mathbb{P}^n]=\frac{(\mathbb{T}-1)^n-1}{\mathbb{T}}$	Proof. Notice that for $g(\mathbb{T})=[\mathbb{P}^n]=\frac{(\mathbb{T}-1)^n-1}{\mathbb{T}}$, equation (13) gives
cardona-qft-methods_F03626	364	■ 1.000	$\left[\Sigma_n\right]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=g(\mathbb{T})$	$[\Sigma_n]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=g(\mathbb{T})$	Since $[\Sigma_n]=\frac{(\mathbb{T}+1)^n-1-\mathbb{T}^n}{\mathbb{T}}=g(\mathbb{T})$, using the previous result gives that
cardona-qft-methods_F03627	364	■ 1.000	$\mathcal{S}_n$	$\mathcal{S}_n$	On the other hand, consider the variety $\mathcal{S}_n$ generated by the ideal
cardona-qft-methods_F03628	364	■ 1.000	Σ_{n-1}	Σ_{n-1}	the union of its coordinate hyperplanes Σ_{n-1} . Thus, to obtain the class of
cardona-qft-methods_F03629	365	■ 0.995	$\left[X_{\Gamma^\vee}\right]$	$[X_{\Gamma^\vee}]$	Proof. Writing $[X_{\Gamma^\vee}]$ as
cardona-qft-methods_F03630	365	■ 0.989	$\left[X_{\Gamma^\vee}\backslash\Sigma\right]$	$[X_{\Gamma^\vee}\backslash\Sigma]$	and using the previous results for $[X_{\Gamma^\vee}\backslash\Sigma]$ and $[\mathcal{S}_n]$ we find the desired
cardona-qft-methods_F03631	365	■ 0.989	$\left[\mathcal{S}_n\right]$	$[\mathcal{S}_n]$	and using the previous results for $[X_{\Gamma^\vee}\backslash\Sigma]$ and $[\mathcal{S}_n]$ we find the desired
cardona-qft-methods_F03632	366	■ 1.000	$p(t)$	$p(t)$	for $p(t)$ and $q(t)$ non constant polynomials. Now, let $t_i = x$ be variable and
cardona-qft-methods_F03633	366	■ 1.000	$q(t)$	$q(t)$	for $p(t)$ and $q(t)$ non constant polynomials. Now, let $t_i = x$ be variable and
cardona-qft-methods_F03634	366	■ 1.000	$t_i=x$	$t_i = x$	for $p(t)$ and $q(t)$ non constant polynomials. Now, let $t_i = x$ be variable and
cardona-qft-methods_F03635	366	■ 1.000	$t_j, j \neq i$	$t_j, j \neq i$	fix all other $t_j, j \neq i$. Therefore $\Psi_\Gamma(x)$ is a linear polynomial, and hence
cardona-qft-methods_F03636	366	■ 1.000	$\Psi_\Gamma(x)$	$\Psi_\Gamma(x)$	fix all other $t_j, j \neq i$. Therefore $\Psi_\Gamma(x)$ is a linear polynomial, and hence
cardona-qft-methods_F03637	366	■ 1.000	$p(x)$	$p(x)$	either $p(x)$ or $q(x)$ is a constant for any choice of t_j . Without loss of gen-
cardona-qft-methods_F03638	366	■ 1.000	$q(x)$	$q(x)$	either $p(x)$ or $q(x)$ is a constant for any choice of t_j . Without loss of gen-
cardona-qft-methods_F03639	366	■ 1.000	t_j	t_j	either $p(x)$ or $q(x)$ is a constant for any choice of t_j . Without loss of gen-
cardona-qft-methods_F03640	367	■ 1.000	$P \cap Q=\emptyset, P \cup Q$	$P \cap Q = \emptyset, P \cup Q$	Since $P \cap Q = \emptyset, P \cup Q$ are all the variables appearing in Ψ_Γ , and no
cardona-qft-methods_F03641	367	■ 1.000	Ψ_Γ	Ψ_Γ	Since $P \cap Q = \emptyset, P \cup Q$ are all the variables appearing in Ψ_Γ , and no
cardona-qft-methods_F03642	368	■ 1.000	$\tilde{m}g$	$\tilde{m}g$	Let G be a real Lie group with Lie algebra $\tilde{m}g$. Given a unitary representation π
cardona-qft-methods_F03643	368	■ 0.960	$\mathbb{I} \pi$	$\mathbb{I} \pi$	of G , one obtains by differentiation a representation $\mathbb{I} D \pi$ of $\tilde{m}g$ by unbounded,
cardona-qft-methods_F03644	368	■ 1.000	$g \mapsto \langle \xi, \pi(g) \eta \rangle$	$g \mapsto \langle \xi, \pi(g) \eta \rangle$	of $\tilde{m}g$ by derivations on the algebra of representative functions $g \mapsto \langle \xi, \pi(g) \eta \rangle$,
cardona-qft-methods_F03645	368	■ 1.000	$M=G / H$	$M = G/H$	which are naturally defined on the homogeneous space $M = G/H$, where H is
cardona-qft-methods_F03646	368	■ 1.000	$\mathcal{E}(\tilde{m}g)$	$\mathcal{E}(\tilde{m}g)$	universal enveloping algebra $\mathcal{E}(\tilde{m}g)$ to generalize this to representations which
cardona-qft-methods_F03647	368	■ 1.000	$\mathbb{C}[G]$	$\mathbb{C}[G]$	that the <i>group algebra</i> $\mathbb{C}[G]$ is the involutive algebra consisting of finite,
cardona-qft-methods_F03648	368	■ 1.000	$g \otimes h \mapsto gh$	$g \otimes h \mapsto gh$	obtained by linearly extending the maps $g \otimes h \mapsto gh$ and $g \mapsto g^{-1}$, respec-
cardona-qft-methods_F03649	368	■ 1.000	$g \mapsto g^{-1}$	$g \mapsto g^{-1}$	obtained by linearly extending the maps $g \otimes h \mapsto gh$ and $g \mapsto g^{-1}$, respec-
cardona-qft-methods_F03650	368	■ 1.000	$\bar{g}=g$	$\bar{g} = g$	extending $\bar{g} = g$. The infinitesimal version of $\mathbb{C}[G]$ is the <i>universal en-</i>
cardona-qft-methods_F03651	368	■ 1.000	$\mathcal{E}=\mathcal{E}(\tilde{m}g)$	$\mathcal{E} = \mathcal{E}(\tilde{m}g)$	<i>veloping algebra</i> $\mathcal{E} = \mathcal{E}(\tilde{m}g)$, defined as the quotient of the complex tensor

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cardona-qft-methods_ F03652	369	■ 1.000	$\mathcal{T}(\tilde{m}g)=\left(\bigoplus_{n\in\mathbb{N}}\tilde{m}g^{\otimes n}\right)\otimes\mathbb{C}$	$\mathcal{T}(\tilde{m}g)=\left(\bigoplus_{n\in\mathbb{N}}\tilde{m}g^{\otimes n}\right)\otimes\mathbb{C}$	algebra $\mathcal{T}(\tilde{m}g)=\left(\bigoplus_{n\in\mathbb{N}}\tilde{m}g^{\otimes n}\right)\otimes\mathbb{C}$ by the ideal generated by the elements
cardona-qft-methods_ F03653	369	■ 1.000	$\left(X_1\cdots X_n\right)^{\dagger}=(-1)^nX_n\cdots X_1$	$(X_1\cdots X_n)^{\dagger}=(-1)^nX_n\cdots X_1$	by linear extension from $(X_1\cdots X_n)^{\dagger}=(-1)^nX_n\cdots X_1$, and anti-linear
cardona-qft-methods_ F03654	369	■ 1.000	$\overline{X_1\cdots X_n}=X_1\cdots X_n$	$\overline{X_1\cdots X_n}=X_1\cdots X_n$	extension from $\overline{X_1\cdots X_n}=X_1\cdots X_n$, respectively. The subalgebra of real
cardona-qft-methods_ F03655	369	■ 1.000	$\mathcal{E}_{\mathbb{R}}(\tilde{m}g)$	$\mathcal{E}_{\mathbb{R}}=\mathcal{E}_{\mathbb{R}}(\tilde{m}g)$	elements with respect to this conjugation will be denoted by $\mathcal{E}_{\mathbb{R}}=\mathcal{E}_{\mathbb{R}}(\tilde{m}g)$.
cardona-qft-methods_ F03656	369	■ 1.000	$\xi\in\mathcal{H}$	$\xi\in\mathcal{H}$	$\mathcal{H}$. Given $g\in G$ and $\xi\in\mathcal{H}$, it is sometimes convenient to write $g\xi$ instead
cardona-qft-methods_ F03657	369	■ 1.000	$g\xi$	$g\xi$	$\mathcal{H}$. Given $g\in G$ and $\xi\in\mathcal{H}$, it is sometimes convenient to write $g\xi$ instead
cardona-qft-methods_ F03658	369	■ 1.000	$\pi(g)\xi$	$\pi(g)\xi$	of $\pi(g)\xi$, and consequently we can also think of $\mathcal{H}$ as a $\mathbb{C}[G]$ -module. In
cardona-qft-methods_ F03659	369	■ 1.000	$C(G)$	$C(G)$	We want to understand $\mathcal{H}$ as a submodule of $C(G)$, which is naturally
cardona-qft-methods_ F03660	369	■ 1.000	$\nu\in\mathcal{H}$	$\nu\in\mathcal{H}$	In order to do so, suppose that there is a distinguished cyclic vector $\nu\in\mathcal{H}$,
cardona-qft-methods_ F03661	369	■ 0.693	$\mathcal{H}=\overline{\mathbb{C}[G]\nu}$	$\mathcal{H}=\overline{\mathbb{C}[G]\nu}$	i.e. $\mathcal{H}=\overline{\mathbb{C}[G]\nu}$, and an anti-unitary operator $J:\mathcal{H}\rightarrow\mathcal{H}$ whose action
cardona-qft-methods_ F03662	369	■ 1.000	$ge_{\xi}=e_{g\xi}$	$ge_{\xi}=e_{g\xi}$	Since $ge_{\xi}=e_{g\xi}$, we see that $\xi\mapsto e_{\xi}$ exposes $\mathcal{H}$ as a submodule of $C(G)$.
cardona-qft-methods_ F03663	369	■ 1.000	$\xi\mapsto e_{\xi}$	$\xi\mapsto e_{\xi}$	Since $ge_{\xi}=e_{g\xi}$, we see that $\xi\mapsto e_{\xi}$ exposes $\mathcal{H}$ as a submodule of $C(G)$.
cardona-qft-methods_ F03664	369	■ 1.000	$g\mapsto\langle g\nu,\xi\rangle$	$g\mapsto\langle g\nu,\xi\rangle$	the anti-holomorphic “representative functions” $g\mapsto\langle g\nu,\xi\rangle$. Instead of us-
cardona-qft-methods_ F03665	369	■ 0.996	$\{e_{\xi}\mid\xi\in\mathcal{H}\}$	$\{e_{\xi}\mid\xi\in\mathcal{H}\}$	Consider, then, the subalgebra of $C(G)$ generated by $\{e_{\xi}\mid\xi\in\mathcal{H}\}$. It
cardona-qft-methods_ F03666	369	■ 1.000	e_{ξ}	e_{ξ}	does not separate points of G , because each e_{ξ} is constant along the cosets
cardona-qft-methods_ F03667	369	■ 1.000	gH	gH	gH , where $H\subseteq G$ is the stabilizer of ν . Thus, it is essentially an algebra of
cardona-qft-methods_ F03668	370	■ 1.000	$\mathcal{H}^{\infty}=\mathcal{H}^{\infty}(\pi)\subseteq\mathcal{H}$	$\mathcal{H}^{\infty}=\mathcal{H}^{\infty}(\pi)\subseteq\mathcal{H}$	Now, let $\mathcal{H}^{\infty}=\mathcal{H}^{\infty}(\pi)\subseteq\mathcal{H}$ be the subspace of <i>smooth vectors</i> for π ,
cardona-qft-methods_ F03669	370	■ 0.964	$g\in G\mapsto\pi(g)\xi\in\mathcal{H}$	$g\in G\mapsto\pi(g)\xi\in\mathcal{H}$	i.e. those $\xi\in\mathcal{H}$ such that the map $g\in G\mapsto\pi(g)\xi\in\mathcal{H}$ is smooth. ^a Then,
cardona-qft-methods_ F03670	370	■ 1.000	$X\in\tilde{m}g$	$X\in\tilde{m}g$	given $X\in\tilde{m}g$ and $\xi\in\mathcal{H}^{\infty}$, we can define
cardona-qft-methods_ F03671	370	■ 1.000	$\xi\in\mathcal{H}^{\infty}$	$\xi\in\mathcal{H}^{\infty}$	given $X\in\tilde{m}g$ and $\xi\in\mathcal{H}^{\infty}$, we can define
cardona-qft-methods_ F03672	370	■ 1.000	$\exp:\tilde{m}g\rightarrow G$	$\exp:\tilde{m}g\rightarrow G$	where $\exp:\tilde{m}g\rightarrow G$ is the exponential map. This way, the unitary repre-
cardona-qft-methods_ F03673	370	■ 1.000	$\mathbb{D}\pi$	$\mathbb{D}\pi$	sentation π of G gives rise to a representation $\mathbb{D}\pi$ of $\tilde{m}g$ by unbounded,
cardona-qft-methods_ F03674	370	■ 1.000	$\mathcal{H}^{\infty}$	$\mathcal{H}^{\infty}$	skew-adjoint operators, thus naturally turning $\mathcal{H}^{\infty}$ into an $\mathcal{E}$ -module. We
cardona-qft-methods_ F03675	370	■ 1.000	$\{e_{\xi}\mid\xi\in\mathcal{H}^{\infty}\}$	$\{e_{\xi}\mid\xi\in\mathcal{H}^{\infty}\}$	it with $\{e_{\xi}\mid\xi\in\mathcal{H}^{\infty}\}$. Now, the action of $\tilde{m}g$ is by derivation along the
cardona-qft-methods_ F03676	370	■ 1.000	$gH\in M\mapsto\epsilon^{tX}gH\in M$	$gH\in M\mapsto\epsilon^{tX}gH\in M$	vector fields which generate the flows $gH\in M\mapsto\epsilon^{tX}gH\in M$.
cardona-qft-methods_ F03677	370	■ 1.000	$X\mapsto\mathbb{D}\pi(X) _{\mathcal{F}}$	$X\mapsto\mathbb{D}\pi(X) _{\mathcal{F}}$	$X\mapsto\mathbb{D}\pi(X) _{\mathcal{F}}$, where π is a unitary representation of G , and $\mathcal{F}\subseteq\mathcal{H}^{\infty}(\pi)$

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cardona-qft-methods_ F03678	370	■ 1.000	$\mathcal{F} \subseteq \mathcal{H}^{\infty}(\pi)$	$\mathcal{F} \subseteq \mathcal{H}^{\infty}(\pi)$	$X \mapsto \mathbb{D}\pi(X) _{\mathcal{F}}$, where π is a unitary representation of G , and $\mathcal{F} \subseteq \mathcal{H}^{\infty}(\pi)$
cardona-qft-methods_ F03679	370	■ 0.998	$\left.\mathbb{1} \pi(X)\right _{\mathcal{F}}$	$\mathbb{1}\pi(X) _{\mathcal{F}}$	is “telling enough,” in the sense that the closure of $\mathbb{D}\pi(X) _{\mathcal{F}}$ coincides with
cardona-qft-methods_ F03680	370	■ 1.000	$\mathbb{D} \pi(X)$	$\mathbb{D}\pi(X)$	the closure of $\mathbb{D}\pi(X)$, for each $X \in \tilde{m}g$.
cardona-qft-methods_ F03681	370	■ 1.000	$\left.\mathbb{D} \pi(X)\right _{\mathcal{F}}$	$\mathbb{D}\pi(X) _{\mathcal{F}}$	is densely defined; thus, the operators $\mathbb{D}\pi(X)$ and $\mathbb{D}\pi(X) _{\mathcal{F}}$ above are
cardona-qft-methods_ F03682	370	■ 1.000	$\{ \}^{\mathrm{a}} \mathcal{H}^{\infty}$	${}^{\mathrm{a}}\mathcal{H}^{\infty}$	${}^{\mathrm{a}}\mathcal{H}^{\infty}$ is also called the <i>Gårding space</i> of π . As Harish-Chandra pointed out, ² the space of
cardona-qft-methods_ F03683	371	■ 1.000	$\mathcal{L}^{\dagger}(\mathcal{F})$	$\mathcal{L}^{\dagger}(\mathcal{F})$	Definition 1.2. We write $\mathcal{L}^{\dagger}(\mathcal{F})$ for the algebra of linear operators A :
cardona-qft-methods_ F03684	371	■ 1.000	$\mathcal{F} \rightarrow \mathcal{F}$	$\mathcal{F} \rightarrow \mathcal{F}$	$\mathcal{F} \rightarrow \mathcal{F}$ such that $\mathcal{F} \subseteq \mathrm{dom}(A^*)$. It comes equipped with the involution
cardona-qft-methods_ F03685	371	■ 1.000	$\mathcal{F} \subseteq \mathrm{dom}(A^*)$	$\mathcal{F} \subseteq \mathrm{dom}(A^*)$	$\mathcal{F} \rightarrow \mathcal{F}$ such that $\mathcal{F} \subseteq \mathrm{dom}(A^*)$. It comes equipped with the involution
cardona-qft-methods_ F03686	371	■ 1.000	$A^{\dagger} = A^* _{\mathcal{F}}$	$A^{\dagger} = A^* _{\mathcal{F}}$	$A^{\dagger} = A^* _{\mathcal{F}}$.
cardona-qft-methods_ F03687	371	■ 1.000	$\mathcal{C}^*(\mathcal{F})$	$\mathcal{C}^*(\mathcal{F})$	Definition 1.3. We write $\mathcal{C}^*(\mathcal{F})$ for the set of closed operators A on $\mathcal{H}$
cardona-qft-methods_ F03688	371	■ 1.000	$A _{\mathcal{F}} \in \mathcal{L}^{\dagger}(\mathcal{F})$	$A _{\mathcal{F}} \in \mathcal{L}^{\dagger}(\mathcal{F})$	which satisfy $A _{\mathcal{F}} \in \mathcal{L}^{\dagger}(\mathcal{F})$.
cardona-qft-methods_ F03689	371	■ 1.000	$\pi : \mathcal{A} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$	$\pi : \mathcal{A} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$	gebra $\mathcal{A}$ is an involutive algebra morphism $\pi : \mathcal{A} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$, where $\mathcal{F}$ sits
cardona-qft-methods_ F03690	371	■ 1.000	$\nu \in \mathcal{F}$	$\nu \in \mathcal{F}$	a so-called <i>cyclic vector</i> $\nu \in \mathcal{F}$ such that $\mathcal{F} = \mathcal{A}\nu$.
cardona-qft-methods_ F03691	371	■ 1.000	$\mathcal{F} = \mathcal{A}\nu$	$\mathcal{F} = \mathcal{A}\nu$	a so-called <i>cyclic vector</i> $\nu \in \mathcal{F}$ such that $\mathcal{F} = \mathcal{A}\nu$.
cardona-qft-methods_ F03692	371	■ 1.000	$\mathcal{A}^{\dagger}A$	$A^{\dagger}A$	$\mathcal{A}$: the one with positive cone generated by the elements of the form $A^{\dagger}A$.
cardona-qft-methods_ F03693	371	■ 1.000	$\mathcal{A}^{\dagger}$	$\mathcal{A}^{\dagger}$	written $\mathcal{A}^{\dagger}$: an $\omega \in \mathcal{A}^{\dagger}$ is positive if, and only if, it is positive on positive
cardona-qft-methods_ F03694	371	■ 1.000	$\omega \in \mathcal{A}^{\dagger}$	$\omega \in \mathcal{A}^{\dagger}$	written $\mathcal{A}^{\dagger}$: an $\omega \in \mathcal{A}^{\dagger}$ is positive if, and only if, it is positive on positive
cardona-qft-methods_ F03695	371	■ 1.000	$\ \nu\ =1$	$\ \nu\ =1$	any vector $\nu \in \mathcal{F}$ with $\ \nu\ =1$ by the formula
cardona-qft-methods_ F03696	371	■ 0.999	$\pi _{\mathcal{A}\nu}$	$\pi _{\mathcal{A}\nu}$	The GNS construction allows for a recovery of both $\pi _{\mathcal{A}\nu}$ (modulo unitary
cardona-qft-methods_ F03697	372	■ 1.000	$\mathcal{I} = \{A \in \mathcal{A} \mid \omega(A^{\dagger}A) = 0\}$	$\mathcal{I} = \{A \in \mathcal{A} \mid \omega(A^{\dagger}A) = 0\}$	Let ω be a state of $\mathcal{A}$ and consider the set $\mathcal{I} = \{A \in \mathcal{A} \mid \omega(A^{\dagger}A) = 0\}$.
cardona-qft-methods_ F03698	372	■ 1.000	$\mathcal{I}$	$\mathcal{I}$	Using the Cauchy-Schwarz inequality it is easily seen that $\mathcal{I}$ is a left ideal.
cardona-qft-methods_ F03699	372	■ 1.000	$\mathcal{F} = \mathcal{A}/\mathcal{I}$	$\mathcal{F} = \mathcal{A}/\mathcal{I}$	The quotient $\mathcal{F} = \mathcal{A}/\mathcal{I}$ is densely embedded in its Hilbert space completion
cardona-qft-methods_ F03700	372	■ 1.000	$\nu = [1] \in \mathcal{F}$	$\nu = [1] \in \mathcal{F}$	of $\mathcal{F}$. It admits the cyclic vector $\nu = [1] \in \mathcal{F}$.
cardona-qft-methods_ F03701	372	■ 1.000	$\pi : \mathcal{E} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$	$\pi : \mathcal{E} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$	haviour, in the following sense. Consider a representation $\pi : \mathcal{E} \rightarrow \mathcal{L}^{\dagger}(\mathcal{F})$.
cardona-qft-methods_ F03702	372	■ 0.903	$E \in \mathcal{E}$	$E \in \mathcal{E}$	Denote the closure of (the image under π of) $E \in \mathcal{E}$ by $cl(E)$. Then, inte-
cardona-qft-methods_ F03703	372	■ 0.903	$\mathrm{cl}(E)$	$\mathrm{cl}(E)$	Denote the closure of (the image under π of) $E \in \mathcal{E}$ by $cl(E)$. Then, inte-
cardona-qft-methods_ F03704	372	■ 0.611	$cl(E+F) = cl(E) + cl(F)$	$cl(E+F) = cl(E) + cl(F)$	(i) $cl(E+F) = cl(E) + cl(F)$,

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cardona-qft-methods_ F03705	372	■ 0.887	$\operatorname{cl}\{E\}F=\operatorname{cl}\{E\}\operatorname{cl}\{F\}$	$\operatorname{cl}(EF) = \operatorname{cl}(E) \operatorname{cl}(F)$	(ii) $cl(EF) = cl(E)cl(F)$,
cardona-qft-methods_ F03706	372	■ 0.644	$\operatorname{cl}\{E\}\left(E^{\dagger}\right)=\operatorname{cl}\{E\}E^{*}$	$\operatorname{cl}\left(E^{\dagger}\right)=\operatorname{cl}(E)^{*}$	(iii) $cl(E^{\dagger}) = cl(E)^{*}$,
cardona-qft-methods_ F03707	372	■ 0.963	$c\left(1+E^{\dagger} E\right)$	$cl\left(1+E^{\dagger} E\right)$	actually implies the first two). Hence, integrability implies that $cl(1+E^{\dagger}E)$
cardona-qft-methods_ F03708	372	■ 1.000	$\mathcal{E} \backslash \mathcal{S}^{-1}$	$\mathcal{E} \mathcal{S}^{-1}$	tion $\mathcal{E} \mathcal{S}^{-1}$.
cardona-qft-methods_ F03709	373	■ 1.000	$\tilde{\mathcal{E}} \supseteq \mathcal{E}$	$\tilde{\mathcal{E}} \supseteq \mathcal{E}$	with $\tilde{\mathcal{E}} \supseteq \mathcal{E}$ and $\tilde{\mathcal{F}} \supseteq \mathcal{F}$. The image of $\tilde{\mathcal{E}}$ under the upper arrow must be
cardona-qft-methods_ F03710	373	■ 1.000	$\tilde{\mathcal{F}} \supseteq \mathcal{F}$	$\tilde{\mathcal{F}} \supseteq \mathcal{F}$	with $\tilde{\mathcal{E}} \supseteq \mathcal{E}$ and $\tilde{\mathcal{F}} \supseteq \mathcal{F}$. The image of $\tilde{\mathcal{E}}$ under the upper arrow must be
cardona-qft-methods_ F03711	373	■ 1.000	$\tilde{\mathcal{E}}$	$\tilde{\mathcal{E}}$	with $\tilde{\mathcal{E}} \supseteq \mathcal{E}$ and $\tilde{\mathcal{F}} \supseteq \mathcal{F}$. The image of $\tilde{\mathcal{E}}$ under the upper arrow must be
cardona-qft-methods_ F03712	373	■ 1.000	$\mathcal{C}^{*}(\tilde{\mathcal{F}})$	$\mathcal{C}^{*}(\tilde{\mathcal{F}})$	a subset of $\mathcal{C}^{*}(\tilde{\mathcal{F}})$ which is an algebra with respect to the strong opera-
cardona-qft-methods_ F03713	373	■ 1.000	$\tilde{\mathcal{F}} \subseteq \mathcal{H}$	$\tilde{\mathcal{F}} \subseteq \mathcal{H}$	must commute. We do not necessarily require that $\tilde{\mathcal{F}} \subseteq \mathcal{H}$. Again under
cardona-qft-methods_ F03714	373	■ 0.943	$\tilde{\mathcal{E}}=\mathcal{E}$	$\tilde{\mathcal{E}}=\mathcal{E}$	resentations always admit an integrable extension with $\tilde{\mathcal{E}}=\mathcal{E}$, see. ⁸
cardona-qft-methods_ F03715	373	■ 1.000	$\mathcal{E}^{\dagger}$	$\mathcal{E}^{\dagger}$	2.1. Algebraic structure of $\mathcal{E}^{\dagger}$
cardona-qft-methods_ F03716	373	■ 1.000	$\Delta: \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{E}$	$\Delta: \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{E}$	to products, so that a coproduct is a linear function $\Delta: \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{E}$ which
cardona-qft-methods_ F03717	373	■ 1.000	$\mathcal{T}(\tilde{m} g)$	$\mathcal{T}(\tilde{m} g)$	tensor algebra $\mathcal{T}(\tilde{m} g)$, by extending the linear map
cardona-qft-methods_ F03718	373	■ 1.000	$\mathcal{I} \subseteq \mathcal{T}(\tilde{m} g)$	$\mathcal{I} \subseteq \mathcal{T}(\tilde{m} g)$	universal property of tensor algebras). Now, the ideal $\mathcal{I} \subseteq \mathcal{T}(\tilde{m} g)$ generated
cardona-qft-methods_ F03719	373	■ 1.000	$X Y-Y X-[X, Y]$	$X Y-Y X-[X, Y]$	by the elements $X Y-Y X-[X, Y]$, with $X, Y \in \tilde{m} g$, is also a <i>coideal</i> , i.e.
cardona-qft-methods_ F03720	373	■ 1.000	$X, Y \in \tilde{m} g$	$X, Y \in \tilde{m} g$	by the elements $X Y-Y X-[X, Y]$, with $X, Y \in \tilde{m} g$, is also a <i>coideal</i> , i.e.
cardona-qft-methods_ F03721	373	■ 1.000	$\mathcal{E}=\mathcal{T}(\tilde{m} g) / \mathcal{I}$	$\mathcal{E}=\mathcal{T}(\tilde{m} g) / \mathcal{I}$	Thus, it can be readily seen that Δ passes to the quotient $\mathcal{E}=\mathcal{T}(\tilde{m} g) / \mathcal{I}$.
cardona-qft-methods_ F03722	374	■ 1.000	$\left(X_1, \ldots, X_n\right)$	$\left(X_1, \ldots, X_n\right)$	$\left(X_1, \ldots, X_n\right)$ of $\tilde{m} g$ and use the multi-index notation $X^{\alpha}=X_1^{\alpha_1} \cdots X_n^{\alpha_n}$.
cardona-qft-methods_ F03723	374	■ 1.000	$X^{\alpha}=X_1^{\alpha_1} \cdots X_n^{\alpha_n}$	$X^{\alpha}=X_1^{\alpha_1} \cdots X_n^{\alpha_n}$	$\left(X_1, \ldots, X_n\right)$ of $\tilde{m} g$ and use the multi-index notation $X^{\alpha}=X_1^{\alpha_1} \cdots X_n^{\alpha_n}$.
cardona-qft-methods_ F03724	374	■ 0.998	$\binom{\alpha}{\beta}=\frac{\alpha !}{\beta !(\alpha-\beta) !}=\binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_d}{\beta_d}$	$\binom{\alpha}{\beta}=\frac{\alpha !}{\beta !(\alpha-\beta) !}=\binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_d}{\beta_d}$	where $\binom{\alpha}{\beta}=\frac{\alpha !}{\beta !(\alpha-\beta) !}=\binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_d}{\beta_d}$.
cardona-qft-methods_ F03725	374	■ 1.000	$E \otimes F \mapsto F \otimes E$	$E \otimes F \mapsto F \otimes E$	where τ is obtained by linear extension from $E \otimes F \mapsto F \otimes E$, with $E, F \in \mathcal{E}$.
cardona-qft-methods_ F03726	374	■ 1.000	$E, F \in \mathcal{E}$	$E, F \in \mathcal{E}$	where τ is obtained by linear extension from $E \otimes F \mapsto F \otimes E$, with $E, F \in \mathcal{E}$.
cardona-qft-methods_ F03727	374	■ 1.000	$\tilde{m} g \rightarrow \operatorname{der}\left(\mathcal{E}^{\dagger}\right)$	$\tilde{m} g \rightarrow \operatorname{der}\left(\mathcal{E}^{\dagger}\right)$	Prop 2.1. The action (1) defines a morphism $\tilde{m} g \rightarrow \operatorname{der}\left(\mathcal{E}^{\dagger}\right)$, where
cardona-qft-methods_ F03728	374	■ 0.999	$\operatorname{der}\left(\mathcal{E}^{\dagger}\right)$	$\operatorname{der}\left(\mathcal{E}^{\dagger}\right)$	$\operatorname{der}\left(\mathcal{E}^{\dagger}\right)$ stands for the Lie algebra of linear maps $\delta: \mathcal{E}^{\dagger} \rightarrow \mathcal{E}^{\dagger}$ such that
cardona-qft-methods_ F03729	374	■ 0.999	$\delta: \mathcal{E}^{\dagger} \rightarrow \mathcal{E}^{\dagger}$	$\delta: \mathcal{E}^{\dagger} \rightarrow \mathcal{E}^{\dagger}$	$\operatorname{der}\left(\mathcal{E}^{\dagger}\right)$ stands for the Lie algebra of linear maps $\delta: \mathcal{E}^{\dagger} \rightarrow \mathcal{E}^{\dagger}$ such that
cardona-qft-methods_ F03730	374	■ 1.000	$a, b \in \mathcal{E}^{\dagger}$	$a, b \in \mathcal{E}^{\dagger}$	Proof. Indeed, let $X \in \tilde{m} g$ and $a, b \in \mathcal{E}^{\dagger}$. One has that

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cardona-qft-methods_ F03731	375	■ 1.000	$X_{\{1\}}, \ldots, X_{\{n\}}$	$X_1, \dots, X_n$	Prop 2.2. Let $X_1, \dots, X_n$ be a basis of $\tilde{m}g$. One has that $\mathcal{E}^\dagger \cong$
cardona-qft-methods_ F03732	375	■ 1.000	$\mathcal{E}^\dagger \cong$	$\mathcal{E}^\dagger \cong$	Prop 2.2. Let $X_1, \dots, X_n$ be a basis of $\tilde{m}g$. One has that $\mathcal{E}^\dagger \cong$
cardona-qft-methods_ F03733	375	■ 1.000	$\mathbb{C}[\llbracket x_{\{1\}}, \ldots, x_{\{n\}} \rrbracket]$	$\mathbb{C}[[x_1, \dots, x_n]]$	$\mathbb{C}[[x_1, \dots, x_n]]$, with algebra isomorphism given by
cardona-qft-methods_ F03734	375	■ 1.000	$1 \in \mathcal{E}$	$1 \in \mathcal{E}$	Thus, evaluation at $1 \in \mathcal{E}$ is a character of $\mathcal{E}^\dagger$, which we will denote by χ_0 ,
cardona-qft-methods_ F03735	375	■ 1.000	$\chi_{\{0\}}$	χ_0	Thus, evaluation at $1 \in \mathcal{E}$ is a character of $\mathcal{E}^\dagger$, which we will denote by χ_0 ,
cardona-qft-methods_ F03736	375	■ 1.000	$a \in \mathcal{E}^\dagger$	$a \in \mathcal{E}^\dagger$	Prop 2.3. For all $a \in \mathcal{E}^\dagger$ and $E \in \mathcal{E}$, $(Ea)^\dagger = \overline{Ea}^\dagger$.
cardona-qft-methods_ F03737	375	■ 1.000	$E \in \mathcal{E}, (Ea)^\dagger = \overline{Ea}^\dagger$	$E \in \mathcal{E}, (Ea)^\dagger = \overline{Ea}^\dagger$	Prop 2.3. For all $a \in \mathcal{E}^\dagger$ and $E \in \mathcal{E}$, $(Ea)^\dagger = \overline{Ea}^\dagger$.
cardona-qft-methods_ F03738	375	■ 1.000	$E, F \in \mathcal{E}_{\mathbb{R}}$	$E, F \in \mathcal{E}_{\mathbb{R}}$	Proof. For $E, F \in \mathcal{E}_{\mathbb{R}}$, one has $(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$, and the
cardona-qft-methods_ F03739	375	■ 1.000	$(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$	$(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$	Proof. For $E, F \in \mathcal{E}_{\mathbb{R}}$, one has $(Ea)^\dagger(F) = \overline{a(E^\dagger F)} = Ea^\dagger(F)$, and the
cardona-qft-methods_ F03740	376	■ 1.000	$J\nu = \nu$	$J\nu = \nu$	$J : \mathcal{H} \rightarrow \mathcal{H}$ commuting with π and such that $J\nu = \nu$, even though this
cardona-qft-methods_ F03741	376	■ 1.000	$\{e_\xi \mid \xi \in \mathcal{F}\}$	$\{e_\xi \mid \xi \in \mathcal{F}\}$	and let $\mathcal{A}$ be the involutive subalgebra of $\mathcal{E}^\dagger$ generated by $\{e_\xi \mid \xi \in \mathcal{F}\}$. We
cardona-qft-methods_ F03742	376	■ 1.000	$\chi_{\{0\}} \in M$	$\chi_0 \in M$	Note that $\chi_0 \in M$.
cardona-qft-methods_ F03743	376	■ 1.000	$\xi \in \mathcal{H} \setminus \mathcal{F}$	$\xi \in \mathcal{H} \setminus \mathcal{F}$	regard the functions e_ξ with $\xi \in \mathcal{H} \setminus \mathcal{F}$ as not belonging to $\mathcal{A}$, for they are
cardona-qft-methods_ F03744	376	■ 1.000	$E \in \mathcal{E}, Ee_\xi$	$E \in \mathcal{E}, Ee_\xi$	not smooth in the sense that, given $E \in \mathcal{E}$, Ee_ξ might not be of the form
cardona-qft-methods_ F03745	376	■ 1.000	e_η	e_η	e_η , for any $\eta \in \mathcal{H}$.
cardona-qft-methods_ F03746	376	■ 1.000	$\eta \in \mathcal{H}$	$\eta \in \mathcal{H}$	e_η , for any $\eta \in \mathcal{H}$.
cardona-qft-methods_ F03747	376	■ 1.000	$Ee_\xi = e_{E\xi}$	$Ee_\xi = e_{E\xi}$	Prop 2.4. One has that $Ee_\xi = e_{E\xi}$, for all $\xi \in \mathcal{F}$ and $E \in \mathcal{E}$. In particular,
cardona-qft-methods_ F03748	376	■ 1.000	$\xi \in \mathcal{F}$	$\xi \in \mathcal{F}$	Prop 2.4. One has that $Ee_\xi = e_{E\xi}$, for all $\xi \in \mathcal{F}$ and $E \in \mathcal{E}$. In particular,
cardona-qft-methods_ F03749	376	■ 1.000	$F \in \mathcal{E}$	$F \in \mathcal{E}$	Proof. Indeed, given $F \in \mathcal{E}$,
cardona-qft-methods_ F03750	376	■ 1.000	$\omega \in \mathcal{E}^\dagger$	$\omega \in \mathcal{E}^\dagger$	under the action of $\mathcal{E}$. Thus, the fundamental object is $\omega \in \mathcal{E}^\dagger$, and the
cardona-qft-methods_ F03751	377	■ 1.000	$\mathcal{E}^\dagger \cong \mathbb{C}[\llbracket x_{\{1\}}, \ldots, x_{\{n\}} \rrbracket]$	$\mathcal{E}^\dagger \cong \mathbb{C}[[x_1, \dots, x_n]]$	one minimal prime ideal), as follows from the fact that $\mathcal{E}^\dagger \cong \mathbb{C}[[x_1, \dots, x_n]]$
cardona-qft-methods_ F03752	377	■ 1.000	$\chi_{\{0\}}(a) = a(1)$	$\chi_0(a) = a(1)$	where $\chi_0(a) = a(1)$. Here, we have only the derivation X , which will be,
cardona-qft-methods_ F03753	377	■ 1.000	$\omega \cong \cos(x)$	$\omega \cong \cos(x)$	and so on. Thus, $\omega \cong \cos(x)$, as one might have anticipated, and $\mathcal{A}$ is the
cardona-qft-methods_ F03754	377	■ 1.000	$X\omega$	$X\omega$	algebra of trigonometric polynomials. Writing u for ω and v for $X\omega$, we see
cardona-qft-methods_ F03755	377	■ 1.000	$\mathcal{A} \cong \mathbb{C}[u, v] / (u^2 + v^2 - 1)$	$\mathcal{A} \cong \mathbb{C}[u, v] / (u^2 + v^2 - 1)$	that $\mathcal{A} \cong \mathbb{C}[u, v] / (u^2 + v^2 - 1)$, and M is a circle.